Skill Navora

TRACK 1

Advanced Excel

From the Excel window to dashboards: formatting, 100+ functions, lookups, pivot tables, advanced charts, Power Query, Power Pivot and AI in Excel.

16 lessons | Version 2026.10 | Updated October 2026

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ADVANCED EXCEL · LESSON 1

Getting Started with Excel for Data Analysis

Learn the Excel window, the words analysts use every day, how to lay out data so it can be analysed, the five data types Excel understands, and the shortcuts that save hours.

Live classes 1–2 · ✦ Excel for Microsoft 365, 2026 · Beginner · ⏱ 90 min

By the end of this lesson you can

- Name every part of the Excel window and explain what it does
- Use the core vocabulary (workbook, worksheet, cell, range, reference, formula, function) correctly
- Recognise structured data and fix a badly laid-out sheet
- Identify the five data types Excel stores, and spot text that only looks like a number or date
- Navigate and select data quickly with keyboard shortcuts

Why Excel still matters in 2026

Even with Python, SQL, Power BI and AI assistants, Excel is still the first tool almost every analyst opens. It is where business teams keep budgets, targets, trackers and exports from other systems. If you can clean, analyse and present data in Excel quickly, you can add value from day one in any data role.

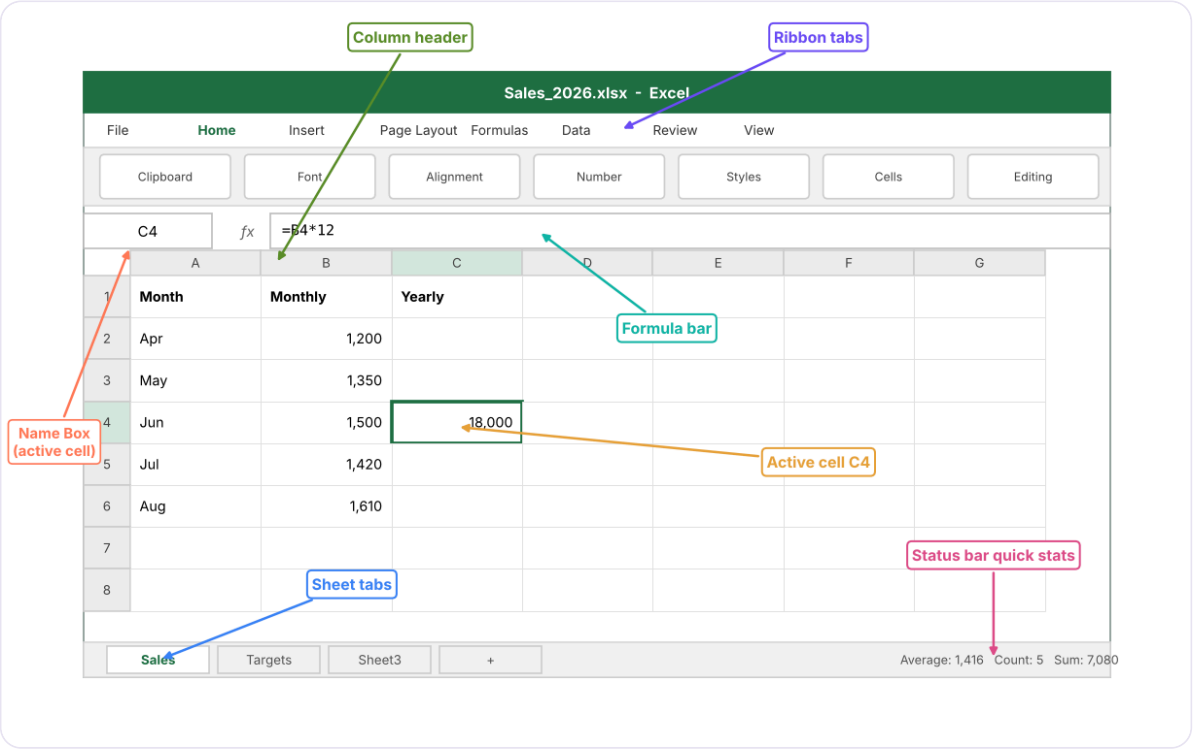
Excel is also the easiest place to *learn* data thinking. Every idea you meet here comes back later:

Idea in Excel	Same idea later in the course
A table with one header row	A SQL table, a pandas DataFrame, a Power BI table
SUMIFS, COUNTIFS	`GROUP BY` in SQL, `groupby()` in pandas, DAX measures
VLOOKUP / XLOOKUP	SQL `JOIN`, pandas `merge()`, Power BI relationships
Pivot table	SQL aggregation, `pivot_table()` in pandas, Power BI matrix
Conditional formatting	Highlighting in dashboards

LIVE CLASS **What was covered in class**

Live class 1 was the programme introduction. Live class 2 covered the basics of Excel, terminology, shortcuts, Excel properties, data types, data validation and formatting, and we started working on the first sheet. This lesson covers the first half; data validation and formatting continue in Lessons 2 and 3.

The Excel window, part by part



Month	Monthly	Yearly
Apr	1,200	
May	1,350	
Jun	1,500	18,000
Jul	1,420	
Aug	1,610	

The main parts of the Excel window. The active cell is C4, and the formula bar shows the formula inside it.

Part	What it does	Tip
Ribbon	Commands grouped into tabs: Home, Insert, Formulas, Data, Review, View	Press `Alt` to see a key letter on every command
Name Box	Shows the address of the active cell or the name of a selected range	Type `D500` and press Enter to jump straight there
Formula bar	Shows what is really inside the cell: a value or a formula	The cell shows the *result*; the bar shows the *recipe*
Column headers	Letters A, B, C … XFD (16,384 columns)	Double-click the border between headers to auto-fit width
Row headers	Numbers 1 to 1,048,576	Click a row number to select the whole row
Sheet tabs	Each tab is one worksheet inside the workbook	Right-click a tab to rename, colour, move or copy it
Status bar	Quick sum, average and count of the selected cells	Right-click it to add Min, Max and Numerical Count

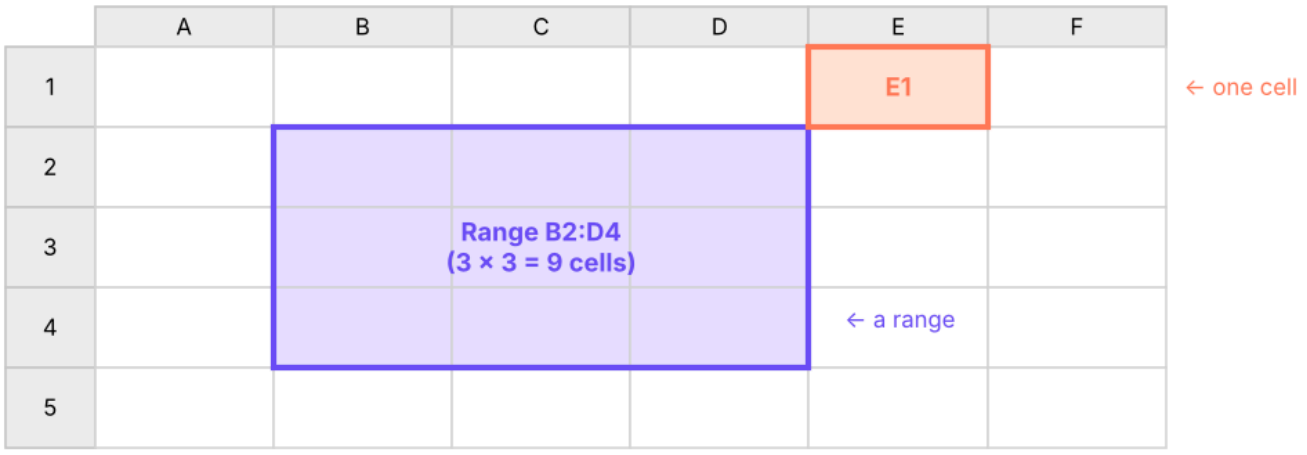
TIP **The status bar is a free calculator**

Select any numbers and look at the bottom-right corner. Excel shows the Average, Count and Sum without you writing a formula. It is the fastest way to sanity-check a total someone sends you.

The vocabulary every analyst uses

DEFINITION **Core terms**

Workbook: the file (`.xlsx`). One workbook holds many worksheets. **Worksheet (sheet)**: one grid of rows and columns inside a workbook. **Cell**: one box where a row and a column meet, for example `C4` . **Cell reference (address)**: column letter + row number, such as `C4` . Used inside formulas to point at cells. **Range**: a rectangle of cells written as `first:last` , for example `A1:C10` (3 columns × 10 rows = 30 cells). **Formula**: anything that starts with `=` and calculates a result, such as `=B4*12` . **Function**: a built-in, named formula that takes inputs (arguments), such as `=SUM(B2:B6)` . **Argument**: an input you give a function, separated by commas: `=ROUND(B2, 1)` has two arguments.



How cells, ranges and references relate. A range is always written from its top-left cell to its bottom-right cell.

Excel limits worth knowing

Item	Limit in Excel 365
Rows per sheet	1,048,576
Columns per sheet	16,384 (A to XFD)
Characters in one cell	32,767
Characters in a formula	8,192
Arguments in one function	255
Decimal precision	15 significant digits

WATCH OUT **The 15-digit trap**

Excel keeps only 15 significant digits. If you type a 16-digit card number or a long ID, the last digits turn into zeros. Store long IDs, phone numbers, PIN codes and account numbers as **text** (format the column as Text first, or start the entry with an apostrophe `'`).

Structured data: the single most important habit

Excel will do almost anything, but it can only *analyse* data that is laid out as a clean table. Analysts call this **structured** or **tidy** data.

STEP BY STEP **The six rules of a structured table**

1. **One header row** at the top, with a unique, short name for every column.
2. **One record per row**, for example one order, one employee or one day.
3. **One kind of information per column**: dates only in the date column, numbers only in the amount column.
4. **No blank rows or blank columns** inside the table.
5. **No merged cells** and no totals typed in the middle of the data.
6. **Raw values, not decorations**: write `Delhi`, not `Delhi ✔ (checked)`.

Here is a typical "report-style" sheet that looks fine to a human but is hard to analyse:

Before: a report, not a table

	A	B	C	D
1	**Sales Report – Q1**			
2	*Region: North*			
3	Name	Jan	Feb	Mar
4	Asha Rao	12,000	15,500	n/a
5	Ravi Verma	9,800		11,250
6				
7	*Region: South*			
8	Name	Jan	Feb	Mar
9	Meera Iyer	14,250	13,900	16,100
10	**TOTAL**	36,050	29,400	27,350

Problems: a title row above the headers, the region written as a heading instead of a column, a blank row, the header row repeated, numbers stored as text with commas, `n/a` mixed into a number column, and a total row inside the data.

The same information as a structured table:

After: a structured table (long format)

	A	B	C	D
1	**Region**	**Name**	**Month**	**Sales**
2	North	Asha Rao	Jan	12000
3	North	Asha Rao	Feb	15500
4	North	Ravi Verma	Jan	9800
5	North	Ravi Verma	Mar	11250
6	South	Meera Iyer	Jan	14250
7	South	Meera Iyer	Feb	13900
8	South	Meera Iyer	Mar	16100
9	**Total**			92800

```
D9   =SUM(D2:D8)
```

Now a pivot table, a SUMIFS, a chart or Power BI can use the data immediately. The total sits **below** the table, not inside it, and is calculated, not typed.

> **TIP** **Long format beats wide format**
>
> In the "before" sheet each month is a separate column (wide format). In the "after" sheet the month is a value in one `Month` column (long format). Long format is what pivot tables, SQL and Power BI expect. In Lesson 12 you will see how Power Query's **Unpivot** turns wide data into long data in one click.

The five data types Excel understands

Every cell holds exactly one type of value. Knowing the type explains most "my formula doesn't work" problems.

Type	Examples	Default alignment	Notes
Number	`42`, `3.14`, `-500`, `18%`	Right	Percentages and currency are numbers with a format
Text (string)	`Delhi`, `E101`, `'00451`	Left	Anything Excel cannot read as a number
Date / time	`27-04-2026`, `14:30`	Right	Stored as a serial number: days since 1 Jan 1900
Logical (Boolean)	`TRUE`, `FALSE`	Centre	Result of comparisons like `=B2>100`
Error	`#N/A`, `#DIV/0!`, `#VALUE!`	Centre	Tells you a formula has a problem

Excel's own functions can tell you the type of a value. Each formula in column B below checks the value in column A:

Checking data types with functions

	A	B	C	D
1	Value	ISNUMBER	ISTEXT	TYPE (1=number, 2=text, 4=logical)
2	42	TRUE	FALSE	1
3	Delhi	FALSE	TRUE	2
4	00451	FALSE	TRUE	2
5	27-04-2026	TRUE	FALSE	1
6	TRUE	FALSE	FALSE	4
7	1500	FALSE	TRUE	2

```
B2   =ISNUMBER(A2)
C2   =ISTEXT(A2)
D2   =TYPE(A2)
```

Each pattern is copied down/across to the other cells in its column.

Notice two important results:

- The **date** in row 5 is a number. Excel stores 27-04-2026 as the serial number 46139, which is why you can subtract one date from another to get days.
- `'1500` in row 7 is **text**, even though it looks like a number. `SUM` ignores it. Imported data often has "numbers stored as text"; Excel flags them with a small green triangle.

Dates are numbers underneath

	A	B	C
1	Date	Same cell shown as a number	Days until 31-Dec-2026
2	27-04-2026	46139	248
3	01-01-2026	46023	364

```
B2   =A2
C2   =DATE(2026,12,31)-A2
B3   =A3
C3   =DATE(2026,12,31)-A3
```

COMMON MISTAKES **Fixing numbers stored as text**

- Select the cells, click the yellow warning icon, choose **Convert to Number**.
- Or use **Data → Text to Columns → Finish** on the column.
- Or in a formula: `=VALUE(A7)` or `=A7*1`.
- Remove hidden spaces first with `=TRIM(A7)` (and `=CLEAN()` for line breaks from web exports).

Error values and what they mean

Error	Meaning	Typical cause
`#DIV/0!`	Division by zero	`=A2/B2` where B2 is empty or 0
`#N/A`	Value not available	A lookup could not find the value
`#VALUE!`	Wrong type of argument	Doing maths with text, such as `="abc"*2`
`#REF!`	Invalid reference	You deleted a row or column a formula used
`#NAME?`	Excel does not recognise a name	A typo in a function name, or text without quotes
`#NUM!`	Invalid number	`=SQRT(-1)`
`#SPILL!`	A dynamic array has no room	Cells below the formula are not empty (Lesson 6)
`#####`	Not an error	The column is too narrow to show the value

Excel properties: what makes a workbook "smart"

These behaviours make Excel more than a calculator:

1. **Automatic recalculation**: change an input and every dependent formula updates instantly (Formulas → Calculation Options can switch this to Manual for huge files; press F9 to recalculate).
2. **Relative references**: copy `=B2*12` down and it becomes `=B3*12`, `=B4*12` automatically (Lesson 4 covers `$` to lock references).
3. **AutoFill**: drag the small square (fill handle) at the bottom-right of a selection to continue patterns: `Jan` → `Feb` → `Mar`, `1, 2, 3`, `Mon, Tue`, dates and formulas.
4. **Flash Fill** (`Ctrl + E`): type one or two examples of a pattern (such as first names taken from full names) and Excel fills the rest.
5. **Excel Tables** (`Ctrl + T`): turn a range into a named table that expands automatically, keeps formatting, and allows readable formulas like `=SUM(Sales[Amount])`.
6. **File properties**: File → Info shows the author, size, last-modified date and lets you inspect the workbook for hidden data before sharing.

2026 UPDATE **What's new in Excel (2025–2026)**

- **Copilot in Excel** can explain formulas, suggest formula columns, highlight insights and create charts from a plain-English request (Microsoft 365 Copilot licence required).
- **Python in Excel** lets you type `=PY(` and run pandas code inside a cell; the code runs in Microsoft's cloud.
- **Dynamic array functions** such as `UNIQUE`, `SORT`, `FILTER`, `XLOOKUP`, `TEXTSPLIT`, `GROUPBY` and `PIVOTBY` are now standard in Microsoft 365.
- **Checkbox cells** (Insert → Checkbox) store TRUE/FALSE and are useful for trackers. You will use each of these later in the Excel track.

Keyboard shortcuts that save hours

You do not need to memorise 100 shortcuts. Learn these 25 and you will already be faster than most office users. (Mac users: replace `Ctrl` with `Cmd` for most of them.)

Task	Shortcut
Jump to the edge of the data	`Ctrl + Arrow key`
Select to the edge of the data	`Ctrl + Shift + Arrow key`
Select the whole current table	`Ctrl + A` (inside the data)
Go to cell A1 / last used cell	`Ctrl + Home` / `Ctrl + End`
Select a whole column / row	`Ctrl + Space` / `Shift + Space`
Insert / delete rows or columns	`Ctrl + Shift + +` / `Ctrl + -`
Edit the active cell	`F2`
Lock a reference ($A$1 → A$1 → $A1)	`F4` while editing a formula
Fill down / fill right	`Ctrl + D` / `Ctrl + R`
Enter the same value in all selected cells	type, then `Ctrl + Enter`
AutoSum	`Alt + =`
Today's date / current time	`Ctrl + ;` / `Ctrl + Shift + ;`
Create an Excel Table	`Ctrl + T`
Toggle filters	`Ctrl + Shift + L`
Format Cells dialog	`Ctrl + 1`
Show formulas instead of results	`Ctrl + ` `
Flash Fill	`Ctrl + E`
Paste Special	`Ctrl + Alt + V`
Find / Replace	`Ctrl + F` / `Ctrl + H`
New sheet	`Shift + F11`
Move between sheets	`Ctrl + Page Up / Page Down`
Hide rows / columns	`Ctrl + 9` / `Ctrl + 0`
Undo / Redo	`Ctrl + Z` / `Ctrl + Y`
Repeat last action	`F4` (outside a formula)
Show key tips on the ribbon	`Alt`

TIP **Practise with real data**

Open `employees_small.csv` (download it from the top of this lesson). Click any cell inside the data, press `Ctrl + A`, then `Ctrl + T` to make it a Table. Press `Ctrl + Down` to jump to the last row and `Ctrl + Home` to come back. Select the salary column and read the Sum and Average in the status bar.

Your first worksheet, step by step

Let's build a small salary sheet with the 20-row employee file. The first rows look like this:

```python
import pandas as pd
emp = pd.read_csv("employees_small.csv")
emp.head(6)
```

OUTPUT

	emp_id	name	department	city	join_date	salary	rating
0	E101	Kavya Iyer	HR	Mumbai	2016-02-28	97000	4
1	E102	Imran Reddy	Finance	Delhi	2019-12-14	157000	3
2	E103	Harpreet Kulkarni	Finance	Mumbai	2024-10-05	131000	4
3	E104	Diya Iyer	HR	Bengaluru	2024-01-02	90000	4
4	E105	Simran Iyer	HR	Bengaluru	2017-08-03	139000	4
5	E106	Saanvi Rao	HR	Delhi	2025-02-18	83000	4

6 rows × 7 columns

STEP BY STEP **Build it**

1. **Open** the CSV in Excel (File → Open, or Data → From Text/CSV).
2. Press `Ctrl + T` to convert it to a Table, and name it `Emp` (Table Design → Table Name).
3. In a new column `annual_salary`, type = then click the salary cell and type *12. Because it is a Table, Excel fills the whole column with `=[@salary]*12`.
4. Below the data, add summary formulas: total salary, average salary, number of employees.
5. Format the salary columns as `#,##0` with `Ctrl + 1` (Lesson 3 explains every format).
6. Freeze the header row: View → Freeze Panes → Freeze Top Row.

First worksheet: summary formulas (data on sheet Emp)

	A	B	C
1	Measure	Formula result	What the formula does
2	Employees	20	Counts non-empty IDs
3	Total monthly salary	1,978,000	Adds the salary column
4	Average monthly salary	98,900	Total ÷ count
5	Highest salary	157,000	Largest value
6	Lowest salary	45,000	Smallest value

```
B2   =COUNTA(Emp!A2:A21)
B3   =SUM(Emp!F2:F21)
B4   =AVERAGE(Emp!F2:F21)
```

```
B5   =MAX(Emp!F2:F21)
B6   =MIN(Emp!F2:F21)
```

The results above were calculated from the real file. Check them yourself in Python:

Python

```
s = emp["salary"]
print("Employees:", s.count())
print("Total:", f"{s.sum():,}", "| Average:", f"{s.mean():,.0f}", "| Max:", f"{s.max():,}", "| Min:",
f"{s.min():,}")
```

OUTPUT

```
Employees: 20
Total: 1,978,000 | Average: 98,900 | Max: 157,000 | Min: 45,000
```

Saving and sharing the right way

Format	Use it when
`.xlsx`	Normal workbooks (default)
`.xlsm`	Workbooks that contain macros (VBA)
`.xlsb`	Very large workbooks; binary format opens and saves faster
`.csv`	Plain data to move between systems (Python, SQL, Power BI). Keeps only one sheet and no formatting or formulas
`.pdf`	Read-only reports for people who should not edit

WATCH OUT **CSV and leading zeros**

When you open a CSV directly, Excel guesses types and may drop leading zeros (`00451` → `451`) or turn codes like `1-2` into dates. Use **Data → From Text/CSV** and set the column type to Text in the preview to keep values exactly as they are.

Summary

KEY TAKEAWAYS

- The **cell** shows the result; the **formula bar** shows what is really inside.
- A **range** is written top-left to bottom-right, for example `B2:D4`.
- Analyse only **structured data**: one header row, one record per row, one type per column, no blanks, no merged cells, no totals inside.
- Excel has five value types: **number, text, date/time (a number underneath), logical, error**.
- Long IDs and codes must be stored as **text** to avoid the 15-digit and leading-zero traps.
- Learn the navigation shortcuts first: `Ctrl + Arrow`, `Ctrl + Shift + Arrow`, `Ctrl + T`, `Ctrl + 1`, `F2`, `F4`.

Quick quiz

6 questions · tap an answer to check it

1. A cell shows `18,000`, but the formula bar shows `=B4*12`. What is inside the cell?

A The number 18,000

B A formula whose result is 18,000

C Text that looks like a number

D An error

Answer: B. The cell displays the result; the formula bar shows what is stored. This cell stores a formula.

2. How many cells are in the range `B2:D5`?

A 8

B 9

C 12

D 15

Answer: C. Columns B, C, D = 3 columns; rows 2 to 5 = 4 rows; 3 × 4 = 12 cells.

3. Which of these breaks the rules of structured data?

- **A** One header row with unique names
- **B** A merged "Region" heading above each group of rows
- **C** One record per row
- **D** A date column that contains only dates

Answer: B. Region should be its own column, with the value repeated on every row. Merged headings make the data impossible to sort, filter or pivot.

4. Why does `SUM` ignore some numbers imported from a CSV file?

- **A** SUM only works on Excel Tables
- **B** The values are stored as text
- **C** The numbers have too many decimals
- **D** The column is too narrow

Answer: B. Numbers stored as text are skipped by SUM. Convert them with Text to Columns, VALUE() or Convert to Number.

5. What does Excel store when you type the date 27-04-2026?

- **A** The text "27-04-2026"
- **B** A serial number counting days from 1 January 1900
- **C** Three separate numbers for day, month and year
- **D** A logical value

Answer: B. Dates are numbers formatted to look like dates. That is why date arithmetic works.

6. Which shortcut turns a range into an Excel Table?

A Ctrl + E

B Ctrl + 1

C Ctrl + T

D Ctrl + Shift + L

Answer: C. Ctrl + T creates a Table. Ctrl + E is Flash Fill, Ctrl + 1 opens Format Cells, Ctrl + Shift + L toggles filters.

EXERCISE **Clean a messy sheet**

Look at the "Before" report in this lesson. Write down, in order, the steps you would take in Excel to turn it into the "After" table.

Show solution (worked answer)

1. Delete the title row and the blank row.
2. Insert a new first column called `Region` and fill it with North or South for each person; delete the "Region:" heading rows and the repeated header row.
3. Delete the TOTAL row (totals belong below or in a pivot table).
4. Replace `n/a` with an empty cell (Ctrl + H, replace `n/a` with nothing).
5. Convert the Jan–Mar columns to real numbers (Text to Columns or Convert to Number).
6. Unpivot the month columns into `Month` and `Sales` (Power Query → Unpivot Other Columns, Lesson 12) so each row is one person-month.
7. Press Ctrl + T to make it a Table and add the total with `=SUM()` below it.

EXERCISE **Spot the data type**

For each entry, say which data type Excel stores: `₹1,250`, `'2026`, `15:45`, `=5>3`, `#N/A`, `Pune`.

Show solution (worked answer)

`₹1,250` is a number (currency format). `'2026` is text because of the apostrophe. `15:45` is a time, which is a number (0.65625 of a day). `=5>3` returns the logical value TRUE. `#N/A` is an error. `Pune` is text.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between a formula and a function? A formula is any expression that starts with = (for example `=A2*B2`). A function is a named, built-in calculation used inside formulas (for example `SUM`, `VLOOKUP`). Every function is used in a formula, but not every formula uses a function.

Q: How does Excel store dates, and why does it matter? As serial numbers (days since 1-Jan-1900; times are fractions of a day). It means you can subtract dates to get durations, add days to a date, and sort dates correctly. If a date is stored as text, none of that works.

Q: What makes data "analysis-ready"? One header row, one record per row, one data type per column, no blank rows or merged cells, no subtotals mixed into the data, and consistent spelling of categories.

ADVANCED EXCEL · LESSON 2

Data Entry, Data Validation & Data Cleaning

Stop bad data at the door with data validation and drop-down lists, then clean what already exists with Flash Fill, Text to Columns, Find & Replace, Go To Special, Remove Duplicates, Paste Special and Excel Tables.

Live class 2 · ✦ Excel for Microsoft 365, 2026 · Beginner · ⏱ 90 min

By the end of this lesson you can

- Enter data quickly and consistently with AutoFill, Flash Fill and custom lists
- Build drop-down lists, number and date limits and custom-formula rules with Data Validation
- Create dependent (cascading) drop-downs
- Clean messy text, split and combine columns, and remove duplicates safely
- Use Paste Special and Excel Tables as everyday cleaning tools

Garbage in, garbage out

Most time in real analytics is not spent on fancy charts; it is spent fixing data: extra spaces, inconsistent spellings (`Bangalore`, `Bengaluru`, `bengaluru`), numbers stored as text, duplicates and blanks. Surveys of data teams regularly report that cleaning and preparing data takes well over half of an analyst's time.

There are two ways to win this fight:

Prevent problems at entry time, and clean whatever slips through. Both are needed.

Fast, consistent data entry

AutoFill and the fill handle

Select a cell and drag the small square at its bottom-right corner (the **fill handle**). Excel continues the pattern it recognises.

You type	Drag gives you
`Jan`	Feb, Mar, Apr ...
`Monday`	Tuesday, Wednesday ...
`1` , `2` (select both)	3, 4, 5 ...
`Q1`	Q2, Q3, Q4, Q1 ...
`01-04-2026`	02-04-2026, 03-04-2026 ... (right-drag for months or years)
`Item 1`	Item 2, Item 3 ...
`=B2*12`	`=B3*12` , `=B4*12` ...

Double-click the fill handle to fill a formula down as far as the column next to it has data. This is the fastest way to copy a formula to 10,000 rows.

> **TIP** **Custom lists**
>
> File → Options → Advanced → **Edit Custom Lists** lets you add your own sequences, for example `North, South, East, West` or your company's department order. AutoFill and custom sort then follow that order.

Flash Fill (Ctrl + E)

Flash Fill learns a pattern from one or two examples you type. It is perfect for splitting names, extracting codes or reformatting text.

Flash Fill: type the first answer in each column, then press Ctrl + E

	A	B	C	D	E
1	**Full name**	**Email**	**First name**	**Last name**	**Username**
2	Kavya Iyer	kavya.iyer@abc.in	Kavya	Iyer	kiyer
3	Imran Reddy	imran.reddy@abc.in	Imran	Reddy	ireddy
4	Harpreet Kulkarni	harpreet.k@abc.in	Harpreet	Kulkarni	hkulkarni
5	Diya Iyer	diya.iyer@abc.in	Diya	Iyer	diyer

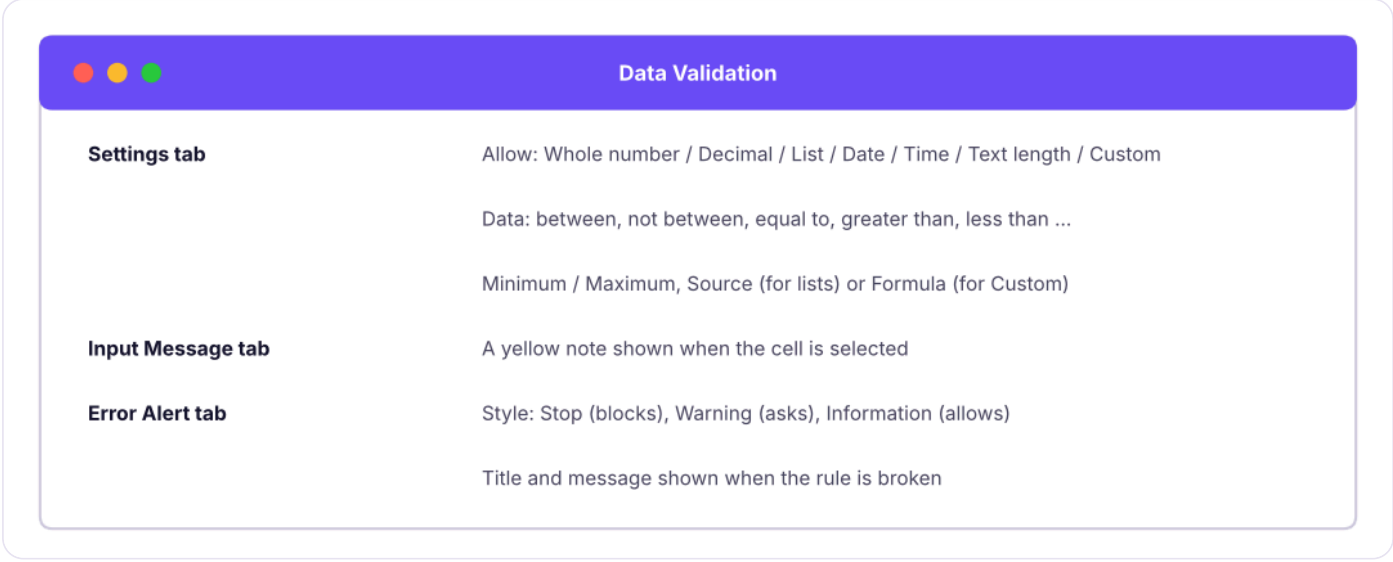
Purple cells are typed by you; green cells are filled by Flash Fill.

WATCH OUT **Flash Fill is not a formula**

Flash Fill writes **static values**. If the source changes, the results do not update. Use formulas (Lesson 5) or Power Query (Lesson 12) when the data will be refreshed.

Data Validation: rules for what can be typed

Data → Data Validation (shortcut `Alt, A, V, V`) attaches a rule to cells. Anyone who types something that breaks the rule is stopped or warned.



The three tabs of the Data Validation dialog.

The validation types you will use most

Allow	Example rule	Business use
Whole number	between 18 and 65	Age of an employee
Decimal	between 0 and 0.5	Discount rate
List	`Sales,Finance,IT,HR,Operations`	Department drop-down
Date	between `01-04-2026` and `31-03-2027`	Entries inside the financial year
Time	between 09:00 and 18:00	Office hours
Text length	equal to 6	PIN code, employee code
Custom	`=COUNTIF($A$2:$A$500,A2)=1`	No duplicate IDs

Building a drop-down list, step by step

STEP BY STEP **Department drop-down**

1. On a sheet called `Lists`, type the allowed values in a column (`Sales`, `Finance`, `IT`, `HR`, `Operations`) and convert it to a Table (`Ctrl + T`) named `tblDept`. A Table grows automatically when you add a department.
2. Select the cells where people will enter a department (for example `C2:C500`).
3. Data → Data Validation → Allow: **List**.
4. Source: `=INDIRECT("tblDept[Department]")` or simply select the range `=Lists!$A$2:$A$6`.
5. Keep **In-cell dropdown** ticked. Add an Input Message ("Choose a department") and an Error Alert (Stop, "Please pick a department from the list").

Data entry sheet with validation rules (rules shown in row 2)

	A	B	C	D	E	F	G
1	**Emp ID**	**Name**	**Department**	**Age**	**Join date**	**Discount**	**PIN code**
2	*Text length = 4*	*Any text*	*List: tblDept*	*Whole 18–65*	*Date ≥ 01-01-2000*	*Decimal 0–0.5*	*Text length = 6*
3	E101	Kavya Iyer	HR	29	28-02-2016	0.1	400001
4	E102	Imran Reddy	Fin	17	14-12-2019	0.75	11002

Row 3 passes every rule (green). In row 4, `Fin` is not in the list, 17 is under the minimum age, 0.75 is above the maximum discount and `11002` has only 5 characters, so Excel would block each of these entries (red).

Custom formula rules

The **Custom** option accepts any formula that returns TRUE (allow) or FALSE (block). Write the formula for the **first** selected cell; Excel adjusts it for the others.

Rule	Custom formula (cells A2:A500 selected)
No duplicates	`=COUNTIF($A$2:$A$500,A2)=1`
Must start with "E" and be 4 characters	`=AND(LEFT(A2,1)="E",LEN(A2)=4)`
Numbers only (for a text-formatted column)	`=ISNUMBER(--A2)`
No spaces	`=ISERROR(FIND(" ",A2))`
Weekdays only (dates)	`=WEEKDAY(A2,2)<6`
Total of a budget column cannot exceed 1,00,000	`=SUM($B$2:$B$50)⩽100000`
Email must contain "@" and "."	`=AND(ISNUMBER(FIND("@",A2)),ISNUMBER(FIND(".",A2)))`

Dependent (cascading) drop-downs

A second list that changes with the first: choose a **Category**, and the **Sub-category** list shows only matching items.

Lists sheet: each column is a named range (Formulas → Create from Selection → Top row)

	A	B	C
1	**Furniture**	**Office_Supplies**	**Technology**
2	Chairs	Binders	Phones
3	Tables	Paper	Accessories
4	Bookcases	Storage	Machines
5	Furnishings	Art	Copiers
6		Appliances	

STEP BY STEP **Make it work**

1. Select the lists (with headers) → Formulas → **Create from Selection** → Top row. Excel creates the names `Furniture`, `Office_Supplies` and `Technology`.
2. Category cell `A2`: Data Validation → List → Source: `Furniture,Office_Supplies,Technology`.
3. Sub-category cell `B2`: Data Validation → List → Source: `=INDIRECT(A2)`.
4. `INDIRECT` turns the text in A2 into a reference to the named range with that name.

2026 UPDATE **Dynamic-array version (Microsoft 365)**

With a two-column mapping table (`Category`, `SubCategory`), put `=FILTER(Map[SubCategory],Map[Category]=A2)` in a helper cell and point the validation source to it with the spill operator, for example `=$H$2#`. No named ranges are needed, and new items appear automatically.

Finding data that already breaks the rules

Validation only checks **new** typing. Data pasted in, or entered before the rule existed, is not checked. Use **Data → Data Validation → Circle Invalid Data** to draw red circles around existing violations.

COMMON MISTAKES **Validation is not security**

Pasting over a validated cell replaces the rule. Use Paste Special → Values when pasting into validated areas, and protect the sheet (Lesson 11) if people must not change rules.

Cleaning text with formulas

These functions fix most text problems. Each formula below is written for row 2 and copied down.

Common text-cleaning formulas

	A	B	C	D	E
1	Raw value	=TRIM(A2)	=PROPER(TRIM(A2))	=SUBSTITUTE(A2,"-","")	=UPPER(A2)
2	bengaluru	bengaluru	Bengaluru	bengaluru	BENGALURU
3	NEW DELHI	NEW DELHI	New Delhi	NEW DELHI	NEW DELHI
4	pune-411001	pune-411001	Pune-411001	pune411001	PUNE-411001
5	mUMBAI	mUMBAI	Mumbai	mUMBAI	MUMBAI

```
B2   =TRIM(A2)
C2   =PROPER(TRIM(A2))
D2   =SUBSTITUTE(A2,"-","")
E2   =UPPER(A2)
```

Each pattern is copied down/across to the other cells in its column.

- `TRIM` removes spaces at the start and end and squeezes repeated inner spaces to one.
- `PROPER` capitalises the first letter of each word; `UPPER` and `LOWER` change the case of everything.
- `SUBSTITUTE(text, old, new)` replaces every occurrence of `old`.
- `CLEAN` removes non-printing characters such as line breaks from copied web data.
- The non-breaking space (character 160) from websites is not removed by TRIM. Use `=TRIM(SUBSTITUTE(A2,CHAR(160)," "))`.

Text to Columns

Data → Text to Columns splits one column into several, either by a **delimiter** (comma, space, `-`, `|`) or by **fixed width**.

Before: one column with a delimiter

	A	B	C
1	Order info		
2	IN-2025-00042	Pune	Technology
3	IN-2025-00043	Delhi	Furniture

STEP BY STEP **Split by a delimiter**

1. Select the column. Data → Text to Columns.
2. Choose **Delimited** → Next → tick **Other** and type `|` (or comma, tab, space).
3. In step 3, set each column's type: keep IDs and codes as **Text** to protect leading zeros; set dates to **Date: DMY**.
4. Choose the destination cell (so the original is not overwritten) → Finish.

Text to Columns is also a quick fix for numbers stored as text: select the column, Text to Columns, and click **Finish** immediately. Excel re-reads every value as a number.

2026 UPDATE **TEXTSPLIT, TEXTBEFORE, TEXTAFTER (Microsoft 365)**

Formula versions that update automatically: `=TEXTSPLIT(A2,"|")` spills the parts across columns; `=TEXTBEFORE(A2,"-")` and `=TEXTAFTER(A2,"-",-1)` return the text before the first or after the last hyphen. See Lesson 5.

Find & Replace and wildcards

`Ctrl + H` replaces text in the selection (or the whole sheet). Click **Options** to match case, match the entire cell, or search in formulas versus values.

Wildcard	Meaning	Example
`*`	Any number of characters	`Bang*` finds Bangalore, Bangaluru
`?`	Exactly one character	`E10?` finds E101 to E109
`~`	Treat the next character literally	`~*` finds a real asterisk

Example: replace `Bang*` with `Bengaluru` (with **Match entire cell contents** ticked) to standardise every spelling in one step.

Go To Special: select blanks, constants, formulas, errors

`F5 → Special` (or `Ctrl + G → Special`) selects cells by type. The most useful trick in data cleaning is **filling blanks with the value above**:

Before: category written only once per block

	A	B	C
1	**Category**	**Product**	**Sales**
2	Furniture	Chair A	12000
3		Table B	18500
4		Bookcase C	9400
5	Technology	Phone X	25200
6		Router Y	4300

STEP BY STEP **Fill blanks with the value above**

1. Select `A2:A6`. Press `F5 → Special → Blanks → OK`. Only the empty cells are selected (yellow).
2. Type = and press the **Up arrow** (the formula becomes =A2).
3. Press `Ctrl + Enter` to put the formula in every selected blank at once.
4. Select the column, copy, and Paste Special → **Values** to freeze the result.

Other Go To Special options: **Constants** (typed values only, great for clearing inputs but keeping formulas), **Formulas** (to check or protect them), **Errors**, **Visible cells only** (copy a filtered list without hidden rows; shortcut `Alt + ;`), **Row differences**.

Removing duplicates

Data → Remove Duplicates deletes rows that repeat in the columns you choose and keeps the **first** occurrence.

Duplicate check before removing

	A	B	C	D
1	Emp ID	Name	City	Is duplicate?
2	E101	Kavya Iyer	Mumbai	First
3	E102	Imran Reddy	Delhi	First
4	E101	Kavya Iyer	Mumbai	Duplicate
5	E103	Harpreet Kulkarni	Mumbai	First
6	E102	Imran Reddy	Delhi	Duplicate

```
D2   =IF(COUNTIF($A$2:A2,A2)>1,"Duplicate","First")
D3   =IF(COUNTIF($A$2:A3,A3)>1,"Duplicate","First")
D4   =IF(COUNTIF($A$2:A4,A4)>1,"Duplicate","First")
D5   =IF(COUNTIF($A$2:A5,A5)>1,"Duplicate","First")
D6   =IF(COUNTIF($A$2:A6,A6)>1,"Duplicate","First")
```

The expanding range `$A$2:A2` counts how many times the ID has appeared **so far**, so the first appearance says "First" and later ones say "Duplicate". This lets you review duplicates before deleting anything.

> **WATCH OUT** **Always work on a copy**
>
> Remove Duplicates permanently deletes rows. Copy the sheet first, or use the formula above, conditional formatting (Home → Conditional Formatting → Highlight Duplicate Values) or `=UNIQUE()` (Lesson 6) to inspect duplicates without losing data.

Paste Special: the cleaning Swiss-army knife

`Ctrl + Alt + V` opens Paste Special.

Option	What it does	Typical use
Values (V)	Pastes results, not formulas	Freeze formula results; break links to other files
Formats (T)	Copies only formatting	Apply a header style to another table
Transpose (E)	Rows become columns	Turn a wide header row into a list
Multiply / Divide / Add / Subtract	Combines the copied number with each target cell	Convert rupees to lakhs: copy `100000`, select the amounts, Paste Special → Divide
Skip blanks	Empty copied cells do not overwrite targets	Merge two partially filled columns
Paste Link	Creates `=Sheet1!A1` references	Live link to a source cell

Excel Tables: structure that maintains itself

Convert any clean range with `Ctrl + T`. Tables are the best habit you can build in Excel:

- **Auto-expand**: new rows and columns join the table, so formulas, charts, pivot tables and validation lists include them automatically.
- **Structured references**: formulas read like English, for example `=SUM(tblSales[Amount])` or `=[@Price]*[@Qty]`.
- **Calculated columns**: type a formula once and it fills the whole column.
- **Total Row**: Table Design → Total Row adds a row with drop-down summaries that respect filters (they use `SUBTOTAL`).
- **Banded rows, filter buttons and a frozen header** come for free.

An Excel Table named tblOrders with a calculated column and Total Row

	A	B	C	D	E
1	OrderID	Product	Price	Qty	Amount
2	1001	Chair	3500	4	14,000
3	1002	Desk	12500	1	12,500
4	1003	Lamp	1200	6	7,200
5	1004	Shelf	5400	2	10,800
6	Total			13	44,500

```
E2   =C2*D2
E3   =C3*D3
E4   =C4*D4
E5   =C5*D5
D6   =SUBTOTAL(109,D2:D5)
E6   =SUBTOTAL(109,E2:E5)
```

In a real Table, the Amount formula appears as `=[@Price]*[@Qty]`. The Total Row uses `SUBTOTAL(109, …)`, which sums only visible rows, so totals change when you filter.

DO IT WITH AI **Do it with AI**

Copilot in Excel or any chat assistant can draft cleaning steps. A good prompt: *"I have a column of Indian city names with inconsistent spelling, extra spaces and mixed case (examples: ' bengaluru', 'BANGALORE', 'Bengaluru '). Give me an Excel formula to standardise them to proper case, and a list of Find & Replace steps to map old names to new ones."* Always test the answer on a copy of your data.

Summary

KEY TAKEAWAYS

- **Prevent** bad data with Data Validation (lists, limits, custom formulas) and **cure** the rest with cleaning tools.
- Validation lists should point to a Table so they grow automatically; use `INDIRECT` or `FILTER` for dependent drop-downs.
- Validation does not check pasted or old data: use **Circle Invalid Data**.
- `TRIM`, `CLEAN`, `PROPER`, `SUBSTITUTE` fix most text; Text to Columns and Flash Fill split data.
- **Go To Special → Blanks** then `=↑` and `Ctrl + Enter` fills blank cells with the value above.
- Inspect duplicates with `COUNTIF` before using Remove Duplicates; always keep a copy.
- Turn every dataset into an **Excel Table** (`Ctrl + T`).

Quick quiz

6 questions · tap an answer to check it

1. Which validation setting stops users entering a duplicate employee ID in A2:A500?

A List with source A2:A500

B Custom: `=COUNTIF($A$2:$A$500,A2)=1`

C Text length equal to 4

D Whole number between 1 and 500

Answer: B. COUNTIF counts how many times the value appears; allowing only a count of 1 blocks duplicates.

2. You pasted 2,000 rows into a validated column. How do you find values that break the rule?

A Re-create the validation rule

B Data Validation → Circle Invalid Data

C Press F9

D Use Remove Duplicates

Answer: B. Validation only checks typing. Circle Invalid Data marks existing violations.

3. What does `=TRIM(" New Delhi ")` return?

A "New Delhi"

B "New Delhi"

C "NewDelhi"

D " New Delhi "

Answer: B. TRIM removes leading and trailing spaces and reduces repeated inner spaces to one.

4. Which feature fills blank cells in a column with the value from the cell above in a few keystrokes?

A Flash Fill

B Remove Duplicates

C Go To Special → Blanks, then = + Up arrow + Ctrl + Enter

D Text to Columns

Answer: C. Go To Special selects only the blanks; Ctrl + Enter writes the relative formula into all of them.

5. What is the main downside of Flash Fill?

A It only works on numbers

B Results are static values that do not update when the source changes

C It needs a macro

D It cannot split text

Answer: B. Use formulas or Power Query for refreshable cleaning.

6. Why should a drop-down list's source be an Excel Table?

- **A** The list grows automatically when new items are added
- **B** Tables are required for validation
- **C** It makes the drop-down faster
- **D** It prevents duplicates

Answer: A. Tables expand automatically, so new departments appear in the drop-down without editing the rule.

EXERCISE **Validation set-up**

Using `employees_small.csv`, add validation so that: department must be one of Sales, Finance, IT, HR, Operations; rating must be a whole number from 1 to 5; join date cannot be in the future; and employee IDs are unique.

Show solution (worked answer)

- Department (C2:C21): List, source `Sales,Finance,IT,HR,Operations`.
- Rating (G2:G21): Whole number, between 1 and 5.
- Join date (E2:E21): Date, less than or equal to `=TODAY()`.
- Emp ID (A2:A21): Custom, `=COUNTIF($A$2:$A$21,A2)=1`. Add an Error Alert (Stop) for each with a helpful message, then run Circle Invalid Data to check the existing rows.

EXERCISE **Clean the city column**

A column contains: `mumbai`, `MUMBAI`, `Bombay`, `Mumbai`, `mumbai city`. Produce one clean value, `Mumbai`, for all five.

Show solution (worked answer)

1. Helper formula `=PROPER(TRIM(A2))` gives `Mumbai`, `Mumbai`, `Bombay`, `Mumbai`, `Mumbai City`.
2. Paste Special → Values over the original.
3. Find & Replace with Match entire cell contents: `Bombay` → `Mumbai`, `Mumbai City` → `Mumbai`. For data that refreshes every week, build the same steps in Power Query so they repeat automatically.

INTERVIEW CORNER

Questions you may be asked

Q: How would you stop users from entering invalid data in a shared tracker? Use Data Validation (drop-down lists from a Table, number and date limits, custom formulas for uniqueness), add input messages, set error alerts to Stop, and protect the sheet so the rules cannot be removed. Check older data with Circle Invalid Data.

Q: How do you remove duplicates without losing information? First flag them with `COUNTIF` or conditional formatting and review which record to keep; decide the key columns that define a duplicate; work on a copy; then use Remove Duplicates (or `UNIQUE`) on those columns.

Q: What is the difference between TRIM and CLEAN? TRIM removes extra spaces; CLEAN removes non-printing characters such as line breaks. Web data often needs both, plus `SUBSTITUTE(…,CHAR(160),"")` for non-breaking spaces.

ADVANCED EXCEL · LESSON 3

Number, Custom & Conditional Formatting

Make numbers readable and insights jump out. Learn cell and number formats, write your own custom number formats (including lakhs and crores, K and M, and colour codes), and use conditional formatting with built-in rules and your own formulas.

Live class 2 · ✦ Excel for Microsoft 365, 2026 · Beginner · ⏱ 100 min

By the end of this lesson you can

- Apply cell formatting (fonts, alignment, borders, wrap, merge alternatives) the professional way
- Choose the right built-in number format and know that formatting never changes the stored value
- Write custom number formats with sections, conditions, colours, text and scaling
- Highlight insights with conditional formatting: rules, Top/Bottom, data bars, colour scales and icon sets
- Write custom-formula conditional formatting, such as highlighting whole rows

Formatting changes how a value looks, not what it is

This is the most important idea in this lesson. When you format `0.256` as `26%`, the cell still contains `0.256`. Every formula that uses the cell sees `0.256`.

X Same stored value, different formats

	A	B	C
1	Stored value	Format applied	What you see
2	0.256	Percentage, 0 decimals	26%
3	0.256	Percentage, 1 decimal	25.6%
4	1,234,568	Number with separator	1,234,567.89
5	46139	Date	27-Apr-2026
6	46139	Long date	Monday, 27 April 2026
7	0.6875	Time	16:30

```
C2   =TEXT(A2,"0%")
C3   =TEXT(A3,"0.0%")
C4   =TEXT(A4,"#,##0.00")
C5   =TEXT(A5,"dd-mmm-yyyy")
C6   =TEXT(A6,"dddd, d mmmm yyyy")
C7   =TEXT(A7,"hh:mm")
```

Column C uses the TEXT function to show exactly what each format produces. (TEXT returns text, so use it for labels, not for numbers you still need to calculate with.)

WATCH OUT **Rounding by format vs rounding by formula**

Formatting `2.456` with 1 decimal *shows* `2.5` but keeps `2.456`. Adding three such cells can show `7.4` while the visible numbers add up to `7.5`. If the rounded value must be used in calculations (invoices, tax), round it with a formula: `=ROUND(A2,1)`.

Cell formatting the professional way

Open **Format Cells** with `Ctrl + 1`. It has six tabs: Number, Alignment, Font, Border, Fill, Protection.

Goal	How
Bold header with fill colour	Home → Bold (`Ctrl + B`) and Fill Color, or Cell Styles → Heading
Wrap long text	Home → Wrap Text (`Alt + H, W`)
Centre a title across columns **without merging**	`Ctrl + 1` → Alignment → Horizontal: **Center Across Selection**
Rotate header text	Alignment → Orientation
Thin borders around a table	Home → Borders → All Borders
Copy formatting	Format Painter (double-click it to paint many places)
Clear only formatting	Home → Clear → Clear Formats
Consistent look across a workbook	Page Layout → Themes, and Cell Styles

COMMON MISTAKES **Avoid merged cells**

Merged cells break sorting, filtering, copying, selecting columns and pivot tables. Use **Center Across Selection** instead: it looks the same and keeps every cell independent.

Built-in number formats

Format	Example display of 1234.5	Use for
General	1234.5	Default, no fixed format
Number	1,234.50	Quantities, amounts
Currency	₹1,234.50	Money, symbol next to the number
Accounting	₹ 1,234.50	Money in financial statements: symbol aligned at the left edge, zero shown as a dash
Percentage	123450%	Rates; type 12% or format `0.12`
Fraction	1234 1/2	Measurements
Scientific	1.23E+03	Very large or small numbers
Text	1234.5 (as text)	IDs, codes, phone numbers
Date / Time	Various	Dates and times (stored as serial numbers)

Shortcuts: `Ctrl + Shift + 1` number with 2 decimals, `Ctrl + Shift + 4` currency, `Ctrl + Shift + 5` percentage, `Ctrl + Shift + 3` date.

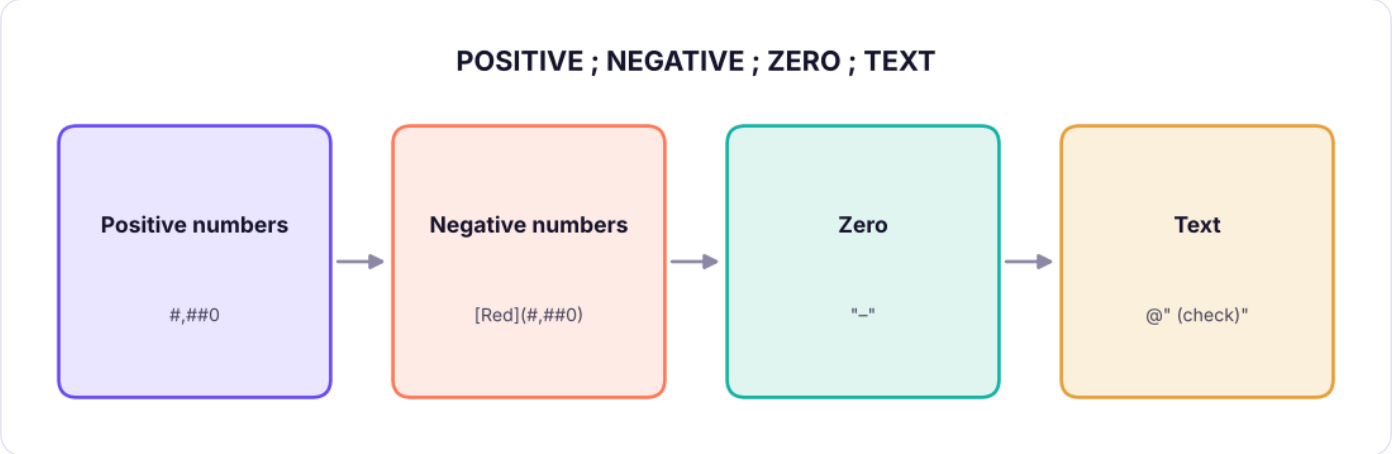
Custom number formats

`Ctrl + 1` → Number → **Custom** lets you write your own format code. Once you understand the structure, you can make dashboards that are much easier to read.

The four sections

A format code can have up to four parts, separated by semicolons:

```text
POSITIVE ; NEGATIVE ; ZERO ; TEXT
#,##0    ; [Red]-#,##0 ; "-" ; @
```



A custom number format has up to four sections separated by semicolons.

Placeholder characters

Code	Meaning	Value 7.5 →
`0`	Digit, shows a zero if none	`000.00` → 007.50
`#`	Digit, shows nothing if none	`#.##` → 7.5
`?`	Digit, shows a space (aligns decimals)	`?.??` → 7.5␣
`,` (in the middle)	Thousands separator	`#,##0`
`,` (at the end)	Divide by 1,000 for each comma	`#,##0,` shows 12,500 as 13
`%`	Multiply by 100 and add %	`0%`
`"text"`	Show literal text	`0" units"`
`@`	The cell's text	`"Dept: "@`
`[Red]`, `[Blue]`, `[Color10]`	Colour	`[Red]-0`
`[⩾1000]`	Condition	`[⩾1000]#,##0,"K";0`

Custom formats you will reuse

Each result in column C was produced by Excel's TEXT function with the format code in column B.

Custom format cookbook

	A	B	C
1	**Value**	**Format code**	**Result**
2	12500	#,##0	12,500
3	12500	#,##0,"K"	13K
4	12500	#,##0.0,"K"	12.5K
5	8450000	#,##0.0,,"M"	8.5M
6	-3200	#,##0;(#,##0)	(3,200)
7	0	#,##0;-#,##0;"–"	–
8	45	0" units"	45 units
9	0.0725	0.0%	7.3%
10	7	000	007
11	9876543210	00000 00000	98765 43210
12	1500	[>=1000]#,##0.0,"K";0	1.5K
13	850	[>=1000]#,##0.0,"K";0	850
14	0.25	▲ 0%;▼ 0%	▲ 25%
15	-0.12	▲ 0%;▼ 0%	▼ 12%

C2 `=TEXT(A2,B2)`

Each pattern is copied down/across to the other cells in its column.

TIP **Up and down arrows in KPIs**

`▲ 0%;▼ 0%` (add `[Color10]` for green and `[Red]` for red: `[Color10]▲ 0%;[Red]▼ 0%`) is the classic format for "change versus last month" cells in dashboards. The arrow and colour change automatically with the sign.

Indian number format: lakhs and crores

Indian reports group digits as 12,34,56,789 (crore, lakh, thousand). If your Windows region is set to English (India), the standard `#,##0` already groups this way. Otherwise use a conditional format:

Text

```
[≥10000000]##\,##\,##\,##0;[≥100000]##\,##\,##0;##,##0
```

X Lakhs and crores

	A	B	C	D
1	Value	Indian grouping	In lakhs	In crores
2	1234567	12,34,567	12.35 L	0.12 Cr
3	98765432	9,87,65,432	987.65 L	9.88 Cr
4	450000	4,50,000	4.50 L	0.05 Cr

```
C2   =TEXT(A2/100000,"0.00")&" L"
D2   =TEXT(A2/10000000,"0.00")&" Cr"
C3   =TEXT(A3/100000,"0.00")&" L"
D3   =TEXT(A3/10000000,"0.00")&" Cr"
C4   =TEXT(A4/100000,"0.00")&" L"
D4   =TEXT(A4/10000000,"0.00")&" Cr"
```

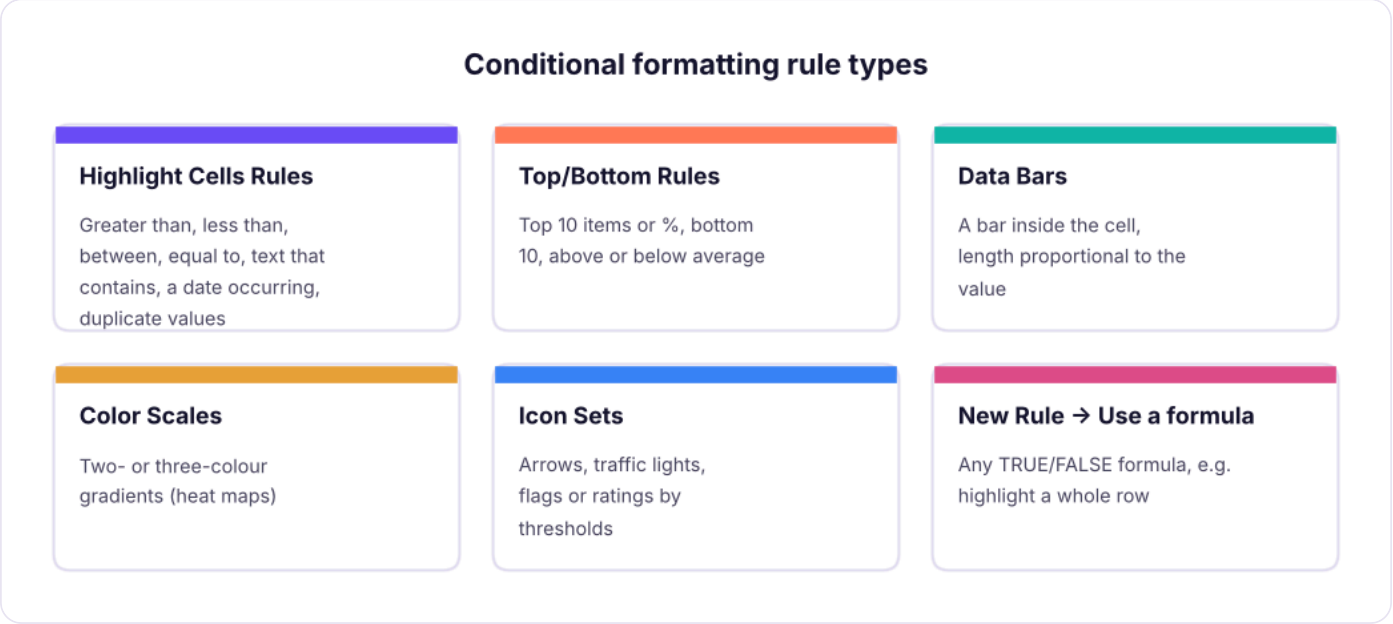
There is no "divide by 1 lakh" comma trick in Excel (each trailing comma divides by 1,000), so for lakhs either divide in a helper column (as above) or use Power BI's display units.

Date and time format codes

Code	Shows (for 27-04-2026 14:05)
`d, dd`	27
`ddd, dddd`	Mon, Monday
`m, mm`	4, 04
`mmm, mmmm`	Apr, April
`yy, yyyy`	26, 2026
`hh:mm`	14:05
`hh:mm AM/PM`	02:05 PM
`[h]:mm`	Total hours beyond 24, for timesheets (e.g. 37:30)
`"Q"` with a formula	Quarter labels need a formula: `="Q"&ROUNDUP(MONTH(A2)/3,0)`

Conditional formatting: let insights jump out

Home → Conditional Formatting changes a cell's look based on its value. It is how you turn a wall of numbers into something a manager understands in five seconds.



The five families of conditional formatting rules.

Highlight rules and Top/Bottom

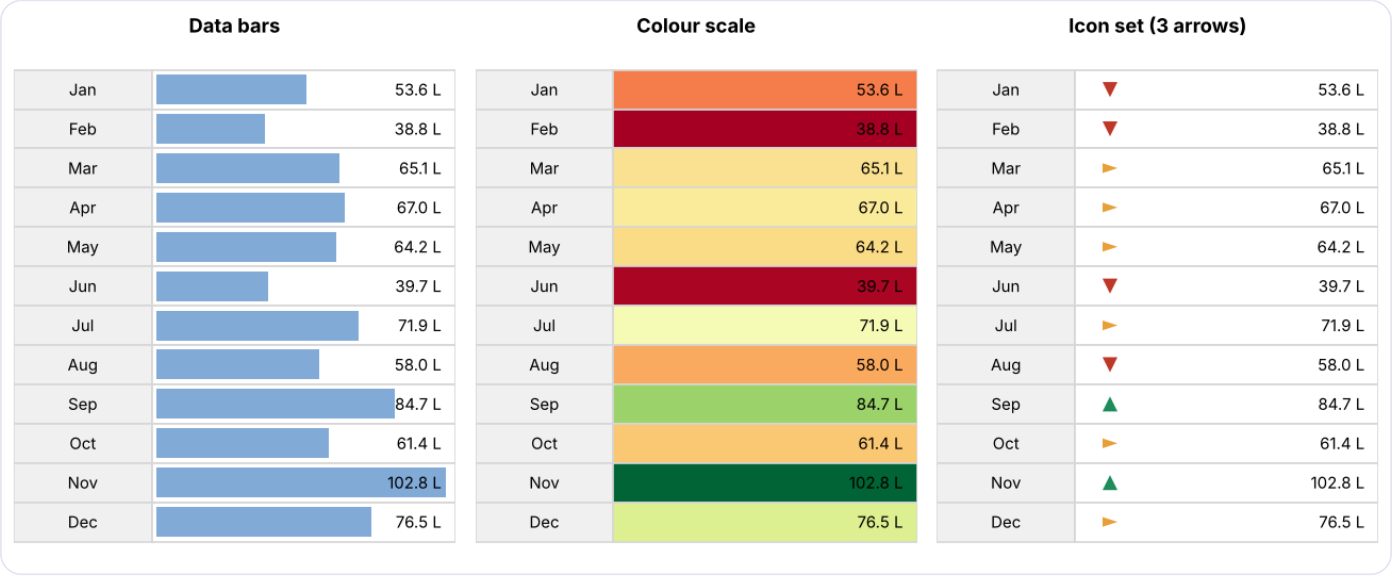
In the table below, green marks sales **above 1,50,000** and red marks **profit below 0** (both "Highlight Cells Rules"):

Highlight rules: sales > 1,50,000 (green), profit < 0 (red)

	A	B	C	D
1	Region	Category	Sales	Profit
2	North	Furniture	128,400	-3,250
3	North	Technology	212,950	28,400
4	South	Office Supplies	96,300	14,820
5	South	Technology	187,200	19,950
6	East	Furniture	74,100	-8,900
7	West	Technology	158,700	22,310

Data bars, colour scales and icon sets

These visual rules are best seen on real data. Here they are applied to monthly sales from the Superstore dataset:



Data bars (left), a green–yellow–red colour scale (middle) and icon sets (right) applied to the same monthly sales.

Rule	When to use	Watch out
Data bars	Compare magnitudes inside a table	Negative values draw leftwards; set a fixed minimum of 0 if needed
Colour scales	Heat maps: sales by region × month, attendance grids	Red/green is hard for colour-blind readers; consider blue–white–orange
Icon sets	Status at a glance: on track, at risk, off track	Default thresholds are percentages of the range (67% / 33%); set them to numbers or percentiles that match your business targets

Under **Manage Rules → Edit Rule** you can change thresholds, choose "Show Bar Only" or "Show Icon Only", and reverse the icon order.

Custom formula conditional formatting

New Rule → Use a formula to determine which cells to format is the most powerful option. Write a formula for the **top-left cell** of your selection that returns TRUE for cells you want formatted. Dollar signs matter:

- $ before the column ($D2) locks the rule to column D, so **every cell in the row** checks column D → the whole row is highlighted.
- No $ before the row number lets the row change as Excel moves down the selection.

Highlight whole rows where Department = Finance AND Salary > 1,00,000 — rule: =AND($C2="Finance",$F2>100000)

	A	B	C	D	E	F	G
1	Emp ID	Name	Department	City	Join date	Salary	Rating
2	E101	Kavya Iyer	HR	Mumbai	28-02-2016	97,000	4
3	E102	Imran Reddy	Finance	Delhi	14-12-2019	157,000	3
4	E103	Harpreet Kulkarni	Finance	Mumbai	05-10-2024	131,000	4
5	E104	Diya Iyer	HR	Bengaluru	02-01-2024	90,000	4
6	E110	Vikram Joshi	Finance	Chennai	19-10-2022	75,000	5
7	E119	Rohan Das	Finance	Delhi	19-07-2018	97,000	5

Rows E102 and E103 meet both conditions. E110 and E119 are in Finance but earn less than 1,00,000, so they stay white.

More custom-formula recipes

Goal	Select	Formula
Zebra stripes (every other row)	the table	`=MOD(ROW(),2)=0`
Overdue tasks (due date before today, status not Done)	the table	`=AND($E2<TODAY(),$F2<>"Done")`
Values above the column average	`C2:C100`	`=C2>AVERAGE($C$2:$C$100)`
Duplicate IDs	`A2:A100`	`=COUNTIF($A$2:$A$100,A2)>1`
Weekend dates	`A2:A100`	`=WEEKDAY(A2,2)>5`
Row of the employee picked in a drop-down (H1)	the table	`=$B2=$H$1`
Missing values	the table	`=ISBLANK(A2)` or `=A2=""`
Top 3 per column	`C2:C100`	`=C2⩾LARGE($C$2:$C$100,3)`

STEP BY STEP **Create a whole-row highlight**

1. Select the data **without** the header (for example `A2:G21`). The active cell must be in row 2.
2. Home → Conditional Formatting → New Rule → **Use a formula to determine which cells to format**.
3. Formula: `=AND($C2="Finance",$F2>100000)`.
4. Click **Format...** → Fill → choose a colour → OK.
5. To edit later: Conditional Formatting → **Manage Rules** (choose "This Worksheet" to see all rules). Rules higher in the list win; tick **Stop If True** to prevent lower rules from also applying.

COMMON MISTAKES **The three most common conditional-formatting bugs**

1. **Wrong anchor**: writing the formula for row 1 while the selection starts at row 2 shifts every result by one row.
2. **Missing $**: `=C2="Finance"` (no $) checks a different column for each cell, so only part of the row lights up.
3. **Copying cells copies rules**: pasting formatted cells many times creates hundreds of duplicate rules and slows the workbook. Clean up in Manage Rules, or paste values only.

Formatting a report: before and after

A clean report follows a few simple rules:

1. **Right-align numbers**, left-align text, and use the same number of decimals in a column.
2. Use **thousands separators**; show large values in K/M or lakh/crore with clear labels.
3. **Bold** headers and totals only; avoid using many colours.
4. Use **one highlight colour** for "needs attention" and keep everything else neutral.
5. **Freeze** the header row; make the column widths fit the content.
6. Add a title, the date range and the data source.

DO IT WITH AI **Do it with AI**

Ask an assistant: *"Write an Excel custom number format that shows values above 1 crore as '1.25 Cr', values above 1 lakh as '4.50 L', and smaller values with Indian comma grouping."* Then test the answer with `=TEXT(value, "format")` on a few numbers before applying it. (A pure format can only scale by thousands, so a good answer will suggest a helper column or Power BI for lakhs and crores.)

Summary

KEY TAKEAWAYS

- Formatting changes the **display**, never the stored value. Round with `ROUND` when it matters.
- Avoid merged cells; use **Center Across Selection**.
- Custom formats have four sections, `positive;negative;zero;text`, and support colours, conditions, text and scaling with trailing commas.
- Conditional formatting families: highlight rules, Top/Bottom, data bars, colour scales and icon sets.
- For whole-row highlights use a formula with $ before the column: `=$C2="Finance"`.
- Manage Rules shows priority; rule order and Stop If True decide which format wins.

Quick quiz

6 questions · tap an answer to check it

1. A cell contains 0.4567 formatted as 0%. What value does `=A1*100` return?

A 46

B 45.67

C 0.46

D 46%

Answer: B. The stored value is still 0.4567, so the formula returns 45.67. Formatting only changes the display.

2. Which custom format shows 8,450,000 as 8.5M?

A `#,##0,"M"`

B `#,##0.0,,"M"`

C `0.0"M"`

D `#,##0.0%`

Answer: B. Two trailing commas divide by 1,000,000, and 0.0 shows one decimal.

3. In the format `#,##0;[Red](#,##0);"-"`, how is -2500 shown?

A -2,500 in black

B (2,500) in red

C –

D 2,500 in red

Answer: B. The second section is for negative numbers: red, in brackets, without a minus sign.

4. You want to highlight entire rows where Status (column F) is "Late". Rows start at 2. Which rule is correct?

A `=F2="Late"`

B `=$F2="Late"`

C `=$F$2="Late"`

D `=F$2="Late"`

Answer: B. Locking the column with $ makes every cell in the row test column F; leaving the row free lets each row test itself.

5. What does "Center Across Selection" replace?

A Wrap Text

B Merge & Center

C Format Painter

D Freeze Panes

Answer: B. It centres text across several cells without merging them, so sorting and selecting still work.

6. Which conditional formatting type is best for a sales heat map of region × month?

A Icon sets

B Data bars

C Colour scales

D Highlight duplicates

Answer: C. Colour scales shade each cell by value, which creates a heat map.

EXERCISE **Dashboard number formats**

Write custom formats for: (a) revenue shown in thousands with "K" and no decimals; (b) month-on-month change shown as a green up arrow with one decimal percentage when positive, and a red down arrow when negative; (c) a 6-digit PIN code that must keep leading zeros.

Show solution (worked answer)

(a) `#,##0,"K"` (b) `[Color10]▲ 0.0%;[Red]▼ 0.0%;"-"` (c) `000000` (or store the PIN as text).

EXERCISE **Conditional formatting practice**

Using `employees_small.csv`: (1) colour-scale the salary column; (2) add an icon set to rating with green for 5, amber for 3–4 and red for 1–2; (3) highlight the whole row of anyone who joined after 1-Jan-2024.

Show solution (worked answer)

(1) Select F2:F21 → Color Scales → Green–Yellow–Red. (2) Select G2:G21 → Icon Sets → 3 traffic lights → Manage Rules → Edit: green when ≥ 5 (Number), amber when ≥ 3 (Number), otherwise red. (3) Select A2:G21 → New Rule → formula `=$E2>DATE(2024,1,1)` → choose a fill.

INTERVIEW CORNER **Questions you may be asked**

Q: What is the difference between formatting a number and rounding it? Formatting changes only how the value looks; the full value is used in calculations. `ROUND` changes the stored value. Use `ROUND` when the rounded figure must flow into other calculations, such as invoices.

Q: How would you highlight the top 10% of customers by revenue? Conditional Formatting → Top/Bottom Rules → Top 10% (change 10 to any value), or a formula rule `=C2 ⩾ PERCENTILE.INC($C$2:$C$500,0.9)` for more control.

Q: Why might a conditional format highlight the wrong rows? The formula was written for a different starting row than the active cell of the selection, or the $ signs are wrong.

ADVANCED EXCEL · LESSON 4

Formulas, Cell References & Aggregate Functions

Write formulas with confidence. Understand operators and their order, relative, absolute and mixed references, and every aggregate function you need, from SUM and COUNTA to SMALL, LARGE, SUMIFS, COUNTIFS, MAXIFS, SUBTOTAL and AGGREGATE.

Live class 3 · ✦ Excel for Microsoft 365, 2026 · Beginner · ⏱ 110 min

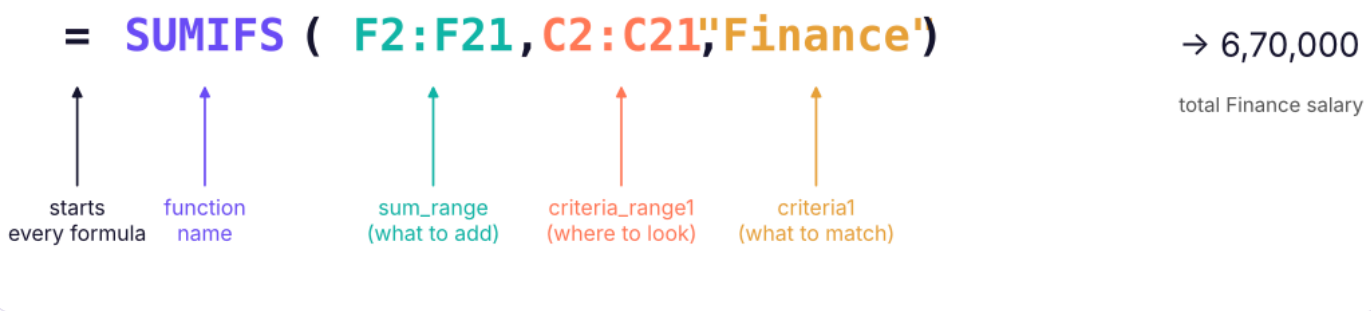
By the end of this lesson you can

- Explain the difference between a formula and a function and read any function's syntax
- Predict how a formula changes when copied, and lock references with $ (F4)
- Summarise data with SUM, COUNT, COUNTA, COUNTBLANK, AVERAGE, MIN, MAX, SMALL, LARGE, MEDIAN and MODE
- Answer conditional questions with COUNTIF(S), SUMIF(S), AVERAGEIF(S), MAXIFS and MINIFS
- Use SUBTOTAL and AGGREGATE for filtered data and data with errors

Formula vs function

DEFINITION **Two words people mix up**

A **formula** is any calculation that starts with `=`. It can use cell references, numbers, operators and functions: `=B2*C2`, `=(B2-C2)/C2`, `=SUM(B2:B10)*1.18`. A **function** is a named, ready-made calculation inside a formula: `SUM`, `AVERAGE`, `IF`. A function has a name, brackets and **arguments** separated by commas.



Anatomy of a function call. Arguments in square brackets are optional.

When you type a function name and `(`, Excel shows a tooltip with its syntax, for example `SUMIFS(sum_range, criteria_range1, criteria1, [criteria_range2, criteria2], …)`. Press `Ctrl + A` after the bracket to open the **Function Arguments** dialog, which explains every argument and previews the result.

Operators and order of calculation

Operator	Meaning	Example	Result
`+ - * /`	Add, subtract, multiply, divide	`=10+4*2`	18
`^`	Power	`=2^10`	1024
`%`	Percent	`=500*18%`	90
`&`	Join text	`="Q"&2`	Q2
`= <> > < ⩾ ⩽`	Comparison, returns TRUE/FALSE	`=B2⩾100`	TRUE or FALSE
`:`	Range	`A1:A10`	10 cells
`,`	Union (inside functions)	`=SUM(A1:A3,C1:C3)`	Adds both ranges

Excel follows the usual maths order, remembered as **BODMAS/PEMDAS**: brackets, then `^`, then `*` and `/`, then `+` and `-`, then `&`, then comparisons.

Order of operations

	A	B	C
1	Formula	Result	Why
2	=10+4*2	18	Multiply first: 10 + 8
3	=(10+4)*2	28	Brackets first: 14 × 2
4	=2^3^2	64	Excel works left to right: (2^3)^2 = 64
5	=-2^2	4	Negation before power in Excel: (-2)^2 = 4
6	=100*(1+18%)	118	Add 18% GST

```
B2    =10+4*2
B3    =(10+4)*2
B4    =2^3^2
B5    =-2^2
B6    =100*(1+18%)
```

WATCH OUT **`-2^2` is 4 in Excel**

In maths -2^2 is −4, but Excel applies the minus sign first. Write `=-(2^2)` if you mean −4. This catches people in finance formulas.

Cell references: relative, absolute and mixed

When you copy a formula, Excel adjusts its references **relative** to where you paste it. Usually that is exactly what you want. Sometimes a reference must stay fixed, for example a tax rate in one cell. A $ sign locks the part after

it.

Reference	Copy down	Copy right	Name
B2	B3, B4 …	C2, D2 …	Relative
$B$2	stays $B$2	stays $B$2	Absolute
B$2	stays row 2	C$2, D$2 …	Mixed (row locked)
$B2	B3, B4 …	stays column B	Mixed (column locked)

Press **F4** while the cursor is on a reference to cycle `B2` → `$B$2` → `B$2` → `$B2` → `B2`.

Absolute reference: one rate for every row

GST rate in one input cell (G2) — formula in D2 is =B2*C2*$G$2, copied down

	A	B	C	D	E	F	G
1	Item	Qty	Price	GST amount	Total		GST rate
2	Notebook	10	45	81.00	531.00		18%
3	Pen pack	25	60	270.00	1,770.00		
4	Stapler	4	180	129.60	849.60		

```
D2   =B2*C2*$G$2
E2   =B2*C2+D2
D3   =B3*C3*$G$2
E3   =B3*C3+D3
D4   =B4*C4*$G$2
E4   =B4*C4+D4
```

The GST rate lives in one input cell (yellow). Because `$G$2` is absolute, every copied formula still points to it. Change the rate once and every total updates.

Mixed references: the multiplication table test

A classic interview question: write **one** formula in B2 that can be copied across and down to fill a multiplication table.

One formula, =$A2*B$1, filled across and down

	A	B	C	D	E	F
1	×	1	2	3	4	5
2	1	1	2	3	4	5
3	2	2	4	6	8	10
4	3	3	6	9	12	15
5	4	4	8	12	16	20
6	5	5	10	15	20	25

```
B2   =$A2*B$1
C2   =$A2*C$1
D2   =$A2*D$1
```

```
E2    =$A2*E$1
F2    =$A2*F$1
```
Each pattern is copied down/across to the other cells in its column.

$A2 always uses column A (the row header) and B$1 always uses row 1 (the column header). The same idea is used for price × discount grids, sales × commission tables, and loan EMI tables.

The same table in formula view

Formula view (Ctrl + `)

	A	B	C	D
1	×	1	2	3
2	1	=$A2*B$1	=$A2*C$1	=$A2*D$1
3	2	=$A3*B$1	=$A3*C$1	=$A3*D$1
4	3	=$A4*B$1	=$A4*C$1	=$A4*D$1

References to other sheets and workbooks

Reference	Meaning
`=Sales!B2`	Cell B2 on the sheet named Sales
`='Sales 2026'!B2`	Quotes are needed when the sheet name has spaces
`=SUM(Jan:Mar!B2)`	A **3D reference**: B2 on every sheet from Jan to Mar
`=[Budget.xlsx]Plan!C5`	Another workbook (an external link; Data → Edit Links manages them)

Aggregate functions

Aggregate functions turn many values into one number. The examples below use the 20-row employee table stored on a sheet named Emp (columns: A emp_id, B name, C department, D city, E join_date, F salary, G rating).

Aggregate functions on Emp (salary in F2:F21, rating in G2:G21)

	A	B	C
1	Question	Formula	Result
2	Total salary	=SUM(Emp!F2:F21)	1,978,000
3	How many numbers in salary	=COUNT(Emp!F2:F21)	20
4	How many non-empty names	=COUNTA(Emp!B2:B21)	20
5	Empty cells in B2:B25	=COUNTBLANK(Emp!B2:B25)	4
6	Average salary	=AVERAGE(Emp!F2:F21)	98,900
7	Median salary	=MEDIAN(Emp!F2:F21)	97,000
8	Most common rating	=MODE.SNGL(Emp!G2:G21)	4
9	Lowest / highest salary	=MIN(…) / =MAX(…)	45,000 / 157,000
10	2nd highest salary	=LARGE(Emp!F2:F21,2)	157,000
11	3rd lowest salary	=SMALL(Emp!F2:F21,3)	57,000
12	Average rating (rounded)	=ROUND(AVERAGE(Emp!G2:G21),2)	3.55

```
C2    =SUM(Emp!F2:F21)
C3    =COUNT(Emp!F2:F21)
```

```
C4   =COUNTA(Emp!B2:B21)
C5   =COUNTBLANK(Emp!B2:B25)
C6   =AVERAGE(Emp!F2:F21)
C7   =MEDIAN(Emp!F2:F21)
C8   =MODE.SNGL(Emp!G2:G21)
C9   =TEXT(MIN(Emp!F2:F21),"#,##0")&" / "&TEXT(MAX(Emp!F2:F21),"#,##0")
C10  =LARGE(Emp!F2:F21,2)
C11  =SMALL(Emp!F2:F21,3)
C12  =ROUND(AVERAGE(Emp!G2:G21),2)
```

Function	Counts or uses	Ignores
`COUNT`	Numbers (and dates)	Text, blanks, logical values typed in cells
`COUNTA`	Anything not empty (numbers, text, errors, `""` from formulas)	Truly empty cells
`COUNTBLANK`	Empty cells and cells with `""`	Everything else
`SUM`, `AVERAGE`, `MIN`, `MAX`	Numbers	Text and blanks in ranges
`MEDIAN`	Middle value of sorted numbers	Text and blanks
`MODE.SNGL`	Most frequent number	Text
`SMALL(range,k)`, `LARGE(range,k)`	k-th smallest or largest	Text and blanks

TIP **AVERAGE ignores blanks but not zeros**

A blank cell is left out of the average; a zero is counted. If "no data" is stored as 0, your average will be too low. Leave missing values blank, or use `AVERAGEIF(range,"<>0")`.

2026 UPDATE **MEDIAN vs AVERAGE in salary data**

Salaries are usually skewed by a few high earners, so the **median** describes a "typical" employee better than the average. You will meet this again in Statistics (ML track, Lesson 1).

Conditional aggregation: the IFS family

These are the most used analysis functions in Excel. They answer questions such as *"total sales in the North region for Technology in 2025"*.

Function	Syntax
COUNTIF	COUNTIF(range, criteria)
COUNTIFS	COUNTIFS(range1, criteria1, range2, criteria2, …)
SUMIF	SUMIF(range, criteria, [sum_range])
SUMIFS	SUMIFS(sum_range, range1, criteria1, range2, criteria2, …)
AVERAGEIF(S)	Same pattern as SUMIF(S)
MAXIFS, MINIFS	MAXIFS(max_range, range1, criteria1, …)

Note that SUMIF puts the sum range **last** while SUMIFS puts it **first**. Learn the SUMIFS pattern and use it even for one condition.

Conditional aggregation on Emp

	A	B	C
1	Question	Result	Formula
2	Employees in Finance	7	=COUNTIF(Emp!C2:C21,"Finance")
3	Finance employees in Delhi	2	=COUNTIFS(Emp!C2:C21,"Finance",Emp!D2:D21,"Delhi")
4	Total Finance salary	670,000	=SUMIFS(Emp!F2:F21,Emp!C2:C21,"Finance")
5	Salary of people rated 4 or more	1,102,000	=SUMIFS(Emp!F2:F21,Emp!G2:G21,">=4")
6	Average salary in HR	96,000	=AVERAGEIFS(Emp!F2:F21,Emp!C2:C21,"HR")
7	Highest salary in Sales	156,000	=MAXIFS(Emp!F2:F21,Emp!C2:C21,"Sales")
8	Joined in 2024 or later	6	=COUNTIFS(Emp!E2:E21,">="&DATE(2024,1,1))
9	Names starting with "A"	4	=COUNTIF(Emp!B2:B21,"A*")
10	Salary between 80k and 1.2 lakh	9	=COUNTIFS(Emp!F2:F21,">=80000",Emp!F2:F21,"<=120000")

```
B2   =COUNTIF(Emp!C2:C21,"Finance")
B3   =COUNTIFS(Emp!C2:C21,"Finance",Emp!D2:D21,"Delhi")
B4   =SUMIFS(Emp!F2:F21,Emp!C2:C21,"Finance")
B5   =SUMIFS(Emp!F2:F21,Emp!G2:G21,"⩾4")
B6   =AVERAGEIFS(Emp!F2:F21,Emp!C2:C21,"HR")
B7   =MAXIFS(Emp!F2:F21,Emp!C2:C21,"Sales")
B8   =COUNTIFS(Emp!E2:E21,"⩾"&DATE(2024,1,1))
B9   =COUNTIF(Emp!B2:B21,"A*")
B10  =COUNTIFS(Emp!F2:F21,"⩾80000",Emp!F2:F21,"⩽120000")
```

Let's confirm two of these with pandas, using the same file:

Python

```python
import pandas as pd
emp = pd.read_csv("employees_small.csv", parse_dates=["join_date"])
print("Finance headcount:", (emp.department == "Finance").sum())
print("Total Finance salary:", f"{emp.loc[emp.department == 'Finance', 'salary'].sum():,}")
print("Joined 2024 or later:", (emp.join_date ⩾ "2024-01-01").sum())
```

OUTPUT

```
Finance headcount: 7
Total Finance salary: 670,000
Joined 2024 or later: 6
```

Writing criteria correctly

Criteria	Meaning
`"Finance"`	Equal to Finance (not case-sensitive)
`"<>Finance"`	Not equal to Finance
`"⩾4"`	Greater than or equal to 4 (operator and number inside quotes)
`"⩾"&H1`	Compare with the value in H1 (join with &)
`"⩾"&DATE(2024,1,1)`	Compare with a date
`"*pune*"`	Contains "pune"
`"A*"` / `"*a"`	Starts with A / ends with a
`"?????"`	Exactly five characters
`""`	Blank cells
`"<>"`	Non-blank cells

TIP **Build a summary table with one formula**

Put departments in A2:A6 and the formula `=SUMIFS(Emp!$F$2:$F$21,Emp!$C$2:$C$21,$A2)` in B2, then copy down. The $ signs keep the data ranges fixed while $A2 moves to each department. In Microsoft 365 you can produce the department list itself with `=UNIQUE(Emp!C2:C21)` (Lesson 6).

X Department summary built with SUMIFS and COUNTIFS

	A	B	C	D	E
1	Department	Headcount	Total salary	Average salary	% of payroll
2	Finance	7	670,000	95,714	33.9%
3	HR	6	576,000	96,000	29.1%
4	IT	1	112,000	112,000	5.7%
5	Operations	2	261,000	130,500	13.2%
6	Sales	4	359,000	89,750	18.1%
7	**Total**	**20**	**1,978,000**	**98,900**	**100.0%**

```
B2   =COUNTIFS(Emp!$C$2:$C$21,$A2)
C2   =SUMIFS(Emp!$F$2:$F$21,Emp!$C$2:$C$21,$A2)
D2   =C2/B2
E2   =C2/$C$7
B7   =SUM(B2:B6)
C7   =SUM(C2:C6)
E7   =SUM(E2:E6)
```

Each pattern is copied down/across to the other cells in its column.

This little table is exactly what a pivot table does (Lesson 8), but with formulas you control every cell, which is useful for fixed report layouts.

SUBTOTAL and AGGREGATE: totals for filtered data

`SUM` adds every row, including rows hidden by a filter. `SUBTOTAL` adds only **visible** rows, so the total changes as you filter.

`=SUBTOTAL(function_num, range)`

function_num (ignores filtered rows)	101–111 also ignore manually hidden rows	Function
1	101	AVERAGE
2	102	COUNT
3	103	COUNTA
4	104	MAX
5	105	MIN
9	109	SUM

`AGGREGATE` does the same and can also **ignore error values**: `=AGGREGATE(9, 6, D2:D100)` sums D2:D100 ignoring errors (option 6). `=AGGREGATE(14, 6, D2:D100, 2)` returns the 2nd largest value ignoring errors (14 = LARGE).

AGGREGATE ignores the #DIV/0! error that breaks SUM

	A	B
1	Value	
2	120	
3	#DIV/0!	
4	80	
5	SUM →	#DIV/0!
6	AGGREGATE(9,6,...) →	200

```
A3   =10/0
B5   =SUM(A2:A4)
B6   =AGGREGATE(9,6,A2:A4)
```

Common formula errors and how to fix them

Error	Example	Fix
`#DIV/0!`	`=C2/B2` with B2 = 0	`=IFERROR(C2/B2,0)` or `=IF(B2=0,"",C2/B2)`
`#VALUE!`	`=A2+B2` where A2 is text	Convert text to numbers; check for spaces
`#REF!`	Deleted a referenced row	Undo, or rewrite the formula
`#NAME?`	`=SUMIFS(…,Finance)` without quotes	Put text criteria in quotes
Circular reference	`=SUM(A1:A10)` typed in A10	Move the total outside the range
Wrong total after sorting	Mixed relative/absolute references	Lock ranges with $

Formulas → Formula Auditing helps: **Trace Precedents** draws arrows to the cells a formula uses, **Trace Dependents** to cells that use it, **Show Formulas** (`Ctrl + `` `) shows every formula, and **Evaluate Formula** steps through a calculation (Lesson 11).

DO IT WITH AI

Do it with AI

Paste a formula into Copilot or a chat assistant and ask: *"Explain this Excel formula step by step and tell me what would make it return an error: `=SUMIFS(Sales!$F:$F,Sales!$C:$C,$A2,Sales!$E:$E,"≥"&$B$1)`."* Use AI to explain and draft formulas, then check the result on a few rows you can verify by hand.

Summary

KEY TAKEAWAYS

- Formula = any calculation starting with =; function = a named calculation such as `SUM`.
- Order: brackets, power, multiply/divide, add/subtract, join, compare. In Excel `-2^2 = 4`.
- $ locks a reference; **F4** cycles relative → absolute → mixed. Mixed references build 2-D tables with one formula.
- `COUNT` counts numbers; `COUNTA` counts non-empty; `COUNTBLANK` counts empty.
- `SUMIFS(sum_range, criteria_range, criteria, …)` and `COUNTIFS` answer most business questions.
- Criteria with operators go in quotes and join cell values with &: `"≥"&H1`.
- `SUBTOTAL(109,…)` sums only visible rows; `AGGREGATE` can also ignore errors.

Quick quiz

6 questions · tap an answer to check it

1. You copy `=B2*$E$1` from C2 to C5. What is the formula in C5?

A `=B5*$E$4`

B `=B5*$E$1`

C `=B2*$E$1`

D `=E5*$B$1`

Answer: B. B2 is relative and moves to B5; $E$1 is absolute and stays.

2. Which formula, typed in B2 and copied across and down, builds a multiplication table with headers in row 1 and column A?

A `=A2*B1`

B `=$A$2*$B$1`

C `=$A2*B$1`

D `=A$2*$B1`

Answer: C. Column A is locked for the row header, and row 1 is locked for the column header.

3. A range contains 10, "abc", blank, 25, TRUE (typed). What does COUNT return?

A 2

B 3

C 4

D 5

Answer: A. COUNT counts only numbers in a range: 10 and 25.

4. Which formula totals Sales (D) where Region (B) is "North" AND Category (C) is "Technology"?

A `=SUMIF(B:B,"North",D:D)`

B `=SUMIFS(D:D,B:B,"North",C:C,"Technology")`

C `=SUMIFS(B:B,"North",C:C,"Technology",D:D)`

D `=COUNTIFS(B:B,"North",C:C,"Technology")`

Answer: B. SUMIFS takes the sum range first, then pairs of criteria range and criteria.

5. How do you count values in A2:A100 greater than the value in H1?

A `=COUNTIF(A2:A100,>H1)`

B `=COUNTIF(A2:A100,">H1")`

C `=COUNTIF(A2:A100,">"&H1)`

D `=COUNTIF(A2:A100,H1>)`

Answer: C. The operator is text and is joined to the cell value with &.

6. You filter a table to show only Pune. Which total updates to show only Pune's sales?

A `=SUM(D2:D500)`

B `=SUBTOTAL(109,D2:D500)`

C `=SUMIF(D2:D500,">0")`

D `=AGGREGATE(9,0,D2:D500)` with option 0

Answer: B. SUBTOTAL 9/109 ignores rows hidden by a filter.

EXERCISE **Superstore questions with formulas**

Open `superstore_sales.csv` in Excel. Using only aggregate and IFS functions, find: (1) total sales; (2) number of order lines in the West region; (3) total profit for Technology in 2025; (4) average discount for Furniture; (5) the largest single order-line sale in the South.

Show solution (worked answer)

Assuming columns: J region, L category, O sales, Q discount, R profit and B order_date (check your own column letters): (1) `=SUM(O:O)` (2) `=COUNTIFS(J:J,"West")` (3) `=SUMIFS(R:R,L:L,"Technology",B:B,"≥"&DATE(2025,1,1),B:B,"≤"&DATE(2025,12,31))` (4) `=AVERAGEIFS(Q:Q,L:L,"Furniture")` (5) `=MAXIFS(O:O,J:J,"South")` Check your answers in Python with pandas (Python track, Lesson 22) or with a pivot table (Lesson 8): they must match exactly.

INTERVIEW CORNER **Questions you may be asked**

Q: What is the difference between COUNT, COUNTA and COUNTBLANK? COUNT counts numeric cells, COUNTA counts every non-empty cell (including text and formulas returning ""), COUNTBLANK counts empty cells and cells containing "".

Q: When would you use a mixed reference? When a formula is copied in two directions and must keep either the row or the column fixed, for example a price × quantity grid or a commission table: `=$A2*B$1`.

Q: Why use SUMIFS instead of SUMIF? SUMIFS supports multiple conditions, has a consistent argument order (sum range first), and works for one condition too, so one pattern covers every case.

Q: How do you total only visible rows after filtering? Use `SUBTOTAL(109, range)` or `AGGREGATE(9, 5, range)`; the Table Total Row does this automatically.

ADVANCED EXCEL · LESSON 5

Text Functions, Logical Functions & IF

Extract, clean, combine and reformat text with LEFT, RIGHT, MID, LEN, FIND, SEARCH, CONCAT, TEXTJOIN, TEXT and the new TEXTSPLIT/TEXTBEFORE/TEXTAFTER. Then make decisions with AND, OR, NOT, IF, nested IF, IFS, SWITCH and IFERROR.

Live classes 3–4 | ✦ Excel for Microsoft 365, 2026 | Beginner | ⏱ 120 min

By the end of this lesson you can

- Extract parts of codes, names and emails with LEFT, RIGHT, MID, FIND and SEARCH
- Combine text with &, CONCAT and TEXTJOIN and format numbers inside text with TEXT
- Split text with Text to Columns and the TEXTSPLIT, TEXTBEFORE and TEXTAFTER functions
- Build conditions with AND, OR, NOT and decisions with IF, nested IF, IFS and SWITCH
- Handle errors cleanly with IFERROR and IFNA

Why text functions matter

Real data arrives as text: order IDs like `IN-2025-00042`, product codes, email addresses, "Last, First" names, addresses and free-text comments. Text functions let you pull out the part you need (the year, the region code, the domain) so you can analyse it.

Measuring and extracting text

Function	What it returns	Example	Result
`LEN(text)`	Number of characters	`=LEN("Pune")`	4
`LEFT(text, n)`	First n characters	`=LEFT("IN-2025-00042",2)`	IN
`RIGHT(text, n)`	Last n characters	`=RIGHT("IN-2025-00042",5)`	00042
`MID(text, start, n)`	n characters from position start	`=MID("IN-2025-00042",4,4)`	2025
`FIND(find, text, [start])`	Position of text (case-sensitive, no wildcards)	`=FIND("-","IN-2025-00042")`	3
`SEARCH(find, text, [start])`	Position (not case-sensitive, wildcards allowed)	`=SEARCH("pune","PUNE-411001")`	1

Extracting parts of an order ID

	A	B	C	D	E	F
1	Order ID	Country =LEFT(A2,2)	Year =MID(A2,4,4)	Order no. =RIGHT(A2,5)	Length	Year as number
2	IN-2025-00042	IN	2025	00042	13	2025
3	IN-2023-01519	IN	2023	01519	13	2023
4	IN-2024-02988	IN	2024	02988	13	2024

```
B2   =LEFT(A2,2)
C2   =MID(A2,4,4)
D2   =RIGHT(A2,5)
E2   =LEN(A2)
F2   =VALUE(MID(A2,4,4))
```

Each pattern is copied down/across to the other cells in its column.

Text functions always return **text**. `MID(A2,4,4)` returns the text `"2025"`, not the number 2025. Wrap with `VALUE()` (or multiply by 1) to get a number you can compare or add.

When the position changes: FIND + MID

Codes rarely have fixed lengths. Combine `FIND` with `LEFT`, `MID` and `RIGHT` to cut at a delimiter wherever it is.

Splitting emails at the @ sign

	A	B	C	D
1	Email	Position of @	User name	Domain
2	kavya.iyer@abc.in	11	kavya.iyer	abc.in
3	imran@datacorp.com	6	imran	datacorp.com
4	harpreet.k@skillnavora.in	11	harpreet.k	skillnavora.in

```
B2   =FIND("@",A2)
C2   =LEFT(A2,FIND("@",A2)-1)
D2   =MID(A2,FIND("@",A2)+1,100)
```

Each pattern is copied down/across to the other cells in its column.

Using a large number like 100 in `MID` simply means "take everything to the end". Alternatively use `=RIGHT(A2,LEN(A2)-FIND("@",A2))`.

First and last name from 'First Last'

	A	B	C	D
1	Full name	First name	Last name	Initials
2	Kavya Iyer	Kavya	Iyer	KI
3	Harpreet Kulkarni	Harpreet	Kulkarni	HK
4	Aarav Kulkarni	Aarav	Kulkarni	AK

```
B2   =LEFT(A2,FIND(" ",A2)-1)
C2   =MID(A2,FIND(" ",A2)+1,50)
D2   =LEFT(A2,1)&LEFT(C2,1)
```

Each pattern is copied down/across to the other cells in its column.

FIND vs SEARCH

	FIND	SEARCH
Case-sensitive	Yes: `FIND("a","Apple")` → #VALUE!	No: `SEARCH("a","Apple")` → 1
Wildcards * ?	No	Yes
Not found	`#VALUE!`	`#VALUE!`

A common pattern to test whether text **contains** a word: `=ISNUMBER(SEARCH("chair",B2))` returns TRUE or FALSE.

Changing and cleaning text

Function	Purpose	Example → Result
`UPPER, LOWER, PROPER`	Change case	`=PROPER("new delhi")` → New Delhi
`TRIM`	Remove extra spaces	`=TRIM(" a b ")` → a b
`CLEAN`	Remove non-printing characters	Line breaks from web data
`SUBSTITUTE(text, old, new, [n])`	Replace by content	`=SUBSTITUTE("2025-04-27","-","/")` → 2025/04/27
`REPLACE(text, start, n, new)`	Replace by position	`=REPLACE("E101",1,1,"EMP")` → EMP101
`REPT(text, n)`	Repeat text	`=REPT("★",4)` → ★★★★
`EXACT(a, b)`	Case-sensitive comparison	`=EXACT("abc","ABC")` → FALSE

> **TIP** **Count how many times a character appears**
>
> `=LEN(A2)-LEN(SUBSTITUTE(A2,",",""))` counts commas in A2 (the length difference after removing them). Add 1 to count items in a comma-separated list.

Joining text

Three ways to combine text

	A	B	C	D	E	F
1	First	Last	City	With &	CONCAT	TEXTJOIN
2	Kavya	Iyer	Mumbai	Kavya Iyer (Mumbai)	Kavya Iyer	Kavya, Iyer, Mumbai
3	Imran		Delhi	Imran (Delhi)	Imran	Imran, Delhi

```
D2   =A2&" "&B2&" ("&C2&")"
E2   =CONCAT(A2," ",B2)
```

```
F2   =TEXTJOIN(", ",TRUE,A2:C2)
D3   =A3&" "&B3&" ("&C3&")"
E3   =CONCAT(A3," ",B3)
F3   =TEXTJOIN(", ",TRUE,A3:C3)
```

- & is the quickest way to join a few pieces.
- `CONCAT` joins ranges as well as single cells (it replaces the older `CONCATENATE`).
- `TEXTJOIN(delimiter, ignore_empty, …)` puts a separator between items and can **skip blanks**, which is why row 3 shows "Imran, Delhi" without a double comma.

Numbers and dates inside text: TEXT

Joining a number or date directly shows its raw value: `="Sales: "&A2` gives `Sales: 1234567.891`, and a date shows as `46139`. Use `TEXT(value, format)` to control how it appears.

TEXT() inside labels

	A	B
1	**Value**	**Label formula result**
2	1,234,568	Revenue: ₹1,234,568
3	27-04-2026	Report date: 27 Apr 2026
4	0.1834	Growth: 18.3%
5	27-04-2026	Day: Monday

```
B2   ="Revenue: ₹"&TEXT(A2,"#,##0")
B3   ="Report date: "&TEXT(A3,"dd mmm yyyy")
B4   ="Growth: "&TEXT(A4,"0.0%")
B5   ="Day: "&TEXT(A5,"dddd")
```

Splitting text

Text to Columns (Lesson 2) splits once into static values. Microsoft 365 adds formula versions that update automatically:

Function	Example	Result
`TEXTBEFORE(text, delimiter, [instance])`	`=TEXTBEFORE("kavya.iyer@abc.in","@")`	kavya.iyer
`TEXTAFTER(text, delimiter, [instance])`	`=TEXTAFTER("IN-2025-00042","-",-1)`	00042 (−1 = last hyphen)
`TEXTSPLIT(text, col_delim, [row_delim])`	`=TEXTSPLIT("North,South,East",",")`	spills North / South / East across three cells

TEXTSPLIT spills into neighbouring cells (Microsoft 365)

	A	B	C	D	E
1	**Input**				
2	IN-2025-00042	IN			
3	Furniture	Chairs	Apex Chair 412		
4	Furniture > Chairs > Apex Chair 412				

```
B2   =TEXTSPLIT(A2,"-")
A4   =TEXTJOIN(" > ",TRUE,A3:C3)
```

The blue cells are a **spill range**: one formula fills several cells. If any of those cells already contain something, the formula shows `#SPILL!` (Lesson 6).

2026 UPDATE **Regular expressions in Excel (2025)**

Microsoft 365 has added `REGEXTEST`, `REGEXEXTRACT` and `REGEXREPLACE`. For example `=REGEXEXTRACT(A2,"\d{6}")` pulls a 6-digit PIN code out of an address, and `=REGEXTEST(A2,"^[\w.]+@[\w.]+\.\w+$")` checks that an email looks valid. They are covered with Python regex in the Python track.

Logical functions: TRUE or FALSE

Comparisons such as `=B2>100` return TRUE or FALSE. Logical functions combine them:

Function	Returns TRUE when	Example
`AND(c1, c2, …)`	**All** conditions are TRUE	`=AND(B2⩾4, C2="Sales")`
`OR(c1, c2, …)`	**At least one** condition is TRUE	`=OR(D2="Delhi", D2="Mumbai")`
`NOT(c)`	The condition is FALSE	`=NOT(E2="Done")`
`XOR(c1, c2)`	An odd number of conditions are TRUE	Rarely needed

AND, OR and NOT on employee data

	A	B	C	D	E	F	G	H
1	**Name**	**Department**	**City**	**Rating**	**Salary**	**Sales & rating≥4**	**Metro city**	**Not HR**
2	Kavya Iyer	HR	Mumbai	4	97000	FALSE	TRUE	FALSE
3	Isha Mehta	Sales	Delhi	3	57000	FALSE	TRUE	TRUE
4	Aadhya Nair	Sales	Bengaluru	5	45000	TRUE	FALSE	TRUE
5	Krishna Khan	Sales	Mumbai	3	156000	FALSE	TRUE	TRUE

```
F2   =AND(B2="Sales",D2⩾4)
G2   =OR(C2="Delhi",C2="Mumbai")
H2   =NOT(B2="HR")
```

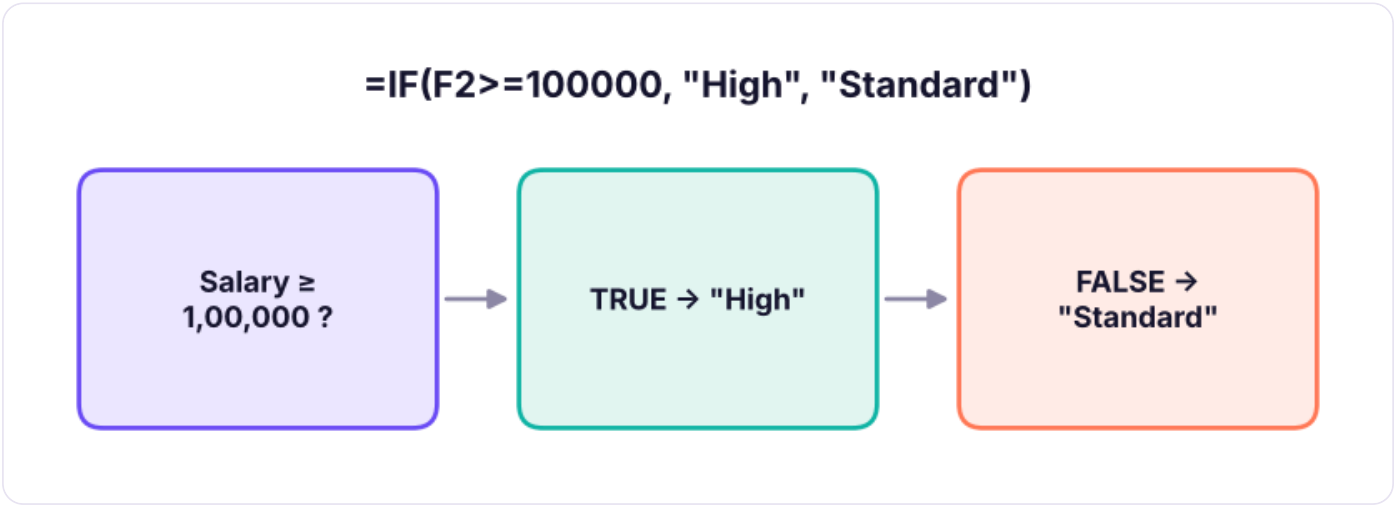
Each pattern is copied down/across to the other cells in its column.

TIP **TRUE = 1, FALSE = 0**

In arithmetic Excel treats TRUE as 1 and FALSE as 0. `=SUM(--(D2:D21≥4))` counts ratings of 4 or more (the double minus converts TRUE/FALSE to 1/0). This trick powers `SUMPRODUCT` in Lesson 6.

IF: making decisions

`=IF(logical_test, value_if_true, value_if_false)`



IF checks a condition and returns one of two results.

IF in practice

	A	B	C	D	E	F
1	Name	Salary	Rating	Band	Bonus (10% if rating ≥ 4)	Pass/Fail target 80k
2	Kavya Iyer	97000	4	Standard	9,700	Pass
3	Imran Reddy	157000	3	High	0	Pass
4	Isha Mehta	57000	3	Standard	0	Fail
5	Vikram Joshi	75000	5	Standard	7,500	Fail

```
D2   =IF(B2≥100000,"High","Standard")
E2   =IF(C2≥4,B2*10%,0)
F2   =IF(B2≥80000,"Pass","Fail")
```

Each pattern is copied down/across to the other cells in its column.

IF can return text, numbers, another formula's result, or "" (an empty-looking cell).

Combining IF with AND and OR

IF + AND / OR

	A	B	C	D	E	F
1	Name	Dept	Rating	Years	Promotion: rating ≥4 AND years ≥3	Review: rating ≤2 OR years <1
2	Kavya Iyer	HR	4	10	Eligible	OK
3	Saanvi Rao	HR	4	1	Not yet	OK
4	Aarav Kulkarni	HR	2	1	Not yet	Review
5	Neha Menon	Sales	3	0	Not yet	Review

```
E2   =IF(AND(C2≥4,D2≥3),"Eligible","Not yet")
F2   =IF(OR(C2≤2,D2<1),"Review","OK")
```

Each pattern is copied down/across to the other cells in its column.

Nested IF vs IFS

For more than two outcomes, you can put an IF inside another IF (**nested IF**), or use **IFS**, which is easier to read.

Grading marks: nested IF and IFS give the same result

	A	B	C
1	Marks	Nested IF	IFS
2	92	A	A
3	78	B	B
4	60	C	C
5	41	Fail	Fail

```
B2   =IF(A2≥90,"A",IF(A2≥75,"B",IF(A2≥60,"C","Fail")))
C2   =IFS(A2≥90,"A",A2≥75,"B",A2≥60,"C",TRUE,"Fail")
```

Each pattern is copied down/across to the other cells in its column.

COMMON MISTAKES **Order matters in nested IF and IFS**

Excel stops at the **first** TRUE condition. If you test `A2≥60` first, a mark of 92 returns "C". Always test from the highest threshold to the lowest (or lowest to highest with <). In IFS, end with `TRUE, "default"` so values that match nothing do not return `#N/A`.

SWITCH: matching exact values

When you compare one value against a list of exact matches, `SWITCH` is cleaner than nested IFs.

SWITCH maps codes to names

	A	B
1	Region code	Region name
2	N	North
3	W	West
4	X	Unknown

```
B2   =SWITCH(A2,"N","North","S","South","E","East","W","West","Unknown")
B3   =SWITCH(A3,"N","North","S","South","E","East","W","West","Unknown")
B4   =SWITCH(A4,"N","North","S","South","E","East","W","West","Unknown")
```

For long mappings, a lookup table with XLOOKUP (Lesson 7) is better than a long formula, because the mapping can be edited without touching formulas.

Handling errors: IFERROR and IFNA

IFERROR replaces any error with a value you choose

	A	B	C	D
1	Sales	Orders	Average order	With IFERROR
2	120000	40	3,000	3,000
3	0	0	#DIV/0!	0
4	95000	25	3,800	3,800

```
C2   =A2/B2
D2   =IFERROR(A2/B2,0)
C3   =A3/B3
D3   =IFERROR(A3/B3,0)
C4   =A4/B4
D4   =IFERROR(A4/B4,0)
```

- `IFERROR(value, value_if_error)` catches **every** error type.
- `IFNA(value, value_if_na)` catches only `#N/A`, typically from lookups, so real problems such as `#REF!` stay visible.

WATCH OUT **Do not hide problems**

Wrapping every formula in IFERROR can hide genuine mistakes (a deleted column, a typo). Use it where an error is expected and meaningful (division by zero, value not found), and prefer IFNA for lookups.

Case: building columns for the Superstore data

Here is how analysts typically enrich the Superstore order table with text and logical formulas before analysis. The same logic in pandas gives us the real results on all 6,323 rows:

Python

```
import pandas as pd
s = pd.read_csv("superstore_sales.csv")
s["order_year"] = s["order_id"].str[3:7].astype(int)            # =VALUE(MID(A2,4,4))
s["product_brand"] = s["product_name"].str.split(" ").str[0]     # =TEXTBEFORE(N2," ")
s["profit_flag"] = s["profit"].apply(lambda p: "Loss" if p < 0 else "Profit")       #
=IF(R2<0,"Loss","Profit")
s["discount_band"] = pd.cut(s["discount"], [-0.01, 0, 0.2, 1], labels=["No discount", "Up to 20%",
"Above 20%"])  # =IFS(...)
s[["order_id", "order_year", "product_name", "product_brand", "discount", "discount_band", "profit",
"profit_flag"]].head(6)
```

OUTPUT

	order_id	order_year	product_name	product_brand	discount	discount_band	profit	pr
0	IN-2023-00001	2023	Nova Phone 223	Nova	0.0	No discount	3,472.55	Pr
1	IN-2023-00001	2023	Orbit Appliance 798	Orbit	0.0	No discount	1,169.54	Pr
2	IN-2023-00001	2023	Orbit Table 248	Orbit	0.0	No discount	6,044.47	Pr
3	IN-2023-00002	2023	Crest Storage 946	Crest	0.2	Up to 20%	450.16	Pr
4	IN-2023-00003	2023	Crest Bookcase 170	Crest	0.3	Above 20%	-5,512.33	Lo
5	IN-2023-00003	2023	Zenith Table 111	Zenith	0.3	Above 20%	-1,598.43	Lo

6 rows × 8 columns

Python

```
pd.crosstab(s["discount_band"], s["profit_flag"], margins=True)
```

OUTPUT

profit_flag	Loss	Profit	All
discount_band			
No discount	8	3472	3480
Up to 20%	776	1167	1943
Above 20%	898	2	900
All	1682	4641	6323

4 rows × 3 columns

The crosstab already tells a business story: almost every order-line with a discount above 20% makes a loss. You will build this same view as a pivot table in Lesson 8.

New column	Excel formula (row 2)
Order year	`=VALUE(MID(B2,4,4))`
Brand	`=TEXTBEFORE(N2," ")` or `=LEFT(N2,FIND(" ",N2)-1)`
Profit flag	`=IF(S2<0,"Loss","Profit")`
Discount band	`=IFS(Q2=0,"No discount",Q2⩽0.2,"Up to 20%",TRUE,"Above 20%")`

DO IT WITH AI

Do it with AI

Prompt: *"Column B has order IDs like IN-2025-00042. Write an Excel formula to extract the year as a number, and another to flag orders where discount (Q) is above 20% and profit (S) is negative as 'Review'. Explain each part."* Expected answer: `=VALUE(MID(B2,4,4))` and `=IF(AND(Q2>0.2,S2<0),"Review","")`.

Summary

KEY TAKEAWAYS

- `LEFT`, `RIGHT`, `MID` extract text; `FIND`/`SEARCH` locate a character so you can cut at a delimiter.
- Text functions return text: wrap with `VALUE()` to get numbers.
- Join with &, `CONCAT` or `TEXTJOIN` (which can skip blanks); format numbers inside labels with `TEXT`.
- `TEXTSPLIT`, `TEXTBEFORE` and `TEXTAFTER` split text with formulas that update.
- `AND` needs all conditions, `OR` needs any; `TRUE` counts as 1 in arithmetic.
- Use `IFS` (with a final `TRUE`) instead of deep nested IFs; order conditions from strictest to loosest.
- Use `IFERROR` for expected errors and `IFNA` for lookups.

Quick quiz

6 questions · tap an answer to check it

1. What does `=MID("SKU-7781-BLU",5,4)` return?

- **A** SKU-
- **B** 7781
- **C** 781-
- **D** BLU

Answer: B. MID starts at position 5 ("7") and takes 4 characters: 7781.

2. A2 contains `ravi.verma@xyz.com`. Which formula returns `ravi.verma`?

- **A** `=LEFT(A2,FIND("@",A2)-1)`
- **B** `=LEFT(A2,FIND("@",A2))`
- **C** `=RIGHT(A2,FIND("@",A2))`
- **D** `=MID(A2,1,FIND(".",A2))`

Answer: A. FIND returns 11 (the @ position); take 10 characters from the left.

3. Which function joins A2:D2 with ", " and skips empty cells?

- **A** `=CONCAT(A2:D2)`
- **B** `=TEXTJOIN(", ",TRUE,A2:D2)`
- **C** `=A2&", "&B2&", "&C2&", "&D2`
- **D** `=TEXT(A2:D2,", ")`

Answer: B. TEXTJOIN's second argument TRUE ignores empty cells.

4. `=IF(A2≥60,"C",IF(A2≥75,"B","A"))` is used to grade a mark of 85. What is the result?

A C

B B

C A

D **VALUE!**

Answer: A. The first test (85 >= 60) is TRUE, so "C" is returned immediately. The conditions are in the wrong order.

5. Why end an IFS formula with `TRUE,"Other"`?

A It makes IFS case-insensitive

B It returns a default value instead of #N/A when no condition matches

C It is required syntax

D It speeds up the calculation

Answer: B. TRUE always matches, so it acts as the "else" branch.

6. What does `="Total: "&TEXT(1234.5,"#,##0")` return?

A Total: 1234.5

B Total: 1,235

C Total: 1,234

D Total: #,##0

Answer: B. TEXT formats the number with separators and no decimals, rounding 1234.5 to 1,235.

EXERCISE **Product codes**

Codes look like `FUR-CH-10001234` (category-subcategory-number). Write formulas for: (a) the category code; (b) the sub-category code; (c) the number as a value; (d) a flag "Furniture" when the category is FUR, otherwise "Other".

Show solution (worked answer)

(a) `=LEFT(A2,3)` or `=TEXTBEFORE(A2,"-")` (b) `=MID(A2,5,2)` or `=TEXTBEFORE(TEXTAFTER(A2,"-"),"-")` (c) `=VALUE(RIGHT(A2,8))` or `=VALUE(TEXTAFTER(A2,"-",-1))` (d) `=IF(LEFT(A2,3)="FUR","Furniture","Other")`

EXERCISE **Commission rules**

Sales reps get: 10% commission if sales ≥ 5,00,000; 7% if ≥ 3,00,000; 4% if ≥ 1,00,000; otherwise 0. Reps in the "Trainee" grade get half. Write one formula (sales in B2, grade in C2).

Show solution (worked answer)

`=B2*IFS(B2⩾500000,10%,B2⩾300000,7%,B2⩾100000,4%,TRUE,0)*IF(C2="Trainee",0.5,1)`

INTERVIEW CORNER **Questions you may be asked**

Q: Difference between FIND and SEARCH? FIND is case-sensitive and does not allow wildcards; SEARCH is not case-sensitive and supports `*` and `?`. Both return `#VALUE!` when the text is not found.

Q: How would you extract the domain from an email address? `=MID(A2,FIND("@",A2)+1,LEN(A2))` or, in Microsoft 365, `=TEXTAFTER(A2,"@")`.

Q: Nested IF or IFS? IFS for readability when testing several conditions in order; SWITCH for exact matches of one value; a lookup table when the mapping is long or changes often.

ADVANCED EXCEL · LESSON 6

Sorting, Filtering & Dynamic Arrays

Organise data with single and multi-level sorts and custom orders, find what matters with AutoFilter and Advanced Filter, and then let formulas do it for you with the dynamic array functions UNIQUE, SORT, SORTBY, FILTER and SEQUENCE, plus the power tool SUMPRODUCT.

Live class 4 · ✦ Excel for Microsoft 365, 2026 · Beginner–Intermediate · ⏱ 110 min

By the end of this lesson you can

- Sort by one or many levels, by custom lists, by colour and left to right
- Filter text, numbers and dates, and use Advanced Filter with AND/OR criteria ranges
- Explain spill ranges, the
- Build live lists with UNIQUE, SORT, SORTBY, FILTER and SEQUENCE
- Answer multi-condition questions with SUMPRODUCT

The sample table

We will use this small sales table (sheet `Reps`, range `A1:F9`) so every result is easy to check by hand. Revenue is `=D2*E2`.

Reps: sample sales table

	A	B	C	D	E	F
1	Rep	Region	Product	Units	Price	Revenue
2	Asha	North	Laptop	5	55,000	275,000
3	Ravi	West	Phone	12	18,000	216,000
4	Meera	South	Laptop	3	55,000	165,000
5	Kabir	North	Tablet	8	25,000	200,000
6	Zoya	East	Phone	15	18,000	270,000
7	Farhan	West	Laptop	4	55,000	220,000
8	Neha	South	Tablet	10	25,000	250,000
9	Rohan	North	Phone	9	18,000	162,000

F2 `=D2*E2`

Each pattern is copied down/across to the other cells in its column.

Sorting

Quick sort

Click any cell in a column and use **Data → Sort A to Z** or **Sort Z to A** (or Home → Sort & Filter). Excel detects the table around the cell and keeps each row together.

> **WATCH OUT** **Never sort a single selected column**
>
> If you select only one column and sort, Excel asks whether to "expand the selection". Choosing "Continue with the current selection" sorts that column alone and scrambles your data: names no longer match their revenue. Always click a single cell inside the table, or select the whole table.

Multi-level (level-wise) sort

Data → Sort opens the Sort dialog, where you add levels. Excel sorts by the first level, then breaks ties with the next.

The Sort dialog with two levels: Region A to Z, then Revenue largest to smallest.

Result: Region A→Z, then Revenue high→low

	A	B	C	D
1	**Rep**	**Region**	**Product**	**Revenue**
2	Zoya	East	Phone	270,000
3	Asha	North	Laptop	275,000
4	Kabir	North	Tablet	200,000
5	Rohan	North	Phone	162,000
6	Neha	South	Tablet	250,000
7	Meera	South	Laptop	165,000
8	Farhan	West	Laptop	220,000
9	Ravi	West	Phone	216,000

Other sort options

Option	Where	Example
Custom list	Order → Custom List	Sort months Jan, Feb, Mar instead of alphabetically; sort sizes S, M, L, XL; sort priority High, Medium, Low
Sort by colour	Sort On → Cell Color / Font Color / Conditional Formatting Icon	Put all red-flagged rows on top
Left to right	Options → Sort left to right	Reorder columns by a header row
Case sensitive	Options	Rarely needed
Random order	Add a column `=RAND()` and sort by it	Draw a random sample

> **TIP** **Keep the original order**
>
> Before sorting raw data, add an `Index` column with 1, 2, 3 ... (type 1 and 2, double-click the fill handle). You can always sort by Index to get back to the original order.

Filtering

AutoFilter

Click inside the data and press `Ctrl + Shift + L` (Data → Filter). Every header gets a drop-down arrow.

Filter type	Options
Text filters	Equals, does not equal, begins with, ends with, contains, does not contain, custom (with `*` and `?`)
Number filters	Greater than, between, Top 10 (items or percent), above/below average
Date filters	Today, this week, last month, next quarter, year to date, between, and grouped tick-boxes by year → month → day
By colour	Filter by cell colour, font colour or icon
Search box	Type part of a value to tick only matching items

Filters on several columns combine with **AND**: Region = North **and** Product = Laptop. Within one column, ticking several items is **OR**: North **or** South.

> **TIP** **Filtered totals**
>
> When you filter, the status bar shows "8 of 6,323 records found". Use `SUBTOTAL(109, …)` (Lesson 4) or the Table Total Row for totals that follow the filter. To copy only the visible rows, select them, press `Alt + ;` (visible cells only), then copy.

Here is the same idea on the full Superstore file: filter to Technology orders in the West with a discount above 20%.

Python

```
import pandas as pd
s = pd.read_csv("superstore_sales.csv")
f = s[(s.category == "Technology") & (s.region == "West") & (s.discount > 0.2)]
print(f"{len(f)} of {len(s):,} records found  |  visible sales total: {f.sales.sum():,.0f}  |  visible
profit total: {f.profit.sum():,.0f}")
f[["order_id", "city", "sub_category", "sales", "discount", "profit"]].head(5)
```

OUTPUT

```
36 of 6,323 records found  |  visible sales total: 1,725,761  |  visible profit total: -323,079
```

	order_id	city	sub_category	sales	discount	profit
14	IN-2023-00007	Mumbai	Phones	23,904	0.4	-4,153.37
37	IN-2023-00021	Ahmedabad	Accessories	2,821	0.3	-281.53
76	IN-2023-00037	Ahmedabad	Accessories	4,668	0.4	-1,569.18
77	IN-2023-00038	Ahmedabad	Copiers	53,082	0.4	-11,596.45
134	IN-2023-00066	Pune	Machines	22,155	0.3	-4,687.77

5 rows × 6 columns

Advanced Filter: complex AND/OR in one go

Data → Advanced filters with a **criteria range**: a small table with the same headers as your data.

- Conditions on the **same row** are combined with **AND**.
- Conditions on **different rows** are combined with **OR**.

X Criteria range: (North AND Revenue > 1,80,000) OR (Product = Phone)

	A	B	C
1	**Region**	**Product**	**Revenue**
2	North		>180000
3		Phone	

Rows that match: Asha (North, 2,75,000), Kabir (North, 2,00,000), and every Phone row: Ravi, Zoya and Rohan. Rohan is in the North but has only 1,62,000; he still appears because he sells phones.

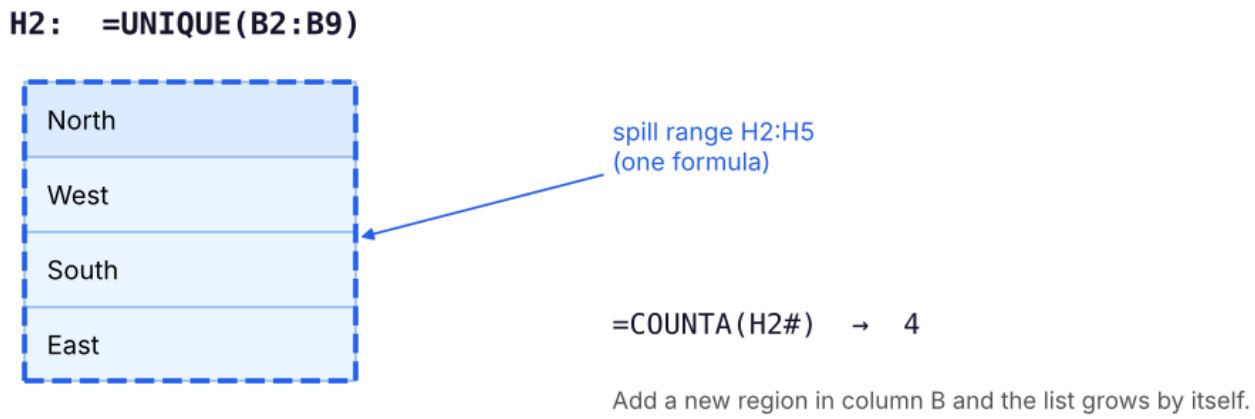
STEP BY STEP **Run an Advanced Filter**

1. Type the criteria range above your data or on another sheet (headers must match exactly).
2. Click inside the data → Data → **Advanced**.
3. List range: your data. Criteria range: the criteria table including headers.
4. Choose **Copy to another location** and pick an output cell to keep the original intact.
5. Tick **Unique records only** to get a list without duplicates (a quick alternative to Remove Duplicates).

Advanced Filter results are static: run it again after the data changes. For live results use `FILTER`.

Dynamic arrays: formulas that spill

In Microsoft 365, a formula can return **many values**. They **spill** into the cells below and to the right. The whole result is called the **spill range**; it has a blue border, and you can refer to it with the # operator.



One formula in H2 spills into H2:H5. Refer to the whole result with H2#.

WATCH OUT **#SPILL!**

If any cell in the area the result needs is not empty, the formula returns `#SPILL!`. Click the error to see which cell is blocking, and clear it. Spilling formulas also cannot be used inside an Excel Table; put them next to the table instead.

UNIQUE

```
=UNIQUE(array, [by_col], [exactly_once])
```

UNIQUE examples (formulas in H2 and J2)

	A	B	C	D	E	F
1				Region list =UNIQUE(B2:B9)		Product list =UNIQUE(C2:C9)
2				North		Laptop
3				West		Phone
4				South		Tablet
5				East		

```
D2   =UNIQUE(Reps!B2:B9)
F2   =UNIQUE(Reps!C2:C9)
```

- `=COUNTA(UNIQUE(C2:C9))` → **3** distinct products. (Before dynamic arrays this needed `=SUMPRODUCT(1/COUNTIF(C2:C9,C2:C9))`.)
- `=UNIQUE(B2:C9)` returns unique **combinations** of Region and Product.
- `=UNIQUE(B2:B9,,TRUE)` returns values that appear **exactly once**: East.

SORT and SORTBY

`=SORT(array, [sort_index], [sort_order], [by_col])` sorts by a column **position** in the array. `-1` means descending. `=SORTBY(array, by_array1, [order1], …)` sorts by **other ranges**, which do not have to be shown.

Top performers: =SORT(A2:F9,6,-1) and =SORTBY(A2:A9,F2:F9,-1)

	A	B	C	D	E	F	G	H
1	Rep	Region	Product	Units	Price	Revenue		SORTBY names
2	Asha	North	Laptop	5	55000	275000		Asha
3	Zoya	East	Phone	15	18000	270000		Zoya
4	Neha	South	Tablet	10	25000	250000		Neha
5	Farhan	West	Laptop	4	55000	220000		Farhan
6	Ravi	West	Phone	12	18000	216000		Ravi
7	Kabir	North	Tablet	8	25000	200000		Kabir
8	Meera	South	Laptop	3	55000	165000		Meera
9	Rohan	North	Phone	9	18000	162000		Rohan

```
A2   =SORT(Reps!A2:F9,6,-1)
H2   =SORTBY(Reps!A2:A9,Reps!F2:F9,-1)
```

Combine functions: `=SORT(UNIQUE(B2:B9))` gives the regions alphabetically: East, North, South, West. `=TAKE(SORT(A2:F9,6,-1),3)` returns only the top 3 rows.

FILTER

`=FILTER(array, include, [if_empty])`

The `include` argument is a TRUE/FALSE test with one result per row. Combine conditions with `*` for AND and `+` for OR.

FILTER examples

	A	B	C	D	E	F	G
1	**North reps**		**Laptops with ≥4 units**		**North OR Phone**		**No match**
2	Asha		Asha		Asha		None found
3	Kabir		Farhan		Ravi		
4	Rohan				Kabir		
5					Zoya		
6					Rohan		

```
A2  =FILTER(Reps!A2:A9,Reps!B2:B9="North")
C2  =FILTER(Reps!A2:A9,(Reps!C2:C9="Laptop")*(Reps!D2:D9≥4))
E2  =FILTER(Reps!A2:A9,(Reps!B2:B9="North")+(Reps!C2:C9="Phone"))
G2  =FILTER(Reps!A2:A9,Reps!B2:B9="Central","None found")
```

Logic	Write it as	Why it works
AND	`(cond1)*(cond2)`	TRUE×TRUE = 1; anything × FALSE = 0
OR	`(cond1)+(cond2)`	Any TRUE makes the sum ≥ 1
NOT	`(cond)=FALSE` or `NOT(cond)`	Reverses the test

TIP **An interactive report in two cells**

Put a Data Validation drop-down of regions in `H1` (source `=SORT(UNIQUE(B2:B9))`), then `=FILTER(A2:F9,B2:B9=H1)` below it. Choosing a region instantly shows its reps. This is the formula version of a slicer.

SEQUENCE and friends

Function	Example	Result
`SEQUENCE(rows, [cols], [start], [step])`	`=SEQUENCE(5)`	1, 2, 3, 4, 5 down a column
	`=SEQUENCE(1,12,1,1)`	1 to 12 across a row
	`=SEQUENCE(7,1,DATE(2026,4,27),1)`	Seven consecutive dates
`RANDARRAY(rows, cols, min, max, integer)`	`=RANDARRAY(5,1,1,100,TRUE)`	Five random whole numbers
`TAKE(array, rows)` / `DROP(array, rows)`	`=TAKE(SORT(F2:F9,,-1),3)`	Top 3 revenues
`VSTACK` / `HSTACK`	`=VSTACK(Jan!A2:C50,Feb!A2:C50)`	Stack two ranges into one
`CHOOSECOLS(array, 1, 6)`		Only columns 1 and 6
`GROUPBY(row_fields, values, function)`	`=GROUPBY(B2:B9,F2:F9,SUM)`	A summary by region, like a pivot table, in one formula (newer Microsoft 365)

SUMPRODUCT: the original array function

`SUMPRODUCT` multiplies arrays element by element and adds the results. It works in **every** version of Excel and is the classic way to do multi-condition sums before SUMIFS existed, plus things SUMIFS cannot do.

SUMPRODUCT on the Reps table

	A	B	C
1	Question	Formula	Result
2	Total revenue (units × price)	=SUMPRODUCT(D2:D9,E2:E9)	1,758,000
3	Revenue in North	=SUMPRODUCT((B2:B9="North")*D2:D9*E2:E9)	637,000
4	Phone units in West or East	=SUMPRODUCT((C2:C9="Phone")*((B2:B9="West")+(B2:B9="East"))*D2:D9)	27
5	Rows where units > 8	=SUMPRODUCT(--(D2:D9>8))	4
6	Weighted average price	=SUMPRODUCT(D2:D9,E2:E9)/SUM(D2:D9)	26,636.36

```
C2  =SUMPRODUCT(Reps!D2:D9,Reps!E2:E9)
C3  =SUMPRODUCT((Reps!B2:B9="North")*Reps!D2:D9*Reps!E2:E9)
C4  =SUMPRODUCT((Reps!C2:C9="Phone")*((Reps!B2:B9="West")+(Reps!B2:B9="East"))*Reps!D2:D9)
C5  =SUMPRODUCT(--(Reps!D2:D9>8))
C6  =SUMPRODUCT(Reps!D2:D9,Reps!E2:E9)/SUM(Reps!D2:D9)
```

Check: Phone units in West or East = Ravi 12 + Zoya 15 = 27. Rows with more than 8 units: Ravi, Zoya, Neha, Rohan = 4.

NOTE **Why the double minus `--`?**

`(D2:D9>8)` produces TRUE/FALSE values. SUMPRODUCT ignores logical values unless they are converted to numbers. `--` (minus minus) turns TRUE into 1 and FALSE into 0. Multiplying by another array (`*`) also converts them, which is why the other formulas do not need it.

Putting it together: a self-updating region summary

Region summary with UNIQUE + SUMIFS + SORT

	A	B	C
1	Region =SORT(UNIQUE(B2:B9))	Revenue =SUMIFS(F2:F9,B2:B9,H2#)	Share
2	East	270,000	15.4%
3	North	637,000	36.2%
4	South	415,000	23.6%
5	West	436,000	24.8%

```
A2   =SORT(UNIQUE(Reps!B2:B9))
B2   =SUMIFS(Reps!F2:F9,Reps!B2:B9,"East")
C2   =B2/SUM(Reps!F2:F9)
```

In Microsoft 365 the second column is a single formula, `=SUMIFS(F2:F9,B2:B9,H2#)`: because H2# refers to the whole spilled list, SUMIFS spills one total per region. Add a new region to the data and the summary grows automatically.

Let's confirm the numbers with pandas:

Python

```
reps = pd.DataFrame({"rep": ["Asha","Ravi","Meera","Kabir","Zoya","Farhan","Neha","Rohan"],
                     "region": ["North","West","South","North","East","West","South","North"],
                     "units": [5,12,3,8,15,4,10,9], "price":
[55000,18000,55000,25000,18000,55000,25000,18000]})
reps["revenue"] = reps.units * reps.price
summary = reps.groupby("region")["revenue"].sum().to_frame()
summary["share"] = (summary.revenue / summary.revenue.sum()).round(3)
summary
```

OUTPUT

	revenue	share
region		
East	270000	0.15
North	637000	0.36
South	415000	0.24
West	436000	0.25

4 rows × 2 columns

Summary

KEY TAKEAWAYS

- Click **one cell** inside the table before sorting; use the Sort dialog for multi-level, custom-list and colour sorts.
- AutoFilter: AND across columns, OR within a column. Advanced Filter: AND on one row, OR across rows of a criteria range.
- Dynamic arrays **spill**; refer to the result with #; clear blocking cells to fix `#SPILL!`.
- `UNIQUE`, `SORT`, `SORTBY`, `FILTER`, `SEQUENCE`, `TAKE`, `VSTACK`, `GROUPBY` replace many manual steps with live formulas.
- In FILTER and SUMPRODUCT, * means AND and + means OR.
- `SUMPRODUCT` handles weighted sums and complex conditions in any Excel version.

Quick quiz

6 questions · tap an answer to check it

1. You select only the Revenue column and sort it, choosing "Continue with the current selection". What happens?

A The whole table is sorted correctly

B Only Revenue is reordered, so revenues no longer match their reps

C Excel refuses to sort

D The table is sorted by Rep name

Answer: B. Sorting a single column separates it from the rest of the row. Always sort the whole table.

2. In an Advanced Filter criteria range, two conditions on the same row mean:

- **A** AND
- **B** OR
- **C** NOT
- **D** XOR

Answer: A. Same row = AND; different rows = OR.

3. `=FILTER(A2:A9,(B2:B9="North")+(C2:C9="Phone"))` returns reps who are:

- **A** In the North and sell phones
- **B** In the North or sell phones (or both)
- **C** Not in the North
- **D** In the North but not selling phones

Answer: B. Adding conditions works as OR.

4. A formula returns #SPILL!. What is the most likely cause?

- **A** A typo in the function name
- **B** A cell in the spill area is not empty
- **C** The data has duplicates
- **D** The workbook is in manual calculation mode

Answer: B. Dynamic arrays need empty cells to spill into.

5. What does `=SUMPRODUCT(--(D2:D9>8))` calculate?

- **A** The sum of values greater than 8
- **B** The number of values greater than 8
- **C** The average of values greater than 8
- **D** The largest value

Answer: B. -- turns TRUE/FALSE into 1/0, and SUMPRODUCT adds them up: a count.

6. Which formula lists each region once, sorted A to Z?

- **A** `=UNIQUE(SORT(B2:B9),TRUE)`
- **B** `=SORT(UNIQUE(B2:B9))`
- **C** `=SORTBY(B2:B9,B2:B9)`
- **D** `=FILTER(B2:B9,B2:B9<>"")`

Answer: B. UNIQUE removes duplicates and SORT orders the result.

EXERCISE **Superstore with dynamic arrays**

In a copy of `superstore_sales.csv` (as a Table named S): (1) list the distinct sub-categories, sorted; (2) show all orders from Kochi with profit below zero; (3) show the 5 largest order lines by sales (order ID, city, sales); (4) count distinct customers.

Show solution (worked answer)

(1) `=SORT(UNIQUE(S[sub_category]))` (2) `=FILTER(S,(S[city]="Kochi")*(S[profit]<0),"None")` (3) `=TAKE(SORTBY(CHOOSECOLS(S,2,9,16),S[sales],-1),5)` (check the column numbers in your file: here 2 = order_id, 9 = city, 16 = sales) (4) `=COUNTA(UNIQUE(S[customer_id]))`

INTERVIEW CORNER

Questions you may be asked

Q: How would you get a list of unique values in Excel? `UNIQUE` in Microsoft 365; otherwise Remove Duplicates on a copy, Advanced Filter with "Unique records only", or a pivot table.

Q: Explain AND/OR logic in FILTER or SUMPRODUCT. Each condition becomes an array of TRUE/FALSE. Multiplying arrays gives 1 only when all are TRUE (AND); adding gives at least 1 when any is TRUE (OR).

Q: How do you sort by a custom order such as High, Medium, Low? Sort dialog → Order → Custom List, and add the list; or use `SORTBY` with `MATCH(range,{"High","Medium","Low"},0)` as the sort key.

ADVANCED EXCEL · LESSON 7

Lookup Functions: VLOOKUP, XLOOKUP and INDEX-MATCH

Bring data together from different tables. Master VLOOKUP and HLOOKUP (exact and approximate match), the modern XLOOKUP, the flexible INDEX + MATCH combination (including two-way and reverse lookups), OFFSET, consolidating several sheets, CHOOSE, and how to troubleshoot

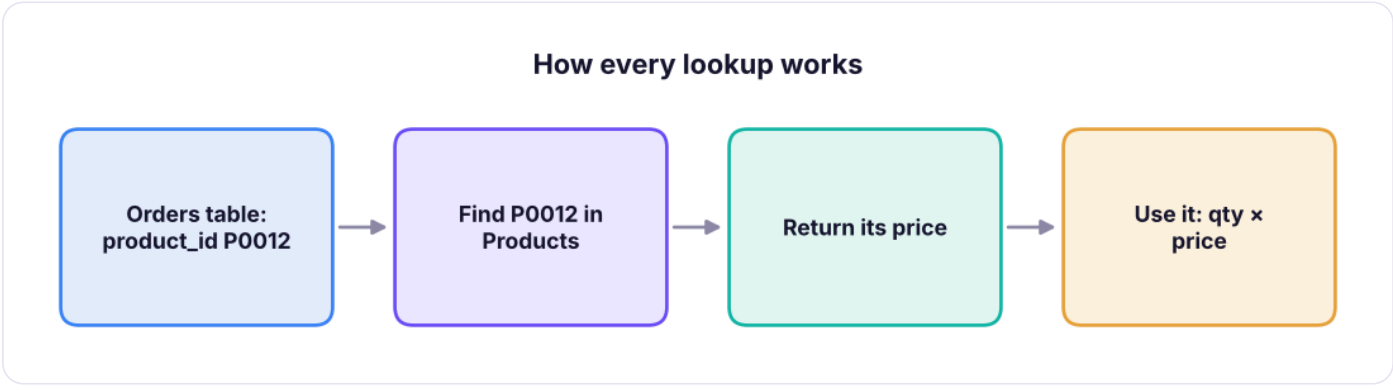
Live classes 4–5 · ✦ Excel for Microsoft 365, 2026 · Intermediate · ⏱ 120 min

By the end of this lesson you can

- Explain why lookups are the Excel equivalent of a SQL join
- Use VLOOKUP and HLOOKUP with exact and approximate match, and know their limits
- Use XLOOKUP for left lookups, default values, last-match search and multiple return columns
- Combine INDEX and MATCH for reverse (left) and two-way (matrix) lookups
- Use CHOOSE, and fix the common causes of

Why lookups matter

Business data is split across tables: orders in one, products in another, customers in a third. A **lookup** fetches a value from one table using a key that both tables share, such as a product ID or employee ID.



A lookup takes a key from one table, finds it in another table, and returns a value from the same row.

TIP **You are already learning SQL joins**

A lookup is a **left join** for one column at a time: keep every row of the main table and bring in matching information from the other table. You will see the same idea as `JOIN` in SQL (SQL Lesson 8), `merge()` in pandas, and relationships in Power BI.

We will use this lookup table on a sheet called `Prod`:

Prod: product lookup table (A1:E7)

	A	B	C	D	E
1	Product ID	Product	Category	Price	Stock
2	P101	Office Chair	Furniture	4500	32
3	P102	Standing Desk	Furniture	18500	8
4	P103	Laser Printer	Technology	12900	15
5	P104	Notebook Pack	Office Supplies	240	540
6	P105	Smartphone	Technology	21999	27
7	P106	Bookcase	Furniture	7600	0

VLOOKUP

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

- **lookup_value**: what you are looking for (the key).
- **table_array**: the table to search; the key **must be in its first column**.
- **col_index_num**: which column of that table to return (1 = first column).
- **range_lookup**: FALSE (or 0) for an **exact** match. Use TRUE only for approximate matching on sorted bands.

Order lines looking up product details with VLOOKUP

	A	B	C	D	E	F
1	Order	Product ID	Qty	Product =VLOOKUP(B2,Prod!$A$2:$E$7,2,FALSE)	Price	Amount
2	1001	P103	2	Laser Printer	12,900	25,800
3	1002	P101	6	Office Chair	4,500	27,000
4	1003	P105	1	Smartphone	21,999	21,999
5	1004	P104	20	Notebook Pack	240	4,800
6	1005	P109	3	#N/A	#N/A	#N/A

D2 `=VLOOKUP(B2,Prod!$A$2:$E$7,2,FALSE)`

F2 `=C2*E2`

Each pattern is copied down/across to the other cells in its column.

Row 6 shows `#N/A` because P109 does not exist in the product table. That is the correct answer: the key is not there.

COMMON MISTAKES **The four classic VLOOKUP mistakes**

1. **Forgetting FALSE**: the default is approximate match, which silently returns a *wrong* row for unsorted data.
2. **Not locking the table**: `A2:E7` copied down becomes `A3:E8`, so the last rows are missed. Use `$A$2:$E$7` or a Table name.
3. **Hard-coded column numbers**: inserting a column inside the table changes which column 4 means. Use `MATCH` for the column number, or XLOOKUP.
4. **Key not in the first column**: VLOOKUP cannot look to the left (use INDEX + MATCH or XLOOKUP).

Approximate match: bands and slabs

With TRUE, VLOOKUP finds the **largest value less than or equal to** the lookup value. The first column must be sorted ascending. This is perfect for tax slabs, commission tiers, discount bands and grades.

Approximate match for commission tiers

	A	B	C	D	E	F	G
1	Sales from	Commission		Rep	Sales	Rate	Commission
2	0	0%		Asha	85,000	0%	0
3	100000	3%		Ravi	240,000	3%	7,200
4	250000	5%		Meera	250,000	5%	12,500
5	500000	7%		Kabir	610,000	7%	42,700
6	1000000	10%		Zoya	1,200,000	10%	120,000

```
F2   =VLOOKUP(E2,$A$2:$B$6,2,TRUE)
G2   =E2*F2
```

Each pattern is copied down/across to the other cells in its column.

Ravi's 2,40,000 falls between 1,00,000 and 2,50,000, so he gets the 1,00,000 row: 3%. Meera's 2,50,000 matches the 2,50,000 row exactly: 5%.

HLOOKUP

HLOOKUP is VLOOKUP turned sideways: it searches the **first row** and returns a value from a row below. Use it when data runs across columns (for example months as headers).

HLOOKUP across a horizontal table

	A	B	C	D	E	F	G	H
1	Month	Jan	Feb	Mar	Apr		Find month	Mar
2	Target	120	130	145	150		Target →	145
3	Actual	118	136	139	162		Actual →	139

```
H2   =HLOOKUP(H1,$A$1:$E$3,2,FALSE)
H3   =HLOOKUP(H1,$A$1:$E$3,3,FALSE)
```

XLOOKUP: the modern lookup (Microsoft 365 / Excel 2021+)

`=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`

XLOOKUP fixes every VLOOKUP limitation:

Feature	VLOOKUP	XLOOKUP
Default match	Approximate (risky)	**Exact**
Look to the left	No	**Yes**: lookup and return arrays are separate
Column inserted in the table	Breaks the index number	**Safe**
Custom "not found" message	Needs IFERROR	Built-in 4th argument
Return several columns	One at a time	**Spills** many columns
Search last to first	No	`search_mode = -1`
Wildcards	Yes	`match_mode = 2`
Next smaller / larger match	Sorted data only	`match_mode = -1` or 1, any order

XLOOKUP examples (lookup table on sheet Prod)

	A	B	C
1	**Need**	**Formula**	**Result**
2	Price of P105	=XLOOKUP("P105",Prod!A2:A7,Prod!D2:D7)	21,999
3	ID of "Bookcase" (left lookup)	=XLOOKUP("Bookcase",Prod!B2:B7,Prod!A2:A7)	P106
4	Missing ID with a message	=XLOOKUP("P109",Prod!A2:A7,Prod!B2:B7,"Not found")	Not found
5	Product, category and price at once	=XLOOKUP("P103",Prod!A2:A7,Prod!B2:D7)	Laser Printer
6	Last Furniture item in the list	=XLOOKUP("Furniture",Prod!C2:C7,Prod!B2:B7,,0,-1)	Bookcase
7	Product name containing "Desk"	=XLOOKUP("*Desk*",Prod!B2:B7,Prod!D2:D7,,2)	18,500

```
C2  =XLOOKUP("P105",Prod!A2:A7,Prod!D2:D7)
C3  =XLOOKUP("Bookcase",Prod!B2:B7,Prod!A2:A7)
C4  =XLOOKUP("P109",Prod!A2:A7,Prod!B2:B7,"Not found")
C5  =XLOOKUP("P103",Prod!A2:A7,Prod!B2:D7)
C6  =XLOOKUP("Furniture",Prod!C2:C7,Prod!B2:B7,,0,-1)
C7  =XLOOKUP("*Desk*",Prod!B2:B7,Prod!D2:D7,,2)
```

The fourth row returns three columns at once: the result spills across C5:E5.

XLOOKUP for bands (no sorting needed)

`match_mode = -1` returns the exact match or the **next smaller** item, like VLOOKUP's TRUE, but the table does not need to be sorted:

`=XLOOKUP(E2, $A$2:$A$6, $B$2:$B$6, , -1)` gives the same commission rates as the approximate VLOOKUP above.

> **TIP** **XMATCH**
>
> `XMATCH` is the modern version of MATCH with the same match and search modes as XLOOKUP. `=XMATCH("P105",Prod!A2:A7)` → 5.

INDEX and MATCH

Before XLOOKUP, professionals used **INDEX + MATCH**, and you will still see it everywhere. It works in every Excel version and is very flexible.

- `=INDEX(array, row_num, [col_num])` returns the value **at a position**.
- `=MATCH(lookup_value, lookup_array, [match_type])` returns the **position** of a value. Use `0` for exact match.

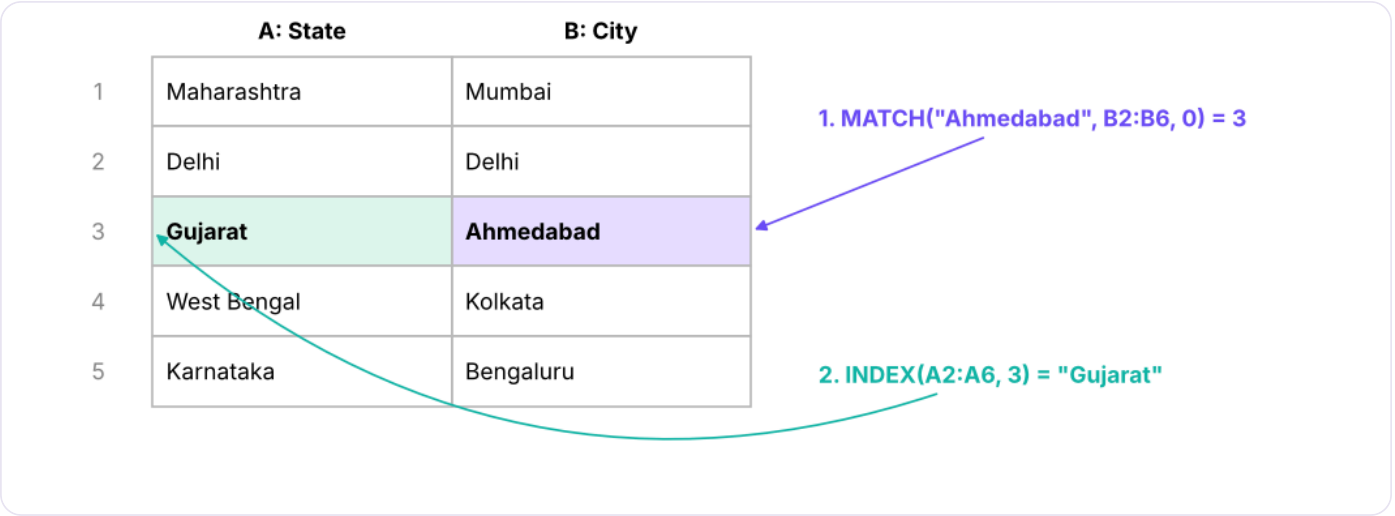
INDEX returns a value at a position; MATCH returns the position of a value

	A	B	C	D	E	F
1	State	City	Population 2011 (mn)		Formula	Result
2	Maharashtra	Mumbai	18.4		=INDEX(A2:C6,2,1)	Delhi
3	Delhi	Delhi	16.3		=MATCH("Ahmedabad",B2:B6,0)	3
4	Gujarat	Ahmedabad	6.4		=MATCH("Population 2011 (mn)",A1:C1,0)	3
5	West Bengal	Kolkata	14.1		=INDEX(C2:C6,MATCH("Kolkata",B2:B6,0))	14.1
6	Karnataka	Bengaluru	8.5		=INDEX(A2:A6,MATCH("Ahmedabad",B2:B6,0))	Gujarat

```
F2   =INDEX(A2:C6,2,1)
F3   =MATCH("Ahmedabad",B2:B6,0)
F4   =MATCH("Population 2011 (mn)",A1:C1,0)
F5   =INDEX(C2:C6,MATCH("Kolkata",B2:B6,0))
F6   =INDEX(A2:A6,MATCH("Ahmedabad",B2:B6,0))
```

Reverse (left) lookup

In the table above, City is in column B and State is in column A. VLOOKUP cannot return a column to the **left** of the key, but INDEX + MATCH can: `=INDEX(A2:A6, MATCH("Ahmedabad", B2:B6, 0))` → **Gujarat**.



INDEX + MATCH: MATCH finds the row number of the city; INDEX returns the state on that row.

Two-way (matrix) lookup

INDEX can take a row **and** a column number. Use one MATCH for each to look up the intersection of a row and a column, for example an employee's salary from a table where both the employee and the field are chosen in drop-downs.

Two-way lookup: =INDEX($B$2:$E$6, MATCH($G3,$A$2:$A$6,0), MATCH(H$2,$B$1:$E$1,0))

	A	B	C	D	E	F	G	H	I
1	EmpName	DepartmentID	JobTitle	Salary	City				
2	Employee_1	D10	Developer	82000	Pune		Employee	JobTitle	Salary
3	Employee_2	D20	Analyst	68000	Delhi		Employee_4	Manager	125000
4	Employee_3	D10	Tester	54000	Pune		Employee_2	Analyst	68000
5	Employee_4	D30	Manager	125000	Mumbai				
6	Employee_5	D20	Developer	79000	Delhi				

```
H3   =INDEX($B$2:$E$6,MATCH($G3,$A$2:$A$6,0),MATCH(H$2,$B$1:$E$1,0))
I3   =INDEX($B$2:$E$6,MATCH($G3,$A$2:$A$6,0),MATCH(I$2,$B$1:$E$1,0))
H4   =INDEX($B$2:$E$6,MATCH($G4,$A$2:$A$6,0),MATCH(H$2,$B$1:$E$1,0))
I4   =INDEX($B$2:$E$6,MATCH($G4,$A$2:$A$6,0),MATCH(I$2,$B$1:$E$1,0))
```

Because the headers in row 2 (JobTitle, Salary) are matched by MATCH, you can change a header to `City` and the formula returns cities without editing it. The XLOOKUP version nests one XLOOKUP inside another: `=XLOOKUP($G3,$A$2:$A$6,XLOOKUP(H$2,$B$1:$E$1,$B$2:$E$6))`.

INDEX + MATCH with multiple conditions

To match on two columns at once (for example Region **and** Product), multiply the conditions inside MATCH:

`=INDEX(F2:F9, MATCH(1, (B2:B9="North")*(C2:C9="Tablet"), 0))` → returns the revenue for the North + Tablet row. In Microsoft 365 this works as a normal formula; in older Excel press `Ctrl + Shift + Enter`. With XLOOKUP: `=XLOOKUP(1,(B2:B9="North")*(C2:C9="Tablet"),F2:F9)`.

OFFSET and consolidating worksheets

In class we also used lookups to **consolidate** data: pull values from several tables or sheets into one summary. Three tools do this job.

OFFSET: a value or range by position

`=OFFSET(reference, rows, cols, [height], [width])` starts at `reference`, moves down `rows` and right `cols`, and returns that cell (or a range of `height` × `width`). Combined with MATCH it does the same two-way lookup as INDEX + MATCH. The table below holds month-end share prices (illustrative numbers for fictional companies).

Two-way lookup of a share price with INDEX + MATCH and with OFFSET + MATCH

	A	B	C	D	E	F	G	H
1	Date	Asha Foods	Kiran Motors	Meru Steel		Company	Date	INDEX + MATCH
2	31-01-2026	412	1180	96		Kiran Motors	31-03-2026	1262
3	28-02-2026	398	1215	101		Meru Steel	31-05-2026	112
4	31-03-2026	425	1262	99		Asha Foods	28-02-2026	398
5	30-04-2026	440	1240	108				Last 3 months avg (Asha)
6	31-05-2026	436	1298	112				

```
H2   =INDEX($B$2:$D$6,MATCH(G2,$A$2:$A$6,0),MATCH(F2,$B$1:$D$1,0))
```

```
I2    =OFFSET($A$1,MATCH(G2,$A$2:$A$6,0),MATCH(F2,$B$1:$D$1,0))
I5    =AVERAGE(OFFSET($B$1,COUNT($B$2:$B$6)-2,0,3,1))
```

Each pattern is copied down/across to the other cells in its column.

The last formula shows OFFSET's real strength: `OFFSET($B$1, COUNT(B2:B6)-2, 0, 3, 1)` is a range of the **last three** prices, so the average updates automatically when a new month is added. The same idea builds dynamic named ranges and charts (Lesson 9).

> COMMON MISTAKES **OFFSET is volatile**
>
> OFFSET (like INDIRECT, TODAY and RAND) recalculates every time anything in the workbook changes. A few hundred OFFSETs are fine; tens of thousands slow a workbook down. For lookups prefer INDEX + MATCH or XLOOKUP, and for growing ranges prefer an Excel Table.

Consolidating several sheets into one

Situation	Best tool	How
Same layout on every sheet (Jan, Feb, Mar…) and you need totals	3-D reference	`=SUM(Jan:Mar!B2)` adds cell B2 on every sheet from Jan to Mar (Lesson 4)
Same kind of data, but rows in a different order on each sheet	**Data → Consolidate**	Choose Sum, add each sheet's range, tick *Top row* and *Left column*, optionally *Create links to source data*
Different tables that share a key (city code, product ID)	Lookups	Bring each column in with XLOOKUP or INDEX + MATCH, as in the class tasks below
Many files every month	Power Query	*Get Data → From Folder*, then Append (Lesson 12)

Class practice: city tables

These are the consolidation tasks from the live class. Table 1 maps cities to states and population; Table 2 maps city codes to cities.

Consolidation tasks: state from city, population from city code, grade from marks

	A	B	C	D	E	F	G
1	City Code	City	State	Population 2011 (mn)		Task	Input
2	C1	Mumbai	Maharashtra	12.48		State of this city	Pune
3	C2	Delhi	Delhi	11.01		Population of city code	C31
4	C3	Ahmedabad	Gujarat	5.57		Population of city code	C4
5	C4	Surat	Gujarat	4.47		City code of this city	Allahabad
6	C31	Allahabad	Uttar Pradesh	1.11			
7	C21	Pune	Maharashtra	3.12		Marks	Grade (approx. match)
8						0	D
9						58	C
10						91	A
11						Min marks	Grade
12						0	D
13						40	C
14						60	B
15						80	A

```
H2   =IFNA(INDEX($C$2:$C$7,MATCH(G2,$B$2:$B$7,0)),"Not found")
H3   =IFNA(INDEX($D$2:$D$7,MATCH(G3,$A$2:$A$7,0)),"Not found")
H5   =IFNA(INDEX($A$2:$A$7,MATCH(G5,$B$2:$B$7,0)),"Not found")
G8   =VLOOKUP(F8,$F$12:$G$15,2,TRUE)
```

Each pattern is copied down/across to the other cells in its column.

The state and city-code answers use INDEX + MATCH wrapped in IFNA; `=XLOOKUP(G2,B2:B7,C2:C7,"Not found")` gives the same result in Microsoft 365. The grade lookup uses **approximate match** (TRUE): VLOOKUP finds the largest "Min marks" that is less than or equal to the marks, so the grade table must be sorted ascending. This is the classic use of approximate match: tax slabs, commission bands and grades.

CHOOSE, and VLOOKUP with CHOOSE

`=CHOOSE(index_num, value1, value2, …)` returns the item at a position.

CHOOSE examples

	A	B	C
1	Input	Formula	Result
2	2	=CHOOSE(A2,"Bronze","Silver","Gold")	Silver
3	27-04-2026	=CHOOSE(WEEKDAY(A3,2),"Mon","Tue","Wed","Thu","Fri","Sat","Sun")	Mon
4	15-08-2026	="Q"&CHOOSE(MONTH(A4),4,4,4,1,1,1,2,2,2,3,3,3)	Q2

```
C2   =CHOOSE(A2,"Bronze","Silver","Gold")
C3   =CHOOSE(WEEKDAY(A3,2),"Mon","Tue","Wed","Thu","Fri","Sat","Sun")
C4   ="Q"&CHOOSE(MONTH(A4),4,4,4,1,1,1,2,2,2,3,3,3)
```

The last example gives the **Indian financial-year quarter** (April–June = Q1), which a plain `ROUNDUP(MONTH()/3)` cannot do.

VLOOKUP + CHOOSE was the old trick for a left lookup: `CHOOSE({1,2}, B2:B6, A2:A6)` builds a virtual two-column table with City first and State second, so VLOOKUP can search it:

`=VLOOKUP("Ahmedabad", CHOOSE({1,2}, B2:B6, A2:A6), 2, FALSE)` → Gujarat.

Today, prefer XLOOKUP or INDEX + MATCH: they are easier to read and faster.

Troubleshooting #N/A

Cause	How to check	Fix
The key really is missing	Search the lookup column with `Ctrl + F`	Return a message: `IFNA(…,"Not found")` or XLOOKUP's 4th argument
Extra spaces	`=LEN(A2)` is longer than expected	`TRIM` both sides, or `XLOOKUP(TRIM(B2),TRIM(Prod!A2:A7),…)`
Number vs text (`101` vs `"101"`)	`=ISTEXT(B2)` differs from the lookup column	Convert one side: `VALUE()` or `TEXT(…,"0")`
Approximate match on unsorted data	Wrong but no error	Use exact match (FALSE / 0)
Range not locked	Errors appear only in lower rows	Add $ or use a Table
Invisible characters	`=CODE(RIGHT(B2))` returns 160 or 10	`SUBSTITUTE` and `CLEAN`

2026 UPDATE **Lookups in 2026**

- **XLOOKUP** and **XMATCH** are now the standard; use VLOOKUP only when sharing files with very old Excel versions.
- **FILTER** returns **all** matching rows, not just the first (Lesson 6).
- For joining large tables regularly, **Power Query → Merge Queries** (Lesson 12) is faster and repeatable, and it is exactly a SQL join.

Case: pricing the Superstore order lines

Suppose the order lines arrive without a price and you must look it up from the product master (`products.csv`). This is a lookup on 6,323 rows. pandas does the same as XLOOKUP with `merge`:

Python

```python
import pandas as pd
orders = pd.read_csv("superstore_sales.csv")[["order_id", "product_id", "quantity"]]
products = pd.read_csv("products.csv")
priced = orders.merge(products[["product_id", "product_name", "list_price"]], on="product_id",
how="left")   # = XLOOKUP for every row
priced["gross_amount"] = priced.quantity * priced.list_price
print("Rows:", len(priced), "| unmatched product IDs:", priced.list_price.isna().sum())
priced.head(5)
```

OUTPUT

```
Rows: 6323 | unmatched product IDs: 0
```

	order_id	product_id	quantity	product_name	list_price	gross_amount
0	IN-2023-00001	P0059	2	Nova Phone 223	18,000	36,000
1	IN-2023-00001	P0050	1	Orbit Appliance 798	3,850	3,850
2	IN-2023-00001	P0011	4	Orbit Table 248	15,520	62,080
3	IN-2023-00002	P0037	2	Crest Storage 946	3,260	6,520
4	IN-2023-00003	P0018	3	Crest Bookcase 170	11,710	35,130

5 rows × 6 columns

Zero unmatched IDs tells you every order line found its product: the same check you should do in Excel with `=COUNTIF(price_column,"#N/A")` or `=SUM(--ISNA(price_column))`.

Summary

KEY TAKEAWAYS

- Lookups bring columns from one table into another through a shared key, like a SQL left join.
- VLOOKUP: key in the first column, column number counted from 1, **FALSE for exact match**, lock the table with $.
- Approximate match (TRUE / match_mode −1) is for sorted bands: tax slabs, commissions, grades.
- XLOOKUP is exact by default, looks left, has a built-in "not found" value, returns several columns and can search from the end.
- INDEX returns a value at a position; MATCH finds the position. Together they do reverse and two-way lookups in any Excel version.
- Most #N/A errors come from missing keys, spaces, or number-vs-text mismatches.

Quick quiz

6 questions · tap an answer to check it

1. `=VLOOKUP("P103",A2:E7,4,FALSE)` returns a value from which column of the range?

- **A** A
- **B** C
- **C** D
- **D** E

Answer: C. Column index 4 counts from the first column of the table array: A=1, B=2, C=3, D=4.

2. Why does VLOOKUP return the wrong price (not #N/A) for unsorted data?

- **A** The fourth argument was omitted, so it used approximate match
- **B** The table was not locked with $
- **C** The column number was too large
- **D** VLOOKUP cannot handle text keys

Answer: A. Without FALSE, VLOOKUP uses approximate match, which assumes sorted data.

3. Which formula returns the State for the city in H2, where State is column A and City is column B?

- **A** `=VLOOKUP(H2,A2:B6,1,FALSE)`
- **B** `=INDEX(A2:A6,MATCH(H2,B2:B6,0))`
- **C** `=MATCH(H2,A2:A6,0)`
- **D** `=HLOOKUP(H2,A2:B6,1,FALSE)`

Answer: B. MATCH finds the row of the city in column B; INDEX returns column A on that row.

4. Sales of 2,40,000 with tiers at 0, 1,00,000, 2,50,000 and 5,00,000 (approximate match). Which tier is used?

A 0

B 1,00,000

C 2,50,000

D 5,00,000

Answer: B. Approximate match returns the largest tier less than or equal to the value.

5. What does XLOOKUP's 4th argument do?

A Sets the match mode

B Returns a value when nothing is found

C Searches from last to first

D Returns several columns

Answer: B. `if_not_found` replaces #N/A with your own value or message.

6. A lookup returns #N/A even though the ID is visible in both tables. Which is NOT a likely cause?

A One side has trailing spaces

B One side is a number and the other is text

C The workbook is saved as .xlsx

D There is an invisible non-breaking space

Answer: C. File format does not affect lookups. Spaces, invisible characters and type mismatches are the usual culprits.

EXERCISE **Grade lookup**

Create a grade table: 0 → F, 40 → D, 55 → C, 70 → B, 85 → A. Write a formula that returns the grade for a mark in B2 using (a) VLOOKUP, (b) XLOOKUP and (c) INDEX + MATCH.

Show solution (worked answer)

With the table in E2:F6 (marks ascending): (a) `=VLOOKUP(B2,$E$2:$F$6,2,TRUE)` (b) `=XLOOKUP(B2,$E$2:$E$6,$F$2:$F$6,,-1)` (c) `=INDEX($F$2:$F$6,MATCH(B2,$E$2:$E$6,1))` (match_type 1 = largest value ≤ B2, needs ascending order).

EXERCISE **Two-way lookup**

Make a table of 4 regions (rows) × 4 quarters (columns) with sales figures. Add drop-downs for region (H1) and quarter (H2) and return the matching sales value.

Show solution (worked answer)

With regions in A2:A5, quarters in B1:E1 and values in B2:E5: `=INDEX($B$2:$E$5,MATCH($H$1,$A$2:$A$5,0),MATCH($H$2,$B$1:$E$1,0))` or `=XLOOKUP($H$1,$A$2:$A$5,XLOOKUP($H$2,$B$1:$E$1,$B$2:$E$5))`. Make the drop-downs with Data Validation → List, source `$A$2:$A$5` and `$B$1:$E$1`.

INTERVIEW CORNER **Questions you may be asked**

Q: What are the limitations of VLOOKUP and how do you overcome them? It cannot look left, it defaults to approximate match, it breaks when columns are inserted (hard-coded index), and it returns only the first match. Use XLOOKUP or INDEX + MATCH, and FILTER for multiple matches.

Q: Why is INDEX + MATCH considered more flexible than VLOOKUP? The lookup and return ranges are separate, so it can look left, it is not affected by inserted columns, it supports two-way lookups, and it can be faster on large sheets because it only reads the columns it needs.

Q: How would you look up a value using two criteria? `XLOOKUP(1,(range1=x)*(range2=y),return_range)`, `INDEX/MATCH` with the same multiplied condition, or a helper column that concatenates the two keys.

ADVANCED EXCEL · LESSON 8

Pivot Tables from Basic to Advanced

Summarise thousands of rows in seconds. Build pivot tables, understand the pivot cache and the four areas, change value field settings, show values as % of total, MoM and YoY change and running totals, group dates and numbers into bins, add calculated fields, slicers and timelines, and solve the Video Game Sales case study.

● Live classes 5–7 ✦ Excel for Microsoft 365, 2026 Intermediate ⏱ 150 min

By the end of this lesson you can

- Build a pivot table and explain the pivot cache, the four areas and Refresh vs Refresh All
- Change the summary function and number format, and show values as % of total, % of row and running totals
- Calculate month-on-month (MoM) and year-on-year (YoY) change
- Group dates by year, quarter and month, group items manually, and group numbers into bins
- Add calculated fields, slicers, timelines and pivot charts, and answer a full case study

What a pivot table does

A pivot table **groups** rows by one or more fields and **summarises** a number for each group (sum, count, average...). It is the fastest way to answer "how much, by what?" questions, with no formulas.

Raw data: 6,323 rows

order_id	region	category	sales
IN-2023-00001	South	Technology	36000.0
IN-2023-00001	South	Office Supplies	3850.0
IN-2023-00001	South	Furniture	62080.0
IN-2023-00002	West	Office Supplies	5216.0
IN-2023-00003	West	Furniture	24591.0
IN-2023-00003	West	Furniture	8288.0
IN-2023-00003	West	Office Supplies	2100.0
IN-2023-00003	West	Technology	69280.0
IN-2023-00004	East	Office Supplies	840.0

Pivot: sum of sales (₹ lakh)

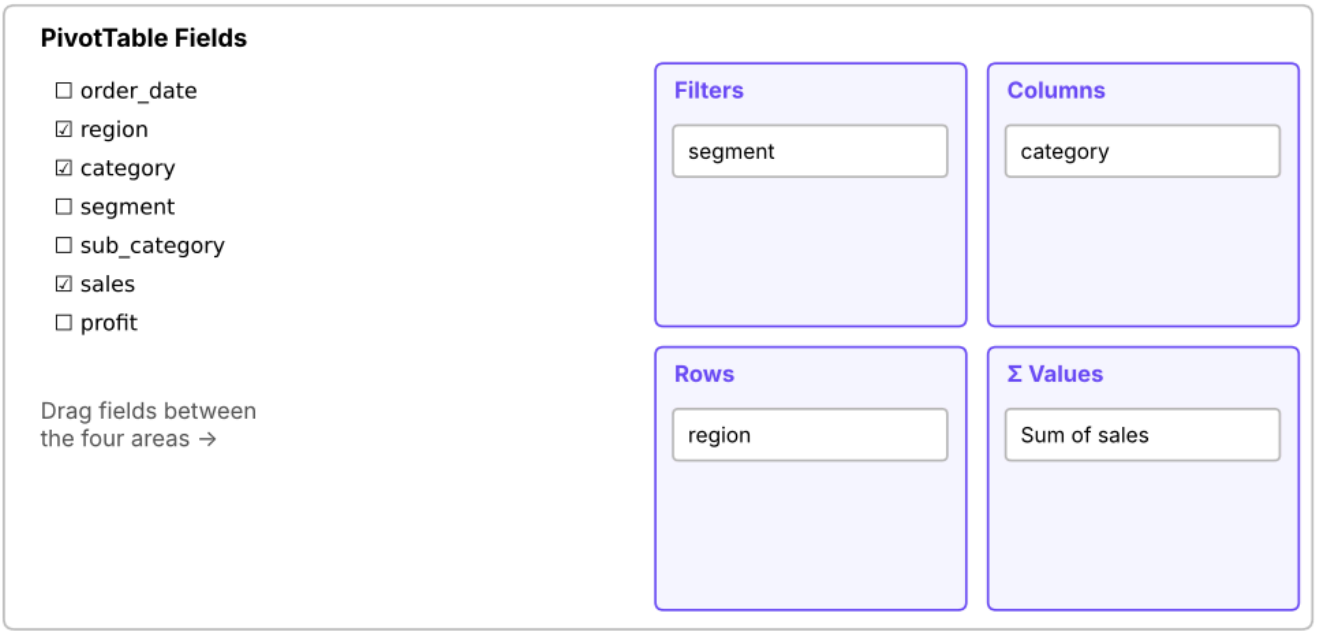
Region	Furniture	Office Supplies	Technology	Total
East	79.4	27.2	358.2	464.8
North	79.8	22.2	388.2	490.2
South	95.9	24.7	418.9	539.5
West	93.3	27.3	399.0	519.6
Grand Total	**348.4**	**101.3**	**1,564.4**	**2,014.1**

A pivot table turns thousands of order lines into a compact summary: rows grouped by Region, columns by Category, values summed.

Creating a pivot table

STEP BY STEP **Step by step**

1. Make sure the data is **structured** (Lesson 1): one header row, no blank rows, no merged cells. Ideally convert it to a Table (`Ctrl + T`) so new rows are included automatically.
2. Click any cell in the data → **Insert → PivotTable** (or `Alt, N, V`).
3. Choose **New Worksheet** → OK. An empty pivot appears with the **PivotTable Fields** pane on the right.
4. Drag fields into the four areas: drag `region` to **Rows**, `category` to **Columns**, `sales` to **Values**.
5. Right-click any number → **Number Format** → `#,##0` so the format sticks when the pivot refreshes.

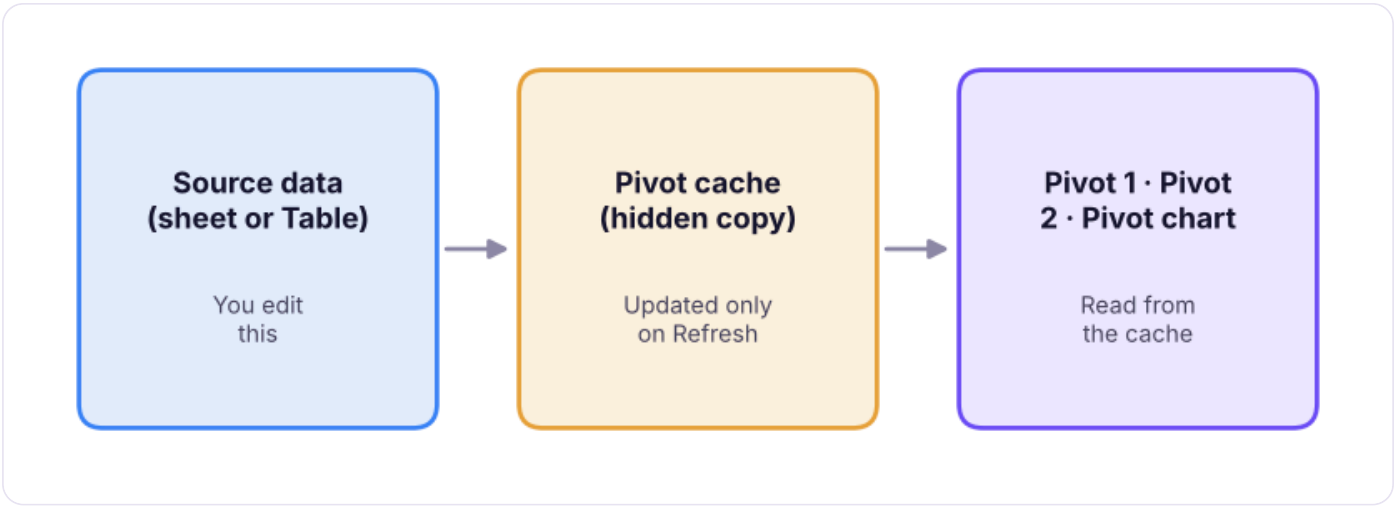


The PivotTable Fields pane: a list of fields at the top, four areas at the bottom.

Area	What it does	Example
Rows	One row per unique value	Region, Category, Order Date
Columns	One column per unique value	Year, Segment
Values	The numbers being summarised	Sum of Sales, Count of Order ID
Filters	Filter the whole pivot by a field	Segment = Corporate

The pivot cache, Refresh and Refresh All

A pivot table does not read your sheet directly. When you create it, Excel copies the data into a hidden **pivot cache**, and the pivot reads from that cache. This is why pivots are fast, and also why they do **not** update automatically when the data changes.



Every pivot reads from a cache. Pivots made from the same data share one cache.

Action	Shortcut	Effect
Refresh	`Alt + F5`	Re-reads the source into this pivot's cache. Every pivot that **shares the same cache** updates too
Refresh All	`Ctrl + Alt + F5`	Refreshes every pivot cache, query and data connection in the workbook
Change Data Source	PivotTable Analyze tab	Point the pivot to a bigger range (unnecessary if the source is a Table)
Refresh on open	PivotTable Options → Data	Refreshes automatically when the file opens

> **TIP** **Use a Table as the source**
>
> If the source is a Table, new rows are included the next time you refresh. If the source is a fixed range like `A1:R6324`, new rows below are ignored until you change the data source.

Value field settings

Click a field in Values → **Value Field Settings** (or right-click a number).

Summarize values by

Sum, Count, Average, Max, Min, Product, Count Numbers, StdDev, Var. Text fields in Values are counted automatically. **Distinct Count** appears only when the pivot is added to the **Data Model** (tick "Add this data to the Data Model" when creating it; Lesson 12).

Show values as

This tab turns a plain sum into an analytical view without writing formulas.

Show values as	Answers	Example
% of Grand Total	What share of the whole?	Furniture = 22% of all sales
% of Column Total	Share within each column	Within 2025, North = 31%
% of Row Total	Mix within each row	In the West, 46% of sales are Technology
% of Parent Row Total	Share within a parent group	Each sub-category's share of its category
Difference From (previous)	Change from the previous item	MoM or YoY change in ₹
% Difference From (previous)	Growth rate	MoM % or YoY %
Running Total In	Cumulative total	Year-to-date sales
Rank Largest to Smallest	Position	Top-selling region = 1
Index	Relative importance	Above 1 = over-represented

Here are three of these views on the Superstore data (computed with pandas to show the exact numbers a pivot would give):

Python

```python
import pandas as pd
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
pv = s.pivot_table(index="region", columns="category", values="sales", aggfunc="sum")
pct_total = (pv / pv.values.sum() * 100).round(1)
pct_row = (pv.div(pv.sum(axis=1), axis=0) * 100).round(1)
print("Sum of sales (₹):"); print(pv.round(0).astype(int).to_string())
print("\n% of grand total:"); print(pct_total.to_string())
print("\n% of row total (category mix inside each region):"); print(pct_row.to_string())
```

OUTPUT

```
Sum of sales (₹):
category  Furniture  Office Supplies  Technology
region
East        7942092          2719513    35822514
North       7975868          2219542    38822750
South       9593302          2468888    41890579
West        9330590          2725360    39902570

% of grand total:
category  Furniture  Office Supplies  Technology
region
East            3.9              1.4        17.8
North           4.0              1.1        19.3
South           4.8              1.2        20.8
West            4.6              1.4        19.8

% of row total (category mix inside each region):
category  Furniture  Office Supplies  Technology
region
East           17.1              5.9        77.1
```

```
North          16.3            4.5         79.2
South          17.8            4.6         77.6
West           18.0            5.2         76.8
```

MoM and YoY

Month-on-month (MoM) compares each month with the previous month; **year-on-year (YoY)** compares a period with the same period last year (which removes seasonality, such as the festive Q4 peak).

STEP BY STEP **MoM % in a pivot**

1. Put `order_date` in Rows and group it by **Years** and **Months** (see Grouping below).
2. Add `sales` to Values **twice**.
3. On the second copy: Value Field Settings → Show Values As → **% Difference From** → Base field: `order_date` (Months) → Base item: **(previous)**.
4. Rename the field "MoM %".

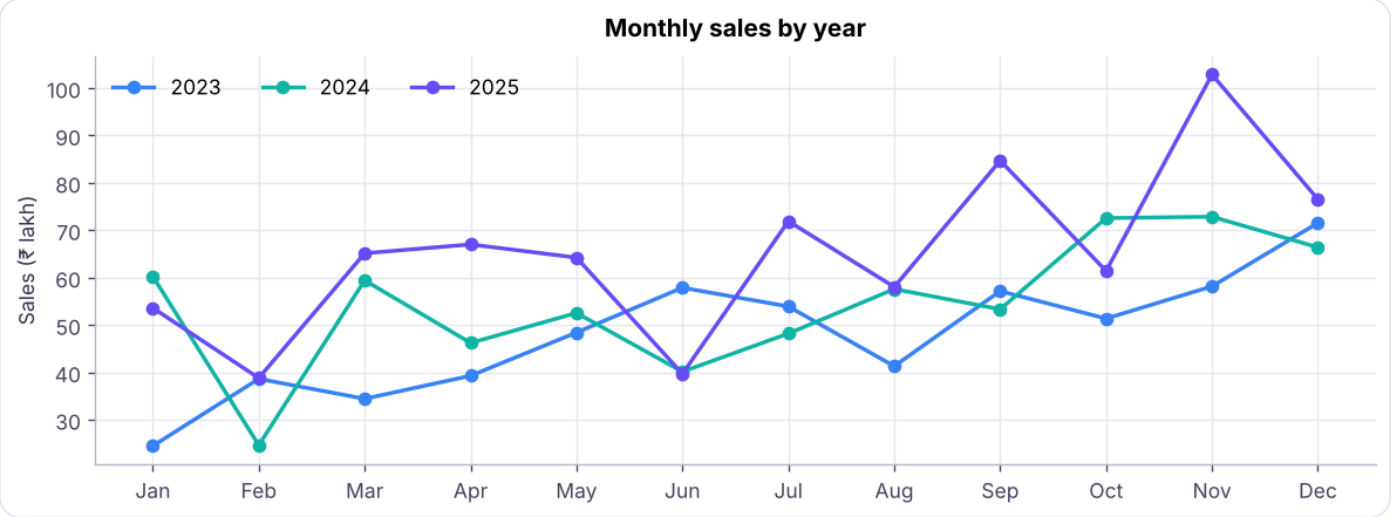
Python

```python
m = s.set_index("order_date").resample("MS")["sales"].sum().to_frame("sales")
m["MoM %"] = (m.sales.pct_change() * 100).round(1)
m["YoY %"] = (m.sales.pct_change(12) * 100).round(1)
m.index = m.index.strftime("%b-%Y")
m.tail(8)
```

OUTPUT

order_date	sales	MoM %	YoY %
May-2025	6,421,274	-4.2	22.2
Jun-2025	3,968,572	-38.2	-1.1
Jul-2025	7,190,087	81.2	49.2
Aug-2025	5,802,184	-19.3	0.8
Sep-2025	8,468,060	45.9	58.9
Oct-2025	6,140,290	-27.5	-15.4
Nov-2025	10,284,516	67.5	41.2
Dec-2025	7,652,266	-25.6	15.3

8 rows × 3 columns



Monthly sales with YoY growth. The festive season (Oct–Dec) peaks every year, so YoY is a fairer comparison than MoM.

In formulas, the same calculations are: MoM % `=(C3-C2)/C2`, YoY % `=(C14-C2)/C2` when months are listed one per row.

Grouping

Group dates

Right-click a date in Rows → **Group** → choose Years, Quarters, Months (and Days, Hours if needed). Excel 365 groups dates automatically when you drag a date field to Rows; you can ungroup with right-click → Ungroup.

> **TIP** **Indian financial year**
>
> Pivot grouping uses calendar quarters (Jan–Mar = Q1). For an April–March financial year, add helper columns to the data first: `FY = YEAR(date)+(MONTH(date)>=4)` and `FQ = "Q"&ROUNDUP(MOD(MONTH(date)-4,12)+1)/3,0)`, or use `CHOOSE` (Lesson 7), and put those in the pivot.

Group items manually

Select several labels (hold `Ctrl`) → right-click → **Group**. Excel creates a new field such as `Product2` with a group called `Group1`; rename it (for example "Low Sales Products"). This is how you combine the three lowest-selling products into one line, or merge small cities into "Others".

Groups vs bins

	Groups	Bins
What	Segregate items into chunks based on a common behaviour or criteria	Split a **numeric** field into equal-width intervals
Size	Can be any size	Every bin has the same width
Works on	Text, dates and numbers	Numbers
Example	"Metro" vs "Non-metro" cities	Order value 0–5,000, 5,000–10,000 …

To bin numbers: put the number field (for example `sales`) in **Rows** → right-click → Group → Starting at 0, Ending at 50,000, By 5,000. Then put `order_id` in Values as a Count.

Python

```
bins = pd.cut(s.sales, range(0, 55000, 5000), right=False)
dist = s.groupby(bins, observed=False).size().to_frame("Count of order lines")
dist.index = [f"{int(i.left):,}-{int(i.right - 1):,}" for i in dist.index]
dist
```

OUTPUT

	Count of order lines
0–4,999	2530
5,000–9,999	829
10,000–14,999	473
15,000–19,999	423
20,000–24,999	238
25,000–29,999	172
30,000–34,999	177
35,000–39,999	202
40,000–44,999	61
45,000–49,999	92

10 rows × 1 columns

Most order lines are small; a few very large ones (machines, copiers) form the long tail. This is a frequency distribution, the pivot-table version of a histogram (Lesson 10).

Layout, subtotals and grand totals

The **Design** tab controls how the pivot looks:

Setting	Options	Tip
Report Layout	Compact (default), Outline, **Tabular**	Tabular + **Repeat All Item Labels** gives a flat table you can copy and reuse
Subtotals	Don't show, at bottom, at top	Turn off for exports
Grand Totals	On for rows/columns, off	Off when sharing a percentage view
Blank Rows	Insert after each item	Easier to read printed reports
Expand / Collapse	+/- buttons, or right-click → Expand/Collapse entire field	Drill from Category down to Sub-category
Show Details	Double-click any number	Opens a new sheet with the rows behind that number: great for auditing

Parent total: when two fields are in Rows (Category → Sub-category), "% of Parent Row Total" shows each sub-category's share of its own category.

Sorting and filtering inside a pivot

- **Sort**: right-click a value → Sort → Largest to Smallest.
- **Label filter**: Row Labels drop-down → Label Filters → Begins with, contains ...
- **Value filter / Top 10**: Row Labels drop-down → Value Filters → **Top 10** → change to Top 5 Items by Sum of sales. This is how you answer "Find the top 5 categories".
- **Slicers**: PivotTable Analyze → **Insert Slicer** → pick fields (Region, Segment). Slicers are clickable buttons that filter the pivot and show what is selected. Right-click a slicer → **Report Connections** to control several pivots with one slicer (they must share the same cache).
- **Timeline**: Insert Timeline on a date field to filter by years, quarters, months or days with a slider.

Calculated fields

A **calculated field** creates a new value field from other fields in the pivot: PivotTable Analyze → Fields, Items & Sets → **Calculated Field**.

Example: `Profit Margin = profit / sales`, formatted as a percentage.

> **WATCH OUT** **Calculated fields work on totals**
>
> Excel calculates the field on the **sums**: Profit Margin for the North = total North profit ÷ total North sales. That is correct for ratios. But a calculated field like `=quantity * price` would multiply the *sum* of quantity by the *sum* of price, which is wrong. Add such row-level calculations as a new column in the source data instead.

Python

```python
pm = s.groupby("category")[["sales", "profit"]].sum()
pm["Profit Margin"] = (pm.profit / pm.sales * 100).round(1).astype(str) + "%"
pm.round(0)
```

OUTPUT

	sales	profit	Profit Margin
category			
Furniture	34,841,853	-818,153	-2.3%
Office Supplies	10,133,304	1,514,794	14.9%
Technology	156,438,412	17,655,524	11.3%

3 rows × 3 columns

GETPIVOTDATA and pivot charts

When you type = and click a cell inside a pivot, Excel writes `GETPIVOTDATA`, for example `=GETPIVOTDATA("sales",$A$3,"region","North")`. It keeps returning North's sales even if the pivot layout changes. Turn it off in PivotTable Analyze → Options → Generate GetPivotData if you prefer normal references.

PivotChart (PivotTable Analyze → PivotChart) creates a chart linked to the pivot: filtering the pivot or clicking a slicer updates the chart. This is the basis of the dashboard in Lesson 11.

Case study: Video Game Sales (regional sales trends)

You are an analyst at a game publisher. The file `video_game_sales.csv` has 2,400 fictional titles with platform, release year, genre, publisher and sales in millions of units by region: North America (`na_sales`), Europe (`eu_sales`), Japan (`jp_sales`), other regions and global. Answer the questions with pivot tables. The answers below were computed from the file so you can check your pivots.

Python

```
g = pd.read_csv("video_game_sales.csv")
print(g.shape); g.head(4)
```

OUTPUT

```
(2400, 11)
```

	rank	name	platform	year	genre	publisher	na_sales	eu_sales	jp_sales	other
0	1	Hidden Kingdom	XSX	2023	Sports	Red Lantern Studios	3.69	2.62	0.29	0.87
1	2	Galactic Kingdom	PS5	2020	Adventure	Saffron Works	3.49	1.79	0.74	0.94
2	3	Galactic League	Mobile	2017	Strategy	Red Lantern Studios	3.08	2.44	0.57	0.67
3	4	Shadow League	Mobile	2025	Role-Playing	Northwind Play	1.66	2.12	2.33	0.63

4 rows × 11 columns

CASE STUDY **Q1. Which genres sell the most worldwide?**

Pivot: Rows = genre, Values = Sum of global_sales, sorted largest to smallest.

Python

```
q1 = g.pivot_table(index="genre", values="global_sales", aggfunc="sum").sort_values("global_sales",
ascending=False)
q1["% of total"] = (q1.global_sales / q1.global_sales.sum() * 100).round(1)
q1.round(1).head(6)
```

OUTPUT

	global_sales	% of total
genre		
Adventure	136.7	10.6
Sports	133.8	10.4
Action	116.3	9.0
Racing	113.6	8.8
Strategy	111.5	8.6
Simulation	108.4	8.4

6 rows × 2 columns

CASE STUDY **Q2. Who are the top 5 publishers, and how many titles did each release?**

Pivot: Rows = publisher, Values = Sum of global_sales and Count of name; Value Filter → Top 5 by global sales.

Python

```
q2 = g.groupby("publisher").agg(global_sales=("global_sales", "sum"), titles=("name",
"count")).sort_values("global_sales", ascending=False).head(5)
q2["sales per title"] = (q2.global_sales / q2.titles).round(2)
q2.round(1)
```

OUTPUT

	global_sales	titles	sales per title
publisher			
Starforge Interactive	168.0	295	0.6
Blue Comet Games	161.4	316	0.5
Northwind Play	144.6	266	0.5
Red Lantern Studios	144.5	268	0.5
Pixel Harbor	143.1	267	0.5

5 rows × 3 columns

CASE STUDY **Q3. How does regional preference differ by genre?**

Pivot: Rows = genre, Values = Sum of na_sales, eu_sales, jp_sales, other_sales; Show Values As → **% of Row Total**.

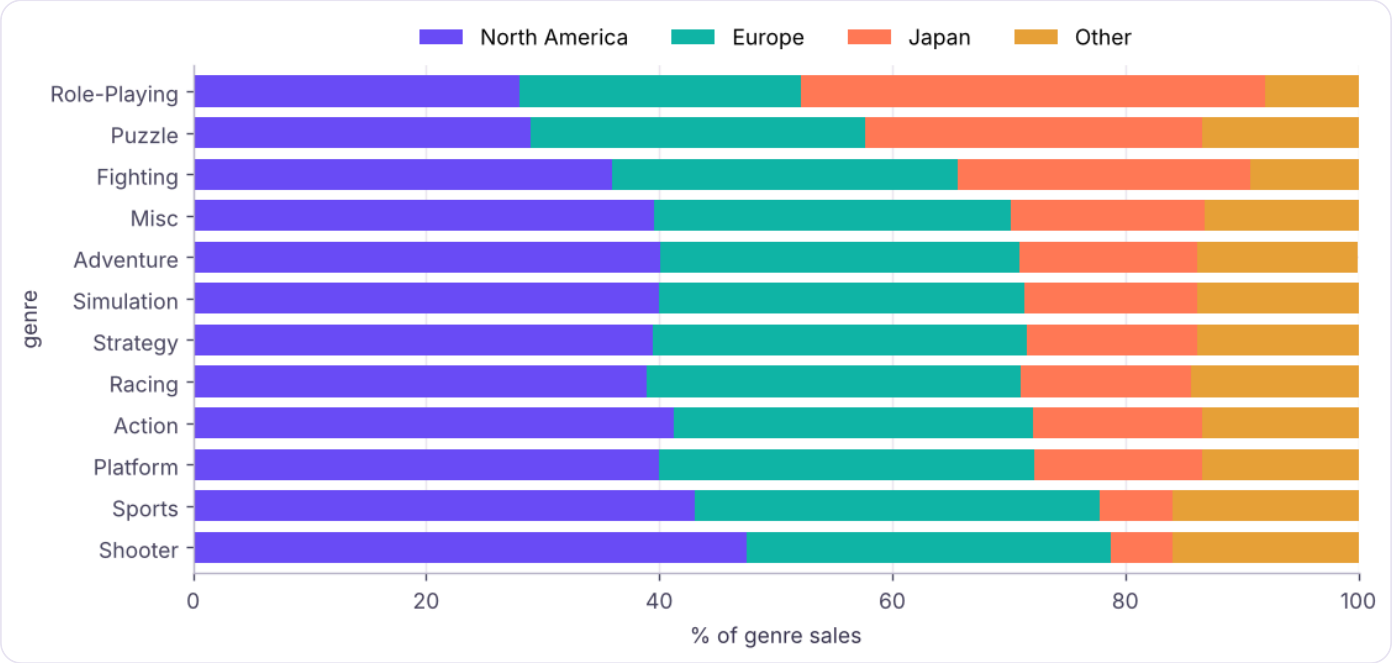
Python

```
reg = ["na_sales", "eu_sales", "jp_sales", "other_sales"]
q3 = g.groupby("genre")[reg].sum()
q3 = (q3.div(q3.sum(axis=1), axis=0) * 100).round(1).sort_values("jp_sales", ascending=False)
q3.head(6)
```

OUTPUT

	na_sales	eu_sales	jp_sales	other_sales
genre				
Role-Playing	28.0	24.2	39.8	8.1
Puzzle	29.0	28.7	28.9	13.4
Fighting	36.0	29.6	25.1	9.3
Misc	39.6	30.6	16.6	13.2
Adventure	40.1	30.8	15.3	13.7
Simulation	40.0	31.3	14.9	13.8

6 rows × 4 columns



Regional mix by genre (% of each genre's sales). Role-Playing and Puzzle games lean much more on Japan than Shooter or Sports titles do.

CASE STUDY **Q4. How have sales changed over the years?**

Pivot: Rows = year, Values = Sum of global_sales, plus a second copy shown as % Difference From (previous) for YoY growth. Insert a PivotChart (line).

Python

```
q4 = g.pivot_table(index="year", values="global_sales", aggfunc="sum")
q4["YoY %"] = (q4.global_sales.pct_change() * 100).round(1)
q4.round(1).tail(8)
```

OUTPUT

year	global_sales	YoY %
2018	87.2	19.0
2019	59.4	-31.9
2020	89.2	50.2
2021	78.0	-12.6
2022	71.9	-7.9
2023	76.4	6.3
2024	80.9	5.9
2025	87.0	7.4

8 rows × 2 columns

CASE STUDY **Q5. Which platform leads in each region?**

Pivot: Rows = platform, Values = the four regional sales columns; sort each column, or use conditional formatting → Top 1 per column.

Python

```
q5 = g.groupby("platform")[reg].sum().round(1)
q5.loc["Leader"] = q5.idxmax()
q5
```

OUTPUT

	na_sales	eu_sales	jp_sales	other_sales
platform				
3DS	45.90	40.40	37.60	17.20
Mobile	63.30	50	22.60	21.10
PC	43.40	34.20	14.90	14.90
PS3	48.10	38.50	18.80	17
PS4	55.40	43.10	16.90	19.20
PS5	58.50	45.40	20.80	19.70
Switch	40	34.40	38	13
X360	42.20	32.60	15.20	14.60
XOne	50.10	39.20	17.70	17.40
XSX	53.80	40.40	18	17.20
Leader	Mobile	Mobile	Switch	Mobile

11 rows × 4 columns

KEY TAKEAWAYS **Insights to present**

Write 3–5 sentences of insight, not just tables. For example: which genres to prioritise for the Japanese market, which publishers are efficient (high sales per title), and how platform transitions show in the year trend. Keep every number traceable to a pivot.

Summary

KEY TAKEAWAYS

- Pivot tables summarise structured data by dragging fields into **Rows, Columns, Values and Filters**.
- They read from a **pivot cache**: Refresh updates one cache; Refresh All updates everything. Use a Table as the source.
- Value Field Settings: change the function (Sum, Count, Average…) and **Show Values As** (% of total, % of row, difference from previous, running total, rank).
- MoM = change from the previous month; YoY = change from the same period last year.
- Group dates (years/quarters/months), group items manually, and group numbers into equal **bins**.
- Calculated fields work on totals, so use them for ratios like margin; add row-level maths to the source data.
- Slicers and timelines make pivots interactive; double-click a number to see the rows behind it.

Quick quiz

6 questions · tap an answer to check it

1. You add 500 rows to the source range, but the pivot does not change. Why?

A Pivot tables only read the first 1,000 rows

B The pivot reads from its cache, which updates only when you refresh (and the range must include the new rows)

C The new rows need a pivot formula

D Pivot tables cannot be updated

Answer: B. Refresh re-reads the source. If the source is a fixed range, also change the data source, or use a Table.

2. Which "Show Values As" option gives month-on-month growth?

A % of Grand Total

B Running Total In

C % Difference From → (previous)

D Rank Largest to Smallest

Answer: C. % Difference From the previous item compares each month with the one before it.

3. What is the main difference between groups and bins?

A Bins are equal-width numeric intervals; groups can be any size and any data type

B Groups only work with dates

C Bins only work with text

D There is no difference

Answer: A. Bins split numbers into equal intervals. Groups are custom collections of items.

4. A calculated field `Revenue = Qty * Price` gives inflated numbers. Why?

- **A** Calculated fields cannot multiply
- **B** It multiplies the sum of Qty by the sum of Price for each group
- **C** The pivot needs refreshing
- **D** Price must be in Rows

Answer: B. Calculated fields work on aggregated values. Compute Qty × Price in the source data instead.

5. How do you see the individual rows behind a number in a pivot table?

- **A** Press F9
- **B** Double-click the number (Show Details)
- **C** Use GETPIVOTDATA
- **D** Insert a slicer

Answer: B. Double-clicking opens a new sheet with the underlying records.

6. Two pivots were built from the same data. You click Refresh on one. What happens to the other?

- **A** It also updates, because they share the same pivot cache
- **B** Nothing; you must refresh it separately
- **C** It is deleted
- **D** It shows an error

Answer: A. Pivots from the same source share a cache; refreshing the cache updates all of them.

EXERCISE **Superstore pivots**

Build pivot tables from `superstore_sales.csv` to answer: (1) sales and profit by category and sub-category, with profit margin; (2) yearly sales by region with YoY %; (3) the 5 cities with the highest profit; (4) the number of order lines by discount (0%, 10%, ... 40%) and their total profit.

Show solution (worked answer)

(1) Rows: category, sub_category; Values: Sum of sales, Sum of profit; Calculated Field `Margin = profit/sales` formatted as %. (2) Rows: region; Columns: order_date grouped by Years; Values: Sum of sales, plus a second copy shown as % Difference From (previous) Years. (3) Rows: city; Values: Sum of profit; Value Filters → Top 10 → Top 5 Items by Sum of profit. (4) Rows: discount; Values: Count of order_id and Sum of profit. You will see profit turn negative at the higher discount levels, matching the Lesson 5 crosstab.

INTERVIEW CORNER **Questions you may be asked**

Q: What is a pivot cache and why does it matter? A hidden copy of the source data that pivots read from. It makes pivots fast and lets several pivots share data, but pivots only update on refresh, and many separate caches make files large.

Q: How do you show each item's share of the total in a pivot? Value Field Settings → Show Values As → % of Grand Total (or % of Column/Row Total).

Q: How would you count unique customers in a pivot? Create the pivot with "Add this data to the Data Model" ticked, then choose Distinct Count in Value Field Settings.

Q: When would you use formulas (SUMIFS) instead of a pivot? When the report needs a fixed layout, mixes data from several sources, or must update without refreshing; pivots are better for exploration and flexible summaries.

ADVANCED EXCEL · LESSON 9

Date, Time & Number Functions and Named Ranges

Work confidently with dates and times (DATE, EOMONTH, EDATE, WORKDAY, NETWORKDAYS, WEEKDAY, WEEKNUM, DATEDIF, fiscal quarters), clean up numbers with INT, MOD, ROUND, CEILING and FLOOR, and make formulas readable with static and dynamic named ranges and LET.

● Live classes 7–8 ✦ Excel for Microsoft 365, 2026 Intermediate ⏱ 120 min

By the end of this lesson you can

- Explain how Excel stores dates and times, and do date arithmetic correctly
- Extract parts of dates, find month ends, add months, and count working days with holidays
- Calculate age, tenure and fiscal-year quarters (Indian April–March and US October–September)
- Use INT, MOD, ROUND, ROUNDUP, ROUNDDOWN, MROUND, CEILING.MATH, FLOOR.MATH, SQRT, POWER and ABS
- Create static and dynamic named ranges and simplify long formulas with LET

Dates and times are numbers

Excel stores a date as the number of days since **1 January 1900** (that day is 1). A time is a **fraction of a day**: 12:00 noon is 0.5 and 06:00 is 0.25. A date with a time is simply a whole number plus a fraction.

X What is really stored

	A	B	C
1	You see	Stored as	Explanation
2	27-04-2026	46139	Days since 1-Jan-1900
3	12:00	0.5	Half a day
4	06:30	0.2708	6.5 hours ÷ 24
5	27-04-2026 18:00	46139.75	Date + time

```
B2    =A2
B3    =TIME(12,0,0)
B4    =TIME(6,30,0)
B5    =DATE(2026,4,27)+TIME(18,0,0)
```

This is why date arithmetic works:

Date arithmetic

	A	B	C	D	E
1	Order date	Ship date	Days to ship =B2-A2	Due date (+30 days)	Overdue? (as of 15-Jun-2026)
2	27-04-2026	02-05-2026	5	27-05-2026	Overdue
3	20-05-2026	21-05-2026	1	19-06-2026	OK
4	01-06-2026	08-06-2026	7	01-07-2026	OK

C2 `=B2-A2`

D2 `=A2+30`

E2 `=IF(DATE(2026,6,15)>D2,"Overdue","OK")`

Each pattern is copied down/across to the other cells in its column.

TIP **TODAY() and NOW()**

`=TODAY()` returns today's date and `=NOW()` the date and time. They are **volatile**: they recalculate every time the workbook changes, so results move every day. For reports that must not change, type the "as of" date into a cell and refer to it, as in the overdue example above. Shortcuts: `Ctrl + ;` types today's date as a fixed value.

Building and taking apart dates

Function	Returns	Example (A2 = 27-04-2026)	Result
`DATE(y, m, d)`	A date from parts	`=DATE(2026,4,27)`	27-04-2026
`YEAR, MONTH, DAY`	Parts of a date	`=MONTH(A2)`	4
`WEEKDAY(date, 2)`	1 = Monday ... 7 = Sunday	`=WEEKDAY(A2,2)`	1
`WEEKNUM(date, 2)`	Week of the year (weeks start Monday)	`=WEEKNUM(A2,2)`	18
`ISOWEEKNUM(date)`	ISO 8601 week number	`=ISOWEEKNUM(A2)`	18
`TEXT(date, "mmm")`	Month name as text	`=TEXT(A2,"mmmm")`	April
`EOMONTH(date, n)`	Last day of the month n months away	`=EOMONTH(A2,0)`	30-04-2026
`EDATE(date, n)`	Same day n months later	`=EDATE(A2,3)`	27-07-2026
`DATEVALUE(text)`	Converts date text to a real date	`=DATEVALUE("27-Apr-2026")`	46139

Date parts and month boundaries

	A	B	C	D	E	F	G	H
1	Date	Year	Month	Day	Weekday (Mon=1)	Month end	First of month	+3 months
2	27-04-2026	2026	4	27	1	30-04-2026	01-04-2026	27-07-2026
3	31-01-2026	2026	1	31	6	31-01-2026	01-01-2026	28-02-2026
4	29-02-2024	2024	2	29	4	29-02-2024	01-02-2024	28-02-2025

B2 `=YEAR(A2)`

```
C2   =MONTH(A2)
D2   =DAY(A2)
E2   =WEEKDAY(A2,2)
F2   =EOMONTH(A2,0)
G2   =EOMONTH(A2,-1)+1
H2   =EDATE(A2,3)
```

Each pattern is copied down/across to the other cells in its column.

Notice that EDATE handles month ends correctly: 31-Jan + 1 month = 28-Feb (2026 is not a leap year), and 29-Feb-2024 + 12 months = 28-Feb-2025.

Working days

Function	Use
`NETWORKDAYS(start, end, [holidays])`	Count Mon–Fri days between two dates (inclusive) minus holidays
`NETWORKDAYS.INTL(start, end, weekend, [holidays])`	Choose weekend days: 11 = Sunday only, `"0000011"` = Sat + Sun
`WORKDAY(start, days, [holidays])`	The date after adding n working days
`WORKDAY.INTL(start, days, weekend, [holidays])`	Same with a custom weekend

SLA and delivery planning (holidays in G2:G4)

	A	B	C	D	E	F
1	Ticket	Opened	Closed	Working days =NETWORKDAYS(B2,C2,$G$2:$G$4)	Due (+5 working days)	
2	T-101	10-08-2026	18-08-2026	7	17-08-2026	
3	T-102	16-10-2026	23-10-2026	5	26-10-2026	
4	T-103	06-11-2026	10-11-2026	4	12-11-2026	

```
D2   =NETWORKDAYS(B2,C2,$G$2:$G$4)
E2   =WORKDAY(B2,5,$G$2:$G$4)
D3   =NETWORKDAYS(B3,C3,$G$2:$G$4)
E3   =WORKDAY(B3,5,$G$2:$G$4)
D4   =NETWORKDAYS.INTL(B4,C4,11,$G$2:$G$4)
E4   =WORKDAY.INTL(B4,5,11,$G$2:$G$4)
```

T-103 uses a **Sunday-only weekend** (code 11), common in Indian factories and retail. The holiday on 15-Aug-2026 falls on a Saturday, so it does not change T-101's count of working days.

Age, tenure and durations

`DATEDIF(start, end, unit)` is a hidden function (no tooltip) that returns complete years, months or days between two dates.

Unit	Returns
`"y"`	Complete years
`"m"`	Complete months
`"d"`	Days
`"ym"`	Months after removing complete years
`"md"`	Days after removing complete months

Tenure as of 31-Mar-2026 (in F1)

	A	B	C	D	E	F
1	Employee	Join date	Years	Months after years	Tenure label	31-03-2026
2	Kavya Iyer	28-02-2016	10	1	10 yrs 1 mths	
3	Imran Reddy	14-12-2019	6	3	6 yrs 3 mths	
4	Saanvi Rao	18-02-2025	1	1	1 yrs 1 mths	

```
C2   =DATEDIF(B2,$F$1,"y")
D2   =DATEDIF(B2,$F$1,"ym")
E2   =C2&" yrs "&D2&" mths"
```

Each pattern is copied down/across to the other cells in its column.

(Cell F1 holds the date 31-03-2026.) For fractional years use `=YEARFRAC(B2,$F$1)`, for example 10.09 years.

Fiscal years and quarters

Calendar quarter: `="Q"&ROUNDUP(MONTH(A2)/3,0)`. Many organisations use a different financial year:

Year type	Q1	Formula for the quarter number
Calendar	Jan–Mar	`=ROUNDUP(MONTH(A2)/3,0)`
India (FY Apr–Mar)	Apr–Jun	`=ROUNDUP(MOD(MONTH(A2)-4,12)/3+0.01,0)` or `=CHOOSE(MONTH(A2),4,4,4,1,1,1,2,2,2,3,3,3)`
US federal (FY Oct–Sep)	Oct–Dec	`=CHOOSE(MONTH(A2),2,2,2,3,3,3,4,4,4,1,1,1)`

Calendar, Indian FY and US FY quarters

	A	B	C	D	E
1	Date	Calendar Q	India FY quarter	India FY label	US FY quarter
2	10-02-2026	Q1	Q4	FY26	Q2
3	27-04-2026	Q2	Q1	FY27	Q3
4	03-11-2026	Q4	Q3	FY27	Q1

```
B2   ="Q"&ROUNDUP(MONTH(A2)/3,0)
C2   ="Q"&CHOOSE(MONTH(A2),4,4,4,1,1,1,2,2,2,3,3,3)
```

```
D2   ="FY"&(YEAR(A2)+(MONTH(A2)>=4))-2000
```

Each pattern is copied down/across to the other cells in its column.

In the Indian convention, 27-Apr-2026 falls in **FY27** (April 2026 – March 2027), quarter Q1.

Times

Function	Example	Result
`TIME(h, m, s)`	`=TIME(14,30,0)`	14:30
`HOUR`, `MINUTE`, `SECOND`	`=HOUR(A2)`	14
Duration in hours	`=(end-start)*24`	8.5
Duration beyond 24 h	Format `[h]:mm`	37:30
Shifts past midnight	`=MOD(end-start,1)`	Works when end is the next day

A very common clean-up task: a system exports times as plain numbers like `1430` meaning 14:30. Convert them with `INT` and `MOD`:

Convert 1430-style numbers into real times

	A	B	C	D	E
1	Raw	Hours =INT(A2/100)	Minutes =MOD(A2,100)	Time =TIME(INT(A2/100),MOD(A2,100),0)	Shift length (to B6)
2	930	9	30	09:30	8.75
3	1430	14	30	14:30	
4	2245	22	45	22:45	
5	5	0	5	00:05	
6	1815	18	15	18:15	

```
B2   =INT(A2/100)
C2   =MOD(A2,100)
D2   =TIME(INT(A2/100),MOD(A2,100),0)
E2   =MOD(D6-D2,1)*24
```

Each pattern is copied down/across to the other cells in its column.

Number functions

Function	What it does	Example	Result
`INT(n)`	Rounds **down** to the integer	`=INT(-7.6)`	`-8`
`TRUNC(n, [d])`	Cuts off decimals (towards zero)	`=TRUNC(-7.6)`	`-7`
`ROUND(n, d)`	Normal rounding	`=ROUND(7.456,2)`	7.46
`ROUNDUP(n, d)`	Always away from zero	`=ROUNDUP(7.401,0)`	8
`ROUNDDOWN(n, d)`	Always towards zero	`=ROUNDDOWN(7.99,0)`	7
`ROUND(n, -3)`	Round to thousands	`=ROUND(123456,-3)`	123000
`MROUND(n, m)`	Nearest multiple	`=MROUND(17,5)`	15
`CEILING.MATH(n, m)`	Up to the next multiple	`=CEILING.MATH(23,7)`	28
`FLOOR.MATH(n, m)`	Down to the previous multiple	`=FLOOR.MATH(23,7)`	21
`MOD(n, d)`	Remainder	`=MOD(28,3)`	1
`QUOTIENT(n, d)`	Integer part of a division	`=QUOTIENT(28,3)`	9
`ABS(n)`	Absolute value	`=ABS(-450)`	450
`SIGN(n)`	−1, 0 or 1	`=SIGN(-450)`	`-1`
`SQRT(n)`	Square root	`=SQRT(144)`	12
`POWER(n, p)`	Power (same as ^)	`=POWER(25,2)`	625

Rounding functions side by side

	A	B	C	D	E	F	G	H
1	Value	INT	ROUND 0	ROUNDUP 0	ROUNDDOWN 0	ROUND 2	CEILING.MATH to 5	FLOOR.MATH to 5
2	7.456	7	7	8	7	7.46	10	5
3	-7.456	-8	-7	-8	-7	-7.46	-5	-10
4	23.5	23	24	24	23	23.5	25	20

```
B2   =INT(A2)
C2   =ROUND(A2,0)
D2   =ROUNDUP(A2,0)
E2   =ROUNDDOWN(A2,0)
G2   =CEILING.MATH(A2,5)
H2   =FLOOR.MATH(A2,5)
```

Each pattern is copied down/across to the other cells in its column.

Practical uses

- **Packaging**: boxes needed for 523 items at 24 per box = `=CEILING.MATH(523,24)/24` or `=ROUNDUP(523/24,0)` → 22.
- **Price points**: round prices up to the next 99: `=CEILING.MATH(A2,100)-1` turns 1,234 into 1,299.
- **Split profit and loss** into two columns with ABS:

Profit and Loss columns

	A	B	C	D
1	**Store**	**Profit / Loss**	**Profit**	**Loss**
2	Pune	45200	45200	0
3	Delhi	-12800	0	12800
4	Kochi	3300	3300	0
5	Patna	-7600	0	7600
6	**Total**	**28100**	**48500**	**20400**

```
C2   =IF(B2>0,B2,0)
D2   =IF(B2<0,ABS(B2),0)
B6   =SUM(B2:B5)
C6   =SUM(C2:C5)
D6   =SUM(D2:D5)
```

Each pattern is copied down/across to the other cells in its column.

Named ranges

A **name** is a label for a cell, a range, a constant or a formula. `=SUM(Sales)` is easier to read and audit than `=SUM(Data!$O$2:$O$6324)`.

Static named ranges

Method	How
Name Box	Select the range, click the Name Box (left of the formula bar), type `GST_Rate`, press Enter
Define Name	Formulas → Define Name: set the name, scope (workbook or one sheet) and comment
Create from Selection	Select a table with headers → Formulas → Create from Selection (`Ctrl + Shift + F3`) → Top row: one name per column
Name Manager	Formulas → Name Manager (`Ctrl + F3`) to edit, delete or find broken names

Rules: names start with a letter or underscore, contain no spaces (use _), cannot look like a cell address (`Q1` or FY26 are not allowed) and are not case-sensitive.

Formulas with names (GST_Rate = F1, Qty and Price are named columns)

	A	B	C	D	E	F
1	**Qty**	**Price**	**Amount =Qty*Price**	**With GST =Qty*Price*(1+GST_Rate)**		18%
2	4	1250	5000	5,900.00		
3	10	299	2990	3,528.20		
4	2	8999	17998	21,237.64		

```
C2   =A2*B2
D2   =A2*B2*(1+$F$1)
C3   =A3*B3
D3   =A3*B3*(1+$F$1)
C4   =A4*B4
```

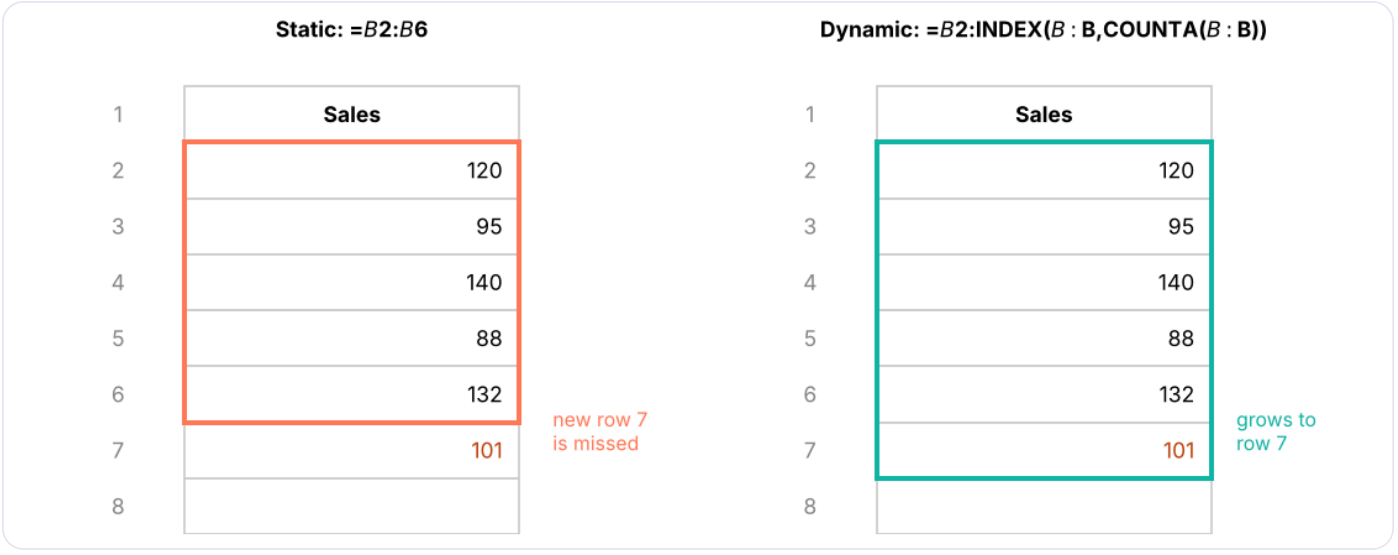
```
D4   =A4*B4*(1+$F$1)
```

(F1, the yellow cell, is named `GST_Rate`. With names defined, the formula in D2 reads `=Qty*Price*(1+GST_Rate)`.)

Dynamic named ranges

A static name like `=$B$2:$B$100` does not grow when you add row 101. A **dynamic** named range expands automatically, which keeps charts, validation lists and formulas up to date.

Method	Refers to	Notes
Excel Table (best)	`=tblSales[Amount]`	Automatic, no formula needed
`OFFSET + COUNTA`	`=OFFSET(Data!$B$2,0,0,COUNTA(Data!$B:$B)-1,1)`	Classic; volatile (recalculates often)
`INDEX + COUNTA`	`=Data!$B$2:INDEX(Data!$B:$B,COUNTA(Data!$B:$B))`	Not volatile; preferred over OFFSET
Spill reference	`=Data!$H$2#`	Points to a dynamic-array result



A static range stays fixed; a dynamic range grows with the data.

LET: names inside one formula

LET (Microsoft 365) assigns names to values **inside** a formula. It makes long formulas readable and faster, because each part is calculated once.

Without LET: `=IF(SUMIFS(F:F,C:C,"Finance")/COUNTIFS(C:C,"Finance")>90000,"High","Normal")`

With LET:

Excel formula

```
=LET(
    total, SUMIFS(F:F, C:C, "Finance"),
    heads, COUNTIFS(C:C, "Finance"),
    avg,   total / heads,
    IF(avg > 90000, "High", "Normal")
)
```

LAMBDA goes one step further and lets you create your own reusable functions; see Lesson 12.

Case: date analysis on Superstore

How long do orders take to ship, by ship mode, and which weekday gets the most orders? In Excel you would add columns `=ship_date-order_date` and `=TEXT(order_date,"ddd")` and pivot them. Here are the real answers:

Python

```
import pandas as pd
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date", "ship_date"])
orders = s.drop_duplicates("order_id").copy()
orders["days_to_ship"] = (orders.ship_date - orders.order_date).dt.days            # =C2-B2
orders["weekday"] = orders.order_date.dt.day_name().str[:3]                         # =TEXT(B2,"ddd")
orders["fy"] = "FY" + (orders.order_date.dt.year + (orders.order_date.dt.month ≥
4)).astype(str).str[2:]   # Indian FY
print(orders.groupby("ship_mode")["days_to_ship"].agg(["mean", "min", "max"]).round(1).to_string())
print()
print(orders.groupby("fy").size().rename("orders").to_string())
```

OUTPUT

```
                mean  min  max
ship_mode
First Class      1.5    1    2
Same Day         0.0    0    0
Second Class     3.0    2    4
Standard Class   5.5    4    7

fy
FY23     169
FY24     952
FY25    1114
FY26     965
```

Summary

KEY TAKEAWAYS

- Dates are whole numbers (days since 1-Jan-1900); times are fractions of a day. Subtract dates to get days; multiply time differences by 24 for hours.
- `EOMONTH` and `EDATE` handle month ends correctly; `NETWORKDAYS` and `WORKDAY` (with `.INTL` for custom weekends) handle working days and holidays.
- `DATEDIF` gives complete years/months; build fiscal quarters with `CHOOSE(MONTH(…),…)`.
- `INT` rounds down, `TRUNC` cuts towards zero, `ROUNDUP`/`ROUNDDOWN` force direction, `CEILING.MATH`/`FLOOR.MATH` round to multiples, `MOD` gives remainders.
- Named ranges make formulas readable; use Tables or `INDEX`-based names for dynamic ranges, and `LET` for names inside a formula.

Quick quiz

6 questions · tap an answer to check it

1. A1 contains 27-04-2026 and B1 contains 02-05-2026. What does `=B1-A1` return?

A 5 days as text

B 5

C 0.05

D **VALUE!**

Answer: B. Dates are numbers, so subtracting gives the number of days.

2. Which formula returns the last day of the month for the date in A2?

A `=DATE(YEAR(A2),MONTH(A2),31)`

B `=EOMONTH(A2,0)`

C `=EDATE(A2,1)-1`

D `=MONTH(A2)+30`

Answer: B. EOMONTH(date,0) returns the month end; the other options fail for 30-day months or February.

3. What does `=INT(-3.2)` return?

A −3

B −4

C 3

D −3.2

Answer: B. INT always rounds down (towards minus infinity). TRUNC(-3.2) would give −3.

4. Which formula converts the number 1745 into the time 17:45?

A `=TIME(1745)`

B `=1745/24`

C `=TIME(INT(A2/100),MOD(A2,100),0)`

D `=TEXT(1745,"hh:mm")`

Answer: C. INT(1745/100) = 17 hours and MOD(1745,100) = 45 minutes.

5. Why is an INDEX-based dynamic named range often preferred to OFFSET?

A It works only in Excel 365

B OFFSET is volatile and recalculates on every change, which can slow large workbooks

C INDEX cannot refer to whole columns

D OFFSET does not grow with the data

Answer: B. Both grow, but OFFSET is volatile.

6. In the Indian financial year, which quarter is 15-January?

A Q1

B Q2

C Q3

D Q4

Answer: D. FY runs April–March, so January–March is Q4.

EXERCISE **HR date calculations**

Using `employees_small.csv` and an "as of" date of 31-03-2026 in a cell: (1) tenure in complete years; (2) the probation end date (6 months after joining); (3) the first working day after the probation end; (4) the joining quarter in the Indian FY.

Show solution (worked answer)

With join date in E2 and the as-of date in $J$1: (1) `=DATEDIF(E2,$J$1,"y")` (2) `=EDATE(E2,6)` (3) `=WORKDAY(EDATE(E2,6)-1,1)` (the first working day on or after the probation end) (4) `="Q"&CHOOSE(MONTH(E2),4,4,4,1,1,1,2,2,2,3,3,3)`

INTERVIEW CORNER **Questions you may be asked**

Q: How does Excel store dates and times? Dates as serial numbers counting days from 1-Jan-1900; times as fractions of a day. This makes arithmetic, sorting and formatting consistent.

Q: How do you count working days between two dates excluding holidays? `NETWORKDAYS(start, end, holidays)` or `NETWORKDAYS.INTL` with a custom weekend code.

Q: What is a dynamic named range and when do you need one? A name whose range grows with the data, built with an Excel Table, `INDEX/COUNTA` or `OFFSET/COUNTA`. It keeps charts, validation lists and formulas current as data is added.

ADVANCED EXCEL · LESSON 10

Charts and Advanced Charts

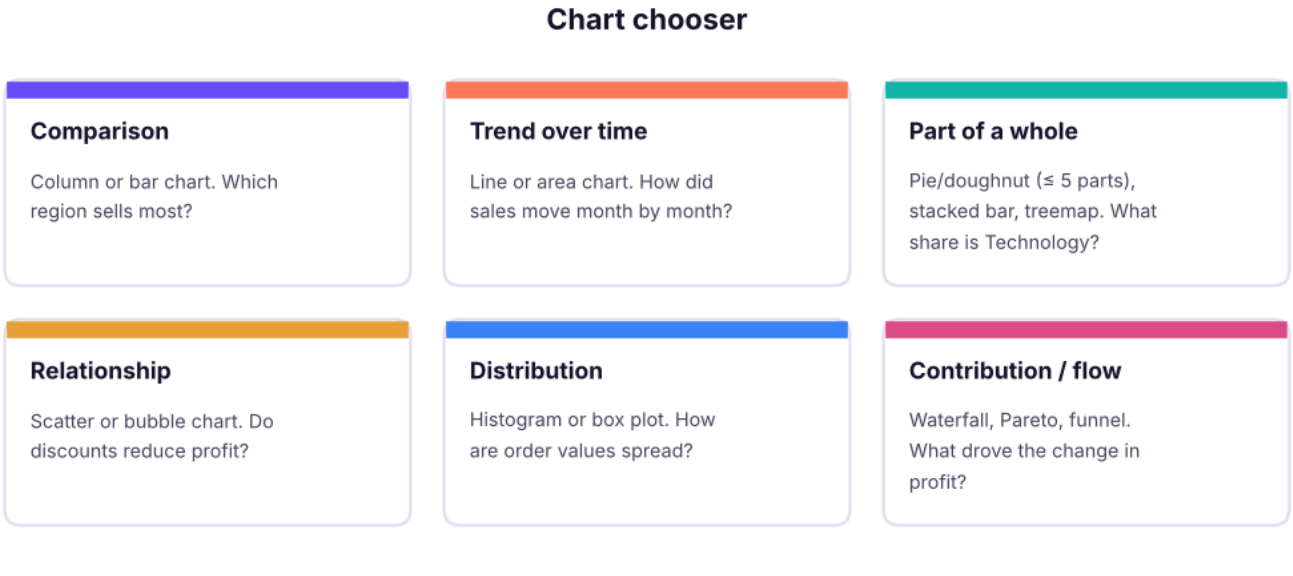
Choose the right chart for every question, build and format column, bar, line, area, pie, scatter and combo charts, add sparklines, and create the advanced business charts taught in class: thermometer, Pareto, population pyramid and waterfall, plus histogram, box plot, treemap and funnel.

Live classes 7–8 · ✦ Excel for Microsoft 365, 2026 · Intermediate · ⏱ 130 min

By the end of this lesson you can

- Pick the right chart type from the question you want to answer
- Insert, edit and format charts quickly (elements, axes, labels, colours, templates)
- Build combo charts with a secondary axis and add sparklines
- Build thermometer, Pareto, population pyramid and waterfall charts step by step
- Use the newer statistical and hierarchy charts and apply chart design best practices

Start with the question, not the chart



Choose a chart by the question you are answering.

Creating and editing a chart

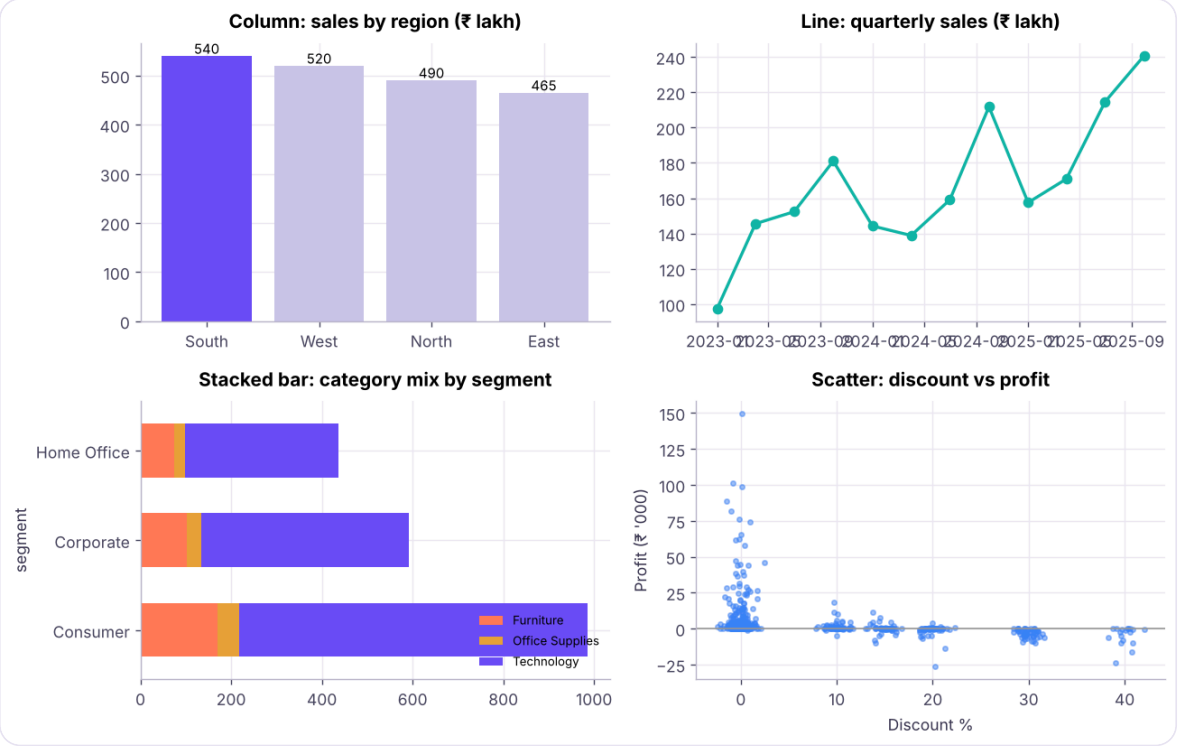
STEP BY STEP **The fast way**

1. Select the data **including headers** (or a summary table / pivot table). For a quick chart press `Alt + F1` (embedded) or `F11` (new chart sheet).
2. Or use **Insert → Recommended Charts** to preview options.
3. Click the chart: three buttons appear at its top-right: **Chart Elements** (+), **Chart Styles** (brush) and **Chart Filters** (funnel).
4. **Chart Design** tab: Switch Row/Column, Select Data, Change Chart Type, Move Chart, Quick Layout.
5. **Format** pane: double-click any element (axis, series, label) or press `Ctrl + 1` with it selected.

Element	Good practice
Title	Say the insight: "Technology drives 77% of sales", not "Sales by Category"
Axes	Start bar/column axes at zero; format numbers (K, L, %); remove unnecessary decimals
Gridlines	Light grey or none
Data labels	Use them instead of an axis when there are few bars
Legend	Only if there is more than one series; better: label lines directly
Colour	One highlight colour for the point you want to make, grey for the rest
Template	Right-click → Save as Template to reuse your style

The core chart types

Here are the core charts built from Superstore data. The Excel steps follow each one.



Column (comparison), line (trend), clustered and stacked bar (comparison of parts), and scatter (relationship).

Chart	Excel path	Use when
Clustered column / bar	Insert → Column or Bar	Comparing categories; use **bar** when labels are long or there are many categories
Stacked / 100% stacked	Same menu	Showing parts of a total across groups
Line	Insert → Line (with markers)	Trends over time; up to ~5 lines
Area	Insert → Line → Area	Cumulative trends; volume over time
Pie / doughnut	Insert → Pie	2–5 parts of a whole that add to 100%
Scatter (XY)	Insert → Scatter	Relationship between two numbers; add a **trendline** (right-click series → Add Trendline, tick Display R^2)
Bubble	Insert → Scatter → Bubble	Scatter with a third measure as bubble size

COMMON MISTAKES

Chart crimes to avoid

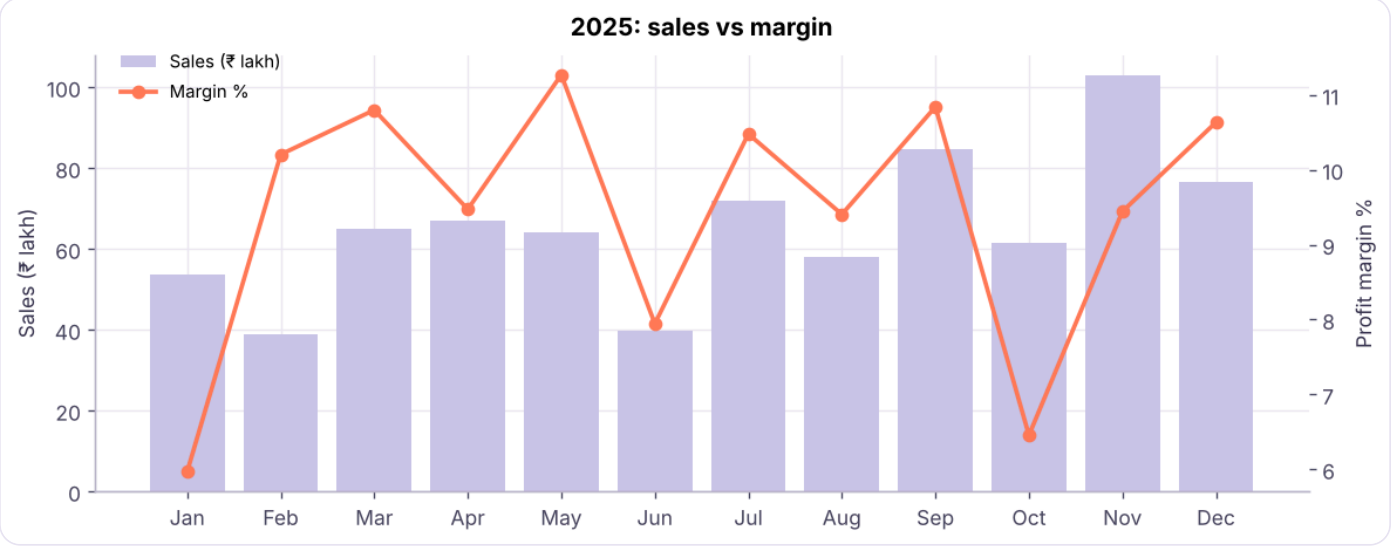
3D charts, pie charts with 10 slices, two different units on one axis, truncated bar axes that exaggerate differences, rainbow colours, and charts without units. If you need to squint, the chart has failed.

Combo charts and the secondary axis

When two measures have very different scales (sales in lakhs and profit margin in %), plot one as columns and the other as a line on a **secondary axis**.

STEP BY STEP **Build a combo chart**

1. Make a summary table: Month, Sales, Margin %.
2. Select it → Insert → **Combo Chart** → Create Custom Combo Chart.
3. Sales: Clustered Column. Margin %: Line with Markers, tick **Secondary Axis**.
4. Add axis titles so readers know which axis belongs to which series.



Combo chart: monthly sales (columns, left axis) and profit margin (line, right axis) for 2025.

Sparklines

Sparklines are tiny charts **inside a cell**: Insert → **Sparklines** → Line, Column or Win/Loss. Select the data range (for example 12 months in a row) and the location cell. Use **Sparkline → Show → High Point / Low Point** to mark extremes. They are ideal in dashboard tables, one row per product or region.

Thermometer chart (target vs actual)

A thermometer shows progress towards a single target, such as "₹ 7.4 crore of the ₹ 9 crore annual target".

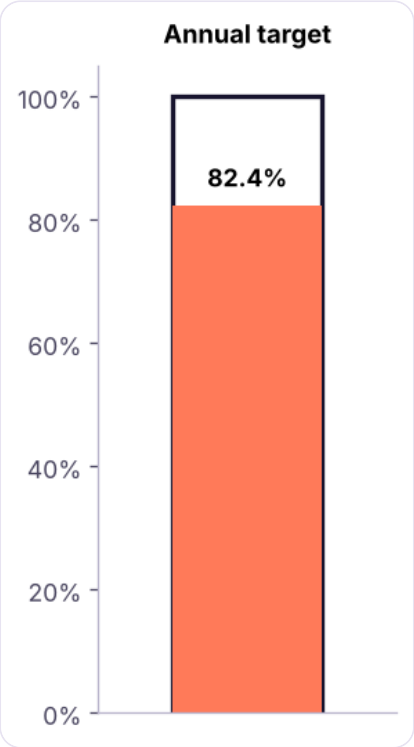
Thermometer data

	A	B
1	Metric	Value
2	Target	900
3	Achieved	742
4	% achieved	82.4%

```
B4    =B3/B2
```

STEP BY STEP **Build it**

1. Select only the **% achieved** cell (and its label) → Insert → **Clustered Column**.
2. Format the vertical axis: Minimum **0**, Maximum **1** (100%). Delete the horizontal axis and gridlines.
3. Format the series: Gap Width **0%** so the column fills the width; choose a strong fill colour.
4. Optional: add a second series with value 1 (100%) on the secondary axis, no fill and a thick border to draw the "glass" of the thermometer; set the secondary axis to 0–1 as well.
5. Resize the chart to be tall and narrow; add a data label showing 82.4%.



Thermometer chart: 82.4% of the annual target achieved.

Pareto chart (the 80/20 rule)

The Pareto principle says that roughly 80% of effects come from 20% of causes: a few sub-categories drive most sales, a few complaint types cause most tickets. A Pareto chart shows sorted bars plus a **cumulative %** line.

Python

```python
pareto = s.groupby("sub_category")["sales"].sum().sort_values(ascending=False).to_frame()
pareto["share %"] = (pareto.sales / pareto.sales.sum() * 100).round(1)
pareto["cumulative %"] = pareto["share %"].cumsum().round(1)
pareto["sales"] = (pareto.sales / 1e5).round(1)
pareto.rename(columns={"sales": "sales (₹ lakh)"})
```

OUTPUT

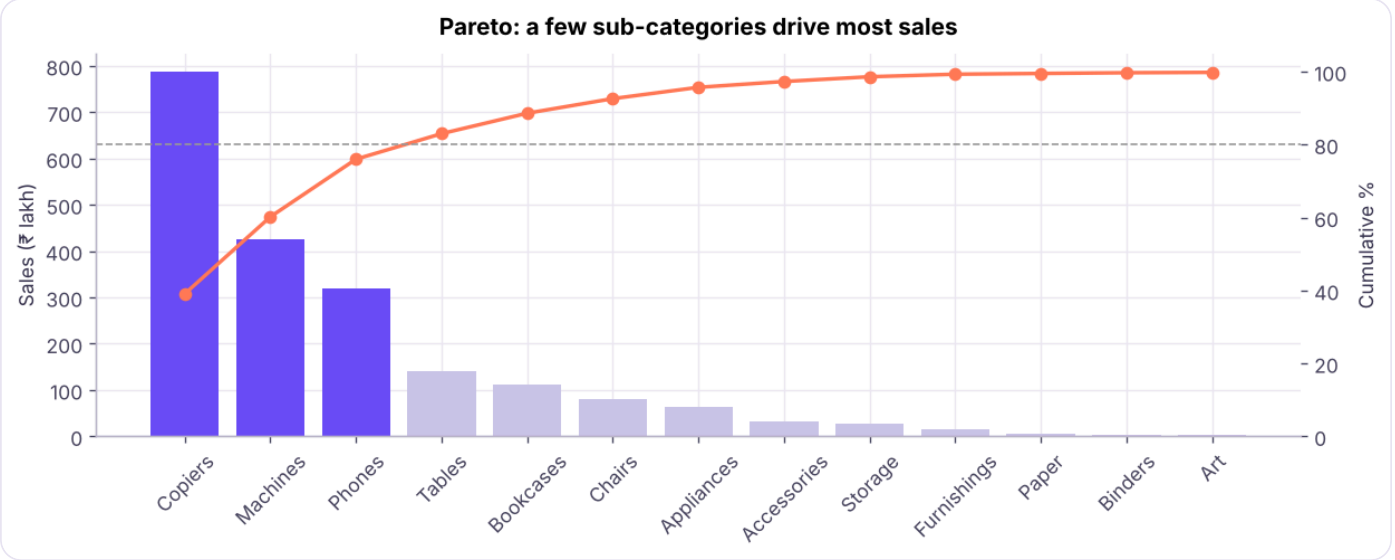
	sales (₹ lakh)	share %	cumulative %
sub_category			
Copiers	787.5	39.1	39.1
Machines	425.9	21.1	60.2
Phones	319.2	15.8	76.0
Tables	140.5	7.0	83.0
Bookcases	112.7	5.6	88.6
Chairs	80.9	4.0	92.6
Appliances	63.3	3.1	95.7
Accessories	31.8	1.6	97.3
Storage	26.6	1.3	98.6
Furnishings	14.3	0.7	99.3
Paper	4.9	0.2	99.5
Binders	4.0	0.2	99.7

13 rows × 3 columns (first 12 shown)

STEP BY STEP

Build it in Excel

1. Summarise sales by sub-category (pivot or SUMIFS) and **sort largest to smallest**.
2. Add `Cumulative %`: in C2 `=SUM($B$2:B2)/SUM($B$2:$B$13)`, copy down.
3. Select the three columns → Insert → **Combo**: Sales = Clustered Column, Cumulative % = Line on the **Secondary Axis**.
4. Set the secondary axis maximum to 100%. Optionally add a horizontal 80% reference line.
5. Excel 2016+ also has a built-in **Insert → Statistical → Pareto** chart that sorts and accumulates for you from raw data.



Pareto chart of sales by sub-category. Copiers, Machines and Phones alone make up 76% of sales; adding Tables passes the 80% line.

Population pyramid

A population pyramid compares two groups (typically male and female) across age bands, with one group's bars going left and the other's going right. In business it works for any two-sided comparison: this year vs last year by product, or two regions by age band.

Python

```
hr = pd.read_csv("hr_employees.csv")
hr["age_band"] = pd.cut(hr.age, [20, 25, 30, 35, 40, 45, 50, 55, 60], labels=["21-25", "26-30", "31-35", "36-40", "41-45", "46-50", "51-55", "56-60"])
pyr = hr.pivot_table(index="age_band", columns="gender", values="emp_id", aggfunc="count", observed=False)
pyr["Male (negative)"] = -pyr["Male"]
pyr
```

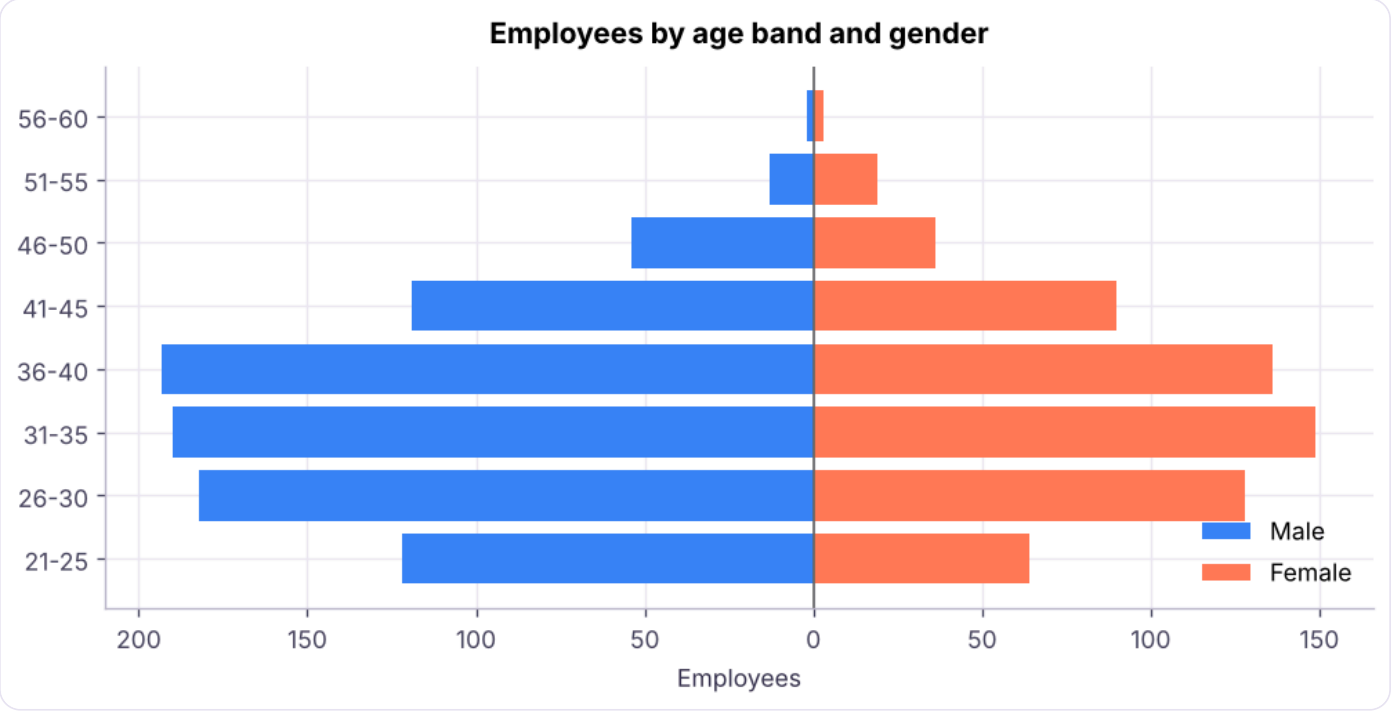
OUTPUT

gender	Female	Male	Male (negative)
age_band			
21-25	64	122	-122
26-30	128	182	-182
31-35	149	190	-190
36-40	136	193	-193
41-45	90	119	-119
46-50	36	54	-54
51-55	19	13	-13
56-60	3	2	-2

8 rows × 3 columns

STEP BY STEP **Build it**

1. Count employees by age band and gender (pivot table or COUNTIFS).
2. Make the male counts **negative** (`=-C2`) in a helper column so they plot to the left.
3. Select age bands, the negative male column and the female column → Insert → **Clustered Bar**.
4. Format series: Series Overlap **100%**, Gap Width **10–30%**.
5. Format the horizontal axis numbers with the custom format `#,##0;#,##0` so negative values show without a minus sign.
6. Vertical axis → Labels → Label Position: **Low** so the age labels sit on the left edge.



Population pyramid of the 1,500 employees in the HR dataset.

Waterfall chart

A waterfall shows how a starting value becomes an ending value through positive and negative steps: revenue to profit, last year's sales to this year's, or budget to actual.

Waterfall data: from revenue to net profit (₹ lakh)

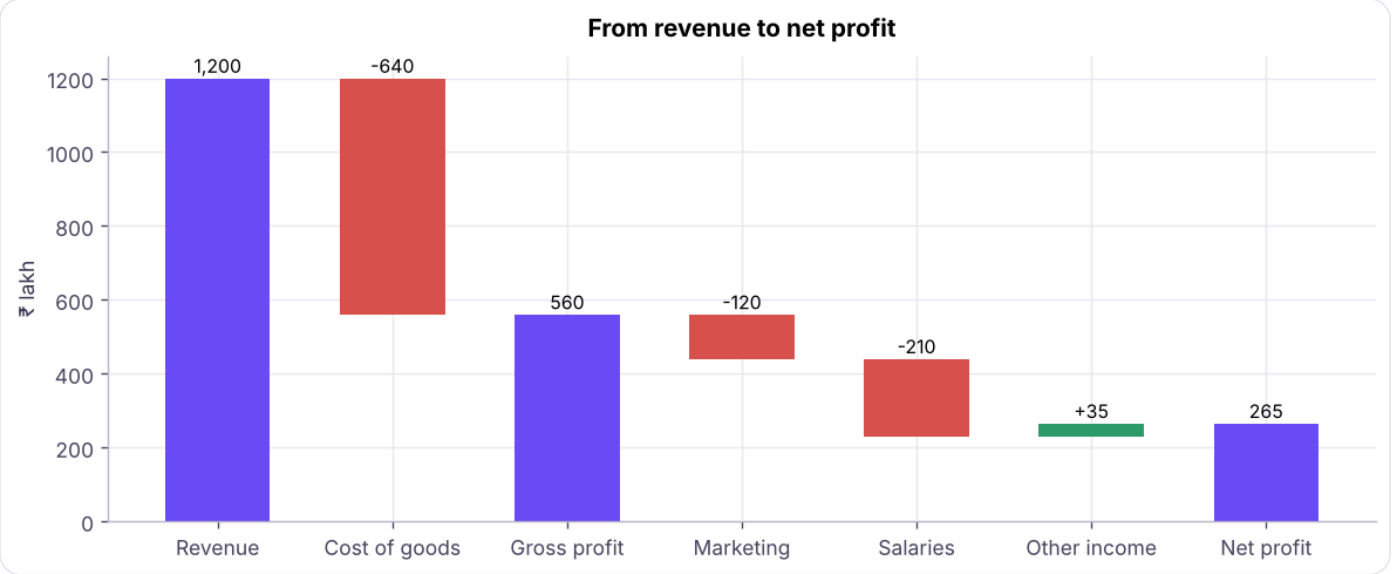
	A	B
1	**Step**	**Amount**
2	Revenue	1200
3	Cost of goods	-640
4	Gross profit	560
5	Marketing	-120
6	Salaries	-210
7	Other income	35
8	Net profit	265

B4 `=B2+B3`
B8 `=B4+B5+B6+B7`

STEP BY STEP **Build it (Excel 2016 and later)**

1. Select the table → Insert → **Waterfall** (Insert → Charts → Waterfall, Funnel...).
2. Click the **Gross profit** bar twice (to select only that point) → right-click → **Set as total**. Do the same for **Net profit**. Totals then start from zero instead of floating.
3. Colour increases, decreases and totals differently (Format Data Point) and add data labels.

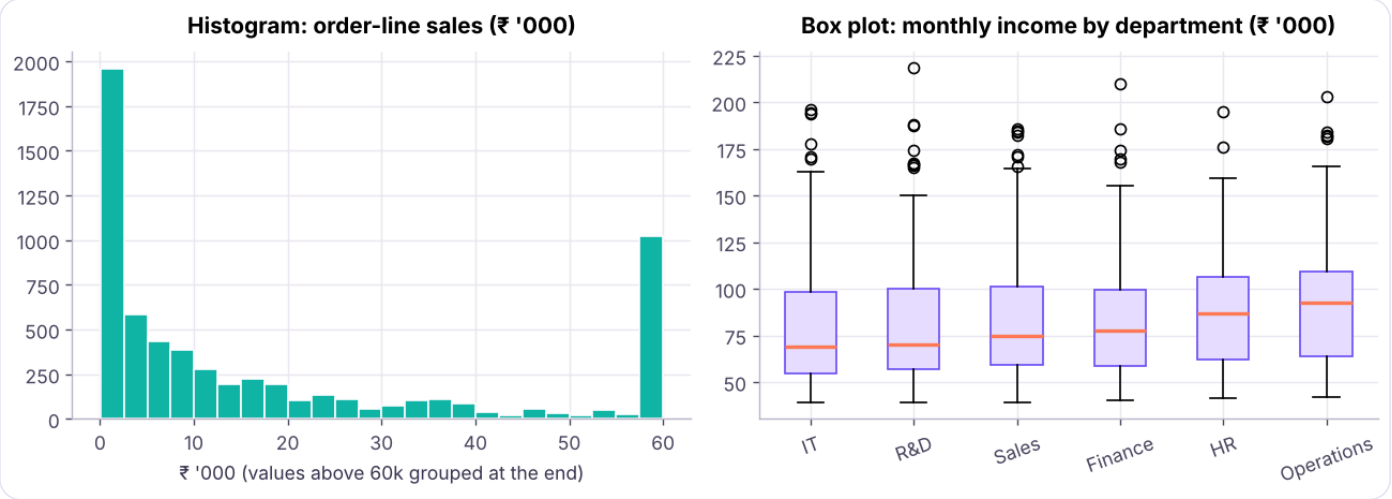
In older Excel versions you build the same chart as a **stacked column** with an invisible "base" series, an "up" series and a "down" series computed with formulas.



Waterfall from revenue to net profit. Totals are anchored at zero; increases are green and decreases are red.

Statistical and hierarchy charts (Excel 2016+)

Chart	Insert path	Shows	Example question
Histogram	Insert → Statistical → Histogram	Frequency distribution; set bin width in Format Axis	How are order values distributed?
Box & Whisker	Insert → Statistical → Box and Whisker	Median, quartiles and outliers by group	Do salaries vary more in Sales than in IT?
Treemap	Insert → Hierarchy → Treemap	Nested rectangles sized by value	Category → sub-category share of sales
Sunburst	Insert → Hierarchy → Sunburst	Rings for hierarchy levels	Region → state → city
Funnel	Insert → Waterfall → Funnel	Drop-off between stages	Visitors → sign-ups → orders
Filled map	Insert → Maps	Values by country or state	Sales by state (needs internet)



Histogram of order-line sales values and box plots of monthly income by department (HR data).

Dynamic charts

Charts become interactive when their source updates:

- **Table source**: new rows are added to the chart automatically.
- **Named range / dynamic named range** (Lesson 9) as the series values: `=Book1.xlsx!SalesLast12`.
- **Drop-down + formulas**: a Data Validation drop-down picks a region, formulas (XLOOKUP/FILTER) pull that region's numbers into a helper table, and the chart reads the helper table.
- **Pivot charts + slicers** (Lesson 8) for click-to-filter dashboards (Lesson 11).

2026 UPDATE **Charts in Excel 365 (2026)**

Copilot can suggest and insert charts from a sentence ("show monthly sales by region as a line chart"), and **Analyze Data** (Home → Analyze Data) proposes ready-made pivot charts with insights. Python in Excel can create seaborn or matplotlib charts inside the grid with `=PY()`.

Summary

KEY TAKEAWAYS

- Choose the chart from the question: comparison (bar/column), trend (line), part-of-whole (pie for ≤5, stacked bar, treemap), relationship (scatter), distribution (histogram, box), contribution (waterfall, Pareto).
- Titles should state the insight; start bar axes at zero; use one highlight colour.
- Combo charts with a secondary axis show two measures of different scales.
- **Thermometer**: one % column, axis 0–100%, gap width 0.
- **Pareto**: sorted bars + cumulative % line on a secondary axis; look for the ~80% point.
- **Population pyramid**: one series negative, overlap 100%, custom format `#,##0;#,##0`.
- **Waterfall**: Insert → Waterfall, then "Set as total" for subtotal bars.

Quick quiz

6 questions · tap an answer to check it

1. Which chart best shows how monthly revenue changed over two years?

- **A** Pie
- **B** Line
- **C** Treemap
- **D** Scatter

Answer: B. Lines show trends over time.

2. In a Pareto chart, what does the line represent?

- **A** The average value
- **B** The cumulative percentage of the total
- **C** The target
- **D** The trend

Answer: B. The line accumulates each bar's share to show where ~80% is reached.

3. How do you make the male bars go to the left in a population pyramid?

- **A** Make the male values negative and format the axis to hide the minus sign
- **B** Use a pie chart
- **C** Reverse the category order
- **D** Put males on a secondary axis

Answer: A. Negative values plot to the left; `#,##0;#,##0` shows them as positive numbers.

4. In Excel's waterfall chart, why must Gross profit be "Set as total"?

A To change its colour

B So it starts from zero instead of floating on the previous step

C To hide it

D To make it a line

Answer: B. Totals should be anchored to the axis; otherwise Excel treats them as another step.

5. When should you use a secondary axis?

A Whenever there are two series

B When two series have very different scales or units, such as sales and margin %

C When the chart is a pie

D Never

Answer: B. A secondary axis lets both series be readable; label both axes clearly.

6. Which chart shows median, quartiles and outliers for several groups?

A Histogram

B Box and whisker

C Funnel

D Sunburst

Answer: B. Box plots summarise distributions and highlight outliers.

EXERCISE **Build four charts from Superstore**

(1) A Pareto chart of profit by sub-category (watch out for negative profit: what does it mean?). (2) A combo chart of quarterly sales (columns) and order count (line, secondary axis). (3) A waterfall from 2024 total sales to 2025 total sales, with one step per region showing the change. (4) A box plot of discount by category.

Show solution (worked answer)

(1) Sort profit descending; negative sub-categories (such as Tables and Bookcases here) reduce the cumulative total, so the cumulative line rises above 100% and then falls; a Pareto chart works best on positive values, so plot losses separately. (2) Pivot by quarter with Sum of sales and Count of order_id → Combo chart with the count on the secondary axis. (3) Table: 2024 total; North change; South change; East change; West change; 2025 total. Insert → Waterfall; set the first and last bars as totals. (4) Select category and discount columns → Insert → Box and Whisker.

INTERVIEW CORNER **Questions you may be asked**

Q: How do you choose a chart type? From the question: comparison, trend, composition, relationship, distribution or contribution. Then simplify: remove clutter, highlight the key point and write an insight title.

Q: What is a Pareto analysis used for? To find the vital few causes (products, customers, complaint types) that produce most of the effect, so that effort is focused where it matters.

Q: How would you show actual vs target for 10 sales reps? A bar chart of actuals with a target marker (a line or a thin overlapping bar per rep), or a bullet chart; a thermometer works for one overall target.

ADVANCED EXCEL · LESSON 11

Interactive Dashboard: Superstore Case Study

Build a complete, interactive sales dashboard from the Superstore data, from business questions and KPIs to pivot charts, slicers, a timeline, KPI cards and a dynamic title. Then protect it (sheet, workbook, allowed ranges), document it with comments and notes, and audit formulas with Evaluate Formula.

● Live classes 8–9 ✦ Excel for Microsoft 365, 2026 Intermediate–Advanced ⏱ 150 min

By the end of this lesson you can

- Turn a business brief into KPIs and questions before building anything
- Prepare data, create one pivot per visual and connect them with slicers and a timeline
- Build KPI cards and a dynamic title with GETPIVOTDATA and TEXT
- Lay out a clean dashboard and apply design rules
- Protect sheets and workbooks, use comments and notes, and audit formulas with Evaluate Formula

The brief

CASE STUDY **Superstore sales review**

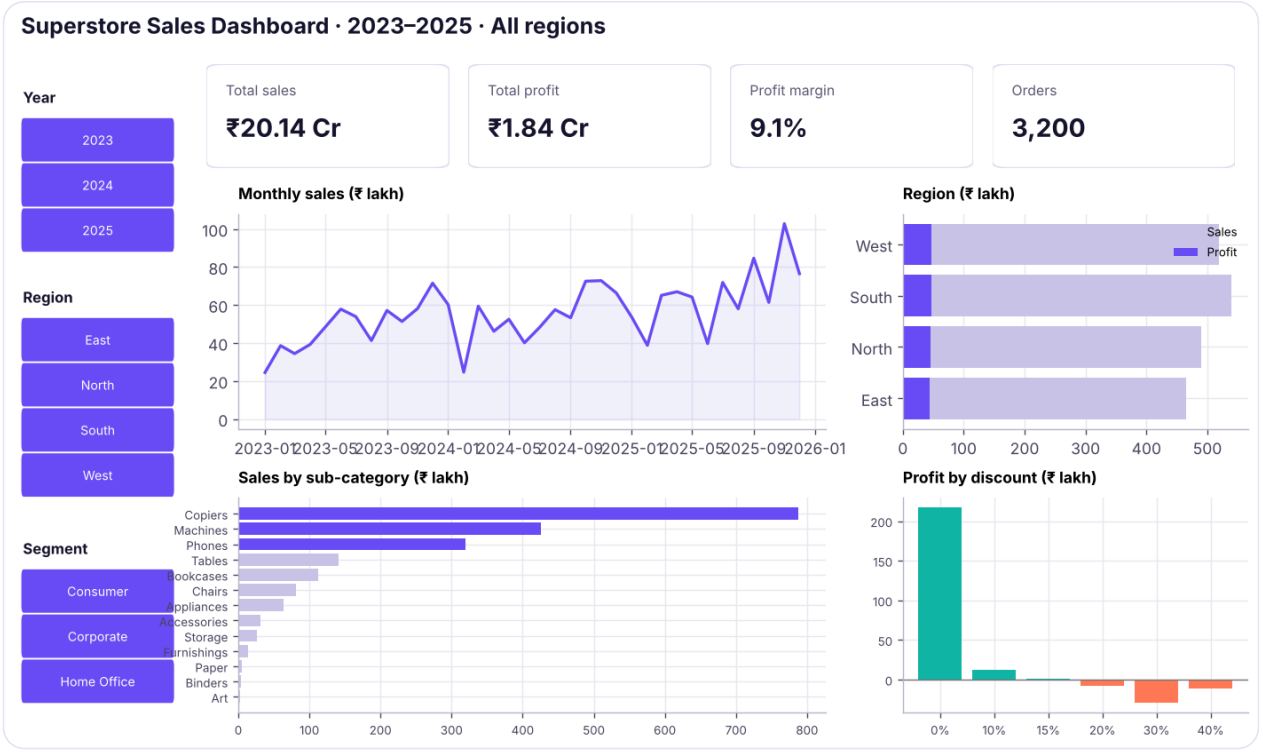
The Head of Sales wants a one-page Excel dashboard to review performance for 2023–2025. They want to see, at a glance: total sales, profit, profit margin and orders; how sales are trending; which regions, categories and sub-categories perform best and worst; and the effect of discounts on profit. They want to filter by year, region and segment. The dashboard will be emailed every month, so it must be protected against accidental changes.

Step 0: questions and KPIs first

KPI / question	Calculation	Visual
Total sales	Sum of sales	KPI card
Total profit and margin	Sum of profit; profit ÷ sales	KPI cards
Number of orders	Distinct count of order_id	KPI card
Sales trend	Sales by month	Line chart
Regional performance	Sales and profit by region	Bar chart
Product mix	Sales by category and sub-category	Bar chart / treemap
Discount impact	Profit by discount level	Column chart
Filters	Year, Region, Segment	Timeline + slicers

Agreeing on this table before building saves hours of rework. It is also your "requirements document" in interviews.

Here is the finished result we are aiming for (drawn from the real data):



The finished Superstore dashboard: KPI cards on top, trend and breakdown charts below, slicers on the left.

Step 1: prepare the data

1. Import `superstore_sales.csv` with **Data → From Text/CSV** (Power Query, Lesson 12), or open it and convert the range to a Table named `tblSales` (`Ctrl + T`).
2. Add helper columns that the pivots will need:

Column	Formula (structured reference)	Why
`year`	`=YEAR([@order_date])`	Year slicer (or use a Timeline)
`month`	`=TEXT([@order_date],"mmm")`	Labels
`margin`	`=[@profit]/[@sales]`	Row-level margin for analysis
`discount_band`	`=IFS([@discount]=0,"0%",[@discount]⩽0.2,"1-20%",TRUE,">20%")`	Simpler grouping
`days_to_ship`	`=[@ship_date]-[@order_date]`	Operations KPI

3. Check data quality: blanks (`COUNTBLANK`), duplicates of `row_id` , negative quantities, dates outside 2023–2025.

Python

```
s.describe(include="all").T[["count", "unique", "min", "max"]].head(12)
```

OUTPUT

	count	unique	min	max
row_id	6,323	NaN	1	6,323
order_id	6323	3200	NaN	NaN
order_date	6323	NaN	2023-01-01	2025-12-31
ship_date	6323	1040	NaN	NaN
ship_mode	6323	4	NaN	NaN
customer_id	6323	785	NaN	NaN
customer_name	6323	424	NaN	NaN
segment	6323	3	NaN	NaN
city	6323	16	NaN	NaN
state	6323	15	NaN	NaN
region	6323	4	NaN	NaN
product_id	6323	78	NaN	NaN

12 rows × 4 columns

Step 2: one pivot table per visual

Create a sheet called **Calc** (it will be hidden later). Insert one pivot table for each chart, all from `tblSales` so they share **one pivot cache**:

Pivot	Rows	Values	Notes
P1 KPIs	(none)	Sum of sales, Sum of profit, Distinct count of order_id	Tick "Add this data to the Data Model" to get Distinct Count
P2 Trend	order_date grouped by Years + Months	Sum of sales	Line chart
P3 Region	region	Sum of sales, Sum of profit	Bar chart
P4 Sub-category	sub_category	Sum of sales (sorted)	Bar chart
P5 Discount	discount	Sum of profit	Column chart

Give each pivot a meaningful name (PivotTable Analyze → PivotTable Name: `ptTrend`, `ptRegion` ...). Names make GETPIVOTDATA formulas and maintenance easier.

Step 3: pivot charts

Click each pivot → PivotTable Analyze → **PivotChart** → choose the type → cut and paste the chart onto a sheet called **Dashboard**.

Clean each chart:

- Right-click the field buttons → **Hide All Field Buttons on Chart**.
- Delete gridlines and legends that are not needed; add data labels where there are few bars.
- Apply the same font and a two-colour palette (one highlight, one neutral).
- Write insight titles, for example "Technology drives most sales" or "Discounts above 20% destroy profit".

Step 4: slicers and a timeline that control everything

1. Select any pivot → PivotTable Analyze → **Insert Slicer** → tick `region` and `segment`.
2. **Insert Timeline** → `order_date` → switch the timeline to **YEARS**.
3. Right-click each slicer → **Report Connections** → tick **all five** pivots. Now one click filters every chart and KPI.
4. Format slicers: Slicer tab → Columns = 1 or 2, a matching style, size them to align with the grid.

> **WATCH OUT** **Report Connections are greyed out?**
>
> Slicers can only connect pivots that share the same pivot cache (or the same Data Model). If you built a pivot from a different range, rebuild it from `tblSales`.

Step 5: KPI cards and a dynamic title

KPI cards are simply cells or shapes linked to the KPI pivot. `GETPIVOTDATA` keeps returning the right number even when slicers change the pivot.

Calc sheet: KPI formulas (values shown for the unfiltered data)

	A	B	C
1	KPI	Formula	Value
2	Total sales	=GETPIVOTDATA("Sum of sales",Calc!$A$3)	201,413,569
3	Total profit	=GETPIVOTDATA("Sum of profit",Calc!$A$3)	18,352,164
4	Profit margin	=C3/C2	9.1%
5	Orders	=GETPIVOTDATA("Distinct Count of order_id",Calc!$A$3)	3,200
6	Sales label	="₹"&TEXT(C2/10^7,"0.00")&" Cr"	₹20.14 Cr

```
C4   =C3/C2
C6   ="₹"&TEXT(C2/10^7,"0.00")&" Cr"
```

Let's verify these numbers with pandas:

Python

```python
print(f"Sales  : {s.sales.sum():,.0f}")
print(f"Profit : {s.profit.sum():,.0f}")
print(f"Margin : {s.profit.sum() / s.sales.sum():.1%}")
print(f"Orders : {s.order_id.nunique():,}")
```

OUTPUT

```
Sales  : 201,413,569
Profit : 18,352,164
Margin : 9.1%
Orders : 3,200
```

To show a KPI card on the dashboard: Insert → Shapes → Rounded Rectangle, click the shape, type `=Calc!C6` in the **formula bar**, and press Enter. The shape now displays the live value.

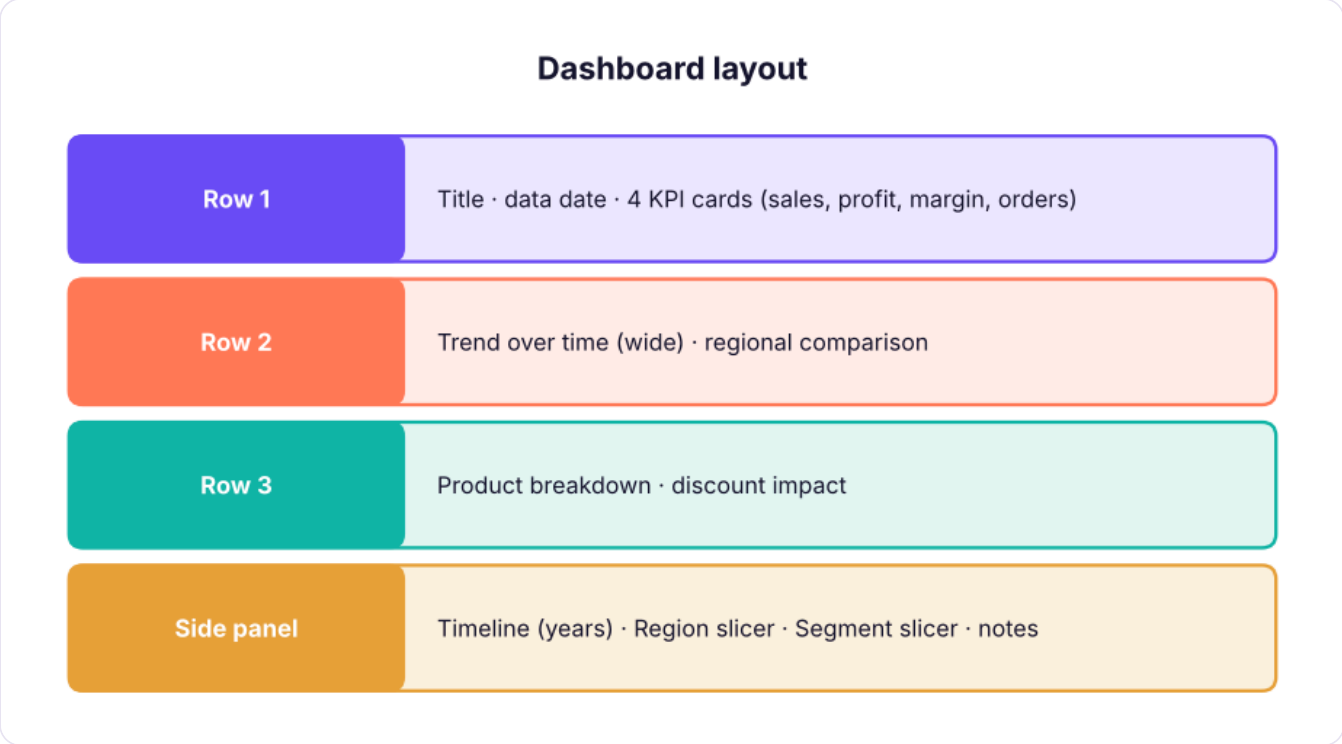
Dynamic title: build the title text in a cell and link a text box to it:

Excel formula

```
="Superstore Sales Dashboard · " & TEXTJOIN(", ", TRUE, Calc!H2:H5)
```

where `H2:H5` is a small pivot with `region` in Rows that is also connected to the slicers, so it lists only the selected regions.

Step 6: layout and design rules



A simple dashboard grid. Most important information top-left; filters on one side; consistent spacing.

1. **Z-pattern**: people read top-left first: put the headline KPIs there.
2. **Align to the grid**: hold `Alt` while dragging or resizing a chart to snap it to cell borders; set every chart to the same height.
3. **Limit colours**: one brand colour, one highlight, red only for bad news.
4. **Remove clutter**: View → untick Gridlines and Headings; hide the Calc sheet (right-click tab → Hide).
5. **Label the date and the source**: "Data: Superstore orders, Jan 2023 – Dec 2025. Refreshed 05-Jan-2026."
6. **Test it**: click every slicer combination; check that totals match the raw data.

Step 7: protect the dashboard

Protection	Where	What it does
Lock / unlock cells	`Ctrl + 1` → Protection → Locked	All cells are locked by default; **unlock** the input cells users may change. Locking only takes effect when the sheet is protected
Hide formulas	`Ctrl + 1` → Protection → Hidden	Formula bar shows nothing for those cells once the sheet is protected
Protect Sheet	Review → Protect Sheet	Prevents editing locked cells; tick what users may still do: **Use PivotTable & PivotChart**, **Edit objects** (needed for slicers), sort, filter, format
Allow Edit Ranges	Review → Allow Edit Ranges	Lets specific ranges be edited, optionally with their own password
Protect Workbook	Review → Protect Workbook	Protects the **structure**: no adding, deleting, renaming, moving or unhiding sheets
Encrypt with password	File → Info → Protect Workbook → Encrypt	Password needed to **open** the file
Mark as Final	File → Info → Protect Workbook	A read-only hint (not security)

STEP BY STEP **Protect so slicers still work**

1. Unlock any cells users need to type in (for example a target input).
2. Review → **Protect Sheet** → tick "Select unlocked cells", "Use PivotTable & PivotChart" and "Edit objects" → set a password.
3. Review → **Protect Workbook** (structure) so the hidden Calc sheet cannot be unhidden.
4. Test with a colleague before emailing.

WATCH OUT **Sheet protection is not strong security**

Sheet and structure passwords stop accidents, not determined users. Never rely on them to protect confidential data: use file encryption, SharePoint permissions or share only what is needed (for example a PDF export).

Comments and notes

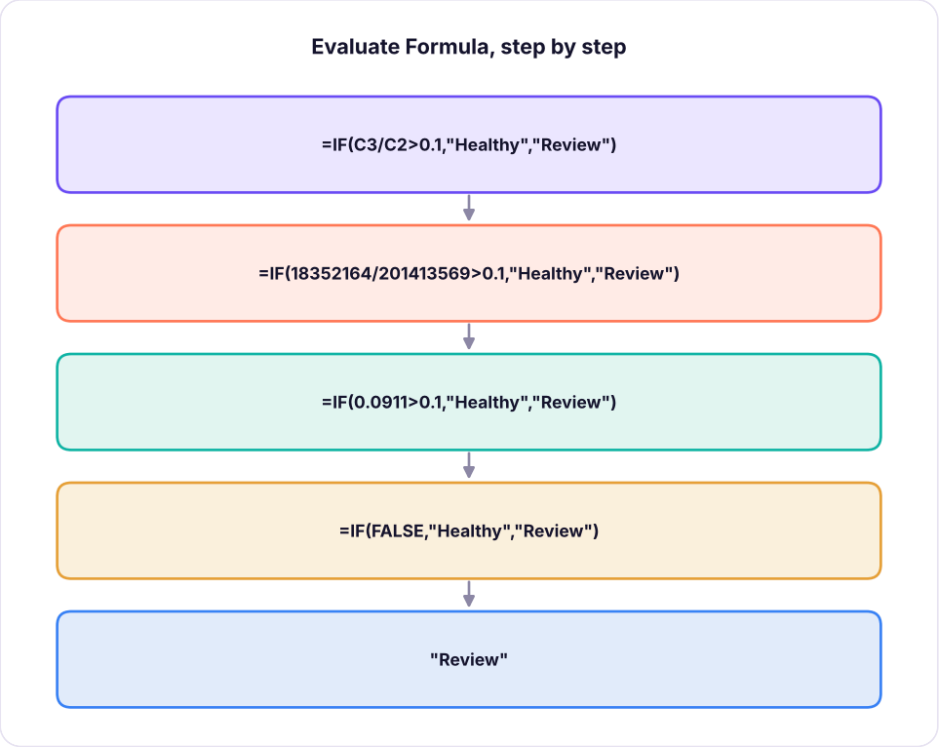
Feature	Insert	Use
Note (old "comment")	Review → Notes → New Note (`Shift + F2`)	A sticky note explaining a cell: "Target approved by CFO on 12-Jan"
Comment (threaded)	Review → New Comment (`Ctrl + Alt + M`)	A conversation: @mention a colleague, reply, resolve

Notes are best for permanent documentation; comments for review discussions. Add a small **"How to use"** note on the dashboard explaining the slicers and data date.

Auditing formulas: Evaluate Formula and friends

When a KPI looks wrong, use **Formulas → Formula Auditing**:

Tool	What it shows
Evaluate Formula	Steps through a formula one calculation at a time: press Evaluate to see each part replaced by its value
Trace Precedents / Dependents	Arrows to the cells a formula uses, or that use it
Show Formulas (`Ctrl + ``)	Every formula instead of its result
Error Checking	Walks through cells with errors and suggests fixes
Watch Window	Keeps an eye on key cells while you work elsewhere
F9 inside the formula bar	Select part of a formula and press F9 to see its value (press Esc to undo)



Evaluate Formula replaces one part at a time until only the result remains.

Delivering the dashboard

- **Refresh**: Data → Refresh All before sending.
- **Export**: File → Export → PDF (select the Dashboard sheet, landscape, fit to one page) for people who only need to read it.
- **Share live**: save to OneDrive/SharePoint and share a link; viewers can use slicers in Excel for the web.
- **Automate**: Power Query + Refresh All, or Office Scripts / Power Automate to refresh and email on a schedule.

DO IT WITH AI **Do it with AI**

Copilot in Excel can draft the analysis: *"Summarise sales and profit by region and category, identify the three biggest loss-making sub-categories, and suggest charts for a dashboard."* Use its suggestions as a starting point, then build the final dashboard deliberately: you are responsible for every number on it.

Insights from the case

Python

```
sub = s.groupby("sub_category")[["sales", "profit"]].sum()
sub["margin"] = (sub.profit / sub.sales * 100).round(1)
print("Loss-making sub-categories:\n", sub[sub.profit < 0].round(0).to_string())
disc = s.groupby(pd.cut(s.discount, [-0.01, 0, 0.2, 1], labels=["0%", "1-20%", ">20%"]),
observed=False)[["sales", "profit"]].sum()
disc["margin %"] = (disc.profit / disc.sales * 100).round(1)
print("\nMargin by discount band:\n", disc.round(0).to_string())
```

OUTPUT

```
Loss-making sub-categories:
                   sales    profit  margin
sub_category
Bookcases     11274848.0 -422786.0    -4.0
Tables        14046224.0 -816042.0    -6.0

Margin by discount band:
                 sales      profit  margin %
discount
0%        123936290.0  21718963.0      18.0
1-20%      57729097.0    697141.0       1.0
>20%       19748182.0  -4063940.0     -21.0
```

A good dashboard closes with 3–4 written insights, for example:

1. Technology generates about three quarters of sales and most of the profit; furniture is unprofitable overall.
2. Tables and Bookcases lose money; both are heavily discounted.
3. Order lines with discounts above 20% lose money almost every time: review the discount policy.
4. Sales peak every October–December (festive season); plan stock and staffing accordingly.

Summary

KEY TAKEAWAYS

- Start with the brief, KPIs and questions; then data prep; then one pivot per visual.
- Build all pivots from one Table so they share a cache and can be connected to slicers and a timeline (Report Connections).
- KPI cards use GETPIVOTDATA; shapes can display a cell's value by typing `=Cell` in the formula bar.
- Design: top-left headline KPIs, aligned grid, few colours, insight titles, data date and source.
- Protect: unlock inputs, Protect Sheet (allow PivotTables and objects), Protect Workbook structure; use encryption for real security.
- Audit with Evaluate Formula, Trace Precedents/Dependents, Show Formulas and the Watch Window.

Quick quiz

6 questions · tap an answer to check it

1. Why should all dashboard pivots be built from the same Table?

A It makes the file smaller only

B So they share one pivot cache and can all be connected to the same slicers

C Pivot charts require it

D It is needed for conditional formatting

Answer: B. Slicers can only control pivots that share a cache (or data model).

2. After protecting the sheet, slicers stop working. What did you forget?

A To unlock the slicer cells

B To allow "Use PivotTable & PivotChart" and "Edit objects" in Protect Sheet

C To protect the workbook structure

D To refresh the pivots

Answer: B. Slicers are objects that change pivots, so both permissions must be allowed.

3. Which tool lets you see each step of a complex formula's calculation?

- **A** Trace Dependents
- **B** Evaluate Formula
- **C** Watch Window
- **D** Name Manager

Answer: B. Evaluate Formula replaces one part at a time with its value.

4. What does Protect Workbook (structure) prevent?

- **A** Opening the file without a password
- **B** Editing any cell
- **C** Adding, deleting, renaming, moving, hiding or unhiding sheets
- **D** Using slicers

Answer: C. Structure protection guards the sheets, not the cells.

5. How do you count distinct orders in a pivot table?

- **A** Count of order_id
- **B** Add the data to the Data Model and use Distinct Count
- **C** Sum of order_id
- **D** Use a calculated field

Answer: B. Distinct Count is available only for Data Model pivots.

6. What is the difference between a note and a comment in Microsoft 365?

- **A** Notes are simple sticky notes; comments are threaded conversations with replies and mentions
- **B** They are identical
- **C** Comments cannot be printed
- **D** Notes can only be added to formulas

Answer: A. Use notes for documentation and comments for review discussions.

EXERCISE **Build it yourself**

Build the dashboard in this lesson from `superstore_sales.csv`. Then add: (1) a **Ship mode** slicer; (2) a KPI card for **average days to ship**; (3) a chart of the top 10 customers by sales that responds to all slicers; (4) sheet protection that still allows slicers.

Show solution (worked answer)

(1) Insert Slicer → ship_mode → Report Connections → all pivots. (2) Add `days_to_ship` to tblSales; add Average of days_to_ship to the KPI pivot; link a shape to its GETPIVOTDATA cell. (3) New pivot from tblSales: Rows customer_name, Values Sum of sales, Value Filter Top 10; PivotChart (bar); connect to the slicers. (4) Review → Protect Sheet with "Use PivotTable & PivotChart" and "Edit objects" ticked; protect workbook structure.

INTERVIEW CORNER **Questions you may be asked**

Q: Walk me through how you would build an Excel dashboard. Clarify the audience and questions; define KPIs; clean and structure the data as a Table; create one pivot per visual; build pivot charts; connect slicers and a timeline; add KPI cards with GETPIVOTDATA; design the layout; validate numbers against the source; protect and share.

Q: How do you make sure dashboard numbers are correct? Reconcile totals with the raw data (SUM vs pivot), test filters, check distinct counts, use Evaluate Formula and trace precedents for KPI formulas, and have a second person review.

Q: Excel dashboard or Power BI? Excel for small data, quick turnaround and audiences who live in Excel; Power BI for larger data, many users, scheduled refresh, row-level security and richer interactivity.

ADVANCED EXCEL · LESSON 12

Power Query, Power Pivot, Analysis ToolPak & What-If Analysis

Automate cleaning with Power Query (import, transform, unpivot, merge, append, group), model several tables with Power Pivot and DAX, get descriptive statistics and regression from the Analysis ToolPak, plan with Goal Seek, Data Tables, Scenario Manager and Solver, and meet LAMBDA, Python in Excel and macros.

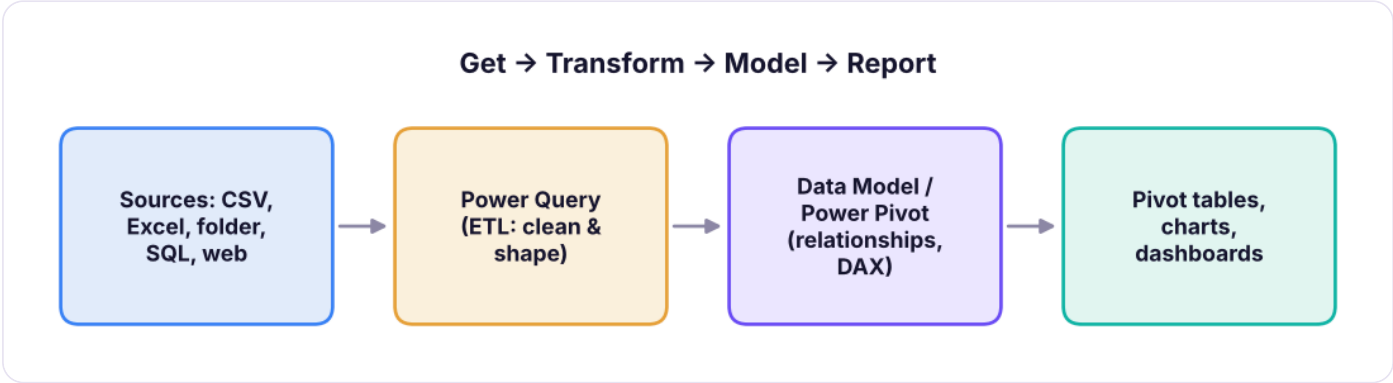
Live class 9 · ✦ Excel for Microsoft 365, 2026 · Advanced · ⏱ 150 min

By the end of this lesson you can

- Build repeatable ETL in Power Query and explain the Applied Steps and M code
- Unpivot wide data, merge (join) and append (union) tables, and group data in Power Query
- Load tables to the Data Model, create relationships and write simple DAX measures in Power Pivot
- Produce descriptive statistics, histograms, correlation and regression with the Analysis ToolPak
- Answer what-if questions with Goal Seek, Data Tables, Scenario Manager and Solver
- Know when to use LAMBDA, Python in Excel, VBA macros and Office Scripts

Why "Power" tools?

Formulas and pivots are great until the same cleaning has to be repeated every month, the data has several related tables, or it grows past a million rows. The Power tools solve exactly these problems, and they are the **same engines used in Power BI**, so everything you learn here transfers directly.



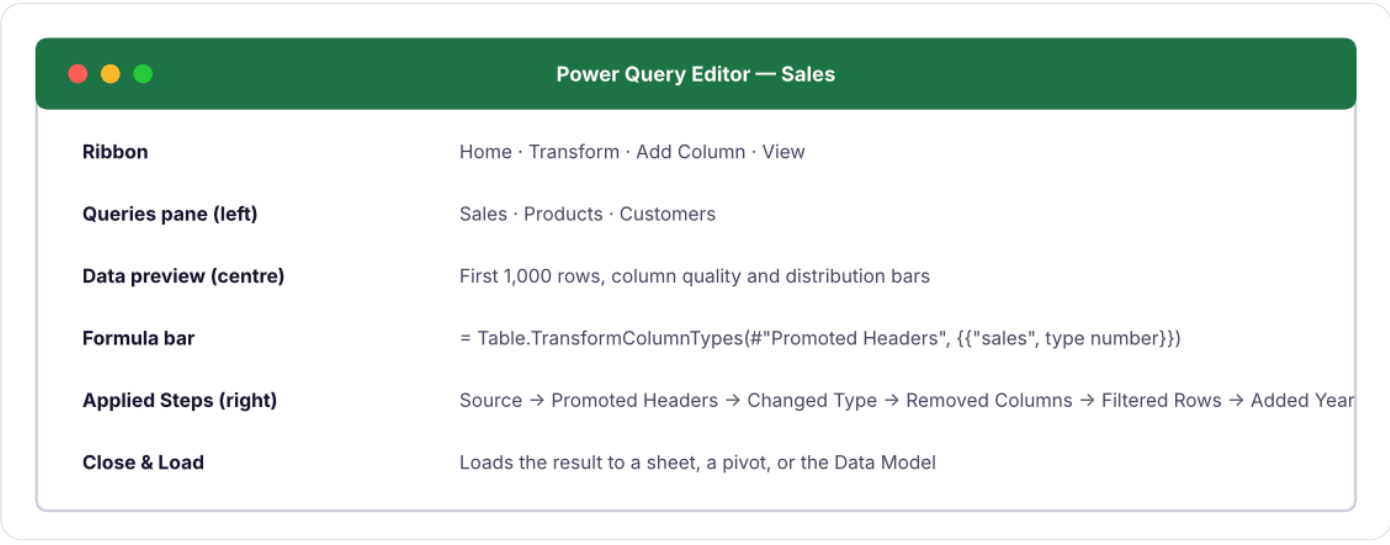
The modern Excel analytics stack. Power Query and the Data Model are shared with Power BI.

Power Query: repeatable cleaning

Power Query (Data → **Get & Transform Data**) imports data and records every cleaning step you click. Next month, put the new file in place and click **Refresh**: all steps run again automatically.

STEP BY STEP **Your first query**

1. Data → **Get Data → From File → From Text/CSV** → choose `superstore_sales.csv` → **Transform Data**. The Power Query Editor opens.
2. Check **data types** in each column header (ABC = text, 123 = whole number, 1.2 = decimal, calendar = date). Fix any wrong ones (right-click header → Change Type).
3. Remove columns you do not need (select → right-click → Remove, or **Choose Columns**).
4. Filter rows with the header drop-downs (for example remove test orders).
5. Add a column: **Add Column → Date → Year**, or **Custom Column** with a formula such as `[sales] - [profit]`.
6. Rename the query (right panel) to `Sales` → **Home → Close & Load To...** → Table, PivotTable report, or **Only Create Connection** + **Add to Data Model**.



The Power Query Editor. Every action you take is recorded in Applied Steps on the right.

Transformations you will use most

Transformation	Where	Example
Remove / keep rows	Home → Remove Rows / Keep Rows	Remove top 3 title rows, remove blank rows, remove duplicates
Use first row as headers	Home → Use First Row as Headers	After importing a report
Split column	Home → Split Column → By Delimiter	`IN-2025-00042` → country, year, number
Replace values	Transform → Replace Values	`Bangalore` → `Bengaluru`
Trim / Clean / Format case	Transform → Format	Same as TRIM, CLEAN, PROPER
Fill down	Transform → Fill → Down	Fill blank category cells (Lesson 2's Go To Special trick, but repeatable)
Conditional column	Add Column → Conditional Column	`if [discount] > 0.2 then "High" else "Normal"`
Column from examples	Add Column → Column From Examples	Like Flash Fill, but repeatable
Group By	Home → Group By	Sales by region (like a pivot)
Unpivot	Transform → Unpivot Columns	Turn month columns into rows
Merge Queries	Home → Merge Queries	A join (lookup) between two tables
Append Queries	Home → Append Queries	Stack tables with the same columns (Jan + Feb + Mar)
From Folder	Get Data → From File → From Folder	Combine every file in a folder automatically

Unpivot: from wide to long

Reports often arrive **wide**: one column per month or per region. Pivot tables, Power BI and Python all prefer **long** data (one row per observation). Unpivot fixes it in one click.

Wide (before): one column per month

	A	B	C	D
1	**Region**	**Jan**	**Feb**	**Mar**
2	North	120	135	150
3	South	98	101	110

STEP BY STEP **Unpivot in Power Query**

1. Load the wide table into Power Query.
2. Select the **Region** column (the one that should stay as it is).
3. Transform → **Unpivot Columns → Unpivot Other Columns**.
4. Rename `Attribute` to `Month` and `Value` to `Sales`.

Long (after): one row per region and month

	A	B	C
1	**Region**	**Month**	**Sales**
2	North	Jan	120
3	North	Feb	135
4	North	Mar	150
5	South	Jan	98
6	South	Feb	101
7	South	Mar	110

"Unpivot **Other** Columns" is better than selecting the month columns, because new months (Apr, May…) added later are unpivoted automatically. The same operation is `melt()` in pandas and `UNPIVOT` in SQL.

Python

```
import pandas as pd
wide = pd.DataFrame({"Region": ["North", "South"], "Jan": [120, 98], "Feb": [135, 101], "Mar": [150,
110]})
wide.melt(id_vars="Region", var_name="Month", value_name="Sales")
```

OUTPUT

	Region	Month	Sales
0	North	Jan	120
1	South	Jan	98
2	North	Feb	135
3	South	Feb	101
4	North	Mar	150
5	South	Mar	110

6 rows × 3 columns

Merge (join) and Append (union)

- **Merge Queries** = a lookup or SQL **JOIN**. Choose the two tables, click the matching key column in each, and choose the join kind: Left Outer (all rows from the first table, matching rows from the second: like

XLOOKUP), Inner (only matches), Full Outer, Left Anti (rows with **no** match: great for finding missing products). Then expand the new column to pick which fields to bring in.

- **Append Queries** = SQL **UNION ALL**: stack January, February and March files that share the same columns.

The M language

Every step is written in **M**, Power Query's formula language. View it in Home → **Advanced Editor**:

Power Query M

```
let
    Source = Csv.Document(File.Contents("C:\Data\superstore_sales.csv"), [Delimiter=",",
Encoding=65001]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Typed = Table.TransformColumnTypes(Promoted, {{"order_date", type date}, {"sales", type number},
{"profit", type number}}),
    AddedYear = Table.AddColumn(Typed, "Year", each Date.Year([order_date]), Int64.Type),
    HighDiscount = Table.AddColumn(AddedYear, "Discount band", each if [discount] > 0.2 then "High"
else "Normal"),
    Grouped = Table.Group(HighDiscount, {"region", "Year"}, {{"Sales", each List.Sum([sales]), type
number}})
in
    Grouped
```

You rarely type M from scratch, but reading it helps you debug, and you can ask an AI assistant to explain or write a step.

> **TIP** **Change the file path once**
>
> Put the file path in a cell, name it `SourcePath`, and in M use `Excel.CurrentWorkbook(){[Name="SourcePath"]}[Content]{0}[Column1]`. Colleagues can then point the query to their own folder without opening the editor.

Power Pivot and the Data Model

The **Data Model** stores tables inside the workbook in a compressed, columnar engine (VertiPaq, the same engine as Power BI). It handles millions of rows and lets one pivot table use **several related tables**, without VLOOKUPs.

STEP BY STEP **Build a small model**

1. Load `Sales`, `Products` and `Customers` with Power Query → Close & Load To → **Only Create Connection** + **Add this data to the Data Model**.
2. Data → **Manage Data Model** (Power Pivot window) → **Diagram View**.
3. Drag `Sales[product_id]` onto `Products[product_id]` and `Sales[customer_id]` onto `Customers[customer_id]` to create **one-to-many relationships** (Lesson 2 of the SQL track explains why).
4. Insert → PivotTable → **From Data Model**. You can now put `Products[category]` in Rows and `Sales[sales]` in Values: the relationship connects them.

DAX measures in Power Pivot

Measures are formulas that calculate on the fly for whatever is in the pivot (filter context). In Power Pivot → Measures → New Measure (or type in the calculation area):

```DAX
Total Sales := SUM ( Sales[sales] )
Total Profit := SUM ( Sales[profit] )
Profit Margin := DIVIDE ( [Total Profit], [Total Sales] )
Orders := DISTINCTCOUNT ( Sales[order_id] )
Technology Sales := CALCULATE ( [Total Sales], Products[category] = "Technology" )
```

Unlike pivot calculated fields (Lesson 8), `Profit Margin` here is always correct at every level, and `DISTINCTCOUNT` is built in. DAX is covered in depth in the Power BI track (Lesson 4).

Analysis ToolPak: statistics without formulas

The **Analysis ToolPak** is a free add-in: File → Options → Add-ins → Manage: Excel Add-ins → Go → tick **Analysis ToolPak**. It then appears as **Data → Data Analysis**.

Tool	What it produces
Descriptive Statistics	Mean, standard error, median, mode, standard deviation, variance, kurtosis, skewness, range, min, max, sum, count (tick "Summary statistics")
Histogram	Frequency table (and chart) for bins you specify
Correlation	Correlation matrix of several columns
Regression	Linear regression with coefficients, R^2, p-values and residuals
t-Test, z-Test, ANOVA, F-Test	Hypothesis tests (ML track, Lessons 3–5)
Moving Average, Exponential Smoothing	Simple forecasting
Random Number Generation, Sampling	Simulation and sampling

Here is what **Descriptive Statistics** returns for the order-line `sales` column (computed with Python to match the ToolPak's definitions):

Python

```
import numpy as np
from scipy import stats
s = pd.read_csv("superstore_sales.csv")
x = s["sales"]
out = pd.Series({
    "Mean": x.mean(), "Standard Error": x.std() / np.sqrt(len(x)), "Median": x.median(), "Mode":
x.mode().iloc[0],
    "Standard Deviation": x.std(), "Sample Variance": x.var(), "Kurtosis": stats.kurtosis(x,
bias=False), "Skewness": stats.skew(x, bias=False),
    "Range": x.max() - x.min(), "Minimum": x.min(), "Maximum": x.max(), "Sum": x.sum(), "Count":
len(x)}, name="sales")
out.round(2).to_frame()
```

OUTPUT

	sales
Mean	31,854.11
Standard Error	759.76
Median	8,430.00
Mode	420.00
Standard Deviation	60,414.17
Sample Variance	3,649,872,212.91
Kurtosis	16.52
Skewness	3.61
Range	619,248.00
Minimum	42.00
Maximum	619,290.00
Sum	201,413,569.00

13 rows × 1 columns (first 12 shown)

How to read it: the mean (about 31,900) is far above the median (about 8,400), and skewness is strongly positive, so a few very large orders pull the average up. Report the **median** as the typical order line.

The same numbers with Excel functions: `AVERAGE`, `MEDIAN`, `MODE.SNGL`, `STDEV.S`, `VAR.S`, `KURT`, `SKEW`, `MIN`, `MAX`, `SUM`, `COUNT`.

NOTE **ToolPak output is static**

The Analysis ToolPak writes numbers, not formulas. If the data changes, run the tool again. For live results use the equivalent functions.

What-if analysis

Data → What-If Analysis answers "what would happen if..." questions.

Goal Seek: work backwards to a target

"What price gives a profit of exactly ₹ 2,00,000?" Set up the model with formulas, then Data → What-If Analysis → **Goal Seek**: Set cell = profit cell, To value = 200000, By changing cell = price.

Goal Seek model: which price makes profit = 2,00,000?

	A	B
1	**Input / output**	**Value**
2	Units sold	5000
3	Price	90.40
4	Variable cost per unit	42
5	Fixed costs	42000
6	Profit =B2*(B3-B4)-B5	200,000

```
B3   =50+40.4
B6   =B2*(B3-B4)-B5
```

Goal Seek finds the price ₹90.40: 5,000 × (90.40 − 42) − 42,000 = 2,00,000.

Data Tables: many scenarios at once

A one-variable or two-variable **Data Table** recalculates a model for a list of inputs. Example: profit for prices 80–100 (rows) and units 4,000–6,000 (columns).

STEP BY STEP **Two-variable data table**

1. Put the profit formula in the top-left corner cell of the grid (for example E2: =B6).
2. Prices down the column below it (E3:E7), units across the row to its right (F2:H2).
3. Select the whole grid (E2:H7) → What-If Analysis → **Data Table** → Row input cell = B2 (units), Column input cell = B3 (price).

Python

```python
prices = [80, 85, 90, 95, 100]; units = [4000, 5000, 6000]
grid = pd.DataFrame({u: [u * (p - 42) - 42000 for p in prices] for u in units}, index=prices)
grid.index.name = "Price ↓ / Units →"
grid
```

OUTPUT

Price ↓ / Units →	4000	5000	6000
80	110000	148000	186000
85	130000	173000	216000
90	150000	198000	246000
95	170000	223000	276000
100	190000	248000	306000

5 rows × 3 columns

Scenario Manager

Save named sets of inputs (Best case, Base case, Worst case) and switch between them or create a **Scenario Summary** report: What-If Analysis → Scenario Manager → Add.

Solver: optimisation

Solver (another free add-in) finds the best value of a target cell by changing several inputs, subject to constraints: for example, the product mix that maximises profit given limited machine hours and a minimum quantity for each product. Choose Simplex LP for linear problems.

LAMBDA: your own functions

`LAMBDA` (Microsoft 365) turns a formula into a reusable, named function. Define it once in Name Manager:

- Name: `GSTINCL`
- Refers to: `=LAMBDA(amount, rate, amount * (1 + rate))`

Then use it like any built-in function: `=GSTINCL(B2, 18%)`. Helper functions such as `MAP`, `REDUCE`, `SCAN`, `BYROW` and `BYCOL` apply a LAMBDA to every row or element of an array.

Python in Excel

Type `=PY(` in a cell (Microsoft 365, Formulas → Insert Python) and write pandas code. Python runs securely in Microsoft's cloud with Anaconda libraries (pandas, matplotlib, seaborn, scikit-learn, statsmodels):

```python
df = xl("tblSales[#All]", headers=True)
df.groupby("region")["sales"].sum().sort_values(ascending=False)
```

The result can be returned as an Excel value or a Python object, and charts appear as images in the cell. It is a great bridge to the Python track.

Macros, VBA and Office Scripts

Tool	Where it runs	Use for
Macro recorder	View → Macros → Record Macro	Recording repetitive formatting steps; saves as VBA
VBA	Desktop Excel (`Alt + F11`)	Custom automation, user forms; file must be `.xlsm`; macros can carry security risk, so enable only from trusted sources
Office Scripts	Excel for the web (Automate tab), TypeScript	Cloud automation; can be scheduled with **Power Automate**
Power Automate	Cloud	Refresh, email or save reports on a schedule without opening Excel

2026 UPDATE

Excel AI in 2026

Copilot in Excel can generate formulas, highlight patterns, create pivot tables and charts, and run Python analysis from a plain-English prompt. Agentic "office agents" can plan multi-step tasks such as cleaning a file and building a summary. Use them to move faster, and always validate the output (AI for Data Analysis track, Lesson 4).

Summary

KEY TAKEAWAYS

- **Power Query** records cleaning steps and repeats them on Refresh: change types, split, replace, fill down, conditional columns, group by, **unpivot**, **merge** (join) and **append** (union).
- Its language is **M**; read it in the Advanced Editor.
- **Power Pivot / Data Model** stores related tables in a fast columnar engine; relationships replace VLOOKUP; **DAX measures** (SUM, DIVIDE, DISTINCTCOUNT, CALCULATE) compute correctly at every level.
- **Analysis ToolPak**: descriptive statistics, histogram, correlation, regression and tests (static output).
- **What-if**: Goal Seek (one input, target output), Data Tables (grid of scenarios), Scenario Manager (named sets), Solver (optimisation with constraints).
- **LAMBDA** creates custom functions; **Python in Excel** brings pandas into cells; macros/VBA and Office Scripts automate.

Quick quiz

6 questions · tap an answer to check it

1. Which Power Query command turns columns Jan, Feb, Mar into rows with Month and Sales columns?

A Pivot Column

B Unpivot Other Columns (with Region selected)

C Transpose

D Group By

Answer: B. Unpivot converts wide columns into attribute/value rows; "Other Columns" keeps working when new months are added.

2. Merge Queries in Power Query is equivalent to:

A A SQL JOIN or an XLOOKUP for every row

B A SQL UNION

C A pivot table

D Remove Duplicates

Answer: A. Merge joins two tables on matching key columns. Append is the UNION.

3. What is the main benefit of Power Query over manual cleaning?

A It is the only way to remove duplicates

B Steps are recorded and repeated automatically on Refresh

C It uses VBA

D It works only with CSV files

Answer: B. The recorded steps make the process repeatable and auditable.

4. Which tool finds the input value that produces a given result for one formula?

- **A** Goal Seek
- **B** Scenario Manager
- **C** Data Table
- **D** Descriptive Statistics

Answer: A. Goal Seek works backwards from a target output to one input.

5. Why is a DAX measure `DIVIDE([Total Profit],[Total Sales])` better than a pivot calculated field for margin?

- **A** It is faster to type
- **B** It is evaluated correctly in the current filter context at every level and handles division by zero
- **C** Calculated fields cannot divide
- **D** DAX works without a data model

Answer: B. Measures are context-aware and DIVIDE returns blank instead of an error for zero denominators.

6. The Analysis ToolPak reports mean 31,900 and median 8,400 for order values. What does this suggest?

- **A** The data is symmetric
- **B** The distribution is right-skewed by a few large orders
- **C** There is an error in the data
- **D** The mode is larger than the mean

Answer: B. When the mean is much larger than the median, a long right tail is pulling the mean up.

EXERCISE **Power Query practice**

Using `video_game_sales.csv`: (1) load it with Power Query and set the correct types; (2) unpivot the four regional sales columns into `Region` and `Sales`; (3) group by Genre and Region to get total sales; (4) load the result to a pivot table and show % of row total.

Show solution (worked answer)

(1) Data → From Text/CSV → Transform Data; check that year is whole number and sales are decimal. (2) Select every column except na_sales, eu_sales, jp_sales and other_sales (or select those four and choose Unpivot Columns); rename Attribute → Region, Value → Sales; remove the global_sales column first so it is not unpivoted. (3) Home → Group By → Advanced: group by genre and Region, new column Sales = Sum of Sales. (4) Close & Load To → PivotTable Report; Rows genre, Columns Region, Values Sales → Show Values As % of Row Total. You should get the same mix as Lesson 8, Q3.

INTERVIEW CORNER **Questions you may be asked**

Q: What is Power Query and why use it? Excel's (and Power BI's) ETL tool. It connects to many sources, applies recorded, repeatable transformations (types, splits, merges, unpivot, grouping) and refreshes with one click, which reduces manual errors.

Q: What is the difference between Merge and Append? Merge adds columns from another table by matching keys (a join); Append adds rows from tables with the same structure (a union).

Q: What is the Data Model and when do you need it? An in-memory columnar database inside Excel for related tables and large data. Use it when you have several tables, need distinct counts, or exceed normal sheet limits; it also supports DAX measures.

Q: How would you find the break-even price? Build the profit formula and use Goal Seek to set profit to 0 by changing price, or solve algebraically: price = variable cost + fixed costs ÷ units.

ADVANCED EXCEL · LESSON 13

Flash Fill, Text to Columns and Standardising Messy Data

Clean messy exports fast. Use Flash Fill to learn a pattern from examples, Text to Columns to split values, and formulas to standardise case, spacing, phone numbers and dates, then check every result before you rely on it.

Gap-fill lesson | ✦ Excel for Microsoft 365, 2026 | Intermediate | ⏱ 120 min

By the end of this lesson you can

- Use Flash Fill to split, join and reformat values from one example
- Split a column with Text to Columns using a delimiter or fixed widths
- Standardise case, spacing, phone numbers and dates with formulas
- Check cleaned data with simple tests before it reaches a report

Why cleaning comes first

Most analysis time goes into cleaning. A CRM export has names in capitals, phone numbers in four formats, dates stored as text and cities with extra spaces. Every report built on that data inherits its errors. Excel has three quick tools for standardising data: **Flash Fill**, **Text to Columns** and **text formulas**. This lesson uses all three and ends with checks.

CASE STUDY **A messy customer list**

A sales operations team receives a list of 200 customers from a partner. Names are in one column, phone numbers are mixed, cities have stray spaces and the join date is text. The CRM import rejects the file. The task is to fix it in under an hour, and to show the steps so the next import can be repeated.

Flash Fill: learn the pattern from one example

Flash Fill watches what you type and fills the rest of the column when it spots a pattern. Type the result for the first row, then start typing the second row. When Excel shows the grey preview of the remaining rows, press **Enter**. Or press **Ctrl+E** on a selected column.

Typical uses:

- Split a full name into first and last names.
- Join first name and last name with a space.
- Pull a code out of a longer text, such as the three-letter city code from "BLR-2025-0042".

- Reformat phone numbers into one pattern.

Flash Fill example: names

	A	B	C
1	**Full name as received**	**First name (typed once)**	**Last name (typed once)**
2	asha RAO	Asha	Rao
3	Vikram Singh	Vikram	Singh
4	PRIYA mehta	Priya	Mehta

Type the first and second rows yourself, then use Ctrl+E on the empty cells. Excel learns that the first word is the first name and the last word is the last name.

::::warn Flash Fill is a guess Flash Fill infers a rule from your examples and applies it everywhere. If the data has mixed patterns, one unusual row can produce a wrong result without any error message. Always scan the output, and sort or filter to look at the unusual cases. ::::

Text to Columns: split one column into several

Text to Columns is for splitting a column in a fixed rule-based way. It is faster than formulas for one-off splits.

1. Select the column.
2. Go to **Data > Text to Columns**.
3. Choose **Delimited** (a comma, space, semicolon or other character) or **Fixed width** (positions).
4. Set the destination so that the original column is not overwritten by mistake.

Example: an address stored as "Plot 12, Indiranagar, Bengaluru, Karnataka" splits cleanly on commas. A product code "FUR-CH-10001234" splits on the hyphen into category, sub-category and number.

::::tip Keep the original Text to Columns overwrites the column to the right when you press Finish. Copy the source column first, or set the destination cell to an empty column. ::::

Standardising case and spacing

Three functions fix most text problems:

Problem	Formula	Result from " rAVI kUMAR "
Extra spaces inside and at the ends	`=TRIM(A2)`	"rAVI kUMAR"
Consistent capitals	`=PROPER(A2)`	"Ravi Kumar"
Upper or lower case	`=UPPER(A2)`, `=LOWER(A2)`	"RAVI KUMAR"

Combine them so that one formula cleans the value: `=PROPER(TRIM(A2))`. Nest `TRIM` inside `PROPER` so that the spaces are removed before the capitals are set.

Standardise names and cities

	A	B	C	D	E
1	Raw name	Raw city	Clean name	Clean city	Check
2	rAVI kUMAR	bengaluru	Ravi Kumar	Bengaluru	ok
3	PRIYA mehta	Mumbai	Priya Mehta	Mumbai	ok
4	sanjay Reddy	delhi	Sanjay Reddy	Delhi	ok

```
C2   =PROPER(TRIM(A2))
D2   =PROPER(TRIM(B2))
E2   =IF(C2=PROPER(TRIM(A2)),"ok","check")
```

Each pattern is copied down/across to the other cells in its column.

PROPER capitalises the first letter of each word and lowers the rest, so "bengaluru" becomes "Bengaluru". For acronyms such as "IT" or "ID", PROPER gives "It" or "Id"; keep those in a lookup list and replace them afterwards.

::::note CLEAN() removes hidden characters Text pasted from web pages can contain non-breaking spaces and line breaks that TRIM does not remove. Use `=TRIM(CLEAN(SUBSTITUTE(A2,CHAR(160)," ")))` when the data comes from a website or a PDF. ::::

Standardising phone numbers

Phone numbers from different sources arrive as "98450 12345", "+91-98450-12345" and "(080) 2345 6789". A standard form, such as ten digits, makes matching and de-duplication reliable.

Step 1 removes the common separators with nested SUBSTITUTE:

Keep only the digits of a phone number

	A	B	C	D
1	Raw number	Without spaces and dashes	Without brackets	Last 10 digits
2	+91-98450-12345	+919845012345	+919845012345	9845012345
3	(080) 2345 6789	(080)23456789	08023456789	8023456789
4	98450 12345	9845012345	9845012345	9845012345

```
B2   =SUBSTITUTE(SUBSTITUTE(A2,"-",""),"  ","")
C2   =SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(B2,"(",""),")",""),"  ","")
D2   =RIGHT(C2,10)
```

Each pattern is copied down/across to the other cells in its column.

`RIGHT(..., 10)` keeps the last ten digits, which removes the country code. Check the results: a number with fewer than ten digits after cleaning is an error to review.

Converting text dates into real dates

A date stored as text, such as "15/03/2025" or "15-Mar-25", cannot be sorted or used in a timeline. Convert it with DATE (for numeric parts) or DATEVALUE (for recognised text), and check the result format.

Text dates to real dates

	A	B	C	D	E	F
1	Text as received	Day	Month	Year	Real date	Days since joining
2	15/03/2025	15	3	2025	15-03-2025	606
3	01/11/2024	1	11	2024	01-11-2024	740
4	28/02/2025	28	2	2025	28-02-2025	621

```
B2   =VALUE(LEFT(A2,2))
C2   =VALUE(MID(A2,4,2))
D2   =VALUE(RIGHT(A2,4))
E2   =DATE(D2,C2,B2)
F2   =TODAY()-E2
```

Each pattern is copied down/across to the other cells in its column.

Splitting the parts yourself is safer than relying on the locale. A text "03/04/2025" means different months in India and the United States; explicit day, month and year parts remove that doubt.

::::warn Locale changes date meaning Excel reads "03/04/2025" as 3 April in a day-first locale and as 4 March in a month-first locale. Use explicit parts, as above, whenever the source format is not confirmed. ::::

Checking the clean data

Cleaning is not finished until you test it. Three simple checks catch most mistakes:

1. **Count check**: the number of rows before and after cleaning must match, unless you deliberately removed duplicates.
2. **Blank check**: `=COUNTBLANK(range)` should be zero for required columns.
3. **Pattern check**: `=SUMPRODUCT(--(LEN(range)<>10))` counts phone numbers that are not ten characters.

For duplicates, use **Data > Remove Duplicates** on a copy, or `=COUNTIF(range, A2)>1` to flag repeats before removing them.

Quality flags on a cleaned list

	A	B	C	D
1	Customer	Phone (clean)	Duplicate?	Phone ok?
2	Ravi Kumar	9845012345	duplicate	ok
3	Priya Mehta	9812345678	unique	ok
4	Ravi Kumar	98450	duplicate	check

```
C2   =IF(COUNTIF(A$2:A$4,A2)>1,"duplicate","unique")
D2   =IF(LEN(B2)=10,"ok","check")
C3   =IF(COUNTIF(A$2:A$4,A3)>1,"duplicate","unique")
D3   =IF(LEN(B3)=10,"ok","check")
C4   =IF(COUNTIF(A$2:A$4,A4)>1,"duplicate","unique")
D4   =IF(LEN(B4)=10,"ok","check")
```

The third row is flagged twice: its name repeats and its number is too short. Both are reasons to contact the partner before import.

A repeatable cleaning routine

1. Copy the raw data to a new sheet and never edit the original.
2. Trim, proper-case and remove hidden characters.
3. Split combined fields with Text to Columns or Flash Fill, checking each result.
4. Standardise phone numbers and dates with formulas, so the logic can be re-run.
5. Run the count, blank and pattern checks, and record what you found.
6. Save the final sheet with a date in the file name.

::::key What to remember - Flash Fill (Ctrl+E) learns a pattern from examples and must be checked. - Text to Columns splits by delimiter or position; copy the source first. - `PROPER(TRIM(...))` is the standard text clean-up; add `CLEAN` for pasted text. - Standardise phone numbers and dates with formulas, using explicit date parts. - Test the result with count, blank and pattern checks before import. ::::

Quick quiz

5 questions · tap an answer to check it

1. Which shortcut runs Flash Fill on the selected column?

- **A** Ctrl+F
- **B** Ctrl+E
- **C** Ctrl+R
- **D** Alt+F4

Answer: B. Ctrl+E is the Flash Fill shortcut.

2. What does `=PROPER(TRIM(" rAVI kUMAR "))` return?

- **A** rAVI kUMAR
- **B** Ravi Kumar
- **C** RAVI KUMAR
- **D** Ravi Kumar

Answer: B. TRIM removes extra spaces, then PROPER capitalises each word.

3. Text to Columns overwrites which cells by default?

A The column on the left

B The columns to the right of the selected column

C The first row only

D Nothing; it always creates a new sheet

Answer: B. Choose an empty destination to keep the original data.

4. Why use explicit DATE(year, month, day) for text dates?

A DATE is faster than typing the value

B It removes the risk that the locale reads day and month in the wrong order

C It creates text dates

D It changes the time zone

Answer: B. Explicit parts avoid day-month confusion.

5. Which formula counts rows whose required column is blank?

A =COUNT(range)

B =COUNTA(range)

C =COUNTBLANK(range)

D =SUM(range)

Answer: C. COUNTBLANK counts empty cells.

EXERCISE **Clean a partner list**

Using a sheet of 30 rows you create, with names in capitals, extra spaces, phone numbers in mixed formats and join dates as text in dd/mm/yyyy: (a) clean the names with `PROPER(TRIM())`; (b) produce a 10-digit phone column; (c) convert the dates to real dates; (d) add a flag column that marks duplicate names; (e) write a one-line summary showing how many rows failed the phone check.

Show solution (worked answer)

Hints: (a) `=PROPER(TRIM(A2))`. (b) nest `SUBSTITUTE` for "-", " ", "(" and ")", then `=RIGHT(x,10)`. (c) `=DATE(RIGHT(C2,4), MID(C2,4,2), LEFT(C2,2))`. (d) `=IF(COUNTIF(A$2:A$31,A2)>1,"duplicate","unique")`. (e) `=SUMPRODUCT(--(LEN(B2:B31)<>10))`.

ADVANCED EXCEL · LESSON 14

Excel Macros and VBA: Recording and Writing Automation

Automate a repeated monthly report. Record a macro, read the code it writes, change it to work on any data, add a loop over several sheets, and keep macro files safe from untrusted code.

Gap-fill lesson | ✦ Excel for Microsoft 365, 2026 | Advanced | ⏱ 150 min

By the end of this lesson you can

- Record a macro and explain the code it creates
- Replace hard-coded cell addresses with ranges that adapt to the data
- Write a VBA loop that processes several sheets
- Save macro-enabled workbooks safely and recognise risky macros

What a macro is

A **macro** is a saved set of steps. Excel can record your actions and store them as code in the **VBA** language (Visual Basic for Applications). Running the macro repeats the steps exactly. VBA is the older but still widely used way to automate Excel on the desktop.

Use a macro when:

- the same steps happen every week or month,
- the steps are mechanical (format, filter, copy, save as PDF),
- the same person or team would otherwise repeat them by hand.

CASE STUDY **The monthly regional pack**

Every month, a finance analyst receives a sales sheet with a different number of rows. The steps are always the same: remove blank rows, format the header, add a total row, split the data into one sheet per region and save each as a PDF. It takes three hours by hand. A tested macro does it in under a minute and does it the same way every time.

Step 1: enable the Developer tab

The Developer tab is hidden by default. Go to **File > Options > Customize Ribbon** and tick **Developer**. The tab then shows **Record Macro**, **Visual Basic** and **Macros**.

Step 2: record a simple macro

1. Go to **Developer > Record Macro**.
2. Name it `FormatSalesTable`, store it in **This Workbook**, and give it a shortcut such as Ctrl+Shift+F.
3. Select a table, make the header bold with a fill colour, add a border, and set the number format on the sales column.
4. Click **Stop Recording**.

Now open the code with **Alt+F11** (the Visual Basic Editor) and find the module called `Module1`. It will look something like this:

```vba
Sub FormatSalesTable()
    Range("A1:F1").Select
    With Selection
        .Font.Bold = True
        .Interior.Color = RGB(30, 142, 90)
    End With
    Range("E2:E100").Select
    Selection.NumberFormat = "#,##0.00"
End Sub
```

The recorder wrote the exact addresses you clicked. That works once, on this data. It breaks when the data grows or moves.

::::warn Recorded macros are fragile A recorded macro uses cell addresses, `Select` and `Selection`. These depend on where the data was when you recorded. Read the code and rewrite it to use ranges and objects, as shown below. ::::

Step 3: make the macro adapt to the data

The same task, written to work on any table size, finds the last row and column itself:

```vba
Sub FormatSalesTable()
    Dim ws As Worksheet
    Dim lastRow As Long
    Dim lastCol As Long

    Set ws = ActiveSheet
    lastRow = ws.Cells(ws.Rows.Count, 1).End(xlUp).Row
    lastCol = ws.Cells(1, ws.Columns.Count).End(xlToLeft).Column

    With ws.Range(ws.Cells(1, 1), ws.Cells(1, lastCol))
        .Font.Bold = True
        .Interior.Color = RGB(30, 142, 90)
        .Font.Color = RGB(255, 255, 255)
    End With

    ws.Range(ws.Cells(2, 1), ws.Cells(lastRow, lastCol)).Borders.LineStyle = xlContinuous
End Sub
```

What changed:

- `Set ws = ActiveSheet` stores the sheet in a variable, so later lines refer to it clearly.
- `End(xlUp)` moves up from the bottom of column 1 to the last filled row.
- No `Select` is needed. `Range` and `Cells` work on the object directly, which is faster and less error-prone.

Step 4: remove blank rows and add a total

Blank rows in exported data break sorts and totals. The macro below deletes any row where the first column is empty, working from the bottom up so that row numbers do not shift during the loop.

```vba
Sub RemoveBlankRows()
    Dim ws As Worksheet
    Dim r As Long

    Set ws = ActiveSheet
    For r = ws.Cells(ws.Rows.Count, 1).End(xlUp).Row To 2 Step -1
        If Application.WorksheetFunction.CountA(ws.Rows(r)) = 0 Then
            ws.Rows(r).Delete
        End If
    Next r
End Sub
```

::::tip Loop backwards when deleting Deleting rows changes the row numbers below them. Looping from the bottom up (`Step -1`) avoids skipping rows. Looping from the top is a common bug that leaves some blanks in place. ::::

Step 5: loop over several sheets

The monthly pack needs the same formatting on every regional sheet. A `For Each` loop applies the macro to each sheet in the workbook.

```vba
Sub FormatAllSheets()
    Dim ws As Worksheet
    Application.ScreenUpdating = False
    For Each ws In ThisWorkbook.Worksheets
        If ws.Name <> "Instructions" Then
            ws.Activate
            Call FormatSalesTable
        End If
    Next ws
    Application.ScreenUpdating = True
    MsgBox "Formatting complete."
End Sub
```

`Application.ScreenUpdating = False` makes the macro much faster on large workbooks, because Excel stops redrawing the screen while it works. Always set it back to `True` at the end; the last line does this here.

Step 6: error handling

Macros fail in the middle of a run, which can leave a workbook half-changed. Add an error handler so that the user sees a clear message and screen updating is restored:

```vba
Sub SafeRun()
    On Error GoTo Fail
    Application.ScreenUpdating = False
    Call FormatAllSheets
    Application.ScreenUpdating = True
    Exit Sub
Fail:
    Application.ScreenUpdating = True
    MsgBox "The macro stopped: " & Err.Description, vbExclamation
End Sub
```

Before running any macro that changes data, save a copy. A macro cannot always be undone.

Saving and running safely

- Save a workbook that contains macros as **.xlsm** (Macro-Enabled Workbook). A plain .xlsx file cannot hold macros.
- Excel blocks macros by default. Enable them only for files you trust, from people you know.
- Macros can run any action your Windows account can, including deleting files. Treat a macro from an email attachment or a download with the same caution as an unknown program.

- Keep the code in the workbook you use, or in a shared personal macro workbook, and document what each macro does at the top of its code.

::::warn Risky macros A macro that calls `Shell`, `Kill`, or downloads files from the internet is doing something beyond formatting. Read the code line by line before you enable it, or ask someone who can. ::::

Macros versus other tools

Need	Best tool
One repeated desktop task	Macro (VBA)
Automation on Excel for the web	Office Scripts (next lesson)
Repeatable data cleaning and combining files	Power Query
Scheduled, cross-app workflows	Power Automate or n8n

::::key What to remember - Record a macro, then read and rewrite the code: replace `Select` with object references and fixed addresses with `End(xlUp)`. - Loop backwards when deleting rows. - Use `ScreenUpdating = False` for speed and always restore it, with an error handler. - Save macros in .xlsm files and enable them only for trusted sources. ::::

Quick quiz

5 questions · tap an answer to check it

1. Which key combination opens the Visual Basic Editor?

A Ctrl+E

B Ctrl+Shift+F

C Alt+F11

D F2

Answer: C. Alt+F11 opens the VBA editor.

2. Why is a recorded macro often fragile?

- **A** It uses fixed addresses and Select, which break when the data changes size or position
- **B** It cannot be saved
- **C** It runs slower than a formula only
- **D** It works only in Excel for the web

Answer: A. Recorded code refers to the exact cells selected.

3. When deleting rows in a loop, why loop from the bottom up?

- **A** It is faster in every case
- **B** Deleting shifts the row numbers below, so going up avoids skipping rows
- **C** Excel requires it
- **D** It sorts the data

Answer: B. Going up keeps the row numbers you have yet to process unchanged.

4. Which file extension is needed to save a workbook with macros?

- **A** .xlsx
- **B** .csv
- **C** .xlsm
- **D** .pdf

Answer: C. Only macro-enabled workbooks keep VBA code.

5. What does `Application.ScreenUpdating = False` do?

- **A** It turns off the monitor
- **B** It stops Excel redrawing the screen, so the macro runs faster
- **C** It locks the workbook
- **D** It deletes errors

Answer: B. Restore it to True at the end of the macro.

EXERCISE

A tidy-up macro

Write a macro called `TidyPack` that, for every sheet except one called "Summary": (a) deletes rows that are completely blank; (b) makes the header row bold; (c) freezes the header row; and (d) shows a message box with the number of sheets it processed. Test it on a copy of a workbook with three sheets and some blank rows.

Show solution (worked answer)

Hints: loop over `ThisWorkbook.Worksheets` with `For Each`, reuse the backwards loop from Step 4, use `ws.Activate` then `ActiveWindow.FreezePanes` after setting `ActiveWindow.SplitRow = 1` and `ActiveWindow.FreezePanes = True`, and count the sheets in a variable that you display at the end.

ADVANCED EXCEL · LESSON 15

Office Scripts, Power Automate and AI-Assisted Excel Formulas

Automate Excel on the web with Office Scripts written in TypeScript, run them on a schedule with Power Automate, and use AI assistants to draft formulas, with a checklist for testing every formula before you rely on it.

Gap-fill lesson | ✦ Excel for Microsoft 365, 2026 | Advanced | ⏱ 120 min

By the end of this lesson you can

- Explain how Office Scripts differ from VBA macros
- Record and edit an Office Script in Excel on the web
- Trigger a script from Power Automate on a schedule
- Test an AI-written formula on edge cases before using it

Office Scripts: automation for Excel on the web

Office Scripts are automations written in **TypeScript** that run in Excel on the web and in Microsoft 365. They do not need a desktop macro, so they work on any device with a browser, and they can be started by Power Automate. Use them when the file lives in SharePoint or OneDrive and the team works in the browser.

	VBA macro	Office Script
Language	VBA	TypeScript
Runs in	Excel desktop	Excel on the web and desktop (Microsoft 365)
Started by	Button, shortcut, Alt+F8	Button, Power Automate, Excel on the web
File type	.xlsm	Normal .xlsx
Good for	Desktop-only tasks	Shared team files and scheduled runs

CASE STUDY **A Monday summary for the team**

A team lead wants a summary sheet rebuilt every Monday from the shared sales file: totals by region, a sort by sales and a timestamp. The file is in SharePoint and edited by five people. A script runs at 7 a.m. through Power Automate, so the summary is ready before the team arrives.

Step 1: record and open a script

1. Open the workbook in Excel on the web.
2. Go to **Automate > Record Actions**.
3. Format a table, add a total row and sort it. Click **Stop**.
4. Open **Automate > Code Editor** to see the script.

The recorded script uses the Office Scripts API. It is written as a function, with the workbook passed in as `workbook`:

```typescript
function main(workbook: ExcelScript.Workbook) {
    let sheet = workbook.getActiveWorksheet();
    let range = sheet.getRange("A1:F1");
    range.getFormat().getFont().setBold(true);
    range.getFormat().getFill().setColor("#1E8E5A");
}
```

As with VBA, recorded code uses fixed addresses. Replace them with values that come from the data.

Step 2: make the script adapt

This version finds the used range and totals the sales column, whatever its size:

```typescript
function main(workbook: ExcelScript.Workbook) {
    const sheet = workbook.getActiveWorksheet();
    const used = sheet.getUsedRange();
    const rowCount = used.getRowCount();

    // Column E holds sales; rows 2 to the last data row
    const sales = sheet.getRange(`E2:E${rowCount}`);
    const total = sales.getValues().reduce((sum, row) => sum + Number(row[0] || 0), 0);

    // Write the total in a labelled cell below the data
    sheet.getRange(`D${rowCount + 2}:E${rowCount + 2}`).setValues([["Total sales", total]]);
    sheet.getRange(`E${rowCount + 2}`).getFormat().setColumnWidth(110);
}
```

The `reduce` call adds up the numbers in JavaScript. Writing a total as a value is fine for a report; if the total must update as data changes, write a formula with `setFormula("=SUM(E2:E...)")` instead.

::::tip Prefer formulas for live totals A script that writes a value stops being correct the moment the data changes. A script that writes a formula stays correct. Use values for a snapshot (for example, a Monday report that must not change later) and formulas for anything that people will keep editing. ::::

Step 3: run it on a schedule with Power Automate

Office Scripts can be called from a Power Automate flow:

1. Create a flow with the trigger **Recurrence**, set to every Monday at 7:00.
2. Add the action **Excel Online (Business) > Run script**.
3. Choose the workbook from SharePoint or OneDrive, and the script name.
4. Optional: add **Send an email** to notify the team after the run.

Test the flow once by running it manually, and open the workbook to check the result before you rely on the schedule.

::::warn Scheduled runs need care A scheduled script runs without anyone watching. Make sure it cannot overwrite the source data, works on a copy if the file is shared, and sends a notice when it fails. Keep the raw data on a separate sheet from the output. ::::

AI-assisted formulas

AI tools inside Excel, such as Copilot, and general AI assistants can write formulas from a plain-language request. This is useful for drafting, and dangerous when the result is trusted without checking.

A good request names the columns, the logic and the output:

> "In column F, show 'High' if sales in column E are above 5000 and the region in column C is 'North' or 'South', otherwise 'Standard'. Use one formula."

A reasonable answer is:

```
=IF(AND(E2>5000, OR(C2="North", C2="South")), "High", "Standard")
```

Read the formula before you use it. Ask: does it match the rule, the column letters and the case of the text?

Testing an AI-written formula

Every formula from an AI tool should pass four tests before it enters a report:

1. **Known answer**: pick three rows where you know the right result by hand and check them.
2. **Boundary values**: test exactly 5000 (is it "above"?), blank cells and zeros.
3. **Unexpected text**: test "north" in lower case, extra spaces and misspellings.
4. **Totals**: compare the total of the new column with an independent calculation, such as a pivot table.

Test cases for the High/Standard formula

	A	B	C	D	E
1	Sales	Region	Expected	Formula result	Pass?
2	6000	North	High	High	yes
3	5000	North	Standard	Standard	yes
4	7000	north	High	High	yes

```
D2    =IF(AND(A2>5000,OR(B2="North",B2="South")),"High","Standard")
E2    =IF(D2=C2,"yes","NO")
D3    =IF(AND(A3>5000,OR(B3="North",B3="South")),"High","Standard")
E3    =IF(D3=C3,"yes","NO")
D4    =IF(AND(A4>5000,OR(B4="North",B4="South")),"High","Standard")
E4    =IF(D4=C4,"yes","NO")
```

All three rows pass. The third row passes because Excel's text comparison (=) ignores case, so "north" matches "North". That is the behaviour you want here, but you should know it is a choice, not an accident. If a future rule needed case-sensitive matching, the test would catch the difference only if you had written a case-sensitive expectation.

::::note Case sensitivity in Excel Excel comparisons with = ignore case, so "north" equals "North". Use `EXACT(a, b)` only when case must matter. ::::

Choosing the right tool

- **Repeated steps on a shared file in the browser**: Office Scripts, run by hand or by Power Automate.
- **Repeated steps on a desktop file**: a VBA macro.
- **Combining and cleaning many files**: Power Query, which refreshes with one click.
- **A one-off formula**: write it yourself, or check an AI draft with the four tests.

::::key What to remember - Office Scripts are TypeScript automations for Excel on the web; they run without desktop macros. - Power Automate can run a script on a schedule; test the flow manually first. - Prefer formulas over values for totals that must keep updating. - Test every AI-written formula with a known answer, boundary values, unexpected text and a total check. ::::

Quick quiz

5 questions · tap an answer to check it

1. What language are Office Scripts written in?

A VBA

B Python

C TypeScript

D C#

Answer: C. Office Scripts use TypeScript and the Office Scripts API.

2. Which tool can run an Office Script every Monday without someone opening the file?

A Power Query

B Power Automate with a Recurrence trigger

C Goal Seek

D Conditional Formatting

Answer: B. A Recurrence trigger starts the flow on a schedule, and the flow runs the script.

3. Why should a script that writes a total use a formula rather than a value if people keep editing the data?

A Values cannot be stored

B The value stops being correct when the data changes, while a formula updates

C Formulas are faster to write

D Scripts cannot write values

Answer: B. Formulas recalculate; a pasted value does not.

4. An AI-written formula gives the right result for the rows you tested. What should you do next?

A Use it in the report immediately

B Test boundary values, blanks and unexpected text before relying on it

C Delete it

D Convert it to a macro

Answer: B. Passing one set of rows does not prove the formula handles every case.

5. Does = in Excel distinguish "North" from "north"?

- **A** No, text comparison with = ignores case
- **B** Yes, always
- **C** Only in pivot tables
- **D** Only on Excel for the web

Answer: A. Use EXACT if case must matter.

EXERCISE **A scheduled summary**

Record an Office Script that takes a sheet of sales lines, creates a new sheet called "Summary" with total sales by region using formulas (`=SUMIF`), and writes the run time in a labelled cell. Then write the Power Automate flow that runs it every weekday at 8:00. List the four tests you would run on the summary before the team sees it.

Show solution (worked answer)

Hints: in the script, use `workbook.addWorksheet("Summary")` and `setFormula` with `=SUMIF(Data!C:C, A2, Data!E:E)`. For the timestamp, write `=NOW()` or the script's `new Date()` value with a label. Test with a known total, a region that has no sales, a blank row in the data, and a changed source value to see whether the summary updates.

ADVANCED EXCEL · LESSON 16

Excel Business Projects: Complaints, Finance, Productivity and Segmentation

Four portfolio-ready Excel projects, each from a business brief. Analyse customer complaints, build a financial performance tracker, measure employee productivity, and segment customers with a sales performance dashboard, using the skills from the course.

Gap-fill lesson | Excel for Microsoft 365, 2026 | Advanced | 180 min

By the end of this lesson you can

- Turn a business question into a clear analysis plan with a metric and a deliverable
- Clean, model and analyse a dataset with formulas, pivot tables and Power Query
- Build a one-page dashboard that answers the brief
- Write a short recommendation that a manager can act on

How to run a business project

Every project in this lesson follows the same five steps. Use them for any brief you meet in an interview or on the job.

1. **Brief**: write the business question in one sentence and name the decision it supports.
2. **Data**: list the sources and check their quality (see the cleaning lesson).
3. **Analyse**: calculate the metrics, using pivot tables and formulas.
4. **Visualise**: put the answer on one dashboard sheet with at most four charts.
5. **Recommend**: write three sentences: what you found, what to do, and how to measure it.

CASE STUDY **Four briefs, one method**

Each of the four briefs below came from a real team. Work through them in order. For each, produce the workbook, a dashboard sheet and a half-page note, which together make a portfolio piece.

Project 1: Customer complaint analysis

Brief: The customer experience head wants to know which complaint themes cost the most customer trust, and where to fix them first.

Data: a list of reviews with a rating (1 to 5) and a theme tag. The sample below is small so you can see the logic.

Complaint table with a severity score

	A	B	C	D	E
1	Theme	Average rating	Mentions	Share of mentions	Priority score
2	Delivery delay	2	42	0.35	126
3	Damaged item	1	18	0.15	72
4	Wrong product	2	25	0.2083	75
5	Poor support	3	35	0.2917	70

```
D2    =C2/SUM(C$2:C$5)
E2    =C2*(5-B2)
```

Each pattern is copied down/across to the other cells in its column.

Method:

- Count mentions per theme with a pivot table, and average the rating per theme.
- Priority score combines frequency and severity: many complaints with low ratings score highest. The formula weights each mention by how far the rating falls below 5.
- Sort by priority score and show the top two themes on the dashboard.

Deliverable: the top two themes, their share of mentions, and one fix for each. Say how you would measure whether the fix worked: for example, delivery-delay mentions per 100 orders next quarter.

::::tip Frequency is not severity A theme with many mentions and high ratings is annoying; a theme with fewer mentions and one-star ratings can cost more customers. The priority score keeps both in view. Explain the weighting to the manager, because it is a judgement. ::::

Project 2: Financial performance tracker

Brief: The finance manager wants one sheet that tracks budget against actual spend by month, flags overspend, and shows the year-to-date position.

Method:

- Set up three tables: budget by category, actual spend by month, and a monthly summary.
- Use `SUMIFS` to total actuals by category and month, so the tracker updates when new rows are added.
- Flag overspend with conditional formatting, and show the variance in both rupees and percent.

Budget against actual for one quarter

	A	B	C	D	E	F
1	Category	Budget	Actual	Variance	Variance %	Status
2	Salaries	300000	298500	1500	0.005	Within
3	Software	60000	66200	-6200	-0.1033	Over
4	Travel	45000	38900	6100	0.1356	Within
5	Marketing	90000	97400	-7400	-0.0822	Over
6	Total	495000	501000	-6000	-0.0121	Over

```
D2    =B2-C2
E2    =D2/B2
```

```
F2   =IF(C2>B2,"Over","Within")
B6   =SUM(B2:B5)
C6   =SUM(C2:C5)
```

Each pattern is copied down/across to the other cells in its column.

Deliverable: the categories over budget, the total variance, and two actions (for example, review the software licences and set a marketing cap). Show the year-to-date position and a forecast for the rest of the year using the monthly run rate, labelled as an estimate.

::::warn Variance sign Decide the sign convention before you build the tracker and state it on the sheet. Here, a positive variance means under budget. Mixing conventions is a common reason a manager misreads a finance report. ::::

Project 3: Employee productivity dashboard

Brief: The HR director wants to understand what drives performance and attrition across departments, so that retention plans target the right teams. The data is the HR employee file.

Method:

- Load the file into Power Query, check for blanks and set the data types.
- Use a pivot table to compare departments on average performance rating, average monthly income, overtime share and attrition rate.
- Calculate the attrition rate with `COUNTIFS` (employees who left divided by all employees in each group).
- Compare overtime and satisfaction, which often explain attrition, using a pivot chart.

Attrition and overtime by department (sample rows)

	A	B	C	D	E
1	Department	Employees	Left	Attrition rate	Overtime share
2	Operations	300	84	0.28	0.42
3	IT	280	42	0.15	0.18
4	Sales	310	77	0.2484	0.36
5	Finance	180	27	0.15	0.15

```
D2   =C2/B2
D3   =C3/B3
D4   =C4/B4
D5   =C5/B5
```

Deliverable: the department with the highest attrition, whether its overtime share is the likely driver, and a retention action with a measurable target (for example, cut overtime above 40 hours per month by half).

::::note Correlation is not a cause Overtime and attrition can move together for several reasons. Say "associated with" rather than "caused by", and suggest a test: pilot a change in one team and compare it with another. ::::

Project 4: Customer segmentation and sales performance dashboard

Brief: The commercial head wants to know which customer groups generate the most profit, which are at risk, and where the sales team should focus next quarter. The data is the retail sales file.

Method:

- Load the superstore data into Power Query, keep one row per order line, and build a customer summary: total sales, total profit, number of orders and last order date.
- Segment customers with a formula: for example, "Champion" when recent and high spend, "At risk" when no order in the last 180 days, and "Regular" otherwise.
- Build a pivot table of sales and profit by segment and region, and a slicer for year.
- Add a KPI row: total sales, profit margin, number of active customers and the share of profit from the top 20% of customers.

Segment rules applied to a customer summary

	A	B	C	D	E
1	Customer	Lifetime profit	Days since last order	Orders	Segment
2	C0765	15200	40	9	Champion
3	C0112	1800	210	2	At risk
4	C0498	9400	95	5	Regular

```
E2   =IF(AND(B2>10000,C2⩽180),"Champion",IF(C2>180,"At risk","Regular"))
E3   =IF(AND(B3>10000,C3⩽180),"Champion",IF(C3>180,"At risk","Regular"))
E4   =IF(AND(B4>10000,C4⩽180),"Champion",IF(C4>180,"At risk","Regular"))
```

Deliverable: the profit share of each segment, the number of customers at risk with their combined past profit, and one campaign for the at-risk group with its target and measurement. Be clear that the thresholds (10,000 profit, 180 days) are choices you made and can change.

::::tip Make the thresholds visible Put segment thresholds in input cells, not inside formulas. A manager can then change 180 days to 120 and see the result immediately, without editing any formula. ::::

Presenting the four projects

For a portfolio or interview, each project needs:

- the brief in one sentence,
- a dashboard with the answer in the first screen,
- the method in three or four bullet points,
- the recommendation with a measure of success,
- the limits of the data, stated plainly.

Stating the limits is what makes a project credible. Say which data was missing, which assumptions were made and what would change the answer.

::::key What to remember - Start every project with the brief and the decision it supports. - Build priority, budget and segment logic in formulas with visible thresholds. - Use pivot tables and slicers for interactive answers; keep the dashboard to the question. - Distinguish association from cause, and state the limits of the data. - Every recommendation needs a target and a way to measure it. ::::

Quick quiz

5 questions · tap an answer to check it

1. **In the complaint priority score, why multiply frequency by (5 minus the average rating)?**

A To make the number bigger

B To give more weight to themes that are frequent and have low ratings

C To remove duplicates

D To convert ratings into percentages

Answer: B. The score is high when a theme is common and poorly rated.

2. **Which function totals actual spend for a category and month that can change as rows are added?**

A SUM on a fixed range

B SUMIFS over a range that covers the data

C COUNTA

D AVERAGE

Answer: B. SUMIFS adds values that meet several conditions.

3. **A department's attrition rate is 84 leavers out of 300 employees. The rate is:**

A 84%

B 28%

C 36%

D 300%

Answer: B. 84 divided by 300 is 0.28, or 28%.

4. Why put segment thresholds in input cells rather than in formulas?

A Formulas cannot use numbers

B A manager can change a threshold without editing every formula

C Input cells are faster

D It hides the logic

Answer: B. Visible inputs make the model easy to test and change.

5. Which statement is the most careful about overtime and attrition?

A Overtime causes attrition

B Overtime is associated with attrition; a pilot would show whether reducing it helps

C Attrition causes overtime

D Overtime has no relationship with attrition

Answer: B. Association does not prove cause; test the change.

EXERCISE **Pick one project and present it**

Choose one of the four projects and build it end to end: a cleaned data sheet, the analysis in formulas and pivot tables, a dashboard sheet with at most four visuals, and a half-page note with the brief, the answer, one recommendation with a target, and the limits of the data.

Show solution (worked answer)

Hints: start with the brief and write the recommendation first, then build only what the recommendation needs. Check that every number on the dashboard can be traced back to a formula or pivot table on the data sheet, and that the thresholds sit in visible input cells.

Skill Navora

TRACK 2

SQL for Data Analysis

Relational databases, ER diagrams and keys, DDL/DML, filtering, aggregation, joins, set operators, subqueries, window functions, CTEs, views, procedures and performance.

20 lessons | Version 2026.10 | Updated October 2026

Created by **Varun Kumar**
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Contents

Every entry is a clickable link. The PDF also has bookmarks for each lesson and section.

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07 GROUP BY, HAVING, CASE WHEN, ROLLUP, CUBE & COALESCE

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SQL FOR DATA ANALYSIS · LESSON 1

Databases, RDBMS & SQL Basics

Understand what a database is, how relational databases organise data into tables, the vocabulary of SQL, the four families of SQL statements (DDL, DML, DCL, TCL), how SQL compares with Excel, and how to install SQL Server and SSMS and run your first query.

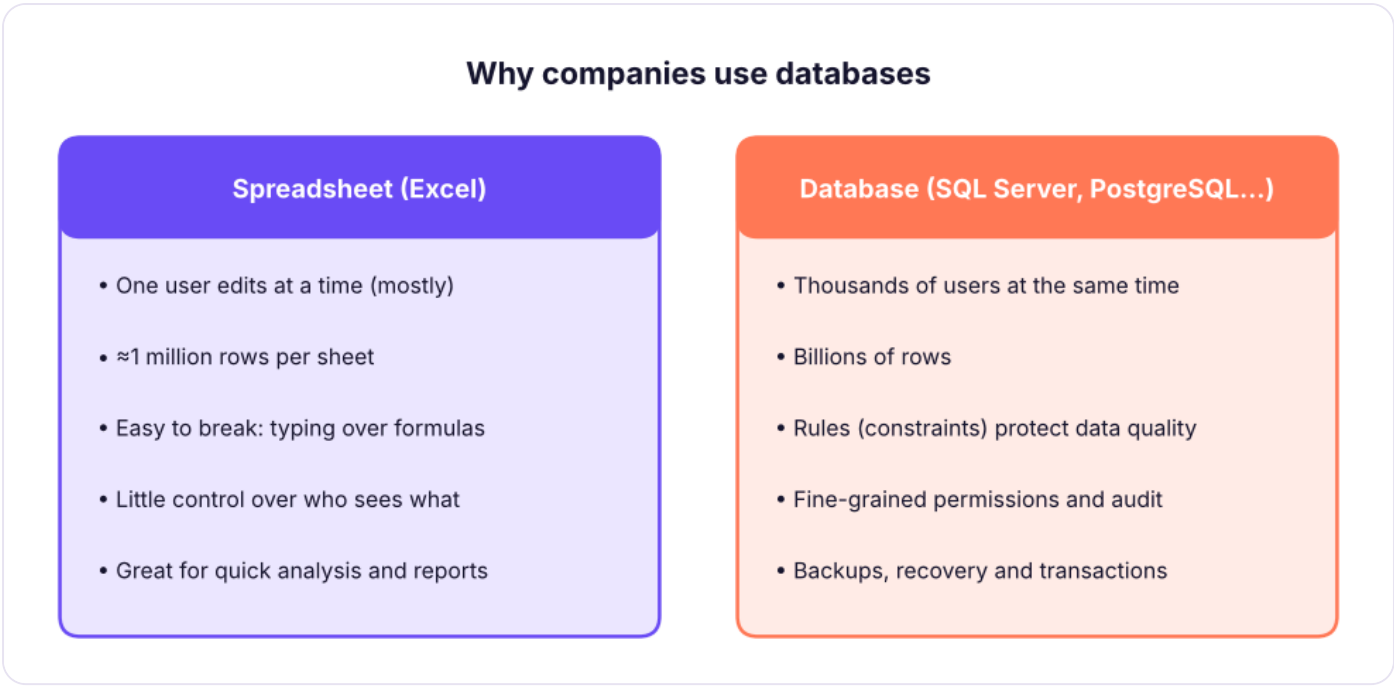
SQL class 1 · SQL Server 2022 / SSMS 21, 2026 · Beginner · 100 min

By the end of this lesson you can

- Explain what a database, a DBMS and an RDBMS are and why companies use them instead of spreadsheets
- Use the core vocabulary (table, row/record, column/field, schema, query, result set)
- Classify SQL statements into DDL, DML, DQL, DCL and TCL
- Compare Excel and SQL and know when to use each
- Install SQL Server Developer edition and SSMS, connect, create a database and run a first SELECT

What is a database?

A **database** is an organised collection of data stored so that it can be found, updated and analysed efficiently and safely by many people at the same time. A **DBMS** (database management system) is the software that manages it. Your bank balance, every Swiggy order, every Aadhaar-linked record and every hospital appointment lives in a database.



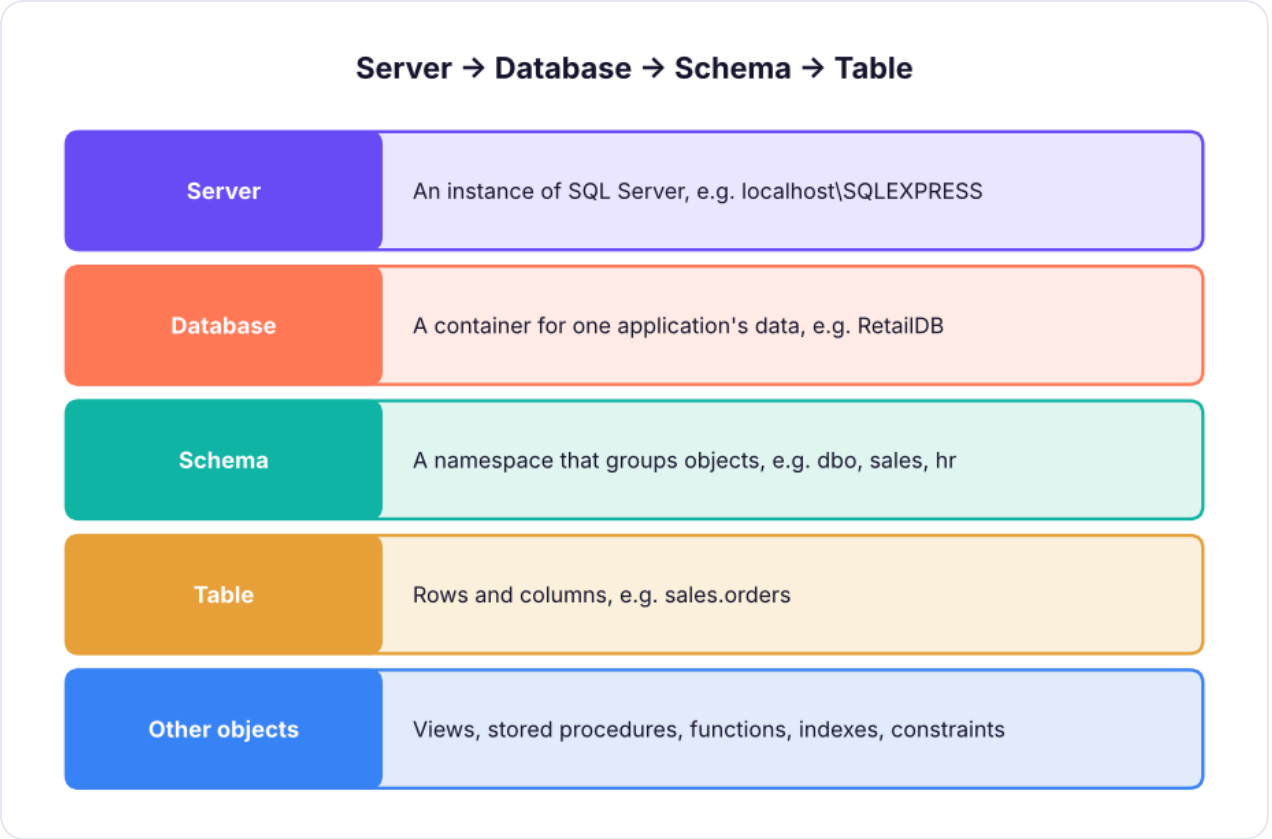
Spreadsheets are personal tools; databases are shared systems built for scale, safety and many users.

Types of databases

Type	Organises data as	Examples	Typical use
Relational (RDBMS)	Tables with rows and columns, linked by keys	SQL Server, PostgreSQL, MySQL, Oracle, SQLite	Orders, payments, HR, finance: most business data
Document	JSON-like documents	MongoDB, Cosmos DB	Product catalogues, app content
Key-value	Key → value pairs	Redis, DynamoDB	Caching, sessions
Column-family / warehouse	Columns, built for analytics	Snowflake, BigQuery, Redshift, Databricks SQL	Analytics on huge data
Graph	Nodes and edges	Neo4j	Social networks, fraud rings
Vector	Number vectors (embeddings)	pgvector, Pinecone	AI search and RAG (GenAI track)

This track focuses on **relational databases** and **SQL**, the language almost every data role uses daily. You will practise in **Microsoft SQL Server** with **SSMS**, as in class. The SQL you learn transfers to PostgreSQL, MySQL, Snowflake and BigQuery with small differences in functions.

How a relational database organises data



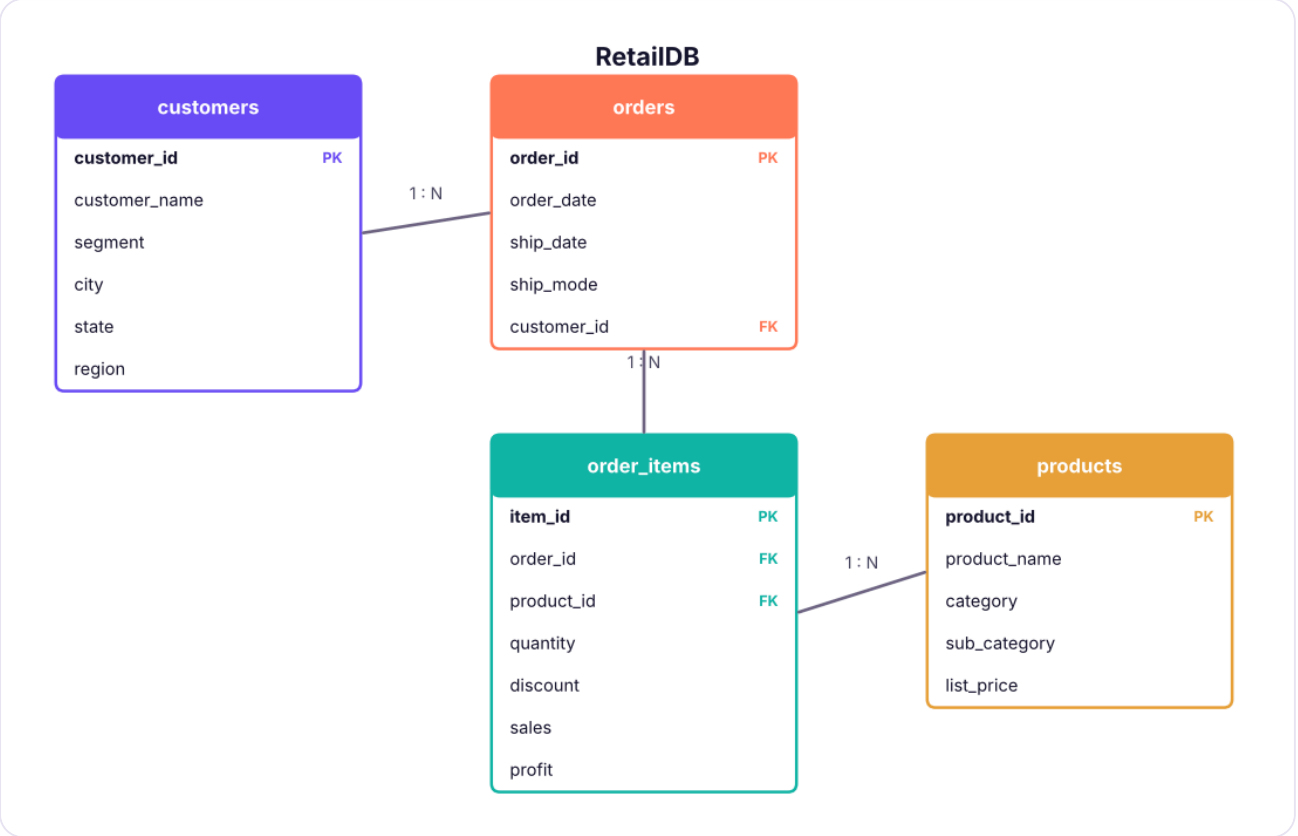
The hierarchy inside SQL Server: a server holds databases, a database holds schemas, a schema holds tables and other objects.

The vocabulary

DEFINITION **Terms you will use every day**

Table (relation): a set of rows about one kind of thing, such as customers or orders. **Row (record, tuple)**: one item, such as one customer. **Column (field, attribute)**: one property, such as `city`. Every value in a column has the same data type. **Primary key**: the column (or columns) that uniquely identifies each row, such as `customer_id`. **Foreign key**: a column that refers to the primary key of another table, linking them. **Schema**: (1) the structure of the database (tables, columns, types, relationships); (2) in SQL Server, a named container of objects such as `dbo`. **Query**: a request for data written in SQL. Its output is a **result set**. **SQL**: Structured Query Language, the standard language for relational databases (ANSI/ISO standard since 1986–87). Pronounced "S-Q-L" or "sequel". **T-SQL**: Transact-SQL, Microsoft SQL Server's dialect with extra features (variables, TOP, procedures).

Here is the course's retail database. The order data is split into related tables instead of one wide sheet:



The course retail database. Each order belongs to one customer; each order has many order items; each item refers to one product.

Your first queries

A `SELECT` statement asks for columns `FROM` a table. Here are the first rows of the `customers` table:

SQL

```sql
SELECT TOP 6 customer_id, customer_name, segment, city, region
FROM customers;
```

RESULT

customer_id	customer_name	segment	city	region
C0001	Saanvi Khan	Corporate	Chandigarh	North
C0002	Meera Bose	Corporate	Delhi	North
C0003	Riya Shah	Consumer	Kolkata	East
C0004	Diya Fernandes	Corporate	Indore	West
C0005	Neha Sharma	Corporate	Hyderabad	South
C0006	Vikram Mehta	Consumer	Chandigarh	North

6 rows × 5 columns

How many rows are in each table?

SQL

```sql
SELECT 'customers' AS table_name, COUNT(*) AS row_count FROM customers
UNION ALL SELECT 'orders', COUNT(*) FROM orders
UNION ALL SELECT 'order_items', COUNT(*) FROM order_items
UNION ALL SELECT 'products', COUNT(*) FROM products;
```

RESULT

table_name	row_count
customers	800
orders	3200
order_items	6323
products	78

4 rows × 2 columns

A taste of what is coming: total sales by region in one statement (Lesson 7 explains `GROUP BY`):

SQL

```sql
SELECT c.region,
       COUNT(DISTINCT o.order_id) AS orders,
       ROUND(SUM(oi.sales), 0)     AS total_sales
FROM customers AS c
JOIN orders AS o       ON o.customer_id = c.customer_id
JOIN order_items AS oi ON oi.order_id = o.order_id
GROUP BY c.region
ORDER BY total_sales DESC;
```

RESULT

region	orders	total_sales
South	831	53,952,769
West	811	51,958,521
North	792	49,018,161
East	766	46,484,119

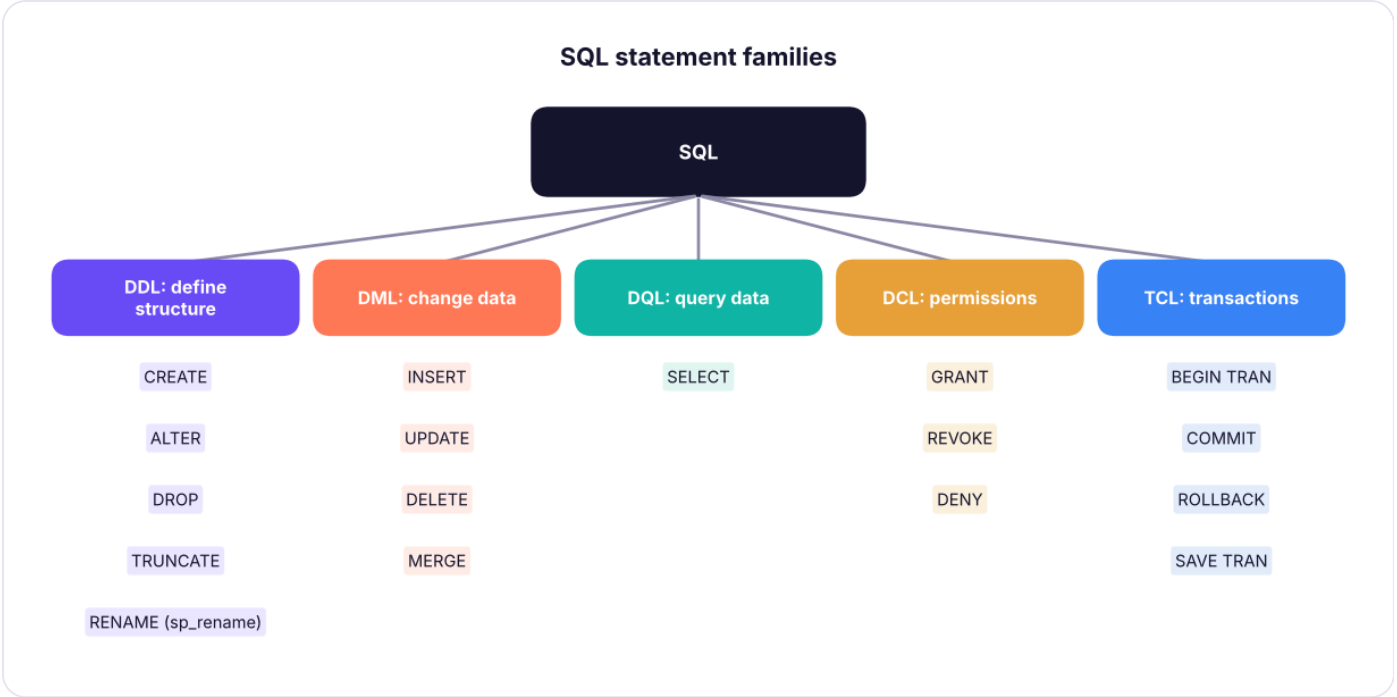
4 rows × 3 columns

> **TIP** **SQL reads almost like English**
>
> "Select the region, the number of distinct orders and the total sales, from customers joined to orders and order items, grouped by region, ordered by total sales descending." That readability is why SQL has lasted 50 years.

Properties of SQL

1. **Declarative**: you say *what* you want, not *how* to get it. The database's **query optimiser** decides the fastest way (which index, which join order).
2. **Set-based**: statements work on whole sets of rows at once, not one row at a time like a loop.
3. **Not case-sensitive for keywords**: `select`, `SELECT` and `Select` are the same. By convention, keywords are written in UPPERCASE. In SQL Server, text comparisons are usually case-insensitive too (default collation).
4. **Whitespace does not matter**: you can split a query over many lines; indent it for readability.
5. **Statements end with a semicolon ;** (optional in SQL Server for most statements, but a good habit).
6. **Standard, with dialects**: the core is an ANSI/ISO standard; each product adds its own functions (T-SQL, PL/pgSQL, MySQL).

The families of SQL statements



SQL statements fall into five families.

Family	Full name	What it does	Example
DDL	Data Definition Language	Creates and changes the **structure**: databases, tables, columns	`CREATE TABLE customers (...)`
DML	Data Manipulation Language	Adds, changes and removes **rows**	`INSERT INTO customers VALUES (...)`
DQL	Data Query Language	Reads data	`SELECT * FROM customers`
DCL	Data Control Language	Gives and removes **permissions**	`GRANT SELECT ON customers TO analyst`
TCL	Transaction Control Language	Groups changes so they succeed or fail together	`BEGIN TRAN … COMMIT`

Many books put SELECT inside DML; others separate it as DQL. Both are fine in interviews if you explain them.

Excel vs SQL

Task	Excel	SQL
See the data	Open the sheet	`SELECT * FROM table`
Filter rows	AutoFilter	`WHERE`
Sort	Sort dialog	`ORDER BY`
New calculated column	Formula in a new column	Expression in `SELECT`
Totals by group	Pivot table / SUMIFS	`GROUP BY` with `SUM`
Filter groups	Pivot value filter	`HAVING`
Lookup from another table	VLOOKUP / XLOOKUP	`JOIN`
Stack two tables	Copy-paste / Power Query Append	`UNION ALL`
Remove duplicates	Remove Duplicates	`DISTINCT`
Rank / running total	RANK, SUM($B$2:B2)	Window functions `RANK() OVER`, `SUM() OVER`
Data size	~1 million rows per sheet	Billions of rows
Repeatability	Manual steps (unless Power Query)	Saved queries, views, procedures
Sharing	Files get copied	One source of truth with permissions

Use Excel for quick, small, ad-hoc analysis and presentation; use SQL to get, combine and summarise data at source. In practice analysts use both: SQL to extract the right data, Excel or Power BI to present it.

Installing SQL Server and SSMS

STEP BY STEP

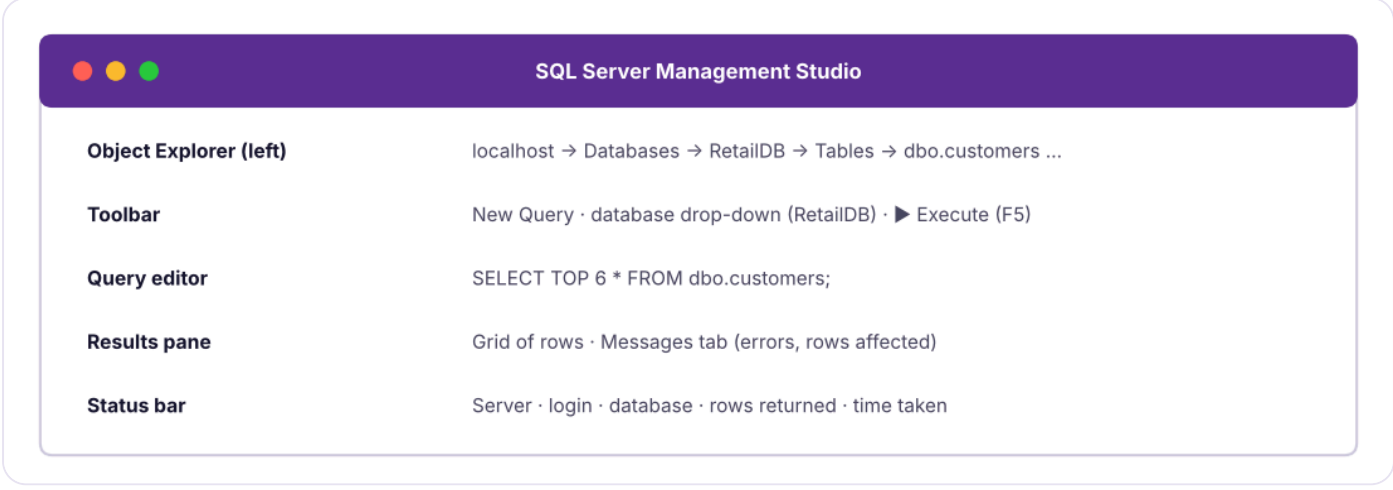
Install on Windows (free)

1. Download **SQL Server 2022 Developer** edition (free, full-featured, for development and learning) from Microsoft's SQL Server downloads page. Choose **Basic** installation. Express edition is also free but limited to 10 GB per database.
2. Note the instance name shown at the end (for example `MSSQLSERVER` = default, or `SQLEXPRESS`).
3. Download and install **SQL Server Management Studio (SSMS)** from Microsoft Learn ("Download SSMS"). SSMS is the editor you write queries in.
4. Open SSMS → **Connect to Server**: Server type *Database Engine*; Server name `localhost` (or `localhost\SQLEXPRESS`); Authentication *Windows Authentication*; tick **Trust server certificate** → Connect.
5. In Object Explorer you will see Databases, Security and more. Click **New Query** (`Ctrl + N`).

NOTE

Mac or Linux?

SQL Server runs in **Docker** on Mac and Linux (`mcr.microsoft.com/mssql/server` image). Connect with **VS Code + the MSSQL extension** or Azure Data Studio's successor tools. Alternatively, practise the same SQL in **PostgreSQL** (pgAdmin) or **SQLite** (DB Browser); only a few functions differ.



The SSMS window: Object Explorer on the left, the query editor in the middle, results below.

Create a database and load data

```sql
-- 1. Create the database and switch to it
CREATE DATABASE RetailDB;
GO
USE RetailDB;
GO
-- 2. Check which database you are in
SELECT DB_NAME() AS current_database;
```

To load the course CSV files: right-click the database → **Tasks → Import Flat File** → choose `customers.csv` → check the column names and data types in the preview (make IDs `nvarchar`, dates `date`, amounts `decimal(12,2)`) → Finish. Repeat for the other files. Lesson 4 shows how to create tables with SQL and load them with `BULK INSERT`.

Running queries in SSMS

Action	Shortcut
Execute the whole window (or the selected text only)	F5 or `Ctrl + E`
New query window	`Ctrl + N`
Comment / uncomment selected lines	`Ctrl + K, Ctrl + C` / `Ctrl + K, Ctrl + U`
Show/hide results pane	`Ctrl + R`
IntelliSense refresh (after creating tables)	`Ctrl + Shift + R`
Results to grid / text / file	`Ctrl + D` / `Ctrl + T` / `Ctrl + Shift + F`
Estimated execution plan	`Ctrl + L`

> **TIP** **Select a part, run a part**
>
> Highlight just one statement and press F5 to run only that statement. This is how you test a long script step by step.

Summary

KEY TAKEAWAYS

- A **database** stores shared data safely; an **RDBMS** stores it in related **tables** of **rows** and **columns**.
- **Primary keys** identify rows; **foreign keys** link tables.
- SQL is declarative and set-based; keywords are not case-sensitive; statements end with `;`.
- Statement families: **DDL** (structure), **DML** (rows), **DQL** (SELECT), **DCL** (permissions), **TCL** (transactions).
- Excel is for quick analysis and presentation; SQL is for getting, combining and summarising data at scale.
- Practise in SQL Server Developer + SSMS (or Docker / PostgreSQL on Mac).

Quick quiz

6 questions · tap an answer to check it

1. Which statement family does `CREATE TABLE` belong to?

A DDL

B DML

C DCL

D TCL

Answer: A. DDL defines structure: CREATE, ALTER, DROP, TRUNCATE.

2. In a relational database, what links the orders table to the customers table?

A A merged cell

B A foreign key in orders that refers to the customer's primary key

C A pivot cache

D A shared file name

Answer: B. orders.customer_id is a foreign key that refers to customers.customer_id.

3. What does "SQL is declarative" mean?

A SQL must declare every variable

B You describe what result you want and the database decides how to get it

C SQL statements must be in capitals

D SQL can only define tables

Answer: B. The optimiser chooses the execution plan; you write the logic.

4. Which is NOT an advantage of a database over a spreadsheet?

A Many users at the same time

B Permissions on who sees what

C Constraints that protect data quality

D It is always faster to build a quick one-off chart

Answer: D. For quick one-off charts, Excel is usually faster. Databases win on scale, safety and sharing.

5. `GRANT SELECT ON orders TO analyst_role` is an example of:

A DDL

B DML

C DCL

D TCL

Answer: C. DCL controls permissions.

6. In SSMS, what does pressing F5 with part of the script highlighted do?

- **A** Runs only the highlighted statement(s)
- **B** Runs the whole script
- **C** Formats the code
- **D** Opens the execution plan

Answer: A. Highlighting limits execution to the selected text.

EXERCISE

Map the vocabulary

For the `orders` table in RetailDB, name: (a) one row; (b) one column; (c) the primary key; (d) the foreign key and the table it refers to; (e) the statement family of `INSERT INTO orders …`.

Show solution (worked answer)

(a) One order, for example the row for `IN-2023-00001`. (b) Any field, for example `ship_mode`. (c) `order_id`. (d) `customer_id`, which refers to `customers.customer_id`. (e) DML.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between DBMS and RDBMS? A DBMS manages data in any form; an RDBMS stores data in tables with relationships defined by keys, enforces integrity with constraints and uses SQL. Examples: SQL Server, PostgreSQL, MySQL, Oracle.

Q: Explain DDL, DML, DCL and TCL with examples. DDL changes structure (CREATE, ALTER, DROP, TRUNCATE); DML changes rows (INSERT, UPDATE, DELETE); DCL manages permissions (GRANT, REVOKE); TCL manages transactions (COMMIT, ROLLBACK, SAVE TRANSACTION). SELECT is sometimes called DQL.

Q: Why would a company keep data in SQL Server instead of Excel files? Concurrency, scale, data integrity (constraints, transactions), security and auditing, backup and recovery, and a single source of truth for reports and applications.

SQL FOR DATA ANALYSIS · LESSON 2

ER Diagrams, Relationships, Keys & Normalisation

Design databases the way professionals do. Draw ER diagrams, read crow's-foot notation, model one-to-one, one-to-many and many-to-many relationships, master every type of key (super, candidate, primary, alternate, composite, foreign, surrogate) and normalise tables to 3NF.

SQL class 1 · 2026 · Beginner · 110 min

By the end of this lesson you can

- Read and draw an ER diagram with entities, attributes and relationships
- Identify 1:1, 1:N and M:N relationships and resolve M:N with a junction table
- Define super, candidate, primary, alternate, composite, foreign, surrogate and unique keys, with examples
- Explain referential integrity and what happens on delete/update of a parent row
- Normalise a flat table into 1NF, 2NF and 3NF, and know when to denormalise

Why design matters

A database that is designed badly is slow, wastes space and, worst of all, lets data contradict itself. Imagine the customer's city stored in every order row: if the customer moves, you must update thousands of rows, and missing one creates two different "truths". Good design stores **each fact once** and connects tables with **keys**.

Entity–Relationship (ER) diagrams

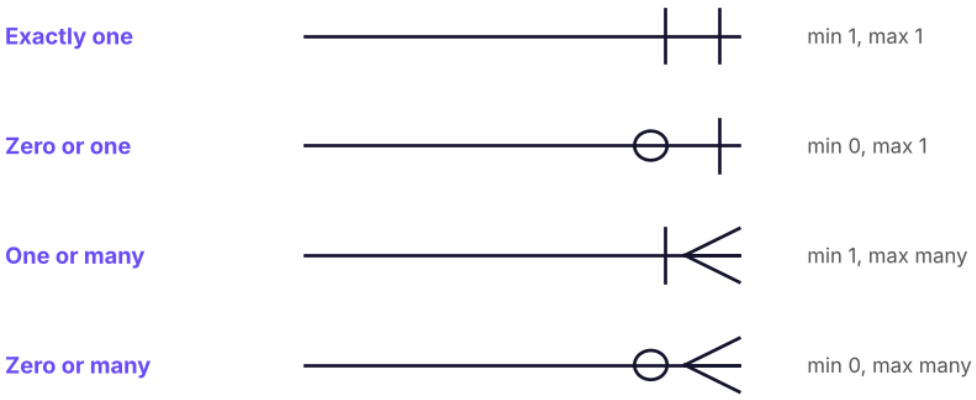
An **ER diagram** is a picture of the data model before any table is created.

DEFINITION **Building blocks**

Entity: a real-world thing you store data about: Customer, Order, Product, Employee. Becomes a **table**. **Attribute**: a property of an entity: name, city, price. Becomes a **column**. **Relationship**: how entities are associated: a Customer *places* Orders. Implemented with **foreign keys**. **Cardinality**: how many of one entity relate to another: one-to-one, one-to-many, many-to-many. **Participation (optionality)**: whether the relationship is mandatory (every order must have a customer) or optional (a customer may have zero orders).

Crow's-foot notation

The most common notation in industry draws symbols at the ends of each relationship line:



Crow's-foot symbols. Read each end as "minimum ... maximum".

Reading example: Customer `||—o<` Order means "each order belongs to **exactly one** customer; each customer places **zero or many** orders".

Types of relationships

One-to-one (1:1)

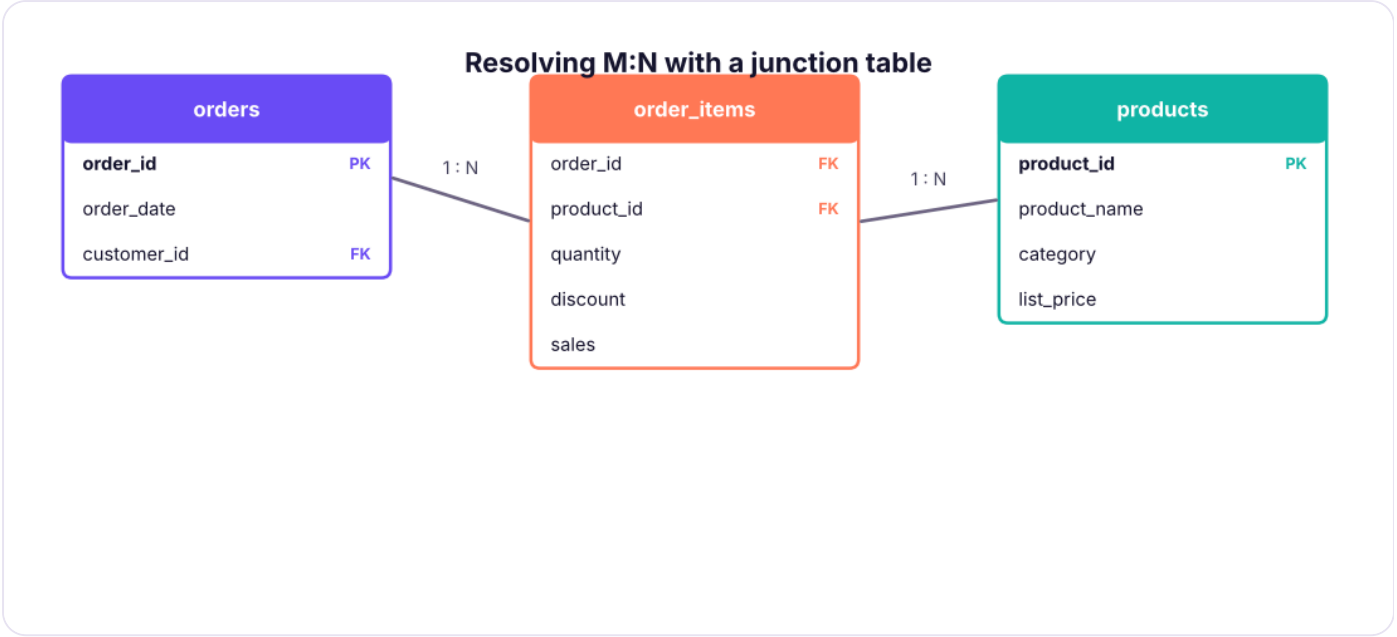
Each row in A matches at most one row in B. Example: Employee ↔ Employee_Bank_Details. Often used to split sensitive or rarely used columns into a separate, more protected table. Implemented with a foreign key that is also **unique** (often the same as the primary key).

One-to-many (1:N) — the most common

One row in A matches many rows in B; each B row matches one A row. Examples: Customer → Orders, Department → Employees, Category → Products. The **foreign key goes on the "many" side** (`orders.customer_id`).

Many-to-many (M:N)

Many rows in A match many in B: Students ↔ Courses, Orders ↔ Products, Actors ↔ Movies. Relational tables cannot store this directly, so we add a **junction (bridge, associative) table** that holds a pair of foreign keys, turning one M:N into two 1:N relationships.



An order contains many products and a product appears in many orders. The junction table order_items resolves the many-to-many relationship and stores facts about each pair (quantity, discount).

Relationship	Example	Where the foreign key goes
1 : 1	Employee – Passport	Either side, marked UNIQUE
1 : N	Department – Employees	On the N side (`employees.dept_id`)
M : N	Students – Courses	In a junction table `enrolments(student_id, course_id)`
Self (recursive)	Employee – Manager	In the same table: `employees.manager_id` → `employees.emp_id`

Let's check the real cardinalities in our retail data with SQL. How many orders does each customer have, and how many items does each order have?

SQL

```sql
SELECT MIN(n_orders) AS min_orders_per_customer,
       MAX(n_orders) AS max_orders_per_customer,
       ROUND(AVG(n_orders * 1.0), 2) AS avg_orders_per_customer
FROM (SELECT customer_id, COUNT(*) AS n_orders FROM orders GROUP BY customer_id) AS t;
```

RESULT

min_orders_per_customer	max_orders_per_customer	avg_orders_per_customer
1	10	4.08

1 rows × 3 columns

SQL

```sql
SELECT items_in_order, COUNT(*) AS number_of_orders
FROM (SELECT order_id, COUNT(*) AS items_in_order FROM order_items GROUP BY order_id) AS t
GROUP BY items_in_order
ORDER BY items_in_order;
```

RESULT

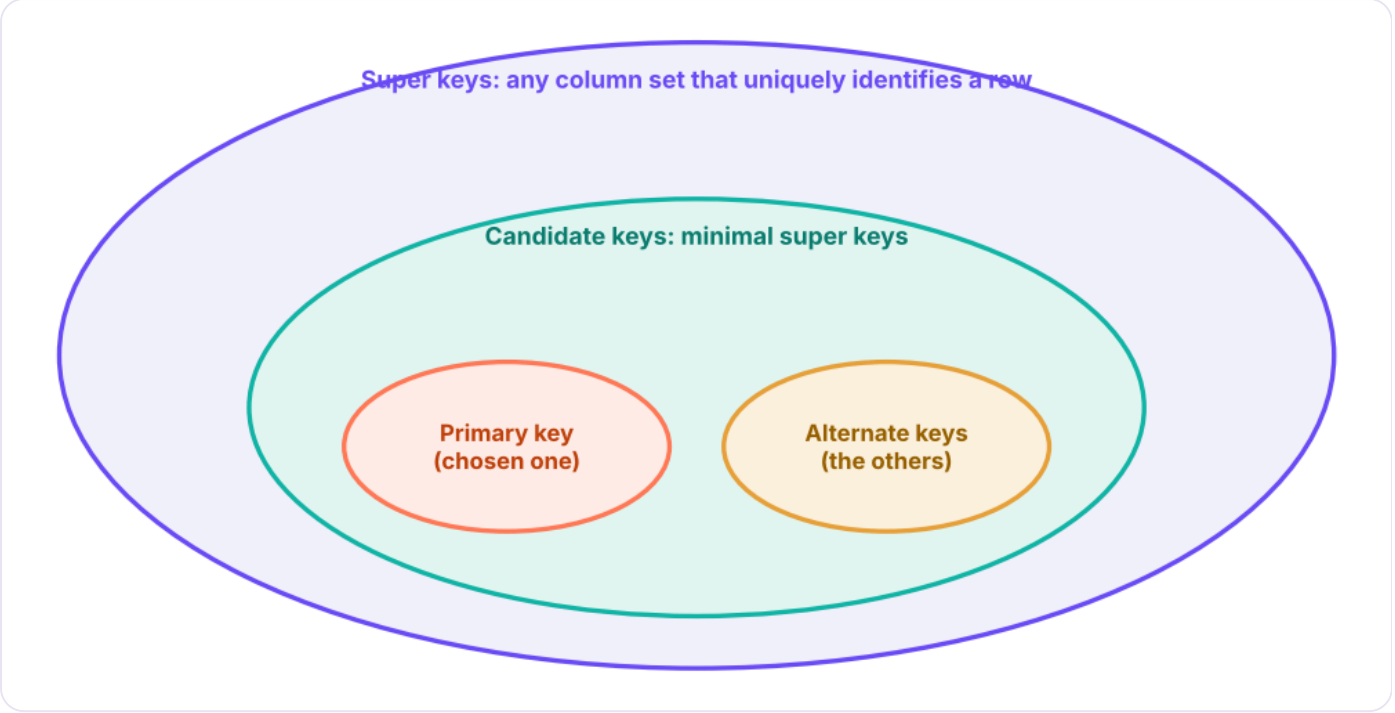
items_in_order	number_of_orders
1	1394
2	920
3	455
4	431

4 rows × 2 columns

So Customer → Orders is 1:N, and Order → Order items is 1:N (1 to 4 items per order here).

Keys

Keys are columns (or sets of columns) with special meaning. They are among the most frequently asked interview topics.



How the main key types relate: every candidate key is a minimal super key; one candidate is chosen as the primary key and the rest become alternate keys.

Example table `employees(emp_id, email, pan_number, name, dept_id, phone)`, where `emp_id`, `email` and `pan_number` are each unique:

Key type	Definition	Example
Super key	Any set of columns that uniquely identifies a row (may include extra columns)	`{emp_id}`, `{emp_id, name}`, `{email, dept_id}`
Candidate key	A **minimal** super key: remove any column and it stops being unique	`{emp_id}`, `{email}`, `{pan_number}`
Primary key (PK)	The candidate key chosen to identify rows. Unique, **not NULL**, one per table	`emp_id`
Alternate key	Candidate keys not chosen as primary; enforce them with `UNIQUE`	`email`, `pan_number`
Composite key	A key made of two or more columns	`(order_id, product_id)` in order_items; `(student_id, course_id)` in enrolments
Foreign key (FK)	A column that refers to a PK (or unique key) in another table	`employees.dept_id` → `departments.dept_id`
Unique key	A UNIQUE constraint: no duplicates, but (in SQL Server) one NULL is allowed	`phone`
Surrogate key	An artificial key with no business meaning, often an auto-number	`customer_key INT IDENTITY(1,1)`
Natural key	A key with business meaning	PAN number, GSTIN, ISBN, email

> **TIP** **Surrogate or natural?**
>
> Natural keys can change (people change email, products get recoded) and can be long. Most warehouses and Power BI models use **surrogate integer keys** for joins, and keep natural keys as alternate (unique) keys.

Primary key vs unique key

	Primary key	Unique key
Duplicates allowed	No	No
NULL allowed	No	Yes (SQL Server allows one NULL; PostgreSQL allows many)
Per table	Exactly one	Many
Default index (SQL Server)	Clustered	Non-clustered

Referential integrity

A foreign key guarantees that every child row points to a parent that exists: you cannot insert an order for customer `C9999` if that customer does not exist. What happens when a **parent** row is deleted or its key is updated is controlled by the FK's actions:

Action	On DELETE of the parent	Use when
`NO ACTION` (default)	The delete is blocked if children exist	Most business data
`CASCADE`	Children are deleted too	Order → order items
`SET NULL`	Children's FK becomes NULL	Employee → manager who leaves
`SET DEFAULT`	Children get the column's default	Rare

```sql
CREATE TABLE departments (
    dept_id   INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL UNIQUE
);
CREATE TABLE employees (
    emp_id     INT PRIMARY KEY,
    email      VARCHAR(100) NOT NULL UNIQUE,          -- alternate key
    name       VARCHAR(100) NOT NULL,
    dept_id    INT NOT NULL REFERENCES departments(dept_id),   -- foreign key
    manager_id INT NULL REFERENCES employees(emp_id)           -- self-referencing FK
);
```

Lesson 4 shows these constraints in action.

Normalisation

Normalisation organises columns and tables to reduce redundancy and avoid **anomalies**:

- **Update anomaly**: the same fact stored in many rows gets changed in some but not others.
- **Insert anomaly**: you cannot add a fact without another fact (cannot add a new product until someone orders it).
- **Delete anomaly**: deleting one fact removes another (deleting the only order of a product deletes the product's information).

Start with the flat Superstore file, which repeats customer and product details on every order line:

SQL

```sql
SELECT TOP 5 order_id, order_date, customer_id, customer_name, city, product_id,
product_name, category, quantity, sales
FROM superstore;
```

RESULT

order_id	order_date	customer_id	customer_name	city	product_id	product_n
IN-2023-00001	2023-01-01	C0765	Ishaan Gupta	Bengaluru	P0059	Nova Pho 223
IN-2023-00001	2023-01-01	C0765	Ishaan Gupta	Bengaluru	P0050	Orbit Appliance
IN-2023-00001	2023-01-01	C0765	Ishaan Gupta	Bengaluru	P0011	Orbit Tabl 248
IN-2023-00002	2023-01-03	C0791	Kabir Singh	Ahmedabad	P0037	Crest Stor 946
IN-2023-00003	2023-01-03	C0049	Sneha Khan	Ahmedabad	P0018	Crest Bookcase

5 rows × 10 columns

First normal form (1NF)

- Each cell holds **one atomic value** (no lists like `"Chair, Table"`).
- No repeating groups of columns (`product1`, `product2`, `product3`).
- Each row is unique (a key exists).

The flat file is already in 1NF: one product per row, and every row has a unique `row_id`. The natural key would be `(order_id, product_id)`, but 70 lines repeat the same product within the same order, which is exactly why real systems add a surrogate key such as `row_id` or `item_id`.

Second normal form (2NF)

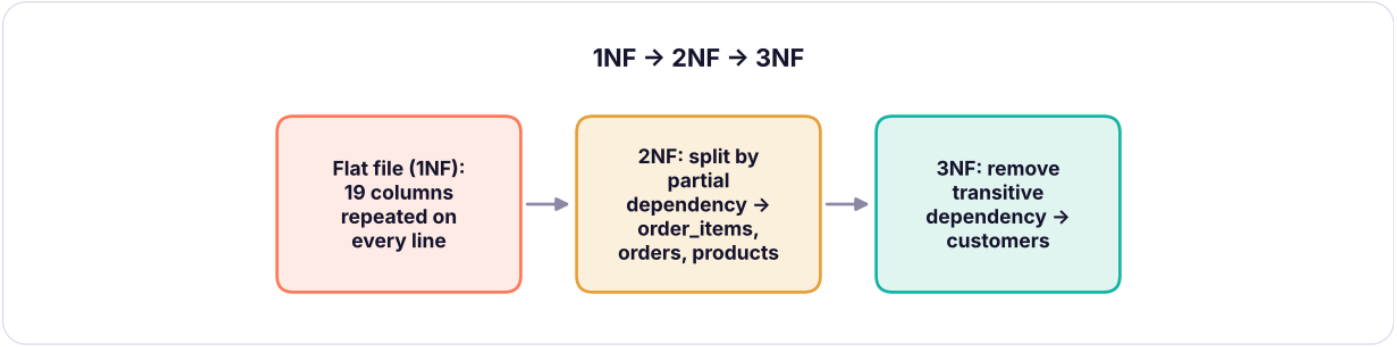
1NF **and** every non-key column depends on the **whole** key, not part of it (this matters only with composite keys). With key `(order_id, product_id)`:

- `quantity`, `discount`, `sales` depend on both: they stay.
- `order_date`, `customer_id`, `ship_mode` depend only on `order_id` → move to an `orders` table.
- `product_name`, `category` depend only on `product_id` → move to a `products` table.

Third normal form (3NF)

2NF **and** no **transitive** dependency: non-key columns must depend only on the key, not on other non-key columns. In `orders`, `customer_name`, `city` and `region` depend on `customer_id`, not on `order_id` → move them to a `customers` table.

> "Every non-key attribute must provide a fact about the key, the whole key, and nothing but the key." (Bill Kent)



Normalising the flat Superstore file to 3NF gives the four-table RetailDB design.

How much repetition did normalisation remove? Count the distinct customers and products versus rows:

```sql
SELECT COUNT(*)                   AS order_lines,
       COUNT(DISTINCT order_id)     AS orders,
       COUNT(DISTINCT customer_id)  AS customers,
       COUNT(DISTINCT product_id)   AS products
FROM superstore;
```

RESULT

order_lines	orders	customers	products
6323	3200	785	78

1 rows × 4 columns

The flat file stores customer details 6,323 times; the normalised design stores them once per customer (fewer than 800 rows).

Normal form	Rule in plain words
1NF	One value per cell; no repeating columns; a key exists
2NF	No column depends on only part of a composite key
3NF	No column depends on another non-key column
BCNF	Every determinant is a candidate key (stricter 3NF)
4NF / 5NF	Remove multi-valued and join dependencies (rare in practice)

Denormalisation and the star schema

Transactional systems (OLTP) are normalised for safe, fast writes. Analytical systems (OLAP, data warehouses, Power BI) are often **denormalised** into a **star schema**: one central **fact table** (sales events with numbers) surrounded by **dimension tables** (customer, product, date, region) to make reading and aggregating fast and simple. You will build one in the Power BI track (Lesson 3).

	OLTP (normalised)	OLAP / warehouse (star schema)
Purpose	Run the business: place orders, update stock	Analyse the business
Operations	Many small inserts and updates	Large reads and aggregations
Design	3NF, many tables	Fact + dimension tables
Examples	App database in SQL Server/PostgreSQL	Snowflake, BigQuery, Fabric, Power BI model

Summary

KEY TAKEAWAYS

- ER diagrams show **entities** (tables), **attributes** (columns) and **relationships** with **cardinality** (1:1, 1:N, M:N).
- The foreign key sits on the **many** side; M:N needs a **junction table** with a composite key.
- Super key ⊇ candidate keys (minimal); one candidate becomes the **primary key** (unique, not NULL), the others are **alternate keys** (UNIQUE).
- Composite keys combine columns; surrogate keys are artificial integers; natural keys carry business meaning.
- Foreign keys enforce **referential integrity**; ON DELETE CASCADE / SET NULL / NO ACTION decide what happens to children.
- Normalise to **3NF** to remove redundancy and anomalies; denormalise into a **star schema** for analytics.

Quick quiz

6 questions · tap an answer to check it

1. Students can enrol in many courses and courses have many students. How do you model this?

- **A** Put course_id in the students table
- **B** Put student_id in the courses table
- **C** Create a junction table enrolments(student_id, course_id)
- **D** Use one table with a column per course

Answer: C. A junction table resolves M:N into two 1:N relationships.

2. In a 1:N relationship between departments and employees, where does the foreign key go?

- **A** In departments
- **B** In employees
- **C** In both tables
- **D** In a third table

Answer: B. The FK goes on the "many" side: employees.dept_id.

3. Which statement about a primary key is FALSE?

A It must be unique

B It cannot be NULL

C A table can have several primary keys

D It can consist of more than one column

Answer: C. A table has one primary key, which may be composite.

4. `email` is unique in the employees table but `emp_id` was chosen as the primary key. What is email?

A A foreign key

B An alternate (candidate) key

C A surrogate key

D A super key only

Answer: B. Unchosen candidate keys are alternate keys, enforced with UNIQUE.

5. In `orders(order_id, customer_id, customer_city)`, customer_city depends on customer_id. Which normal form is violated?

A 1NF

B 2NF

C 3NF

D None

Answer: C. A non-key column depending on another non-key column is a transitive dependency, violating 3NF.

6. What does ON DELETE CASCADE on order_items.order_id do?

- **A** Deleting an order also deletes its order items
- **B** Deleting an order is blocked if it has items
- **C** Order items get a NULL order_id
- **D** Nothing

Answer: A. CASCADE propagates the delete to child rows.

EXERCISE **Design a library database**

A library has members, books and loans. A member can borrow many books over time; a book (copy) can be borrowed many times; each loan has a borrow date, due date and return date. Books have an ISBN, title and author; authors can write many books and a book can have several authors. Draw the entities, keys and relationships.

Show solution (worked answer)

- `members(member_id PK, name, email UNIQUE, phone)`
- `books(book_id PK, isbn UNIQUE, title, publisher, year)`
- `authors(author_id PK, name)`
- `book_authors(book_id FK, author_id FK, PRIMARY KEY (book_id, author_id))`: junction for M:N.
- `copies(copy_id PK, book_id FK, shelf)`: one book has many physical copies.
- `loans(loan_id PK, copy_id FK, member_id FK, borrow_date, due_date, return_date NULL)`. Relationships: members 1:N loans; copies 1:N loans; books 1:N copies; books M:N authors (via book_authors).

INTERVIEW CORNER

Questions you may be asked

Q: Difference between primary key, unique key and foreign key? PK uniquely identifies rows, no NULLs, one per table. A unique key also prevents duplicates but allows a NULL (one in SQL Server) and a table can have many. A foreign key references a PK/unique key in another table to enforce relationships.

Q: What is a candidate key? A composite key? A candidate key is a minimal set of columns that uniquely identifies rows; one is chosen as the PK. A composite key uses two or more columns together, such as (order_id, product_id).

Q: What is normalisation and why denormalise? Normalisation removes redundancy and anomalies (1NF–3NF). Data warehouses denormalise into star schemas to make analytical queries simpler and faster.

SQL FOR DATA ANALYSIS · LESSON 3

Data Types, Operators, NULLs, Variables & Comments

Choose the right SQL Server data type for every column, use arithmetic, comparison, logical and string operators correctly, understand NULL and three-valued logic, declare variables, follow naming rules and write readable, commented SQL.

SQL class 2 · SQL Server 2022, 2026 · Beginner · 100 min

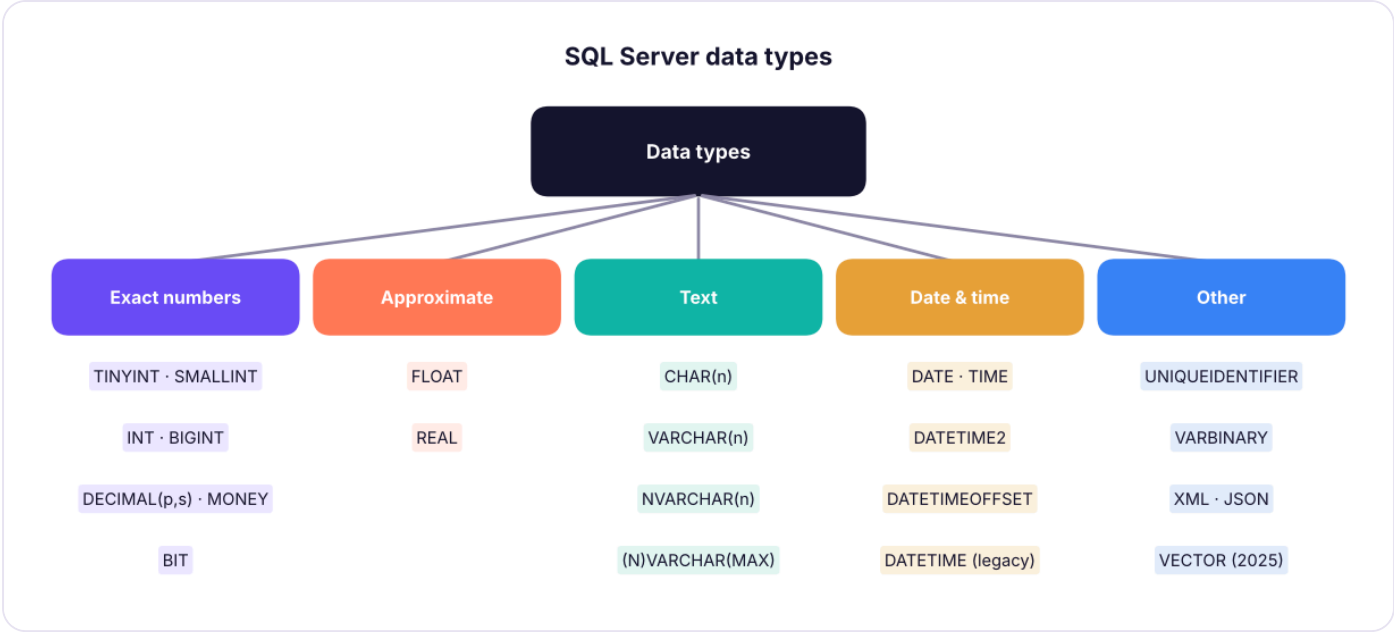
By the end of this lesson you can

- Pick suitable numeric, text, date/time and other data types, and know why the choice matters
- Use arithmetic, comparison, logical and concatenation operators, including integer division and modulo
- Explain NULL, three-valued logic, IS NULL and COALESCE
- Declare and use T-SQL variables with DECLARE, SET and SELECT
- Follow identifier naming rules and conventions, and comment code

Why data types matter

Every column has exactly one **data type**. The type decides what values are allowed, how much storage they use, how they sort and compare, and which functions work on them. A column of dates stored as text sorts "01-02-2026" before "31-01-2025" and cannot be used for date arithmetic, the same problem you met in Excel.

SQL Server data types



The main families of SQL Server data types.

Numbers

Type	Range / precision	Storage	Use for
`TINYINT`	0 to 255	1 byte	Ratings, small codes
`SMALLINT`	−32,768 to 32,767	2 bytes	Small counts
`INT`	about ±2.1 billion	4 bytes	IDs, quantities (default choice)
`BIGINT`	about ±9.2 × 10^{18}	8 bytes	Very large IDs, event logs
`DECIMAL(p, s)` / `NUMERIC`	exact, p digits with s after the point	5–17 bytes	**Money**, rates: `DECIMAL(12,2)` holds up to 9,999,999,999.99
`MONEY`	exact, 4 decimals	8 bytes	Currency (DECIMAL is more portable)
`FLOAT`, `REAL`	**approximate** binary floating point	8 / 4 bytes	Scientific measurements, ML scores
`BIT`	0, 1 or NULL	1 bit	Yes/No flags

WATCH OUT **Never store money in FLOAT**

`FLOAT` cannot represent many decimal fractions exactly (0.1 + 0.2 is not exactly 0.3), so totals drift by tiny amounts. Use `DECIMAL(p, s)` for currency.

Text

Type	Description	Example
`CHAR(n)`	Fixed length, padded with spaces	`CHAR(2)` for state codes `MH`
`VARCHAR(n)`	Variable length, up to n bytes (n ≤ 8000)	`VARCHAR(100)` for names in English
`NVARCHAR(n)`	Variable length **Unicode** (Hindi, Tamil, emoji)	`NVARCHAR(100)` for names in any language
`VARCHAR(MAX)`, `NVARCHAR(MAX)`	Up to 2 GB	Long descriptions, JSON

Prefix Unicode string literals with `N`: `N'नमस्ते'`. Without `N`, non-Latin characters can turn into `?`.

Dates and times

Type	Stores	Example
`DATE`	Date only	`2026-04-27`
`TIME`	Time only	`14:30:00`
`DATETIME2(p)`	Date + time, precise (preferred)	`2026-04-27 14:30:15.1234567`
`DATETIME`	Older date + time (3.33 ms precision)	Legacy tables
`DATETIMEOFFSET`	Date + time + time zone	`2026-04-27 14:30 +05:30`
`SMALLDATETIME`	Date + time to the minute	Low-precision logs

Always write date literals in the unambiguous **ISO format** `'YYYY-MM-DD'` (`'2026-04-27'`). `'04/05/2026'` means April 5 or May 4 depending on the server's language settings.

Others

`UNIQUEIDENTIFIER` (GUIDs), `VARBINARY` (files, images), `XML`, JSON stored in `NVARCHAR(MAX)` with JSON functions, `GEOGRAPHY` (maps), and from SQL Server 2025 a native `VECTOR` type for AI embeddings.

Choosing types for the retail tables

Column	Good type	Why
`customer_id`	`VARCHAR(10)`	Codes like C0001 keep leading letters/zeros
`customer_name`	`NVARCHAR(100)`	Names in any language
`order_date`	`DATE`	No time needed
`quantity`	`INT` (or `SMALLINT`)	Whole numbers
`discount`	`DECIMAL(4,2)`	0.00 to 0.99 exactly
`sales`, `profit`	`DECIMAL(12,2)`	Money
`is_returned`	`BIT`	Yes/No

Here are the types DuckDB inferred when loading the CSV (SSMS's Import Flat File wizard does the same kind of guessing; always review its suggestions):

OUTPUT

	column_name	column_type
0	row_id	BIGINT
1	order_id	VARCHAR
2	order_date	DATE
3	ship_date	DATE
4	ship_mode	VARCHAR
5	customer_id	VARCHAR
6	customer_name	VARCHAR
7	segment	VARCHAR
8	city	VARCHAR
9	state	VARCHAR
10	region	VARCHAR
11	product_id	VARCHAR

19 rows × 2 columns (first 12 shown)

Operators

Arithmetic

Operator	Meaning	Example	Result in SQL Server
+	Add	`SELECT 7 + 2`	9
-	Subtract	`SELECT 7 - 2`	5
*	Multiply	`SELECT 7 * 2`	14
/	Divide	`SELECT 7 / 2`	**3** (integer ÷ integer = integer)
/	Divide	`SELECT 7 / 2.0`	3.500000
%	Modulo (remainder)	`SELECT 7 % 2`	1

COMMON MISTAKES

Integer division

In SQL Server, dividing two integers drops the decimals: `SELECT 1/3` returns 0, and `COUNT(...)/COUNT(...)` returns 0 or 1 instead of a percentage. Multiply by `1.0` or `CAST` one side: `SELECT 100.0 * returned / total` or `CAST(returned AS DECIMAL(10,2)) / total`. (PostgreSQL behaves the same way; MySQL and DuckDB return decimals.)

Calculated columns are written right in the SELECT list. Here is unit price and margin for a few order lines:

SQL

```sql
SELECT TOP 6
       order_id,
       quantity,
       sales,
       profit,
       ROUND(sales / quantity, 2)          AS unit_price,
       ROUND(100.0 * profit / sales, 1)    AS margin_pct
FROM superstore
ORDER BY row_id;
```

RESULT

order_id	quantity	sales	profit	unit_price	margin_pct
IN-2023-00001	2	36,000	3,472.55	18,000	9.6
IN-2023-00001	1	3,850	1,169.54	3,850	30.4
IN-2023-00001	4	62,080	6,044.47	15,520	9.7
IN-2023-00002	2	5,216	450.16	2,608	8.6
IN-2023-00003	3	24,591	-5,512.33	8,197	-22.4
IN-2023-00003	1	8,288	-1,598.43	8,288	-19.3

6 rows × 6 columns

Comparison

Operator	Meaning
=	Equal
<> or ≠	Not equal
> < ≥ ≤	Greater / less (than or equal)
BETWEEN a AND b	Inclusive range
IN (…)	Matches any value in a list
LIKE	Pattern match with % and _
IS NULL / IS NOT NULL	Tests for missing values

Logical

AND (both true), OR (at least one true), NOT (reverses). **Precedence**: NOT first, then AND, then OR. Use brackets to make intent obvious.

SQL

```sql
-- Without brackets, AND is evaluated first:
SELECT COUNT(*) AS rows_without_brackets
FROM superstore
WHERE region = 'North' OR region = 'South' AND category = 'Technology';
```

RESULT

rows_without_brackets
2025

1 rows × 1 columns

SQL

```
-- With brackets: (North or South) and Technology
SELECT COUNT(*) AS rows_with_brackets
FROM superstore
WHERE (region = 'North' OR region = 'South') AND category = 'Technology';
```

RESULT

rows_with_brackets
952

1 rows × 1 columns

The first query returns all North rows plus South Technology rows: probably not what was meant. Brackets fix it.

String concatenation

Method	Example	NULL behaviour
+	first_name + ' ' + last_name	Any NULL makes the whole result NULL
CONCAT()	CONCAT(first_name, ' ', last_name)	Treats NULL as empty text
CONCAT_WS()	CONCAT_WS(', ', city, state, 'India')	Separator; skips NULLs

SQL

```
SELECT TOP 4 customer_name,
       CONCAT(city, ', ', state)           AS city_state,
       CONCAT_WS(' | ', segment, region)   AS segment_region
FROM customers;
```

RESULT

customer_name	city_state	segment_region
Saanvi Khan	Chandigarh, Punjab	Corporate | North
Meera Bose	Delhi, Delhi	Corporate | North
Riya Shah	Kolkata, West Bengal	Consumer | East
Diya Fernandes	Indore, Madhya Pradesh	Corporate | West

4 rows × 3 columns

NULL: the missing value

`NULL` means **unknown or missing**. It is not zero and not an empty string. Any comparison with NULL returns **UNKNOWN**, not TRUE or FALSE. SQL uses **three-valued logic**:

Expression	Result
`NULL = NULL`	UNKNOWN
`NULL <> 5`	UNKNOWN
`5 + NULL`	NULL
`TRUE AND UNKNOWN`	UNKNOWN
`FALSE AND UNKNOWN`	FALSE
`TRUE OR UNKNOWN`	TRUE
`NOT UNKNOWN`	UNKNOWN

AND

	TRUE	FALSE	UNKNOWN
TRUE	TRUE	FALSE	UNKNOWN
FALSE	FALSE	FALSE	FALSE
UNKNOWN	UNKNOWN	FALSE	UNKNOWN

OR

	TRUE	FALSE	UNKNOWN
TRUE	TRUE	TRUE	TRUE
FALSE	TRUE	FALSE	UNKNOWN
UNKNOWN	TRUE	UNKNOWN	UNKNOWN

Three-valued logic. Any comparison involving NULL is UNKNOWN; WHERE keeps only TRUE rows.

`WHERE` keeps only rows where the condition is TRUE, so rows where it is UNKNOWN disappear silently. That is why you must test with `IS NULL`:

SQL

```sql
WITH t AS (
    SELECT 'A' AS item, 10 AS score UNION ALL
    SELECT 'B', NULL UNION ALL
    SELECT 'C', 30
)
SELECT
    (SELECT COUNT(*) FROM t WHERE score = NULL)   AS equals_null_rows,     -- always 0!
    (SELECT COUNT(*) FROM t WHERE score IS NULL)  AS is_null_rows,
    (SELECT COUNT(score) FROM t)                  AS count_score,          -- ignores NULL
    (SELECT COUNT(*) FROM t)                      AS count_star,
    (SELECT AVG(score * 1.0) FROM t)              AS avg_ignores_null,     -- (10+30)/2
    (SELECT AVG(COALESCE(score, 0) * 1.0) FROM t) AS avg_null_as_zero;     -- (10+0+30)/3
```

RESULT

equals_null_rows	is_null_rows	count_score	count_star	avg_ignores_null	avg_null_as_
0	1	2	3	20	13.33

1 rows × 6 columns

Function	What it does
`ISNULL(x, y)`	T-SQL: return y when x is NULL
`COALESCE(x, y, z, …)`	ANSI: the first non-NULL value; works in every database
`NULLIF(a, b)`	NULL if a = b, otherwise a. Classic use: `sales / NULLIF(qty, 0)` avoids divide-by-zero errors

Variables in T-SQL

T-SQL lets you store a value in a variable for the length of a batch. Variable names start with @.

```sql
DECLARE @target_region VARCHAR(20) = 'West';     -- declare and assign
DECLARE @min_sales     DECIMAL(12,2);
SET @min_sales = 50000;                           -- assign with SET

SELECT order_id, city, sales
FROM superstore
WHERE region = @target_region AND sales ⩾ @min_sales;

DECLARE @total DECIMAL(14,2);
SELECT @total = SUM(sales) FROM superstore;       -- assign from a query
PRINT CONCAT('Total sales: ', @total);            -- message in the Messages tab
```

Variables make scripts reusable (change `'West'` once) and are the basis of stored procedures (Lesson 13). Other databases use different syntax (PostgreSQL `DO` blocks or `\set` in psql).

Identifiers: naming rules and conventions

Rules (SQL Server): - Start with a letter, _, @ (variables) or # (temporary tables). - Up to 128 characters; letters, digits, _, @, $, #. - Reserved words (`ORDER`, `GROUP`, `USER`) and names with spaces must be wrapped in `[square brackets]` (or `"double quotes"`): `[Order Details]`, `[order]`.

Conventions used in this handbook (pick one style and be consistent): - `snake_case`, lowercase table and column names: `order_items`, `customer_id`. - Plural table names (`customers`) or singular (`customer`): be consistent. - Primary key `id` or `<table>_id`; foreign keys named after the referenced key (`customer_id`). - Prefix views `vw_`, procedures `usp_`, indexes `ix_`. - Avoid spaces, special characters and reserved words, so you never need brackets.

Comments and formatting

```sql
-- A single-line comment: everything after -- is ignored

/* A multi-line comment:
   Purpose : Top 10 customers by 2025 sales
   Author  : Varun Kumar
   Updated : 27-Apr-2026                          */

SELECT TOP 10
       customer_id,
       SUM(sales) AS total_sales          -- revenue in rupees
FROM   superstore
WHERE  order_date ⩾ '2025-01-01'
  AND  order_date <  '2026-01-01'         -- half-open range: safe with times
GROUP  BY customer_id
ORDER  BY total_sales DESC;
```

TIP **Readable SQL checklist**

One clause per line (`SELECT`, `FROM`, `WHERE`...), one column per line in long queries, consistent indentation, keywords in capitals, meaningful aliases, and a header comment explaining the purpose. Tools such as the "Format SQL" extension or SQLFluff format code automatically.

Summary

KEY TAKEAWAYS

- Choose types deliberately: `INT` for IDs and counts, `DECIMAL(p,s)` for money, `NVARCHAR` for names, `DATE`/`DATETIME2` for dates; never `FLOAT` for money.
- Write dates as `'YYYY-MM-DD'`.
- `7/2 = 3` in SQL Server: use `1.0 *` or `CAST` for decimal division; `%` gives the remainder.
- Precedence: NOT, AND, OR; use brackets.
- `NULL` is unknown: compare with `IS NULL`; aggregates ignore NULLs; `COALESCE`, `ISNULL` and `NULLIF` handle them.
- `CONCAT` treats NULL as empty; `+` with a NULL returns NULL.
- Variables: `DECLARE @x type = value`; name things consistently and comment your code.

Quick quiz

6 questions · tap an answer to check it

1. Which data type should store a product price in rupees and paise?

A FLOAT

B INT

C DECIMAL(10,2)

D VARCHAR(10)

Answer: C. DECIMAL stores exact decimal values, which money needs.

2. In SQL Server, what does `SELECT 9 / 4` return?

A 2

B 2.25

C 2.0

D 3

Answer: A. Integer divided by integer gives an integer (the fraction is dropped).

3. What does `WHERE discount = NULL` return?

A Rows where discount is missing

B No rows

C All rows

D An error

Answer: B. Comparisons with NULL are UNKNOWN, so WHERE keeps nothing. Use IS NULL.

4. Which expression returns 'Pune' when city is NULL?

A `city + 'Pune'`

B `COALESCE(city, 'Pune')`

C `NULLIF(city, 'Pune')`

D `city = 'Pune'`

Answer: B. COALESCE returns the first non-NULL value.

5. `WHERE a = 1 OR b = 2 AND c = 3` **is evaluated as:**

A `(a = 1 OR b = 2) AND c = 3`

B `a = 1 OR (b = 2 AND c = 3)`

C Left to right with equal precedence

D An error

Answer: B. AND has higher precedence than OR.

6. Why use NVARCHAR instead of VARCHAR for customer names?

A It is faster

B It stores Unicode characters such as Devanagari or Tamil correctly

C It allows numbers

D It is required for primary keys

Answer: B. NVARCHAR stores Unicode text; VARCHAR may lose characters outside the code page.

EXERCISE

Fix the query

A colleague wrote: `SELECT order_id, profit/sales*100 AS margin FROM superstore WHERE category = 'Furniture' OR category = 'Technology' AND discount > 0.2 AND ship_mode = NULL;` List every problem and rewrite it.

Show solution (worked answer)

1. `AND` binds before `OR`, so the discount and ship-mode filters apply only to Technology: add brackets.
2. `ship_mode = NULL` never matches: use `IS NULL` (or remove it if not needed).
3. If profit and sales were integers, `profit/sales` would be integer division; use `100.0 * profit / sales` and guard against zero with `NULLIF(sales, 0)`. Rewritten: `SELECT order_id, ROUND(100.0 * profit / NULLIF(sales, 0), 1) AS margin_pct FROM superstore WHERE category IN ('Furniture', 'Technology') AND discount > 0.2;`

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between CHAR, VARCHAR and NVARCHAR? CHAR is fixed length (padded), VARCHAR is variable length non-Unicode, NVARCHAR is variable length Unicode (two bytes per character for most scripts).

Q: How does SQL treat NULL? As unknown: comparisons return UNKNOWN, aggregates (except COUNT(*)) ignore NULLs, arithmetic with NULL returns NULL. Use IS NULL, COALESCE/ISNULL and NULLIF.

Q: Why does `SELECT 100 * returned / total` give 0? Integer division when both values are integers; cast to decimal or multiply by 100.0.

Q: DECIMAL vs FLOAT? DECIMAL is exact (money, rates); FLOAT is approximate binary (scientific values) and can introduce rounding errors.

SQL FOR DATA ANALYSIS · LESSON 4

DDL, DML & Constraints: CREATE, ALTER, INSERT, UPDATE, DELETE

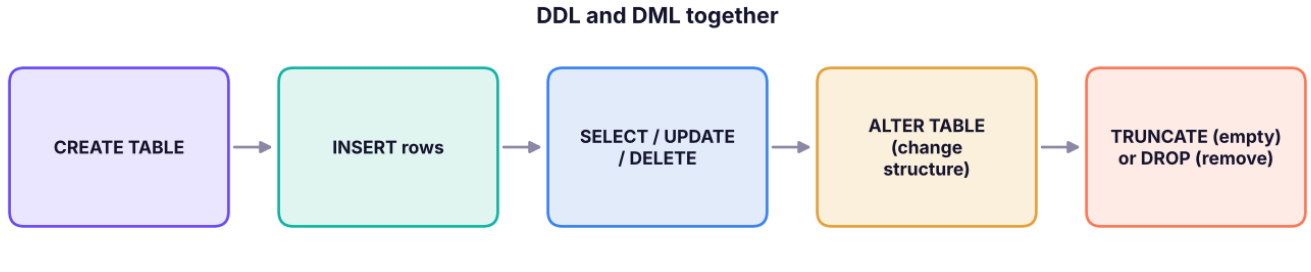
Create and change tables with DDL (CREATE, ALTER, DROP, TRUNCATE), add, change and remove rows with DML (INSERT, UPDATE, DELETE, MERGE), protect data quality with constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT, IDENTITY) and understand DELETE vs TRUNCATE vs DROP.

SQL classes 2–3 · SQL Server 2022, 2026 · Beginner · 120 min

By the end of this lesson you can

- Create databases and tables with appropriate data types and constraints
- Modify table structure with ALTER TABLE (add, change, drop columns and constraints)
- Insert single and multiple rows, insert from a query, update and delete rows safely
- Explain every constraint type and read the error each one raises
- Compare DELETE, TRUNCATE and DROP, and use MERGE for upserts

The life cycle of a table



DDL creates and changes structure; DML fills and changes the rows inside it.

CREATE TABLE

SQL

```sql
CREATE TABLE departments (
    dept_id    INT          PRIMARY KEY,
    dept_name  VARCHAR(50)  NOT NULL UNIQUE,
    location   VARCHAR(50)  DEFAULT 'Bengaluru'
);

CREATE TABLE staff (
    emp_id     INT            PRIMARY KEY,
    full_name  VARCHAR(100)   NOT NULL,
    email      VARCHAR(100)   NOT NULL UNIQUE,
    dept_id    INT            NOT NULL REFERENCES departments(dept_id),
    salary     DECIMAL(10,2)  CHECK (salary > 0),
    join_date  DATE           DEFAULT CURRENT_DATE,
    is_active  BOOLEAN        DEFAULT TRUE
);
```

RESULT

```
Commands completed successfully.
Commands completed successfully.
```

In SQL Server you would usually write `IDENTITY(1,1)` for an auto-numbered key, `BIT` for the flag and `DEFAULT GETDATE()` for today. Here is the same `staff` table in full T-SQL style, with named constraints (names make error messages and later changes much easier):

SQL

```sql
CREATE TABLE dbo.staff (
    emp_id     INT IDENTITY(1,1)  NOT NULL,
    full_name  NVARCHAR(100)      NOT NULL,
    email      VARCHAR(100)       NOT NULL,
    dept_id    INT                NOT NULL,
    salary     DECIMAL(10,2)      NULL,
    join_date  DATE               NOT NULL CONSTRAINT df_staff_join  DEFAULT
(CAST(GETDATE() AS DATE)),
    is_active  BIT                NOT NULL CONSTRAINT df_staff_active DEFAULT (1),
    CONSTRAINT pk_staff         PRIMARY KEY (emp_id),
    CONSTRAINT uq_staff_email   UNIQUE (email),
    CONSTRAINT fk_staff_dept    FOREIGN KEY (dept_id) REFERENCES dbo.departments(dept_id),
    CONSTRAINT ck_staff_salary  CHECK (salary > 0)
);
```

Constraints at a glance

Constraint	Guarantees	Example
`PRIMARY KEY`	Unique and not NULL; identifies each row	`emp_id`
`FOREIGN KEY` / `REFERENCES`	Value must exist in the parent table	`dept_id` → `departments`
`UNIQUE`	No duplicate values (NULL allowed once in SQL Server)	`email`
`NOT NULL`	A value is required	`full_name`
`CHECK`	Value must satisfy a condition	`salary > 0`, `discount BETWEEN 0 AND 0.9`
`DEFAULT`	Value used when none is supplied	`is_active = 1`
`IDENTITY(seed, step)`	Auto-numbering (SQL Server)	`IDENTITY(1,1)`

INSERT: adding rows

SQL

```sql
INSERT INTO departments (dept_id, dept_name, location) VALUES
    (10, 'Finance', 'Mumbai'),
    (20, 'IT',      'Pune'),
    (30, 'HR',      DEFAULT),
    (40, 'Sales',   'Delhi');

INSERT INTO staff (emp_id, full_name, email, dept_id, salary, join_date) VALUES
    (1, 'Kavya Iyer',        'kavya@abc.in',    30,  97000, '2016-02-28'),
    (2, 'Imran Reddy',       'imran@abc.in',    10, 157000, '2019-12-14'),
    (3, 'Harpreet Kulkarni', 'harpreet@abc.in', 10, 131000, '2024-10-05'),
    (4, 'Rahul Iyer',        'rahul@abc.in',    20, 112000, '2017-02-06'),
    (5, 'Isha Mehta',        'isha@abc.in',     40,  57000, '2021-09-07');

SELECT * FROM departments;
```

RESULT

```
(4 rows affected)
(5 rows affected)
```

dept_id	dept_name	location
10	Finance	Mumbai
20	IT	Pune
30	HR	Bengaluru
40	Sales	Delhi

4 rows × 3 columns

SQL

```
SELECT emp_id, full_name, dept_id, salary, join_date, is_active FROM staff;
```

RESULT

emp_id	full_name	dept_id	salary	join_date	is_active
1	Kavya Iyer	30	97,000	2016-02-28	True
2	Imran Reddy	10	157,000	2019-12-14	True
3	Harpreet Kulkarni	10	131,000	2024-10-05	True
4	Rahul Iyer	20	112,000	2017-02-06	True
5	Isha Mehta	40	57,000	2021-09-07	True

5 rows × 6 columns

`DEFAULT` filled HR's location with Bengaluru and every employee's `is_active` with TRUE.

> **TIP** **Always list the columns**
>
> `INSERT INTO staff VALUES (…)` without a column list depends on the column order and breaks the day someone adds a column. Always write `INSERT INTO staff (col1, col2, …) VALUES (…)`.

INSERT … SELECT and SELECT … INTO

Copy rows from a query into a table:

SQL

```sql
-- Into an existing table
INSERT INTO staff_archive (emp_id, full_name, dept_id, salary)
SELECT emp_id, full_name, dept_id, salary
FROM staff
WHERE is_active = 0;

-- SQL Server: create a NEW table from a query (structure + data, no constraints)
SELECT emp_id, full_name, salary
INTO   staff_backup_2026
FROM   staff;
```

The portable equivalent of `SELECT … INTO` is `CREATE TABLE new_table AS SELECT …` (PostgreSQL, MySQL, Snowflake, DuckDB).

What constraints do when rules are broken

Constraints reject bad data **at the database**, whatever app or person sends it. These statements would all fail in SQL Server with the messages shown (abridged):

Statement	Broken rule	SQL Server message
`INSERT INTO departments VALUES (10,'Ops','Pune')`	PRIMARY KEY	Violation of PRIMARY KEY constraint 'pk_departments'. Cannot insert duplicate key … The duplicate key value is (10).
`INSERT INTO staff (…, email, …) VALUES (…, 'kavya@abc.in', …)`	UNIQUE	Violation of UNIQUE KEY constraint 'uq_staff_email'. Cannot insert duplicate key …
`INSERT INTO staff (…, dept_id, …) VALUES (…, 99, …)`	FOREIGN KEY	The INSERT statement conflicted with the FOREIGN KEY constraint "fk_staff_dept".
`INSERT INTO staff (emp_id, email, dept_id) VALUES (9,'x@abc.in',10)`	NOT NULL	Cannot insert the value NULL into column 'full_name' … column does not allow nulls. INSERT fails.
`UPDATE staff SET salary = -500 WHERE emp_id = 1`	CHECK	The UPDATE statement conflicted with the CHECK constraint "ck_staff_salary".
`DELETE FROM departments WHERE dept_id = 10`	FOREIGN KEY (children exist)	The DELETE statement conflicted with the REFERENCE constraint "fk_staff_dept".

Our practice engine raises the same kind of error. For example, trying to add an employee to a department that does not exist:

OUTPUT

```
Error: Constraint Error: Violates foreign key constraint because key "dept_id: 99"
does not exist in the referenced table
```

UPDATE: changing rows

SQL

```sql
-- Give everyone in Finance a 5% raise
UPDATE staff
SET    salary = salary * 1.05
WHERE  dept_id = 10;

-- Change two columns for one person
UPDATE staff
SET    dept_id = 20, salary = 60000
WHERE  emp_id = 5;

SELECT emp_id, full_name, dept_id, salary FROM staff ORDER BY emp_id;
```

RESULT

```
(2 rows affected)
(1 row affected)
```

emp_id	full_name	dept_id	salary
1	Kavya Iyer	30	97,000
2	Imran Reddy	10	164,850
3	Harpreet Kulkarni	10	137,550
4	Rahul Iyer	20	112,000
5	Isha Mehta	20	60,000

5 rows × 4 columns

WATCH OUT **UPDATE or DELETE without WHERE changes EVERY row**

`UPDATE staff SET salary = 0;` sets every salary to zero. Before any UPDATE or DELETE: 1. Write the `WHERE` first and run it as a `SELECT` to see exactly which rows will change. 2. Wrap the change in a transaction (`BEGIN TRAN` … then check, then `COMMIT` or `ROLLBACK`) (Lesson 14). 3. On production data, take a backup or work in a copy.

UPDATE with a join (SQL Server)

SQL

```sql
-- Mark staff in Mumbai departments as 'on-site'
UPDATE s
SET    s.work_mode = 'on-site'
FROM   staff AS s
JOIN   departments AS d ON d.dept_id = s.dept_id
WHERE  d.location = 'Mumbai';
```

DELETE: removing rows

SQL

```sql
DELETE FROM staff
WHERE emp_id = 5;

SELECT COUNT(*) AS staff_left FROM staff;
```

RESULT

(1 row affected)

staff_left
4

1 rows × 1 columns

ALTER TABLE: changing structure

SQL

```sql
ALTER TABLE staff ADD COLUMN phone VARCHAR(15);
ALTER TABLE staff ADD COLUMN work_mode VARCHAR(10) DEFAULT 'hybrid';
SELECT emp_id, full_name, phone, work_mode FROM staff;
```

RESULT

emp_id	full_name	phone	work_mode
1	Kavya Iyer	NULL	hybrid
2	Imran Reddy	NULL	hybrid
3	Harpreet Kulkarni	NULL	hybrid
4	Rahul Iyer	NULL	hybrid

4 rows × 4 columns

The SQL Server forms of the common changes:

```sql
ALTER TABLE staff ADD phone VARCHAR(15) NULL;                          -- add a column (no COLUMN keyword in T-SQL)
ALTER TABLE staff ALTER COLUMN full_name NVARCHAR(150) NOT NULL;       -- change type / nullability
ALTER TABLE staff DROP COLUMN phone;                                   -- remove a column
ALTER TABLE staff ADD CONSTRAINT ck_staff_mode CHECK (work_mode IN ('remote','hybrid','on-site'));
ALTER TABLE staff DROP CONSTRAINT ck_staff_mode;                       -- remove a constraint by name
EXEC sp_rename 'staff.full_name', 'employee_name', 'COLUMN';           -- rename a column
EXEC sp_rename 'staff', 'employees_master';                            -- rename a table
```

NOTE **Changing a column that has data**

Making a column smaller (`VARCHAR(100)` → `VARCHAR(10)`) fails if existing values do not fit; adding `NOT NULL` fails if NULLs exist. Clean the data first, or add a `DEFAULT`.

DELETE vs TRUNCATE vs DROP

This is one of the most common SQL interview questions.

	DELETE	TRUNCATE TABLE	DROP TABLE
Family	DML	DDL	DDL
Removes	Some or all **rows**	**All rows**	The **whole table**: rows, structure, indexes, constraints, permissions
WHERE allowed	Yes	No	No
Speed on large tables	Slower: logs each row	Very fast: deallocates pages	Fast
Logging	Every row logged	Minimal (page deallocations)	Minimal
Fires DELETE triggers	Yes	No	No
Resets IDENTITY	No	**Yes**, back to the seed	Table is gone
Works when referenced by a FK	Yes (if no child rows violate it)	**No**	**No** (drop the FK first)
Can be rolled back in a transaction	Yes	Yes in SQL Server (inside an explicit transaction)	Yes in SQL Server (inside an explicit transaction)

SQL

```sql
CREATE TABLE temp_log (id INT, msg VARCHAR(50));
INSERT INTO temp_log VALUES (1, 'start'), (2, 'load'), (3, 'end');
DELETE FROM temp_log WHERE id = 2;           -- removes one row
SELECT COUNT(*) AS after_delete FROM temp_log;
```

RESULT

```
Commands completed successfully.
(3 rows affected)
(1 row affected)
```

after_delete
2

1 rows × 1 columns

```sql
TRUNCATE TABLE temp_log;                    -- removes all rows, keeps the table
SELECT COUNT(*) AS after_truncate FROM temp_log;
```

RESULT

after_truncate
0

1 rows × 1 columns

```sql
DROP TABLE temp_log;                        -- removes the table itself
SELECT COUNT(*) AS tables_named_temp_log FROM information_schema.tables WHERE table_name =
'temp_log';
```

RESULT

tables_named_temp_log
0

1 rows × 1 columns

In SQL Server, `DROP TABLE IF EXISTS temp_log;` avoids an error when the table is already gone.

MERGE: insert or update in one statement (upsert)

Data pipelines often receive a file of changes: some rows are new, others update existing rows. `MERGE` handles both:

```sql
MERGE dbo.staff AS target
USING dbo.staff_updates AS src
      ON target.emp_id = src.emp_id
WHEN MATCHED THEN
     UPDATE SET target.salary = src.salary, target.dept_id = src.dept_id
WHEN NOT MATCHED BY TARGET THEN
     INSERT (emp_id, full_name, email, dept_id, salary)
     VALUES (src.emp_id, src.full_name, src.email, src.dept_id, src.salary)
WHEN NOT MATCHED BY SOURCE THEN
     UPDATE SET target.is_active = 0;          -- in staff but not in the file
```

Loading CSV files with SQL

```sql
CREATE TABLE dbo.employees_small (
    emp_id VARCHAR(10) PRIMARY KEY, name NVARCHAR(100), department VARCHAR(30),
    city VARCHAR(30), join_date DATE, salary DECIMAL(10,2), rating TINYINT
);

BULK INSERT dbo.employees_small
FROM 'C:\Data\employees_small.csv'
WITH (FORMAT = 'CSV', FIRSTROW = 2, FIELDTERMINATOR = ',', ROWTERMINATOR = '0x0a');
```

The SSMS **Import Flat File** wizard is easier for one-off loads; `BULK INSERT`, `bcp` or an ETL tool (SSIS, Azure Data Factory, Fabric pipelines) are used for repeatable loads.

DO IT WITH AI

Do it with AI

Ask: *"Write SQL Server CREATE TABLE statements with constraints for a hospital database: patients, doctors, appointments (a patient can see many doctors; a doctor sees many patients). Include primary keys, foreign keys, CHECK constraints for appointment status, and sensible data types."* Then review it: are the keys right, is money DECIMAL, are dates DATE/DATETIME2, are the constraints named?

Summary

KEY TAKEAWAYS

- **DDL**: `CREATE`, `ALTER`, `DROP`, `TRUNCATE` change structure. **DML**: `INSERT`, `UPDATE`, `DELETE`, `MERGE` change rows.
- Constraints (PK, FK, UNIQUE, NOT NULL, CHECK, DEFAULT, IDENTITY) protect data quality at the source; name them.
- Always list columns in `INSERT`; always test the `WHERE` of an `UPDATE`/`DELETE` with a `SELECT` first, inside a transaction.
- `DELETE` removes chosen rows (logged, triggers fire); `TRUNCATE` empties the table fast and resets IDENTITY; `DROP` removes the table.
- `MERGE` performs upserts; `BULK INSERT` and the Import wizard load files.

Quick quiz

6 questions · tap an answer to check it

1. Which statement removes all rows but keeps the table structure and resets the IDENTITY counter?

- **A** DELETE FROM t
- **B** TRUNCATE TABLE t
- **C** DROP TABLE t
- **D** ALTER TABLE t

Answer: B. TRUNCATE empties the table quickly and resets IDENTITY to its seed.

2. You run `UPDATE staff SET salary = 50000;` by mistake. What happened?

- **A** Only the first row changed
- **B** Every row's salary is now 50,000
- **C** Nothing, because there is no WHERE
- **D** An error is raised

Answer: B. Without WHERE, UPDATE changes every row. Use transactions and test with SELECT first.

3. Which constraint stops a salary of −500?

- **A** UNIQUE
- **B** DEFAULT
- **C** CHECK (salary > 0)
- **D** FOREIGN KEY

Answer: C. CHECK validates values against a condition.

4. Inserting an employee with dept_id 99 fails with a FOREIGN KEY error. Why?

A dept_id must be unique

B No department with id 99 exists in the parent table

C dept_id cannot be a number

D The employee table is full

Answer: B. A foreign key value must exist in the referenced table.

5. Which is TRUE about DELETE?

A It can remove selected rows with WHERE and fires DELETE triggers

B It is DDL

C It resets IDENTITY

D It removes the table structure

Answer: A. DELETE is DML, works row by row with a WHERE, and fires triggers.

6. How do you add a column `phone` to `staff` in SQL Server?

A `ALTER TABLE staff ADD COLUMN phone VARCHAR(15)` only

B `ALTER TABLE staff ADD phone VARCHAR(15)`

C `UPDATE staff ADD phone`

D `INSERT COLUMN phone INTO staff`

Answer: B. T-SQL uses ADD without the COLUMN keyword (PostgreSQL/MySQL accept ADD COLUMN).

EXERCISE **Build and populate a small schema**

Create `projects(project_id PK, name UNIQUE NOT NULL, budget DECIMAL(12,2) CHECK budget ≥ 0, start_date DATE, status with CHECK IN ('planned','active','closed') DEFAULT 'planned')` and `assignments(project_id FK, emp_id FK, role, hours, PRIMARY KEY (project_id, emp_id))`. Insert 3 projects and 4 assignments, raise the budget of active projects by 10%, close one project, and delete assignments of closed projects.

Show solution (worked answer)

SQL

```
CREATE TABLE projects (
  project_id INT PRIMARY KEY, name VARCHAR(80) NOT NULL UNIQUE,
  budget DECIMAL(12,2) CHECK (budget ⩾ 0), start_date DATE,
  status VARCHAR(10) DEFAULT 'planned' CHECK (status IN
('planned','active','closed')));
CREATE TABLE assignments (
  project_id INT REFERENCES projects(project_id), emp_id INT REFERENCES staff(emp_id),
  role VARCHAR(30), hours INT CHECK (hours > 0), PRIMARY KEY (project_id, emp_id));
INSERT INTO projects (project_id, name, budget, start_date, status) VALUES
  (1,'CRM revamp',2500000,'2026-01-10','active'), (2,'Payroll audit',600000,'2026-03-
01','active'), (3,'Data lake',4000000,'2026-06-01','planned');
INSERT INTO assignments VALUES (1,1,'Analyst',120),(1,4,'Developer',300),
(2,2,'Lead',80),(3,3,'Architect',40);
UPDATE projects SET budget = budget * 1.10 WHERE status = 'active';
UPDATE projects SET status = 'closed' WHERE project_id = 2;
DELETE FROM assignments WHERE project_id IN (SELECT project_id FROM projects WHERE
status = 'closed');
```

INTERVIEW CORNER

Questions you may be asked

Q: DELETE vs TRUNCATE vs DROP? DELETE (DML) removes chosen rows with WHERE, logs each row, fires triggers and keeps IDENTITY. TRUNCATE (DDL) removes all rows quickly, minimal logging, resets IDENTITY, no WHERE, not allowed when referenced by a FK. DROP removes the whole table and its definition.

Q: What are constraints and why use them instead of app validation? Rules enforced by the database (PK, FK, UNIQUE, NOT NULL, CHECK, DEFAULT). They protect data no matter which app, script or person writes to the table.

Q: How do you safely run an UPDATE on production? Run the WHERE as a SELECT first, wrap in BEGIN TRAN, check @@ROWCOUNT and the result, then COMMIT or ROLLBACK; have a backup.

SQL FOR DATA ANALYSIS · LESSON 5

SELECT, WHERE, ORDER BY & Filtering Operators

Retrieve exactly the rows and columns you need. Master SELECT, aliases with AS, DISTINCT, TOP, WHERE with comparison and logical operators, IN, BETWEEN, LIKE wildcards, IS NULL, ORDER BY, and the logical order in which SQL processes a query.

SQL class 3 | SQL Server 2022, 2026 | Beginner | 120 min

By the end of this lesson you can

- Select columns, calculated expressions and aliases, and remove duplicates with DISTINCT
- Filter rows with comparison, logical, IN, NOT IN, BETWEEN, LIKE, NOT LIKE, IS NULL and IS NOT NULL
- Sort results by one or more columns, expressions and positions; limit rows with TOP
- Explain the logical processing order of a query and why it matters for aliases
- Translate business questions into correct SELECT statements

The shape of a SELECT statement

SQL

```sql
SELECT   column_list          -- what to show
FROM     table_name           -- where it comes from
WHERE    row_conditions       -- which rows to keep
GROUP BY grouping_columns     -- (Lesson 7)
HAVING   group_conditions     -- (Lesson 7)
ORDER BY sort_columns;        -- how to sort the result
```

Only `SELECT` is required (`SELECT 1 + 1;` works). Everything else is optional but must appear in this order.

SELECT: columns, expressions and aliases

SQL

```sql
SELECT TOP 5
       order_id,
       order_date,
       customer_name,
       sales,
       profit,
       sales - profit                    AS cost,
       ROUND(100.0 * profit / sales, 1)  AS margin_pct,
       'INR'                             AS currency
FROM superstore;
```

RESULT

order_id	order_date	customer_name	sales	profit	cost	margin_pct	cu
IN-2023-00001	2023-01-01	Ishaan Gupta	36,000	3,472.55	32,527.45	9.6	IN
IN-2023-00001	2023-01-01	Ishaan Gupta	3,850	1,169.54	2,680.46	30.4	IN
IN-2023-00001	2023-01-01	Ishaan Gupta	62,080	6,044.47	56,035.53	9.7	IN
IN-2023-00002	2023-01-03	Kabir Singh	5,216	450.16	4,765.84	8.6	IN
IN-2023-00003	2023-01-03	Sneha Khan	24,591	-5,512.33	30,103.33	-22.4	IN

5 rows × 8 columns

- `SELECT *` returns every column. Use it to explore, never in production code (it breaks when columns change and reads more data than needed).
- An **alias** renames a column or table in the result: `AS margin_pct`. `AS` is optional (`sales - profit cost`) but makes code clearer. Aliases with spaces need brackets or quotes: `AS [Margin %]`.
- Expressions can use operators, functions and literals.

Table aliases

```sql
SELECT s.order_id, s.sales
FROM superstore AS s;          -- 's' is a short name for the table
```

Table aliases become essential with joins (Lesson 8).

DISTINCT: unique values

```sql
SELECT DISTINCT region
FROM superstore
ORDER BY region;
```

RESULT

region
East
North
South
West

4 rows × 1 columns

```sql
SELECT DISTINCT category, sub_category
FROM superstore
ORDER BY category, sub_category;
```

RESULT

category	sub_category
Furniture	Bookcases
Furniture	Chairs
Furniture	Furnishings
Furniture	Tables
Office Supplies	Appliances
Office Supplies	Art
Office Supplies	Binders
Office Supplies	Paper
Office Supplies	Storage
Technology	Accessories
Technology	Copiers
Technology	Machines

13 rows × 2 columns (first 12 shown)

`DISTINCT` applies to the **whole row** of selected columns: the second query returns each category–sub-category combination once. To count unique values: `COUNT(DISTINCT customer_id)`.

TOP: limit the rows

SQL

```sql
SELECT TOP 5 order_id, customer_name, city, sales
FROM superstore
ORDER BY sales DESC;
```

RESULT

order_id	customer_name	city	sales
IN-2024-01857	Arjun Fernandes	Delhi	619,290
IN-2023-00564	Krishna Reddy	Hyderabad	551,530
IN-2024-01040	Aditya Kulkarni	Chennai	551,530
IN-2025-02880	Farhan Khan	Kochi	530,820
IN-2023-00809	Rohan Singh	Jaipur	526,396

5 rows × 4 columns

Dialect	Syntax for "first 5 rows"
SQL Server	`SELECT TOP 5 … ORDER BY …` (also `TOP 10 PERCENT`, `TOP 5 WITH TIES`)
SQL Server 2012+, ANSI	`ORDER BY … OFFSET 0 ROWS FETCH NEXT 5 ROWS ONLY` (Lesson 11)
PostgreSQL, MySQL, SQLite, DuckDB	`… ORDER BY … LIMIT 5`
Oracle	`FETCH FIRST 5 ROWS ONLY`

WATCH OUT **TOP without ORDER BY**

Without `ORDER BY`, the database returns rows in no guaranteed order, so "top 5" may differ between runs. Always sort when you use TOP.

WHERE: filtering rows

Comparison operators

SQL

```sql
SELECT COUNT(*) AS lines_over_50k
FROM superstore
WHERE sales > 50000;
```

RESULT

lines_over_50k
1126

1 rows × 1 columns

SQL

```sql
SELECT order_id, city, category, sales, discount
FROM superstore
WHERE region = 'West'
  AND category = 'Furniture'
  AND discount ⩾ 0.3
ORDER BY sales DESC;
```

RESULT

order_id	city	category	sales	discount
IN-2025-02904	Ahmedabad	Furniture	76,048	0.3
IN-2025-02783	Pune	Furniture	69,762	0.3
IN-2024-01002	Indore	Furniture	58,135	0.3
IN-2023-00431	Ahmedabad	Furniture	54,474	0.3
IN-2025-02016	Ahmedabad	Furniture	54,138	0.3
IN-2024-01490	Indore	Furniture	49,728	0.3

139 rows × 5 columns (first 6 shown)

Text values go in **single quotes** (`'West'`). Double quotes are for identifiers in standard SQL. Dates are written as text in ISO format: `order_date ⩾ '2025-01-01'`.

IN and NOT IN

`IN` is shorthand for several `OR` conditions on the same column.

SQL

```sql
SELECT city, COUNT(*) AS order_lines
FROM superstore
WHERE city IN ('Pune', 'Mumbai', 'Ahmedabad')
GROUP BY city
ORDER BY city;
```

RESULT

city	order_lines
Ahmedabad	373
Mumbai	283
Pune	529

3 rows × 2 columns

SQL

```sql
SELECT COUNT(*) AS non_metro_lines
FROM superstore
WHERE city NOT IN ('Delhi', 'Mumbai', 'Bengaluru', 'Chennai', 'Kolkata', 'Hyderabad');
```

RESULT

non_metro_lines
3918

1 rows × 1 columns

COMMON MISTAKES **NOT IN with a NULL**

If the list (or subquery) inside `NOT IN` contains a NULL, `NOT IN` returns **no rows at all**, because x `<>` `NULL` is UNKNOWN. Use `NOT EXISTS` (Lesson 10) or filter NULLs out of the subquery.

BETWEEN

`BETWEEN a AND b` is **inclusive** of both ends.

SQL

```sql
SELECT COUNT(*) AS lines_between_10k_and_20k
FROM superstore
WHERE sales BETWEEN 10000 AND 20000;
```

RESULT

lines_between_10k_and_20k
896

1 rows × 1 columns

For dates, prefer a **half-open range**: `order_date ⩾ '2025-01-01' AND order_date < '2026-01-01'`. It works whether the column is `DATE` or `DATETIME2`; `BETWEEN '2025-01-01' AND '2025-12-31'` silently misses orders placed after midnight on 31-Dec when the column has times.

SQL

```sql
SELECT COUNT(DISTINCT order_id) AS orders_in_2025
FROM superstore
WHERE order_date ⩾ '2025-01-01' AND order_date < '2026-01-01';
```

RESULT

orders_in_2025
1232

1 rows × 1 columns

LIKE: pattern matching

Wildcard	Meaning	Example	Matches
%	Any sequence of characters (including none)	'Apex%'	Apex Chair 412, Apex Phone 118
_	Exactly one character	'C00_1'	C0011, C0021 …
[ ] (SQL Server)	One character from a set or range	'[A-C]%'	starts with A, B or C
[^] (SQL Server)	One character not in the set	'[^A-M]%'	starts with N–Z

SQL

```sql
SELECT DISTINCT product_name
FROM superstore
WHERE product_name LIKE 'Apex Ch%'
ORDER BY product_name;
```

RESULT

product_name
Apex Chair 969

1 rows × 1 columns

SQL

```sql
SELECT COUNT(DISTINCT customer_id) AS customers_named_with_khan
FROM superstore
WHERE customer_name LIKE '%Khan%';
```

RESULT

customers_named_with_khan
44

1 rows × 1 columns

SQL

```sql
SELECT DISTINCT city
FROM superstore
WHERE city NOT LIKE '%a%'          -- city names without a lowercase 'a'
ORDER BY city;
```

RESULT

city
Delhi
Indore
Kochi
Lucknow
Pune

5 rows × 1 columns

> **NOTE** **Case sensitivity**
>
> SQL Server's default collation is **case-insensitive**, so `LIKE 'apex%'` also matches "Apex...". PostgreSQL is case-sensitive (use `ILIKE`), and so is the practice engine used for the outputs in this handbook. Write patterns in the same case as the data to be safe everywhere.

Leading wildcards (`'%Khan'`) cannot use an index and scan the whole table; fine for analysis, slow for apps on big tables.

IS NULL and IS NOT NULL

SQL

```sql
SELECT COUNT(*) AS homes_missing_area
FROM houses
WHERE area_sqft IS NULL;
```

RESULT

homes_missing_area
25

1 rows × 1 columns

(The `houses` table is loaded from `house_prices.csv`; 25 houses have a missing area, which you will clean in the Python track.)

Combining conditions

SQL

```sql
SELECT order_id, region, category, ship_mode, sales, profit
FROM superstore
WHERE (region = 'North' OR region = 'East')
  AND category <> 'Office Supplies'
  AND ship_mode IN ('First Class', 'Same Day')
  AND profit < 0
ORDER BY profit;
```

RESULT

order_id	region	category	ship_mode	sales	profit
IN-2023-00320	North	Technology	Same Day	337,316	-67,392.8
IN-2024-01213	East	Technology	Same Day	317,128	-35,733.2
IN-2024-01808	North	Technology	First Class	155,328	-31,737.4
IN-2023-00406	North	Technology	First Class	159,390	-27,928.8
IN-2023-00358	North	Technology	Same Day	77,664	-18,678.2
IN-2024-01289	East	Technology	First Class	149,088	-18,075.8
IN-2025-02849	East	Technology	First Class	81,522	-17,598.3
IN-2024-01055	East	Technology	First Class	86,968	-16,505.7

128 rows × 6 columns (first 8 shown)

ORDER BY: sorting

SQL

```sql
SELECT region, city, COUNT(*) AS lines, ROUND(SUM(sales), 0) AS sales
FROM superstore
GROUP BY region, city
ORDER BY region ASC, sales DESC;
```

RESULT

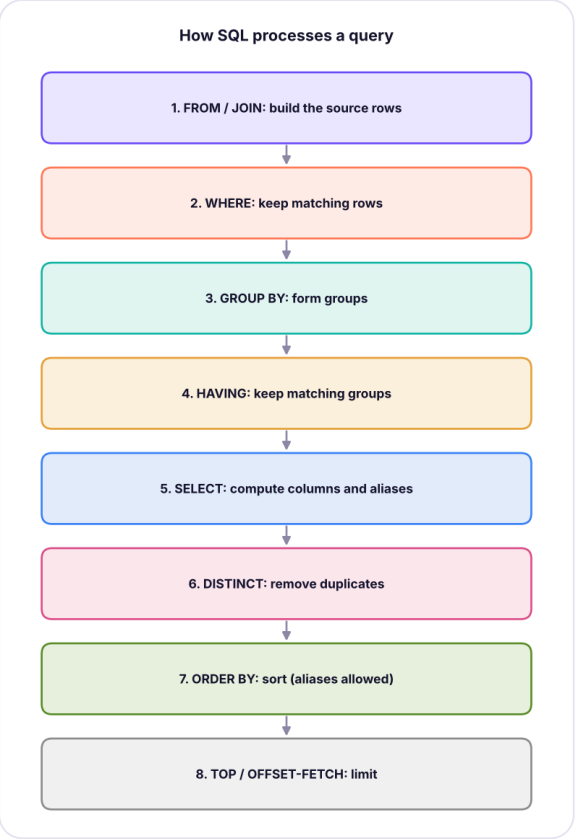
region	city	lines	sales
East	Kolkata	430	13,888,466
East	Patna	375	12,444,393
East	Bhubaneswar	383	10,482,685
East	Guwahati	335	9,668,576
North	Delhi	453	15,657,610
North	Chandigarh	361	11,541,792
North	Lucknow	422	11,328,882
North	Jaipur	294	10,489,878

16 rows × 4 columns (first 8 shown)

- `ASC` (default) or `DESC` for each column separately.
- Sort by an expression or alias: `ORDER BY sales - profit DESC`, `ORDER BY margin_pct`.
- Sort by column position (`ORDER BY 2 DESC`) works but is fragile; avoid it in saved code.
- NULLs sort **first** in ascending order in SQL Server (last in PostgreSQL/Oracle by default).
- Custom order with CASE: `ORDER BY CASE ship_mode WHEN 'Same Day' THEN 1 WHEN 'First Class' THEN 2 ELSE 3 END`.

The logical processing order

You *write* `SELECT` first, but SQL *processes* the clauses in a different order. This explains several common errors.



Logical processing order of a SELECT statement. Each step works on the output of the previous one.

Consequences:

- You **cannot** use a column alias in `WHERE`, because WHERE runs before SELECT: `WHERE margin_pct > 10` fails. Repeat the expression, or use a subquery/CTE.
- You **can** use an alias in `ORDER BY`, because it runs after SELECT.
- `WHERE` cannot contain aggregate functions (`WHERE SUM(sales) > 100`); use `HAVING` (Lesson 7).

SQL

```sql
-- Fails: Invalid column name 'margin_pct'
SELECT order_id, 100.0 * profit / sales AS margin_pct
FROM superstore
WHERE margin_pct > 10;

-- Works: repeat the expression in WHERE, use the alias in ORDER BY
SELECT order_id, 100.0 * profit / sales AS margin_pct
FROM superstore
WHERE 100.0 * profit / sales > 10
ORDER BY margin_pct DESC;
```

From business question to query

STEP BY STEP **Translate a question**

Question: *"Which Technology orders in the South had a discount but still made more than ₹10,000 profit in 2025? Show the biggest first."* 1. **Table**: superstore. 2. **Rows** (WHERE): category = Technology, region = South, discount > 0, profit > 10000, order date in 2025. 3. **Columns** (SELECT): order_id, order_date, city, sub_category, discount, profit. 4. **Order**: profit descending.

SQL

```sql
SELECT order_id, order_date, city, sub_category, discount, profit
FROM superstore
WHERE category = 'Technology'
  AND region = 'South'
  AND discount > 0
  AND profit > 10000
  AND order_date ⩾ '2025-01-01' AND order_date < '2026-01-01'
ORDER BY profit DESC;
```

RESULT

order_id	order_date	city	sub_category	discount	profit
IN-2025-02318	2025-04-26	Kochi	Machines	0.1	21,209.4
IN-2025-02908	2025-10-29	Bengaluru	Phones	0.1	16,734.2
IN-2025-02157	2025-03-08	Bengaluru	Copiers	0.1	12,678.4
IN-2025-03094	2025-12-08	Bengaluru	Machines	0.1	11,292.0
IN-2025-02514	2025-07-09	Bengaluru	Machines	0.1	10,891.9
IN-2025-03146	2025-12-20	Bengaluru	Copiers	0.1	10,679.1
IN-2025-02547	2025-07-20	Kochi	Phones	0.1	10,558.6

7 rows × 6 columns

DO IT WITH AI

Do it with AI

Give an assistant your table's column list and a question, and ask for SQL Server syntax: *"Table superstore(order_id, order_date, region, city, category, sub_category, sales, discount, profit). Write a T-SQL query for: Furniture orders in Kolkata or Patna in 2024 with negative profit, sorted by profit."* Read the query before running it: check the date range, the brackets around OR, and that it uses the column names you gave.

Summary

KEY TAKEAWAYS

- `SELECT` columns and expressions; name them with `AS`; avoid `SELECT *` in saved queries.
- `DISTINCT` removes duplicate rows of the selected columns; `TOP n` with `ORDER BY` limits results.
- Filter with `=`, `<>`, `>`, `<`, `>=`, `<=`, `AND/OR/NOT` (use brackets), `IN`, `BETWEEN` (inclusive), `LIKE` with `%` and `_`, and `IS NULL`.
- For dates use half-open ranges: `>= start AND < next_start`.
- `ORDER BY` sorts ascending by default; can use aliases.
- Logical order: FROM → WHERE → GROUP BY → HAVING → SELECT → DISTINCT → ORDER BY → TOP. That is why aliases fail in WHERE.

Quick quiz

6 questions · tap an answer to check it

1. Which query returns each distinct region exactly once?

A `SELECT region FROM superstore`

B `SELECT DISTINCT region FROM superstore`

C `SELECT UNIQUE(region) FROM superstore`

D `SELECT TOP 1 region FROM superstore`

Answer: B. DISTINCT removes duplicate rows from the result.

2. `WHERE sales BETWEEN 100 AND 200` includes which values?

A 101 to 199

B 100 to 200, both included

C 100 to 199

D 101 to 200

Answer: B. BETWEEN is inclusive of both boundaries.

3. Which pattern finds product names that contain "Desk" anywhere?

A `LIKE 'Desk%'`

B `LIKE '%Desk'`

C `LIKE '%Desk%'`

D `LIKE '_Desk_'`

Answer: C. % on both sides matches any characters before and after.

4. Why does `WHERE margin_pct > 10` fail when margin_pct is an alias defined in SELECT?

- **A** WHERE is processed before SELECT, so the alias does not exist yet
- **B** Aliases can only be used with GROUP BY
- **C** margin_pct must be in brackets
- **D** You must use HAVING for all aliases

Answer: A. Logical order: WHERE comes before SELECT.

5. What is wrong with `WHERE order_date BETWEEN '2025-01-01' AND '2025-12-31'` on a DATETIME2 column?

- **A** Nothing
- **B** It misses rows on 31-Dec after 00:00:00
- **C** It includes 2026 dates
- **D** BETWEEN does not work with dates

Answer: B. '2025-12-31' means midnight; later times that day are excluded. Use < '2026-01-01'.

6. `SELECT TOP 3 * FROM orders` without ORDER BY returns:

- **A** The 3 newest orders
- **B** The 3 largest orders
- **C** Any 3 rows, in no guaranteed order
- **D** An error

Answer: C. Without ORDER BY, row order is not defined.

EXERCISE

Superstore filters

Write queries for: (1) all Same Day orders in the West with sales over ₹50,000; (2) customers whose names start with A, B or C (distinct, sorted); (3) order lines with a 40% discount in the Furniture category, showing the 10 biggest losses; (4) the number of order lines in 2024 for the Corporate segment in Pune or Indore.

Show solution (worked answer)

SQL

```sql
-- (1)
SELECT order_id, order_date, city, sales FROM superstore
WHERE ship_mode = 'Same Day' AND region = 'West' AND sales > 50000 ORDER BY sales
DESC;
-- (2) SQL Server
SELECT DISTINCT customer_name FROM superstore WHERE customer_name LIKE '[A-C]%' ORDER
BY customer_name;
-- (3)
SELECT TOP 10 order_id, product_name, sales, profit FROM superstore
WHERE discount = 0.4 AND category = 'Furniture' ORDER BY profit ASC;
-- (4)
SELECT COUNT(*) AS lines FROM superstore
WHERE segment = 'Corporate' AND city IN ('Pune', 'Indore')
  AND order_date ⩾ '2024-01-01' AND order_date < '2025-01-01';
```

INTERVIEW CORNER

Questions you may be asked

Q: What is the order of execution of a SQL query? FROM/JOIN, WHERE, GROUP BY, HAVING, SELECT, DISTINCT, ORDER BY, then TOP/OFFSET-FETCH.

Q: Difference between WHERE and HAVING? WHERE filters rows before grouping and cannot use aggregates; HAVING filters groups after aggregation.

Q: How do you find employees whose name's second letter is "a"? `WHERE name LIKE '_a%'`.

Q: IN vs EXISTS? IN compares a value with a list or subquery result; EXISTS checks whether a correlated subquery returns any row. EXISTS is safer with NULLs and often efficient for large subqueries (Lesson 10).

SQL FOR DATA ANALYSIS · LESSON 6

SQL Functions: Aggregate, Text, Date & Number

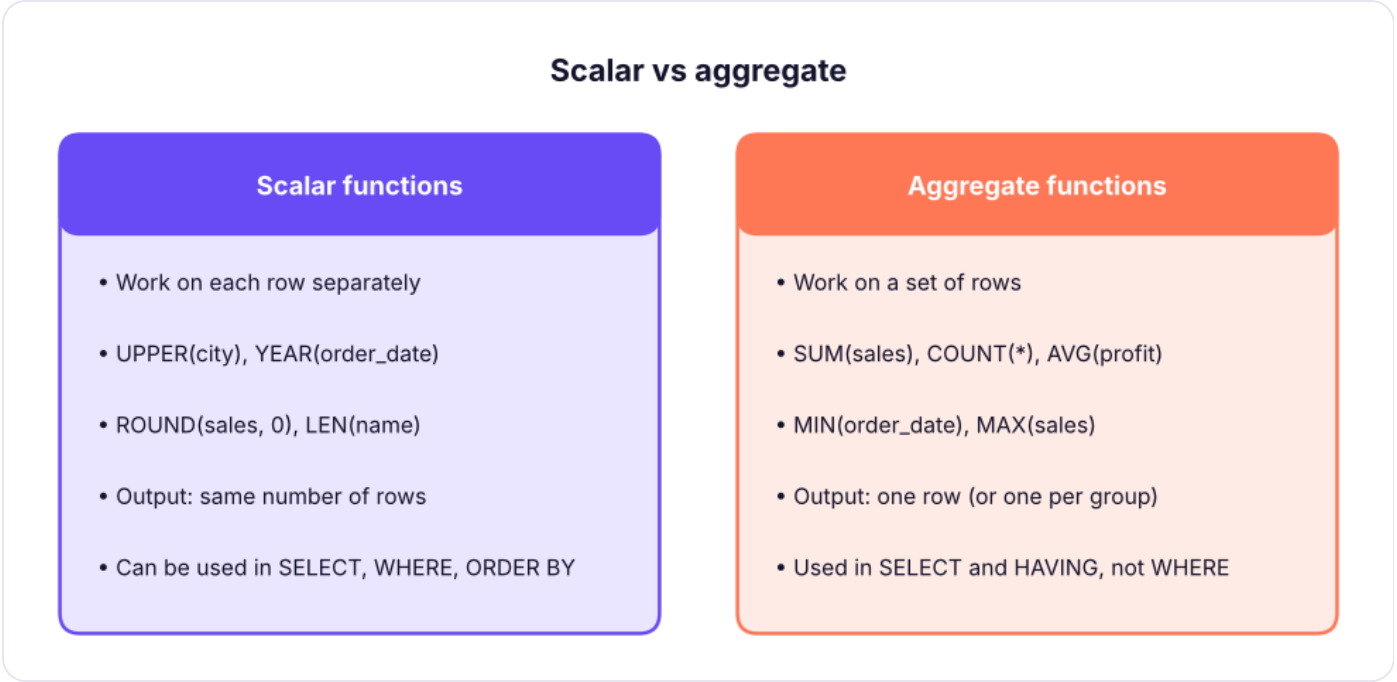
Use SQL Server's built-in functions to summarise, clean and reshape data. Aggregate functions (COUNT, SUM, AVG, MIN, MAX, STRING_AGG), text functions (LEN, LEFT, RIGHT, SUBSTRING, CHARINDEX, REPLACE, TRIM, UPPER, CONCAT), date functions (GETDATE, YEAR, MONTH, DATEPART, DATENAME, DATEADD, DATEDIFF, EOMONTH) and number functions (ROUND, CEILING, FLOOR, ABS, POWER, SQRT), plus CAST and CONVERT.

SQL class 4 · SQL Server 2022, 2026 · Beginner–Intermediate · 120 min

By the end of this lesson you can

- Distinguish scalar functions (one value per row) from aggregate functions (one value per group)
- Summarise data with COUNT(*), COUNT(column), COUNT(DISTINCT), SUM, AVG, MIN, MAX and STRING_AGG
- Clean and extract text with LEN, LEFT, RIGHT, SUBSTRING, CHARINDEX, REPLACE, TRIM, UPPER, LOWER and CONCAT
- Work with dates using YEAR, MONTH, DATEPART, DATENAME, DATEADD, DATEDIFF, EOMONTH and DATEFROMPARTS
- Round and convert numbers with ROUND, CEILING, FLOOR, ABS, POWER, SQRT, CAST, CONVERT and TRY_CAST

Two kinds of functions



Scalar functions return one value per row. Aggregate functions collapse many rows into one value (per group).

Aggregate functions

Function	Returns	NULL handling
`COUNT(*)`	Number of rows	Counts every row
`COUNT(col)`	Number of non-NULL values	Skips NULLs
`COUNT(DISTINCT col)`	Number of distinct non-NULL values	Skips NULLs
`SUM(col)`	Total	Skips NULLs
`AVG(col)`	Average of non-NULL values	Skips NULLs (they are not treated as 0)
`MIN(col)` , `MAX(col)`	Smallest / largest (numbers, text, dates)	Skips NULLs
`STDEV(col)` , `VAR(col)`	Sample standard deviation / variance	Skips NULLs
`STRING_AGG(col, sep)`	Values joined into one string	Skips NULLs

SQL

```sql
SELECT COUNT(*)                       AS order_lines,
       COUNT(DISTINCT order_id)       AS orders,
       COUNT(DISTINCT customer_id)    AS customers,
       ROUND(SUM(sales), 0)           AS total_sales,
       ROUND(AVG(sales), 0)           AS avg_line_sales,
       MIN(order_date)                AS first_order,
       MAX(order_date)                AS last_order,
       ROUND(STDEV(sales), 0)         AS sd_sales
FROM superstore;
```

RESULT

order_lines	orders	customers	total_sales	avg_line_sales	first_order	last_order
6323	3200	785	201,413,569	31,854	2023-01-01	2025-12-31

1 rows × 8 columns

SQL

```sql
SELECT category,
       STRING_AGG(sub_category, ', ') AS sub_categories
FROM (SELECT DISTINCT category, sub_category FROM superstore) AS t
GROUP BY category
ORDER BY category;
```

RESULT

category	sub_categories
Furniture	Tables, Furnishings, Bookcases, Chairs
Office Supplies	Storage, Appliances, Art, Binders, Paper
Technology	Copiers, Phones, Machines, Accessories

3 rows × 2 columns

In SQL Server add `WITHIN GROUP (ORDER BY sub_category)` after `STRING_AGG(...)` to control the order of the joined values.

WATCH OUT **AVG of integers**

`AVG` on an `INT` column returns an integer in SQL Server (`AVG` of 1, 2 = 1). Use `AVG(quantity * 1.0)` or `AVG(CAST(quantity AS DECIMAL(10,2)))` for decimals.

Text (string) functions

Function	Example	Result
`LEN(s)`	`LEN('Pune ')`	4 (trailing spaces not counted)
`UPPER(s)`, `LOWER(s)`	`UPPER('pune')`	PUNE
`LEFT(s, n)`, `RIGHT(s, n)`	`LEFT('IN-2025-00042', 2)`	IN
`SUBSTRING(s, start, n)`	`SUBSTRING('IN-2025-00042', 4, 4)`	2025
`CHARINDEX(find, s)`	`CHARINDEX('-', 'IN-2025-00042')`	3
`REPLACE(s, old, new)`	`REPLACE('2025-04-27', '-', '/')`	2025/04/27
`TRIM(s)`, `LTRIM`, `RTRIM`	`TRIM(' Delhi ')`	Delhi
`CONCAT(a, b, …)`	`CONCAT(city, ', ', state)`	Pune, Maharashtra
`REVERSE(s)`	`REVERSE('abc')`	cba
`REPLICATE(s, n)`	`REPLICATE('*', 3)`	***
`STRING_SPLIT(s, sep)`	`SELECT value FROM STRING_SPLIT('a,b,c', ',')`	3 rows
`FORMAT(value, fmt)`	`FORMAT(1234567, 'N0', 'en-IN')`	12,34,567

SQL

```sql
SELECT TOP 6
       order_id,
       LEFT(order_id, 2)                                  AS country,
       SUBSTRING(order_id, 4, 4)                          AS order_year,
       RIGHT(order_id, 5)                                 AS order_no,
       product_name,
       LEFT(product_name, CHARINDEX(' ', product_name) - 1) AS brand,
       UPPER(city)                                        AS city_upper,
       LEN(customer_name)                                 AS name_length
FROM superstore
ORDER BY row_id;
```

RESULT

order_id	country	order_year	order_no	product_name	brand	city_upper	name_
IN-2023-00001	IN	2023	00001	Nova Phone 223	Nova	BENGALURU	12
IN-2023-00001	IN	2023	00001	Orbit Appliance 798	Orbit	BENGALURU	12
IN-2023-00001	IN	2023	00001	Orbit Table 248	Orbit	BENGALURU	12
IN-2023-00002	IN	2023	00002	Crest Storage 946	Crest	AHMEDABAD	11
IN-2023-00003	IN	2023	00003	Crest Bookcase 170	Crest	AHMEDABAD	10
IN-2023-00003	IN	2023	00003	Zenith Table 111	Zenith	AHMEDABAD	10

6 rows × 8 columns

The brand is everything before the first space: `CHARINDEX(' ', product_name)` finds the space position, and `LEFT(…, position - 1)` cuts there. This is the SQL version of Excel's `LEFT(A2, FIND(" ", A2) - 1)` from Excel Lesson 5.

Cleaning text

```sql
WITH raw AS (
    SELECT '  bengaluru ' AS city UNION ALL
    SELECT 'BENGALURU'           UNION ALL
    SELECT 'Bangalore'           UNION ALL
    SELECT 'Bengaluru'
)
SELECT city                                                    AS raw_value,
       CONCAT(UPPER(LEFT(TRIM(city), 1)), LOWER(SUBSTRING(TRIM(city), 2, 100))) AS
proper_case,
       CASE WHEN LOWER(TRIM(city)) IN ('bangalore', 'bengaluru') THEN 'Bengaluru' ELSE
TRIM(city) END AS standardised
FROM raw;
```

RESULT

raw_value	proper_case	standardised
bengaluru	Bengaluru	Bengaluru
BENGALURU	Bengaluru	Bengaluru
Bangalore	Bangalore	Bengaluru
Bengaluru	Bengaluru	Bengaluru

4 rows × 3 columns

(SQL Server has no `PROPER` function; joining `UPPER(LEFT(…,1))` and `LOWER(SUBSTRING(…,2,…))` capitalises the first letter of a single word. Standardising spellings is best done with `CASE` or a mapping table joined in.)

> **NOTE** **`+` vs CONCAT for text**
>
> In SQL Server `'a' + 'b'` joins text, but `'a' + NULL` is NULL and `'Q' + 2` fails (it tries to convert 'Q' to a number). `CONCAT('Q', 2)` gives Q2 and treats NULL as empty. Prefer `CONCAT`.

Date and time functions

Function	Example (d = '2026-04-27')	Result
`GETDATE()`, `SYSDATETIME()`		Current date and time
`CAST(GETDATE() AS DATE)`		Today's date
`YEAR(d)`, `MONTH(d)`, `DAY(d)`	`MONTH(d)`	4
`DATEPART(part, d)`	`DATEPART(quarter, d)`	2
`DATENAME(part, d)`	`DATENAME(month, d)`, `DATENAME(weekday, d)`	April, Monday
`DATEADD(part, n, d)`	`DATEADD(month, 3, d)`	2026-07-27
`DATEDIFF(part, start, end)`	`DATEDIFF(day, '2026-04-01', d)`	26
`EOMONTH(d, [n])`	`EOMONTH(d)`	2026-04-30
`DATEFROMPARTS(y, m, d)`	`DATEFROMPARTS(2026, 4, 1)`	2026-04-01
`DATETRUNC(part, d)` (2022+)	`DATETRUNC(month, d)`	2026-04-01
`FORMAT(d, 'MMM-yyyy')`		Apr-2026

Date parts you can use: `year`, `quarter`, `month`, `dayofyear`, `day`, `week`, `weekday`, `hour`, `minute`, `second`.

SQL

```sql
SELECT TOP 6
       order_id,
       order_date,
       ship_date,
       YEAR(order_date)                    AS yr,
       DATEPART(quarter, order_date)       AS qtr,
       DATENAME(month, order_date)         AS month_name,
       DATENAME(weekday, order_date)       AS weekday_name,
       DATEDIFF(day, order_date, ship_date) AS days_to_ship,
       EOMONTH(order_date)                 AS month_end
FROM orders
ORDER BY order_date DESC;
```

RESULT

order_id	order_date	ship_date	yr	qtr	month_name	weekday_name	days_to_sh
IN-2025-03197	2025-12-31	2026-01-04	2025	4	December	Wednesday	4
IN-2025-03198	2025-12-31	2026-01-04	2025	4	December	Wednesday	4
IN-2025-03199	2025-12-31	2026-01-02	2025	4	December	Wednesday	2
IN-2025-03200	2025-12-31	2026-01-06	2025	4	December	Wednesday	6
IN-2025-03194	2025-12-30	2026-01-03	2025	4	December	Tuesday	4
IN-2025-03195	2025-12-30	2026-01-06	2025	4	December	Tuesday	7

6 rows × 9 columns

SQL

```sql
SELECT ship_mode,
       COUNT(*)                                      AS orders,
       AVG(DATEDIFF(day, order_date, ship_date) * 1.0)  AS avg_days_to_ship,
       MAX(DATEDIFF(day, order_date, ship_date))        AS max_days
FROM orders
GROUP BY ship_mode
ORDER BY avg_days_to_ship;
```

RESULT

ship_mode	orders	avg_days_to_ship	max_days
Same Day	176	0.00	0
First Class	467	1.53	2
Second Class	621	3.00	4
Standard Class	1936	5.54	7

4 rows × 4 columns

WATCH OUT **DATEDIFF counts boundaries**

`DATEDIFF(year, '2025-12-31', '2026-01-01')` returns **1**, because one year boundary is crossed, even though only one day passed. For exact age use `DATEDIFF(day, …)/365.25` or compare month and day as well.

Useful date patterns

SQL

```sql
-- Orders in the last 30 days
WHERE order_date ⩾ DATEADD(day, -30, CAST(GETDATE() AS DATE))

-- First day of the month
SELECT DATEFROMPARTS(YEAR(order_date), MONTH(order_date), 1)   -- or DATETRUNC(month,
order_date)

-- Indian financial year (April-March)
SELECT CASE WHEN MONTH(order_date) ⩾ 4 THEN YEAR(order_date) + 1 ELSE YEAR(order_date) END
AS fy

-- Month label for reports
SELECT FORMAT(order_date, 'MMM-yyyy') AS month_label
```

Number functions

Function	Example	Result
ROUND(n, d)	ROUND(1234.567, 1)	1234.600
ROUND(n, -2)	ROUND(1234.567, -2)	1200.000
CEILING(n)	CEILING(4.1)	5
FLOOR(n)	FLOOR(4.9)	4
ABS(n)	ABS(-250)	250
POWER(n, p)	POWER(2, 10)	1024
SQRT(n), SQUARE(n)	SQRT(144)	12
SIGN(n)	SIGN(-3)	-1
n % m	17 % 5	2
RAND()		Random float 0–1

SQL

```sql
SELECT sub_category,
       ROUND(SUM(sales), -3)                AS sales_to_thousand,
       CEILING(SUM(quantity) / 12.0)        AS boxes_of_12,
       ROUND(100.0 * SUM(profit) / SUM(sales), 2) AS margin_pct,
       ABS(SUM(profit))                     AS abs_profit,
       SIGN(SUM(profit))                    AS profit_sign
FROM superstore
GROUP BY sub_category
ORDER BY margin_pct;
```

RESULT

sub_category	sales_to_thousand	boxes_of_12	margin_pct	abs_profit	profit_sign
Tables	14,046,000	103	-5.81	816,042	-1
Bookcases	11,275,000	101	-3.75	422,786	-1
Chairs	8,091,000	110	4.23	342,043	1
Furnishings	1,429,000	94	5.50	78,632	1
Copiers	78,750,000	99	10.71	8,436,141	1
Phones	31,919,000	110	11.48	3,663,117	1
Accessories	3,179,000	98	11.63	369,554	1
Machines	42,590,000	106	12.18	5,186,711	1
Art	246,000	103	14.85	36,526	1
Appliances	6,334,000	106	14.92	944,927	1
Storage	2,659,000	107	14.94	397,228	1
Paper	493,000	124	15.21	75,040	1

13 rows × 6 columns (first 12 shown)

Converting types: CAST, CONVERT, TRY_CAST

Function	Example	Notes
`CAST(x AS type)`	`CAST('2026-04-27' AS DATE)`	ANSI standard
`CONVERT(type, x, style)`	`CONVERT(VARCHAR(10), order_date, 103)` → 27/04/2026	SQL Server; style codes format dates (103 = dd/mm/yyyy, 23 = yyyy-mm-dd)
`TRY_CAST(x AS type)`	`TRY_CAST('abc' AS INT)` → NULL	Returns NULL instead of an error: great for dirty data
`TRY_CONVERT`, `TRY_PARSE`		Same idea

SQL

```sql
-- Find rows where a text column does not contain a valid number
SELECT * FROM staging_sales WHERE amount_text IS NOT NULL AND TRY_CAST(amount_text AS DECIMAL(12,2)) IS NULL;
```

Conditional functions (preview)

Function	Example
`COALESCE(a, b, …)`	`COALESCE(phone, email, 'no contact')`
`ISNULL(a, b)`	`ISNULL(discount, 0)`
`NULLIF(a, b)`	`profit / NULLIF(sales, 0)`
`IIF(cond, a, b)`	`IIF(profit < 0, 'Loss', 'Profit')`
`CASE WHEN … END`	Multi-branch logic (Lesson 7)

SQL

```sql
SELECT TOP 6 order_id, sales, profit,
       IIF(profit < 0, 'Loss', 'Profit')          AS result,
       ROUND(profit / NULLIF(sales, 0) * 100, 1)  AS margin_pct
FROM superstore
ORDER BY profit;
```

RESULT

order_id	sales	profit	result	margin_pct
IN-2025-03027	318,492	-75,874	Loss	-23.8
IN-2023-00320	337,316	-67,393	Loss	-20.0
IN-2023-00372	378,192	-59,796	Loss	-15.8
IN-2024-01335	194,160	-56,392	Loss	-29.0
IN-2025-02299	202,475	-46,375	Loss	-22.9
IN-2023-00077	240,940	-45,712	Loss	-19.0

6 rows × 5 columns

DO IT WITH AI

Do it with AI

Prompt: *"In SQL Server, write a query on superstore(order_id, order_date, ship_date, ship_mode) that returns the average and 90th percentile of days to ship by ship mode and quarter."* Expect it to use `DATEDIFF`, `DATEPART(quarter, …)` and `PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY …) OVER (PARTITION BY …)`, which you will meet with window functions in Lesson 12.

Summary

KEY TAKEAWAYS

- **Scalar** functions work per row; **aggregate** functions summarise many rows.
- `COUNT(*)` counts rows; `COUNT(col)` skips NULLs; `COUNT(DISTINCT col)` counts unique values; AVG ignores NULLs.
- Text: `LEN`, `LEFT/RIGHT/SUBSTRING`, `CHARINDEX`, `REPLACE`, `TRIM`, `UPPER/LOWER`, `CONCAT`, `STRING_AGG`, `STRING_SPLIT`.
- Dates: `YEAR/MONTH/DAY`, `DATEPART`, `DATENAME`, `DATEADD`, `DATEDIFF` (counts boundaries), `EOMONTH`, `DATEFROMPARTS`, `DATETRUNC`.
- Numbers: `ROUND` (negative digits round to tens/hundreds), `CEILING`, `FLOOR`, `ABS`, `POWER`, `SQRT`, `%`.
- Convert with `CAST/CONVERT`; use `TRY_CAST` on dirty data.

Quick quiz

6 questions · tap an answer to check it

1. A column has values 10, NULL, 20. What does `AVG(col)` return?

A 10

B 15

C 30

D NULL

Answer: B. AVG ignores NULLs: (10 + 20) / 2 = 15.

2. What does `SUBSTRING('IN-2025-00042', 4, 4)` return?

A IN-2

B 2025

C 0004

D 5-00

Answer: B. Starting at position 4, take 4 characters.

3. Which returns the number of days between order and ship dates?

A `DATEPART(day, ship_date - order_date)`

B `DATEDIFF(day, order_date, ship_date)`

C `DATEADD(day, order_date, ship_date)`

D `DAY(ship_date) - DAY(order_date)`

Answer: B. DATEDIFF(datepart, start, end) counts the boundaries between the two dates.

4. What does `ROUND(45678.9, -3)` return?

A 45678.000

B 45679

C 46000.0

D 45000

Answer: C. −3 rounds to the nearest thousand.

5. Which function returns NULL instead of an error when text cannot be converted?

A CAST

B CONVERT

C TRY_CAST

D ISNULL

Answer: C. TRY_CAST and TRY_CONVERT return NULL on conversion failure.

6. Which counts unique customers?

A `COUNT(customer_id)`

B `COUNT(DISTINCT customer_id)`

C `SUM(DISTINCT customer_id)`

D `DISTINCT COUNT(customer_id)`

Answer: B. COUNT(DISTINCT ...) counts distinct non-NULL values.

EXERCISE **Function practice on Superstore**

Write one query that returns, for each order year: the number of orders, the number of distinct customers, total sales rounded to the nearest lakh (100,000), the average days to ship, and a comma-separated list of the ship modes used.

Show solution (worked answer)

SQL

```sql
WITH modes AS (                                  -- one row per year and ship mode
    SELECT DISTINCT YEAR(order_date) AS order_year, ship_mode FROM orders
),
mode_list AS (
    SELECT order_year, STRING_AGG(ship_mode, ', ') AS ship_modes FROM modes GROUP BY
order_year
)
SELECT YEAR(o.order_date)                                  AS order_year,
       COUNT(DISTINCT o.order_id)                          AS orders,
       COUNT(DISTINCT o.customer_id)                       AS customers,
       ROUND(SUM(oi.sales), -5)                            AS sales_nearest_lakh,
       AVG(DATEDIFF(day, o.order_date, o.ship_date) * 1.0) AS avg_days_to_ship,
       MAX(m.ship_modes)                                   AS ship_modes
FROM orders AS o
JOIN order_items AS oi ON oi.order_id = o.order_id
JOIN mode_list   AS m  ON m.order_year = YEAR(o.order_date)
GROUP BY YEAR(o.order_date)
ORDER BY order_year;
```

`STRING_AGG(DISTINCT …)` is not allowed in SQL Server, so the distinct ship modes are collected in a CTE first (CTEs are covered in Lesson 13). Note that averaging days to ship over order *lines* weights orders with more items more heavily; average over `orders` alone if each order should count once.

INTERVIEW CORNER

Questions you may be asked

Q: COUNT() vs COUNT(column) vs COUNT(DISTINCT column)? COUNT() counts rows; COUNT(column) counts non-NULL values; COUNT(DISTINCT column) counts unique non-NULL values.

Q: How do you extract the domain from an email in SQL Server? `SUBSTRING(email, CHARINDEX('@', email) + 1, LEN(email))` or `RIGHT(email, LEN(email) - CHARINDEX('@', email))`.

Q: How do you get the last day of the previous month? `EOMONTH(GETDATE(), -1)`.

Q: Difference between CAST and CONVERT? CAST is ANSI standard; CONVERT is SQL Server-specific and accepts a style argument for date/number formatting.

SQL FOR DATA ANALYSIS · LESSON 7

GROUP BY, HAVING, CASE WHEN, ROLLUP, CUBE & COALESCE

Turn rows into business summaries. Group with GROUP BY, filter groups with HAVING, create conditional columns and buckets with CASE WHEN, pivot data with conditional aggregation, and add subtotals and grand totals with ROLLUP, CUBE and GROUPING SETS, cleaning the labels with COALESCE and GROUPING().

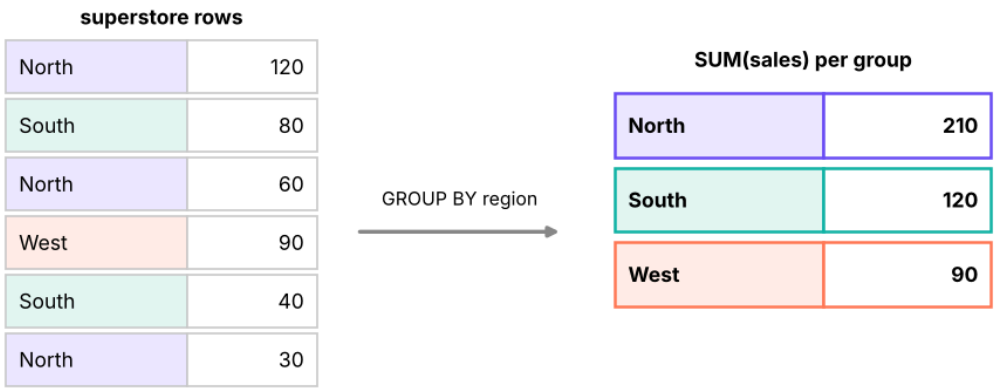
● SQL class 4 · SQL–Power BI bridge ✦ SQL Server 2022, 2026 Intermediate ⏱ 130 min

By the end of this lesson you can

- Summarise data by one or more columns with GROUP BY and aggregate functions
- Filter groups with HAVING and explain WHERE vs HAVING
- Create conditional columns and buckets with CASE WHEN ... THEN ... ELSE ... END
- Build pivot-style reports with conditional aggregation
- Add subtotals and grand totals with ROLLUP, CUBE and GROUPING SETS, and label them with COALESCE and GROUPING()

GROUP BY: one row per group

`GROUP BY` collapses rows that share the same value(s) into one row, and aggregate functions summarise each group. It is the SQL version of a pivot table (Excel Lesson 8) and of `groupby()` in pandas.



GROUP BY region: rows with the same region are collected into one group, then SUM is computed for each group.

SQL

```sql
SELECT region,
       COUNT(DISTINCT order_id)              AS orders,
       ROUND(SUM(sales), 0)                  AS sales,
       ROUND(SUM(profit), 0)                 AS profit,
       ROUND(100.0 * SUM(profit) / SUM(sales), 1) AS margin_pct
FROM superstore
GROUP BY region
ORDER BY sales DESC;
```

RESULT

region	orders	sales	profit	margin_pct
South	831	53,952,769	4,702,012	8.7
West	811	51,958,521	4,719,833	9.1
North	792	49,018,161	4,555,830	9.3
East	766	46,484,119	4,374,490	9.4

4 rows × 5 columns

Grouping by several columns

SQL

```sql
SELECT category,
       sub_category,
       COUNT(*)              AS lines,
       ROUND(SUM(sales), 0)  AS sales,
       ROUND(SUM(profit), 0) AS profit
FROM superstore
GROUP BY category, sub_category
ORDER BY category, sales DESC;
```

RESULT

category	sub_category	lines	sales	profit
Furniture	Tables	505	14,046,224	-816,042
Furniture	Bookcases	477	11,274,849	-422,786
Furniture	Chairs	493	8,091,474	342,043
Furniture	Furnishings	442	1,429,307	78,632
Office Supplies	Appliances	492	6,333,529	944,927
Office Supplies	Storage	507	2,659,340	397,228
Office Supplies	Paper	559	493,232	75,040
Office Supplies	Binders	494	401,219	61,074
Office Supplies	Art	462	245,985	36,526
Technology	Copiers	468	78,749,926	8,436,141
Technology	Machines	491	42,590,335	5,186,711
Technology	Phones	488	31,919,223	3,663,117

13 rows × 5 columns (first 12 shown)

Grouping by an expression

SQL

```sql
SELECT YEAR(order_date)       AS order_year,
       COUNT(DISTINCT order_id) AS orders,
       ROUND(SUM(sales), 0)   AS sales
FROM superstore
GROUP BY YEAR(order_date)
ORDER BY order_year;
```

RESULT

order_year	orders	sales
2023	883	57,640,217
2024	1085	65,396,556
2025	1232	78,376,797

3 rows × 3 columns

WATCH OUT **The golden rule of GROUP BY**

Every column in SELECT must either be **in the GROUP BY** or **inside an aggregate function**. `SELECT region, city, SUM(sales) FROM superstore GROUP BY region` fails in SQL Server: "Column 'city' is invalid in the select list because it is not contained in either an aggregate function or the GROUP BY clause." Which city should it show for the North group? SQL refuses to guess.

HAVING: filtering groups

`WHERE` filters **rows before** grouping; `HAVING` filters **groups after** aggregation.

SQL

```sql
SELECT city,
       COUNT(DISTINCT order_id) AS orders,
       ROUND(SUM(profit), 0)    AS profit
FROM superstore
WHERE category = 'Furniture'          -- 1. keep only furniture rows
GROUP BY city                         -- 2. one row per city
HAVING SUM(profit) < 0                -- 3. keep only loss-making cities
ORDER BY profit;
```

RESULT

city	orders	profit
Chennai	100	-122,258
Delhi	101	-106,170
Chandigarh	92	-92,165
Lucknow	100	-87,616
Guwahati	77	-80,036
Kochi	93	-71,758
Bengaluru	101	-52,961
Bhubaneswar	80	-50,956
Ahmedabad	96	-50,770
Kolkata	111	-39,811
Patna	86	-38,624
Indore	102	-30,482

15 rows × 3 columns (first 12 shown)

	WHERE	HAVING
Runs	Before GROUP BY	After GROUP BY
Filters	Individual rows	Groups
Can use aggregates	No	Yes
Example	`WHERE region = 'West'`	`HAVING COUNT(*) > 100`
Performance	Reduces rows early (faster)	Applied to fewer, summarised rows

TIP **Filter as early as possible**

Put conditions that do not need aggregates in `WHERE`, not `HAVING`: `HAVING region = 'West'` works but makes the database group every region first and discard most of them.

CASE WHEN: conditional columns

`CASE` is SQL's IF/IFS (Excel Lesson 5). It returns a value for the first condition that is true.

SQL

```sql
CASE
    WHEN condition1 THEN result1
    WHEN condition2 THEN result2
    ELSE default_result          -- optional; without ELSE, unmatched rows get NULL
END
```

SQL

```sql
SELECT order_id, sales, discount, profit,
       CASE
           WHEN discount = 0    THEN 'No discount'
           WHEN discount ⩽ 0.2 THEN 'Up to 20%'
           ELSE 'Above 20%'
       END AS discount_band,
       CASE WHEN profit < 0 THEN 'Loss' ELSE 'Profit' END AS result
FROM superstore
ORDER BY row_id;
```

RESULT

order_id	sales	discount	profit	discount_band	result
IN-2023-00001	36,000	0.0	3,472.55	No discount	Profit
IN-2023-00001	3,850	0.0	1,169.54	No discount	Profit
IN-2023-00001	62,080	0.0	6,044.47	No discount	Profit
IN-2023-00002	5,216	0.2	450.16	Up to 20%	Profit
IN-2023-00003	24,591	0.3	-5,512.33	Above 20%	Loss
IN-2023-00003	8,288	0.3	-1,598.43	Above 20%	Loss
IN-2023-00003	2,100	0.0	510.43	No discount	Profit
IN-2023-00003	69,280	0.0	15,852.15	No discount	Profit

6,323 rows × 6 columns (first 8 shown)

The **simple** form compares one expression with values:

```sql
CASE ship_mode
    WHEN 'Same Day'    THEN 1
    WHEN 'First Class' THEN 2
    ELSE 3
END
```

CASE inside GROUP BY: buckets

```sql
SELECT CASE
           WHEN discount = 0    THEN '0%'
           WHEN discount ⩽ 0.2 THEN '1-20%'
           ELSE '>20%'
       END                                      AS discount_band,
       COUNT(*)                                 AS lines,
       ROUND(SUM(sales), 0)                     AS sales,
       ROUND(SUM(profit), 0)                    AS profit,
       ROUND(100.0 * SUM(profit) / SUM(sales), 1) AS margin_pct
FROM superstore
GROUP BY CASE
             WHEN discount = 0    THEN '0%'
             WHEN discount ⩽ 0.2 THEN '1-20%'
             ELSE '>20%'
         END
ORDER BY discount_band;
```

RESULT

discount_band	lines	sales	profit	margin_pct
0%	3480	123,936,290	21,718,963	17.5
1–20%	1943	57,729,097	697,141	1.2
>20%	900	19,748,182	-4,063,940	-20.6

3 rows × 5 columns

The same insight as the Excel crosstab and dashboard: discounts above 20% turn the margin sharply negative.

Conditional aggregation: pivoting with CASE

Put `CASE` inside an aggregate to turn row values into columns, which is how analysts build pivot-style reports in SQL:

SQL

```sql
SELECT region,
       ROUND(SUM(CASE WHEN category = 'Furniture'       THEN sales ELSE 0 END), 0) AS
furniture,
       ROUND(SUM(CASE WHEN category = 'Office Supplies' THEN sales ELSE 0 END), 0) AS
office_supplies,
       ROUND(SUM(CASE WHEN category = 'Technology'      THEN sales ELSE 0 END), 0) AS
technology,
       ROUND(SUM(sales), 0)                                                       AS total
FROM superstore
GROUP BY region
ORDER BY region;
```

RESULT

region	furniture	office_supplies	technology	total
East	7,942,093	2,719,513	35,822,514	46,484,119
North	7,975,869	2,219,543	38,822,750	49,018,161
South	9,593,302	2,468,888	41,890,579	53,952,769
West	9,330,590	2,725,361	39,902,570	51,958,521

4 rows × 5 columns

SQL

```sql
SELECT department,
       COUNT(*)                                                AS employees,
       SUM(CASE WHEN attrition = 'Yes' THEN 1 ELSE 0 END)       AS leavers,
       ROUND(100.0 * SUM(CASE WHEN attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 1) AS
attrition_pct,
       ROUND(AVG(CASE WHEN overtime = 'Yes' THEN 1.0 ELSE 0 END) * 100, 1)             AS
overtime_pct
FROM employees
GROUP BY department
ORDER BY attrition_pct DESC;
```

RESULT

department	employees	leavers	attrition_pct	overtime_pct
Sales	414	126	30.4	26.6
R&D	389	101	26.0	32.1
IT	309	75	24.3	29.8
HR	83	19	22.9	31.3
Finance	154	35	22.7	20.1
Operations	151	33	21.9	32.5

6 rows × 5 columns

`AVG(CASE WHEN … THEN 1.0 ELSE 0 END)` is a neat way to get a percentage of rows meeting a condition.

NOTE **The PIVOT operator**

SQL Server also has a `PIVOT` operator:

```sql
SELECT region, [Furniture], [Office Supplies], [Technology]
FROM (SELECT region, category, sales FROM superstore) AS src
PIVOT (SUM(sales) FOR category IN ([Furniture], [Office Supplies], [Technology])) AS
p;
```

Conditional aggregation is more flexible (several measures, any conditions) and works in every database, so most analysts prefer it. `UNPIVOT` does the reverse, like Power Query's Unpivot (Excel Lesson 12).

Subtotals and grand totals: ROLLUP, CUBE, GROUPING SETS

ROLLUP

`GROUP BY ROLLUP (a, b)` returns the groups for `(a, b)`, **subtotals** for each a, and a **grand total**. Subtotal rows show NULL in the rolled-up columns.

SQL

```sql
SELECT region,
       category,
       ROUND(SUM(sales), 0) AS sales
FROM superstore
GROUP BY ROLLUP (region, category)
ORDER BY region, category;
```

RESULT

region	category	sales
East	Furniture	7,942,093
East	Office Supplies	2,719,513
East	Technology	35,822,514
East	NULL	46,484,119
North	Furniture	7,975,869
North	Office Supplies	2,219,543
North	Technology	38,822,750
North	NULL	49,018,161
South	Furniture	9,593,302
South	Office Supplies	2,468,888
South	Technology	41,890,579
South	NULL	53,952,769
West	Furniture	9,330,590
West	Office Supplies	2,725,361
West	Technology	39,902,570
West	NULL	51,958,521
NULL	NULL	201,413,569

17 rows × 3 columns

Cleaning the labels with COALESCE and GROUPING()

NULLs in subtotal rows are confusing. `COALESCE` replaces them with a label. If the data itself could contain NULL categories, use `GROUPING(col)`, which returns 1 only on rows where the column was rolled up:

SQL

```
SELECT CASE WHEN GROUPING(region) = 1 THEN 'All regions' ELSE region END            AS
region,
       CASE WHEN GROUPING(category) = 1 THEN 'Subtotal' ELSE category END           AS
category,
       ROUND(SUM(sales), 0)                                                         AS
sales
FROM superstore
GROUP BY ROLLUP (region, category)
ORDER BY GROUPING(region), region, GROUPING(category), category;
```

RESULT

region	category	sales
East	Furniture	7,942,093
East	Office Supplies	2,719,513
East	Technology	35,822,514
East	Subtotal	46,484,119
North	Furniture	7,975,869
North	Office Supplies	2,219,543
North	Technology	38,822,750
North	Subtotal	49,018,161
South	Furniture	9,593,302
South	Office Supplies	2,468,888
South	Technology	41,890,579
South	Subtotal	53,952,769
West	Furniture	9,330,590
West	Office Supplies	2,725,361
West	Technology	39,902,570
West	Subtotal	51,958,521
All regions	Subtotal	201,413,569

17 rows × 3 columns

The simple version used in class, `COALESCE(region, 'All regions')`, gives the same labels when the data has no real NULLs.

CUBE

`CUBE (a, b)` returns **every combination** of subtotals: by `(a, b)`, by `a`, by `b`, and the grand total. Use it when you need totals in both directions, like a pivot table with row and column grand totals.

SQL

```sql
SELECT COALESCE(segment, 'All segments')  AS segment,
       COALESCE(region, 'All regions')    AS region,
       COUNT(DISTINCT order_id)           AS orders
FROM superstore
GROUP BY CUBE (segment, region)
ORDER BY GROUPING(segment), segment, GROUPING(region), region;
```

RESULT

segment	region	orders
Consumer	East	355
Consumer	North	402
Consumer	South	385
Consumer	West	412
Consumer	All regions	1554
Corporate	East	245
Corporate	North	224
Corporate	South	269
Corporate	West	217
Corporate	All regions	955
Home Office	East	166
Home Office	North	166
Home Office	South	177
Home Office	West	182
Home Office	All regions	691
All segments	East	766
All segments	North	792
All segments	South	831
All segments	West	811
All segments	All regions	3200

20 rows × 3 columns

GROUPING SETS

`GROUPING SETS` lists exactly the groupings you want in one query:

```sql
SELECT COALESCE(category, '—')   AS category,
       COALESCE(ship_mode, '—')  AS ship_mode,
       ROUND(SUM(sales), 0)      AS sales
FROM superstore
GROUP BY GROUPING SETS ((category), (ship_mode), ())
ORDER BY category, ship_mode;
```

RESULT

category	ship_mode	sales
Furniture	—	34,841,853
Office Supplies	—	10,133,304
Technology	—	156,438,413
—	First Class	31,455,543
—	Same Day	10,082,430
—	Second Class	40,858,316
—	Standard Class	119,017,281
—	—	201,413,569

8 rows × 3 columns

This returns totals by category, totals by ship mode and the grand total, without the category × ship mode detail.

Syntax	Groupings produced for (a, b)
`GROUP BY a, b`	(a, b)
`ROLLUP (a, b)`	(a, b), (a), ()
`CUBE (a, b)`	(a, b), (a), (b), ()
`GROUPING SETS ((a), (b), ())`	exactly (a), (b), ()

TIP **Bridge to Power BI**

Power BI matrices compute these subtotals automatically, but ROLLUP/CUBE are perfect when you must deliver a fixed report table from SQL, feed a paginated report, or check that a dashboard's totals are right.

A complete analysis query

Question: *"For 2025, show each sub-category's sales, profit and margin, a performance label, and only sub-categories with at least ₹10 lakh sales, best margin first."*

SQL

```sql
SELECT sub_category,
       ROUND(SUM(sales), 0)                          AS sales,
       ROUND(SUM(profit), 0)                         AS profit,
       ROUND(100.0 * SUM(profit) / SUM(sales), 1)    AS margin_pct,
       CASE
           WHEN SUM(profit) < 0                       THEN 'Loss-making'
           WHEN 100.0 * SUM(profit) / SUM(sales) < 8  THEN 'Low margin'
           ELSE 'Healthy'
       END                                           AS status
FROM superstore
WHERE order_date ⩾ '2025-01-01' AND order_date < '2026-01-01'
GROUP BY sub_category
HAVING SUM(sales) ⩾ 1000000
ORDER BY margin_pct DESC;
```

RESULT

sub_category	sales	profit	margin_pct	status
Appliances	2,539,536	379,442	14.9	Healthy
Phones	12,935,910	1,708,007	13.2	Healthy
Machines	16,502,707	2,000,026	12.1	Healthy
Accessories	1,221,677	145,007	11.9	Healthy
Copiers	30,984,937	3,349,006	10.8	Healthy
Chairs	3,053,530	132,519	4.3	Low margin
Bookcases	3,918,269	-169,018	-4.3	Loss-making
Tables	5,297,064	-286,188	-5.4	Loss-making

8 rows × 5 columns

Summary

KEY TAKEAWAYS

- `GROUP BY` makes one row per group; every selected column must be grouped or aggregated.
- `WHERE` filters rows before grouping; `HAVING` filters groups after (can use aggregates).
- `CASE WHEN … THEN … ELSE … END` creates conditional columns and buckets; without ELSE, unmatched rows are NULL.
- Conditional aggregation (`SUM(CASE WHEN … THEN x ELSE 0 END)`) pivots rows into columns and computes percentages.
- `ROLLUP` adds hierarchical subtotals + grand total; `CUBE` adds all combinations; `GROUPING SETS` picks exactly the groupings you need.
- Label subtotal rows with `COALESCE` or, more robustly, `GROUPING()`.

Quick quiz

6 questions · tap an answer to check it

1. Why does `SELECT region, city, SUM(sales) FROM superstore GROUP BY region` fail?

A SUM needs a WHERE clause

B city is neither grouped nor aggregated

C region must be aggregated

D GROUP BY needs ORDER BY

Answer: B. Every selected column must appear in GROUP BY or in an aggregate.

2. Which clause keeps only customers with more than 10 orders?

A `WHERE COUNT(order_id) > 10`

B `HAVING COUNT(DISTINCT order_id) > 10`

C `ORDER BY COUNT(*) > 10`

D `GROUP BY COUNT(*) > 10`

Answer: B. Conditions on aggregates go in HAVING.

3. What does a CASE expression return when no WHEN matches and there is no ELSE?

A 0

B An empty string

C NULL

D An error

Answer: C. Without ELSE, unmatched rows return NULL.

4. `GROUP BY ROLLUP (region, category)` produces which groupings?

A (region, category), (region), and the grand total

B Every combination including (category) alone

C Only the grand total

D Only (region, category)

Answer: A. ROLLUP is hierarchical; CUBE would also include (category).

5. What does `AVG(CASE WHEN attrition = 'Yes' THEN 1.0 ELSE 0 END)` compute?

A The number of leavers

B The share (proportion) of employees who left

C The average salary of leavers

D NULL

Answer: B. The average of 1s and 0s is the proportion of 1s.

6. Which function tells you that a row is a subtotal created by ROLLUP?

A COALESCE

B ISNULL

C GROUPING

D NULLIF

Answer: C. GROUPING(col) returns 1 when col was aggregated away in that row.

EXERCISE

HR summary

Using the `employees` table (HR data): (1) headcount, average monthly income and attrition % by department and gender, only for groups with at least 30 employees; (2) a column per education level showing headcount, one row per department; (3) average income by job satisfaction (1–4) with a grand total row using ROLLUP.

Show solution (worked answer)

SQL

```sql
-- (1)
SELECT department, gender, COUNT(*) AS headcount,
       ROUND(AVG(monthly_income * 1.0), 0) AS avg_income,
       ROUND(100.0 * SUM(CASE WHEN attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 1)
AS attrition_pct
FROM employees GROUP BY department, gender HAVING COUNT(*) ⩾ 30 ORDER BY department,
gender;
-- (2)
SELECT department,
       SUM(CASE WHEN education = 'Diploma'    THEN 1 ELSE 0 END) AS diploma,
       SUM(CASE WHEN education = 'Bachelor''s' THEN 1 ELSE 0 END) AS bachelors,
       SUM(CASE WHEN education = 'Master''s'   THEN 1 ELSE 0 END) AS masters,
       SUM(CASE WHEN education = 'PhD'        THEN 1 ELSE 0 END) AS phd
FROM employees GROUP BY department;
-- (3)
SELECT COALESCE(CAST(job_satisfaction AS VARCHAR(5)), 'All') AS satisfaction,
       ROUND(AVG(monthly_income * 1.0), 0) AS avg_income
FROM employees GROUP BY ROLLUP (job_satisfaction) ORDER BY GROUPING(job_satisfaction),
job_satisfaction;
```

Note the doubled apostrophe in `'Bachelor''s'`: that is how you write a quote inside a SQL string.

INTERVIEW CORNER

Questions you may be asked

Q: WHERE vs HAVING? WHERE filters rows before aggregation and cannot use aggregate functions; HAVING filters groups after GROUP BY and can.

Q: How do you pivot rows to columns in SQL? Conditional aggregation `SUM(CASE WHEN col = 'X' THEN value END)` per output column, or SQL Server's PIVOT operator.

Q: What are ROLLUP and CUBE? GROUP BY extensions that add subtotal and grand-total rows: ROLLUP follows the column hierarchy; CUBE produces all combinations. GROUPING() identifies the subtotal rows.

Q: Write a query to find the second-highest salary per department. Use DENSE_RANK() in a subquery/CTE (Lesson 11): `WHERE rnk = 2`.

SQL FOR DATA ANALYSIS · LESSON 8

Joins and a Retail Case Study

Combine tables with INNER, LEFT, RIGHT, FULL OUTER, CROSS and SELF joins, find missing matches with anti-joins, avoid the classic join mistakes (row explosion, filters in ON vs WHERE), and answer a full set of business questions on the retail database.

SQL class 4 · SQL Server 2022, 2026 · Intermediate · 150 min

By the end of this lesson you can

- Explain each join type with a Venn picture and predict how many rows it returns
- Write INNER, LEFT, RIGHT, FULL OUTER, CROSS and SELF joins with table aliases
- Find unmatched rows with LEFT JOIN ... IS NULL (anti-join)
- Avoid row explosion, ambiguous columns, and the ON-vs-WHERE trap in outer joins
- Join three or four tables to answer business questions

Why joins?

In a normalised database (Lesson 2) customer details live in `customers`, products in `products`, and orders in `orders` and `order_items`. To answer "sales by customer segment and product category" you must **join** them back together on their keys. A join is SQL's version of XLOOKUP (Excel Lesson 7) and pandas `merge()`.

A tiny example first

Two small tables make every join type easy to see:

SQL

```
CREATE TABLE students (student_id INT PRIMARY KEY, name VARCHAR(20));
INSERT INTO students VALUES (1, 'Asha'), (2, 'Ravi'), (3, 'Meera'), (4, 'Kabir');

CREATE TABLE enrolments (student_id INT, course VARCHAR(20));
INSERT INTO enrolments VALUES (1, 'SQL'), (1, 'Python'), (2, 'Excel'), (5, 'Power BI');

SELECT 'students' AS t, COUNT(*) AS n FROM students UNION ALL SELECT 'enrolments', COUNT(*)
FROM enrolments;
```

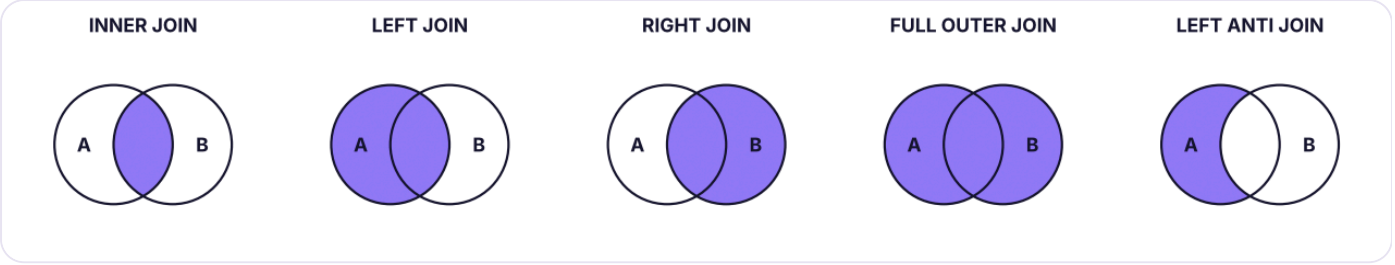
RESULT

```
Commands completed successfully.
(4 rows affected)
Commands completed successfully.
(4 rows affected)
```

t	n
students	4
enrolments	4

2 rows × 2 columns

Asha (1) has two enrolments, Ravi (2) has one, Meera (3) and Kabir (4) have none, and enrolment for student 5 has no matching student (an "orphan" row).



The join types as Venn diagrams. Shaded areas are the rows returned (A = left table, B = right table).

INNER JOIN: only matches

SQL

```
SELECT s.student_id, s.name, e.course
FROM students AS s
INNER JOIN enrolments AS e
        ON e.student_id = s.student_id
ORDER BY s.student_id, e.course;
```

RESULT

student_id	name	course
1	Asha	Python
1	Asha	SQL
2	Ravi	Excel

3 rows × 3 columns

Asha appears **twice** because she has two matching enrolments. Meera, Kabir and the orphan enrolment disappear.

LEFT (OUTER) JOIN: everything from the left

SQL

```sql
SELECT s.student_id, s.name, e.course
FROM students AS s
LEFT JOIN enrolments AS e
       ON e.student_id = s.student_id
ORDER BY s.student_id, e.course;
```

RESULT

student_id	name	course
1	Asha	Python
1	Asha	SQL
2	Ravi	Excel
3	Meera	NULL
4	Kabir	NULL

5 rows × 3 columns

Every student is kept; where there is no match, the right-hand columns are **NULL**.

RIGHT JOIN: everything from the right

SQL

```sql
SELECT e.student_id, s.name, e.course
FROM students AS s
RIGHT JOIN enrolments AS e
        ON e.student_id = s.student_id
ORDER BY e.student_id;
```

RESULT

student_id	name	course
1	Asha	SQL
1	Asha	Python
2	Ravi	Excel
5	NULL	Power BI

4 rows × 3 columns

Every enrolment is kept, including student 5 who has no name. A RIGHT JOIN is just a LEFT JOIN with the tables swapped; most people write LEFT JOINs for consistency.

FULL OUTER JOIN: everything from both

SQL

```sql
SELECT s.student_id AS s_id, s.name, e.student_id AS e_id, e.course
FROM students AS s
FULL OUTER JOIN enrolments AS e
            ON e.student_id = s.student_id
ORDER BY COALESCE(s.student_id, e.student_id), e.course;
```

RESULT

s_id	name	e_id	course
1	Asha	1	Python
1	Asha	1	SQL
2	Ravi	2	Excel
3	Meera	NULL	NULL
4	Kabir	NULL	NULL
NULL	NULL	5	Power BI

6 rows × 4 columns

Matched rows, plus unmatched students, plus unmatched enrolments. Useful for **reconciliation**: comparing two systems' lists to see what is missing on each side.

Anti-join: rows with no match

SQL

```sql
SELECT s.student_id, s.name
FROM students AS s
LEFT JOIN enrolments AS e ON e.student_id = s.student_id
WHERE e.student_id IS NULL;
```

RESULT

student_id	name
3	Meera
4	Kabir

2 rows × 2 columns

"Students not enrolled in anything." The same result with `NOT EXISTS` is shown in Lesson 10.

CROSS JOIN: every combination

SQL

```
CREATE TABLE sizes (size VARCHAR(5));
INSERT INTO sizes VALUES ('S'), ('M'), ('L');
CREATE TABLE colours (colour VARCHAR(10));
INSERT INTO colours VALUES ('Black'), ('White');

SELECT c.colour, z.size
FROM colours AS c
CROSS JOIN sizes AS z
ORDER BY c.colour, z.size;
```

RESULT

```
Commands completed successfully.
(3 rows affected)
Commands completed successfully.
(2 rows affected)
```

colour	size
Black	L
Black	M
Black	S
White	L
White	M
White	S

6 rows × 2 columns

2 colours × 3 sizes = 6 rows: a **Cartesian product**. Use it on purpose (all product–month combinations for a report grid, a calendar × store table); never by accident.

SELF JOIN: a table joined to itself

SQL

```
CREATE TABLE team (emp_id INT, name VARCHAR(20), manager_id INT);
INSERT INTO team VALUES (1, 'Varun', NULL), (2, 'Asha', 1), (3, 'Ravi', 1), (4, 'Meera',
2), (5, 'Kabir', 2);

SELECT e.name AS employee, m.name AS manager
FROM team AS e
LEFT JOIN team AS m ON m.emp_id = e.manager_id
ORDER BY e.emp_id;
```

RESULT

```
Commands completed successfully.
(5 rows affected)
```

employee	manager
Varun	NULL
Asha	Varun
Ravi	Varun
Meera	Asha
Kabir	Asha

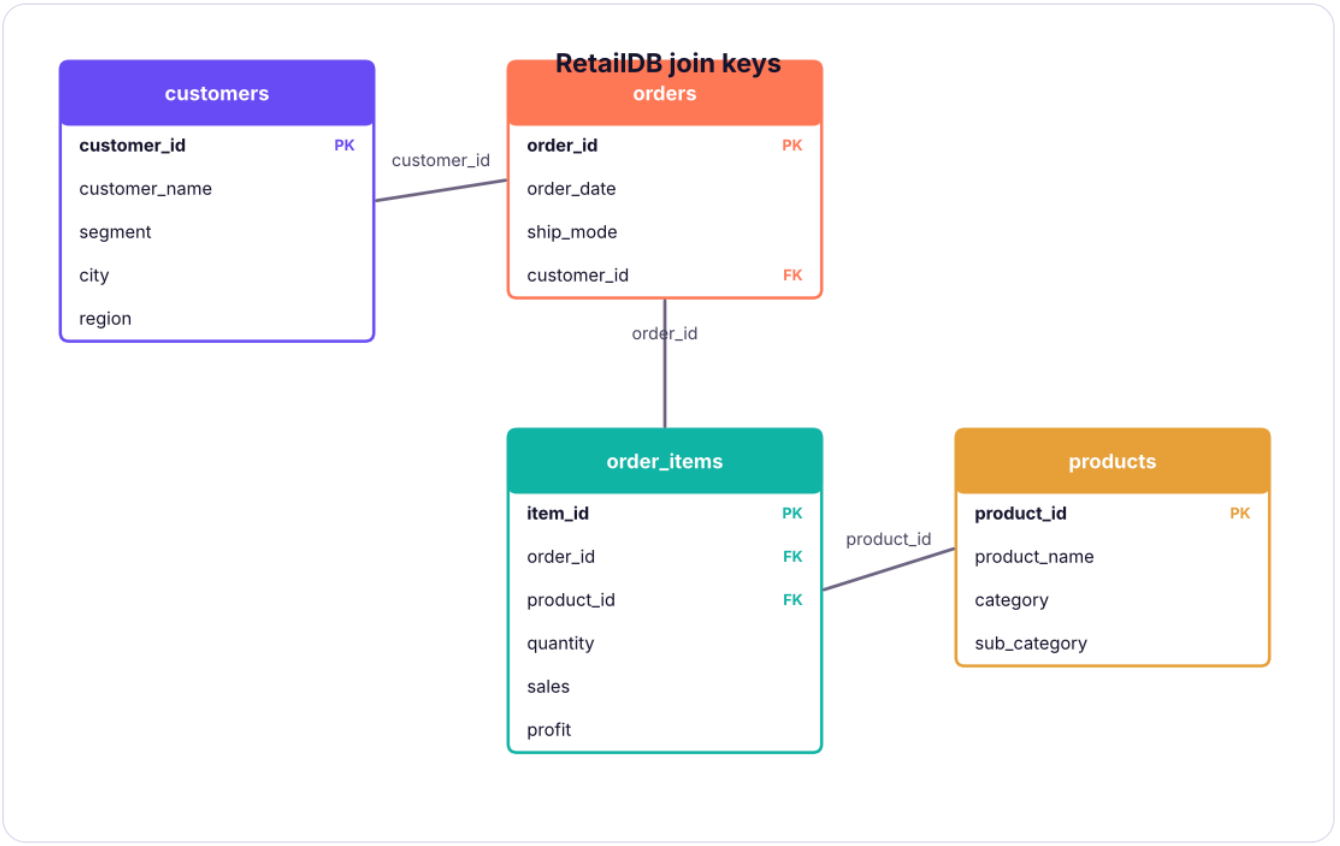
5 rows × 2 columns

The same table appears twice with different aliases (e for employee, `m` for manager). LEFT JOIN keeps Varun, who has no manager.

How many rows will a join return?

Join	Rows returned
INNER	Sum over matches; a key matching m rows on the left and n on the right produces m × n rows
LEFT	Every left row at least once (more if it matches several right rows)
RIGHT	Every right row at least once
FULL	Matches + unmatched left + unmatched right
CROSS	Left rows × right rows

Joins on the retail database



Join paths in RetailDB. Follow the keys: customers → orders → order_items → products.

Two tables

SQL

```sql
SELECT o.order_id, o.order_date, c.customer_name, c.segment, c.city
FROM orders AS o
INNER JOIN customers AS c ON c.customer_id = o.customer_id
ORDER BY o.order_date DESC, o.order_id;
```

RESULT

order_id	order_date	customer_name	segment	city
IN-2025-03197	2025-12-31	Vivaan Patel	Home Office	Bengaluru
IN-2025-03198	2025-12-31	Neha Fernandes	Consumer	Bhubaneswar
IN-2025-03199	2025-12-31	Aarav Patel	Home Office	Jaipur
IN-2025-03200	2025-12-31	Kabir Rao	Home Office	Pune
IN-2025-03192	2025-12-30	Ananya Bose	Consumer	Bengaluru
IN-2025-03193	2025-12-30	Ananya Nair	Consumer	Ahmedabad
IN-2025-03194	2025-12-30	Krishna Iyer	Consumer	Bengaluru
IN-2025-03195	2025-12-30	Ishaan Shah	Corporate	Bhubaneswar

3,200 rows × 5 columns (first 8 shown)

Four tables

SQL

```sql
SELECT c.segment,
       p.category,
       COUNT(DISTINCT o.order_id)  AS orders,
       ROUND(SUM(oi.sales), 0)     AS sales,
       ROUND(SUM(oi.profit), 0)    AS profit
FROM order_items AS oi
JOIN orders      AS o ON o.order_id    = oi.order_id
JOIN customers   AS c ON c.customer_id = o.customer_id
JOIN products    AS p ON p.product_id  = oi.product_id
GROUP BY c.segment, p.category
ORDER BY c.segment, sales DESC;
```

RESULT

segment	category	orders	sales	profit
Consumer	Technology	727	76,891,232	9,074,795
Consumer	Furniture	749	17,031,585	-279,832
Consumer	Office Supplies	909	4,680,475	704,297
Corporate	Technology	452	45,607,235	5,223,370
Corporate	Furniture	443	10,323,095	-285,209
Corporate	Office Supplies	570	3,183,289	488,994
Home Office	Technology	339	33,939,946	3,357,358
Home Office	Furniture	319	7,487,173	-253,113
Home Office	Office Supplies	391	2,269,540	321,503

9 rows × 5 columns

`JOIN` on its own means `INNER JOIN`. Start from the table at the most detailed level (the **grain**: here, one row per order item) and join outward to the lookup tables.

Case study: questions for the Head of Sales

CASE STUDY **Q1. Which customers have never placed an order?**

Marketing wants to send them a welcome-back offer.

SQL

```sql
SELECT c.customer_id, c.customer_name, c.city, c.segment
FROM customers AS c
LEFT JOIN orders AS o ON o.customer_id = c.customer_id
WHERE o.order_id IS NULL
ORDER BY c.customer_id;
```

RESULT

customer_id	customer_name	city	segment
C0057	Meera Patel	Bhubaneswar	Consumer
C0113	Sneha Shah	Guwahati	Home Office
C0153	Nikhil Das	Delhi	Consumer
C0316	Joseph Mehta	Patna	Corporate
C0376	Sneha Kulkarni	Kolkata	Consumer
C0401	Neha Kulkarni	Kolkata	Corporate
C0404	Nikhil Fernandes	Lucknow	Corporate
C0507	Ananya Singh	Hyderabad	Home Office
C0524	Sai Mehta	Pune	Consumer
C0539	Harpreet Sharma	Jaipur	Consumer
C0704	Zoya Patel	Mumbai	Consumer
C0711	Vikram Reddy	Bhubaneswar	Consumer

15 rows × 4 columns (first 12 shown)

CASE STUDY

Q2. Which products have never been sold?

SQL

```sql
SELECT COUNT(*) AS products_never_sold
FROM products AS p
LEFT JOIN order_items AS oi ON oi.product_id = p.product_id
WHERE oi.item_id IS NULL;
```

RESULT

products_never_sold
0

1 rows × 1 columns

CASE STUDY

Q3. Top 10 customers by 2025 sales, with their city and segment

SQL

```sql
SELECT TOP 10
       c.customer_name, c.city, c.segment,
       COUNT(DISTINCT o.order_id) AS orders_2025,
       ROUND(SUM(oi.sales), 0)    AS sales_2025
FROM customers AS c
JOIN orders AS o       ON o.customer_id = c.customer_id
JOIN order_items AS oi ON oi.order_id   = o.order_id
WHERE o.order_date ≥ '2025-01-01' AND o.order_date < '2026-01-01'
GROUP BY c.customer_id, c.customer_name, c.city, c.segment
ORDER BY sales_2025 DESC;
```

RESULT

customer_name	city	segment	orders_2025	sales_2025
Joseph Shah	Pune	Consumer	5	872,896
Aarav Mehta	Bengaluru	Consumer	6	835,933
Kavya Iyer	Delhi	Consumer	3	821,696
Vivaan Verma	Mumbai	Home Office	2	757,989
Sneha Khan	Ahmedabad	Consumer	3	756,586
Pooja Nair	Pune	Consumer	4	700,790
Farhan Khan	Kochi	Corporate	2	648,834
Ananya Iyer	Chennai	Consumer	4	636,502
Meera Fernandes	Chennai	Consumer	4	614,306
Neha Fernandes	Pune	Consumer	2	588,041

10 rows × 5 columns

CASE STUDY

Q4. Which region × category combinations lose money on discounted Furniture?

SQL

```sql
SELECT c.region,
       p.sub_category,
       COUNT(*)                    AS lines,
       ROUND(AVG(oi.discount), 2)  AS avg_discount,
       ROUND(SUM(oi.profit), 0)    AS profit
FROM order_items AS oi
JOIN orders    AS o ON o.order_id    = oi.order_id
JOIN customers AS c ON c.customer_id = o.customer_id
JOIN products  AS p ON p.product_id  = oi.product_id
WHERE p.category = 'Furniture' AND oi.discount > 0
GROUP BY c.region, p.sub_category
HAVING SUM(oi.profit) < 0
ORDER BY profit;
```

RESULT

region	sub_category	lines	avg_discount	profit
South	Tables	100	0.25	-407,431
West	Tables	90	0.26	-369,074
East	Tables	73	0.27	-318,641
North	Tables	74	0.28	-301,656
North	Bookcases	82	0.28	-283,651
South	Bookcases	81	0.27	-263,943
East	Bookcases	80	0.27	-248,275
West	Bookcases	80	0.26	-216,090
North	Chairs	53	0.21	-76,380
West	Chairs	60	0.19	-71,006
South	Chairs	58	0.19	-60,298
East	Chairs	46	0.17	-31,667

16 rows × 5 columns (first 12 shown)

CASE STUDY **Q5. How does actual selling price compare with the list price?**

SQL

```sql
SELECT p.category,
       ROUND(SUM(oi.sales), 0)                                    AS actual_sales,
       ROUND(SUM(oi.quantity * p.list_price), 0)                  AS list_value,
       ROUND(100.0 * (1 - SUM(oi.sales) / SUM(oi.quantity * p.list_price)), 1) AS
avg_discount_given_pct
FROM order_items AS oi
JOIN products AS p ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY avg_discount_given_pct DESC;
```

RESULT

category	actual_sales	list_value	avg_discount_given_pct
Furniture	34,841,853	41,094,510	15.2
Office Supplies	10,133,304	10,963,700	7.6
Technology	156,438,413	168,811,900	7.3

3 rows × 4 columns

Common join mistakes

1. Row explosion (fan-out)

Joining two "many" tables on a shared key multiplies rows. If you join `order_items` to another table with several rows per order (for example a `payments` table with instalments), each item is repeated for each payment and `SUM(sales)` is inflated. Always ask: *what is one row of each table, and is the key unique on one side?* Check with `COUNT(*)` before and after the join, or aggregate each table to the same grain first (Lesson 10).

2. Ambiguous column names

`SELECT order_id FROM orders JOIN order_items ON …` fails: "Ambiguous column name 'order_id'". Prefix columns with the table alias (`o.order_id`). Do it for every column in multi-table queries.

3. Filtering an outer join in WHERE

A condition on the **right** table placed in `WHERE` removes the NULL rows and silently turns a LEFT JOIN into an INNER JOIN:

SQL

```sql
-- Intended: all students, with their SQL course if any
SELECT s.name, e.course
FROM students AS s
LEFT JOIN enrolments AS e ON e.student_id = s.student_id
WHERE e.course = 'SQL';            -- wrong: drops students without SQL
```

RESULT

name	course
Asha	SQL

1 rows × 2 columns

SQL

```sql
SELECT s.name, e.course
FROM students AS s
LEFT JOIN enrolments AS e
       ON e.student_id = s.student_id
      AND e.course = 'SQL'         -- right: condition in ON keeps every student
ORDER BY s.student_id;
```

RESULT

name	course
Asha	SQL
Ravi	NULL
Meera	NULL
Kabir	NULL

4 rows × 2 columns

4. Joining on the wrong or non-unique key

Joining customers to orders on `city` instead of `customer_id` creates nonsense matches. Join on keys, and check the relationship (1:1, 1:N) you expect.

5. NULL keys never match

`NULL = NULL` is UNKNOWN, so rows with a NULL key never match in any join. Clean or default keys first if that matters.

> **TIP** **Join check-list**
>
> 1. What is the grain (one row = ?) of each table? 2. Which key connects them, and is it unique on one side? 3. Do I need unmatched rows (LEFT/FULL) or only matches (INNER)? 4. Did the row count change as expected? 5. Are filters on outer-joined tables in ON or WHERE on purpose?

Older join syntax you may see

SQL

```sql
-- Implicit join (comma join), pre-1992 style: works but avoid
SELECT o.order_id, c.customer_name
FROM orders o, customers c
WHERE o.customer_id = c.customer_id;
```

Forgetting the WHERE turns it into a CROSS JOIN. Use explicit `JOIN … ON`.

Summary

KEY TAKEAWAYS

- **INNER**: only matching rows. **LEFT**: all left rows + matches (NULLs where none). **RIGHT**: the mirror. **FULL**: everything. **CROSS**: every combination. **SELF**: a table with itself (aliases).
- **Anti-join**: `LEFT JOIN … WHERE right.key IS NULL` finds rows with no match.
- Join from the most detailed table outward, on keys, with aliases on every column.
- Watch for row explosion, ambiguous columns, and filters on the right table in WHERE (move them to ON).
- Joins = XLOOKUP / Power Query Merge / pandas merge / Power BI relationships.

Quick quiz

6 questions · tap an answer to check it

1. Table A has 4 rows, B has 3. A CROSS JOIN returns:

- **A** 3 rows
- **B** 4 rows
- **C** 7 rows
- **D** 12 rows

Answer: D. A Cartesian product returns 4 × 3 rows.

2. Which join lists every customer, including those with no orders?

- **A** `customers INNER JOIN orders`
- **B** `customers LEFT JOIN orders`
- **C** `customers RIGHT JOIN orders` with orders on the right
- **D** `customers CROSS JOIN orders`

Answer: B. LEFT JOIN keeps every row from the left table.

3. How do you find products that were never ordered?

- **A** `products INNER JOIN order_items WHERE item_id IS NULL`
- **B** `products LEFT JOIN order_items … WHERE order_items.item_id IS NULL`
- **C** `products FULL JOIN order_items`
- **D** `SELECT DISTINCT product_id FROM order_items`

Answer: B. A left anti-join returns left rows with no match.

4. `FROM students s LEFT JOIN enrolments e ON … WHERE e.course = 'SQL'` **behaves like:**

- **A** A LEFT JOIN
- **B** An INNER JOIN, because rows with NULL course are removed
- **C** A FULL JOIN
- **D** A CROSS JOIN

Answer: B. The WHERE condition on the right table discards the unmatched (NULL) rows.

5. What is a self join used for?

- **A** Relating rows of the same table, such as employees and their managers
- **B** Joining a table to a copy in another database
- **C** Removing duplicates
- **D** Joining without a key

Answer: A. The same table is used twice with different aliases.

6. After joining order_items to a payments table with 2 instalments per order, SUM(sales) doubled. What happened?

- **A** A NULL key
- **B** Row explosion: each item was repeated once per payment row
- **C** Integer division
- **D** A wrong data type

Answer: B. Joining two many-sided tables multiplies rows; aggregate to the same grain first.

EXERCISE **Join practice**

(1) List each order with customer name, city and the number of items in the order. (2) For each customer segment, show the share of sales from each category (%). (3) Find customers who bought Technology but never Furniture. (4) Using `team`, show each manager with the number of direct reports.

Show solution (worked answer)

SQL

```
-- (1)
SELECT o.order_id, c.customer_name, c.city, COUNT(oi.item_id) AS items
FROM orders o JOIN customers c ON c.customer_id = o.customer_id
JOIN order_items oi ON oi.order_id = o.order_id
GROUP BY o.order_id, c.customer_name, c.city;
-- (2)
SELECT c.segment, p.category,
       ROUND(100.0 * SUM(oi.sales) / SUM(SUM(oi.sales)) OVER (PARTITION BY c.segment),
1) AS pct_of_segment
FROM order_items oi JOIN orders o ON o.order_id = oi.order_id
JOIN customers c ON c.customer_id = o.customer_id JOIN products p ON p.product_id =
oi.product_id
GROUP BY c.segment, p.category;            -- the window function is explained in
Lesson 12
-- (3)
SELECT DISTINCT c.customer_id, c.customer_name
FROM customers c JOIN orders o ON o.customer_id = c.customer_id
JOIN order_items oi ON oi.order_id = o.order_id JOIN products p ON p.product_id =
oi.product_id
WHERE p.category = 'Technology'
  AND c.customer_id NOT IN (SELECT o2.customer_id FROM orders o2 JOIN order_items oi2
ON oi2.order_id = o2.order_id
                          JOIN products p2 ON p2.product_id = oi2.product_id WHERE
p2.category = 'Furniture');
-- (4)
SELECT m.name AS manager, COUNT(e.emp_id) AS direct_reports
FROM team m LEFT JOIN team e ON e.manager_id = m.emp_id
GROUP BY m.emp_id, m.name HAVING COUNT(e.emp_id) > 0;
```

INTERVIEW CORNER

Questions you may be asked

Q: Explain the different types of joins. INNER returns matches; LEFT/RIGHT keep all rows from one side with NULLs where unmatched; FULL keeps all rows from both; CROSS returns every combination; SELF joins a table to itself.

Q: Table A has 5 rows with id 1 repeated twice, B has 3 rows with id 1 repeated three times. How many rows does an inner join on id return (other ids unmatched)? 2 × 3 = 6 rows.

Q: How do you find records in one table but not another? LEFT JOIN ... WHERE right.key IS NULL, NOT EXISTS, or EXCEPT (Lesson 9).

Q: What is the difference between putting a condition in ON vs WHERE for a LEFT JOIN? In ON it only limits which right rows match (all left rows stay); in WHERE it filters the final result and removes unmatched left rows.

SQL FOR DATA ANALYSIS · LESSON 9

Set Operators: UNION, UNION ALL, INTERSECT & EXCEPT

Stack and compare query results. Append data with UNION ALL, remove duplicates with UNION, find common rows with INTERSECT and differences with EXCEPT, follow the rules for compatible columns, and know when to use a set operator instead of a join.

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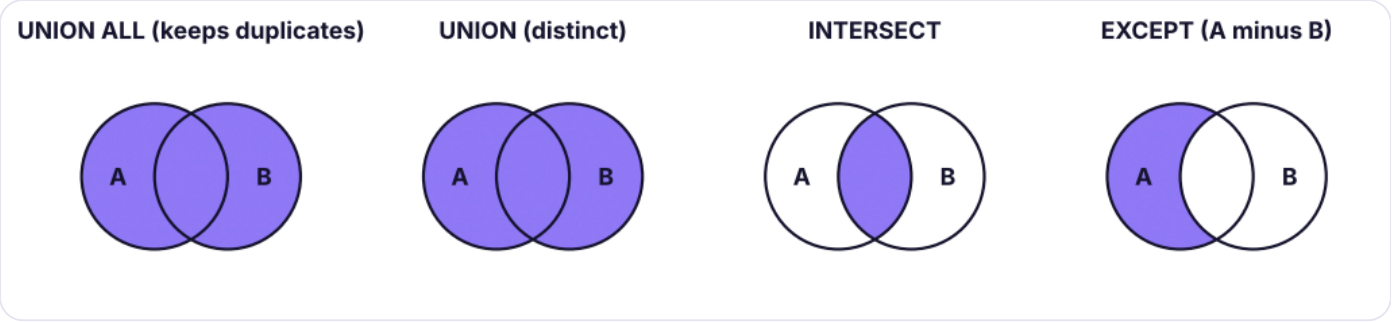
By the end of this lesson you can

- Explain the difference between joins (add columns) and set operators (add or compare rows)
- Append results with UNION ALL and de-duplicate with UNION
- Find common rows with INTERSECT and missing rows with EXCEPT
- Apply the rules: same number of columns, compatible types, names from the first query, one ORDER BY at the end
- Use set operators for reconciliation and data-quality checks

Joins add columns; set operators add rows

A join puts tables side by side (more columns). A set operator stacks or compares the rows of two query results with the same columns.



The four set operators. A and B are the result sets of two queries.

Rules for set operators

1. Both queries must return the **same number of columns**.
2. Corresponding columns must have **compatible data types** (text with text, numbers with numbers).
3. Column **names come from the first query**.
4. Only **one ORDER BY**, at the very end, applies to the combined result.
5. `UNION`, `INTERSECT` and `EXCEPT` remove duplicates; `UNION ALL` does not.

UNION ALL: append

`UNION ALL` stacks results exactly as they are, like Power Query's Append or pasting one table under another. It is the fastest set operator because it does not check for duplicates.

SQL

```sql
SELECT 'Furniture' AS source, COUNT(*) AS lines, ROUND(SUM(sales), 0) AS sales FROM
superstore WHERE category = 'Furniture'
UNION ALL
SELECT 'Technology', COUNT(*), ROUND(SUM(sales), 0) FROM superstore WHERE category =
'Technology'
UNION ALL
SELECT 'Office Supplies', COUNT(*), ROUND(SUM(sales), 0) FROM superstore WHERE category =
'Office Supplies'
ORDER BY sales DESC;
```

RESULT

source	lines	sales
Technology	1892	156,438,413
Furniture	1917	34,841,853
Office Supplies	2514	10,133,304

3 rows × 3 columns

A very common real use: combining monthly or regional tables with the same structure.

SQL

```sql
SELECT * FROM sales_jan_2026
UNION ALL
SELECT * FROM sales_feb_2026
UNION ALL
SELECT * FROM sales_mar_2026;
```

UNION: append and remove duplicates

SQL

```sql
SELECT COUNT(*) AS cities_union_all
FROM (SELECT city FROM customers WHERE segment = 'Corporate'
      UNION ALL
      SELECT city FROM customers WHERE segment = 'Home Office') AS t;
```

RESULT

cities_union_all
407

1 rows × 1 columns

SQL

```
SELECT COUNT(*) AS cities_union
FROM (SELECT city FROM customers WHERE segment = 'Corporate'
      UNION
      SELECT city FROM customers WHERE segment = 'Home Office') AS t;
```

RESULT

cities_union
16

1 rows × 1 columns

`UNION ALL` returned one row per customer; `UNION` returned each city once.

> **TIP** **UNION or UNION ALL?**
>
> Use `UNION ALL` unless you **need** duplicates removed. `UNION` has to sort or hash the whole result to find duplicates, which is slow on large data, and it can silently remove rows that are genuinely repeated (two identical order lines).

INTERSECT: rows in both

"Which customers ordered in **both** 2024 and 2025?" (retained customers)

SQL

```
SELECT COUNT(*) AS customers_in_both_years
FROM (
    SELECT customer_id FROM orders WHERE YEAR(order_date) = 2024
    INTERSECT
    SELECT customer_id FROM orders WHERE YEAR(order_date) = 2025
) AS t;
```

RESULT

customers_in_both_years
467

1 rows × 1 columns

EXCEPT: rows in the first but not the second

"Which 2024 customers did **not** come back in 2025?" (churned customers)

SQL

```sql
SELECT COUNT(*) AS customers_lost_after_2024
FROM (
    SELECT customer_id FROM orders WHERE YEAR(order_date) = 2024
    EXCEPT
    SELECT customer_id FROM orders WHERE YEAR(order_date) = 2025
) AS t;
```

RESULT

customers_lost_after_2024
123

1 rows × 1 columns

"Which 2025 customers are **new** (did not order in 2024 or 2023)?"

SQL

```sql
SELECT COUNT(*) AS new_customers_2025
FROM (
    SELECT customer_id FROM orders WHERE YEAR(order_date) = 2025
    EXCEPT
    SELECT customer_id FROM orders WHERE YEAR(order_date) < 2025
) AS t;
```

RESULT

new_customers_2025
52

1 rows × 1 columns

Order matters for EXCEPT: `A EXCEPT B` ≠ `B EXCEPT A`. Oracle calls it `MINUS`.

Let's put the customer movement together in one table:

SQL

```
SELECT 'Ordered in 2024' AS measure, COUNT(DISTINCT customer_id) AS customers FROM orders
WHERE YEAR(order_date) = 2024
UNION ALL
SELECT 'Ordered in 2025', COUNT(DISTINCT customer_id) FROM orders WHERE YEAR(order_date) =
2025
UNION ALL
SELECT 'Retained (2024 and 2025)', COUNT(*) FROM (SELECT customer_id FROM orders WHERE
YEAR(order_date) = 2024 INTERSECT SELECT customer_id FROM orders WHERE YEAR(order_date) =
2025) AS r
UNION ALL
SELECT 'Lost (2024 only)', COUNT(*) FROM (SELECT customer_id FROM orders WHERE
YEAR(order_date) = 2024 EXCEPT SELECT customer_id FROM orders WHERE YEAR(order_date) =
2025) AS l;
```

RESULT

measure	customers
Ordered in 2024	590
Ordered in 2025	623
Retained (2024 and 2025)	467
Lost (2024 only)	123

4 rows × 2 columns

Retention rate 2024 → 2025 = retained ÷ customers in 2024. This is the core of customer-retention analysis.

Reconciliation and data-quality checks

Set operators are perfect for comparing two versions of the same data, for example a source system extract and the warehouse table:

SQL

```sql
-- Rows in the source extract that are missing or different in the warehouse
SELECT order_id, order_date, customer_id, total FROM staging.orders_extract
EXCEPT
SELECT order_id, order_date, customer_id, total FROM dw.fact_orders;

-- And the reverse: rows in the warehouse not in the source
SELECT order_id, order_date, customer_id, total FROM dw.fact_orders
EXCEPT
SELECT order_id, order_date, customer_id, total FROM staging.orders_extract;
```

If both queries return no rows, the two tables match exactly. Because set operators compare **whole rows**, any difference in any column shows up. (Set operators treat two NULLs as equal, unlike = in a join.)

Set operator or join?

Question	Use
Add columns from another table	JOIN
Stack tables with the same columns	UNION ALL
Distinct list from several sources	UNION
Which keys are in both lists?	INTERSECT, or INNER JOIN / EXISTS
Which keys are in A but not B?	EXCEPT, or LEFT JOIN … IS NULL / NOT EXISTS
Compare entire rows column by column	EXCEPT (handles NULLs as equal)

Summary

KEY TAKEAWAYS

- Joins add **columns**; set operators add or compare **rows**.
- Rules: same number of columns, compatible types, names from the first query, one ORDER BY at the end.
- `UNION ALL` appends everything (fast); `UNION` removes duplicates.
- `INTERSECT` returns rows in both results; `EXCEPT` returns rows in the first but not the second (order matters).
- Use them for combining periods, retention/churn analysis and reconciling systems.

Quick quiz

6 questions · tap an answer to check it

1. Which operator keeps duplicate rows?

- **A** UNION ALL
- **B** UNION
- **C** INTERSECT
- **D** EXCEPT

Answer: A. UNION ALL does not remove duplicates; the others return distinct rows.

2. Where can ORDER BY appear in a UNION query?

- **A** After each SELECT
- **B** Once, at the end of the whole statement
- **C** Only in the first SELECT
- **D** ORDER BY is not allowed

Answer: B. One ORDER BY applies to the combined result.

3. `SELECT a, b FROM t1 UNION SELECT c FROM t2` fails because:

- **A** UNION needs WHERE
- **B** The queries return different numbers of columns
- **C** Column names differ
- **D** t2 must be first

Answer: B. Both queries must return the same number of columns.

4. To find customers who ordered in 2024 but not in 2025, use:

A 2024 UNION 2025

B 2024 INTERSECT 2025

C 2024 EXCEPT 2025

D 2025 EXCEPT 2024

Answer: C. EXCEPT returns rows in the first set that are not in the second.

5. Which column names appear in the result of a UNION?

A Those of the first query

B Those of the last query

C Both, combined

D Generic names like col1

Answer: A. Names (and aliases) come from the first SELECT.

6. Why is UNION ALL usually faster than UNION?

A It uses an index

B It does not need to sort or hash rows to remove duplicates

C It returns fewer rows

D It runs queries in parallel

Answer: B. Duplicate removal is extra work that UNION ALL skips.

EXERCISE

Customer movement by region

For each region, find the number of customers who ordered in 2024, the number retained in 2025, and the retention rate. Hint: build the retained list with INTERSECT, then join to customers for the region.

Show solution (worked answer)

SQL

```
WITH c24 AS (SELECT DISTINCT customer_id FROM orders WHERE YEAR(order_date) = 2024),
     kept AS (SELECT customer_id FROM c24
              INTERSECT
              SELECT customer_id FROM orders WHERE YEAR(order_date) = 2025)
SELECT c.region,
       COUNT(DISTINCT c24.customer_id)  AS customers_2024,
       COUNT(DISTINCT kept.customer_id) AS retained_2025,
       ROUND(100.0 * COUNT(DISTINCT kept.customer_id) / COUNT(DISTINCT
c24.customer_id), 1) AS retention_pct
FROM c24
JOIN customers AS c ON c.customer_id = c24.customer_id
LEFT JOIN kept ON kept.customer_id = c24.customer_id
GROUP BY c.region
ORDER BY retention_pct DESC;
```

(CTEs with `WITH` are explained in Lesson 13.)

INTERVIEW CORNER

Questions you may be asked

Q: UNION vs UNION ALL? UNION removes duplicate rows (slower); UNION ALL keeps all rows (faster). Use UNION ALL unless de-duplication is required.

Q: JOIN vs UNION? JOIN combines columns from related tables on a key; UNION stacks rows from queries with the same column structure.

Q: How would you check that two tables contain exactly the same data? Run `A EXCEPT B` and `B EXCEPT A`; both must return zero rows (also compare row counts to catch duplicates).

SQL FOR DATA ANALYSIS · LESSON 10

Subqueries, Derived Tables & EXISTS

Put queries inside queries. Use scalar subqueries, multi-row subqueries with IN, ANY and ALL, correlated subqueries, EXISTS and NOT EXISTS, and derived tables (inline views) to answer questions that one level of SELECT cannot, and learn when a join, CTE or window function is better.

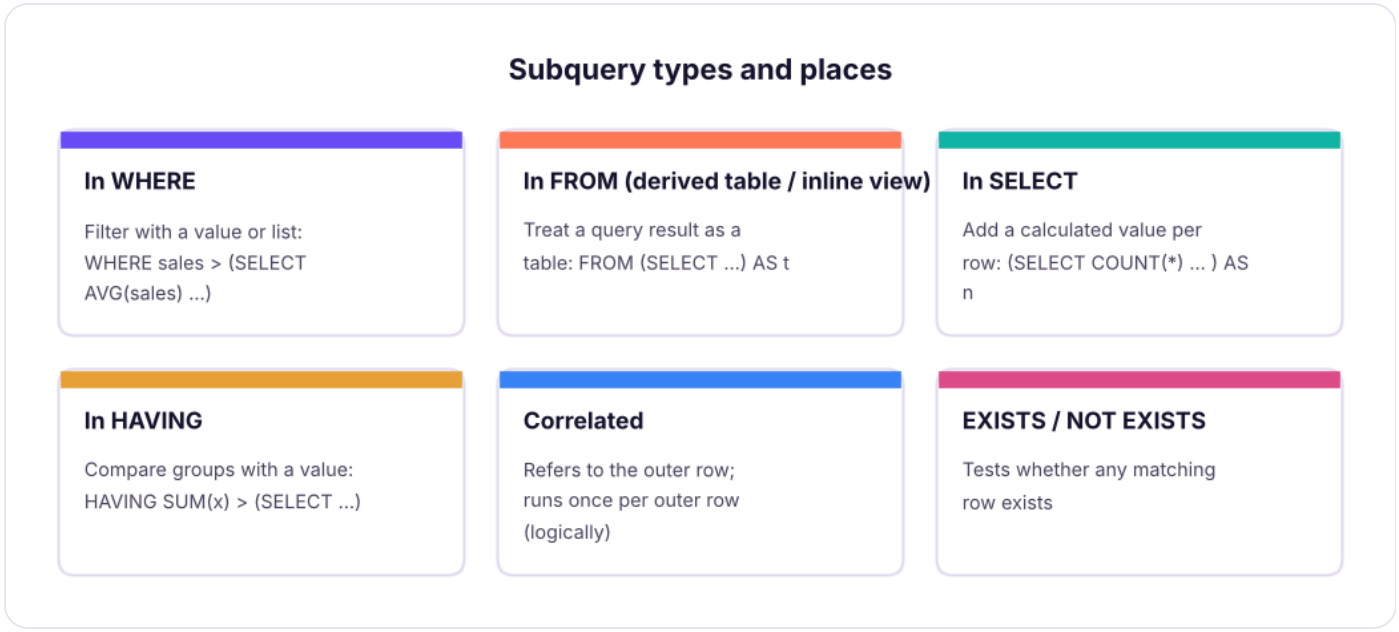
SQL classes 4–5 · SQL Server 2022, 2026 · Intermediate · 120 min

By the end of this lesson you can

- Identify scalar, multi-row, multi-column and table subqueries and where each can be used
- Filter with subqueries using comparison operators, IN, NOT IN, ANY and ALL
- Write correlated subqueries and explain how they are evaluated
- Use EXISTS and NOT EXISTS safely, and know why NOT IN fails with NULLs
- Build derived tables (inline views) to aggregate before joining

What is a subquery?

A **subquery** (inner query, nested query) is a `SELECT` written inside another statement, in brackets. The outer query uses its result.



Where subqueries can appear and what they return.

By what it returns	Example	Used with
Scalar (one value)	`(SELECT AVG(sales) FROM superstore)`	`=, >, <` ... or as a column
Multi-row (one column, many rows)	`(SELECT customer_id FROM orders WHERE …)`	`IN`, `NOT IN`, `ANY`, `ALL`, `EXISTS`
Table (many columns and rows)	`(SELECT region, SUM(sales) … GROUP BY region)`	`FROM` as a derived table

Scalar subqueries

"Which order lines are above the average line value?"

SQL

```sql
SELECT COUNT(*) AS lines_above_average,
       (SELECT ROUND(AVG(sales), 0) FROM superstore) AS average_line_sales
FROM superstore
WHERE sales > (SELECT AVG(sales) FROM superstore);
```

RESULT

lines_above_average	average_line_sales
1603	31,854

1 rows × 2 columns

The inner query runs first and returns one number; the outer query compares each row with it. You cannot write `WHERE sales > AVG(sales)` directly, because aggregates are not allowed in WHERE (Lesson 5's processing order).

"Show each region's sales and its share of the grand total":

```sql
SELECT region,
       ROUND(SUM(sales), 0)                                                    AS sales,
       ROUND(100.0 * SUM(sales) / (SELECT SUM(sales) FROM superstore), 1)       AS
pct_of_total
FROM superstore
GROUP BY region
ORDER BY sales DESC;
```

RESULT

region	sales	pct_of_total
South	53,952,769	26.8
West	51,958,521	25.8
North	49,018,161	24.3
East	46,484,119	23.1

4 rows × 3 columns

Multi-row subqueries: IN, NOT IN, ANY, ALL

"Orders from customers in the Corporate segment" (without a join):

```sql
SELECT COUNT(*) AS corporate_orders
FROM orders
WHERE customer_id IN (SELECT customer_id FROM customers WHERE segment = 'Corporate');
```

RESULT

corporate_orders
955

1 rows × 1 columns

"Employees earning more than **every** employee in HR" and "more than **at least one** Sales manager":

```sql
SELECT COUNT(*) AS earn_more_than_all_hr
FROM employees
WHERE monthly_income > ALL (SELECT monthly_income FROM employees WHERE department = 'HR');
```

RESULT

earn_more_than_all_hr
4

1 rows × 1 columns

```sql
SELECT COUNT(*) AS earn_more_than_some_sales_manager
FROM employees
WHERE monthly_income > ANY (SELECT monthly_income FROM employees WHERE job_title = 'Sales Manager');
```

RESULT

earn_more_than_some_sales_manager
745

1 rows × 1 columns

Operator	Meaning	Equivalent
`> ALL (…)`	Greater than every value	`> (SELECT MAX(…))`
`> ANY (…)` / `> SOME (…)`	Greater than at least one value	`> (SELECT MIN(…))`
`= ANY (…)`	Equal to one of the values	`IN (…)`
`<> ALL (…)`	Different from every value	`NOT IN (…)`

Correlated subqueries

A **correlated** subquery refers to a column of the outer query, so its result changes for every outer row.

"Which employees earn more than the average of **their own** department?"

```sql
SELECT e.emp_id, e.department, e.job_title, e.monthly_income
FROM employees AS e
WHERE e.monthly_income > (SELECT AVG(e2.monthly_income)
                          FROM employees AS e2
                          WHERE e2.department = e.department)     -- refers to the outer
row
ORDER BY e.monthly_income DESC;
```

RESULT

emp_id	department	job_title	monthly_income
E00210	R&D	Research Director	218417
E00519	Finance	Finance Manager	209833
E00328	Operations	Operations Manager	202750
E01187	IT	Support Engineer	196167
E01094	HR	HR Manager	194917
E01473	IT	Tech Lead	194500
E00439	IT	Tech Lead	193917
E00580	R&D	Research Director	188083

638 rows × 4 columns (first 8 shown)

Conceptually, for each employee e, SQL computes the average of e's department and compares. Optimisers usually rewrite this efficiently, but on large tables a window function (Lesson 12: `AVG(…) OVER (PARTITION BY department)`) or a join to a derived table is often clearer and faster.

"Latest order of each customer" (a classic):

```sql
SELECT o.customer_id, o.order_id, o.order_date
FROM orders AS o
WHERE o.order_date = (SELECT MAX(o2.order_date)
                      FROM orders AS o2
                      WHERE o2.customer_id = o.customer_id)
ORDER BY o.customer_id;
```

RESULT

customer_id	order_id	order_date
C0001	IN-2025-02263	2025-04-09
C0002	IN-2025-02871	2025-10-21
C0003	IN-2025-02642	2025-08-18
C0004	IN-2025-02344	2025-05-10
C0005	IN-2025-02982	2025-11-12
C0006	IN-2025-02960	2025-11-08

793 rows × 3 columns (first 6 shown)

EXISTS and NOT EXISTS

`EXISTS (subquery)` is TRUE if the subquery returns **at least one row**. It stops at the first match, and the columns it selects do not matter (`SELECT 1` is conventional).

"Customers who bought at least one Copier":

SQL

```sql
SELECT COUNT(*) AS customers_who_bought_copiers
FROM customers AS c
WHERE EXISTS (SELECT 1
              FROM orders AS o
              JOIN order_items AS oi ON oi.order_id = o.order_id
              JOIN products AS p     ON p.product_id = oi.product_id
              WHERE o.customer_id = c.customer_id
                AND p.sub_category = 'Copiers');
```

RESULT

customers_who_bought_copiers
343

1 rows × 1 columns

"Customers who have never ordered" (anti-join, compare with Lesson 8):

SQL

```sql
SELECT COUNT(*) AS customers_without_orders
FROM customers AS c
WHERE NOT EXISTS (SELECT 1 FROM orders AS o WHERE o.customer_id = c.customer_id);
```

RESULT

customers_without_orders
15

1 rows × 1 columns

Why NOT EXISTS is safer than NOT IN

SQL

```sql
WITH a AS (SELECT 1 AS id UNION ALL SELECT 2 UNION ALL SELECT 3),
     b AS (SELECT 2 AS id UNION ALL SELECT NULL)
SELECT
    (SELECT COUNT(*) FROM a WHERE id NOT IN (SELECT id FROM b))                    AS
not_in_rows,
    (SELECT COUNT(*) FROM a WHERE NOT EXISTS (SELECT 1 FROM b WHERE b.id = a.id))  AS
not_exists_rows;
```

RESULT

not_in_rows	not_exists_rows
0	2

1 rows × 2 columns

`NOT IN` returns **0 rows** because b contains a NULL: `3 NOT IN (2, NULL)` means `3 <> 2 AND 3 <> NULL`, which is UNKNOWN. `NOT EXISTS` correctly returns 2 rows (1 and 3). Prefer `NOT EXISTS`, or add `WHERE id IS NOT NULL` inside the `NOT IN` subquery.

Derived tables (inline views)

A subquery in `FROM` acts as a temporary table for the rest of the query. It **must have an alias**. Derived tables are the standard way to **aggregate first, then filter, join or aggregate again**.

"Average order value by segment" requires two levels: first total each order, then average the orders.

SQL

```sql
SELECT segment,
       COUNT(*)                   AS orders,
       ROUND(AVG(order_total), 0) AS avg_order_value
FROM (
    SELECT o.order_id, c.segment, SUM(oi.sales) AS order_total     -- level 1: one row per order
    FROM orders AS o
    JOIN customers AS c    ON c.customer_id = o.customer_id
    JOIN order_items AS oi ON oi.order_id   = o.order_id
    GROUP BY o.order_id, c.segment
) AS order_totals                                                  -- level 2: average of orders
GROUP BY segment
ORDER BY avg_order_value DESC;
```

RESULT

segment	orders	avg_order_value
Consumer	1554	63,451
Home Office	691	63,237
Corporate	955	61,899

3 rows × 3 columns

If you averaged `oi.sales` directly, you would get the average **line** value, not the average **order** value: a frequent interview trap.

"Cities whose 2025 sales beat their 2024 sales", using two derived tables joined together:

SQL

```sql
SELECT a.city, ROUND(a.sales, 0) AS sales_2024, ROUND(b.sales, 0) AS sales_2025,
       ROUND(100.0 * (b.sales - a.sales) / a.sales, 1) AS growth_pct
FROM (SELECT city, SUM(sales) AS sales FROM superstore WHERE YEAR(order_date) = 2024 GROUP BY city) AS a
JOIN (SELECT city, SUM(sales) AS sales FROM superstore WHERE YEAR(order_date) = 2025 GROUP BY city) AS b
  ON b.city = a.city
WHERE b.sales > a.sales
ORDER BY growth_pct DESC;
```

RESULT

city	sales_2024	sales_2025	growth_pct
Pune	4,469,290	8,808,243	97.1
Jaipur	2,381,435	4,387,920	84.3
Bengaluru	3,955,891	5,909,257	49.4
Bhubaneswar	3,241,529	4,715,977	45.5
Chennai	4,306,703	6,192,082	43.8
Guwahati	3,063,023	4,156,269	35.7
Kolkata	4,204,948	5,388,829	28.2
Kochi	4,419,418	5,550,394	25.6
Indore	4,663,519	5,766,524	23.7
Ahmedabad	3,655,966	4,499,715	23.1
Chandigarh	3,384,407	3,711,620	9.7
Patna	4,459,339	4,649,126	4.3

12 rows × 4 columns

NOTE **Nested (ordinary) subqueries vs inline views**

A subquery in `WHERE` that does not refer to the outer query is an **ordinary (nested) subquery**: it runs once. A subquery in `FROM` is an **inline view** (derived table). One that refers to the outer row is **correlated**. In class these were introduced as inline views and ordinary subqueries; the next step is the **CTE** (Lesson 13), which is a named, more readable inline view.

Subqueries in SELECT

A scalar subquery in the column list adds a per-row value:

SQL

```
SELECT c.customer_id, c.customer_name,
       (SELECT COUNT(*) FROM orders AS o WHERE o.customer_id = c.customer_id)     AS
order_count,
       (SELECT MAX(o.order_date) FROM orders AS o WHERE o.customer_id = c.customer_id) AS
last_order
FROM customers AS c
ORDER BY order_count DESC, c.customer_id;
```

RESULT

customer_id	customer_name	order_count	last_order
C0219	Reyansh Verma	10	2025-04-10
C0396	Saanvi Patel	10	2025-07-18
C0069	Isha Verma	9	2025-10-04
C0110	Riya Das	9	2025-12-07
C0271	Meera Khan	9	2025-12-26
C0291	Meera Rao	9	2025-02-04

800 rows × 4 columns (first 6 shown)

Each correlated scalar subquery runs per customer. For many such columns, a single `LEFT JOIN` to an aggregated derived table is usually faster.

Choosing the right tool

Need	Good options
Compare with an overall aggregate	Scalar subquery, or window function
Filter by membership in another table	`IN`, `EXISTS`, or `JOIN`
Rows with no match	`NOT EXISTS` (safest), `LEFT JOIN … IS NULL`, `EXCEPT`
Compare with the row's own group	Correlated subquery, or `AVG() OVER (PARTITION BY …)`
Aggregate in two steps	Derived table or CTE
Reuse the same subquery several times	CTE (Lesson 13)

Summary

KEY TAKEAWAYS

- Subqueries are queries in brackets inside another query; they can return a value, a list or a table.
- Scalar subqueries compare with or display a single value (e.g. the overall average).
- `IN`, `ANY`, `ALL` work with multi-row subqueries; `> ALL` = greater than the max.
- Correlated subqueries reference the outer row (e.g. above the employee's department average).
- `EXISTS` / `NOT EXISTS` test for matching rows; prefer `NOT EXISTS` over `NOT IN` when NULLs are possible.
- Derived tables (inline views) aggregate first, then filter/join; they need an alias.

Quick quiz

6 questions · tap an answer to check it

1. Why does `WHERE sales > AVG(sales)` fail?

A AVG needs GROUP BY

B Aggregate functions are not allowed in WHERE; use a subquery

C sales must be in brackets

D The comparison must use ALL

Answer: B. WHERE runs before aggregation, so compute the average in a subquery.

2. A subquery in FROM must have:

A An ORDER BY

B An alias

C A WHERE clause

D A GROUP BY

Answer: B. Derived tables must be named.

3. `WHERE x NOT IN (SELECT y FROM t)` **returns no rows even though some x values are missing from t. Most likely cause?**

A t is empty

B t.y contains a NULL

C x has duplicates

D NOT IN cannot be used with subqueries

Answer: B. One NULL in the list makes every NOT IN comparison UNKNOWN.

4. What is a correlated subquery?

A A subquery in FROM

B A subquery that references a column from the outer query

C A subquery with a JOIN

D A subquery that returns one value

Answer: B. It is evaluated in the context of each outer row.

5. `salary > ALL (SELECT salary FROM emp WHERE dept = 'HR')` **is equivalent to:**

A salary > the maximum HR salary

B salary > the minimum HR salary

C salary > the average HR salary

D salary IN the HR salaries

Answer: A. Greater than all values means greater than the largest one.

6. To compute average order value from order lines, you should:

- **A** AVG(sales) on order_items
- **B** Sum sales per order in a derived table, then average those totals
- **C** Divide total sales by the number of lines
- **D** Use COUNT(DISTINCT sales)

Answer: B. Average order value is the mean of order totals, which needs two aggregation levels.

EXERCISE **Subquery practice**

(1) List products whose total sales are above the average product's total sales. (2) For each department, show employees whose income is the highest in that department. (3) Find cities where every order line in 2025 had a discount of at most 20%. (4) Show customers whose first order was in 2025.

Show solution (worked answer)

SQL

```sql
-- (1)
SELECT product_id, SUM(sales) AS sales FROM order_items GROUP BY product_id
HAVING SUM(sales) > (SELECT AVG(t.s) FROM (SELECT SUM(sales) AS s FROM order_items
GROUP BY product_id) AS t);
-- (2)
SELECT e.department, e.emp_id, e.monthly_income FROM employees e
WHERE e.monthly_income = (SELECT MAX(e2.monthly_income) FROM employees e2 WHERE
e2.department = e.department);
-- (3)
SELECT DISTINCT city FROM superstore s
WHERE YEAR(order_date) = 2025
  AND NOT EXISTS (SELECT 1 FROM superstore s2 WHERE s2.city = s.city AND
YEAR(s2.order_date) = 2025 AND s2.discount > 0.2);
-- (4)
SELECT customer_id FROM orders GROUP BY customer_id HAVING MIN(order_date) ⩾ '2025-
01-01';
```

INTERVIEW CORNER

Questions you may be asked

Q: Difference between a correlated and a non-correlated subquery? A non-correlated subquery runs independently once; a correlated subquery references the outer query and is evaluated per outer row.

Q: EXISTS vs IN? IN compares a value with a list; EXISTS checks for the existence of matching rows and stops at the first one. EXISTS handles NULLs safely in the negative form (NOT EXISTS).

Q: Find employees who earn more than their department's average. Correlated subquery on department, or `AVG(income) OVER (PARTITION BY department)` in a CTE and filter.

Q: Subquery vs JOIN: which is faster? It depends; modern optimisers often produce the same plan. Joins are usually clearer for combining columns; EXISTS for existence tests; derived tables/CTEs for multi-step aggregation. Check the execution plan.

SQL FOR DATA ANALYSIS · LESSON 11

OFFSET-FETCH and Window Functions I: OVER, PARTITION BY & Ranking

Page through results with OFFSET and FETCH, then learn window functions, the most powerful analytics feature in SQL. Understand OVER and PARTITION BY, rank rows with ROW_NUMBER, RANK, DENSE_RANK and NTILE, solve top-N-per-group and de-duplication problems, and compare aggregate windows with GROUP BY.

SQL class 5 · SQL Server 2022, 2026 · Intermediate–Advanced · 130 min

By the end of this lesson you can

- Paginate with ORDER BY ... OFFSET ... ROWS FETCH NEXT ... ROWS ONLY
- Explain what a window function is and how OVER, PARTITION BY and ORDER BY define the window
- Use ROW_NUMBER, RANK, DENSE_RANK and NTILE and explain how ties behave
- Solve top-N per group, Nth highest value and duplicate-removal problems
- Use aggregate window functions to show totals and shares beside detail rows

OFFSET and FETCH: pagination

`OFFSET n ROWS FETCH NEXT m ROWS ONLY` skips n rows and returns the next m. It requires `ORDER BY`. It is the standard (ANSI) way to page results, used for "page 3 of results" in applications and for "rows 11 to 20" questions.

SQL

```sql
SELECT customer_name, city, segment
FROM customers
ORDER BY customer_id
OFFSET 10 ROWS FETCH NEXT 5 ROWS ONLY;
```

RESULT

customer_name	city	segment
Sai Iyer	Indore	Consumer
Neha Kulkarni	Bengaluru	Consumer
Diya Khan	Bhubaneswar	Home Office
Kabir Das	Kolkata	Consumer
Reyansh Joshi	Mumbai	Corporate

5 rows × 3 columns

Page	OFFSET	FETCH
1	0	page size
2	page size	page size
page p	(p − 1) × page size	page size

`TOP n` cannot skip rows; OFFSET-FETCH can. PostgreSQL/MySQL write `LIMIT m OFFSET n`. On very deep pages (OFFSET 1,000,000) databases still read and discard all skipped rows, so large systems use **keyset pagination** (`WHERE id > last_seen_id ORDER BY id`).

The "Nth highest value" question can be answered with it too: the second-highest line value is `ORDER BY sales DESC OFFSET 1 ROWS FETCH NEXT 1 ROWS ONLY` on distinct values:

SQL

```sql
SELECT DISTINCT sales AS second_highest_sales
FROM superstore
ORDER BY sales DESC
OFFSET 1 ROWS FETCH NEXT 1 ROWS ONLY;
```

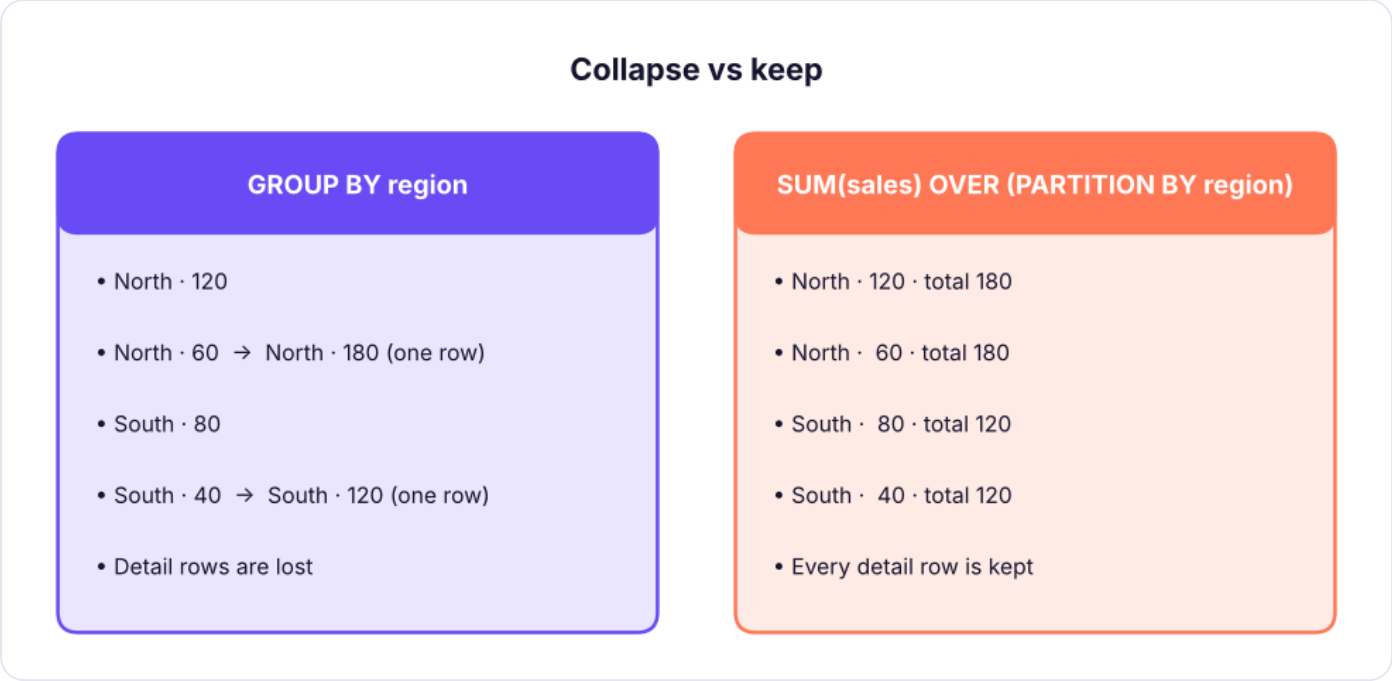
RESULT

second_highest_sales
551,530

1 rows × 1 columns

What is a window function?

A **window function** computes a value for **each row** using a set of related rows (its **window**), **without collapsing the rows** like `GROUP BY` does.



GROUP BY collapses rows into one per group. A window function keeps every row and adds the group's result beside it.

The OVER clause

```sql
function_name(expression) OVER (
    PARTITION BY column_list      -- split rows into groups (optional)
    ORDER BY column_list          -- order inside each group (needed for ranking, running totals)
    ROWS BETWEEN … AND …          -- optional frame (Lesson 12)
)
```

- **`PARTITION BY`** is like GROUP BY for the window: the function restarts for every partition. Without it, the whole result is one partition.
- **`ORDER BY`** inside OVER orders rows *within the partition* (it does not sort the final output; use the normal ORDER BY for that).

```sql
SELECT region, city, order_id, sales,
       SUM(sales) OVER (PARTITION BY region)                        AS region_total,
       ROUND(100.0 * sales / SUM(sales) OVER (PARTITION BY region), 4) AS pct_of_region,
       AVG(sales) OVER ()                                           AS overall_avg
FROM superstore
ORDER BY region, sales DESC;
```

RESULT

region	city	order_id	sales	region_total	pct_of_region	overall_avg
East	Kolkata	IN-2025-03160	472,740	46,484,119	1.02	31,854
East	Kolkata	IN-2025-02134	442,350	46,484,119	0.95	31,854
East	Kolkata	IN-2025-02822	422,640	46,484,119	0.91	31,854
East	Guwahati	IN-2025-02562	413,040	46,484,119	0.89	31,854
East	Patna	IN-2023-00171	393,950	46,484,119	0.85	31,854
East	Patna	IN-2025-02317	388,320	46,484,119	0.84	31,854
East	Bhubaneswar	IN-2025-02981	372,720	46,484,119	0.80	31,854
East	Patna	IN-2024-01792	372,720	46,484,119	0.80	31,854

6,323 rows × 7 columns (first 8 shown)

Each row keeps its own detail but now also shows its region total, its share of the region, and the overall average. With GROUP BY you would need a join to get this.

Ranking functions

All four need `ORDER BY` inside `OVER`.

Function	Behaviour with ties	Example values 90, 80, 80, 70
`ROW_NUMBER()`	Unique numbers; ties broken arbitrarily	1, 2, 3, 4
`RANK()`	Ties share a rank; **gaps** follow	1, 2, 2, 4
`DENSE_RANK()`	Ties share a rank; **no gaps**	1, 2, 2, 3
`NTILE(n)`	Splits rows into n equal buckets	4 rows, NTILE(2): 1, 1, 2, 2

SQL

```sql
WITH scores AS (
    SELECT 'Asha' AS name, 90 AS score UNION ALL SELECT 'Ravi', 80 UNION ALL SELECT
'Meera', 80 UNION ALL SELECT 'Kabir', 70
)
SELECT name, score,
       ROW_NUMBER() OVER (ORDER BY score DESC) AS row_number,
       RANK()       OVER (ORDER BY score DESC) AS rank,
       DENSE_RANK() OVER (ORDER BY score DESC) AS dense_rank,
       NTILE(2)     OVER (ORDER BY score DESC) AS half
FROM scores;
```

RESULT

name	score	row_number	rank	dense_rank	half
Asha	90	1	1	1	1
Meera	80	2	2	2	1
Ravi	80	3	2	2	2
Kabir	70	4	4	3	2

4 rows × 6 columns

Ravi and Meera tie at 80: `RANK` gives both 2 and skips 3; `DENSE_RANK` gives both 2 and continues with 3; `ROW_NUMBER` gives 2 and 3 in arbitrary order (add a tie-breaker column to make it deterministic).

Ranking within groups

"Rank each sub-category by sales within its category":

SQL

```sql
SELECT category, sub_category,
       ROUND(SUM(sales), 0)                                          AS sales,
       RANK() OVER (PARTITION BY category ORDER BY SUM(sales) DESC)  AS
rank_in_category
FROM superstore
GROUP BY category, sub_category
ORDER BY category, rank_in_category;
```

RESULT

category	sub_category	sales	rank_in_category
Furniture	Tables	14,046,224	1
Furniture	Bookcases	11,274,849	2
Furniture	Chairs	8,091,474	3
Furniture	Furnishings	1,429,307	4
Office Supplies	Appliances	6,333,529	1
Office Supplies	Storage	2,659,340	2
Office Supplies	Paper	493,232	3
Office Supplies	Binders	401,219	4
Office Supplies	Art	245,985	5
Technology	Copiers	78,749,926	1
Technology	Machines	42,590,335	2
Technology	Phones	31,919,223	3

13 rows × 4 columns (first 12 shown)

Note the order: `GROUP BY` runs first, then the window function ranks the grouped rows. That is why `SUM(sales)` can appear inside `OVER`.

Top-N per group

You cannot filter on a window function in `WHERE` (windows are computed in the SELECT step, after WHERE). Wrap the query in a CTE or derived table, then filter the rank.

"Top 2 cities by sales in each region":

SQL

```sql
WITH city_sales AS (
    SELECT region, city, SUM(sales) AS sales
    FROM superstore
    GROUP BY region, city
), ranked AS (
    SELECT region, city, sales,
           DENSE_RANK() OVER (PARTITION BY region ORDER BY sales DESC) AS rnk
    FROM city_sales
)
SELECT region, city, ROUND(sales, 0) AS sales, rnk
FROM ranked
WHERE rnk ⩽ 2
ORDER BY region, rnk;
```

RESULT

region	city	sales	rnk
East	Kolkata	13,888,466	1
East	Patna	12,444,393	2
North	Delhi	15,657,610	1
North	Chandigarh	11,541,792	2
South	Kochi	14,713,126	1
South	Bengaluru	13,625,216	2
West	Pune	17,357,038	1
West	Indore	14,408,485	2

8 rows × 4 columns

"Each customer's most recent order" (use ROW_NUMBER to get exactly one row per customer even with ties):

SQL

```sql
WITH ranked AS (
    SELECT customer_id, order_id, order_date,
           ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date DESC, order_id
DESC) AS rn
    FROM orders
)
SELECT customer_id, order_id, order_date
FROM ranked
WHERE rn = 1
ORDER BY customer_id;
```

RESULT

customer_id	order_id	order_date
C0001	IN-2025-02263	2025-04-09
C0002	IN-2025-02871	2025-10-21
C0003	IN-2025-02642	2025-08-18
C0004	IN-2025-02344	2025-05-10
C0005	IN-2025-02982	2025-11-12
C0006	IN-2025-02960	2025-11-08

785 rows × 3 columns (first 6 shown)

Nth highest per group

"Second-highest paid employee in each department" (a top interview question):

SQL

```sql
WITH r AS (
    SELECT department, emp_id, monthly_income,
           DENSE_RANK() OVER (PARTITION BY department ORDER BY monthly_income DESC) AS rnk
    FROM employees
)
SELECT department, emp_id, monthly_income
FROM r
WHERE rnk = 2
ORDER BY department;
```

RESULT

department	emp_id	monthly_income
Finance	E00145	185500
HR	E00450	175750
IT	E01473	194500
Operations	E00378	184167
R&D	E00580	188083
Sales	E01218	184833

6 rows × 3 columns

Use `DENSE_RANK` for "Nth highest **value**" (ties share a position, no gaps) and `ROW_NUMBER` for "Nth **row**".

Removing duplicates with ROW_NUMBER

SQL

```sql
WITH d AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY customer_id, order_date, product_id ORDER BY
item_id) AS rn
    FROM order_items_raw
)
DELETE FROM d WHERE rn > 1;            -- SQL Server lets you DELETE through a CTE
```

Rows with the same business key get numbers 1, 2, 3…; keeping `rn = 1` keeps one copy. Always check with a `SELECT … WHERE rn > 1` first.

Let's count how many lines in our data repeat the same product in the same order (what a de-duplication would touch):

SQL

```sql
WITH d AS (
    SELECT order_id, product_id,
           ROW_NUMBER() OVER (PARTITION BY order_id, product_id ORDER BY row_id) AS rn
    FROM superstore
)
SELECT COUNT(*) AS repeated_lines FROM d WHERE rn > 1;
```

RESULT

repeated_lines
70

1 rows × 1 columns

(These are legitimate repeat lines, not errors; the point is the pattern.)

NTILE: buckets and percentiles

SQL

```sql
WITH cust AS (
    SELECT customer_id, SUM(sales) AS sales FROM superstore GROUP BY customer_id
), q AS (
    SELECT customer_id, sales, NTILE(4) OVER (ORDER BY sales DESC) AS quartile FROM cust
)
SELECT quartile,
       COUNT(*)               AS customers,
       ROUND(SUM(sales), 0)   AS sales,
       ROUND(100.0 * SUM(sales) / SUM(SUM(sales)) OVER (), 1) AS pct_of_total_sales
FROM q
GROUP BY quartile
ORDER BY quartile;
```

RESULT

quartile	customers	sales	pct_of_total_sales
1	197	111,100,409	55.2
2	196	54,386,341	27.0
3	196	28,080,117	13.9
4	196	7,846,704	3.9

4 rows × 4 columns

Quartile 1 (the top 25% of customers) generates a disproportionate share of sales: the **Pareto principle** from the Excel charts lesson, now computed in SQL. NTILE is also how RFM scoring is done (ML track).

Aggregate window functions

Any aggregate (`SUM`, `AVG`, `COUNT`, `MIN`, `MAX`) becomes a window function with `OVER`. Without `ORDER BY` in the window, it uses the whole partition: perfect for totals, shares and comparisons with the

group average.

```sql
SELECT department, emp_id, monthly_income,
       ROUND(AVG(monthly_income) OVER (PARTITION BY department), 0)          AS dept_avg,
       monthly_income - ROUND(AVG(monthly_income) OVER (PARTITION BY department), 0) AS
vs_dept_avg,
       COUNT(*) OVER (PARTITION BY department)                               AS
dept_headcount,
       MAX(monthly_income) OVER (PARTITION BY department)                    AS dept_max
FROM employees
ORDER BY department, monthly_income DESC;
```

RESULT

department	emp_id	monthly_income	dept_avg	vs_dept_avg	dept_headcount	dept_
Finance	E00519	209833	83,681	126,152	154	2098
Finance	E00145	185500	83,681	101,819	154	2098
Finance	E00336	174417	83,681	90,736	154	2098
Finance	E00201	169500	83,681	85,819	154	2098
Finance	E01172	168167	83,681	84,486	154	2098
Finance	E01151	155167	83,681	71,486	154	2098
Finance	E00759	152083	83,681	68,402	154	2098
Finance	E01004	144667	83,681	60,986	154	2098

1,500 rows × 7 columns (first 8 shown)

This replaces the correlated subquery from Lesson 10 ("earns more than the department average") with one pass over the data.

GROUP BY vs window functions

	GROUP BY	Window function
Rows returned	One per group	Same as input
Detail columns	Only grouped columns	All columns
Use for	Summary reports	Rankings, running totals, shares, comparisons with group values
Filter the result	`HAVING`	Wrap in CTE/subquery and `WHERE`
Can combine	Yes: group first, then window over the groups	

Summary

KEY TAKEAWAYS

- `OFFSET n ROWS FETCH NEXT m ROWS ONLY` (with `ORDER BY`) paginates; it can skip rows, unlike `TOP`.
- A **window function** adds a calculation over a set of related rows to each row, without collapsing them. `OVER (PARTITION BY … ORDER BY …)` defines the window.
- `ROW_NUMBER` (unique), `RANK` (ties, gaps), `DENSE_RANK` (ties, no gaps), `NTILE(n)` (equal buckets).
- To filter on a window result, compute it in a CTE or derived table, then use `WHERE`.
- Top-N per group and Nth highest per group: `DENSE_RANK`/`ROW_NUMBER` partitioned by group; de-duplicate with `ROW_NUMBER … WHERE rn > 1`.
- Aggregate windows (`SUM() OVER (PARTITION BY …)`) show totals, shares and group averages beside each row.

Quick quiz

6 questions · tap an answer to check it

1. Scores 100, 90, 90, 80. What does DENSE_RANK() OVER (ORDER BY score DESC) return?

A 1, 2, 3, 4

B 1, 2, 2, 4

C 1, 2, 2, 3

D 1, 1, 2, 3

Answer: C. DENSE_RANK gives tied rows the same rank and continues without gaps.

2. Why can't you write `WHERE ROW_NUMBER() OVER (...) = 1`?

A ROW_NUMBER needs GROUP BY

B Window functions are evaluated after WHERE, so you must filter in an outer query or CTE

C OVER cannot contain ORDER BY

D It is allowed in SQL Server

Answer: B. Logical processing order: WHERE runs before SELECT, where windows are computed.

3. What is the difference between PARTITION BY and GROUP BY?

A None

B PARTITION BY defines the window groups but keeps every row; GROUP BY collapses rows

C PARTITION BY only works with RANK

D GROUP BY keeps all rows

Answer: B. Partitions restart the calculation for each group without reducing the number of rows.

4. Which returns the 3rd highest distinct salary overall?

A DENSE_RANK() OVER (ORDER BY salary DESC) = 3

B ROW_NUMBER() OVER (ORDER BY salary DESC) = 3

C TOP 3 salary

D RANK() OVER (ORDER BY salary) = 3

Answer: A. DENSE_RANK counts distinct values; ROW_NUMBER would count rows even if values repeat.

5. `SUM(sales) OVER (PARTITION BY region)` returns:

A One row per region

B On every row, the total sales of that row's region

C The overall total on every row

D A running total

Answer: B. Without ORDER BY in the window, it sums the whole partition.

6. To fetch rows 21–30 of a sorted result you write:

A `TOP 30`

B `ORDER BY … OFFSET 20 ROWS FETCH NEXT 10 ROWS ONLY`

C `LIMIT 10, 20` in T-SQL

D `WHERE ROW 21 TO 30`

Answer: B. Skip 20, take 10.

EXERCISE **Window practice**

(1) For each customer segment, rank customers by total sales and show the top 3. (2) For each order line, show its sales as a percentage of its order's total. (3) For each department, list employees whose income is within the top 5% of their department (use NTILE(20)). (4) Show products whose total sales rank in the top 10% of their category.

Show solution (worked answer)

SQL

```sql
-- (1)
WITH t AS (SELECT c.segment, c.customer_name, SUM(oi.sales) AS sales
           FROM order_items oi JOIN orders o ON o.order_id = oi.order_id JOIN
customers c ON c.customer_id = o.customer_id
           GROUP BY c.segment, c.customer_id, c.customer_name),
     r AS (SELECT *, DENSE_RANK() OVER (PARTITION BY segment ORDER BY sales DESC) AS
rnk FROM t)
SELECT * FROM r WHERE rnk ⩽ 3 ORDER BY segment, rnk;
-- (2)
SELECT item_id, order_id, sales, ROUND(100.0 * sales / SUM(sales) OVER (PARTITION BY
order_id), 1) AS pct_of_order FROM order_items;
-- (3)
WITH x AS (SELECT department, emp_id, monthly_income, NTILE(20) OVER (PARTITION BY
department ORDER BY monthly_income DESC) AS bucket FROM employees)
SELECT * FROM x WHERE bucket = 1;
-- (4)
WITH p AS (SELECT pr.category, pr.product_id, SUM(oi.sales) AS sales FROM order_items
oi JOIN products pr ON pr.product_id = oi.product_id GROUP BY pr.category,
pr.product_id),
     n AS (SELECT *, NTILE(10) OVER (PARTITION BY category ORDER BY sales DESC) AS
decile FROM p)
SELECT * FROM n WHERE decile = 1;
```

INTERVIEW CORNER

Questions you may be asked

Q: RANK vs DENSE_RANK vs ROW_NUMBER? ROW_NUMBER numbers rows uniquely; RANK gives ties the same rank and skips following ranks; DENSE_RANK gives ties the same rank without gaps.

Q: Find the second-highest salary in each department. CTE with `DENSE_RANK() OVER (PARTITION BY dept ORDER BY salary DESC)`, filter `= 2`.

Q: How do you remove duplicate rows keeping one? `ROW_NUMBER() OVER (PARTITION BY business_key ORDER BY …)` and delete rows where it is greater than 1.

Q: Window function vs GROUP BY? Windows keep detail rows and add group-level values; GROUP BY returns one row per group.

SQL FOR DATA ANALYSIS · LESSON 12

Window Functions II: Running Totals, LEAD, LAG & Frames

Build time-series analytics in SQL. Use window frames (ROWS BETWEEN) for running totals and moving averages, LAG and LEAD for month-on-month and year-on-year growth, FIRST_VALUE and LAST_VALUE, cumulative percentages, PERCENT_RANK, CUME_DIST and PERCENTILE_CONT, and gaps-and-islands patterns.

SQL class 5–6 · SQL Server 2022, 2026 · Advanced · 130 min

By the end of this lesson you can

- Define a window frame with ROWS BETWEEN and explain the default frame when ORDER BY is present
- Calculate running totals, moving averages and cumulative shares
- Compare each row with the previous or next row using LAG and LEAD (MoM, YoY, gaps)
- Use FIRST_VALUE, LAST_VALUE and NTH_VALUE, PERCENT_RANK, CUME_DIST and PERCENTILE_CONT
- Solve streak and gap problems with ROW_NUMBER differences

Window frames

Inside `OVER`, `ORDER BY` defines the order and an optional **frame** defines *which rows around the current row* the function uses.

```sql
SUM(x) OVER (PARTITION BY p ORDER BY d
             ROWS BETWEEN <start> AND <end>)
```

Frame boundary	Meaning
`UNBOUNDED PRECEDING`	From the first row of the partition
`n PRECEDING`	n rows before the current row
`CURRENT ROW`	The current row
`n FOLLOWING`	n rows after
`UNBOUNDED FOLLOWING`	To the last row of the partition

Frames relative to the current row (highlighted). A running total uses everything up to the current row; a 3-row moving average uses the previous two rows plus the current one.

COMMON MISTAKES

The default frame trap

When `ORDER BY` is present and no frame is written, SQL Server uses `RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW`. `RANGE` treats rows with the **same ORDER BY value** as peers and includes them all, which gives surprising results with ties, and `LAST_VALUE` then returns the current row's value. For running totals always write `ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` explicitly; it is also much faster.

Running totals

First build the monthly sales table, then add a running (cumulative) total and a share:

SQL

```sql
WITH monthly AS (
    SELECT YEAR(order_date) AS yr, MONTH(order_date) AS mth, SUM(sales) AS sales
    FROM superstore
    GROUP BY YEAR(order_date), MONTH(order_date)
)
SELECT yr, mth,
       ROUND(sales, 0) AS sales,
       ROUND(SUM(sales) OVER (PARTITION BY yr ORDER BY mth
                              ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW), 0) AS
ytd_sales,
       ROUND(100.0 * SUM(sales) OVER (PARTITION BY yr ORDER BY mth
                              ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)
                  / SUM(sales) OVER (PARTITION BY yr), 1)                          AS
ytd_pct_of_year
FROM monthly
ORDER BY yr, mth;
```

RESULT

yr	mth	sales	ytd_sales	ytd_pct_of_year
2023	1	2,447,151	2,447,151	4.2
2023	2	3,861,852	6,309,002	10.9
2023	3	3,441,306	9,750,308	16.9
2023	4	3,927,557	13,677,865	23.7
2023	5	4,840,130	18,517,994	32.1
2023	6	5,785,712	24,303,706	42.2
2023	7	5,390,409	29,694,115	51.5
2023	8	4,129,386	33,823,500	58.7
2023	9	5,714,731	39,538,231	68.6
2023	10	5,134,331	44,672,561	77.5
2023	11	5,815,308	50,487,869	87.6
2023	12	7,152,348	57,640,217	100.0
2024	1	6,022,738	6,022,738	9.2
2024	2	2,463,147	8,485,885	13.0

36 rows × 5 columns (first 14 shown)

`PARTITION BY yr` restarts the running total every January (year-to-date). The percentage uses a second window (the whole year) as the denominator.

Moving averages

A moving average smooths noisy data and shows the trend. Here is the 7-day average of daily units sold:

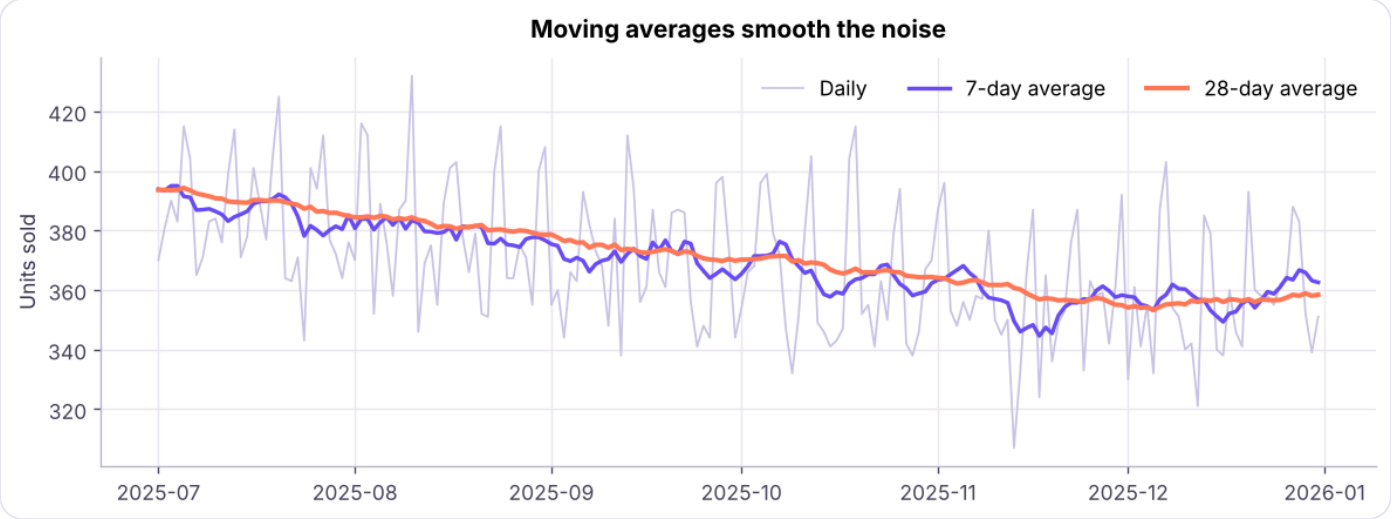
SQL

```sql
SELECT date, units_sold,
       ROUND(AVG(units_sold * 1.0) OVER (ORDER BY date ROWS BETWEEN 6 PRECEDING AND CURRENT
ROW), 1) AS avg_7_day
FROM daily_sales
WHERE date ⩾ '2025-11-01'
ORDER BY date;
```

RESULT

date	units_sold	avg_7_day
2025-11-01	387	387.0
2025-11-02	396	391.5
2025-11-03	353	378.7
2025-11-04	348	371.0
2025-11-05	356	368.0
2025-11-06	350	365.0
2025-11-07	358	364.0
2025-11-08	357	359.7

61 rows × 3 columns (first 8 shown)



Daily units sold and the 7-day and 28-day moving averages (last 6 months of 2025). The weekly pattern (weekend peaks) disappears in the averages; the 28-day line shows the trend.

The first six rows of a 7-row window have fewer than seven rows available: SQL averages whatever exists (partial windows). Check `COUNT(*) OVER (...)` if you need to exclude them.

LAG and LEAD: compare with previous and next rows

Function	Returns
`LAG(col, n, default)`	The value n rows **before** the current row (default n = 1)
`LEAD(col, n, default)`	The value n rows **after**

SQL

```
WITH monthly AS (
    SELECT YEAR(order_date) * 100 + MONTH(order_date) AS ym, SUM(sales) AS sales
    FROM superstore
    GROUP BY YEAR(order_date) * 100 + MONTH(order_date)
)
SELECT ym,
       ROUND(sales, 0)                                                    AS sales,
       ROUND(LAG(sales)  OVER (ORDER BY ym), 0)                           AS prev_month,
       ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY ym)) / LAG(sales) OVER (ORDER BY
ym), 1) AS mom_pct,
       ROUND(LAG(sales, 12) OVER (ORDER BY ym), 0)                        AS
same_month_last_year,
       ROUND(100.0 * (sales - LAG(sales, 12) OVER (ORDER BY ym)) / LAG(sales, 12) OVER
(ORDER BY ym), 1) AS yoy_pct,
       ROUND(LEAD(sales) OVER (ORDER BY ym), 0)                           AS next_month
FROM monthly
ORDER BY ym;
```

RESULT

ym	sales	prev_month	mom_pct	same_month_last_year	yoy_pct	next_mor
202301	2,447,151	NULL	NULL	NULL	NULL	3,861,85
202302	3,861,852	2,447,151	57.8	NULL	NULL	3,441,30
202303	3,441,306	3,861,852	-10.9	NULL	NULL	3,927,55
202304	3,927,557	3,441,306	14.1	NULL	NULL	4,840,13
202305	4,840,130	3,927,557	23.2	NULL	NULL	5,785,71
202306	5,785,712	4,840,130	19.5	NULL	NULL	5,390,40
202307	5,390,409	5,785,712	-6.8	NULL	NULL	4,129,38
202308	4,129,386	5,390,409	-23.4	NULL	NULL	5,714,73
202309	5,714,731	4,129,386	38.4	NULL	NULL	5,134,33
202310	5,134,331	5,714,731	-10.2	NULL	NULL	5,815,30
202311	5,815,308	5,134,331	13.3	NULL	NULL	7,152,34
202312	7,152,348	5,815,308	23.0	NULL	NULL	6,022,73

36 rows × 7 columns (first 12 shown)

- **MoM** compares with `LAG(sales, 1)`; **YoY** compares with `LAG(sales, 12)` when there is exactly one row per month. This is the SQL equivalent of the pivot table's "Difference From" settings (Excel Lesson 8).
- The first row has no previous value, so `LAG` returns NULL (or the default you supply: `LAG(sales, 1, 0)`).

WATCH OUT **Gaps in the data**

`LAG(sales, 12)` assumes every month exists. If a month has no sales it also has no row, so LAG(12) points at the wrong month. Build a calendar table (a list of every month) and LEFT JOIN the sales onto it first.

Time between events

"How many days passed between each customer's consecutive orders?"

SQL

```
WITH o AS (
    SELECT customer_id, order_date,
           LAG(order_date) OVER (PARTITION BY customer_id ORDER BY order_date) AS
prev_order
    FROM orders
)
SELECT COUNT(*)                                          AS repeat_orders,
       ROUND(AVG(DATEDIFF(day, prev_order, order_date) * 1.0), 1) AS
avg_days_between_orders,
       MIN(DATEDIFF(day, prev_order, order_date))        AS shortest_gap,
       MAX(DATEDIFF(day, prev_order, order_date))        AS longest_gap
FROM o
WHERE prev_order IS NOT NULL;
```

RESULT

repeat_orders	avg_days_between_orders	shortest_gap	longest_gap
2415	186.6	0	1001

1 rows × 4 columns

FIRST_VALUE, LAST_VALUE, NTH_VALUE

SQL

```
SELECT department, emp_id, monthly_income,
       FIRST_VALUE(emp_id) OVER (PARTITION BY department ORDER BY monthly_income DESC)
AS top_earner,
       LAST_VALUE(emp_id)  OVER (PARTITION BY department ORDER BY monthly_income DESC
                                 ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING)
AS lowest_earner,
       monthly_income - FIRST_VALUE(monthly_income) OVER (PARTITION BY department ORDER BY
monthly_income DESC) AS gap_to_top
FROM employees
ORDER BY department, monthly_income DESC;
```

RESULT

department	emp_id	monthly_income	top_earner	lowest_earner	gap_to_top
Finance	E00519	209833	E00519	E00322	0
Finance	E00145	185500	E00519	E00322	-24333
Finance	E00336	174417	E00519	E00322	-35416
Finance	E00201	169500	E00519	E00322	-40333
Finance	E01172	168167	E00519	E00322	-41666
Finance	E01151	155167	E00519	E00322	-54666
Finance	E00759	152083	E00519	E00322	-57750
Finance	E01004	144667	E00519	E00322	-65166

1,500 rows × 6 columns (first 8 shown)

`LAST_VALUE` needs the explicit full frame (`UNBOUNDED FOLLOWING`); with the default frame it just returns the current row.

Distribution functions

Function	Meaning
`PERCENT_RANK()`	(rank − 1) ÷ (rows − 1): 0 for the lowest, 1 for the highest
`CUME_DIST()`	Share of rows with a value ≤ the current one
`PERCENTILE_CONT(p) WITHIN GROUP (ORDER BY x) OVER (…)`	Interpolated percentile (e.g. median at p = 0.5)
`PERCENTILE_DISC(p) …`	An actual value from the data

SQL

```sql
SELECT DISTINCT department,
       ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY monthly_income) OVER (PARTITION BY
department), 0) AS median_income,
       ROUND(AVG(monthly_income) OVER (PARTITION BY department), 0)
AS mean_income
FROM employees
ORDER BY department;
```

RESULT

department	median_income	mean_income
Finance	77,709	83,681
HR	86,833	87,805
IT	68,750	79,051
Operations	92,667	92,461
R&D	70,250	80,255
Sales	74,709	82,448

6 rows × 3 columns

In SQL Server, `PERCENTILE_CONT` is a window function (needs `OVER`); PostgreSQL uses it as an aggregate (`GROUP BY`). Mean above median is the signature of a right-skewed distribution (Statistics, ML track).

Cumulative share: the Pareto in SQL

Which products produce the first 80% of sales?

SQL

```sql
WITH p AS (
    SELECT product_id, SUM(sales) AS sales FROM order_items GROUP BY product_id
), c AS (
    SELECT product_id, sales,
           SUM(sales) OVER (ORDER BY sales DESC ROWS BETWEEN UNBOUNDED PRECEDING AND
CURRENT ROW) * 1.0 / SUM(sales) OVER () AS cum_share,
           ROW_NUMBER() OVER (ORDER BY sales DESC) AS rnk
    FROM p
)
SELECT COUNT(*) AS products_for_80pct_of_sales,
       (SELECT COUNT(*) FROM p)  AS total_products
FROM c
WHERE cum_share ⩽ 0.8;
```

RESULT

products_for_80pct_of_sales	total_products
20	78

1 rows × 2 columns

Gaps and islands: streaks

A classic: find runs of consecutive days. The trick is that `date - ROW_NUMBER()` is **constant within a streak**.

SQL

```sql
WITH days AS (
    SELECT DISTINCT order_date AS d FROM orders WHERE order_date ⩾ '2025-12-01'
), g AS (
    SELECT d, DATEADD(day, -ROW_NUMBER() OVER (ORDER BY d), d) AS grp FROM days
)
SELECT MIN(d) AS streak_start, MAX(d) AS streak_end, COUNT(*) AS days_in_streak
FROM g
GROUP BY grp
ORDER BY streak_start;
```

RESULT

streak_start	streak_end	days_in_streak
2025-12-01	2025-12-31	31

1 rows × 3 columns

Each group of consecutive order days becomes one row: all of December 2025 had orders every day, so there is a single streak. The same pattern finds consecutive login days, uninterrupted employment spells and missing-sequence ranges.

Summary

KEY TAKEAWAYS

- **Frames** (`ROWS BETWEEN … AND …`) choose the rows around the current row; always write `ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` for running totals (the default RANGE frame includes ties).
- Running totals, YTD, cumulative % and moving averages use aggregate functions with frames.
- `LAG`/`LEAD` read the previous/next row (MoM, YoY, time between events); fill calendar gaps first.
- `FIRST_VALUE`/`LAST_VALUE` need an explicit full frame for LAST_VALUE.
- `PERCENT_RANK`, `CUME_DIST`, `PERCENTILE_CONT` describe position and median within a distribution.
- Gaps and islands: `value - ROW_NUMBER()` is constant within a consecutive run.

Quick quiz

6 questions · tap an answer to check it

1. Which frame gives a 3-month moving average (current and two previous rows)?

A `ROWS BETWEEN 3 PRECEDING AND CURRENT ROW`

B `ROWS BETWEEN 2 PRECEDING AND CURRENT ROW`

C `ROWS BETWEEN CURRENT ROW AND 2 FOLLOWING`

D `RANGE 3`

Answer: B. Two preceding rows plus the current row make three rows.

2. What does `LAG(sales, 12) OVER (ORDER BY ym)` return for monthly data?

A Sales 12 months in the future

B Sales for the same month last year (if no month is missing)

C The 12th highest sales

D The average of 12 months

Answer: B. LAG reads n rows back; with one row per month that is 12 months earlier.

3. Why does LAST_VALUE often return the current row's value?

A The default frame ends at the current row, so you must extend it to UNBOUNDED FOLLOWING

B LAST_VALUE is broken in SQL Server

C It only works with PARTITION BY

D ORDER BY is missing

Answer: A. Specify ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING.

4. How do you restart a running total every year?

A ORDER BY year

B PARTITION BY year in the OVER clause

C GROUP BY year only

D Use LAG

Answer: B. Each partition has its own running total.

5. In the gaps-and-islands trick, what stays constant inside a streak of consecutive dates?

A The date

B The row number

C The date minus its ROW_NUMBER

D The count

Answer: C. Both increase by 1 per row, so their difference is constant.

6. **Mean income is much higher than median income. What does that suggest?**

A Symmetric data

B A right-skewed distribution with a few very high values

C Missing data

D A data error

Answer: B. A long right tail pulls the mean above the median.

EXERCISE **Time-series practice**

(1) Monthly sales per region with MoM %; (2) for each customer, the days since their previous order and a flag for "returned within 30 days"; (3) a 3-month moving average of sales by category; (4) the month when cumulative 2025 sales first exceeded ₹5 crore.

Show solution (worked answer)

SQL

```sql
-- (1)
WITH m AS (SELECT region, YEAR(order_date)*100+MONTH(order_date) AS ym, SUM(sales) AS
sales FROM superstore GROUP BY region, YEAR(order_date)*100+MONTH(order_date))
SELECT region, ym, ROUND(sales,0) AS sales,
       ROUND(100.0*(sales - LAG(sales) OVER (PARTITION BY region ORDER BY ym)) /
LAG(sales) OVER (PARTITION BY region ORDER BY ym), 1) AS mom_pct FROM m ORDER BY
region, ym;
-- (2)
WITH o AS (SELECT customer_id, order_date, LAG(order_date) OVER (PARTITION BY
customer_id ORDER BY order_date) AS prev FROM orders)
SELECT customer_id, order_date, DATEDIFF(day, prev, order_date) AS days_since_prev,
       CASE WHEN DATEDIFF(day, prev, order_date) <= 30 THEN 1 ELSE 0 END AS
within_30_days FROM o WHERE prev IS NOT NULL;
-- (3)
WITH m AS (SELECT category, YEAR(order_date)*100+MONTH(order_date) AS ym, SUM(sales)
AS sales FROM superstore GROUP BY category, YEAR(order_date)*100+MONTH(order_date))
SELECT category, ym, AVG(sales) OVER (PARTITION BY category ORDER BY ym ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS ma3 FROM m;
-- (4)
WITH m AS (SELECT MONTH(order_date) AS mth, SUM(sales) AS sales FROM superstore WHERE
YEAR(order_date) = 2025 GROUP BY MONTH(order_date)),
     c AS (SELECT mth, SUM(sales) OVER (ORDER BY mth ROWS BETWEEN UNBOUNDED PRECEDING
AND CURRENT ROW) AS cum FROM m)
SELECT TOP 1 mth FROM c WHERE cum > 50000000 ORDER BY mth;
```

INTERVIEW CORNER

Questions you may be asked

Q: How do you calculate a running total in SQL? `SUM(x) OVER (ORDER BY d ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)`, optionally partitioned.

Q: How do you calculate month-on-month growth? Aggregate to months, then `(sales - LAG(sales) OVER (ORDER BY month)) / LAG(sales) OVER (ORDER BY month)`; use `LAG(…, 12)` for YoY.

Q: ROWS vs RANGE in a window frame? ROWS counts physical rows; RANGE includes all rows with the same ORDER BY value as the current row (peers). ROWS is predictable and faster.

Q: How would you find consecutive login days? Gaps and islands: `date - ROW_NUMBER() OVER (PARTITION BY user ORDER BY date)` as a group key, then GROUP BY it.

SQL FOR DATA ANALYSIS · LESSON 13

CTEs, Views, Temporary Tables, Stored Procedures & Functions

Write queries other people can read and reuse. Use CTEs (including recursive CTEs for hierarchies and calendars), create views for reusable logic and security, use temp tables and table variables, and build stored procedures, user-defined functions, parameters and basic error handling in T-SQL.

SQL class 6 · SQL Server 2022, 2026 · Advanced · 130 min

By the end of this lesson you can

- Structure complex queries with one or more CTEs and explain CTE vs subquery vs temp table
- Write recursive CTEs for hierarchies (org charts) and number or date series
- Create, alter and use views, and explain what views are good for
- Choose between temp tables, table variables and CTEs
- Write stored procedures with parameters, scalar and table-valued functions, and TRY... CATCH

CTE: a named, readable subquery

A **Common Table Expression** is a named result set defined with `WITH` that exists for **one statement**. It makes complex logic read top to bottom, like paragraphs, instead of inside-out.

SQL

```
WITH name1 AS (
    SELECT …
),
name2 AS (
    SELECT … FROM name1 …          -- later CTEs can use earlier ones
)
SELECT … FROM name2 …;
```

From nested subqueries to CTEs

Question: *"Which regions had above-average 2025 sales per customer?"*

SQL

```sql
WITH customer_sales AS (                        -- step 1: sales per customer in 2025
    SELECT c.region, c.customer_id, SUM(oi.sales) AS sales
    FROM order_items AS oi
    JOIN orders    AS o ON o.order_id    = oi.order_id
    JOIN customers AS c ON c.customer_id = o.customer_id
    WHERE o.order_date ⩾ '2025-01-01' AND o.order_date < '2026-01-01'
    GROUP BY c.region, c.customer_id
),
region_avg AS (                                 -- step 2: average per region
    SELECT region, COUNT(*) AS customers, AVG(sales) AS avg_sales_per_customer
    FROM customer_sales
    GROUP BY region
)
SELECT region, customers,
       ROUND(avg_sales_per_customer, 0)                      AS avg_sales_per_customer,
       ROUND((SELECT AVG(sales) FROM customer_sales), 0)      AS overall_avg
FROM region_avg
WHERE avg_sales_per_customer > (SELECT AVG(sales) FROM customer_sales)   -- step 3: compare
ORDER BY avg_sales_per_customer DESC;
```

RESULT

region	customers	avg_sales_per_customer	overall_avg
West	156	142,513	125,805
South	160	130,010	125,805

2 rows × 4 columns

`customer_sales` is used **twice**, which with plain subqueries would mean copy-pasting the same block. CTEs also make window-function filters (Lesson 11) clean.

> **TIP** **CTE style**
>
> Name CTEs for what they contain (`customer_sales`, not `cte1`); keep each CTE doing one job; comment each step. If a CTE needs a long explanation, it probably should be two CTEs.

CTE, subquery, view and temp table

	Subquery / derived table	CTE	View	Temp table
Lives for	The statement	The statement	Permanently (a saved query)	The session (`#t`)
Stores data?	No	No (re-evaluated)	No (unless indexed/materialised)	**Yes**
Reusable in other queries	No	No	**Yes**	In the same session
Can be referenced several times	Only by repeating	**Yes**, in the same statement	Yes	Yes
Recursive	No	**Yes**	No	No
Can be indexed	No	No	Indexed views only	**Yes**
Best for	Simple one-off logic	Readability, multi-step logic, recursion	Sharing logic, hiding complexity, security	Large intermediate results used many times

> **NOTE** **A CTE is not a cache**
>
> In SQL Server the CTE's query may run each time it is referenced. If an expensive CTE is used many times, load it into a **temp table** once.

Recursive CTEs

A recursive CTE has an **anchor** (starting rows) and a **recursive member** that refers to the CTE itself, joined with `UNION ALL`. It repeats until no new rows are produced.

Number and date series

SQL

```sql
WITH numbers AS (
    SELECT 1 AS n                       -- anchor
    UNION ALL
    SELECT n + 1 FROM numbers WHERE n < 7      -- recursive member with a stop condition
)
SELECT n FROM numbers;
```

RESULT

n
1
2
3
4
5
6
7

7 rows × 1 columns

A **calendar table** of every day in 2026, handy for filling gaps before LAG/LEAD (Lesson 12):

SQL

```
WITH calendar AS (
    SELECT CAST('2026-01-01' AS DATE) AS d
    UNION ALL
    SELECT CAST(DATEADD(day, 1, d) AS DATE) FROM calendar WHERE d < CAST('2026-12-31' AS
DATE)
)
SELECT COUNT(*) AS days, MIN(d) AS first_day, MAX(d) AS last_day,
       SUM(CASE WHEN DATENAME(weekday, d) IN ('Saturday', 'Sunday') THEN 1 ELSE 0 END) AS
weekend_days
FROM calendar;
```

RESULT

days	first_day	last_day	weekend_days
365	2026-01-01	2026-12-31	104

1 rows × 4 columns

In SQL Server, the default limit is 100 recursions, so a year-long calendar needs `OPTION (MAXRECURSION 400)` after the final SELECT (use `0` for no limit, with care).

Hierarchies: org chart

SQL

```sql
CREATE TABLE org (emp_id INT, name VARCHAR(20), manager_id INT);
INSERT INTO org VALUES (1,'Varun',NULL), (2,'Asha',1), (3,'Ravi',1), (4,'Meera',2),
(5,'Kabir',2), (6,'Zoya',4), (7,'Farhan',3);

WITH tree AS (
    SELECT emp_id, name, manager_id, 0 AS level, CAST(name AS VARCHAR(200)) AS path
    FROM org WHERE manager_id IS NULL                              -- anchor: the top
    UNION ALL
    SELECT o.emp_id, o.name, o.manager_id, t.level + 1, CAST(CONCAT(t.path, ' > ', o.name)
AS VARCHAR(200))
    FROM org AS o
    JOIN tree AS t ON o.manager_id = t.emp_id                      -- walk down one level
)
SELECT level, CONCAT(REPLICATE('    ', level), name) AS org_chart, path
FROM tree
ORDER BY path;
```

RESULT

```
Commands completed successfully.
(7 rows affected)
```

level	org_chart	path
0	Varun	Varun
1	Asha	Varun > Asha
2	Kabir	Varun > Asha > Kabir
2	Meera	Varun > Asha > Meera
3	Zoya	Varun > Asha > Meera > Zoya
1	Ravi	Varun > Ravi
2	Farhan	Varun > Ravi > Farhan

7 rows × 3 columns

Recursive CTEs also solve bill-of-materials, category trees and graph-path problems.

Views

A **view** is a saved SELECT with a name. Querying it runs the stored query; it holds no data of its own.

SQL

```sql
CREATE VIEW vw_order_detail AS
SELECT o.order_id, o.order_date, o.ship_mode,
       c.customer_id, c.customer_name, c.segment, c.city, c.region,
       p.product_id, p.product_name, p.category, p.sub_category,
       oi.quantity, oi.discount, oi.sales, oi.profit
FROM order_items AS oi
JOIN orders    AS o ON o.order_id    = oi.order_id
JOIN customers AS c ON c.customer_id = o.customer_id
JOIN products  AS p ON p.product_id  = oi.product_id;

SELECT category, region, ROUND(SUM(sales), 0) AS sales
FROM vw_order_detail
WHERE order_date ⩾ '2025-01-01'
GROUP BY category, region
ORDER BY category, region;
```

RESULT

category	region	sales
Furniture	East	2,875,816
Furniture	North	2,688,426
Furniture	South	3,759,688
Furniture	West	3,436,316
Office Supplies	East	1,075,076
Office Supplies	North	833,204
Office Supplies	South	1,018,343
Office Supplies	West	1,044,699
Technology	East	14,959,309
Technology	North	12,911,446
Technology	South	16,023,523
Technology	West	17,750,953

12 rows × 3 columns

Now analysts query one friendly object instead of repeating a four-table join. Power BI and Excel can connect to the view too.

Use views to	Notes
Hide complexity	One name for a multi-join query
Provide one definition of a metric	"Net sales" defined once, used everywhere
Secure data	`GRANT SELECT ON vw_sales_summary TO analysts` without granting access to base tables; expose only some columns/rows
Keep apps working when tables change	Change the view, not every report

SQL Server syntax: `CREATE VIEW dbo.vw_x AS …`, `ALTER VIEW`, `DROP VIEW IF EXISTS`, `CREATE OR ALTER VIEW`. Rules: no `ORDER BY` inside a view (unless with TOP); views can usually be `INSERT`/`UPDATE`d only if they map simply to one table. **Indexed (materialised) views** store the results physically for speed (`WITH SCHEMABINDING` + a unique clustered index).

Temporary tables and table variables

SQL

```sql
-- Temp table: lives for the session, stored in tempdb, can be indexed, has statistics
SELECT customer_id, SUM(sales) AS sales
INTO   #cust_sales                    -- the # prefix makes it local temporary
FROM   superstore
GROUP  BY customer_id;

CREATE INDEX ix_cust ON #cust_sales (customer_id);
SELECT TOP 10 * FROM #cust_sales ORDER BY sales DESC;
DROP TABLE IF EXISTS #cust_sales;

-- Table variable: scoped to the batch; for small row counts
DECLARE @top TABLE (customer_id VARCHAR(10), sales DECIMAL(14,2));
INSERT INTO @top SELECT TOP 5 customer_id, SUM(sales) FROM superstore GROUP BY customer_id
ORDER BY SUM(sales) DESC;
SELECT * FROM @top;
```

##name is a **global** temp table visible to other sessions. Temp tables vanish when the session ends.

Stored procedures

A **stored procedure** is a named, saved block of T-SQL that can take parameters, run many statements, and return results. It is compiled and its plan cached, and permissions can be granted on the procedure instead of the tables.

SQL

```sql
CREATE OR ALTER PROCEDURE dbo.usp_sales_by_region
    @from_date DATE,
    @to_date   DATE,
    @category  VARCHAR(30) = NULL          -- optional parameter with a default
AS
BEGIN
    SET NOCOUNT ON;                        -- suppress "(n rows affected)" messages

    SELECT c.region,
           COUNT(DISTINCT o.order_id) AS orders,
           SUM(oi.sales)              AS sales,
           SUM(oi.profit)             AS profit
    FROM dbo.order_items AS oi
    JOIN dbo.orders      AS o ON o.order_id    = oi.order_id
    JOIN dbo.customers   AS c ON c.customer_id = o.customer_id
    JOIN dbo.products    AS p ON p.product_id  = oi.product_id
    WHERE o.order_date ⩾ @from_date
      AND o.order_date <  DATEADD(day, 1, @to_date)
      AND (@category IS NULL OR p.category = @category)     -- optional filter
    GROUP BY c.region
    ORDER BY sales DESC;
END;
GO

EXEC dbo.usp_sales_by_region @from_date = '2025-01-01', @to_date = '2025-12-31';
EXEC dbo.usp_sales_by_region '2025-01-01', '2025-12-31', 'Technology';
```

Output parameters, variables, IF and loops

SQL

```sql
CREATE OR ALTER PROCEDURE dbo.usp_add_order_note
    @order_id VARCHAR(20),
    @note     NVARCHAR(200),
    @rows     INT OUTPUT
AS
BEGIN
    SET NOCOUNT ON;
    IF NOT EXISTS (SELECT 1 FROM dbo.orders WHERE order_id = @order_id)
    BEGIN
        RAISERROR('Order %s does not exist.', 16, 1, @order_id);   -- or THROW 50001,
'Order does not exist.', 1;
        RETURN;
    END;

    UPDATE dbo.orders SET note = @note WHERE order_id = @order_id;
    SET @rows = @@ROWCOUNT;                                        -- rows changed by the
last statement
END;
GO

DECLARE @n INT;
EXEC dbo.usp_add_order_note 'IN-2025-00042', N'Gift wrap', @rows = @n OUTPUT;
SELECT @n AS rows_updated;
```

Control flow	Syntax
Condition	`IF … BEGIN … END ELSE BEGIN … END`
Loop	`WHILE @i ⩽ 10 BEGIN … SET @i += 1; END`
Exit / return	`RETURN`
Row count	`@@ROWCOUNT`
Dynamic SQL	`EXEC sp_executesql @sql, N'@x INT', @x = 5` (use parameters; never concatenate user input: SQL injection)

Error handling: TRY…CATCH

SQL

```sql
BEGIN TRY
    BEGIN TRAN;
        UPDATE dbo.accounts SET balance = balance - 5000 WHERE acct_id = 1;
        UPDATE dbo.accounts SET balance = balance + 5000 WHERE acct_id = 2;
    COMMIT;
END TRY
BEGIN CATCH
    IF @@TRANCOUNT > 0 ROLLBACK;
    SELECT ERROR_NUMBER() AS err_no, ERROR_MESSAGE() AS err_msg, ERROR_LINE() AS err_line;
    THROW;                                      -- re-raise to the caller
END CATCH;
```

User-defined functions (UDFs)

Type	Returns	Example use
Scalar	One value	`dbo.fn_margin(profit, sales)`
Inline table-valued	A table, from one SELECT	`dbo.fn_orders_by_city('Pune')`: like a parameterised view
Multi-statement table-valued	A table built in a body	Complex logic (usually slower)

SQL

```sql
CREATE OR ALTER FUNCTION dbo.fn_margin (@profit DECIMAL(12,2), @sales DECIMAL(12,2))
RETURNS DECIMAL(6,2)
AS
BEGIN
    RETURN CASE WHEN @sales = 0 THEN NULL ELSE 100.0 * @profit / @sales END;
END;
GO

CREATE OR ALTER FUNCTION dbo.fn_sales_by_city (@city VARCHAR(30))
RETURNS TABLE
AS RETURN (
    SELECT order_id, order_date, sales, profit FROM dbo.vw_order_detail WHERE city = @city
);
GO

SELECT order_id, dbo.fn_margin(profit, sales) AS margin_pct FROM
dbo.fn_sales_by_city('Pune');
```

WATCH OUT **Scalar UDFs can be slow**

Older SQL Server versions call a scalar UDF once per row, which can make a query many times slower than writing the expression inline (SQL Server 2019+ can inline some). Prefer inline table-valued functions or views for set-based logic.

Procedure vs function vs view

	View	Stored procedure	Function
Parameters	No	Yes	Yes
Can change data (INSERT/UPDATE)	Limited	**Yes**	No (side-effect free)
Used in SELECT / FROM	Yes	No (EXEC)	**Yes**
Returns	A table	Result sets, output parameters, a status code	A value or a table

Triggers (awareness)

A **trigger** runs automatically on INSERT, UPDATE or DELETE: audit logs, maintaining summary tables. Use sparingly: they hide logic and can slow writes. Modern designs prefer explicit procedures, constraints, temporal tables or change data capture.

DO IT WITH AI **Do it with AI**

Prompt: *"Convert this nested-subquery SQL into readable CTEs with clear names and comments, keep the result identical, and explain each step."* AI is good at refactoring; verify equality by running both versions and comparing with EXCEPT in both directions (Lesson 9).

Summary

KEY TAKEAWAYS

- **CTE** (`WITH name AS (…)`): a named, readable subquery for one statement; can be referenced many times; **recursive CTEs** generate series and walk hierarchies (anchor + `UNION ALL` + recursive member).
- **View**: a saved query for reuse, one definition of a metric, and security. **Temp tables** (`#t`) store intermediate data and can be indexed; table variables suit small sets.
- **Stored procedures** take parameters, run multiple statements, change data and handle errors with `TRY…CATCH`; **functions** return values or tables and can be used inside queries.
- Prefer set-based SQL; avoid row-by-row loops and scalar UDFs on big tables; never concatenate user input into dynamic SQL.

Quick quiz

6 questions · tap an answer to check it

1. A CTE exists for:

A The whole session

B A single statement

C Until the server restarts

D As long as the database

Answer: B. A CTE is defined and used within one statement.

2. What are the two parts of a recursive CTE?

A SELECT and WHERE

B An anchor member and a recursive member joined with UNION ALL

C INSERT and UPDATE

D A view and a function

Answer: B. The anchor gives starting rows; the recursive member repeats until nothing new appears.

3. Which object stores data physically for the session and can be indexed?

A CTE

B View

C Temporary table

D Subquery

Answer: C.

temp tables live in tempdb for the session.

4. Which can modify data and return an OUTPUT parameter?

A View

B Stored procedure

C Scalar function

D CTE

Answer: B. Procedures can run INSERT/UPDATE/DELETE and return output parameters.

5. Why grant access to a view instead of the base table?

A Views are faster

B You can expose only selected columns and rows and hide the underlying tables

C Views cannot be dropped

D Tables cannot be queried

Answer: B. Views are a security and simplification layer.

6. A recursive CTE stops with an error after 100 levels. Fix?

A Add DISTINCT

B Add OPTION (MAXRECURSION n) (and a proper stop condition)

C Use a view

D Use TOP

Answer: B. SQL Server's default recursion limit is 100.

EXERCISE **Build reusable objects**

(1) A view `vw_monthly_sales` with year, month, category and total sales. (2) A CTE query that lists, for each region, the best-selling sub-category and its share of the region's sales. (3) A stored procedure `usp_top_customers (@year INT, @n INT)` that returns the top n customers by sales for a year. (4) A recursive CTE that lists all employees under manager 2 in the `org` table with their level.

Show solution (worked answer)

SQL

```sql
-- (1)
CREATE VIEW vw_monthly_sales AS
SELECT YEAR(order_date) AS yr, MONTH(order_date) AS mth, category, SUM(sales) AS sales
FROM vw_order_detail GROUP BY YEAR(order_date), MONTH(order_date), category;
-- (2)
WITH rs AS (SELECT region, sub_category, SUM(sales) AS sales FROM superstore GROUP BY
region, sub_category),
     r  AS (SELECT *, RANK() OVER (PARTITION BY region ORDER BY sales DESC) AS rnk,
100.0*sales/SUM(sales) OVER (PARTITION BY region) AS pct FROM rs)
SELECT region, sub_category, ROUND(pct,1) AS pct_of_region FROM r WHERE rnk = 1;
-- (3) T-SQL
CREATE OR ALTER PROCEDURE usp_top_customers @year INT, @n INT AS
BEGIN
  SELECT TOP (@n) customer_id, customer_name, SUM(sales) AS sales FROM vw_order_detail
WHERE YEAR(order_date) = @year
  GROUP BY customer_id, customer_name ORDER BY sales DESC;
END;
-- (4)
WITH t AS (SELECT emp_id, name, 1 AS level FROM org WHERE manager_id = 2
           UNION ALL SELECT o.emp_id, o.name, t.level + 1 FROM org o JOIN t ON
o.manager_id = t.emp_id)
SELECT * FROM t;
```

INTERVIEW CORNER

Questions you may be asked

Q: CTE vs temp table vs subquery? Subquery: inline, single use. CTE: named, readable, statement-scoped, supports recursion, may be re-evaluated. Temp table: materialised data for the session, indexable: use for large intermediate results reused many times.

Q: View vs stored procedure? A view is a saved SELECT you query like a table; a procedure is a program with parameters and multiple statements that can change data.

Q: What is a recursive CTE and where do you use it? A CTE that references itself, for hierarchies (org charts, categories), series generation and graph traversal.

Q: How do you prevent SQL injection in dynamic SQL? Use parameterised `sp_executesql` and QUOTENAME for identifiers; never concatenate raw user input.

SQL FOR DATA ANALYSIS · LESSON 14

Transactions, Indexes, Performance & Security

Make SQL safe and fast. Understand transactions and ACID (BEGIN, COMMIT, ROLLBACK, SAVE), isolation and locking, indexes (clustered, non-clustered, covering), how to read an execution plan, a checklist for writing fast queries, and permissions with GRANT, REVOKE and roles (DCL).

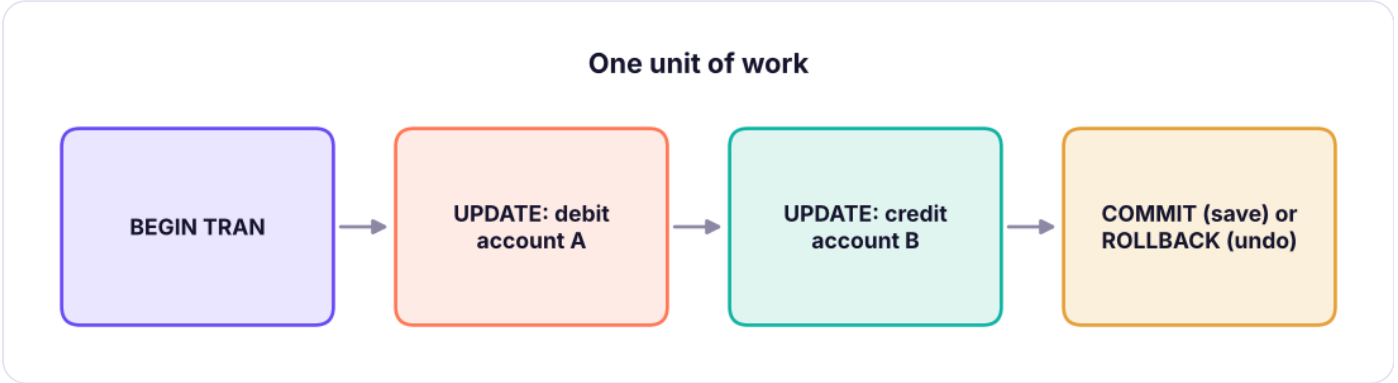
SQL class 6 (extended) · SQL Server 2022, 2026 · Advanced · 110 min

By the end of this lesson you can

- Explain ACID and group statements into transactions that commit or roll back together
- Describe isolation levels, locks, blocking and deadlocks at an analyst level
- Choose between clustered and non-clustered indexes and recognise queries that can or cannot use them
- Read the basics of an execution plan and apply a performance checklist
- Grant and revoke permissions with roles, and apply least privilege

Transactions and ACID

A **transaction** is a group of statements treated as one unit of work: **all succeed (commit) or none do (rollback)**. Moving ₹5,000 between two accounts is two updates; if the server crashes between them, money would vanish without a transaction.



A bank transfer as a transaction. Either both updates are saved, or both are undone.

Property	Meaning	Example
Atomicity	All or nothing	Both account updates happen, or neither
Consistency	Data stays valid (constraints hold)	Total money before = total after; no negative balance if a CHECK forbids it
Isolation	Concurrent transactions do not see each other's half-finished work	Another user does not see the debit without the credit
Durability	Once committed, it survives a crash	Transaction log is written to disk first

Transaction commands

Command	Purpose
`BEGIN TRANSACTION` (`BEGIN TRAN`)	Start a transaction
`COMMIT`	Make all changes permanent
`ROLLBACK`	Undo everything since BEGIN (or since a savepoint)
`SAVE TRANSACTION name`	Create a savepoint inside the transaction
`ROLLBACK TRANSACTION name`	Undo back to the savepoint only
`@@TRANCOUNT`	How many transactions are open (should be 0 when you leave)

By default SQL Server runs in **autocommit** mode: every single statement is its own transaction. Wrap several statements in `BEGIN TRAN … COMMIT` to group them.

Seeing a rollback

SQL

```sql
CREATE TABLE accounts (acct_id INT PRIMARY KEY, holder VARCHAR(20), balance DECIMAL(12,2));
INSERT INTO accounts VALUES (1, 'Asha', 20000), (2, 'Ravi', 5000);

BEGIN TRANSACTION;
UPDATE accounts SET balance = balance - 5000 WHERE acct_id = 1;
UPDATE accounts SET balance = balance + 5000 WHERE acct_id = 2;
SELECT * FROM accounts ORDER BY acct_id;      -- inside the transaction: changes visible to
us
ROLLBACK;                                     -- something went wrong: undo both updates

SELECT * FROM accounts ORDER BY acct_id;      -- balances are back to the originals
```

RESULT

```
Commands completed successfully.
(2 rows affected)
Commands completed successfully.
(1 row affected)
(1 row affected)
Commands completed successfully.
```

RESULT SET 1

acct_id	holder	balance
1	Asha	15,000
2	Ravi	10,000

2 rows × 3 columns

RESULT SET 2

acct_id	holder	balance
1	Asha	20,000
2	Ravi	5,000

2 rows × 3 columns

The safe-update pattern

The pattern that protects you from the "UPDATE without WHERE" disaster (Lesson 4):

SQL

```sql
BEGIN TRAN;

UPDATE dbo.staff SET salary = salary * 1.05 WHERE dept_id = 10;

SELECT @@ROWCOUNT AS rows_changed;                    -- expected: 2?
SELECT emp_id, salary FROM dbo.staff WHERE dept_id = 10;     -- sanity-check

-- If everything looks right:
COMMIT;
-- Otherwise:
-- ROLLBACK;
```

WATCH OUT **Never leave a transaction open**

An open transaction holds locks that block other users and keeps the log from truncating. Always end with COMMIT or ROLLBACK; check `SELECT @@TRANCOUNT;` if a query window seems "stuck".

Concurrency: isolation, locks and deadlocks

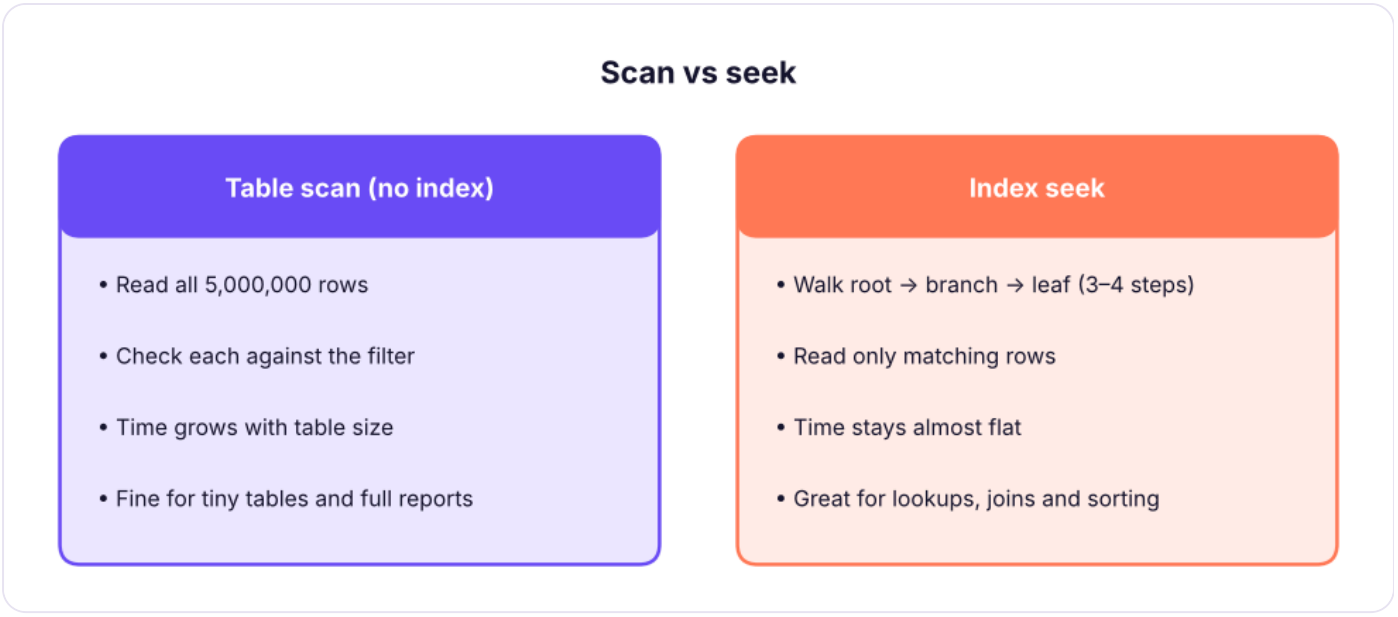
When many users work at once, the database uses **locks** to keep transactions isolated. **Isolation levels** trade consistency for concurrency:

Level	Prevents	Notes
`READ UNCOMMITTED`	Nothing (allows dirty reads; `NOLOCK` hint)	Can read data that is later rolled back; avoid for reports needing accuracy
`READ COMMITTED` (default)	Dirty reads	A row you read can change before you read it again
`REPEATABLE READ`	Dirty and non-repeatable reads	Holds read locks longer
`SERIALIZABLE`	Also phantom rows	Safest, least concurrent
`SNAPSHOT` / RCSI	Readers do not block writers (row versions)	Common in modern systems, uses tempdb

- **Blocking**: one session waits for a lock held by another. Fix: shorter transactions, indexes, row versioning.
- **Deadlock**: two sessions each hold a lock the other needs; SQL Server kills one (the "victim", error 1205). Fix: access tables in the same order, keep transactions short, retry on error 1205.

Indexes: how databases find rows fast

Without an index, a query like `WHERE customer_id = 'C0042'` scans **every row** (table scan). An **index** is a sorted structure (a B-tree) that lets the database jump straight to matching rows, like the index of a book.



A table scan reads every row. An index seek follows a short path through a B-tree.

Index type	What it is	How many per table
Clustered	Defines the physical order of the table's rows; the table *is* the index. Created by default with a PRIMARY KEY	**One**
Non-clustered	A separate sorted copy of selected columns with pointers to the rows	Many (up to 999)
Unique	Enforces uniqueness (created for UNIQUE constraints)	Many
Covering (`INCLUDE`)	A non-clustered index that contains every column a query needs, so the table is never touched	
Filtered	Indexes only rows matching a condition	
Columnstore	Column-wise compressed storage for analytics on huge tables	

SQL

```sql
CREATE CLUSTERED INDEX cx_orders_date ON dbo.orders (order_date);          -- if not already clustered by the PK

CREATE NONCLUSTERED INDEX ix_orders_customer
    ON dbo.orders (customer_id)
    INCLUDE (order_date, ship_mode);                                        -- covering index for "orders per customer" queries

CREATE NONCLUSTERED INDEX ix_items_product ON dbo.order_items (product_id);   -- foreign keys deserve indexes

DROP INDEX ix_orders_customer ON dbo.orders;
```

SQL

```sql
CREATE INDEX ix_super_city ON superstore (city);
SELECT index_name, table_name FROM duckdb_indexes() WHERE table_name = 'superstore';
```

RESULT

index_name	table_name
ix_super_city	superstore

1 rows × 2 columns

Which queries can use an index?

Pattern	Uses an index?	Why
`WHERE customer_id = 'C0042'`	**Yes** (seek)	Equality on the leading column
`WHERE order_date ⩾ '2025-01-01' AND order_date < '2026-01-01'`	**Yes**	Range on an indexed column
`WHERE YEAR(order_date) = 2025`	**No** (scan)	The function must run on every row: **non-SARGable**
`WHERE customer_name LIKE 'Ra%'`	Yes	Prefix match
`WHERE customer_name LIKE '%Rao'`	No	Leading wildcard
`WHERE UPPER(city) = 'PUNE'`	No	Function on the column (the default collation is already case-insensitive)
`WHERE city = 'Pune' OR product_id = 'P0012'`	Often no	OR across different columns
`WHERE CAST(order_id AS VARCHAR) = '1001'`	No	Type conversion on the column

TIP **SARGable predicates**

SARG = Search ARGument able. Keep the **column bare** on one side of the comparison and do the maths on the other side: use `order_date ⩾ '2025-01-01' AND order_date < '2026-01-01'` instead of `YEAR(order_date) = 2025`, and `sales > 100 / 1.18` instead of `sales * 1.18 > 100`.

Index trade-offs

Indexes speed up **reads** but slow down **writes** (every INSERT/UPDATE/DELETE must also update each index) and use disk space. Index columns used in `WHERE`, `JOIN`, `ORDER BY` and `GROUP BY`, and foreign keys; do not index everything. Analytic warehouses (Snowflake, BigQuery) use different techniques (partitioning, clustering, columnar storage) instead of manual indexes.

Execution plans

In SSMS press `Ctrl + L` (estimated plan) or `Ctrl + M` then run (actual plan). The plan is a diagram of operators read **right to left**. Things to look for:

Operator	Meaning	Concern
Index Seek	Direct lookup	Good
Index / Clustered Index Scan	Reads many rows	Fine for big reports, suspicious for single-row lookups
Table Scan	Reads the heap	Usually add an index
Key Lookup	Fetches extra columns from the table per row	Consider a covering index (`INCLUDE`)
Hash / Merge / Nested Loops Join	Join algorithms	Hash for big unsorted inputs; nested loops for small; merge for sorted
Sort	Sorting rows	Expensive; an index in the right order avoids it
Thick arrows, big estimated vs actual row differences	Bad estimates	Update statistics

`SET STATISTICS IO, TIME ON;` shows logical reads (the best measure of work) and CPU time for each query.

Performance checklist for analysts

1. **Select only the columns and rows you need**: avoid `SELECT *`; filter early.
2. **Make filters SARGable** (no functions on columns; use ranges for dates).
3. **Join on indexed keys** with matching data types (an `INT` joined to a `VARCHAR` forces conversions).
4. **Aggregate before joining** when joining big tables (reduce rows first, avoids row explosion).
5. **UNION ALL** instead of UNION; **EXISTS** instead of `IN (huge list)`; avoid `DISTINCT` as a bandage for join duplicates.
6. **Avoid row-by-row logic** (cursors, scalar UDF in WHERE); think in sets.
7. **Test on realistic data**; look at the actual plan; keep statistics updated.
8. **Pull less data into BI tools**: summarise in SQL, or create views and aggregate tables.
9. **Limit while developing**: `SELECT TOP 100` or `WHERE` on a date range, then remove it.

Security: DCL

Data Control Language gives and removes permissions. Follow the **principle of least privilege**: grant only what each person or application needs.

```sql
-- Logins (server level) and users (database level)
CREATE LOGIN priya_analyst WITH PASSWORD = 'Use-A-Strong-Passphrase-1!';
CREATE USER  priya_analyst FOR LOGIN priya_analyst;

-- Roles group permissions
CREATE ROLE analyst_role;
GRANT SELECT ON SCHEMA::dbo        TO analyst_role;          -- read all tables in dbo
DENY  SELECT ON dbo.customers (phone, email) TO analyst_role;     -- but hide personal columns
GRANT EXECUTE ON dbo.usp_sales_by_region TO analyst_role;

ALTER ROLE analyst_role ADD MEMBER priya_analyst;            -- SQL Server 2012+: ALTER ROLE … ADD MEMBER

REVOKE SELECT ON dbo.orders FROM analyst_role;               -- take back a permission
```

Statement	Effect
`GRANT`	Gives a permission
`REVOKE`	Removes a previously granted (or denied) permission
`DENY`	Explicitly blocks a permission (overrides grants)
Fixed roles	`db_datareader` (read all), `db_datawriter`, `db_owner` (full control: avoid)

Other protections: **Row-Level Security** (a region manager sees only their rows), **Dynamic Data Masking** (show `XXXX-1234`), **Always Encrypted** and **TDE** (encryption), **auditing**, and never embedding passwords in scripts (use Windows/Entra ID authentication or a secret store). In India, personal data handling also falls under the **DPDP Act 2023**: collect only what you need and restrict who can see it.

DO IT WITH AI — Do it with AI

Paste a slow query and its actual execution plan (as XML or text) into an assistant: *"Explain what this plan is doing and suggest indexes or rewrites. Do not invent columns."* AI can spot scans, key lookups and non-SARGable filters quickly, but test every suggestion; an index that helps one query can hurt a hundred writes.

Summary

KEY TAKEAWAYS

- A **transaction** groups statements to commit or roll back together; **ACID** = atomicity, consistency, isolation, durability. Use `BEGIN TRAN … COMMIT/ROLLBACK`, check `@@ROWCOUNT`, never leave transactions open.
- Isolation levels and locks keep concurrent users consistent; blocking and deadlocks are managed with short transactions and consistent access order.
- **Indexes** (clustered = row order, one per table; non-clustered = separate structure; covering = `INCLUDE`) speed up reads and slow writes. **SARGable** filters keep the column bare.
- Read execution plans for scans, key lookups and bad estimates; use `SET STATISTICS IO, TIME ON`.
- **DCL**: `GRANT`, `REVOKE`, `DENY`, roles, least privilege; protect personal data.

Quick quiz

6 questions · tap an answer to check it

1. Which ACID property says that a committed transaction survives a power failure?

A Atomicity

B Consistency

C Isolation

D Durability

Answer: D. Durability: committed changes are written to the log/disk.

2. Which filter can use an index on order_date?

A `WHERE YEAR(order_date) = 2025`

B `WHERE order_date ⩾ '2025-01-01' AND order_date < '2026-01-01'`

C `WHERE CONVERT(VARCHAR, order_date, 23) LIKE '2025%'`

D `WHERE DATEPART(year, order_date) = 2025`

Answer: B. The column must be bare (SARGable); functions on the column force a scan.

3. How many clustered indexes can a table have?

A None

B One

C One per column

D Unlimited

Answer: B. The clustered index defines the physical order, so there can be only one.

4. You ran `BEGIN TRAN; UPDATE …;` and closed the window. What is the risk?

A The transaction stays open, holding locks and blocking other users

B The update is committed automatically

C The table is dropped

D Nothing

Answer: A. Open transactions keep locks until committed or rolled back.

5. What does DENY do?

A Removes a permission like REVOKE

B Explicitly blocks a permission, overriding grants

C Deletes a login

D Grants read-only access

Answer: B. DENY takes precedence over GRANT.

6. In an execution plan, a "Key Lookup" repeated for many rows suggests:

A The table is empty

B Adding the needed columns to the index with INCLUDE (a covering index)

C Using UNION instead of UNION ALL

D Dropping the clustered index

Answer: B. Key lookups fetch missing columns from the table row by row; covering the query avoids them.

EXERCISE

Make it faster

This query is slow on 50 million order lines: `SELECT * FROM order_items WHERE YEAR(order_date) = 2025 AND UPPER(region) = 'WEST' ORDER BY sales DESC;` Rewrite it and suggest an index.

Show solution (worked answer)

SQL

```sql
SELECT item_id, order_id, product_id, quantity, sales, profit      -- only the needed columns
FROM dbo.order_items
WHERE order_date ≥ '2025-01-01' AND order_date < '2026-01-01'    -- SARGable range instead of YEAR()
  AND region = 'West'                                             -- default collation is case-insensitive; no UPPER needed
ORDER BY sales DESC;

CREATE NONCLUSTERED INDEX ix_items_date_region
ON dbo.order_items (order_date, region) INCLUDE (order_id, product_id, quantity, sales, profit);
```

Check the actual plan for a seek, compare logical reads with `SET STATISTICS IO ON`, and consider whether the extra index slows nightly loads.

INTERVIEW CORNER

Questions you may be asked

Q: What is ACID? Atomicity (all or nothing), Consistency (valid state), Isolation (concurrent transactions do not interfere), Durability (committed data persists).

Q: Clustered vs non-clustered index? A clustered index sorts and stores the table rows in index order (one per table). A non-clustered index is a separate structure with pointers to rows (many per table).

Q: Why is `WHERE YEAR(date) = 2025` slow? It applies a function to every row, so the index on date cannot be used (non-SARGable). Use a date range.

Q: How do you find and fix a slow query? Look at the actual execution plan and I/O statistics, find scans/lookups/bad estimates, make predicates SARGable, add or adjust indexes, reduce columns/rows early, and re-test.

SQL FOR DATA ANALYSIS · LESSON 15

End-to-End SQL Case Study & Interview Practice

Put everything together. Work through a complete analysis of the retail business (data quality, KPIs, trends, customer segmentation, retention, product and discount analysis) using joins, CTEs and window functions, then practise 25 classic SQL interview questions with model answers.

SQL class 5 (end-to-end case study) · 2026 · Advanced · 180 min

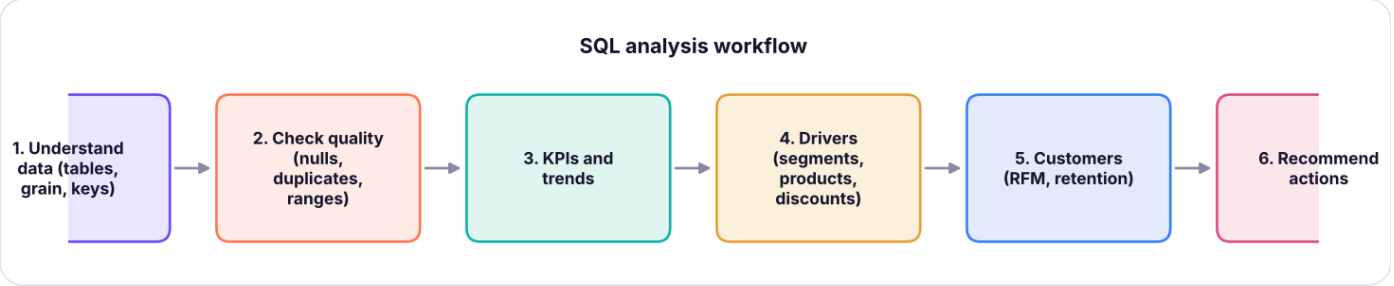
By the end of this lesson you can

- Run a structured SQL analysis: understand the data, check quality, calculate KPIs, find drivers, and recommend actions
- Combine joins, CTEs, aggregation and window functions in multi-step queries
- Build customer segments (RFM), cohort and retention views in SQL
- Answer common SQL interview questions with clear, correct queries
- Explain results in business language

The business problem

CASE STUDY **Retail growth review**

You are the data analyst at an Indian retailer selling Furniture, Office Supplies and Technology through four regions. Sales grew from 2023 to 2025, but the CFO suspects profit is not keeping pace. Management wants answers to five questions: **(1)** Is the data trustworthy? **(2)** How are sales and profit really trending? **(3)** Where are we losing money and why? **(4)** Who are our best customers and are they staying? **(5)** What should we do next quarter?



A repeatable SQL analysis workflow.

Step 1: understand the data

SQL

```
SELECT 'orders'       AS tbl, COUNT(*) AS n, MIN(order_date) AS first_date, MAX(order_date)
AS last_date FROM orders
UNION ALL SELECT 'order_items', COUNT(*), NULL, NULL FROM order_items
UNION ALL SELECT 'customers',   COUNT(*), NULL, NULL FROM customers
UNION ALL SELECT 'products',    COUNT(*), NULL, NULL FROM products;
```

RESULT

tbl	n	first_date	last_date
orders	3200	2023-01-01	2025-12-31
order_items	6323	NULL	NULL
customers	800	NULL	NULL
products	78	NULL	NULL

4 rows × 4 columns

Grain: `order_items` has one row per product line in an order; `orders` one row per order; `customers` one row per customer. KPIs like "orders" must use `COUNT(DISTINCT order_id)`, and "average order value" needs order-level totals (Lesson 10).

Step 2: data-quality checks

Always check before analysing. A compact data-quality report:

SQL

```sql
SELECT
    (SELECT COUNT(*) FROM order_items WHERE sales IS NULL OR profit IS NULL OR quantity IS
NULL)       AS null_measures,
    (SELECT COUNT(*) FROM order_items WHERE sales ⩽ 0 OR quantity ⩽ 0)
AS non_positive,
    (SELECT COUNT(*) FROM order_items WHERE discount < 0 OR discount > 1)
AS bad_discount,
    (SELECT COUNT(*) FROM orders WHERE ship_date < order_date)
AS ship_before_order,
    (SELECT COUNT(*) FROM order_items oi LEFT JOIN orders o ON o.order_id = oi.order_id
WHERE o.order_id IS NULL) AS orphan_items,
    (SELECT COUNT(*) FROM orders o LEFT JOIN customers c ON c.customer_id = o.customer_id
WHERE c.customer_id IS NULL) AS orders_without_customer,
    (SELECT COUNT(*) - COUNT(DISTINCT item_id) FROM order_items)
AS duplicate_item_ids;
```

RESULT

null_measures	non_positive	bad_discount	ship_before_order	orphan_items	orders_wi
0	0	0	0	0	0

1 rows × 7 columns

All zeros: the data is clean, so the numbers can be trusted. In real projects every non-zero here becomes a ticket to fix or a documented exclusion. Also check **business-rule** outliers:

SQL

```sql
SELECT ROUND(MIN(sales), 0) AS min_sales, ROUND(MAX(sales), 0) AS max_sales,
       ROUND(AVG(sales), 0) AS avg_sales,
       MIN(discount) AS min_discount, MAX(discount) AS max_discount,
       MIN(quantity) AS min_qty, MAX(quantity) AS max_qty
FROM order_items;
```

RESULT

min_sales	max_sales	avg_sales	min_discount	max_discount	min_qty	max_qty
42	619,290	31,854	0	0.4	1	7

1 rows × 7 columns

Step 3: KPIs and trends

SQL

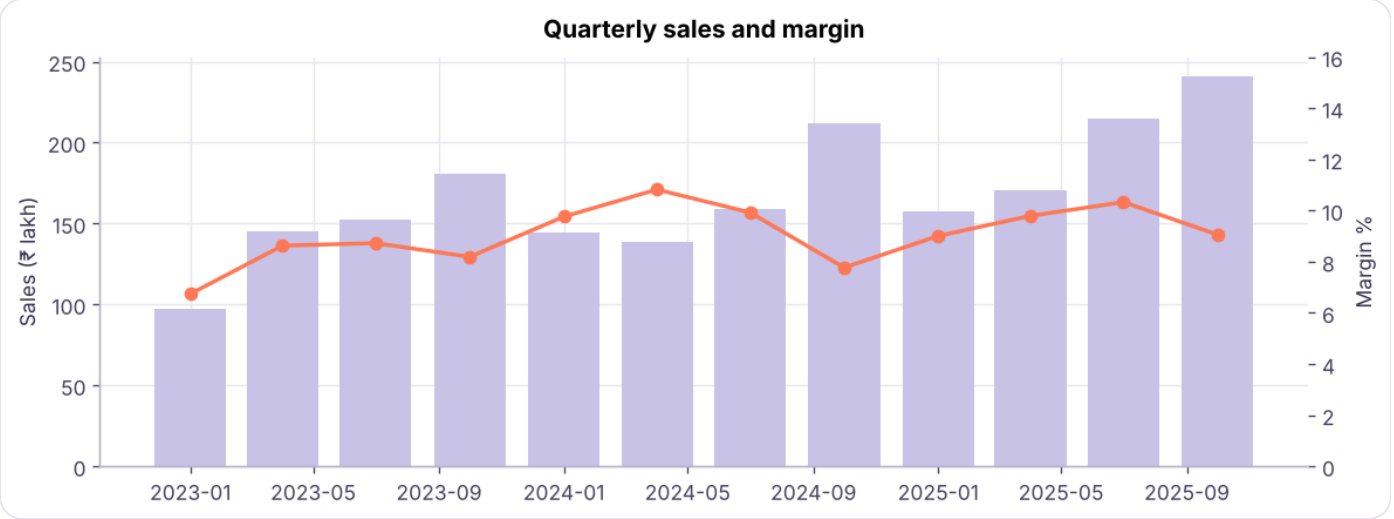
```sql
WITH yearly AS (
    SELECT YEAR(o.order_date)              AS yr,
           COUNT(DISTINCT o.order_id)      AS orders,
           COUNT(DISTINCT o.customer_id)   AS customers,
           SUM(oi.sales)                   AS sales,
           SUM(oi.profit)                  AS profit
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
    GROUP BY YEAR(o.order_date)
)
SELECT yr, orders, customers,
       ROUND(sales, 0)                                          AS sales,
       ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY yr)) / LAG(sales) OVER (ORDER BY
yr), 1)   AS sales_growth_pct,
       ROUND(profit, 0)                                         AS profit,
       ROUND(100.0 * (profit - LAG(profit) OVER (ORDER BY yr)) / LAG(profit) OVER (ORDER BY
yr), 1) AS profit_growth_pct,
       ROUND(100.0 * profit / sales, 1)                         AS margin_pct,
       ROUND(sales / orders, 0)                                 AS avg_order_value
FROM yearly
ORDER BY yr;
```

RESULT

yr	orders	customers	sales	sales_growth_pct	profit	profit_growth_pc
2023	883	540	57,640,217	NULL	4,725,378	NULL
2024	1085	590	65,396,556	13.5	6,135,992	29.9
2025	1232	623	78,376,797	19.8	7,490,793	22.1

3 rows × 9 columns

Finding 1. Sales grew 13.5% in 2024 and 19.8% in 2025, and profit grew even faster (29.9% and 22.1%), so the overall margin edged up from 8.2% to 9.6%. The growth is real, but a 9.6% margin is low compared with the 17–18% earned on lines sold **without** a discount (you will see this in Step 4). The CFO's concern is about the *quality* of profit: let's look at where the money is made and lost.



Quarterly sales and profit margin from the same data. Sales climb steadily and peak each Q4 (festive season); the margin moves in a narrow band below 12%.

Step 4: where do we lose money?

SQL

```sql
SELECT p.category, p.sub_category,
       COUNT(*)                                          AS lines,
       ROUND(SUM(oi.sales), 0)                           AS sales,
       ROUND(SUM(oi.profit), 0)                          AS profit,
       ROUND(100.0 * SUM(oi.profit) / SUM(oi.sales), 1)  AS margin_pct,
       ROUND(AVG(oi.discount) * 100, 1)                  AS avg_discount_pct
FROM order_items oi JOIN products p ON p.product_id = oi.product_id
GROUP BY p.category, p.sub_category
ORDER BY profit;
```

RESULT

category	sub_category	lines	sales	profit	margin_pct	avg_discount_pc
Furniture	Tables	505	14,046,224	-816,042	-5.8	17.5
Furniture	Bookcases	477	11,274,849	-422,786	-3.7	18.3
Office Supplies	Art	462	245,985	36,526	14.8	7.6
Office Supplies	Binders	494	401,219	61,074	15.2	7.4
Office Supplies	Paper	559	493,232	75,040	15.2	7.2
Furniture	Furnishings	442	1,429,307	78,632	5.5	7.4
Furniture	Chairs	493	8,091,474	342,043	4.2	8.3
Technology	Accessories	445	3,178,930	369,554	11.6	7.5
Office Supplies	Storage	507	2,659,340	397,228	14.9	7.6
Office Supplies	Appliances	492	6,333,529	944,927	14.9	7.5
Technology	Phones	488	31,919,223	3,663,117	11.5	7.1
Technology	Machines	491	42,590,335	5,186,711	12.2	6.7

13 rows × 7 columns (first 12 shown)

Finding 2. Tables and Bookcases lose money, and they also carry the heaviest average discounts. Is discounting the cause? Compare margin by discount level and by whether the line is Furniture:

SQL

```
SELECT CASE WHEN p.category = 'Furniture' THEN 'Furniture' ELSE 'Other' END AS group_name,
       CASE WHEN oi.discount = 0 THEN '0%' WHEN oi.discount ⩽ 0.2 THEN '1-20%' ELSE '>20%'
END AS discount_band,
       COUNT(*)                                          AS lines,
       ROUND(SUM(oi.profit), 0)                          AS profit,
       ROUND(100.0 * SUM(oi.profit) / SUM(oi.sales), 1)  AS margin_pct
FROM order_items oi JOIN products p ON p.product_id = oi.product_id
GROUP BY CASE WHEN p.category = 'Furniture' THEN 'Furniture' ELSE 'Other' END,
         CASE WHEN oi.discount = 0 THEN '0%' WHEN oi.discount ⩽ 0.2 THEN '1-20%' ELSE
'>20%' END
ORDER BY group_name, discount_band;
```

RESULT

group_name	discount_band	lines	profit	margin_pct
Furniture	0%	850	1,861,761	11.7
Furniture	1–20%	509	-317,393	-3.9
Furniture	>20%	558	-2,362,521	-21.9
Other	0%	2630	19,857,203	18.4
Other	1–20%	1434	1,014,534	2.0
Other	>20%	342	-1,701,419	-18.9

6 rows × 5 columns

Finding 3. In every group, discounts above 20% destroy margin. Furniture is hit hardest because it is discounted more often. What would profit be if lines discounted above 20% had been sold at a 20% discount instead? A rough "what-if":

SQL

```sql
WITH x AS (
    SELECT oi.sales, oi.profit, oi.discount,
           oi.sales / (1 - oi.discount)           AS list_value,
           (oi.sales / (1 - oi.discount)) * 0.8   AS sales_at_20,
           oi.sales - oi.profit                   AS cost
    FROM order_items oi WHERE oi.discount > 0.2
)
SELECT COUNT(*)                                  AS lines_over_20pct,
       ROUND(SUM(profit), 0)                     AS actual_profit,
       ROUND(SUM(sales_at_20 - cost), 0)         AS profit_if_capped_at_20pct,
       ROUND(SUM(sales_at_20 - cost) - SUM(profit), 0) AS potential_gain
FROM x;
```

RESULT

lines_over_20pct	actual_profit	profit_if_capped_at_20pct	potential_gain
900	-4,063,940	-422,474	3,641,466

1 rows × 4 columns

(This ignores that a lower discount might lose some of those sales. It is an upper bound to size the opportunity, and it must be labelled as such for management.)

Step 5: customers

Top customers and concentration

SQL

```
WITH c AS (
    SELECT o.customer_id, SUM(oi.sales) AS sales, SUM(oi.profit) AS profit, COUNT(DISTINCT
o.order_id) AS orders
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
    GROUP BY o.customer_id
), r AS (
    SELECT *, NTILE(10) OVER (ORDER BY sales DESC) AS decile FROM c
)
SELECT decile,
       COUNT(*)                                      AS customers,
       ROUND(100.0 * SUM(sales) / SUM(SUM(sales)) OVER (), 1) AS pct_of_sales,
       ROUND(100.0 * SUM(profit) / SUM(SUM(profit)) OVER (), 1) AS pct_of_profit
FROM r
GROUP BY decile
ORDER BY decile;
```

RESULT

decile	customers	pct_of_sales	pct_of_profit
1	79	28.8	34.1
2	79	18.7	21.0
3	79	14.6	14.3
4	79	11.3	10.4
5	79	9.0	8.8
6	78	6.7	5.3
7	78	5.2	2.5
8	78	3.4	2.2
9	78	1.9	1.1
10	78	0.5	0.3

10 rows × 4 columns

RFM segmentation

RFM scores each customer by **R**ecency (days since last order), **F**requency (number of orders) and **M**onetary value (total sales). The reference date is the day after the last order in the data.

SQL

```sql
WITH base AS (
    SELECT o.customer_id,
           DATEDIFF(day, MAX(o.order_date), '2026-01-01')  AS recency_days,
           COUNT(DISTINCT o.order_id)                      AS frequency,
           SUM(oi.sales)                                   AS monetary
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
    GROUP BY o.customer_id
), scored AS (
    SELECT *,
           NTILE(4) OVER (ORDER BY recency_days DESC) AS r,        -- 4 = most recent
           NTILE(4) OVER (ORDER BY frequency)         AS f,
           NTILE(4) OVER (ORDER BY monetary)          AS m
    FROM base
)
SELECT customer_id, recency_days, frequency, ROUND(monetary, 0) AS monetary, r, f, m,
       CASE WHEN r ⩾ 3 AND f ⩾ 3 AND m ⩾ 3 THEN 'Champions'
            WHEN r ⩾ 3 AND f ⩾ 2            THEN 'Loyal'
            WHEN r ⩽ 2 AND f ⩾ 3 AND m ⩾ 3  THEN 'At risk (high value)'
            WHEN r = 1                      THEN 'Lost / hibernating'
            ELSE 'Potential' END AS segment
FROM scored
ORDER BY monetary DESC;
```

RESULT

customer_id	recency_days	frequency	monetary	r	f	m	segment
C0741	441	6	1,204,439	1	4	4	At risk (high value)
C0092	11	8	1,140,540	4	4	4	Champions
C0306	278	7	1,126,867	2	4	4	At risk (high value)
C0781	130	5	1,123,527	2	3	4	At risk (high value)
C0443	92	7	1,113,058	3	4	4	Champions
C0720	12	6	1,100,213	4	4	4	Champions
C0493	514	6	1,004,382	1	4	4	At risk (high value)
C0028	266	5	1,000,189	2	3	4	At risk (high value)
C0630	331	5	1,000,175	1	3	4	At risk (high value)
C0262	70	6	952,738	3	4	4	Champions

785 rows × 8 columns (first 10 shown)

SQL

```sql
WITH base AS (
    SELECT o.customer_id, DATEDIFF(day, MAX(o.order_date), '2026-01-01') AS recency_days,
           COUNT(DISTINCT o.order_id) AS frequency, SUM(oi.sales) AS monetary
    FROM orders o JOIN order_items oi ON oi.order_id = o.order_id GROUP BY o.customer_id
), scored AS (
    SELECT *, NTILE(4) OVER (ORDER BY recency_days DESC) AS r, NTILE(4) OVER (ORDER BY
frequency) AS f, NTILE(4) OVER (ORDER BY monetary) AS m FROM base
), seg AS (
    SELECT *, CASE WHEN r ⩾ 3 AND f ⩾ 3 AND m ⩾ 3 THEN 'Champions' WHEN r ⩾ 3 AND f ⩾
2 THEN 'Loyal'
                   WHEN r ⩽ 2 AND f ⩾ 3 AND m ⩾ 3 THEN 'At risk (high value)' WHEN r = 1
THEN 'Lost / hibernating' ELSE 'Potential' END AS segment FROM scored
)
SELECT segment, COUNT(*) AS customers, ROUND(SUM(monetary), 0) AS sales,
       ROUND(100.0 * SUM(monetary) / SUM(SUM(monetary)) OVER (), 1) AS pct_of_sales
FROM seg GROUP BY segment ORDER BY sales DESC;
```

RESULT

segment	customers	sales	pct_of_sales
Champions	178	80,496,072	40.0
At risk (high value)	92	42,616,303	21.2
Potential	198	29,644,898	14.7
Lost / hibernating	169	24,817,833	12.3
Loyal	148	23,838,464	11.8

5 rows × 4 columns

The "At risk (high value)" group is the first win: high-spending customers who have gone quiet deserve a personal outreach before they churn. The same RFM idea is implemented with pandas in the ML track.

Retention and cohorts

A **cohort** groups customers by the year of their first order. **Retention** shows how many come back in later years.

SQL

```sql
WITH first_order AS (
    SELECT customer_id, YEAR(MIN(order_date)) AS cohort_year FROM orders GROUP BY
customer_id
), activity AS (
    SELECT DISTINCT customer_id, YEAR(order_date) AS active_year FROM orders
)
SELECT f.cohort_year,
       COUNT(DISTINCT f.customer_id)                                          AS
cohort_size,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year     THEN a.customer_id END)
AS y0,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year + 1 THEN a.customer_id END)
AS y1,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year + 2 THEN a.customer_id END)
AS y2
FROM first_order f
JOIN activity a ON a.customer_id = f.customer_id
GROUP BY f.cohort_year
ORDER BY f.cohort_year;
```

RESULT

cohort_year	cohort_size	y0	y1	y2
2023	540	540	397	415
2024	193	193	156	0
2025	52	52	0	0

3 rows × 5 columns

Divide y1 by `cohort_size` to get year-1 retention. About 74% of the 2023 cohort (397 of 540) ordered again in 2024, and some who skipped a year returned in 2025: this is a business with sticky customers, so reducing discounts is likely safer than the CFO fears.

Step 6: recommendations

KEY TAKEAWAYS **Findings and actions**

1. **Sales and profit are growing, but the margin is only 9.6%**, far below the 17–18% earned on undiscounted lines.
2. **Technology produces nearly all the profit**; **Furniture is net loss-making** because Tables and Bookcases are sold with heavy discounts.
3. **Discounts above 20% lose money in every category.** Cap discounts at 20% (the what-if sizes the upper-bound upside at about ₹36 lakh over three years; the real gain will be lower if some customers buy less), test in two regions with an A/B design.
4. **Customer base is sticky** (strong year-on-year retention), so price discipline is lower-risk than assumed.
5. **Re-engage "At risk (high value)" customers** with a targeted offer instead of blanket discounts.
6. Build a **monthly SQL view/dashboard** with these KPIs, with data-quality checks as a first step.

SQL interview practice: 25 questions with answers

Run these on the retail database. Each answer is a runnable query.

INTERVIEW CORNER **Q1. Second-highest sales line.**

SQL

```sql
SELECT MAX(sales) AS second_highest FROM order_items WHERE sales < (SELECT MAX(sales)
FROM order_items);
```

RESULT

second_highest
551,530

1 rows × 1 columns

INTERVIEW CORNER

Q2. Customers with at least 8 orders.

SQL

```sql
SELECT customer_id, COUNT(*) AS orders FROM orders GROUP BY customer_id HAVING
COUNT(*) ⩾ 8 ORDER BY orders DESC;
```

RESULT

customer_id	orders
C0219	10
C0396	10
C0291	9
C0664	9
C0110	9
C0384	9
C0596	9
C0419	9
C0271	9
C0527	9
C0719	9
C0069	9

41 rows × 2 columns (first 12 shown)

INTERVIEW CORNER

Q3. Duplicate customer names (possible duplicate records).

SQL

```sql
SELECT customer_name, COUNT(*) AS n FROM customers GROUP BY customer_name HAVING
COUNT(*) > 1 ORDER BY n DESC;
```

RESULT

customer_name	n
Sneha Joshi	6
Ananya Gupta	5
Rahul Shah	5
Asha Rao	5
Ananya Iyer	5
Saanvi Khan	5
Rohan Gupta	5
Reyansh Shah	4
Simran Khan	4
Vikram Kulkarni	4
Arjun Sharma	4
Nikhil Das	4

238 rows × 2 columns (first 12 shown)

INTERVIEW CORNER

Q4. Highest-selling product in each category.

SQL

```sql
WITH t AS (SELECT p.category, p.product_name, SUM(oi.sales) AS sales FROM order_items
oi JOIN products p ON p.product_id = oi.product_id GROUP BY p.category,
p.product_name),
     r AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY category ORDER BY sales DESC) AS
rn FROM t)
SELECT category, product_name, ROUND(sales, 0) AS sales FROM r WHERE rn = 1;
```

RESULT

category	product_name	sales
Furniture	Crest Table 572	3,122,680
Technology	Orbit Copier 746	16,026,341
Office Supplies	Crest Appliance 402	1,909,628

3 rows × 3 columns

INTERVIEW CORNER

Q5. Month-over-month growth for 2025.

SQL

```sql
WITH m AS (SELECT MONTH(order_date) AS mth, SUM(sales) AS sales FROM superstore WHERE YEAR(order_date) = 2025 GROUP BY MONTH(order_date))
SELECT mth, ROUND(sales, 0) AS sales, ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY mth)) / LAG(sales) OVER (ORDER BY mth), 1) AS mom_pct FROM m ORDER BY mth;
```

RESULT

mth	sales	mom_pct
1	5,357,761	NULL
2	3,879,338	-27.6
3	6,512,645	67.9
4	6,699,805	2.9
5	6,421,274	-4.2
6	3,968,572	-38.2
7	7,190,087	81.2
8	5,802,184	-19.3
9	8,468,060	45.9
10	6,140,290	-27.5
11	10,284,516	67.5
12	7,652,267	-25.6

12 rows × 3 columns

INTERVIEW CORNER

Q6. Running total of sales by month in 2025.

SQL

```
WITH m AS (SELECT MONTH(order_date) AS mth, SUM(sales) AS sales FROM superstore WHERE
YEAR(order_date) = 2025 GROUP BY MONTH(order_date))
SELECT mth, ROUND(sales, 0) AS sales, ROUND(SUM(sales) OVER (ORDER BY mth ROWS BETWEEN
UNBOUNDED PRECEDING AND CURRENT ROW), 0) AS running_total FROM m ORDER BY mth;
```

RESULT

mth	sales	running_total
1	5,357,761	5,357,761
2	3,879,338	9,237,099
3	6,512,645	15,749,743
4	6,699,805	22,449,548
5	6,421,274	28,870,822
6	3,968,572	32,839,394
7	7,190,087	40,029,481
8	5,802,184	45,831,665
9	8,468,060	54,299,725
10	6,140,290	60,440,015
11	10,284,516	70,724,530
12	7,652,267	78,376,797

12 rows × 3 columns

INTERVIEW CORNER

Q7. Customers who ordered in every year (2023, 2024 and 2025).

SQL

```sql
SELECT COUNT(*) AS customers_all_three_years
FROM (SELECT customer_id FROM orders GROUP BY customer_id HAVING COUNT(DISTINCT
YEAR(order_date)) = 3) AS t;
```

RESULT

customers_all_three_years
311

1 rows × 1 columns

INTERVIEW CORNER

Q8. Employees earning more than their department's average (HR data).

SQL

```sql
SELECT COUNT(*) AS above_dept_avg
FROM (SELECT emp_id, monthly_income, AVG(monthly_income) OVER (PARTITION BY
department) AS avg_income FROM employees) AS t
WHERE monthly_income > avg_income;
```

RESULT

above_dept_avg
638

1 rows × 1 columns

INTERVIEW CORNER

Q9. Delete duplicates keeping the lowest id (pattern).

SQL

```sql
WITH d AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY customer_name, city ORDER BY
customer_id) AS rn FROM dbo.customers)
DELETE FROM d WHERE rn > 1;
```

INTERVIEW CORNER

Q10. Customers with no orders in the last 6 months of 2025 but orders before that (churn candidates).

SQL

```sql
SELECT COUNT(*) AS churn_candidates
FROM (SELECT customer_id FROM orders WHERE order_date < '2025-07-01' GROUP BY
customer_id
      EXCEPT
      SELECT customer_id FROM orders WHERE order_date ⩾ '2025-07-01') AS t;
```

RESULT

churn_candidates
310

1 rows × 1 columns

INTERVIEW CORNER

Q11. Percentage of orders shipped late (more than 5 days).

SQL

```sql
SELECT ROUND(100.0 * SUM(CASE WHEN DATEDIFF(day, order_date, ship_date) > 5 THEN 1
ELSE 0 END) / COUNT(*), 1) AS pct_late FROM orders;
```

RESULT

pct_late
31.3

1 rows × 1 columns

INTERVIEW CORNER

Q12. Median order-line sales.

SQL

```sql
SELECT DISTINCT PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY sales) OVER () AS
median_sales FROM order_items;
```

RESULT

median_sales
8,430

1 rows × 1 columns

INTERVIEW CORNER

Q13. Categories where more than 30% of lines are loss-making.

SQL

```
SELECT p.category, ROUND(100.0 * AVG(CASE WHEN oi.profit < 0 THEN 1.0 ELSE 0 END), 1)
AS pct_loss_lines
FROM order_items oi JOIN products p ON p.product_id = oi.product_id
GROUP BY p.category HAVING AVG(CASE WHEN oi.profit < 0 THEN 1.0 ELSE 0 END) > 0.3;
```

RESULT

category	pct_loss_lines
Furniture	48.9

1 rows × 2 columns

INTERVIEW CORNER

Q14. Cumulative percentage of sales by customer (Pareto): how many customers make 50% of sales?

SQL

```
WITH c AS (SELECT customer_id, SUM(sales) AS s FROM order_items oi JOIN orders o ON
o.order_id = oi.order_id GROUP BY customer_id),
     p AS (SELECT s, SUM(s) OVER (ORDER BY s DESC ROWS BETWEEN UNBOUNDED PRECEDING AND
CURRENT ROW) * 1.0 / SUM(s) OVER () AS cum FROM c)
SELECT COUNT(*) AS customers_for_half_of_sales, (SELECT COUNT(*) FROM c) AS
total_customers FROM p WHERE cum ⩽ 0.5;
```

RESULT

customers_for_half_of_sales	total_customers
170	785

1 rows × 2 columns

INTERVIEW CORNER

Q15. Swap two values in a column (pattern).

```sql
UPDATE dbo.staff SET gender = CASE gender WHEN 'M' THEN 'F' WHEN 'F' THEN 'M' ELSE
gender END;
```

INTERVIEW CORNER

Q16. Find consecutive order days (streaks) for a customer: see Lesson 12 (gaps and islands).

INTERVIEW CORNER

Q17. Average order value per segment.

```sql
SELECT segment, ROUND(AVG(t.total), 0) AS aov
FROM (SELECT c.segment, o.order_id, SUM(oi.sales) AS total FROM orders o JOIN
customers c ON c.customer_id = o.customer_id JOIN order_items oi ON oi.order_id =
o.order_id GROUP BY c.segment, o.order_id) AS t
GROUP BY segment;
```

RESULT

segment	aov
Consumer	63,451
Corporate	61,899
Home Office	63,237

3 rows × 2 columns

INTERVIEW CORNER

Q18. Products priced above the average of their sub-category.

SQL

```sql
SELECT COUNT(*) AS products_above_subcat_avg
FROM (SELECT product_id, list_price, AVG(list_price) OVER (PARTITION BY sub_category)
AS avg_price FROM products) AS t WHERE list_price > avg_price;
```

RESULT

products_above_subcat_avg
36

1 rows × 1 columns

INTERVIEW CORNER

Q19. Rows in A not in B without NOT IN trap: use NOT EXISTS (Lesson 10).

INTERVIEW CORNER

Q20. Pivot category sales into columns by year.

SQL

```
SELECT category,
       ROUND(SUM(CASE WHEN YEAR(order_date) = 2023 THEN sales ELSE 0 END), 0) AS
y2023,
       ROUND(SUM(CASE WHEN YEAR(order_date) = 2024 THEN sales ELSE 0 END), 0) AS
y2024,
       ROUND(SUM(CASE WHEN YEAR(order_date) = 2025 THEN sales ELSE 0 END), 0) AS y2025
FROM superstore GROUP BY category ORDER BY category;
```

RESULT

category	y2023	y2024	y2025
Furniture	9,952,975	12,128,633	12,760,246
Office Supplies	2,611,874	3,550,109	3,971,321
Technology	45,075,369	49,717,814	61,645,230

3 rows × 4 columns

INTERVIEW CORNER

Q21. First and last order date and the number of days between for each customer (top 3 longest tenures).

SQL

```
SELECT TOP 3 customer_id, MIN(order_date) AS first_order, MAX(order_date) AS
last_order, DATEDIFF(day, MIN(order_date), MAX(order_date)) AS days_active
FROM orders GROUP BY customer_id ORDER BY days_active DESC;
```

RESULT

customer_id	first_order	last_order	days_active
C0079	2023-01-08	2025-12-27	1084
C0192	2023-01-24	2025-12-31	1072
C0070	2023-01-09	2025-12-13	1069

3 rows × 4 columns

INTERVIEW CORNER

Q22. Which month has the highest sales in each year?

SQL

```sql
WITH m AS (SELECT YEAR(order_date) AS yr, MONTH(order_date) AS mth, SUM(sales) AS
sales FROM superstore GROUP BY YEAR(order_date), MONTH(order_date)),
     r AS (SELECT *, RANK() OVER (PARTITION BY yr ORDER BY sales DESC) AS rnk FROM m)
SELECT yr, mth, ROUND(sales, 0) AS sales FROM r WHERE rnk = 1 ORDER BY yr;
```

RESULT

yr	mth	sales
2023	12	7,152,348
2024	11	7,284,664
2025	11	10,284,516

3 rows × 3 columns

INTERVIEW CORNER

Q23. How do you find the number of rows in each table of a database?

SQL

```sql
SELECT t.name AS table_name, SUM(p.rows) AS row_count
FROM sys.tables t JOIN sys.partitions p ON p.object_id = t.object_id AND p.index_id IN
(0, 1)
GROUP BY t.name ORDER BY row_count DESC;
```

INTERVIEW CORNER

Q24. Explain the difference between RANK() and ROW_NUMBER() and when each is right: Lesson 11.

INTERVIEW CORNER

Q25. How would you explain the result of a query to a non-technical manager?

State the answer first, in one sentence with numbers ("Furniture loses ₹8.2 lakh a year, almost entirely on heavily discounted tables and bookcases"), then give the 1–2 facts that support it, the assumption or limitation, and the action you recommend.

Summary

KEY TAKEAWAYS

- A good analysis follows a workflow: understand the grain, check data quality, calculate KPIs, find drivers, study customers, recommend actions.
- Always check data quality first; label what-if estimates as estimates.
- Combine CTEs, joins and window functions to build RFM segments, concentration (Pareto), cohorts and retention.
- In interviews, state your assumptions, clarify the grain, write clean, formatted SQL, and explain the result in business language.

Quick quiz

6 questions · tap an answer to check it

1. Why must "number of orders" use COUNT(DISTINCT order_id) when joining orders to order_items?

A DISTINCT is faster

B Each order appears once per item, so COUNT(*) would count lines

C order_id is a text column

D It avoids NULLs

Answer: B. The join repeats the order for each of its items.

2. In RFM, a customer with recency 400 days, frequency 1 and low monetary value is probably:

- **A** A champion
- **B** Lost or hibernating
- **C** A loyal customer
- **D** A new high-value customer

Answer: B. Long time since the last order, few orders and low spend.

3. A cohort table shows 2023 cohort size 500, year-1 active 350. Year-1 retention is:

- **A** 35%
- **B** 70%
- **C** 50%
- **D** 150%

Answer: B. 350 ÷ 500 = 70%.

4. Which should you do before reporting KPIs?

- **A** Build a dashboard
- **B** Check data quality: nulls, duplicates, orphans, ranges
- **C** Email the CFO
- **D** Add indexes

Answer: B. Untrusted data produces untrusted numbers.

5. **The what-if "profit if discounts were capped at 20%" is best described as:**

A A forecast

B A guarantee

C An upper-bound estimate that ignores possible lost sales

D A data error

Answer: C. It assumes demand would not change, so it overstates the benefit.

6. **NTILE(10) over customers ordered by sales gives:**

A Ten customers

B Ten equal-sized groups (deciles) of customers

C The top 10% only

D The median

Answer: B. NTILE splits rows into n buckets.

EXERCISE

Capstone: your own analysis

Using only SQL on the retail database, prepare a one-page analysis answering: (1) Which three cities have the highest profit and which three the lowest? (2) What is the average days to ship by ship mode, and does a slower mode save money (profit per line)? (3) Which product's sales fell the most from 2024 to 2025? (4) Propose two actions with the numbers that support them.

Show solution (worked answer)

Hints: (1) `GROUP BY city` with `ORDER BY profit` ascending and descending using TOP 3 in two queries or `RANK()` in a CTE. (2) Join orders to order_items, `AVG(DATEDIFF(day, order_date, ship_date))` and `SUM(profit)/COUNT(*)` by ship_mode. (3) Pivot product sales by year with conditional aggregation and compute the difference. (4) Example actions: cap Furniture discounts at 20% in two pilot regions; re-engage at-risk high-value customers. Present each action with a baseline number, an expected impact (labelled as an estimate) and a way to measure success.

SQL FOR DATA ANALYSIS · LESSON 16

Recursive CTEs, Hierarchies and Number Series

Use recursive common table expressions to walk hierarchies such as org charts and category trees, build number and date series without a calendar table, and avoid runaway recursion.

Gap-fill lesson | 2026 | Advanced | 120 min

By the end of this lesson you can

- Explain the anchor member, the recursive member and the stopping condition of a recursive CTE
- Walk an org chart or category tree and report each node's level and full path
- Generate number and date series inside SQL
- Protect queries against infinite recursion

Why hierarchies need recursion

Many business tables describe a tree. An employee reports to a manager who reports to a director. A product category contains sub-categories, which contain products. A bill of materials lists parts that are themselves made of parts. A plain `JOIN` only goes one level deep. A **recursive CTE** keeps joining a table to its own results until no new rows appear, so it can walk any depth.

CASE STUDY **Who reports to whom?**

A regional head wants to know, for every employee, how many management levels sit above them and who their top-level leader is, so that a re-organisation can be planned. The org chart is a single table where each row names its manager. One query has to follow that chain to the top, whatever its length.

Anatomy of a recursive CTE

Every recursive CTE has three parts:

1. **Anchor member**: the starting rows, for example the people with no manager.
2. **Recursive member**: a query that joins back to the CTE itself to find the next level.
3. **Stopping condition**: the recursion ends when the recursive member returns no rows.

```
WITH RECURSIVE name AS (
    SELECT ...            -- anchor: level 0
    UNION ALL
    SELECT ...            -- recursive: joins name to the next level
    FROM table t
    JOIN name n ON ...
)
SELECT * FROM name;
```

`UNION ALL` is the usual choice, because it does not remove duplicates and so runs faster. Use `UNION` only when you need the engine to drop repeated rows, which also protects against cycles (see the warning below). In SQL Server the keyword `RECURSIVE` is not written; in PostgreSQL, DuckDB and SQLite it is.

Example 1: an org chart

The table below is self-contained so you can run it as written. Each person has at most one manager; the CEO has no manager.

SQL

```sql
WITH RECURSIVE org(emp_id, name, manager_id) AS (
    VALUES
        (1, 'Anika',  NULL),
        (2, 'Rohit',  1),
        (3, 'Meera',  1),
        (4, 'Kabir',  2),
        (5, 'Tara',   2),
        (6, 'Dev',    3),
        (7, 'Isha',   4)
),
chain AS (
    -- Anchor: the top of the hierarchy
    SELECT emp_id, name, manager_id, 0 AS level, name AS path
    FROM org
    WHERE manager_id IS NULL
    UNION ALL
    -- Recursive: everyone whose manager is already in the chain
    SELECT o.emp_id, o.name, o.manager_id, c.level + 1, c.path || ' > ' || o.name
    FROM org o
    JOIN chain c ON o.manager_id = c.emp_id
)
SELECT level, name, path
FROM chain
ORDER BY path;
```

RESULT

level	name	path
0	Anika	Anika
1	Meera	Anika > Meera
2	Dev	Anika > Meera > Dev
1	Rohit	Anika > Rohit
2	Kabir	Anika > Rohit > Kabir
3	Isha	Anika > Rohit > Kabir > Isha
2	Tara	Anika > Rohit > Tara

7 rows × 3 columns

Read the result as a tree: level 0 is the CEO, level 1 the direct reports, and so on. The `path` column is built on the way down by appending each name to the parent's path, which is a common pattern for showing a breadcrumb.

Key idea: the recursive member can use anything the anchor produced, here `c.level` and `c.path`. That is how depth and paths are accumulated.

Example 2: team totals across the whole tree

Once you can attach a root and a level to every node, you can roll up numbers. Here every person's team size counts everyone below them.

SQL

```sql
WITH RECURSIVE org(emp_id, name, manager_id) AS (
    VALUES (1,'Anika',NULL),(2,'Rohit',1),(3,'Meera',1),(4,'Kabir',2),(5,'Tara',2),
(6,'Dev',3),(7,'Isha',4)
),
below AS (
    -- Each person paired with every person who reports to them, at any depth
    SELECT manager_id AS boss, emp_id AS report FROM org WHERE manager_id IS NOT NULL
    UNION ALL
    SELECT b.boss, o.emp_id
    FROM below b
    JOIN org o ON o.manager_id = b.report
)
SELECT o.name, COUNT(b.report) AS people_below
FROM org o
LEFT JOIN below b ON b.boss = o.emp_id
GROUP BY o.name
ORDER BY people_below DESC, o.name;
```

RESULT

name	people_below
Anika	6
Rohit	3
Kabir	1
Meera	1
Dev	0
Isha	0
Tara	0

7 rows × 2 columns

Notice that `below` is built by walking down from each boss. `COUNT` then counts every report at every depth, which a single `JOIN` could not do.

Example 3: category trees with depth

Product catalogues often nest categories several levels deep. The example below stores a small tree as a parent-child list and returns each node with its depth, so that a report can indent the categories.

SQL

```sql
WITH RECURSIVE cat(cat_id, cat_name, parent_id) AS (
    VALUES
        (1, 'Technology', NULL),
        (2, 'Phones', 1),
        (3, 'Accessories', 1),
        (4, 'Chargers', 3),
        (5, 'Cables', 3),
        (6, 'Office Supplies', NULL),
        (7, 'Paper', 6)
),
tree AS (
    SELECT cat_id, cat_name, 1 AS depth, cat_name AS full_path
    FROM cat WHERE parent_id IS NULL
    UNION ALL
    SELECT c.cat_id, c.cat_name, t.depth + 1, t.full_path || ' / ' || c.cat_name
    FROM cat c
    JOIN tree t ON c.parent_id = t.cat_id
)
SELECT depth, REPEAT('    ', depth - 1) || cat_name AS indented_name, full_path
FROM tree
ORDER BY full_path;
```

RESULT

depth	indented_name	full_path
1	Office Supplies	Office Supplies
2	Paper	Office Supplies / Paper
1	Technology	Technology
2	Accessories	Technology / Accessories
3	Cables	Technology / Accessories / Cables
3	Chargers	Technology / Accessories / Chargers
2	Phones	Technology / Phones

7 rows × 3 columns

The REPEAT call indents the name by its depth, which is the simplest way to show a tree in a flat table or a report.

Example 4: a number and date series

Recursive CTEs can create rows out of nothing. This is useful for building a list of months so that a month with zero sales still appears in a report, rather than vanishing because no order was placed.

SQL

```sql
WITH RECURSIVE months(month_start) AS (
    SELECT DATE '2024-01-01'
    UNION ALL
    SELECT month_start + INTERVAL 1 MONTH
    FROM months
    WHERE month_start < DATE '2024-12-01'
),
monthly_sales AS (
    SELECT DATE_TRUNC('month', order_date) AS month_start, SUM(sales) AS sales
    FROM superstore
    WHERE order_date ⩾ DATE '2024-01-01' AND order_date < DATE '2025-01-01'
    GROUP BY 1
)
SELECT m.month_start,
       COALESCE(s.sales, 0) AS sales
FROM months m
LEFT JOIN monthly_sales s ON s.month_start = m.month_start
ORDER BY m.month_start;
```

RESULT

month_start	sales
2024-01-01	6,022,738
2024-02-01	2,463,147
2024-03-01	5,940,384
2024-04-01	4,620,479
2024-05-01	5,253,017
2024-06-01	4,012,674
2024-07-01	4,818,164
2024-08-01	5,757,228
2024-09-01	5,330,618
2024-10-01	7,255,348
2024-11-01	7,284,664
2024-12-01	6,638,096

12 rows × 2 columns

Use the same idea with `UNION ALL` and a counter to build any series, such as 1 to 100 for bucketing or a list of working days for a schedule.

Preventing runaway recursion

If the data contains a loop (A reports to B, and B reports to A), a recursive query never ends. Two defences work well:

- **Stop by depth.** Add `WHERE level < 10` (or a similar limit) to the recursive member.
- **Use `UNION` instead of `UNION ALL`** when the same row could be reached twice, so the engine discards repeats.

::::warn Loops in the data A recursive query over a table with a cycle will run until the engine gives up or runs out of memory. Always check for cycles first, and always include a depth limit in production queries. ::::

::::tip Recursion versus window functions Recursion is for hierarchies of unknown depth. If the question is about a fixed number of steps (for example, "the manager's manager"), two self-joins are simpler and faster. ::::

::::key What to remember - A recursive CTE is an anchor, a recursive member that joins to itself, and a stopping point. - Carry depth and path columns down the recursion to report levels and breadcrumbs. - Walk upwards (to the top) or downwards (to all descendants), depending on the join direction. - Add a depth limit to protect against cycles. - Recursion can also generate series of numbers and dates, filling gaps in reports. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is the role of the anchor member in a recursive CTE?

- **A** It joins the CTE to itself
- **B** It stops the recursion
- **C** It supplies the starting rows of the hierarchy
- **D** It removes duplicates

Answer: C. The anchor is the non-recursive first query, such as the top-level employees.

2. Which clause ends the recursion?

- **A** A LIMIT on the outer query
- **B** The recursive member returns no new rows
- **C** A HAVING clause
- **D** The ORDER BY clause

Answer: B. Recursion stops when a pass produces no rows.

3. Why is UNION ALL usually preferred in recursive CTEs?

- **A** It does not remove duplicates, so it is faster
- **B** It is the only syntax allowed
- **C** It sorts the rows
- **D** It guarantees no cycles

Answer: A. UNION ALL skips the duplicate-removal step; the trade-off is that cycles are not stopped automatically.

4. To list every month even when a month has no sales, you should:

- **A** Use an INNER JOIN with the sales table
- **B** Generate the months with a recursive CTE and LEFT JOIN the sales to it
- **C** Use DISTINCT on the order dates
- **D** Filter out zero sales

Answer: B. The generated series supplies the missing months, and the LEFT JOIN keeps them.

5. A query over employee data never finishes. The most likely cause is:

- **A** The table has too many columns
- **B** The data contains a cycle and the recursion has no stopping limit
- **C** The ORDER BY is missing
- **D** The anchor returns no rows

Answer: B. A cycle makes the recursive member produce rows forever.

EXERCISE **Build a reporting line report**

Using the org chart from Example 1, write a query that returns each person's name, their level, their direct manager's name, and the number of people who report directly to them. Order the result by level, then by name.

Show solution (worked answer)

Hint: join the `org` table to itself on `manager_id = emp_id` for the manager's name, and count direct reports with a separate `GROUP BY` on `manager_id`. Use the recursive `chain` CTE from Example 1 for the level.

SQL FOR DATA ANALYSIS · LESSON 17

Percentiles, Medians and Distribution Queries

Go beyond averages. Compute medians and percentiles, split customers into quartiles and deciles, measure how skewed a distribution is, and find outliers with SQL that works on real retail data.

Gap-fill lesson | 2026 | Advanced | 120 min

By the end of this lesson you can

- Explain why the median and percentiles can be more honest than the mean for skewed money data
- Compute medians and arbitrary percentiles with PERCENTILE_CONT and PERCENTILE_DISC
- Bucket rows into quartiles and deciles with NTILE and percent-rank measures
- Flag outliers with the interquartile range

Why averages mislead

Order values in retail are skewed: most orders are small, and a few very large ones pull the average up. The mean order value may look healthy while the typical customer spends much less. The **median** (the middle value) and **percentiles** (values below which a chosen share of rows fall) describe the typical case and the spread.

CASE STUDY **Is the average order really typical?**

The sales director quotes "average order value of about 2,000 rupees". The question is whether that describes a typical order, or whether a few big corporate orders dominate it. Answering needs the median, the quartiles and the top percentile, not one average.

Median and percentiles with PERCENTILE_CONT

`PERCENTILE_CONT(p) WITHIN GROUP (ORDER BY x)` returns the value at fraction `p`, interpolating between rows when needed. `p = 0.5` is the median.

SQL

```sql
WITH order_totals AS (
    SELECT order_id, SUM(sales) AS order_value
    FROM superstore
    GROUP BY order_id
)
SELECT
    ROUND(AVG(order_value), 0)                                          AS
mean_order_value,
    ROUND(PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY order_value), 0)    AS p25,
    ROUND(PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY order_value), 0)    AS median,
    ROUND(PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY order_value), 0)    AS p75,
    ROUND(PERCENTILE_CONT(0.90) WITHIN GROUP (ORDER BY order_value), 0)    AS p90,
    ROUND(MAX(order_value), 0)                                           AS
largest_order
FROM order_totals;
```

RESULT

mean_order_value	p25	median	p75	p90	largest_order
62,942	6,526	26,101	78,458	178,860	622,758

1 rows × 6 columns

If the mean is far above the median, a small share of large orders is pulling it up. Report both, and say which one describes a typical order.

PERCENTILE_CONT versus PERCENTILE_DISC: `PERCENTILE_CONT` interpolates between two values; `PERCENTILE_DISC` returns an actual value from the data. Use `PERCENTILE_DISC` when the answer must be a real observed order, such as "the order value that 90% of orders are below".

Quartiles and deciles with NTILE

`NTILE(n)` assigns each row to one of n equal-sized groups. It is the easiest way to segment customers by spend. The query below ranks customers into quartiles by lifetime sales.

SQL

```sql
WITH customer_sales AS (
    SELECT customer_id, SUM(sales) AS lifetime_sales
    FROM superstore
    GROUP BY customer_id
),
quartiles AS (
    SELECT customer_id, lifetime_sales,
           NTILE(4) OVER (ORDER BY lifetime_sales) AS spend_quartile
    FROM customer_sales
)
SELECT spend_quartile,
       COUNT(*)                          AS customers,
       ROUND(MIN(lifetime_sales), 0)     AS min_spend,
       ROUND(MAX(lifetime_sales), 0)     AS max_spend,
       ROUND(SUM(lifetime_sales), 0)     AS total_spend,
       ROUND(100.0 * SUM(lifetime_sales) / SUM(SUM(lifetime_sales)) OVER (), 1) AS
share_of_sales_pct
FROM quartiles
GROUP BY spend_quartile
ORDER BY spend_quartile;
```

RESULT

spend_quartile	customers	min_spend	max_spend	total_spend	share_of_sales_pct
1	197	140	88,830	7,935,534	3.9
2	196	89,102	200,899	28,192,185	14.0
3	196	201,378	373,460	54,558,902	27.1
4	196	373,864	1,204,439	110,726,949	55.0

4 rows × 6 columns

Quartile 4 is the top 25% of customers by spend. The share column usually shows that a small quartile produces a large share of revenue. That is the Pareto pattern that marketing budgets should reflect.

::::note Ties and buckets `NTILE` splits rows by position, so customers with identical spend can land in different buckets. If you need every customer with the same value to share one bucket, use `PERCENT_RANK()` or `CUME_DIST()` and cut at thresholds instead. ::::

Percent rank and cumulative distribution

`PERCENT_RANK()` gives each row a value from 0 to 1 showing its relative position. `CUME_DIST()` gives the share of rows at or below its value. Both answer "what share of customers spend less than

this one?"

```sql
WITH customer_sales AS (
    SELECT customer_id, SUM(sales) AS lifetime_sales
    FROM superstore
    GROUP BY customer_id
)
SELECT customer_id,
       ROUND(lifetime_sales, 0) AS lifetime_sales,
       ROUND(PERCENT_RANK() OVER (ORDER BY lifetime_sales), 3) AS percent_rank,
       ROUND(CUME_DIST()    OVER (ORDER BY lifetime_sales), 3) AS cume_dist
FROM customer_sales
ORDER BY lifetime_sales DESC
LIMIT 10;
```

RESULT

customer_id	lifetime_sales	percent_rank	cume_dist
C0741	1,204,439	1.00	1.00
C0092	1,140,540	1.00	1.00
C0306	1,126,867	1.00	1.00
C0781	1,123,527	1.00	1.00
C0443	1,113,058	0.99	0.99
C0720	1,100,213	0.99	0.99
C0493	1,004,382	0.99	0.99
C0028	1,000,189	0.99	0.99
C0630	1,000,175	0.99	0.99
C0262	952,738	0.99	0.99

10 rows × 4 columns

Use these measures when you want to say "this customer is in the top 5%" without forcing exact bucket sizes.

Spread: standard deviation and the interquartile range

The median says where the middle is; the spread says how much values vary. Two measures are common:

- **Standard deviation** (STDDEV_SAMP) measures typical distance from the mean. It assumes a roughly symmetric shape.
- **Interquartile range** (IQR = P75 minus P25) measures the spread of the middle half. It is robust to outliers.

The next query computes the IQR for profit per line and counts lines outside the usual outlier fences (1.5 times the IQR beyond the quartiles).

SQL

```
WITH q AS (
    SELECT PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY profit) AS q1,
           PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY profit) AS q3
    FROM order_items
),
fences AS (
    SELECT q1, q3, q3 - q1 AS iqr,
           q1 - 1.5 * (q3 - q1) AS lower_fence,
           q3 + 1.5 * (q3 - q1) AS upper_fence
    FROM q
)
SELECT ROUND(f.q1, 1) AS q1, ROUND(f.q3, 1) AS q3, ROUND(f.iqr, 1) AS iqr,
       ROUND(f.lower_fence, 1) AS lower_fence, ROUND(f.upper_fence, 1) AS upper_fence,
       COUNT(*) FILTER (WHERE i.profit < f.lower_fence OR i.profit > f.upper_fence) AS
outlier_lines,
       COUNT(*) AS all_lines
FROM order_items i
CROSS JOIN fences f
GROUP BY f.q1, f.q3, f.iqr, f.lower_fence, f.upper_fence;
```

RESULT

q1	q3	iqr	lower_fence	upper_fence	outlier_lines	all_lines
-17.9	2,270.1	2,288.1	-3,450.1	5,702.3	1361	6323

1 rows × 7 columns

Outliers are not automatically errors. A very large profitable line may be a genuine bulk deal. Investigate each one before excluding it.

Skewness check in plain SQL

A quick comparison of mean and median shows the direction of skew. If the mean is greater than the median, the distribution has a long right tail; if it is smaller, a long left tail. The query below reports that gap for sales per line by category.

SQL

```sql
SELECT p.category,
       COUNT(*)                                                    AS lines,
       ROUND(AVG(i.sales), 0)                                      AS mean_sales,
       ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY i.sales), 0)  AS median_sales,
       ROUND(AVG(i.sales) - PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY i.sales), 0) AS
mean_minus_median
FROM order_items i
JOIN products p ON p.product_id = i.product_id
GROUP BY p.category
ORDER BY mean_minus_median DESC;
```

RESULT

category	lines	mean_sales	median_sales	mean_minus_median
Technology	1892	82,684	56,789	25,895
Furniture	1917	18,175	12,970	5,205
Office Supplies	2514	4,031	1,152	2,879

3 rows × 5 columns

A large positive gap in a category means a few expensive lines dominate that category's average, which affects how you set targets for it.

::::warn Averages of averages Never average a set of averages (for example, averaging each region's mean order value) and call it the overall mean unless the groups are the same size. Recompute from the underlying rows, or use the median. ::::

::::key What to remember - Report the median and percentiles beside the mean for money data, which is usually skewed. - `PERCENTILE_CONT` interpolates; `PERCENTILE_DISC` returns an actual observed value. - `NTILE` gives equal-sized groups by position; `PERCENT_RANK` and `CUME_DIST` give relative positions. - The IQR and its fences flag outliers without being distorted by them. - Investigate outliers before you remove them. ::::

Quick quiz

5 questions · tap an answer to check it

1. A category has a mean sale of 5,000 and a median of 1,200. What does this suggest?

A Most lines are above average

B A few very large lines pull the mean up; the typical line is much smaller

C The data has no outliers

D The median is wrong

Answer: B. A mean far above the median signals a long right tail.

2. Which function returns the median?

A AVG(x)

B MODE(x) over the whole table

C PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY x)

D COUNT(DISTINCT x)

Answer: C. Fraction 0.5 is the 50th percentile, which is the median.

3. You need an actual observed order value as the 90th percentile. Which is better?

A PERCENTILE_CONT

B PERCENTILE_DISC

C AVG

D STDDEV_SAMP

Answer: B. PERCENTILE_DISC returns a value that exists in the data.

4. NTILE(4) over customers ordered by spend creates:

- **A** Four customers
- **B** Four groups of roughly equal size
- **C** Four groups of equal spend
- **D** The top four customers

Answer: B. NTILE splits rows by position into equal-sized buckets.

5. The interquartile range is calculated as:

- **A** Maximum minus minimum
- **B** Mean minus median
- **C** P75 minus P25
- **D** Standard deviation times 2

Answer: C. IQR spans the middle half of the data.

EXERCISE **Concentration of revenue**

Write one query that returns, for each product category, the median order line sales, the 90th percentile line sales, and the share of total sales coming from lines above the category's 90th percentile.

Show solution (worked answer)

Hint: compute the 90th percentile per category with `PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY sales)` in a CTE grouped by category, join it back to `order_items` through `products`, then use `SUM(CASE WHEN sales > p90 THEN sales END)` divided by the total sales for the category.

SQL FOR DATA ANALYSIS · LESSON 18

Business SQL: Cohorts, Funnels, Churn, Lifetime Value and Retention

Write the SQL behind the business questions that analysts are asked every week: cohort retention, conversion funnels, churn, customer lifetime value and retention rates, using the retail database.

Gap-fill lesson | ✦ 2026 | Advanced | ⏱ 150 min

By the end of this lesson you can

- Build a cohort table and read a retention triangle
- Measure a multi-step funnel with conditional aggregation
- Define churn precisely and calculate a churn rate for a period
- Estimate customer lifetime value from historical orders
- Compute repeat-purchase and retention rates

Business definitions first

Most SQL errors in business reporting come from unclear definitions, not from syntax. Before you query, write the definition in one sentence:

- **Active customer**: placed at least one order in the period.
- **Churned customer**: active in the previous period and not active in this one.
- **Cohort**: customers grouped by the period of their first order.
- **Lifetime value (CLV)**: total gross profit or sales a customer generates over their relationship with the business, in this lesson measured on the data we have.

CASE STUDY **The retention review**

The growth team wants to know whether customers acquired in 2023 still buy in 2024 and 2025, and whether the business is losing customers faster than it gains them. The answers need a cohort table, a churn rate and a lifetime value estimate, all built from the same orders table.

Cohort retention

A cohort table counts how many customers from each first-order year were still active in later years. Each row is a cohort; each column is the year.

SQL

```
WITH first_order AS (
    SELECT customer_id, MIN(EXTRACT(YEAR FROM order_date)) AS cohort_year
    FROM orders
    GROUP BY customer_id
),
activity AS (
    SELECT DISTINCT o.customer_id, EXTRACT(YEAR FROM o.order_date) AS active_year
    FROM orders o
)
SELECT f.cohort_year,
       COUNT(DISTINCT f.customer_id)                                          AS
cohort_size,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year     THEN a.customer_id END)
AS active_y0,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year + 1 THEN a.customer_id END)
AS active_y1,
       COUNT(DISTINCT CASE WHEN a.active_year = f.cohort_year + 2 THEN a.customer_id END)
AS active_y2
FROM first_order f
LEFT JOIN activity a ON a.customer_id = f.customer_id
GROUP BY f.cohort_year
ORDER BY f.cohort_year;
```

RESULT

cohort_year	cohort_size	active_y0	active_y1	active_y2
2023	540	540	397	415
2024	193	193	156	0
2025	52	52	0	0

3 rows × 5 columns

Read the table across a row to see decay: if `active_y1` is well below `cohort_size`, the first renewal is where customers leave. Convert each count to a percentage of `cohort_size` for the retention triangle that stakeholders expect.

SQL

```sql
WITH first_order AS (
    SELECT customer_id, MIN(EXTRACT(YEAR FROM order_date)) AS cohort_year
    FROM orders GROUP BY customer_id
),
activity AS (
    SELECT DISTINCT customer_id, EXTRACT(YEAR FROM order_date) AS active_year FROM orders
),
cohort AS (
    SELECT f.cohort_year, a.active_year - f.cohort_year AS years_since_first,
COUNT(DISTINCT a.customer_id) AS active
    FROM first_order f
    JOIN activity a ON a.customer_id = f.customer_id
    GROUP BY 1, 2
),
sizes AS (
    SELECT cohort_year, COUNT(*) AS cohort_size FROM first_order GROUP BY cohort_year
)
SELECT c.cohort_year, c.years_since_first,
       ROUND(100.0 * c.active / s.cohort_size, 1) AS retention_pct
FROM cohort c
JOIN sizes s ON s.cohort_year = c.cohort_year
ORDER BY c.cohort_year, c.years_since_first;
```

RESULT

cohort_year	years_since_first	retention_pct
2023	0	100.0
2023	1	73.5
2023	2	76.9
2024	0	100.0
2024	1	80.8
2025	0	100.0

6 rows × 3 columns

::::tip Use the right denominator Retention is always measured against the cohort's starting size, not against the previous year's active count. Dividing by the wrong base is one of the most common cohort mistakes. ::::

Conversion funnels

A funnel counts customers who reach each stage. The retail data has no page views, so this example builds a simple funnel from the order history: customers who placed any order, customers who placed a second order, and customers who placed a third order.

SQL

```sql
WITH per_customer AS (
    SELECT customer_id,
           COUNT(DISTINCT order_id) AS orders_placed
    FROM orders
    GROUP BY customer_id
)
SELECT
    COUNT(*)                                                AS stage_1_customers,
    COUNT(*) FILTER (WHERE orders_placed ⩾ 2)               AS stage_2_repeat_buyers,
    COUNT(*) FILTER (WHERE orders_placed ⩾ 3)               AS stage_3_loyal_buyers,
    ROUND(100.0 * COUNT(*) FILTER (WHERE orders_placed ⩾ 2) / COUNT(*), 1) AS
pct_stage_1_to_2,
    ROUND(100.0 * COUNT(*) FILTER (WHERE orders_placed ⩾ 3)
          / NULLIF(COUNT(*) FILTER (WHERE orders_placed ⩾ 2), 0), 1)       AS
pct_stage_2_to_3
FROM per_customer;
```

RESULT

stage_1_customers	stage_2_repeat_buyers	stage_3_loyal_buyers	pct_stage_1_to_2	pct_s
785	729	606	92.9	83.1

1 rows × 5 columns

Each stage is a subset of the one above it, so the counts can only fall. The conversion between stages is the figure to track: a small improvement from stage 1 to stage 2 usually matters more than any change further down.

::::note Funnels need ordered events A real funnel uses ordered events with timestamps (visit, add to cart, checkout, purchase). The pattern above is the same, but each stage must also be restricted to happen before the next, for example with `MIN(event_time)` per stage per user. ::::

Churn

Churn is easy to misreport, so define it in the query. The example below defines a customer as churned in a year if they were active the previous year and did not order in this year.

SQL

```sql
WITH active AS (
    SELECT DISTINCT customer_id, EXTRACT(YEAR FROM order_date) AS y FROM orders
),
years AS (SELECT DISTINCT EXTRACT(YEAR FROM order_date) AS y FROM orders)
SELECT p.y AS year,
       COUNT(*) AS active_last_year,
       COUNT(*) FILTER (WHERE c.customer_id IS NULL) AS churned_this_year,
       ROUND(100.0 * COUNT(*) FILTER (WHERE c.customer_id IS NULL) / COUNT(*), 1) AS
churn_rate_pct
FROM active p
LEFT JOIN active c ON c.customer_id = p.customer_id AND c.y = p.y + 1
WHERE p.y + 1 IN (SELECT y FROM years)
GROUP BY p.y
ORDER BY p.y;
```

RESULT

year	active_last_year	churned_this_year	churn_rate_pct
2023	540	143	26.5
2024	590	123	20.8

2 rows × 4 columns

Read the output carefully: the rate is churn from one year to the next, measured on customers who were active in the earlier year. Customers who were new in the later year are not part of this calculation.

::::warn Churn depends on the window A customer who bought in December 2023 and again in January 2025 is churned in 2024 under the yearly definition used above, then reactivated in 2025. Monthly, quarterly and yearly definitions give different churn rates. State the one you used on every chart. ::::

Customer lifetime value

A simple, honest estimate of customer lifetime value from history is the average gross profit per customer multiplied by the average number of orders per customer, with the period stated. Real CLV models add discounting and predicted future behaviour; this query gives the historical baseline.

SQL

```sql
WITH cust AS (
    SELECT o.customer_id,
           COUNT(DISTINCT o.order_id) AS orders,
           SUM(i.sales)               AS revenue,
           SUM(i.profit)              AS gross_profit
    FROM orders o
    JOIN order_items i ON i.order_id = o.order_id
    GROUP BY o.customer_id
)
SELECT COUNT(*)                              AS customers,
       ROUND(AVG(orders), 2)                 AS avg_orders,
       ROUND(AVG(revenue), 0)                AS avg_revenue,
       ROUND(AVG(gross_profit), 0)           AS avg_profit_to_date,
       ROUND(AVG(gross_profit / NULLIF(orders, 0)), 0) AS avg_profit_per_order
FROM cust;
```

RESULT

customers	avg_orders	avg_revenue	avg_profit_to_date	avg_profit_per_order
785	4.08	256,578	23,379	5,881

1 rows × 5 columns

To find the value of a segment, repeat the query per segment (for example, by `segment` from the customers table). Compare segments on profit, not revenue: a segment with high revenue and thin margins may be worth less.

Retention and repeat rate

The repeat rate is the share of customers with two or more orders. It is simple and widely reported, and it shows whether one-time buyers are the main pattern.

SQL

```sql
WITH counts AS (
    SELECT customer_id, COUNT(DISTINCT order_id) AS orders FROM orders GROUP BY customer_id
)
SELECT COUNT(*)                                                  AS customers,
       COUNT(*) FILTER (WHERE orders = 1)                        AS one_time_buyers,
       COUNT(*) FILTER (WHERE orders ⩾ 2)                        AS repeat_buyers,
       ROUND(100.0 * COUNT(*) FILTER (WHERE orders ⩾ 2) / COUNT(*), 1) AS repeat_rate_pct
FROM counts;
```

RESULT

customers	one_time_buyers	repeat_buyers	repeat_rate_pct
785	56	729	92.9

1 rows × 4 columns

::::key What to remember - Write the definition (active, churned, cohort, CLV) before writing SQL. - Cohorts group customers by first purchase; retention divides by the cohort's starting size. - Funnels are subsets that shrink at each stage; report the stage-to-stage conversion. - Churn depends on the window you choose, so state it. - CLV from history is a baseline; use profit, not revenue, to compare segments. ::::

Quick quiz

5 questions · tap an answer to check it

1. In a cohort table, a customer's cohort is based on:

A Their most recent order

B Their first order

C Their segment

D The product they bought most

Answer: B. A cohort groups customers by the period of their first activity.

2. Retention for year 1 should be divided by:

A The previous year's active count

B Total revenue

C The cohort's starting size

D The number of orders

Answer: C. Retention is measured against the group you started with.

3. A funnel's stage 2 count is larger than stage 1. What is wrong?

- **A** Nothing; funnels can grow
- **B** Stage 2 is a subset of stage 1, so the definition or the join is wrong
- **C** Stage 1 should be ignored
- **D** Stage 2 must use revenue

Answer: B. Each stage must be a subset of the stage before it.

4. Why is comparing segments on profit safer than on revenue?

- **A** Profit is always larger
- **B** Revenue can be high while margins are thin, so profit shows which segment is worth more
- **C** Revenue cannot be queried
- **D** Profit includes discounts

Answer: B. Profit shows what the business keeps.

5. The repeat rate counts customers with:

- **A** One order
- **B** Two or more orders
- **C** No orders
- **D** The highest spend

Answer: B. Repeat buyers placed at least two orders.

EXERCISE

Win-back list

Write a query that lists customers who were active in 2024 but placed no order in 2025. For each, show their segment, their 2024 gross profit and their last order date. Order by 2024 profit, highest first, and return the top 20.

Show solution (worked answer)

Hint: find 2024 customers and 2025 customers in two CTEs, keep 2024 customers with LEFT JOIN to 2025 and WHERE the 2025 side is NULL. Join `customers` for the segment and aggregate `order_items` for profit.

SQL FOR DATA ANALYSIS · LESSON 19

Incremental Loading and Data-Quality SQL

Load only what has changed, using watermarks, MERGE and upserts, and protect the warehouse with reusable data-quality checks for nulls, duplicates, orphans, ranges and freshness.

Gap-fill lesson | 2026 | Advanced | 150 min

By the end of this lesson you can

- Explain full versus incremental loads and when each is appropriate
- Use a watermark (high-water mark) to pick up only new and changed rows
- Write an upsert with MERGE (or INSERT ... ON CONFLICT) that does not create duplicates
- Run a standard set of data-quality checks and report failures with counts

Why incremental loads matter

A full reload copies every row from the source each night. That is fine for ten thousand rows and painful for ten million. An **incremental load** copies only rows that are new or changed since the last run. It is faster, cheaper, and easier to recover from a failed run, provided it is designed correctly.

CASE STUDY **The nightly sales feed**

The order system writes new orders and corrections all day. The warehouse must show yesterday's sales by 8 a.m. without reloading three years of history every night, and without double-counting the orders that were corrected. The engineer's job is to design the load and to prove the numbers match.

The watermark

A **watermark** is the highest value of a column that the last successful load reached, most often a timestamp such as `updated_at`. The next run asks for rows strictly greater than the watermark.

SQL

```sql
-- Pretend the warehouse table already holds orders up to 30 November 2025
CREATE TABLE fact_orders AS
SELECT row_id, order_id, order_date, ship_date, ship_mode, customer_id, sales, profit
FROM superstore
WHERE order_date ⩽ DATE '2025-11-30';

-- Watermark: the latest order date already loaded
SELECT MAX(order_date) AS watermark FROM fact_orders;

-- The incremental batch: only rows after the watermark
SELECT COUNT(*) AS new_rows, MIN(order_date) AS first_new, MAX(order_date) AS last_new
FROM superstore
WHERE order_date > (SELECT MAX(order_date) FROM fact_orders);
```

RESULT

RESULT SET 1

watermark
2025-11-30

1 rows × 1 columns

RESULT SET 2

new_rows	first_new	last_new
262	2025-12-01	2025-12-31

1 rows × 3 columns

The watermark is only as reliable as the column behind it. Two common traps:

- **Late-arriving rows**: a row dated yesterday may arrive today. A watermark on the event date misses it; a watermark on the arrival or `updated_at` timestamp catches it.
- **Corrected rows**: an updated order keeps its old date, so a date-based watermark never sees the change. Use `updated_at`, or re-load a safety window (for example, the last seven days) and let the upsert handle repeats.

::::tip A safety window Many teams reload the last few days on every run and rely on an idempotent upsert. The extra rows cost little, and the design survives late corrections without special logic. ::::

Upserts: insert new rows and update changed ones

An **upsert** inserts a row when its key is new and updates it when the key exists. Without it, a re-run creates duplicates. Standard SQL uses `MERGE`; PostgreSQL and DuckDB offer `INSERT ... ON`

CONFLICT. The example below uses INSERT ... ON CONFLICT on the line-level key, which DuckDB supports.

SQL

```sql
-- Target table keyed on the order line
CREATE TABLE fact_lines (
    row_id     BIGINT PRIMARY KEY,
    order_id   VARCHAR,
    product_id VARCHAR,
    sales      DOUBLE,
    quantity   BIGINT,
    discount   DOUBLE,
    profit     DOUBLE
);

INSERT INTO fact_lines
SELECT row_id, order_id, product_id, sales, quantity, discount, profit
FROM superstore WHERE row_id ⩽ 100;

-- Incoming batch: 50 lines, some of them corrected (same row_id, changed profit)
CREATE TEMP TABLE incoming AS
SELECT row_id, order_id, product_id, sales, quantity, discount,
       CASE WHEN row_id ⩽ 10 THEN profit + 100 ELSE profit END AS profit
FROM superstore WHERE row_id BETWEEN 60 AND 110;

-- Upsert: new keys are inserted, existing keys are updated
INSERT INTO fact_lines
SELECT * FROM incoming
ON CONFLICT (row_id) DO UPDATE SET
    sales = excluded.sales,
    quantity = excluded.quantity,
    discount = excluded.discount,
    profit = excluded.profit;

-- Check: no duplicate keys, and the expected total row count
SELECT COUNT(*) AS total_rows, COUNT(DISTINCT row_id) AS distinct_keys FROM fact_lines;
```

RESULT

```
Commands completed successfully.
(100 rows affected)
Commands completed successfully.
(51 rows affected)
```

total_rows	distinct_keys
110	110

1 rows × 2 columns

After the run, `total_rows` equals `distinct_keys`: the rows that overlapped were updated, not copied. Running the same batch twice produces the same result, which is the property called **idempotency**. Idempotent loads can be re-run safely after a failure.

::::warn Plain INSERT is not an incremental load Inserting the batch without a key check doubles the overlapping rows. If a job failed halfway and was restarted, the totals would be wrong and nobody would see it without a check. ::::

Data-quality checks

Good pipelines run checks after each load and stop or alert when a check fails. The five checks below cover most cases. Each query returns the number of failing rows, so zero means pass.

1. Nulls in required columns

SQL

```sql
SELECT COUNT(*) AS failing_rows
FROM superstore
WHERE order_id IS NULL OR order_date IS NULL OR customer_id IS NULL OR product_id IS NULL;
```

RESULT

failing_rows
0

1 rows × 1 columns

2. Duplicate keys

SQL

```sql
SELECT COUNT(*) AS duplicated_keys
FROM (
    SELECT row_id FROM superstore GROUP BY row_id HAVING COUNT(*) > 1
) d;
```

RESULT

duplicated_keys
0

1 rows × 1 columns

3. Orphans: rows whose parent does not exist

Here every order line must have a product in the product table. An orphan means the product was deleted in the source, or the join key was typed incorrectly.

SQL

```sql
SELECT COUNT(*) AS orphan_lines
FROM superstore s
LEFT JOIN products p ON p.product_id = s.product_id
WHERE p.product_id IS NULL;
```

RESULT

orphan_lines
0

1 rows × 1 columns

4. Range checks

Quantities and discounts have business limits. A discount above 100% or a negative quantity is almost always a source error.

SQL

```sql
SELECT
    SUM(CASE WHEN quantity ⩽ 0 THEN 1 ELSE 0 END)           AS non_positive_quantity,
    SUM(CASE WHEN discount < 0 OR discount > 1 THEN 1 ELSE 0 END) AS discount_out_of_range,
    SUM(CASE WHEN sales < 0 THEN 1 ELSE 0 END)               AS negative_sales
FROM superstore;
```

RESULT

non_positive_quantity	discount_out_of_range	negative_sales
0	0	0

1 rows × 3 columns

5. Freshness

A load that succeeds but brings no recent data is a failure too. This check reports how many days old the newest order is.

SQL

```sql
SELECT MAX(order_date) AS latest_order,
       DATE_DIFF('day', MAX(order_date), DATE '2025-12-31') AS days_behind_reference
FROM superstore;
```

RESULT

latest_order	days_behind_reference
2025-12-31	0

1 rows × 2 columns

The reference date stands in for "today" so that the example is repeatable. In production, use `CURRENT_DATE`.

A reusable check table

Store results so that trends can be watched over time, not only at the moment of a failure.

SQL

```sql
CREATE TABLE IF NOT EXISTS dq_results (
    check_name VARCHAR,
    failing_rows BIGINT,
    checked_on DATE
);

INSERT INTO dq_results
SELECT 'null_required_columns', COUNT(*), DATE '2025-12-31'
FROM superstore WHERE order_id IS NULL OR order_date IS NULL OR customer_id IS NULL OR
product_id IS NULL
UNION ALL
SELECT 'orphan_product_lines', COUNT(*), DATE '2025-12-31'
FROM superstore s LEFT JOIN products p ON p.product_id = s.product_id WHERE p.product_id IS
NULL;

SELECT check_name, failing_rows,
       CASE WHEN failing_rows = 0 THEN 'PASS' ELSE 'FAIL' END AS status
FROM dq_results;
```

RESULT

```
Commands completed successfully.
(2 rows affected)
```

check_name	failing_rows	status
null_required_columns	0	PASS
orphan_product_lines	0	PASS

2 rows × 3 columns

::::key Rules of thumb - Use a watermark on an arrival or update timestamp, and a safety window for late data. - Upsert on a stable key so that re-running a load does not create duplicates. - Check nulls, duplicates, orphans, ranges and freshness after every load, and record the results. - A check that returns a count is easy to alert on; a check that returns rows is easy to investigate. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is a watermark in an incremental load?

A The total row count

B The highest value of a column that the last successful load reached

C The name of the source system

D A password for the warehouse

Answer: B. The next run loads rows beyond the watermark.

2. Why is a date-based watermark risky?

A Dates cannot be compared

B Late-arriving or corrected rows may keep an old date and be missed

C Dates are always duplicated

D It is slower than a full load

Answer: B. Updates to old rows do not move the event date forward.

3. What makes a load idempotent?

A It runs only once

B Running it again with the same input gives the same result

C It deletes the target first

D It uses no keys

Answer: B. Idempotent loads are safe to repeat after a failure.

4. A data-quality check that returns 0 usually means:

A The check failed

B The check passed, because no rows broke the rule

C The table is empty

D The query has a syntax error

Answer: B. Checks in this lesson count failing rows, so zero is a pass.

5. An orphan row is one that:

A Has no date

B Refers to a parent key that does not exist in the parent table

C Is a duplicate

D Has a negative value

Answer: B. Orphans break joins and silently drop rows from reports.

EXERCISE

Freshness and volume alerts

Write a single query over `superstore` that returns one row per month of 2025 with the number of order lines and the sales total. Add a column `alert` that shows 'LOW VOLUME' when the month's line count is less than 80% of the average monthly line count for 2025, otherwise 'OK'.

Show solution (worked answer)

Hint: use a CTE with `DATE_TRUNC('month', order_date)` grouped by month, then a window `AVG(lines) OVER ()` to get the average, and a `CASE` expression comparing each month to `0.8 * average`.

SQL FOR DATA ANALYSIS · LESSON 20

SQL Across Platforms: SQL Server, BigQuery, Snowflake and Databricks SQL

Learn the same analysis on the four platforms most job postings name, see where their SQL dialects differ (dates, string functions, limits, top-N, identifiers), and know which platform concept maps to which.

Gap-fill lesson | 2026 | Advanced | 120 min

By the end of this lesson you can

- Name the main SQL platforms and what each is typically used for
- Translate common operations (top N, dates, string concatenation, casting) between dialects
- Explain the platform concepts of warehouse, dataset or schema and compute cost
- Write portable SQL and know which parts to rewrite when moving platforms

One language, four dialects

SQL is a standard, but every platform adds its own functions, limits and naming rules. An analyst who knows the logic can move between platforms quickly; the work is in translating the syntax. The queries in this course run in DuckDB, which is close to PostgreSQL. The examples below show the same question written for each platform.

Platform	Typical use	Where the company runs it
SQL Server (T-SQL)	Operational databases, enterprise reporting	On-premises, or Azure SQL
BigQuery	Serverless analytics on very large data	Google Cloud
Snowflake	Cloud data warehouse, shared data	Snowflake on AWS, Azure or Google Cloud
Databricks SQL	SQL on a lakehouse (Delta tables)	Databricks on the major clouds

CASE STUDY **One question, four platforms**

A product manager asks for the top five cities by profit in 2025. The analyst writes the same logic on each platform, because the company's warehouse, reporting tool and data team may each use a different one.

Top N rows

The question "top five cities by profit" is written differently on each platform.

SQL

```sql
-- SQL Server (T-SQL)
SELECT TOP 5 city, SUM(profit) AS profit
FROM superstore
WHERE YEAR(order_date) = 2025
GROUP BY city
ORDER BY profit DESC;

-- BigQuery
SELECT city, SUM(profit) AS profit
FROM `project.dataset.superstore`
WHERE EXTRACT(YEAR FROM order_date) = 2025
GROUP BY city
ORDER BY profit DESC
LIMIT 5;

-- Snowflake
SELECT city, SUM(profit) AS profit
FROM superstore
WHERE YEAR(order_date) = 2025
GROUP BY city
ORDER BY profit DESC
LIMIT 5;

-- Databricks SQL (Spark SQL)
SELECT city, SUM(profit) AS profit
FROM superstore
WHERE YEAR(order_date) = 2025
GROUP BY city
ORDER BY profit DESC
LIMIT 5;
```

The logic is identical. The differences are `TOP` against `LIMIT`, and the way each platform names tables and the date parts it accepts.

The same query runs in DuckDB, which is the engine used for the lessons in this course:

SQL

```sql
SELECT city, ROUND(SUM(profit), 0) AS profit
FROM superstore
WHERE EXTRACT(YEAR FROM order_date) = 2025
GROUP BY city
ORDER BY profit DESC
LIMIT 5;
```

RESULT

city	profit
Pune	912,526
Kolkata	739,621
Indore	527,368
Bengaluru	527,164
Chennai	504,760

5 rows × 2 columns

Dates

Date handling is the most common source of dialect errors.

Task	SQL Server	BigQuery	Snowflake	Databricks SQL
Year of a date	`YEAR(d)`	`EXTRACT(YEAR FROM d)`	`YEAR(d)`	`YEAR(d)`
Start of month	`DATEFROMPARTS(YEAR(d), MONTH(d), 1)`	`DATE_TRUNC(d, MONTH)`	`DATE_TRUNC('month', d)`	`DATE_TRUNC('MONTH', d)`
Add 30 days	`DATEADD(day, 30, d)`	`DATE_ADD(d, INTERVAL 30 DAY)`	`DATEADD(day, 30, d)`	`DATE_ADD(d, 30)`
Days between	`DATEDIFF(day, a, b)`	`DATE_DIFF(b, a, DAY)`	`DATEDIFF(day, a, b)`	`DATEDIFF(day, a, b)`

Notice that `DATEDIFF` takes its arguments in a different order on BigQuery: `DATE_DIFF(later, earlier, DAY)`. Argument order is a frequent source of wrong results that pass without error.

Strings and casting

```sql
-- SQL Server: concatenation with +, and NULL makes the result NULL
SELECT customer_name + ' (' + city + ')' FROM customers;

-- Snowflake, BigQuery and Databricks: || or CONCAT
SELECT customer_name || ' (' || city || ')' FROM customers;
SELECT CONCAT(customer_name, ' (', city, ')') FROM customers;

-- Casting
-- SQL Server and Snowflake: CAST(x AS INT)
-- BigQuery: CAST(x AS INT64)
-- Databricks: CAST(x AS INT)
```

Concatenation with + in SQL Server returns NULL if any part is NULL, while `CONCAT` treats NULL as an empty string. Check your platform's behaviour before trusting names with missing middle parts.

Table names and identifiers

- **SQL Server**: `database.schema.table`, with square brackets for unusual names: `[Order Date]`.
- **BigQuery**: `project.dataset.table`, with backticks around the full name.
- **Snowflake**: `database.schema.table`. Unquoted names are converted to upper case, so `order_date` and `ORDER_DATE` are the same column unless quoted.
- **Databricks**: `catalog.schema.table` in Unity Catalog, or `schema.table` in older workspaces.

::::warn Case sensitivity Snowflake folds unquoted names to upper case. A quoted name such as `"Order_Date"` is different from `order_date`. Choose one naming convention and stick to it. ::::

Platform concepts you will hear about

- **Warehouse (Snowflake)**: a compute cluster that you size and pause. Cost depends on how long it runs, so auto-suspend matters.
- **Slots and bytes billed (BigQuery)**: you pay for the bytes a query scans, so selecting only the columns you need reduces cost.
- **Delta tables (Databricks)**: tables stored as Parquet files with a transaction log. They support `MERGE`, time travel and schema evolution, covered in the Power BI and Fabric lessons.
- **Clusters and SQL warehouses (Databricks)**: compute attached to notebooks or SQL endpoints.
- **Indexes and statistics (SQL Server)**: a clustered index orders the table on disk. Columnar warehouses such as Snowflake and BigQuery handle scans differently and rely less on manual indexes.

Writing portable SQL

When the company may move platforms, keep the logic in a layer that is easy to translate:

1. Use ANSI functions where they exist: `CASE`, `COALESCE`, `CAST`, `SUM OVER`, `ROW_NUMBER`, `DATE_TRUNC` (with care about its argument order on each platform).
2. Keep platform-specific date and string code in views or macros, not scattered through reports.
3. Use the same test queries on every platform, such as row counts and totals per month, to check a migration.
4. Note each platform's limits, such as the maximum number of columns in a select or the size of a `GROUP BY`.

::::tip Test the translation with numbers A moved query is correct only when it returns the same totals. Compare the old and new results by month and by category, not just by the grand total, because two errors can cancel out in the total. ::::

::::key What to remember - The logic of SQL transfers; the dialect (top N, dates, strings, identifiers) does not. - Check argument order for date functions on every platform. - Snowflake folds unquoted names to upper case; BigQuery uses backticks and three-part names. - Compute cost is a platform concept: warehouses (Snowflake), bytes scanned (BigQuery) and clusters (Databricks). - Verify any migration by comparing results by period and category. ::::

Quick quiz

5 questions · tap an answer to check it

1. Which clause returns the first five rows in SQL Server?

A LIMIT 5

B TOP 5

C FETCH FIRST 5 ROWS ONLY on every platform

D ROWNUM < 5 only

Answer: B. SQL Server uses TOP; LIMIT is used by BigQuery, Snowflake and Databricks.

2. BigQuery's DATE_DIFF takes its arguments in which order?

A Earlier date then later date

B Later date then earlier date, then the unit

C The unit first

D Dates are not allowed

Answer: B. DATE_DIFF(later, earlier, DAY) gives a positive result for a later first date.

3. In SQL Server, `'Asha' + ' ' + NULL` returns:

A 'Asha '

B NULL

C 'Asha'

D An error only

Answer: B. SQL Server's + operator returns NULL when any part is NULL.

4. Snowflake folds unquoted identifiers to:

A Lower case

B Upper case

C Title case

D Their original case

Answer: B. `order_date` and `ORDER_DATE` are the same column unless quoted.

5. On BigQuery, the most direct way to reduce cost is to:

- **A** Add more indexes
- **B** Select only the columns the query needs
- **C** Turn off caching
- **D** Use more warehouses

Answer: B. BigQuery bills by the bytes scanned.

EXERCISE **Translate a query**

Rewrite this T-SQL for Snowflake and for BigQuery, keeping the same result: `SELECT TOP 10 customer_id, DATEDIFF(day, MIN(order_date), MAX(order_date)) AS days_active FROM orders GROUP BY customer_id ORDER BY days_active DESC;`

Show solution (worked answer)

Snowflake: the same `DATEDIFF(day, ...)` works, with `LIMIT 10` replacing `TOP 10`. BigQuery: `DATE_DIFF(MAX(order_date), MIN(order_date), DAY)` (later date first), `LIMIT 10` at the end, and the table written as `project.dataset.orders`.

Skill Navora

TRACK 3

Power BI

ETL with Power Query, star-schema modelling, DAX measures and time intelligence, interactive reports, case studies, publishing, security and Copilot.

8 lessons | Version 2026.10 | Updated October 2026

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Contents

Every entry is a clickable link. The PDF also has bookmarks for each lesson and section.

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04 DAX Time Intelligence, Ranking, Parameters & Dynamic Measures

05 Visuals, Filters, Drill-through, Tooltips, Buttons & Report Design

06 Power BI Service, Connectivity Modes, Security & Capstone

07 Microsoft Fabric: Lakehouse, Warehouse, Dataflows and Semantic Models

08 Databricks: Notebooks, Delta Lake, Spark Jobs and Unity Catalog

POWER BI · LESSON 1

Getting Started with Power BI & Your First HR Report

Understand what Power BI is and how its parts fit together, install Power BI Desktop, tour the three views, follow the ETL and analytics stages, and build a complete first report on HR data: cards, bar and column charts, a donut, slicers, a year column with DAX, groups and small multiples.

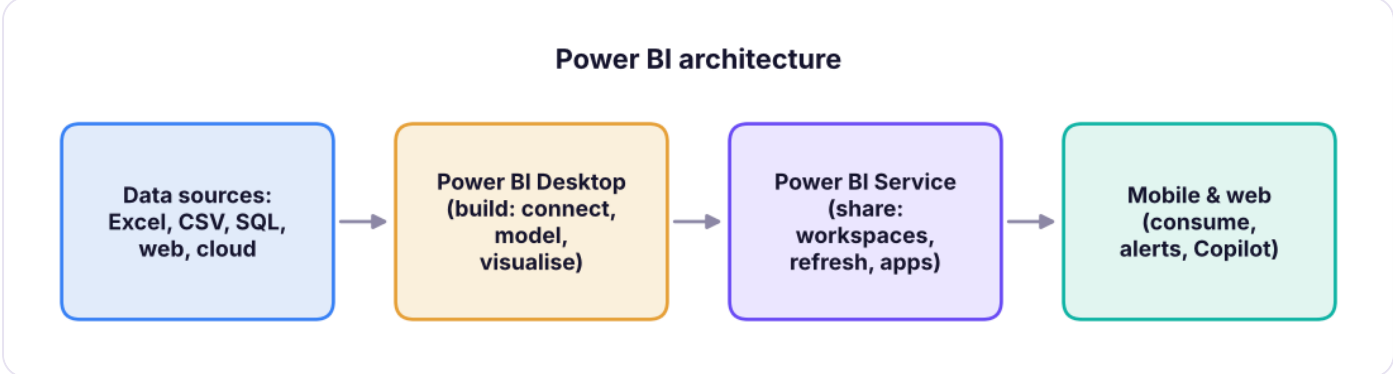
Power BI class 1 | Power BI Desktop 2026 (Copilot, Fabric, PBIP) | Beginner | 120 min

By the end of this lesson you can

- Describe Power BI Desktop, the Service and Mobile, and how data flows between them
- Explain ETL and the stages of the data-analytics process, and map them to Power BI tools
- Navigate Report, Table and Model views, and the Fields, Visualizations and Filters panes
- Load a CSV, check data types and build cards, bar/column charts, a matrix, a donut and slicers
- Add a calculated column with DAX (YEAR), create data groups, and use small multiples

What is Power BI?

Power BI is Microsoft's business-intelligence platform: it connects to data, cleans and models it, and turns it into interactive reports and dashboards that people can share. In 2026 it is part of **Microsoft Fabric**, Microsoft's all-in-one analytics platform, and it is one of the most requested tools in analyst job posts, alongside Excel, SQL and Tableau.



The Power BI components. You build in Desktop, publish to the Service, and consume on the web or mobile.

Component	What it is	Who uses it
Power BI Desktop	Free Windows application to build data models and reports (`.pbix`)	Analysts, developers
Power BI Service (app.powerbi.com)	Cloud site to publish, share, schedule refresh, create dashboards and apps	Analysts and report consumers
Power BI Mobile	iOS/Android apps to view reports and get alerts	Managers on the go
Power BI Report Builder	Paginated, print-ready reports (invoices, statements)	Report developers
On-premises data gateway	Secure bridge so the cloud can refresh data from your own servers	Admins
Power BI Report Server	Self-hosted option for organisations that cannot use the cloud	IT
Copilot	AI assistant that writes DAX, builds report pages and summarises insights (needs a Fabric/Premium capacity)	Everyone

> **TIP** **Report vs dashboard**
>
> In Power BI a **report** is one or more pages of interactive visuals built on a data model. A **dashboard** (Service only) is a single page of **pinned tiles** from different reports. Many teams use reports only, or share them as an **app**.

Licences in brief

Licence	Use
Power BI Desktop	Free: build and save locally
Free (Service)	Personal use in "My workspace"; cannot share with others
Pro	Publish and share with other Pro users (about US$14 per user per month)
Premium Per User / Fabric capacity	Larger models, AI features, share with free users

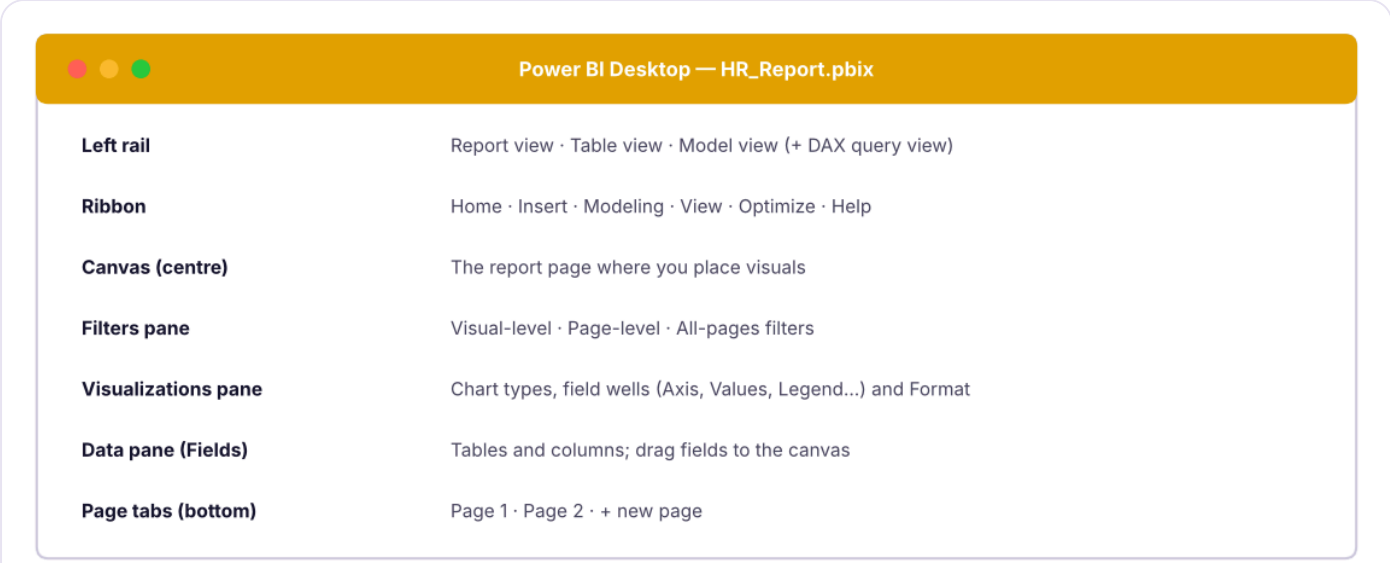
(Prices change; check Microsoft's current pricing page.)

Installing Power BI Desktop

STEP BY STEP **Install**

1. **Easiest:** open the **Microsoft Store**, search "Power BI Desktop" and click Install (it updates itself every month).
2. Or download it from the Microsoft Power BI website → Download → Power BI Desktop.
3. Requirements: Windows 10/11 (64-bit), 4 GB+ RAM (8 GB+ recommended). **There is no Mac version**: Mac users can use the Service in a browser, a Windows virtual machine, or Windows 365.
4. Start the app and sign in with a work or school account (optional for building locally).

The three views

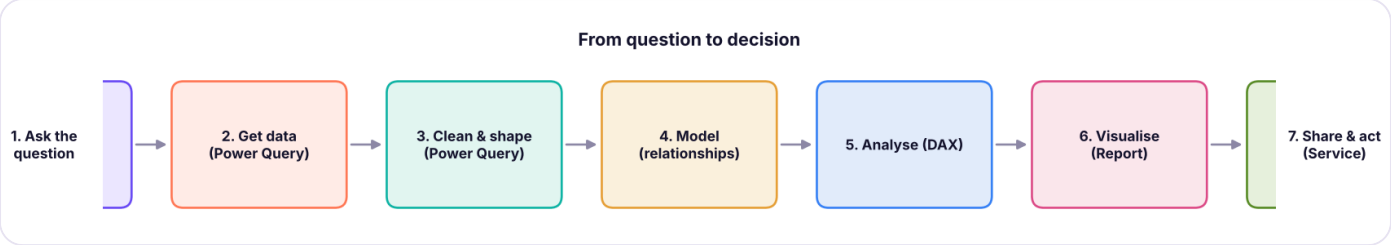


The Power BI Desktop window: left rail with three views, canvas in the middle, panes on the right.

View	Icon	Use it to
Report	bar chart	Design the report pages
Table (Data)	grid	Inspect the data and create calculated columns and measures
Model	diagram	See and edit **relationships** between tables
DAX query view	`{ }`	Run DAX queries and test measures

ETL and the analytics process

ETL = **E**xtract (get data from sources), **T**ransform (clean and reshape it), **L**oad (store it in the model). In Power BI, Extract and Transform are done in **Power Query**, and Load puts the result into the data model. (You met Power Query in Excel Lesson 12: it is the same engine.)



The data analytics stages, mapped to Power BI.

Always start with the **question**: "Why are people leaving?" decides which data and visuals you need. The worst Power BI reports are tours of every chart type with no question.

Your first report: HR attrition

CASE STUDY **The brief**

HR wants a one-page report on the company's 1,500 employees: how many there are, how many left (attrition), where attrition is highest, and whether overtime, job satisfaction or income play a role. Use `hr_employees.csv`.

The data (first rows):

Python

```python
import pandas as pd
hr = pd.read_csv("hr_employees.csv", parse_dates=["hire_date"])
print(hr.shape)
hr.head(5)
```

OUTPUT

```
(1500, 15)
```

	emp_id	gender	age	department	job_title	education	hire_date	years_at_c
0	E00001	Male	39	Operations	Operations Manager	Bachelor's	2025-05-26	0
1	E00002	Female	45	IT	Tester	Bachelor's	2020-10-17	5
2	E00003	Male	41	IT	Tester	Bachelor's	2019-05-06	6
3	E00004	Male	21	R&D	Data Scientist	Bachelor's	2025-08-23	0
4	E00005	Male	41	Sales	Sales Executive	Master's	2015-08-24	10

5 rows × 15 columns

Step 1: get the data

STEP BY STEP **Load the CSV**

1. Home → **Get data → Text/CSV** → choose `hr_employees.csv`.
2. The preview shows the columns and detected data types. Click **Transform Data** (not Load) to check types in Power Query.
3. Confirm: `age`, `years_at_company`, `monthly_income`, `job_satisfaction`, `performance_rating` are whole numbers; `hire_date` is **Date**; the rest are text.
4. Home → **Close & Apply**.

In the Data pane you now see the table `hr_employees` with a **Σ** next to numeric fields (they can be summed) and no symbol next to text fields.

Step 2: KPI cards

Drag fields onto the canvas, then choose the **Card** visual:

- **Employees** = Count of `emp_id`
- **Attrition (leavers)**: use a measure (Lesson 3 explains measures): `Leavers = CALCULATE ( COUNTROWS ( hr_employees ), hr_employees[attrition] = "Yes" )`
- **Attrition %** = `DIVIDE ( [Leavers], COUNTROWS ( hr_employees ) )`, formatted as a percentage
- **Average monthly income** = Average of `monthly_income`

Python

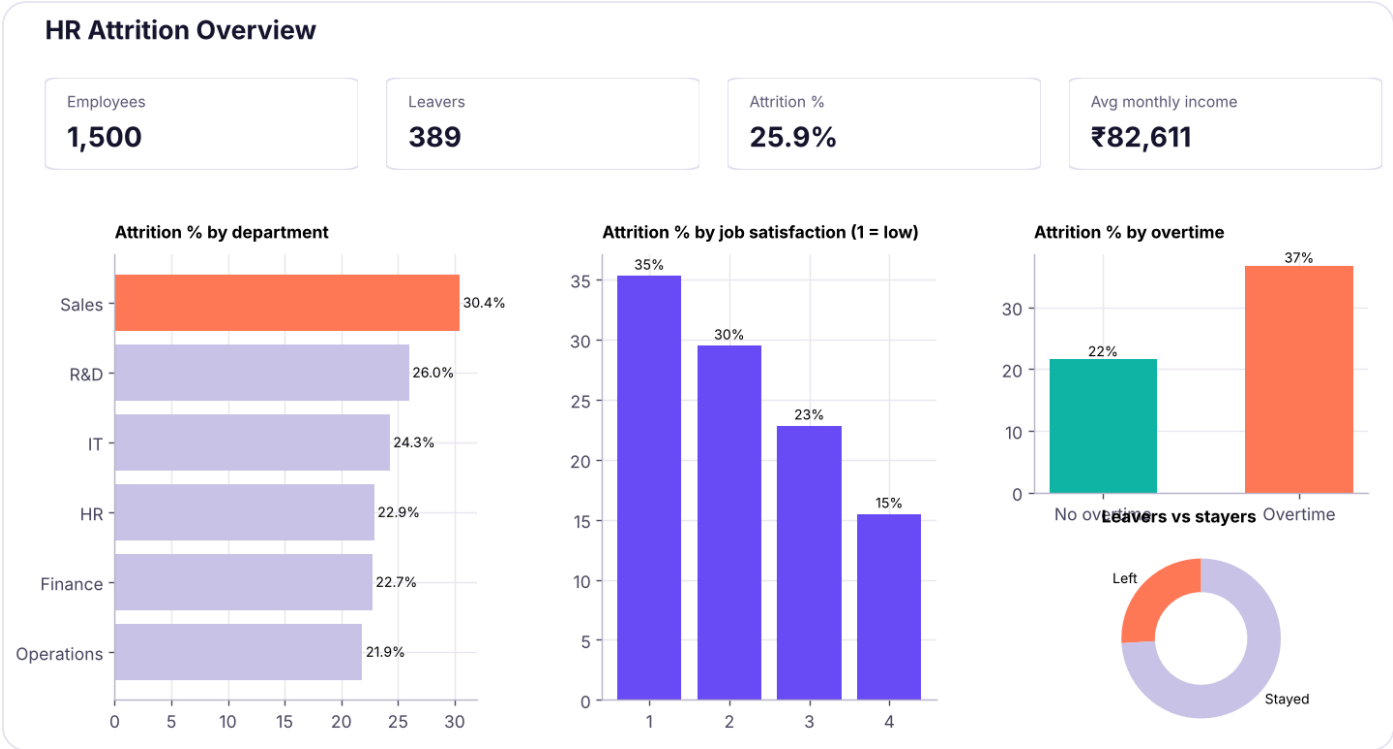
```python
n = len(hr); leavers = (hr.attrition == "Yes").sum()
print(f"Employees      : {n:,}")
print(f"Leavers        : {leavers:,}")
print(f"Attrition %    : {leavers / n:.1%}")
print(f"Avg income (₹) : {hr.monthly_income.mean():,.0f}")
```

OUTPUT

```
Employees      : 1,500
Leavers        : 389
Attrition %    : 25.9%
Avg income (₹) : 82,611
```

Step 3: bar, column and donut charts

- **Clustered bar chart**: Y-axis `department`, X-axis = `Attrition %` → ranks departments.
- **Clustered column chart**: X-axis `job_satisfaction`, Y-axis `Attrition %`.
- **Donut chart**: Legend `attrition`, Values = count of `emp_id`.
- **Matrix**: Rows `department`, Columns `overtime`, Values `Attrition %` (turns into a heat map with conditional formatting, Lesson 5).
- **Slicers**: `gender`, `education`, `city`.



The finished HR page, drawn from the real data: KPI cards, attrition by department, by satisfaction and overtime, and the leaver/stayer donut.

Step 4: a calculated column with DAX

We want to see joining trends by year, but the table has only `hire_date`. Add a **calculated column**: Table view (or Modeling) → **New column**:

DAX

```
Hire Year = YEAR ( hr_employees[hire_date] )
```

A calculated column is evaluated **row by row when the data loads** and stored in the model, just like a new column in Excel. Now a line/column chart of **headcount by hire year** works:

Python

```
hr["hire_year"] = hr.hire_date.dt.year
by_year = hr.groupby("hire_year").agg(hired=("emp_id", "count"), left=("left", "sum"))
by_year["attrition_%"] = (by_year.left / by_year.hired * 100).round(1)
by_year.tail(8)
```

OUTPUT

	hired	left	attrition_%
hire_year			
2019	82	16	19.5
2020	85	20	23.5
2021	120	22	18.3
2022	136	35	25.7
2023	196	60	30.6
2024	234	71	30.3
2025	284	93	32.7
2026	64	17	26.6

8 rows × 3 columns

Step 5: data groups and bins

Groups combine values into custom categories, such as the "age band" below, without changing the source data:

- **Bins** (numbers): right-click the field → **New group** → Group type *Bin* → bin size 5 → creates `age (bins)`.
- **List groups** (text): select several values → right-click → **Group** → rename, for example group *Mumbai* and *Pune* as "Maharashtra".

For flexible logic, a calculated column is more transparent:

DAX

```
Age Band =
SWITCH (
    TRUE (),
    hr_employees[age] < 25, "Under 25",
    hr_employees[age] < 35, "25-34",
    hr_employees[age] < 45, "35-44",
    "45 and over"
)
```

Python

```
bands = pd.cut(hr.age, [0, 24, 34, 44, 120], labels=["Under 25", "25-34", "35-44", "45 and over"])
hr.groupby(bands, observed=False).agg(employees=("emp_id", "count"), attrition_pct=("left", lambda s: round(s.mean() * 100, 1)))
```

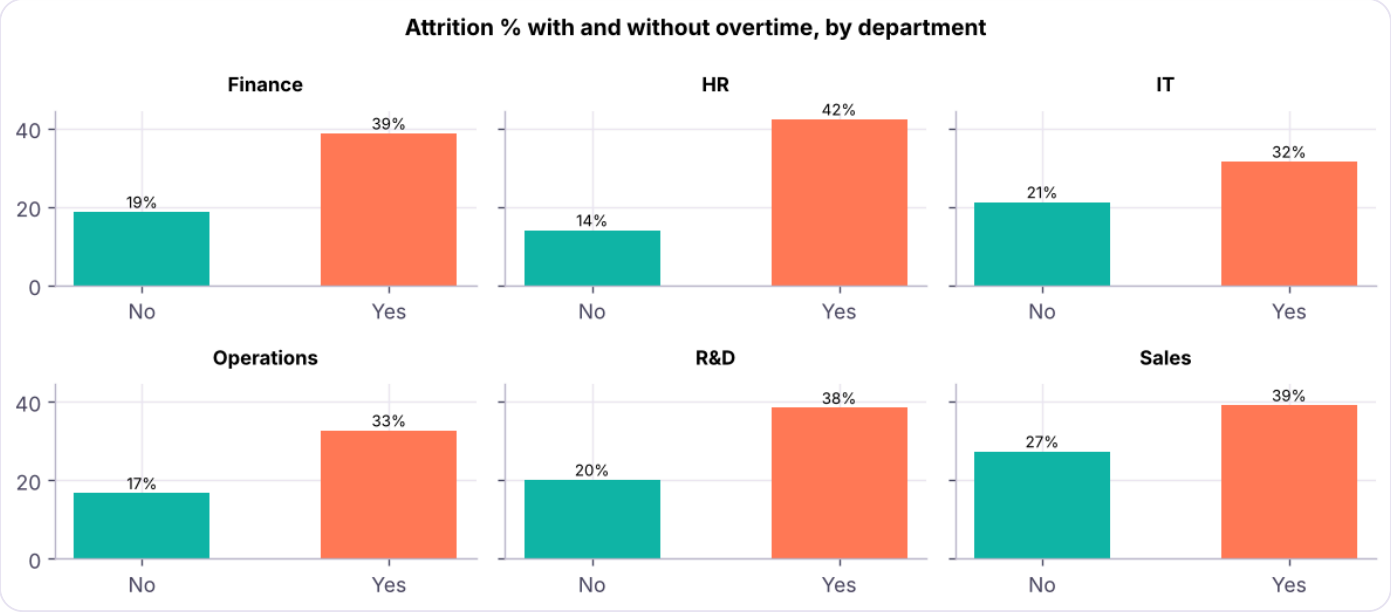
OUTPUT

	employees	attrition_pct
age		
Under 25	148	35.1
25–34	624	26.9
35–44	574	23.5
45 and over	154	22.1

4 rows × 2 columns

Step 6: small multiples

Small multiples split one visual into a grid of identical mini-charts by a field. Drag `department` into the visual's **Small multiples** well: you get one small column chart of attrition by satisfaction for every department. This lets you compare patterns across groups at a glance without a cluttered legend.



Small multiples: attrition % by overtime for each department. The same pattern repeats everywhere, which says overtime is a company-wide issue, not a one-department issue.

What does the report tell us?

Python

```
t = hr.groupby("overtime").left.agg(["count", "mean"]); t["mean"] = (t["mean"] *
100).round(1)
t.columns = ["employees", "attrition_%"]
print(t.to_string()); print()
low = hr[hr.job_satisfaction ⩽ 2].left.mean() * 100; high = hr[hr.job_satisfaction ⩾
3].left.mean() * 100
print(f"Attrition for satisfaction 1-2: {low:.1f}%   vs   3-4: {high:.1f}%")
```

OUTPUT

```
           employees  attrition_%
overtime
No              1067         21.6
Yes              433         36.7

Attrition for satisfaction 1-2: 32.5%   vs   3-4: 19.4%
```

Insight sentences for a manager: *overtime workers leave far more often than others; low satisfaction scores go with higher attrition; Sales has the highest attrition among departments*. Notice the careful wording: these are **associations** in the data, not proof of cause. (The ML track shows how to test them properly.)

Good habits from day one

1. **Name things clearly**: `Attrition %`, not `Measure 3`; rename pages ("HR Overview").

2. **Put measures in a dedicated table** later (Lesson 3) and format them (percentage, thousands separator).
3. **Check data types** in Power Query before loading.
4. **One idea per page**; the title should say the insight.
5. **Save often** (`.pbix`), and use clear version names.
6. **Do not use pie charts** with many slices; prefer bars (ranked) and use donuts only for 2–3 parts.

2026 UPDATE

Power BI in 2026

- **Copilot** can create a report page from a prompt ("Build an attrition overview for HR"), write DAX measures and summarise a page. It needs a paid Fabric/Premium capacity and your data to be well-modelled and named.
- **Power BI Project (PBIP)** format stores the report as text files, so teams can use **Git** and code review.
- **TMDL** (Tabular Model Definition Language) and **DAX query view** make models easier to script and test.
- **Direct Lake** (Fabric) reads Delta tables directly with import-like speed. (See Lesson 6.)

Summary

KEY TAKEAWAYS

- Power BI = Desktop (build) + Service (share) + Mobile (consume), now inside Microsoft Fabric.
- ETL: **Extract and Transform in Power Query, Load into the model**; analytics stages run from question to decision.
- Three views: **Report**, **Table**, **Model**; build with the Fields, Visualizations and Filters panes.
- A **card** shows a KPI; bar/column charts compare categories; a donut shows 2–3 parts; a matrix is a pivot table; slicers filter.
- A DAX **calculated column** (`YEAR`, `SWITCH`) is computed row by row at load time; **groups/bins** categorise without changing the data; **small multiples** repeat a chart per category.

Quick quiz

6 questions · tap an answer to check it

1. In Power BI, where do you perform the Extract and Transform steps of ETL?

- **A** Report view
- **B** Power Query
- **C** The Service
- **D** The Model view

Answer: B. Power Query connects to sources and transforms data; Close & Apply loads it into the model.

2. Which view shows relationships between tables?

- **A** Report view
- **B** Table view
- **C** Model view
- **D** Filters pane

Answer: C. Relationships are created and edited in Model view.

3. Power BI Desktop is available for:

- **A** Windows and Mac
- **B** Windows only
- **C** Mac only
- **D** Browsers only

Answer: B. Desktop is Windows-only; Mac users use the Service or a Windows virtual machine.

4. What does a calculated column do?

A Runs only when a slicer changes

B Computes a value for every row when the data is loaded and stores it in the model

C Replaces a measure

D Changes the source file

Answer: B. Calculated columns are row-by-row and stored; measures calculate on the fly.

5. Which visual repeats the same chart for every value of a field?

A Matrix

B Slicer

C Small multiples

D Card

Answer: C. Small multiples create a grid of mini-charts split by a category.

6. Which licence do you need to share a report with colleagues who also have Pro licences?

A None, Desktop is enough

B Free Service licence

C Power BI Pro (or Premium Per User)

D Gateway licence

Answer: C. Sharing in the Service requires Pro for creators and viewers (unless the workspace is on Premium/Fabric capacity).

EXERCISE

Build the HR page

Build the page shown above in Power BI Desktop with `hr_employees.csv`. Then add: (1) a slicer for `city` and `gender`; (2) a matrix of attrition % by `job_title` and `education`; (3) an `Age Band` column and a column chart of attrition % by band; (4) a title text box that states your main finding.

Show solution (worked answer)

(1) Slicer visual, fields `city`, `gender` (Format → Slicer settings → Style: Tile or Dropdown). (2) Matrix: Rows `job_title`, Columns `education`, Values `Attrition %` (turn on Cell elements → Background colour for a heat map). (3) Use the `SWITCH` formula in this lesson; chart X-axis `Age Band`, Y-axis `Attrition %`; sort the axis in the right order using a sort-by-column. (4) Insert → Text box, for example "Overtime is linked to about 15 more attrition points company-wide." Verify the claim against your visuals before you publish.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between Power BI Desktop and Service? Desktop is the free authoring tool for models and reports on Windows; the Service is the cloud platform to publish, share, refresh on a schedule, build dashboards and apps, and apply security.

Q: What is ETL and where is it done in Power BI? Extract, Transform, Load: Power Query connects to sources and applies transformation steps; the result is loaded into the VertiPaq in-memory model.

Q: Calculated column vs measure? A calculated column is evaluated per row at refresh time and stored (uses memory); a measure is evaluated at query time in the current filter context (uses CPU). Use measures for aggregations such as sums and ratios.

Q: Report vs dashboard? A report is multi-page and interactive, built on one model. A dashboard is a single page of pinned tiles from several reports, available only in the Service.

POWER BI · LESSON 2

Power Query, Star Schema & Data Modelling (Adventure Cycles)

Prepare and model data the professional way. Clean data in Power Query, load a star schema (fact and dimension tables), create one-to-many relationships, understand filter direction, build and mark a date table, create hierarchies, and avoid the most common modelling mistakes, using the Adventure Cycles case study.

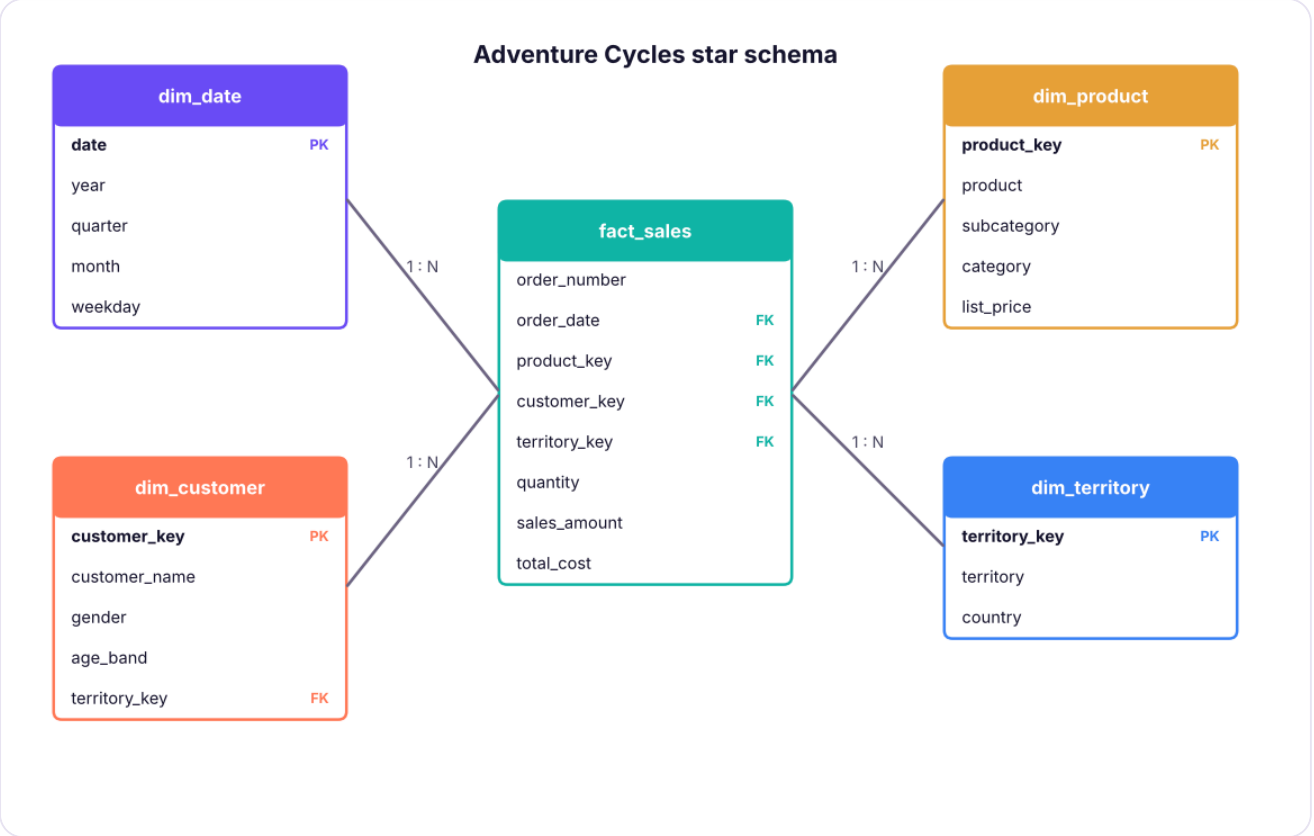
Power BI class 3 | Power BI Desktop 2026 | Intermediate | 140 min

By the end of this lesson you can

- Apply common Power Query steps and read the Applied Steps and M code
- Explain fact tables, dimension tables and the star schema, and why they beat one wide table
- Create relationships with the right cardinality and filter direction, and explain how filters flow
- Build a date table, mark it as a date table and add hierarchies
- Diagnose common modelling problems (many-to-many, ambiguous paths, missing dates)

Why modelling matters more than visuals

Beautiful visuals on a bad model give wrong or slow answers. In Power BI the **data model** decides what numbers mean: how tables relate, which table filters which, and how fast DAX runs. The golden rule: **model as a star schema**.



A star schema: one central fact table of events/numbers surrounded by descriptive dimension tables. Filters flow from dimensions (one side) to the fact (many side).

	Fact table	Dimension table
Describes	Events or transactions (a sale, a click, a shipment)	The "who, what, where, when" of those events
Rows	Many (millions)	Fewer (thousands)
Columns	Keys + numeric measures (quantity, amount, cost)	Keys + descriptive attributes (name, category, city)
Changes	Grows constantly	Changes rarely
Example	`fact_sales`	`dim_product`, `dim_customer`, `dim_date`

Why a star beats one big flat table: smaller file (text stored once), faster visuals, one place to fix a name, correct filtering across several facts, and simpler DAX. A **snowflake** (dimensions normalised into sub-dimensions) works but adds joins; prefer a star in Power BI.

The Adventure Cycles dataset

A fictional bicycle company sells bikes, components, clothing and accessories in India, the UAE and Singapore. The model has one fact table and four dimensions (plus the date table):

Python

```
import pandas as pd
D = "adventure_cycles/"
fact = pd.read_csv(D + "fact_sales.csv", parse_dates=["order_date"]); prod = pd.read_csv(D
+ "dim_product.csv")
cust = pd.read_csv(D + "dim_customer.csv"); terr = pd.read_csv(D + "dim_territory.csv");
dates = pd.read_csv(D + "dim_date.csv", parse_dates=["date"])
for name, df in [("fact_sales", fact), ("dim_product", prod), ("dim_customer", cust),
("dim_territory", terr), ("dim_date", dates)]:
    print(f"{name:14s} {df.shape[0]:>6,} rows × {df.shape[1]} columns")
fact.head(4)
```

OUTPUT

```
fact_sales      9,000 rows × 9 columns
dim_product        45 rows × 6 columns
dim_customer    1,200 rows × 5 columns
dim_territory       6 rows × 3 columns
dim_date        1,096 rows × 6 columns
```

	order_number	order_date	product_key	customer_key	territory_key	quantity	ur
0	SO50387	2023-01-01	11	103	1	3	1:
1	SO55797	2023-01-01	32	798	1	3	2(
2	SO51236	2023-01-01	9	750	5	1	1:
3	SO55082	2023-01-01	13	759	1	1	7!

4 rows × 9 columns

Power Query: prepare each table

Open **Transform Data**. Typical steps for each table (all recorded in **Applied Steps**):

Table	Steps
`fact_sales`	Set types (`order_date` Date; `quantity` Whole number; amounts **Fixed decimal number**), remove blank rows, add `Profit = [sales_amount] - [total_cost]` as a custom column, keep only needed columns
`dim_product`	Trim/clean text, set `list_price` and `standard_cost` as decimals, remove duplicates on `product_key`
`dim_customer`	Replace `M`/`F` with Male/Female, remove duplicates on `customer_key`
`dim_territory`	Check there is one row per `territory_key`
`dim_date`	Set `date` to Date; sort `month` by `month_no` later in the model

Query settings (right pane) shows each step; click any step to see the data at that point, and delete a step to undo it. Rename queries (`Sales`, `Product`…) because those names become table names in the model.

Common transformations (from class and beyond)

Goal	Where	Notes
Promote headers, set types	Home	Always check types: wrong types break relationships
Remove duplicates / blank rows	Home → Remove rows	Needed for dimension keys
Split column by delimiter	Transform → Split Column	e.g. `IN-2025-00042`
Replace values, trim, clean, format	Transform	
Unpivot columns	Transform → Unpivot	Turn Jan, Feb, Mar columns into rows (Excel Lesson 12)
Merge queries	Home → Merge	A join; use to bring a category into the fact (but prefer relationships)
Append queries	Home → Append	Stack monthly files
Group by	Transform → Group By	Pre-aggregate large tables
Conditional / Custom column	Add Column	M formulas
Reference vs Duplicate	Right-click query	*Reference* creates a new query that starts from the output of another (changes flow through); *Duplicate* copies the steps
Parameters	Home → Manage Parameters	A variable such as a file path or a start date, used in queries (Lesson 4)
Disable load	Right-click query	Keep helper queries out of the model

What is "query folding"?

When the source is a database (SQL Server, Snowflake), Power Query **translates your steps into SQL** and lets the database do the work, instead of downloading everything first. A step that cannot be translated "breaks folding" and everything after it runs locally. Check by right-clicking a step → *View Native Query*. Keep foldable steps (filter, select columns, merge, group) **before** non-foldable ones.

Power Query M

```
let
    Source   = Csv.Document(File.Contents("C:\Data\fact_sales.csv"), [Delimiter=",",
Encoding=65001]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Typed    = Table.TransformColumnTypes(Promoted, {{"order_date", type date},
{"quantity", Int64.Type},
                  {"sales_amount", Currency.Type}, {"total_cost", Currency.Type}}),
    Profit   = Table.AddColumn(Typed, "profit", each [sales_amount] - [total_cost],
Currency.Type)
in
    Profit
```

Relationships

A **relationship** links a column in one table to a column in another so that filters can flow between them. In **Model view**, drag `dim_product[product_key]` onto `fact_sales[product_key]`. Power BI usually detects relationships on same-named columns (Home → Manage relationships → Autodetect).

Cardinality

Type	Meaning	Use
One-to-many (1:*)	A key is unique in one table, repeated in the other	The standard: dimension → fact
One-to-one (1:1)	Unique on both sides	Splitting a wide table
Many-to-many (*:*)	Duplicates on both sides	Avoid; use a bridge table

Check uniqueness before creating a one-to-many: if the "one" side has duplicates Power BI will complain.

Python

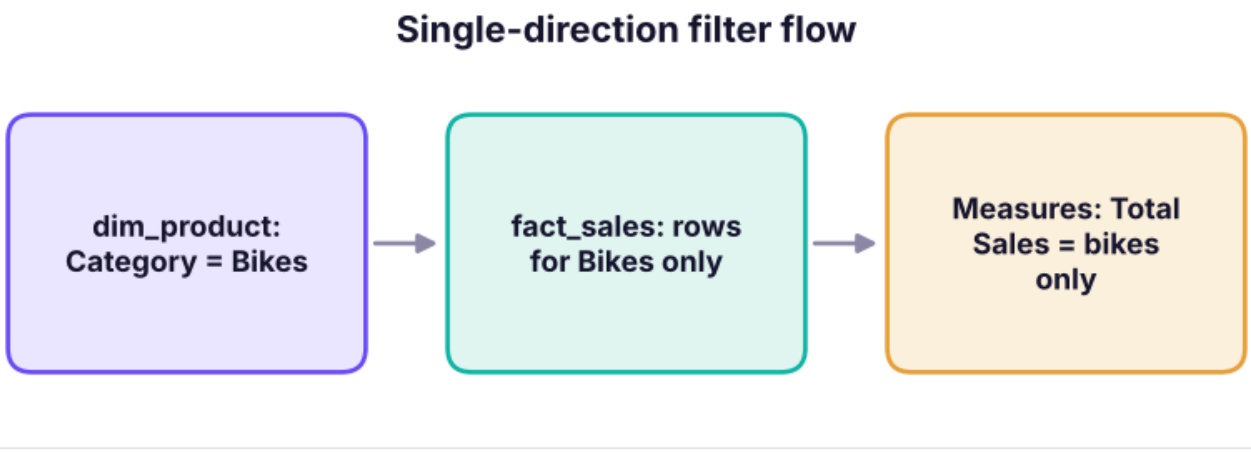
```
checks = {"dim_product[product_key]": prod.product_key, "dim_customer[customer_key]":
cust.customer_key, "dim_territory[territory_key]": terr.territory_key, "dim_date[date]":
dates.date}
for name, col in checks.items(): print(f"{name:30s} unique: {col.is_unique}")
orphans = {"product_key": ~fact.product_key.isin(prod.product_key), "customer_key":
~fact.customer_key.isin(cust.customer_key), "territory_key":
~fact.territory_key.isin(terr.territory_key)}
for k, m in orphans.items(): print(f"fact rows with no matching {k}: {m.sum()}")
```

OUTPUT

```
dim_product[product_key]       unique: True
dim_customer[customer_key]     unique: True
dim_territory[territory_key]   unique: True
dim_date[date]                 unique: True
fact rows with no matching product_key: 0
fact rows with no matching customer_key: 0
fact rows with no matching territory_key: 0
```

Unique keys on the "one" side and no orphans on the "many" side: the model is safe.

Filter direction



Filters flow along relationships from the one side to the many side (single direction). Selecting Bikes in dim_product filters fact_sales, which then filters the other dimensions only if bi-directional filtering is on.

Setting	Behaviour	Advice
Single (dimension → fact)	A selection in a dimension filters the fact	Default and best
Both (bi-directional)	Filters also flow back from fact to dimension	Use only when needed (many-to-many, "show only customers who bought"); can cause ambiguity and slowness
Active vs inactive relationship	Only one active path between two tables; an inactive one can be switched on in DAX with `USERELATIONSHIP`	e.g. Order Date and Ship Date both link to the date table

Seeing the model work

Total sales by product category and by country, using the relationships (a join in pandas):

Python

```python
m = fact.merge(prod, on="product_key").merge(terr, on="territory_key")
m["profit"] = m.sales_amount - m.total_cost
out = m.pivot_table(index="category", columns="country", values="sales_amount",
aggfunc="sum", margins=True, margins_name="Total") / 1e6
out.round(1).rename_axis("Sales (₹ million)")
```

OUTPUT

country	India	Singapore	UAE	Total
Sales (₹ million)				
Accessories	6.7	1.7	1.3	9.8
Bikes	320.6	74.8	71.3	466.8
Clothing	6.0	1.3	1.1	8.5
Components	73.5	18.8	17.2	109.6
Total	406.9	96.7	91.0	594.6

5 rows × 4 columns

In Power BI you get this with a matrix: Rows `category`, Columns `country`, Values `SUM(sales_amount)`. No join code needed: the relationships do it.

The date table

Time intelligence (year-to-date, same period last year, Lesson 4) requires a **proper date table**: one row per day with **no gaps**, covering all dates in the data, and **marked as a date table**.

DAX

```
Date =
VAR MinDate = DATE ( YEAR ( MIN ( fact_sales[order_date] ) ), 1, 1 )
VAR MaxDate = DATE ( YEAR ( MAX ( fact_sales[order_date] ) ), 12, 31 )
RETURN
ADDCOLUMNS (
    CALENDAR ( MinDate, MaxDate ),
    "Year",       YEAR ( [Date] ),
    "Quarter",    "Q" & QUARTER ( [Date] ),
    "Month No",   MONTH ( [Date] ),
    "Month",      FORMAT ( [Date], "mmm" ),
    "Year-Month", FORMAT ( [Date], "yyyy-mm" ),
    "Weekday",    FORMAT ( [Date], "ddd" )
)
```

STEP BY STEP

Set it up

1. Model → **New table** → paste the DAX above (or load `dim_date.csv`).
2. Select the table → **Table tools → Mark as date table** → choose the `Date` column. Power BI validates that dates are unique and continuous.
3. Create the relationship `Date[Date]` → `fact_sales[order_date]` (one-to-many).
4. **Sort `Month` by `Month No`** (select `Month` → Column tools → Sort by column) so April does not come before January.
5. Hide the raw `order_date` in the fact table so report authors use the date table.

Python

```
print("Fact order dates :", fact.order_date.min().date(), "to",
fact.order_date.max().date())
print("Date table       :", dates.date.min().date(), "to", dates.date.max().date(), "|
rows:", len(dates))
full = pd.date_range(dates.date.min(), dates.date.max())
print("Continuous (no gaps):", len(full) == len(dates) and (dates.date.sort_values().values
== full.values).all())
dates.head(4)
```

OUTPUT

```
Fact order dates : 2023-01-01 to 2025-12-30
Date table       : 2023-01-01 to 2025-12-31 | rows: 1096
Continuous (no gaps): True
```

	date	year	quarter	month_no	month	weekday
0	2023-01-01	2023	Q1	1	Jan	Sun
1	2023-01-02	2023	Q1	1	Jan	Mon
2	2023-01-03	2023	Q1	1	Jan	Tue
3	2023-01-04	2023	Q1	1	Jan	Wed

4 rows × 6 columns

WATCH OUT **Do not rely on auto date/time**

Power BI can create hidden date tables automatically ("Auto date/time"). They bloat the model and cannot be customised or shared. Turn it off (File → Options → Data Load → Time intelligence) and always use your own marked date table.

Hierarchies

A **hierarchy** lets users drill down through levels with one click. In the Data pane, right-click `category` in `dim_product` → **Create hierarchy**, then drag `subcategory` and `product` onto it. Visuals with a hierarchy get drill-down buttons (arrows at the top-right of the visual): from category → sub-category → product. The date table's `Year → Quarter → Month → Date` is the classic one.

Common modelling problems

Symptom	Likely cause	Fix
Totals repeat the same number for every category	No relationship between the table you filter and the fact	Create the relationship
"A circular dependency was detected" / ambiguous paths	Two paths between two tables	Make one inactive, or remove the bi-directional filter
Many-to-many warning	Duplicate keys on both sides	Build a clean dimension with unique keys, or a bridge table
Blank row ("(Blank)") appears in visuals	Fact rows whose key is missing in the dimension (orphans)	Add the missing rows to the dimension, or fix keys
YTD / last-year measures give wrong or blank results	Date table not marked, has gaps, or not related	Fix the date table
Slow model, big file	Wide tables, high-cardinality text columns, auto date/time	Star schema, remove unused columns, reduce precision, disable auto date/time
Months sort alphabetically	Missing sort-by-column	Sort `Month` by `Month No`

TIP **Best-practice checklist**

1. Star schema with surrogate integer keys. 2. One-to-many, single-direction relationships. 3. One marked date table. 4. Hide keys and technical columns from report view. 5. Disable auto date/time. 6. Remove columns you do not use. 7. Name tables and columns for business users. 8. Put measures in a separate `_Measures` table (Lesson 3).

DO IT WITH AI **Do it with AI**

Ask Copilot or any assistant: *"Review this list of tables and columns and propose a star schema with a fact table and dimensions, the keys, and the relationships"* and paste your column lists. Compare its answer with the checklist above, and verify key uniqueness yourself in Power Query (Column profile → *Distinct* equals *Count*).

Summary

KEY TAKEAWAYS

- Model as a **star schema**: facts (events and numbers) surrounded by dimensions (descriptions), joined by **one-to-many, single-direction** relationships.
- Power Query prepares each table: types, cleaning, unpivot, merge/append, group, parameters; **query folding** pushes work to the source.
- Filters flow from the **one side** (dimension) to the **many side** (fact); use bi-directional filters sparingly; one active relationship per table pair, others inactive (`USERELATIONSHIP`).
- A continuous, **marked date table** is required for time intelligence; sort month names by month number; turn off auto date/time.
- Hierarchies enable drill-down; most "wrong totals" come from missing relationships, duplicate keys or orphans.

Quick quiz

6 questions · tap an answer to check it

1. In a star schema, which table typically contains the numeric measures such as quantity and sales amount?

A Dimension table

B Fact table

C Date table

D Parameter table

Answer: B. Facts hold events and measures; dimensions hold descriptions.

2. A one-to-many relationship between dim_product and fact_sales means:

A A product appears once in sales

B Each product_key appears once in dim_product and many times in fact_sales

C Both tables have unique keys

D The relationship is bidirectional

Answer: B. The "one" side must have unique keys.

3. Why is a bi-directional relationship risky?

A It is not allowed

B It can create ambiguous filter paths and slow performance

C It deletes data

D It only works with dates

Answer: B. Use it only when you must (for example many-to-many scenarios).

4. What does "Mark as date table" require?

A A column of text

B A unique, continuous Date column with no gaps

C A relationship to every table

D A measure

Answer: B. Time-intelligence functions need a valid contiguous date table.

5. Months show as Apr, Aug, Dec, Feb... in alphabetical order. Fix?

- **A** Rename the months
- **B** Use a pie chart
- **C** Sort the Month column by the Month No column
- **D** Delete the date table

Answer: C. Sort by column controls the display order.

6. What does Power Query's "query folding" do?

- **A** Compresses the PBIX file
- **B** Translates steps into the source system's query (e.g. SQL) so the database does the work
- **C** Hides the query
- **D** Merges two queries

Answer: B. Folding improves refresh performance for database sources.

EXERCISE

Model it yourself

Load the five Adventure Cycles CSV files, set types, create the four relationships, mark `dim_date` as the date table, hide key columns, create a Category → Subcategory → Product hierarchy, and build a matrix of sales by `category` (rows) and `year` (columns).

Show solution (worked answer)

In Power Query set `order_date` and `date` to Date. In Model view drag `product_key`, `customer_key`, `territory_key` from each dimension to the fact, and `date` to `order_date`. Table tools → Mark as date table → `date`. Right-click the key columns → Hide in report view. Create the hierarchy by right-clicking `category`. In Report view add a Matrix with Rows `dim_product[category]`, Columns `dim_date[year]`, Values `Sum of sales_amount`. Compare with the pandas pivot: Bikes should be about ₹46.7 crore of the total ₹59.5 crore.

INTERVIEW CORNER

Questions you may be asked

Q: What is a star schema and why use it in Power BI? One fact table joined to several dimension tables with one-to-many relationships. It compresses well, filters predictably, and makes DAX simple and fast.

Q: How do filters propagate in a Power BI model? From the one side to the many side along active relationships, in the filter direction. Bi-directional relationships also filter from many back to one.

Q: Why do you need a date table? Built-in time-intelligence DAX (YTD, SAMEPERIODLASTYEAR) needs a continuous, marked date table; it also gives you year/quarter/month columns, fiscal calendars and holidays.

Q: How do you handle two dates in the fact table (order date and ship date)? Role-playing dimension: one date table with one active relationship (order date) and one inactive (ship date) activated with `USERELATIONSHIP` in specific measures, or two copies of the date table.

POWER BI · LESSON 3

DAX Fundamentals: Measures, Filter Context & CALCULATE

Learn DAX, the formula language of Power BI. Understand calculated columns vs measures, row context vs filter context, write core aggregations and iterators (SUMX), master CALCULATE and FILTER, ALL/REMOVEFILTERS and ALLSELECTED, handle division safely, use variables, and build a library of business measures.

Power BI classes 2–3 | Power BI Desktop 2026 | Intermediate | 150 min

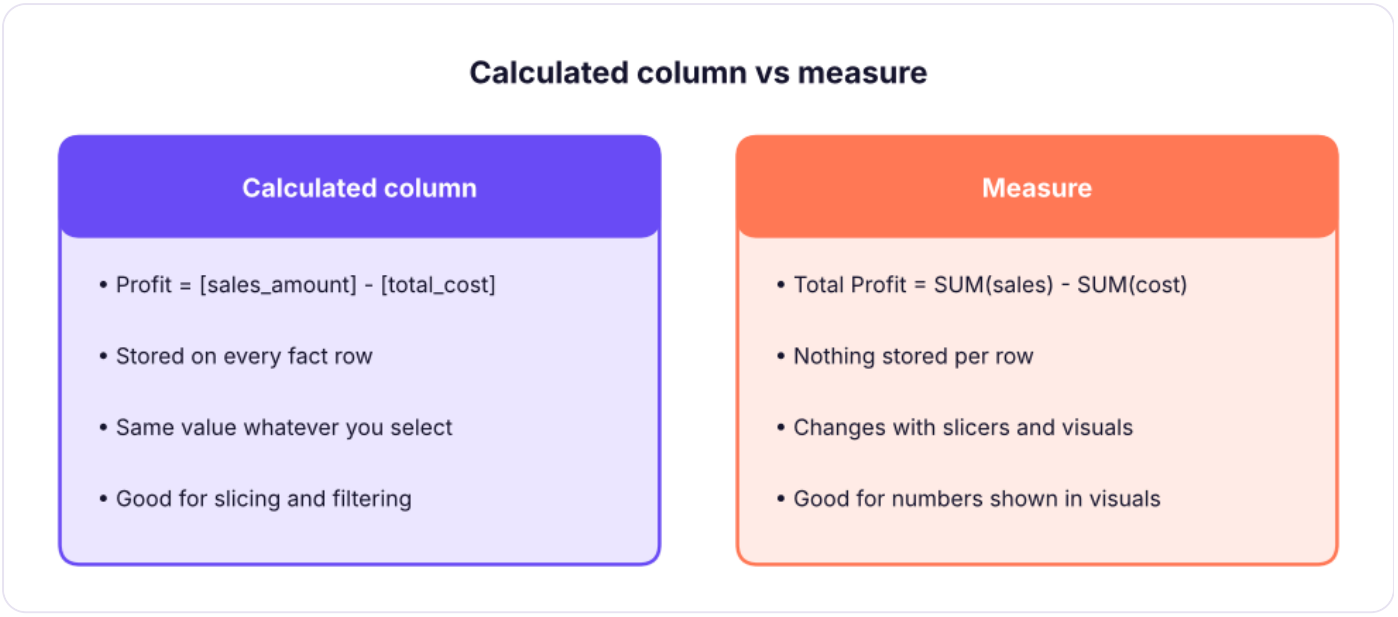
By the end of this lesson you can

- Explain the difference between a calculated column and a measure, and when to use each
- Describe row context and filter context and predict what a measure returns in a visual
- Write aggregation measures and iterators (SUM, COUNTROWS, DISTINCTCOUNT, SUMX, AVERAGEX)
- Use CALCULATE with filter arguments, ALL/REMOVEFILTERS and ALLSELECTED to build ratios and percent-of-total
- Use variables, DIVIDE, IF/SWITCH and display folders to write clean, reusable measures

What is DAX?

DAX (Data Analysis Expressions) is the formula language used in Power BI, Excel's Data Model and Analysis Services. It looks like Excel formulas, but it works on **columns and tables** and recalculates for whatever the user selects in a visual. DAX is the skill that separates a report builder from a Power BI developer.

	Calculated column	Measure
Evaluated	**Row by row** when data is refreshed	**On the fly**, for the current filter selection
Stored in the model	Yes (uses memory)	No (only the formula)
Result depends on slicers?	No	**Yes**
Use for	New attributes to slice by (Age Band, Hire Year, Profit per row)	Totals, ratios, KPIs, growth
Appears in	Rows, columns, slicers, filters	Values area, cards, tooltips



The same table, two ways. A calculated column adds a value to every row. A measure returns one number computed from whichever rows are currently filtered.

> **TIP** **Rule of thumb**
>
> If you want to **put it on an axis or slicer**, make a column. If you want to **put it in the Values well**, make a measure. In a star schema, almost every number is a measure.

DAX syntax in five minutes

```dax
Total Sales = SUM ( fact_sales[sales_amount] )
```

- `Total Sales` is the **measure name** (spaces allowed, no brackets when defining it).
- `=` starts the formula.
- `SUM` is a **function**; `fact_sales[sales_amount]` is a **fully qualified column** (table name, then column in square brackets). Always include the table name for columns; **never** include it for measures: `[Total Sales]`.
- Strings use double quotes; comments use `--` or `//` or `/* … */`.
- Operators: `+ - * /`, `&` (join text), `= <> > < ≥ ≤`, `&&` (AND), `||` (OR), `IN { "A", "B" }`.
- Spaces and line breaks are ignored; format with **daxformatter.com** or the formatting button.

Creating measures

Home → **New measure** (or right-click a table). Put all measures in an empty table named `_Measures` (Home → Enter data → name `_Measures` → Load, then hide its default column) so they are easy to find. Use **display folders** (Model view → Properties) to group them.

Basic aggregations

```dax
Total Sales      = SUM ( fact_sales[sales_amount] )
Total Cost       = SUM ( fact_sales[total_cost] )
Total Profit     = [Total Sales] - [Total Cost]
Profit Margin %  = DIVIDE ( [Total Profit], [Total Sales] )
Units Sold       = SUM ( fact_sales[quantity] )
Orders           = DISTINCTCOUNT ( fact_sales[order_number] )
Order Lines      = COUNTROWS ( fact_sales )
Avg Order Value  = DIVIDE ( [Total Sales], [Orders] )     -- in this data every order has one line, so Orders = Order Lines
Customers        = DISTINCTCOUNT ( fact_sales[customer_key] )
```

Check the values in Python; they are what a card visual with no filters would display:

Python

```python
import pandas as pd
D = "adventure_cycles/"
f = pd.read_csv(D + "fact_sales.csv", parse_dates=["order_date"]); p = pd.read_csv(D +
"dim_product.csv"); t = pd.read_csv(D + "dim_territory.csv")
sales, cost = f.sales_amount.sum(), f.total_cost.sum()
print(f"Total Sales     : {sales:,.0f}")
print(f"Total Cost      : {cost:,.0f}")
print(f"Total Profit    : {sales - cost:,.0f}")
print(f"Profit Margin % : {(sales - cost) / sales:.1%}")
print(f"Units Sold      : {f.quantity.sum():,}")
print(f"Orders          : {f.order_number.nunique():,}")
print(f"Order Lines     : {len(f):,}")
print(f"Avg Order Value : {sales / f.order_number.nunique():,.0f}")
print(f"Customers       : {f.customer_key.nunique():,}")
```

OUTPUT

```
Total Sales     : 594,559,665
Total Cost      : 384,392,790
Total Profit    : 210,166,875
Profit Margin % : 35.3%
Units Sold      : 22,703
Orders          : 9,000
Order Lines     : 9,000
Avg Order Value : 66,062
Customers       : 1,200
```

WATCH OUT **DIVIDE, not /**

`[A] / [B]` returns an error or infinity when B is 0 or blank. `DIVIDE ( [A], [B] )` returns BLANK (or an alternate result: `DIVIDE ( [A], [B], 0 )`). Always use DIVIDE for ratios.

Filter context: the heart of DAX

When you put a measure in a visual, DAX evaluates it **once per cell**, with a **filter context** made of everything that filters the data for that cell: row and column headers, slicers, page/report filters, and cross-filtering from other visuals.

Filter context for one cell of a matrix (Category = Bikes, Country = India, Year = 2025 slicer). DAX applies all of them to the fact table before it sums.

Same measure, different context in each cell. The matrix below is exactly what Power BI shows for `Total Sales` with Category on rows and Country on columns:

Python

```python
m = f.merge(p, on="product_key").merge(t, on="territory_key")
m["year"] = m.order_date.dt.year
print((m.pivot_table(index="category", columns="country", values="sales_amount",
aggfunc="sum") / 1e6).round(1).to_string())
print("\nSame measure, with a Year = 2025 slicer:")
print((m[m.year == 2025].pivot_table(index="category", columns="country",
values="sales_amount", aggfunc="sum") / 1e6).round(1).to_string())
```

OUTPUT

```
country      India  Singapore   UAE
category
Accessories    6.7        1.7   1.3
Bikes        320.6       74.8  71.3
Clothing       6.0        1.3   1.1
Components    73.5       18.8  17.2

Same measure, with a Year = 2025 slicer:
country      India  Singapore   UAE
category
Accessories    2.2        0.7   0.4
Bikes        102.3       24.6  22.0
Clothing       2.0        0.3   0.4
Components    22.9        7.1   5.1
```

Row context

A **row context** exists when DAX goes through a table row by row: in a **calculated column**, and inside **iterator** functions (`SUMX` , `AVERAGEX` , `FILTER` ...). In row context, column references mean "this row's value".

DAX

```
-- Calculated column on fact_sales: row context gives each row's own values
Line Profit = fact_sales[sales_amount] - fact_sales[total_cost]
```

Row context does **not** automatically filter other tables; **CALCULATE** converts a row context into a filter context ("context transition").

Iterators: SUMX, AVERAGEX, MAXX

`SUM ( column )` adds one column. When you need to **calculate per row first** and then aggregate, use an iterator: `SUMX ( table, expression )`.

DAX

```
-- Revenue at list price (unit_price x quantity), calculated row by row
List Value = SUMX ( fact_sales, fact_sales[quantity] * fact_sales[unit_price] )

-- Average margin per order line (not the same as total profit / total sales!)
Avg Line Margin % = AVERAGEX ( fact_sales, DIVIDE ( fact_sales[sales_amount] -
fact_sales[total_cost], fact_sales[sales_amount] ) )

-- Most profitable single line
Max Line Profit = MAXX ( fact_sales, fact_sales[sales_amount] - fact_sales[total_cost] )
```

Python

```
f["line_profit"] = f.sales_amount - f.total_cost
list_value = (f.quantity * f.unit_price).sum()
avg_line_margin = ((f.sales_amount - f.total_cost) / f.sales_amount).mean()
print(f"List Value        : {list_value:,.0f}")
print(f"Realised/List     : {f.sales_amount.sum() / list_value:.1%}  (average price
realisation)")
print(f"Avg Line Margin % : {avg_line_margin:.1%}   vs overall margin
{(f.sales_amount.sum() - f.total_cost.sum()) / f.sales_amount.sum():.1%}")
print(f"Max Line Profit   : {f.line_profit.max():,.0f}")
```

OUTPUT

```
List Value        : 617,201,300
Realised/List     : 96.3%  (average price realisation)
Avg Line Margin % : 34.5%   vs overall margin 35.3%
Max Line Profit   : 174,060
```

The average of line margins (34.5%) differs slightly from the overall margin (35.3%) because large lines count more in the total than in a simple average. Choose the definition the business wants, and

name the measure clearly.

CALCULATE: change the filter context

`CALCULATE ( expression, filter1, filter2, … )` evaluates the expression in a **modified** filter context. It is the most important DAX function.

DAX

```
Bike Sales   = CALCULATE ( [Total Sales], dim_product[category] = "Bikes" )
India Sales  = CALCULATE ( [Total Sales], dim_territory[country] = "India" )
Bike Sales India 2025 =
    CALCULATE ( [Total Sales], dim_product[category] = "Bikes", dim_territory[country] =
"India", dim_date[year] = 2025 )
High Value Lines = CALCULATE ( COUNTROWS ( fact_sales ), fact_sales[sales_amount] > 200000
)
```

Multiple filter arguments are combined with **AND**. A simple column filter like `dim_product[category] = "Bikes"` **replaces** any existing filter on that column.

Python

```
bike = f.merge(p, on="product_key").query("category == 'Bikes'").sales_amount.sum()
print(f"Bike Sales          : {bike:,.0f}")
india_2025_bikes = m[(m.category == "Bikes") & (m.country == "India") & (m.year ==
2025)].sales_amount.sum()
print(f"Bike Sales India 2025: {india_2025_bikes:,.0f}")
print(f"High Value Lines    : {(f.sales_amount > 200000).sum():,}")
```

OUTPUT

```
Bike Sales          : 466,754,465
Bike Sales India 2025: 102,271,520
High Value Lines    : 797
```

Filter functions: ALL, REMOVEFILTERS, ALLSELECTED, ALLEXCEPT, KEEPFILTERS

Function	Effect	Typical use
`ALL ( table or column )` / `REMOVEFILTERS ( … )`	Ignores filters on the table/column	Grand totals, denominators
`ALLSELECTED ( … )`	Ignores filters from the visual's own rows/columns but keeps slicers and page filters	"% of visible total"
`ALLEXCEPT ( table, col1, … )`	Removes all filters on the table except those columns	Subtotals per group
`VALUES ( column )`	The distinct values visible in the current context	Counting / looping over what is filtered
`SELECTEDVALUE ( column, alt )`	The single selected value (or alternate)	Titles, dynamic measures
`KEEPFILTERS ( filter )`	Intersects with, rather than replaces, existing filters	Advanced

Percent of total

DAX

```
Sales % of Total =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], REMOVEFILTERS ( dim_product ) ) )

Sales % of Visible Total =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], ALLSELECTED ( dim_product ) ) )

Share Within Country =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], ALLEXCEPT ( dim_territory,
dim_territory[country] ) ) )
```

Python

```
cat = m.groupby("category").sales_amount.sum()
print((100 * cat / cat.sum()).round(1).rename("Sales % of Total").to_string())
```

OUTPUT

```
category
Accessories     1.6
Bikes          78.5
```

```
Clothing        1.4
Components     18.4
```

FILTER: row-by-row conditions

`FILTER ( table, condition )` is an iterator that returns the rows where the condition is TRUE. Use it when a filter needs a **measure or several columns**; for a simple single-column comparison, the plain `column = value` form inside CALCULATE is faster.

DAX

```dax
-- Sales from products with a list price above 1,00,000
Premium Sales = CALCULATE ( [Total Sales], FILTER ( dim_product, dim_product[list_price] > 100000 ) )

-- Number of products whose total sales beat 1 crore
Strong Products = COUNTROWS ( FILTER ( VALUES ( dim_product[product] ), [Total Sales] > 10000000 ) )
```

Python

```python
prem = f.merge(p, on="product_key").query("list_price > 100000").sales_amount.sum()
ps = f.groupby("product_key").sales_amount.sum()
print(f"Premium Sales  : {prem:,.0f}")
print(f"Strong Products: {(ps > 10_000_000).sum()}  (products with sales above ₹1 crore)")
```

OUTPUT

```
Premium Sales  : 303,639,995
Strong Products: 20  (products with sales above ₹1 crore)
```

Variables and readability

`VAR … RETURN` stores intermediate results: easier to read, easier to debug, and faster (each variable is evaluated once).

DAX

```dax
Profit Margin vs Overall =
VAR CurrentMargin = DIVIDE ( [Total Profit], [Total Sales] )
VAR OverallMargin = CALCULATE ( DIVIDE ( [Total Profit], [Total Sales] ), REMOVEFILTERS () )
RETURN
    CurrentMargin - OverallMargin
```

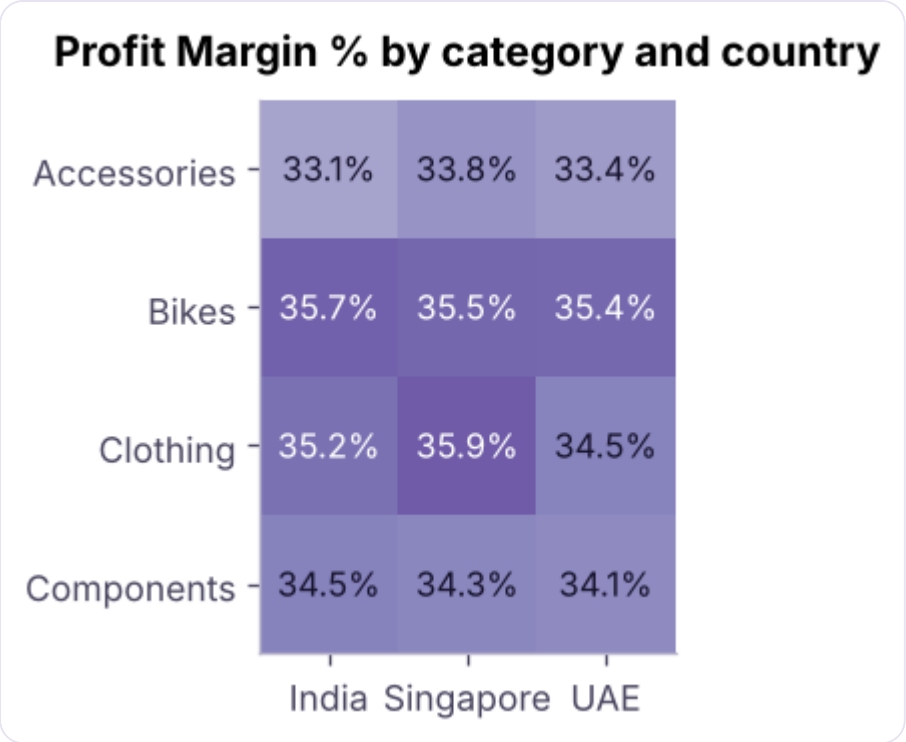
Logical and text measures

```dax
Margin Status =
SWITCH (
    TRUE (),
    ISBLANK ( [Profit Margin %] ), "No data",
    [Profit Margin %] ⩾ 0.4,      "Excellent",
    [Profit Margin %] ⩾ 0.3,      "Good",
    "Review"
)

Selected Country Title =
"Sales for " & SELECTEDVALUE ( dim_territory[country], "all countries" )
```

Function	Purpose
`IF ( test, a, b )`, `SWITCH ( expr, v1, r1, … )`, `SWITCH ( TRUE (), cond1, r1, … )`	Branching
`ISBLANK`, `COALESCE ( x, 0 )`, `BLANK ()`	Handle empty values
`HASONE ( col )`, `ISFILTERED ( col )`	Test the current context
`FORMAT ( value, "0.0%" )`, `CONCATENATEX ( table, col, ", " )`	Text

Measures in action: a profit-margin matrix



Profit margin % by category and country: the matrix you would build with the Profit Margin % measure and conditional formatting (darker = higher).

Common DAX mistakes

Mistake	Fix
Using a calculated column where a measure is needed (results do not respond to slicers)	Create a measure
Dividing with `/`	`DIVIDE`
Writing `SUM ( [Measure] )`	`SUM` takes columns; use `SUMX ( table, [Measure] )` for measures over rows
Averaging ratios (`AVERAGE` of a margin column)	Compute `DIVIDE ( SUM numerator, SUM denominator )` unless you really want the average of ratios
Using `FILTER ( ALL ( table ), … )` when a column filter works	Prefer simple column filters inside CALCULATE
Forgetting that a measure's result depends on context; testing in a card	Test in a table or matrix with the fields you care about
Hard-coding values (`"Bikes"`) in many measures	Use a parameter table or SELECTEDVALUE

DO IT WITH AI

Do it with AI

Copilot and chat assistants write decent DAX, but they invent column names and often misuse context. A reliable prompt gives the **exact table and column names**, the **relationships**, and the **expected result**: *"Tables: fact_sales(order_date, product_key, sales_amount, total_cost), dim_product(product_key, category), dim_date(date). Write a DAX measure for each category's share of total sales that ignores the category filter but respects year slicers. Explain the filter context."* Then test the measure in a table next to your own pandas check.

Summary

KEY TAKEAWAYS

- **Calculated columns** run row by row at refresh and are stored; **measures** run at query time in the current **filter context**.
- Filter context = row/column headers + slicers + filters + cross-filtering; measures are re-evaluated for every cell.
- Row context exists in calculated columns and iterators (`SUMX` , `AVERAGEX` , `FILTER`); `CALCULATE` converts it to filter context.
- `CALCULATE ( expression, filters )` changes the filters; `REMOVEFILTERS` / `ALL` , `ALLSELECTED` , `ALLEXCEPT` remove them for percent-of-total calculations.
- Use `DIVIDE` , variables (`VAR … RETURN`), `SWITCH ( TRUE (), … )` , clear names, a `_Measures` table and display folders.

Quick quiz

6 questions · tap an answer to check it

1. Which is evaluated for every cell of a matrix based on slicers and row/column headers?

- **A** A calculated column
- **B** A measure
- **C** A relationship
- **D** A parameter

Answer: B. Measures respond to filter context; calculated columns are fixed at refresh.

2. Which function lets you compute a total that ignores the Category filter (for a denominator)?

- **A** `SUM`
- **B** `CALCULATE ( [Total Sales], REMOVEFILTERS ( dim_product[category] ) )`
- **C** `FILTER ( [Total Sales] )`
- **D** `COUNTROWS`

Answer: B. REMOVEFILTERS (or ALL) clears the filter inside CALCULATE.

3. Which is the safest way to compute a margin?

- **A** `[Profit] / [Sales]`
- **B** `DIVIDE ( [Profit], [Sales] )`
- **C** `IF ( [Sales] = 0, 0, [Profit] ) / [Sales]`
- **D** `SUM ( [Profit] ) / SUM ( [Sales] )`

Answer: B. DIVIDE handles zero and blank denominators gracefully.

4. `SUMX ( fact_sales, fact_sales[quantity] * fact_sales[unit_price] )` **is needed because:**

- **A** SUM cannot add numbers
- **B** The product must be calculated per row before adding
- **C** SUMX is faster than SUM
- **D** It creates a calculated column

Answer: B. SUM works on one existing column; SUMX evaluates an expression per row.

5. What is row context?

- **A** The current row being evaluated in a calculated column or iterator
- **B** The filters from slicers
- **C** The relationship direction
- **D** A table filter

Answer: A. Row context refers to "this row" during row-by-row evaluation.

6. What does `CALCULATE ( [Total Sales], dim_product[category] = "Bikes" )` **return in a visual that shows each country?**

- **A** Total sales for all products
- **B** Bike sales for each country in the visual
- **C** Bike sales for all countries on every row
- **D** An error

Answer: B. The Category filter is replaced by Bikes, while the Country filter from the visual still applies.

EXERCISE **Write the measures**

Using the Adventure Cycles model, write DAX for: (1) Profit and margin % for **Clothing** only; (2) each country's share of total sales (works in a visual with country on rows); (3) the number of customers who bought more than one product line; (4) a title measure that says "Top category: ..." for the category with the highest sales in the current context.

Show solution (worked answer)

DAX

```
-- (1)
Clothing Profit  = CALCULATE ( [Total Profit], dim_product[category] = "Clothing" )
Clothing Margin  = DIVIDE ( [Clothing Profit], CALCULATE ( [Total Sales],
dim_product[category] = "Clothing" ) )
-- (2)
Country Share = DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], REMOVEFILTERS (
dim_territory ) ) )
-- (3)
Multi-line Customers =
COUNTROWS ( FILTER ( VALUES ( fact_sales[customer_key] ), CALCULATE ( COUNTROWS (
fact_sales ) ) > 1 ) )
-- (4)
Top Category =
VAR t = ADDCOLUMNS ( VALUES ( dim_product[category] ), "@s", [Total Sales] )
VAR top1 = TOPN ( 1, t, [@s], DESC )
RETURN "Top category: " & MAXX ( top1, dim_product[category] )
```

Verify each in a table visual next to a pandas check.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between a calculated column and a measure? A column is computed per row at refresh and stored; a measure is computed at query time for the current filter context and returns a single value.

Q: Explain filter context and row context. Filter context is the set of filters (axes, slicers, filters) applied to the model when a measure is evaluated. Row context is the "current row" while iterating a table (calculated columns, iterators). CALCULATE turns row context into filter context.

Q: What does CALCULATE do? Evaluates an expression in a modified filter context; its filter arguments add, replace or remove filters.

Q: ALL vs ALLSELECTED? ALL removes all filters from the specified table/column (grand total); ALLSELECTED removes only the filters coming from the visual's own rows and columns but keeps outer filters such as slicers, giving a total of what is visible.

Q: SUM vs SUMX? SUM adds a column; SUMX evaluates an expression for each row of a table and sums the results (needed for products of columns, conditional logic per row).

POWER BI · LESSON 4

DAX Time Intelligence, Ranking, Parameters & Dynamic Measures

Make dashboards answer "compared with when?". Build year-to-date, month-to-date, same-period-last-year, month-on-month and year-on-year measures, rolling averages, rankings with RANKX and TOPN, what-if parameters, field parameters and dynamic measure switching, and know visual calculations and calculation groups.

● Power BI classes 3–4

✦ Power BI Desktop 2026 (field parameters, calculation groups, visual calculations)

Advanced

⏱ 150 min

By the end of this lesson you can

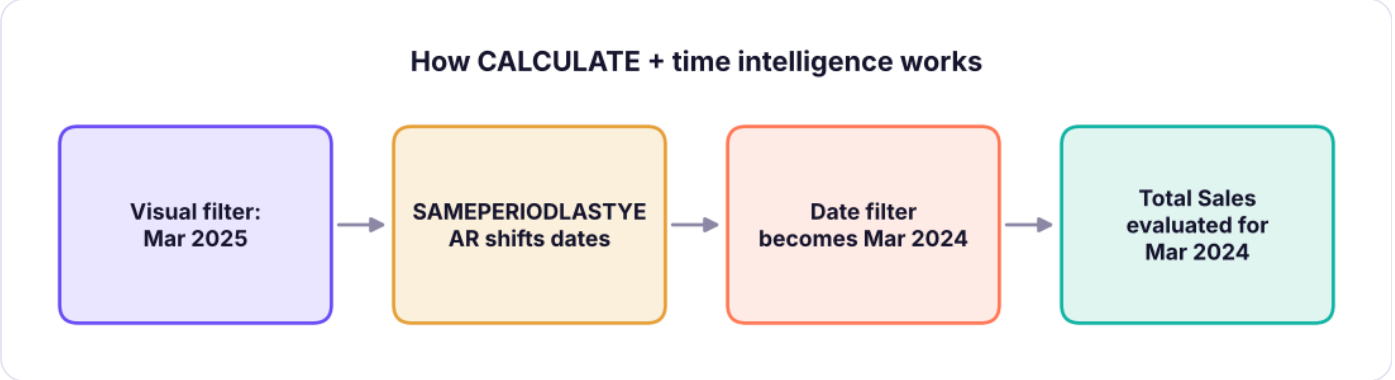
- Explain what time intelligence needs (a marked date table) and use TOTALYTD, DATESYTD, SAMEPERIODLASTYEAR, DATEADD and PARALLELPERIOD
- Build YoY %, MoM %, YTD, rolling 3-month and running-total measures
- Rank items with RANKX and show the Top N with TOPN, including an "Others" bucket
- Create what-if parameters and dynamic measure switching with field parameters
- Describe calculation groups, visual calculations and DAX query view, and when to use them

What time intelligence needs

Time-intelligence functions move or expand the **date filter**: "this year to date", "the same period last year". They only work when:

1. You have a **date table** with every date (Lesson 2),
2. it is **marked as a date table**, and
3. it is **related** to the fact table's date column.

Always use the **date table's** `Date` column inside time functions, never the fact table's date.



Time intelligence changes the date filter, then CALCULATE re-evaluates the measure. SAMEPERIODLASTYEAR shifts the selected dates back exactly one year.

Core measures

DAX

```
Total Sales = SUM ( fact_sales[sales_amount] )

-- Cumulative totals
Sales YTD = TOTALYTD ( [Total Sales], dim_date[date] )
Sales QTD = TOTALQTD ( [Total Sales], dim_date[date] )
Sales MTD = TOTALMTD ( [Total Sales], dim_date[date] )
-- Indian financial year ending 31 March:
Sales FYTD = TOTALYTD ( [Total Sales], dim_date[date], "31-3" )

-- Previous periods
Sales LY    = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( dim_date[date] ) )
Sales PM    = CALCULATE ( [Total Sales], DATEADD ( dim_date[date], -1, MONTH ) )
Sales PQ    = CALCULATE ( [Total Sales], DATEADD ( dim_date[date], -1, QUARTER ) )

-- Growth
YoY %  = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )
MoM %  = DIVIDE ( [Total Sales] - [Sales PM], [Sales PM] )
YoY Δ  = [Total Sales] - [Sales LY]
```

Function	Returns	Notes
`TOTALYTD ( expr, dates, [year_end] )`	Year-to-date total	Shortcut for `CALCULATE ( expr, DATESYTD ( dates ) )`
`DATESYTD / DATESQTD / DATESMTD`	The date set from the period start to the last visible date	Use inside CALCULATE
`SAMEPERIODLASTYEAR ( dates )`	The selected dates shifted back one year	Needs continuous dates
`DATEADD ( dates, n, interval )`	Shift by n days, months, quarters or years	Month/quarter shifts handle month-end well
`PARALLELPERIOD ( dates, n, interval )`	The **full** period shifted (full month, quarter, year)	
`DATESBETWEEN ( dates, start, end )`	Custom ranges	
`DATESINPERIOD ( dates, end, -n, interval )`	Rolling window ending at `end`	Rolling 3/12 months
`PREVIOUSMONTH / PREVIOUSQUARTER / PREVIOUSYEAR`	Whole previous period	
`EOMONTH`, `EDATE`, `DATEDIFF`	Date arithmetic	

Let's compute what these measures return for our data (monthly sales, year-to-date, last year and YoY):

Python

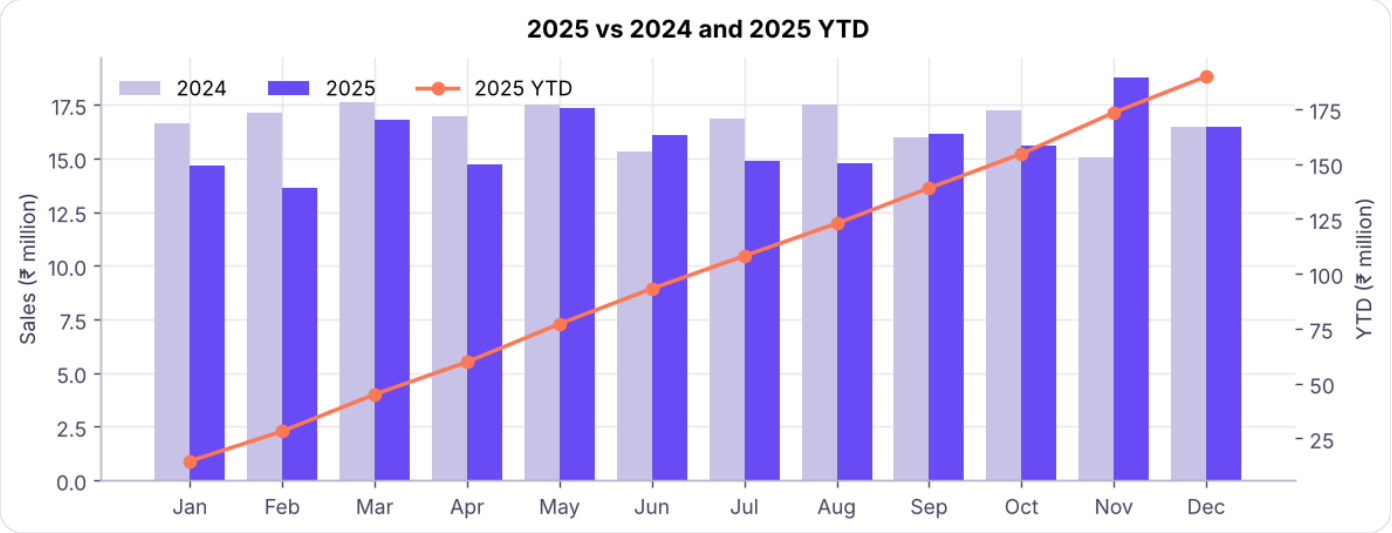
```
import pandas as pd
D = "adventure_cycles/"
f = pd.read_csv(D + "fact_sales.csv", parse_dates=["order_date"]); p = pd.read_csv(D +
"dim_product.csv")
f["ym"] = f.order_date.dt.to_period("M")
mth = f.groupby("ym").sales_amount.sum().to_frame("Total Sales")
mth["Sales PM"] = mth["Total Sales"].shift(1)
mth["MoM %"] = (mth["Total Sales"] / mth["Sales PM"] - 1) * 100
mth["Sales LY"] = mth["Total Sales"].shift(12)
mth["YoY %"] = (mth["Total Sales"] / mth["Sales LY"] - 1) * 100
mth["Sales YTD"] = mth.groupby(mth.index.year)["Total Sales"].cumsum()
out = mth[mth.index.year == 2025].copy()
out["Total Sales"] = (out["Total Sales"] / 1e6).round(1); out["Sales LY"] = (out["Sales
LY"] / 1e6).round(1); out["Sales YTD"] = (out["Sales YTD"] / 1e6).round(1)
out[["Total Sales", "Sales LY", "YoY %", "MoM %", "Sales YTD"]].round(1).rename_axis("2025
month")
```

OUTPUT

	Total Sales	Sales LY	YoY %	MoM %	Sales YTD
2025 month					
2025-01	14.6	16.7	-12.1	-11.2	14.6
2025-02	13.6	17.1	-20.5	-6.9	28.3
2025-03	16.8	17.6	-4.8	23.1	45.1
2025-04	14.7	17.0	-13.2	-12.1	59.8
2025-05	17.4	17.5	-1.0	17.7	77.2
2025-06	16.1	15.3	5.0	-7.4	93.2
2025-07	14.9	16.9	-11.5	-7.2	108.1
2025-08	14.8	17.5	-15.6	-1.0	122.9
2025-09	16.2	16.0	1.0	9.4	139.1
2025-10	15.6	17.3	-9.7	-3.5	154.7
2025-11	18.8	15.1	24.6	20.5	173.4
2025-12	16.5	16.5	0.0	-12.2	189.9

12 rows × 5 columns

(Values in ₹ million.) In a Power BI matrix with `dim_date[month]` on rows and these measures in the Values area you would see exactly this.



Monthly sales for 2025 against the same month in 2024, with year-to-date sales on the right axis. This is the classic "this year vs last year" chart.

COMMON MISTAKES

Common time-intelligence bugs

- **Date table not marked / has gaps**: `SAMEPERIODLASTYEAR` returns blank or wrong values.
- **Using the fact table's date column** in the function: use `dim_date[date]`.
- **Comparing partial periods**: the current month has data for 10 days, last year's month has 31, so YoY looks terrible. Compare "to date" with `DATESYTD` and `SAMEPERIODLASTYEAR`, or restrict to complete months.
- **Fiscal year**: pass `"31-3"` as the year-end argument to `TOTALYTD`, and create `Fiscal Year` and `Fiscal Quarter` columns in the date table.
- **Showing a "last year" value in months with no data** (future months): wrap with `IF ( [Total Sales] <> BLANK (), … )`.

Rolling windows and running totals

DAX

```
-- Rolling 3-month sales ending at the last visible date
Sales Rolling 3M =
CALCULATE ( [Total Sales], DATESINPERIOD ( dim_date[date], MAX ( dim_date[date] ), -3,
MONTH ) )

-- Rolling 3-month average of monthly sales (average over the months in the window)
Avg Monthly Sales R3M =
DIVIDE ( [Sales Rolling 3M], 3 )      -- valid once three full months exist in the window

-- Running total across all dates up to the current one
Sales Running Total =
CALCULATE ( [Total Sales], FILTER ( ALL ( dim_date[date] ), dim_date[date] ≤ MAX (
dim_date[date] ) ) )
```

In SQL this was `SUM() OVER (ORDER BY … ROWS BETWEEN …)` (SQL Lesson 12). DAX does the same thing by redefining the date filter.

Python

```
mth["Rolling 3M"] = mth["Total Sales"].rolling(3).sum()
mth["Rolling 3M avg"] = mth["Total Sales"].rolling(3).mean()
(mth[["Total Sales", "Rolling 3M", "Rolling 3M avg"]].tail(6) /
1e6).round(1).rename_axis("month (₹ million)")
```

OUTPUT

month (₹ million)	Total Sales	Rolling 3M	Rolling 3M avg
2025-07	14.9	48.3	16.1
2025-08	14.8	45.8	15.3
2025-09	16.2	45.8	15.3
2025-10	15.6	46.5	15.5
2025-11	18.8	50.5	16.8
2025-12	16.5	50.9	17.0

6 rows × 3 columns

Ranking: RANKX and TOPN

DAX

```
-- Rank of each product category by sales within the current context
Category Rank = RANKX ( ALL ( dim_product[category] ), [Total Sales], , DESC, DENSE )

-- Show only the top 5 products: sales of a product if it is in the top 5, else blank
Top 5 Product Sales =
VAR top5 = TOPN ( 5, ALL ( dim_product[product] ), [Total Sales], DESC )
RETURN CALCULATE ( [Total Sales], KEEPFILTERS ( top5 ) )

-- Everything outside the top 5, as one "Others" value
Others Sales = [Total Sales] - [Top 5 Product Sales]
```

For simple Top N visuals, you do not need DAX: in the **Filters pane**, choose the visual's field → Filter type **Top N** → Show items Top 5 → By value `Total Sales` → Apply filter. Use DAX when the Top N must respond to slicers in a custom way or you need an "Others" bar.

Python

```
prod_sales = f.merge(p, on="product_key").groupby(["category",
"product"]).sales_amount.sum().sort_values(ascending=False)
top5 = prod_sales.head(5)
print((top5 / 1e6).round(1).to_string())
print(f"\nTop 5 total : {top5.sum() / 1e6:,.1f} M   |   Others: {(prod_sales.sum() -
top5.sum()) / 1e6:,.1f} M   |   Top 5 share: {top5.sum() / prod_sales.sum():.1%}")
```

OUTPUT

```
category  product
Bikes     Road Bike E-331        51.3
          Mountain Bike C-245    42.4
          Touring Bike A-507     39.7
          Mountain Bike D-662    36.2
          Mountain Bike B-168    35.9

Top 5 total : 205.4 M   |   Others: 389.2 M   |   Top 5 share: 34.5%
```

Parameters

What-if parameter

Modeling → New parameter → Numeric range: name `Price Change %`, from −20 to +20, increment 1, default 0, and tick **Add slicer**. Power BI creates a table and a measure `Price Change % Value`. Use it in your own measure to simulate:

DAX

```
Simulated Sales = [Total Sales] * ( 1 + [Price Change % Value] / 100 )
Simulated Profit = [Simulated Sales] - [Total Cost]
```

Dragging the slicer instantly shows what profit would be with a 5% price increase (ignoring demand changes: always say so on the page).

Python

```
base_sales, base_cost = f.sales_amount.sum(), f.total_cost.sum()
rows = [(chg, base_sales * (1 + chg / 100) / 1e6, (base_sales * (1 + chg / 100) -
base_cost) / 1e6) for chg in (-10, -5, 0, 5, 10)]
pd.DataFrame(rows, columns=["Price change %", "Simulated sales (₹ M)", "Simulated profit (₹
M)"]).round(1)
```

OUTPUT

	Price change %	Simulated sales (₹ M)	Simulated profit (₹ M)
0	-10	535.1	150.7
1	-5	564.8	180.4
2	0	594.6	210.2
3	5	624.3	239.9
4	10	654.0	269.6

5 rows × 3 columns

Power Query parameters

Home → Manage parameters creates parameters used inside queries, for example `StartDate` or a `ServerName`. Use them to switch between a test and a production database, or to filter large tables at load time. In the Service, parameters can be changed in dataset settings.

Dynamic measure switching

A very popular pattern: a slicer that lets users choose which measure the chart shows (Sales, Profit, Orders).

The modern way: Field parameters. Modeling → **New parameter → Fields**: add `Total Sales`, `Total Profit`, `Orders`; tick **Add slicer**. Power BI creates a table with the three measures; use that

table's field on a chart's axis or values and the slicer swaps the measure. A field parameter can also switch **axis** fields (Category, Country, Year).

The classic way: a disconnected table + SWITCH.

```DAX
-- Table "Measure Selector" (Enter data): columns Measure Name, Sort
Selected Measure =
SWITCH (
    SELECTEDVALUE ( 'Measure Selector'[Measure Name], "Sales" ),
    "Sales",  [Total Sales],
    "Profit", [Total Profit],
    "Orders", [Orders]
)
```

Format strings can be dynamic too (Power BI 2024+ "dynamic format string"): thousands for sales, percentage for margin.

KPI visuals and forecasting

- **KPI visual**: needs a value (indicator), a trend axis (date) and a target (for example last year's sales). Colours show good/bad.
- **Gauge**: current vs target; min/max/target set in the format pane.
- **Forecasting**: in a **line chart** with a date axis, Analytics pane → **Forecast** → set forecast length (for example 6 months), confidence interval (95%) and seasonality (12 points). Power BI uses exponential smoothing (ETS). It is a quick what-comes-next view, not a replacement for proper forecasting models (ML track).
- **Analytics pane** also offers constant lines, average, min/max, percentile lines, trend lines, and **anomaly detection**.

Newer DAX features worth knowing

Feature	What it is
Calculation groups	A reusable "time-intelligence switcher": define `Current`, `YTD`, `LY`, `YoY %` once and apply them to every measure (created in Tabular Editor or in the model view in recent versions)
Visual calculations	DAX written **inside a visual** that can refer to other fields in the visual (running sum, moving average, % of parent) with simple functions such as `RUNNINGSUM`, `MOVINGAVERAGE`, `PREVIOUS`, `FIRST`
DAX query view	`EVALUATE` queries to test and debug measures; Copilot can write DAX here
User-defined functions (DAX UDFs)	Reusable functions defined with the `FUNCTION` keyword (preview)
INFO.* functions	Query model metadata (`INFO.TABLES()`, `INFO.MEASURES()`)

DAX

```dax
-- Visual calculation (typed in the visual's formula bar)
Running Sales = RUNNINGSUM ( [Total Sales] )
Moving Avg 3  = MOVINGAVERAGE ( [Total Sales], 3 )
```

DO IT WITH AI

Do it with AI

Copilot can write time-intelligence measures from a sentence ("Show sales year to date versus last year"). It works best if the date table is marked, columns and tables have clear names, and you add **descriptions** to measures (Model view → Properties). Check the generated DAX uses your `dim_date[date]` (not the fact table's date), and confirm one value by hand.

Summary

KEY TAKEAWAYS

- Time intelligence needs a **continuous, marked date table**, related to the fact; always reference the date table's column.
- `TOTALYTD`, `SAMEPERIODLASTYEAR`, `DATEADD`, `PARALLELPERIOD`, `DATESINPERIOD` build YTD, last-year, MoM, YoY and rolling windows; `DIVIDE ( current - previous, previous )` gives growth.
- `RANKX` ranks, `TOPN` selects the top items (add an "Others" measure); the Filters-pane **Top N** is the no-code route.
- **What-if parameters** simulate scenarios; **field parameters** let users switch measures or axes; the disconnected table + `SWITCH` pattern does the same in older models.
- Know calculation groups, visual calculations and the DAX query view for advanced work.

Quick quiz

6 questions · tap an answer to check it

1. **Which measure shows year-to-date sales?**

A `SUM ( fact_sales[sales_amount] )`

B `TOTALYTD ( [Total Sales], dim_date[date] )`

C `SAMEPERIODLASTYEAR ( [Total Sales] )`

D `RANKX ( dim_date, [Total Sales] )`

Answer: B. TOTALYTD accumulates from the start of the year to the current date.

2. SAMEPERIODLASTYEAR returns blank for every month. What should you check first?

A The theme

B That the date table is continuous, marked as a date table and related to the fact

C The colour of the visual

D The number of slicers

Answer: B. Time intelligence relies on a valid date table.

3. Which formula gives month-on-month growth?

A `[Total Sales] - [Sales PM]`

B `DIVIDE ( [Total Sales] - [Sales PM], [Sales PM] )`

C `SUM ( [Sales PM] ) / [Total Sales]`

D `DATEADD ( [Total Sales], -1, MONTH )`

Answer: B. Growth rate = (current − previous) ÷ previous, using DIVIDE.

4. What does a field parameter let users do?

A Change the report theme

B Switch which measures or fields a visual shows with a slicer

C Refresh the data

D Create relationships

Answer: B. Field parameters create a slicer that swaps fields in visuals.

5. In the Indian financial year (April–March), how do you build YTD sales?

A `TOTALYTD ( [Total Sales], dim_date[date] )` always works

B `TOTALYTD ( [Total Sales], dim_date[date], "31-3" )`

C `DATESMTD`

D `PARALLELPERIOD ( … )`

Answer: B. The third argument sets the year-end date.

6. What is the correct way to compute "Others" after a Top 5?

A Add a second visual

B Total sales minus the Top 5 measure

C Use FILTER ALL

D Use PREVIOUSMONTH

Answer: B. Others = overall total − Top N total.

EXERCISE

Time-intelligence measures

For Adventure Cycles write measures for: (1) sales for the last 12 months ending at the selected date; (2) YoY % by category; (3) the best month of each year; (4) a what-if parameter "Discount %" from 0 to 30 that reduces `Total Sales` and shows simulated profit margin.

Show solution (worked answer)

DAX

```
-- (1)
Sales L12M = CALCULATE ( [Total Sales], DATESINPERIOD ( dim_date[date], MAX ( dim_date[date] ), -12, MONTH ) )
-- (2) put dim_product[category] on rows and dim_date[year] on columns
YoY % = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )
-- (3)
Best Month =
VAR t = ADDCOLUMNS ( VALUES ( dim_date[month] ), "@s", [Total Sales] )
RETURN MAXX ( TOPN ( 1, t, [@s], DESC ), dim_date[month] )
-- (4)  parameter table 'Discount %' with measure [Discount % Value]
Simulated Margin = DIVIDE ( [Total Sales] * ( 1 - [Discount % Value] / 100 ) - [Total Cost], [Total Sales] * ( 1 - [Discount % Value] / 100 ) )
```

Validate (2) and (3) against pandas group-bys.

INTERVIEW CORNER

Questions you may be asked

Q: What do you need for time-intelligence functions to work? A date table with continuous dates covering the data, marked as a date table, related to the fact table by date.

Q: How would you calculate YoY growth? `Sales LY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( dim_date[date] ) )` then `YoY % = DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )`.

Q: How can users choose which metric a chart shows? Field parameters (modern) or a disconnected table with a `SWITCH ( SELECTEDVALUE (…), … )` measure.

Q: How do you show Top N plus an "Others" bar? A Top N measure using TOPN/KEEPFILTERS and an Others measure = total − Top N.

POWER BI · LESSON 5

Visuals, Filters, Drill-through, Tooltips, Buttons & Report Design

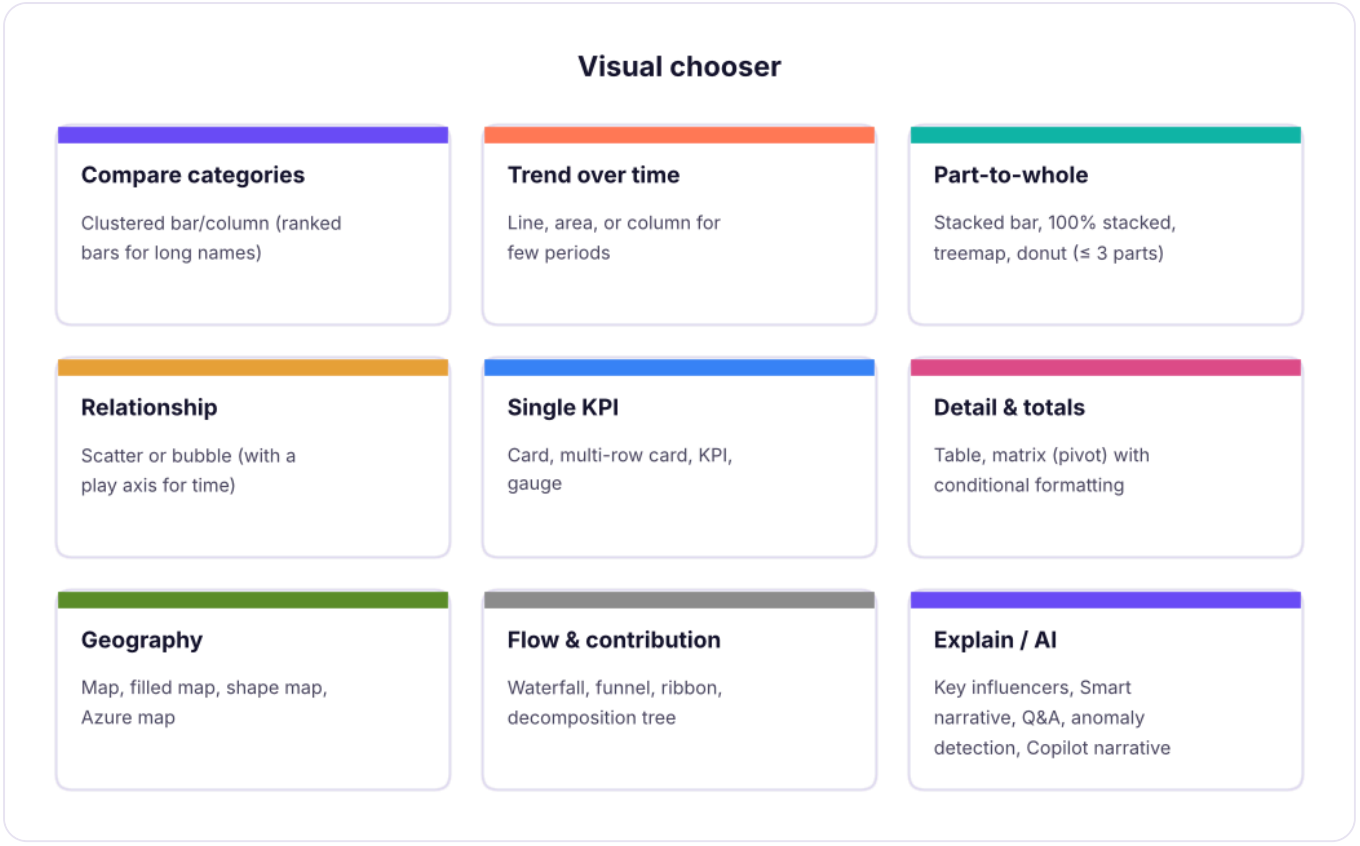
Turn a model into an interactive, well-designed report. Choose visuals, format them (cards, matrix, charts, gauge, KPI, maps, images), use visual, page and report filters, edit interactions, drill-down and drill-through, custom tooltips, bookmarks and buttons, conditional formatting, themes and design principles.

Power BI classes 2–4 | Power BI Desktop 2026 | Intermediate | 150 min

By the end of this lesson you can

- Pick the right visual for each question and configure its field wells and formatting
- Apply visual, page and report-level filters, Top N, relative date filters and slicer sync
- Control cross-filtering with Edit interactions; add drill-down, drill-through and report-page tooltips
- Build navigation with buttons, bookmarks, page navigation and a selection pane
- Apply themes and conditional formatting, and follow layout, colour and accessibility rules

Choosing the visual



Match the visual to the question.

Visual	Fields	Good for	Watch out
Card / multi-row card	Values	One or few headline numbers	Format display units (K, M, Cr) and add a clear label
Clustered bar / column	Axis, Values, Legend	Comparisons	Sort descending, start at zero, ≤ 15 bars
Stacked / 100% stacked	Axis, Legend, Values	Composition	Only the bottom segment is easy to compare
Line / area	X-axis (date), Y-axis, Legend	Trends	Limit to ~5 lines; use a continuous date axis
Line + clustered column	Shared axis, column values, line values	Two measures with different scales (secondary axis)	Label both axes
Waterfall	Category, Y-axis	Bridge from start to end (profit walk)	Needs an ordered category
Scatter / bubble	X, Y, Size, Details, Play axis	Relationships and outliers	Add Details so each point is an entity
Treemap	Group, Values	Hierarchical share of total	Many tiny rectangles are unreadable
Donut / pie	Legend, Values	2–3 parts	More than 4 slices = use a bar
Matrix	Rows, Columns, Values	Pivot tables with drill and subtotals	Use conditional formatting for heat maps
Table	Columns	Exact values, detail list	Keep it short; a table is not a dashboard
KPI / **Gauge**	Value, Target, Trend axis	Progress to a target	Gauges waste space; prefer KPI cards
Funnel	Stage, Values	Drop-off between stages	
Map / Filled map	Location, Size, Legend	Geography	Geocoding needs clean place names (`City, State, Country`)
Slicer	Field	Filter the page	Styles: list, dropdown, tiles, between (range), relative date
Decomposition tree	Analyze, Explain by	Drill into drivers	Interactive exploration

Visual	Fields	Good for	Watch out
Key influencers	Analyze, Explain by	AI finds what drives a metric	Needs a binary/numeric target
Smart narrative	Auto-text	Written summary that updates	Check the wording

Formatting essentials

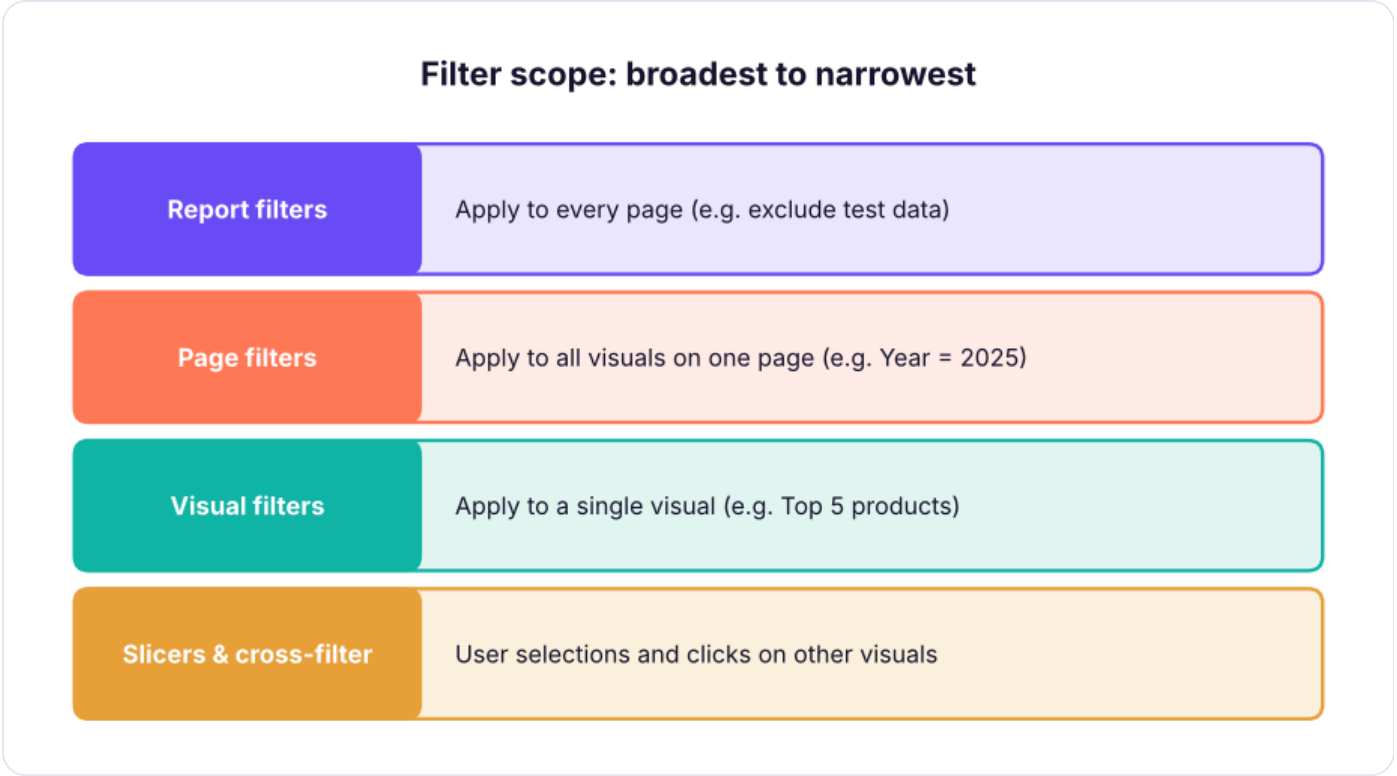
Select a visual → **Format visual** (paint-roller icon): **Visual** tab (axis, labels, colours, legend, gridlines, title) and **General** (title, size, position, background, border, alt text).

Task	Where
Display units and decimals	Format → Callout value / Data labels → Display units: Thousands, Lakhs (not built in; use Millions) or custom format strings
Sort	`…` menu → Sort axis (by value or by another field)
Colour a highlighted series	Format → Colors → per-category colours or conditional
Titles that state the insight	General → Title → fx (dynamic title from a measure)
Accessibility	General → Alt text; keep contrast high; do not rely on colour alone
Copy formatting	Home → **Format painter**
Align / distribute	Format → Arrange → Align and distribute; hold `Shift` to multi-select
Group visuals	Select several → Format → Group

TIP **Dynamic titles**

Create a measure `Title Sales = "Sales by " & SELECTEDVALUE ( dim_territory[country], "all countries" ) & " in " & SELECTEDVALUE ( dim_date[year], "all years" )` and bind it in Title → fx → *Field value*. The title now reflects the user's selections.

Filters: three levels



The three filter levels. Visual filters affect one visual, page filters one page, report filters every page.

Use the **Filters pane** (right side). Filter types:

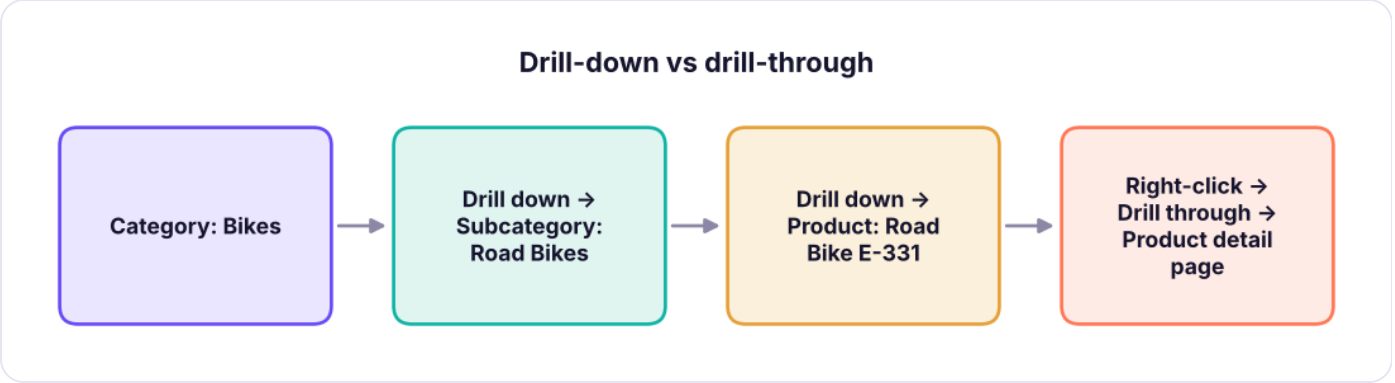
Filter type	Behaviour
Basic	Tick values from a list
Advanced	"contains", "starts with", "is greater than", with AND/OR
Top N	Top/Bottom N items by a measure (even one not in the visual)
Relative date	"Last 6 months", "this year", "next 7 days"
Relative time	Last 4 hours etc.

Tips: **Require single selection** in a slicer; **Sync slicers** across pages (View → Sync slicers); the **"Include" / "Exclude"** options on a selection filter in or out; lock the filters pane or hide it for consumers.

Edit interactions

By default clicking a bar **cross-filters** other visuals. Select the source visual → Format → **Edit interactions** → then on each target visual click the **Filter**, **Highlight** or **None** icon. Example: a KPI card for "overall sales" should ignore the clicks on the country chart (set to None), while the product chart should filter.

Drill-down and drill-through



Drill-down moves through a hierarchy inside one visual. Drill-through jumps to another page that is filtered by the selected item.

- **Drill-down**: needs a hierarchy (Lesson 2). Use the arrow buttons at the top of the visual: ↑ drill up, ↓ drill mode, ⇊ go to the next level in the hierarchy, ⇈ expand all.
- **Drill-through**: create a **detail page** → in the Visualizations pane, drag `product` into **Drill through → Add drill-through fields here** → Power BI adds a back button automatically. Users right-click a product in any visual → **Drill through → Product detail**. Use it for "customer 360", "order detail" and "product detail" pages. **Keep all filters** keeps slicer context.
- **Cross-report drill-through** links to a separate report in the Service.

Report-page tooltips

A **tooltip page** shows a mini-report when hovering over a data point.

STEP BY STEP **Build one**

1. Add a new page; Format page → Page information → **Allow use as tooltip** on; Page size → **Tooltip**.
2. Add visuals (a KPI, a small trend line, top 3 products) using fields that respond to the hovered item.
3. In the main visual: Format → General → **Tooltips** → Type **Report page** → choose the page.
4. Hide the tooltip page from navigation (right-click the tab → Hide page).

Buttons, bookmarks and navigation

Feature	What it does	Use
Buttons (Insert → Buttons)	Back, Bookmark, Page navigation, Q&A, Web URL, Apply all slicers, Clear all slicers	Menus, navigation bars, reset filter
Bookmarks (View → Bookmarks)	Save the current state (filters, slicer selections, visibility, spotlight) of a page	Story-telling views, toggle between visuals
Selection pane (View → Selection)	List, name, show or hide every object on a page	Required for bookmark toggles
Page navigator (Insert → Navigator)	Auto-generated tab buttons for pages or bookmarks	Menu
Conditional visibility	Bookmarks that show/hide visuals	"Chart ⇄ Table" toggle

STEP BY STEP

Chart / table toggle with bookmarks

1. Place a column chart and a table on top of each other.
2. Selection pane: hide the table → View → Bookmarks → **Add** → name it "Chart view"; make the table visible and hide the chart → Add → "Table view". Untick **Data** in each bookmark's options so slicers are not reset.
3. Insert two buttons and set Action → **Bookmark** → choose the matching bookmark.

Conditional formatting

Where	How
Table / matrix cells	Format → Cell elements → Background colour / Font colour / **Data bars** / **Icons** / Web URL → fx → *Gradient*, *Rules* or *Field value*
Column / bar colour	Format → Colors → fx → rules (green above target, red below)
Card text colour or title	fx on the callout/title
Based on a measure	Create a measure returning a hex colour: `Margin Colour = IF ( [Profit Margin %] ⩾ 0.35, "#1E8E5A", "#C0392B" )` then Field value

Subcategory	Sales (₹ M)	Profit (₹ M)	Margin %
Mountain Bikes	161.9	54.9	33.9%
Road Bikes	155.7	56.1	36.0%
Touring Bikes	149.2	55.2	37.0%
Frames	76.1	27.3	35.8%
Wheels	33.4	10.5	31.3%
Helmets	8.2	2.7	33.1%
Jerseys	5.8	2.2	37.2%
Gloves	2.7	0.8	30.8%
Bottles	1.6	0.5	34.0%

A matrix with conditional formatting: a heat-map background on sales and traffic-light colours on margin (rules: ≥ 36% green, otherwise amber/red).

Themes

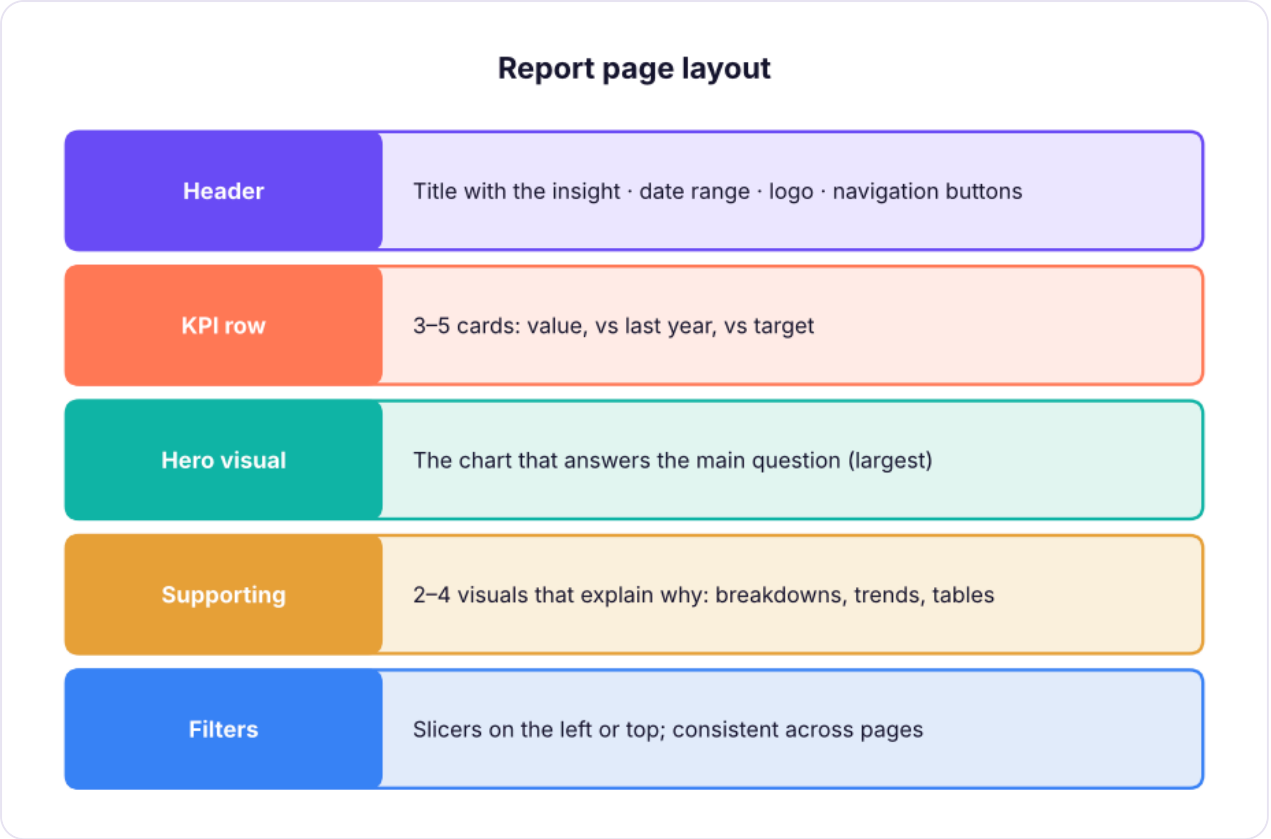
View → Themes offers built-in themes; you can download community themes or **Customize current theme** (colours, fonts, defaults). A **theme JSON file** (View → Themes → Browse for themes) applies one consistent look to every visual:

JSON

```
{
  "name": "SkillNavora",
  "dataColors": ["#6C4CF5", "#FF7A59", "#12B5A6", "#E9A23B", "#3B82F6", "#E04F8A"],
  "background": "#FFFFFF", "foreground": "#17142E", "tableAccent": "#6C4CF5",
  "textClasses": { "title": { "fontFace": "Segoe UI Semibold", "fontSize": 14, "color":
"#17142E" } }
}
```

Apply the company theme to every report so they look consistent, and edit colours in one place.

Report design principles



A layout that works: headline KPIs on top, the main insight chart large, supporting visuals below, filters on one side, consistent spacing.

1. **Start from the audience and decision**, not from the data: what will they do differently?
2. **Hierarchy**: most important top-left; big numbers first; details lower.
3. **Less is more**: 5–8 visuals per page; remove gridlines, borders and chart junk.
4. **Consistent colour logic**: one brand colour plus one highlight; red and green only for bad/good; the same colour always means the same thing.
5. **Align and space** visuals on a grid; same sizes for similar charts.
6. **Label for meaning**: insight titles, units (₹, %), clear axis titles, "as of" dates.
7. **Accessible**: contrast, alt text, tab order (Format → Tab order), do not use colour alone, test on mobile layout (View → Mobile layout).
8. **Performance**: fewer visuals, summarised data, no huge tables; check Optimize → Performance analyzer.

Q&A, images and mobile

- **Q&A visual**: users type "total sales by country in 2025" and get a chart. Improve it with **synonyms** (Model view → Linguistic schema / Q&A setup).

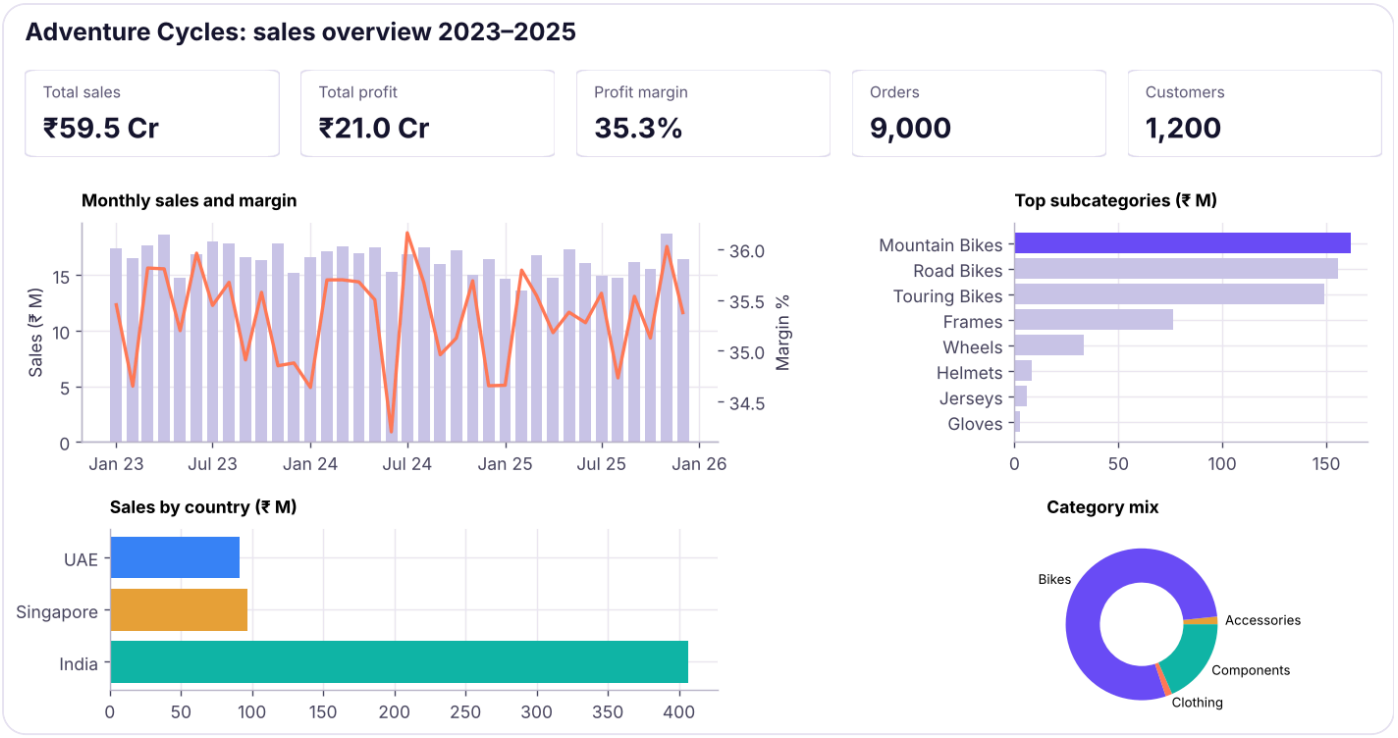
- **Images and logos**: Insert → Image (or text box). For images **in data** (product photos), set the column's **Data category → Image URL**.
- **Mobile layout**: View → Mobile layout; drag key visuals into a phone-shaped canvas.
- **Accessibility and exporting**: export to PDF or PowerPoint (File → Export), analyse in Excel, subscribe in the Service.

> **2026 UPDATE** **Copilot and AI visuals in 2026**
>
> Copilot can generate a report page from a prompt, suggest visuals for your fields, write summaries ("Summarise this page") and answer questions about the data. **Key influencers**, **Decomposition tree** and **Smart narrative** are built-in AI visuals; **anomaly detection** (Analytics pane) flags unusual spikes. Always verify AI-generated visuals and text against real numbers.

A mini case: the Adventure Cycles overview page

Plan the page before you build it:

Element	Fields / measure	Visual
Title	Dynamic title measure	Text box
KPI row	`Total Sales`, `Total Profit`, `Profit Margin %`, `Orders`, each with vs-last-year	4–5 cards
Hero	Monthly sales and profit margin	Line + column
Breakdown 1	Sales by category (Top N subcategories)	Bar
Breakdown 2	Sales by country	Map or bar
Drill-through	Product detail page	Table + KPIs
Filters	Year, Country, Category	Slicers



The Adventure Cycles overview page from the real data: KPI cards, monthly trend, subcategory ranking and country share.

Summary

KEY TAKEAWAYS

- Choose visuals from the question; format with insight titles, units, sorting and accessible colours.
- Filters work at **visual, page and report** level (Basic, Advanced, Top N, Relative date); use slicers, sync them across pages, and **Edit interactions** to control cross-filtering.
- **Drill-down** needs a hierarchy; **drill-through** jumps to a detail page; **tooltip pages** show mini-reports on hover.
- **Buttons + bookmarks + the Selection pane** build navigation and chart/table toggles.
- Use **conditional formatting** (rules, gradients, field value from a measure) and **themes** for consistency.
- Design for the decision: hierarchy, few visuals, consistent colour logic, alignment, labels, accessibility and mobile layout.

Quick quiz

6 questions · tap an answer to check it

1. Which filter level affects every page of a report?

A Visual-level

B Page-level

C Report-level (All pages)

D Slicer-level

Answer: C. Report filters apply to all pages.

2. How do you jump from a product in a chart to a detail page for that product?

A Drill-down

B Drill-through (create a drill-through page with the product field)

C Add a slicer

D Use a tooltip

Answer: B. Drill-through passes the selected value as a filter to another page.

3. What do bookmarks capture?

A Only the page name

B The state of a page: filters, slicers, visibility and more

C The data model

D The theme

Answer: B. Bookmarks save a view that buttons can switch to.

4. Which feature stops a card from reacting when you click a bar chart?

A Sync slicers

B Edit interactions → set the card to None

C Drill-through

D Alt text

Answer: B. Edit interactions controls how visuals filter each other.

5. When is a donut chart appropriate?

A With 12 categories

B With 2–3 parts of a whole

C To show a time trend

D To compare 20 products

Answer: B. More than a few slices are hard to compare; use bars.

6. What is the benefit of a theme JSON file?

A One consistent set of colours and fonts applied to all visuals and reusable across reports

B It refreshes data

C It creates relationships

D It adds measures

Answer: A. Themes standardise the look and make branding quick.

EXERCISE

Build the overview page

Using the Adventure Cycles model: (1) build the overview page above; (2) add a **Year** slicer and a **Country** slicer; (3) create a drill-through page for **Product** with sales, profit, margin and a monthly trend; (4) add a bookmarked toggle between "Sales" and "Profit" using a field parameter or bookmarks; (5) apply conditional formatting to the margin in a matrix and write a dynamic title.

Show solution (worked answer)

(1) Cards for the five measures; line + clustered column for monthly sales and margin; bar for top subcategories; bar for country; donut for category mix. (2) Slicer visuals with `dim_date[year]` (tile style) and `dim_territory[country]` (dropdown). (3) New page → drag `dim_product[product]` into Drill through; add a back button. (4) Modeling → New parameter → Fields → add `Total Sales`, `Total Profit`; use the parameter on the chart values; place its slicer as a toggle. (5) Matrix of `subcategory` × `country` with `Profit Margin %`; Cell elements → Background colour → Rules; title measure with `SELECTEDVALUE`.

INTERVIEW CORNER

Questions you may be asked

Q: How do you design an effective Power BI dashboard? Start from the audience and decision; put headline KPIs top-left; use few visuals with insight titles; consistent colour and spacing; allow drill-down/through; test with users; check performance and accessibility.

Q: Drill-down vs drill-through? Drill-down navigates levels of a hierarchy within a visual; drill-through navigates to another page filtered by the selected item.

Q: What are the filter levels and how do slicers differ? Visual, page and report filters are set by the author in the Filters pane; slicers are on-canvas controls that users change, and they can be synced across pages.

Q: How would you let users toggle between a chart and a table? Two bookmarks that show/hide the visuals through the Selection pane, and two buttons linked to those bookmarks.

POWER BI · LESSON 6

Power BI Service, Connectivity Modes, Security & Capstone

Take a report from Desktop to the people who need it. Understand Import, DirectQuery, Live connection, composite models and Direct Lake, publish to the Service, schedule refresh with a gateway, share through workspaces and apps, secure data with row-level security, tune performance, work with Copilot and Fabric, and complete the Adventure Cycles capstone.

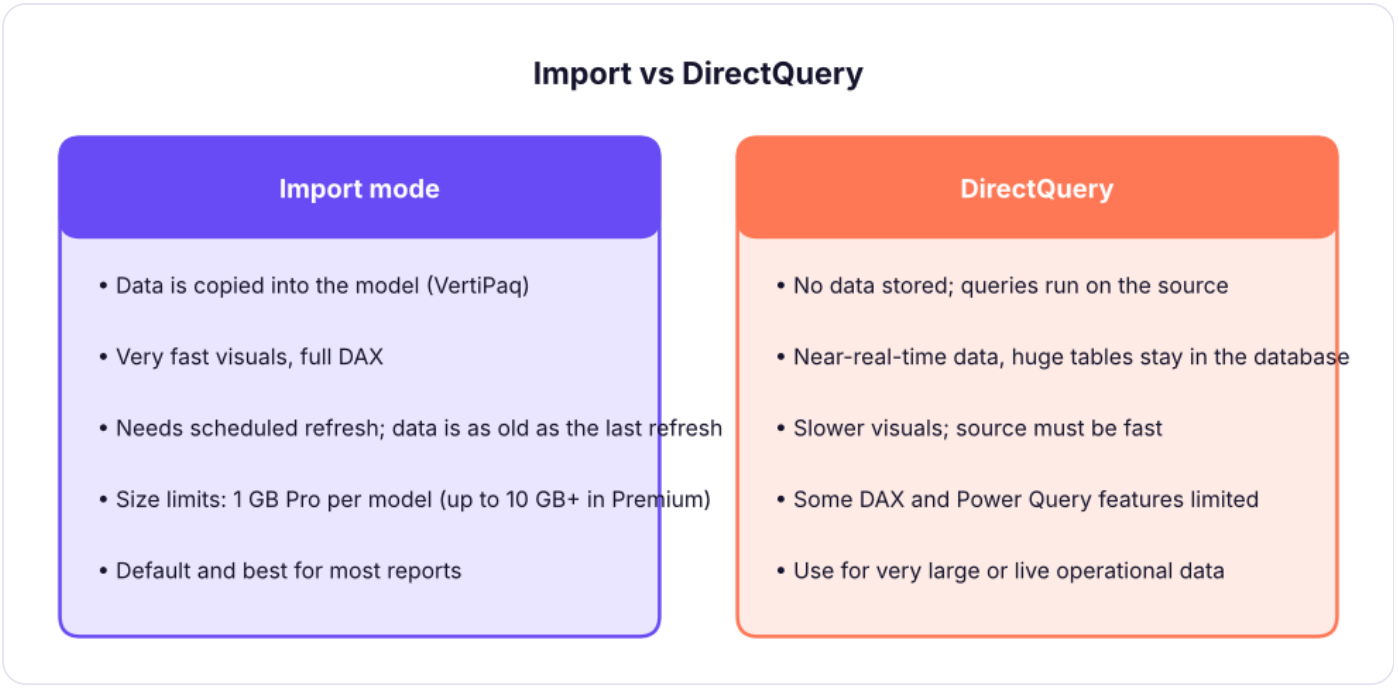
Power BI class 4 | ✦ Power BI Service & Microsoft Fabric 2026 | Advanced | ⏱ 150 min

By the end of this lesson you can

- Compare Import, DirectQuery, Live connection, Composite and Direct Lake modes and choose one
- Publish a report, configure scheduled refresh and a gateway, and share via workspaces, apps and links
- Implement and test row-level security (static and dynamic) with DAX filters
- Use Performance Analyzer and best practices to speed up a model
- Describe Microsoft Fabric, Copilot and governance basics, and deliver an end-to-end capstone

Connectivity modes

How does Power BI get its data: **copy it** into the model (Import) or **ask the source each time** (DirectQuery)?



Import copies data into Power BI's compressed in-memory model; DirectQuery sends a query to the source for every interaction.

Mode	Data stored in Power BI?	Refresh	Typical use
Import	Yes (compressed copy)	Scheduled (up to 8/day Pro, 48/day Premium)	Most reports; Excel, CSV, SQL
DirectQuery	No	Live per interaction	Large SQL databases, near-real-time dashboards
Live connection	No: connects to an existing **semantic model** (Analysis Services or a published Power BI dataset)	Live	One governed model shared by many reports
Composite model	Mix: some tables Import, some DirectQuery, plus other semantic models	Both	Large fact in DirectQuery, small dimensions imported (dual storage mode)
Direct Lake (Fabric)	Reads Delta tables in OneLake directly	Automatic on data change	Import-like speed with no copy; Fabric only

TIP **A practical rule**

Start with **Import**. Move to DirectQuery or Direct Lake only if data volume or freshness requirements force you to. Reduce the data first: only needed columns, aggregated tables, filtered date range.

Publishing and the Service

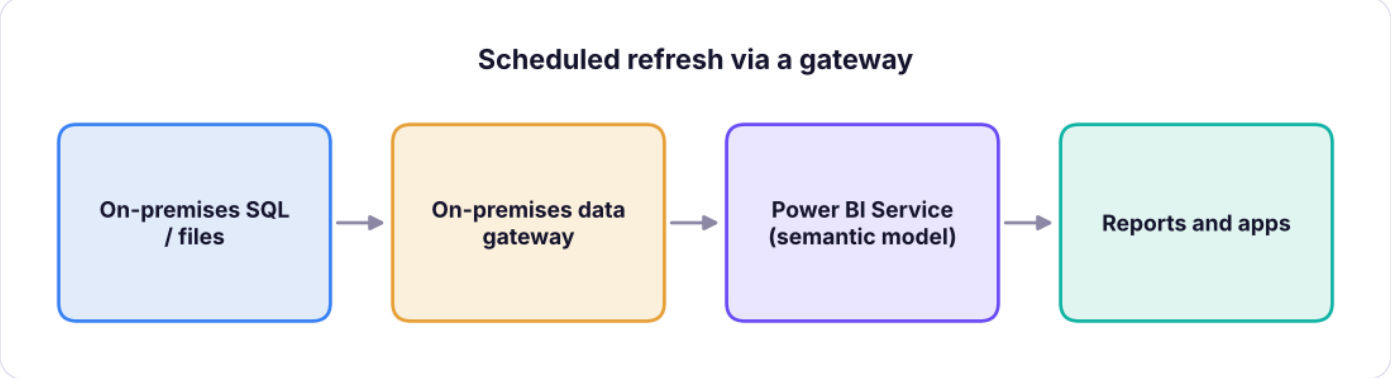
STEP BY STEP **Publish a report**

1. In Desktop: **Home → Publish** → sign in with a work/school account → choose a **workspace** (not "My workspace" if you want to share).
2. The Service now has a **semantic model** (the data and model) and a **report** (the visuals): the same objects in two items.
3. Open the report in the browser, check it, and **Share** with people or add it to an **app**.
4. Configure **scheduled refresh** (Settings → Semantic model → Scheduled refresh) with credentials; for on-premises sources install a **gateway**.

Workspaces, apps and sharing

Concept	Meaning
Workspace	A shared container for semantic models, reports, dashboards. Roles: **Admin, Member, Contributor, Viewer**
App	A packaged, read-only, curated set of reports/dashboards published to a wider audience (best way to distribute)
Share link	Give a person or group access to one report (they need appropriate licence)
Dashboard	One page of **pinned tiles** from several reports; supports alerts and mobile
Subscriptions & alerts	Email snapshots on a schedule; data alerts on card/gauge thresholds
Export / Analyze in Excel / Embed	PDF/PowerPoint export, live pivot tables on the model, secure embedding in portals
Deployment pipelines	Move content Dev → Test → Prod with controlled changes
Endorsement, sensitivity labels, lineage view	Governance: certified datasets, data classification, impact analysis

Refresh and gateways



A scheduled refresh for on-premises data goes through a gateway installed inside the company network.

- **Cloud sources** (Azure SQL, SharePoint Online, OneDrive files) refresh directly without a gateway.
- A **standard gateway** serves many users; a **personal gateway** is for one person's testing.
- **Incremental refresh** (right-click a table → Incremental refresh) loads only new or changed partitions (for example the last 7 days) instead of the whole fact table: essential for big tables. It needs `RangeStart` and `RangeEnd` parameters (Power Query).

- Refresh failures (expired credentials, changed schema, gateway offline) appear in the refresh history; enable failure notifications.

Row-level security (RLS)

RLS restricts the rows a user can see: a country manager sees only their country.

Static RLS

STEP BY STEP **Create roles in Desktop**

1. **Modeling → Manage roles → Create**. Name the role `India`.
2. Select the table `dim_territory` and add a DAX filter: `[country] = "India"`.
3. Create roles `UAE` and `Singapore` the same way. **Save**.
4. **View as** (Modeling → View as) → choose a role to test what that person sees.
5. Publish; in the Service: **Semantic model → Security** → add users or groups to each role.

Because filters flow from `dim_territory` to `fact_sales` through the relationship, the India role sees only India's sales everywhere:

Python

```
import pandas as pd
D = "adventure_cycles/"
f = pd.read_csv(D + "fact_sales.csv"); t = pd.read_csv(D + "dim_territory.csv"); m =
f.merge(t, on="territory_key")
role_view = m.groupby("country").agg(orders=("order_number", "count"), sales=
("sales_amount", "sum"))
role_view["sales"] = (role_view.sales / 1e6).round(1)
role_view.rename(columns={"sales": "sales (₹ M) visible to that role"}).assign(**{"all-
company total (₹ M)": round(f.sales_amount.sum() / 1e6, 1)})
```

OUTPUT

	orders	sales (₹ M) visible to that role	all-company total (₹ M)
country			
India	6135	406.9	594.6
Singapore	1487	96.7	594.6
UAE	1378	91.0	594.6

3 rows × 3 columns

Dynamic RLS

One role for everyone, driven by a **security table** mapping user emails to territories:

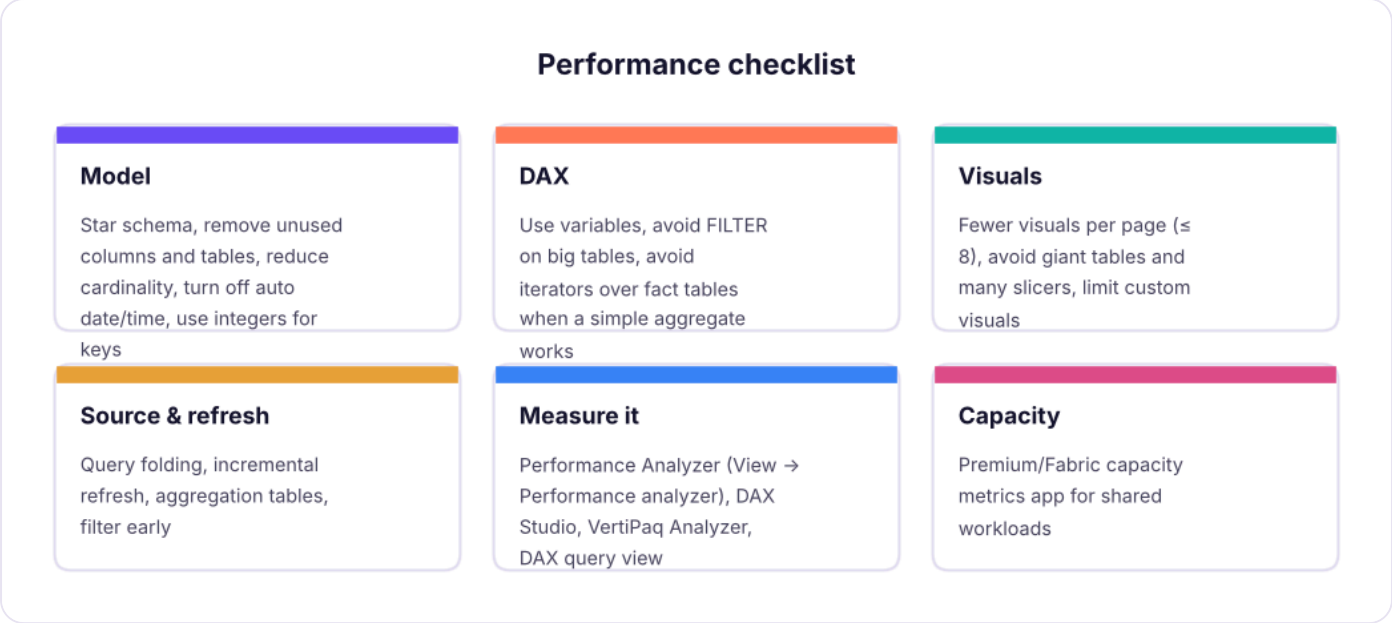
```dax
-- Table 'UserSecurity'(user_email, territory_key); role filter on dim_territory:
[territory_key]
    IN CALCULATETABLE ( VALUES ( UserSecurity[territory_key] ),
                        UserSecurity[user_email] = USERPRINCIPALNAME () )
```

`USERPRINCIPALNAME()` returns the signed-in user's email. Add a person to the table and they get access automatically: no role edits.

WATCH OUT **RLS pitfalls**

- RLS applies to **Viewers** only. Workspace Admins, Members and Contributors can see all data.
- RLS does not hide column names or the existence of tables; use **object-level security** for that.
- Filters must flow along relationships: apply RLS to a **dimension**, not the fact, where possible.
- RLS with DirectQuery depends on the source's behaviour; test with *View as*.
- Do not store secrets or personal data you do not need in the model. In India, handling personal data is governed by the DPDP Act 2023.

Performance tuning



Where Power BI report time goes, and where to look.

Performance Analyzer (View → Performance analyzer → Start recording → refresh visuals) shows, for each visual, the time spent in **DAX query**, **Visual display** and **Other**. If DAX dominates, optimise the measure or model; if display dominates, simplify the visual.

Practical wins:

Problem	Fix
File is huge	Remove columns you do not use; store IDs as integers; split date and time; use `Whole number`/`Fixed decimal` rather than text
Slow card or matrix	Rewrite measures with variables; avoid nested `FILTER ( ALL ( bigTable ) )`
Slow refresh	Query folding, incremental refresh, filter at source
Slow DirectQuery	Aggregations table (Import) in front of the big table; index the source
Many identical measures	Calculation groups or one base measure referenced by others

Microsoft Fabric, Copilot and AI

Microsoft Fabric unifies data engineering, data factory, data warehouse, data science, real-time analytics and Power BI over one data lake called **OneLake** (Delta/Parquet files). Power BI's semantic models can use **Direct Lake** over lakehouse tables. For data professionals: learn SQL (this track), Python (next track), and then Fabric notebooks, pipelines and lakehouses.

Fabric item	Role
Lakehouse / Warehouse	Stores data (files + SQL endpoint)
Dataflow Gen2	Power Query in the cloud
Notebook	Python/Spark for data science and engineering
Pipeline	Orchestrates data movement
Semantic model	The Power BI model (formerly "dataset")
Eventstream / Real-time	Streaming data

Copilot in Power BI (needs a Fabric or Premium capacity): *create a report page* from a prompt, *suggest measures* and *write DAX*, *summarise* a page or report, and *answer questions* about the data. **Prep data for AI** (add descriptions, synonyms, verified answers) improves the answers. Treat it as a fast assistant, not an authority.

Governance and good practice

1. **One certified semantic model per subject area**, many thin reports on top (live connection).
2. Name items consistently (`Sales - Overview`, `Sales_Model`); keep an **owner** and a description.
3. Version control: **PBIP** + Git, deployment pipelines, a test workspace.
4. Document measures (description property) and keep a measure dictionary.
5. Limit **who can export** data; use sensitivity labels.
6. Monitor usage (usage metrics report) and remove unused reports.
7. Train users: the best dashboard is useless if nobody trusts or understands it.

Capstone: Adventure Cycles performance review

CASE STUDY

The brief

The sales director wants a 3-page Power BI report for the Adventure Cycles leadership team and country managers: **(1) Executive summary**, **(2) Products and customers**, **(3) Country detail (drill-through)**. It must show YTD and vs-last-year numbers, let managers filter by year, country and category, and each country manager must see **only their own country**.

Deliverables checklist

Step	What to produce
1. Requirements	Audience, questions, KPIs, refresh frequency, security
2. Data	Load the five CSV tables; clean in Power Query
3. Model	Star schema, relationships, marked date table, hierarchies (Lessons 1–2)
4. Measures	Sales, Profit, Margin %, Orders, AOV, YTD, LY, YoY %, Rank, Top 5 + Others (Lessons 3–4)
5. Report	Three pages, theme, tooltips, drill-through, bookmarks, mobile layout (Lesson 5)
6. Security	RLS roles by country, tested with *View as*
7. Publish	Workspace, scheduled refresh, app, subscriptions (this lesson)
8. Documentation	One-page "how to read this report" and a measure dictionary

The analysis the report must support: verify your page numbers against this independent calculation.

Python

```
m = f.merge(t, on="territory_key").merge(pd.read_csv(D + "dim_product.csv"),
on="product_key"); m["profit"] = m.sales_amount - m.total_cost
m["order_date"] = pd.to_datetime(m.order_date); m["year"] = m.order_date.dt.year
kpi = m.groupby("year").agg(sales=("sales_amount", "sum"), profit=("profit", "sum"),
orders=("order_number", "nunique"))
kpi["margin_%"] = (kpi.profit / kpi.sales * 100).round(1); kpi["YoY_sales_%"] =
(kpi.sales.pct_change() * 100).round(1)
kpi[["sales", "profit"]] = (kpi[["sales", "profit"]] / 1e6).round(1)
kpi
```

OUTPUT

	sales	profit	orders	margin_%	YoY_sales_%
year					
2023	204.2	72.2	3004	35.4	NaN
2024	200.4	70.8	3022	35.3	-1.9
2025	189.9	67.2	2974	35.4	-5.2

3 rows × 5 columns

Python

```
c = m.groupby("country").agg(sales=("sales_amount", "sum"), profit=("profit", "sum"))
c["margin_%"] = (c.profit / c.sales * 100).round(1); c["share_%"] = (c.sales /
c.sales.sum() * 100).round(1); c[["sales", "profit"]] = (c[["sales", "profit"]] /
1e6).round(1)
bike_share = (m[m.category == "Bikes"].sales_amount.sum() / m.sales_amount.sum() *
100).round(1)
print(c.to_string()); print(f"\nBikes are {bike_share}% of total sales; Components
{m[m.category == 'Components'].sales_amount.sum() / m.sales_amount.sum() * 100:.1f}%.")
```

OUTPUT

```
           sales  profit  margin_%  share_%
country
India      406.9   144.1      35.4     68.4
Singapore   96.7    34.1      35.2     16.3
UAE         91.0    32.0      35.1     15.3

Bikes are 78.5% of total sales; Components 18.4%.
```

KEY TAKEAWAYS **What a good leadership summary says**

1. **Sales are shrinking, not growing**: ₹204.2 M (2023) → ₹200.4 M (2024, −1.9%) → ₹189.9 M (2025, −5.2%), while the margin holds steady at about 35%. The problem is **volume or demand**, not pricing or cost.
2. **Bikes** generate 78.5% of sales and **India** 68.4%, so any decline in Indian bike sales moves the whole company: a concentration risk.
3. Singapore (16.3% of sales) and the UAE (15.3%) earn the same ~35% margin as India, so growth there is as profitable as at home.
4. Recommended next steps: find where the 2025 decline comes from (by category, country and month, using the YoY measures from Lesson 4), set recovery targets per country, and track them monthly in the report.

(Verify each statement against the tables above before you present it; the numbers, not the template, are the story.)

Summary

KEY TAKEAWAYS

- **Import** is fast and the default; **DirectQuery** keeps data in the source (slower, near-live); **Live connection** reuses a shared semantic model; **Composite** mixes modes; **Direct Lake** reads Fabric Delta tables directly.
- Publish to a **workspace**, schedule refresh (gateway for on-premises, incremental refresh for big facts), distribute through **apps**, and use subscriptions and alerts.
- **RLS** filters rows by role (static) or by the signed-in user (dynamic, `USERPRINCIPALNAME()`); it applies to Viewers and must be tested with *View as*.
- Tune performance with a star schema, lean columns, efficient DAX and **Performance Analyzer**.
- Fabric unifies the data stack; Copilot speeds up building; governance (certified models, pipelines, labels) keeps it trustworthy.

Quick quiz

6 questions · tap an answer to check it

1. Which mode stores a compressed copy of the data inside the Power BI model?

A Import

B DirectQuery

C Live connection

D Streaming

Answer: A. Import loads the data into the in-memory VertiPaq model.

2. When is an on-premises data gateway needed?

A For any Excel file

B To schedule refresh from data sources inside a company network

C To publish reports

D To use slicers

Answer: B. Gateways let the cloud service reach on-premises sources.

3. A user with the Admin role in the workspace is also in an RLS role. What do they see?

A Only the role's rows

B All data, because RLS does not apply to Admin, Member and Contributor roles

C Nothing

D Only their own data

Answer: B. RLS applies to Viewers; test with View as.

4. What does incremental refresh do?

A Refreshes slicers only

B Refreshes only new or changed partitions of a large table

C Deletes old data

D Converts the model to DirectQuery

Answer: B. It speeds up refresh by loading only recent data.

5. Which tool shows time spent in DAX vs visual display?

A Model view

B Performance Analyzer

C Selection pane

D Bookmarks

Answer: B. Performance Analyzer breaks down each visual's load time.

6. Which function enables dynamic row-level security by user?

A `TODAY ()`

B `SELECTEDVALUE ()`

C `USERPRINCIPALNAME ()`

D `RELATED ()`

Answer: C. It returns the logged-in user's email for comparison with a security table.

EXERCISE

Deploy and secure

(1) Create RLS roles for the three countries and test each with *View as*. (2) Create a dynamic RLS design using a `UserSecurity` table (write the table and the DAX filter). (3) List five changes you would make to a 2 GB model that refreshes in 45 minutes. (4) Write the checklist you would give a colleague before publishing a report to leadership.

Show solution (worked answer)

(1) Manage roles → filter `dim_territory[country]`; View as each role and confirm the totals equal the pandas numbers above. (2) Table: `user_email, territory_key`; filter on `dim_territory`: `[territory_key] IN CALCULATETABLE ( VALUES ( UserSecurity[territory_key] ), UserSecurity[user_email] = USERPRINCIPALNAME () )`. (3) Remove unused columns; star schema and integer keys; incremental refresh on the fact; push transformations to the source (query folding); aggregate to the needed grain; disable auto date/time. (4) Numbers reconciled with source; filters and slicers tested; titles state insights; RLS tested; refresh scheduled and tested; owner and description added; performance checked; accessibility and mobile layout checked; app published with notes.

INTERVIEW CORNER

Questions you may be asked

Q: Import vs DirectQuery? Import copies and compresses data into the model (fast, needs refresh, size limits); DirectQuery queries the source live (fresh, handles huge data, slower, DAX limitations).

Q: What is row-level security and how do you implement it? Restricting the rows a user can see: define roles with DAX filters on tables (static), or use USERPRINCIPALNAME and a security table (dynamic); assign users in the Service; test with View as.

Q: How do you improve a slow Power BI report? Measure with Performance Analyzer; fix the model (star schema, fewer columns), optimise DAX, simplify visuals, use aggregations/incremental refresh, and reduce data volume.

Q: What is a semantic model and why use live connections? The data model (tables, relationships, measures); multiple reports can connect live to one governed model so everyone uses the same definitions.

POWER BI · LESSON 7

Microsoft Fabric: Lakehouse, Warehouse, Dataflows and Semantic Models

Understand Microsoft Fabric as one platform. Load data into a lakehouse, shape it in a warehouse or dataflow, publish a semantic model that Power BI reports read from, and keep the whole chain governed and refreshed.

Gap-fill lesson | Microsoft Fabric 2026 | Advanced | 150 min

By the end of this lesson you can

- Describe the Fabric workspace items: lakehouse, warehouse, dataflow, notebook and semantic model
- Choose between a lakehouse and a warehouse for a given workload
- Load and shape data with a dataflow and with the SQL analytics endpoint
- Publish a semantic model and connect a Power BI report to it
- Explain OneLake, shortcuts and the refresh chain

What Fabric is

Microsoft Fabric is a single platform for data work, built on one storage layer called **OneLake**. Instead of separate tools for storage, pipelines, warehousing and reporting, every item in Fabric reads from the same copy of the data. Power BI is one of the experiences inside Fabric.

For a Power BI analyst, the important items are:

Item	What it is	Typical use
Lakehouse	Files and tables stored in OneLake, read with SQL or Spark	Large data, mixed files and tables
Warehouse	A SQL-first store for structured tables	Star schemas for reporting
Dataflow Gen2	No-code Power Query data preparation	Cleaning and loading without code
Notebook	Python, PySpark or SQL code	Advanced transformations and data science
Semantic model	The model (tables, relationships, DAX measures) that reports use	Shared, governed measures
Report	The Power BI visuals	What people look at

CASE STUDY **One sales model, many reports**

A retail company keeps sales data in OneLake. Finance, marketing and operations each want their own report, but they must all agree on what "revenue" means. The answer is one semantic model in a Fabric workspace that all three reports connect to, so the measure is written once and changed in one place.

OneLake and shortcuts

OneLake is the single logical data lake for the organisation. Each workspace has a lakehouse or warehouse, and the data is stored once. A **shortcut** points to data in another location (another lakehouse, or Azure storage, or Amazon S3) without copying it. Use shortcuts to avoid duplicating large tables across workspaces.

::::tip Shortcut, do not copy A copy drifts out of date and doubles storage. A shortcut always reads the current data. Copy only when you need a snapshot, for example for an audit. ::::

Lakehouse versus warehouse

Both store tables in OneLake. The choice depends on the workload:

- **Lakehouse**: keeps files (CSV, JSON, Parquet) alongside tables. Spark notebooks and SQL both work on it. Good for data engineering and mixed data.
- **Warehouse**: SQL-only tables with full T-SQL support, including stored procedures and fine-grained permissions. Good for structured reporting and teams that think in SQL.

A lakehouse also exposes a **SQL analytics endpoint**, which lets you query its tables with SQL. Many teams therefore land raw data in a lakehouse, then build reporting tables in a warehouse or view them through the endpoint.

Loading data with a dataflow

A dataflow uses Power Query, the same tool as Power BI Desktop, to load and shape data in the service. It is the right choice when the transformation is ordinary cleaning and the team does not write code.

Steps:

1. In the workspace, select **New item > Dataflow Gen2**.
2. Add a source, for example a CSV file uploaded to the lakehouse.
3. Apply transformations: change types, remove blank rows, merge with a customer table.
4. Set the output destination to a lakehouse or warehouse table.
5. Publish, then run the dataflow once to check the output.

The output is a table in OneLake. Any report can read it, which is why dataflows suit shared, reusable cleaned data.

::::warn Keep logic in one place If the same cleaning happens in a dataflow, in a notebook and again in Power BI Desktop, the three versions will disagree. Choose one place for each transformation and document it. ::::

Building a star schema in the warehouse

Reporting works best on a star schema: one fact table of sales and several dimension tables of customers and products. Create the tables in the warehouse with a short script, then load them.

SQL

```sql
-- Dimension: customers (one row per customer)
CREATE TABLE dbo.dim_customer (
    customer_key   INT         NOT NULL,
    customer_name  VARCHAR(100),
    segment        VARCHAR(50),
    territory      VARCHAR(50)
);

-- Fact: sales (one row per order line, keys point to dimensions)
CREATE TABLE dbo.fact_sales (
    order_date     DATE,
    customer_key   INT,
    product_key    INT,
    quantity       INT,
    sales_amount   DECIMAL(12, 2),
    cost_amount    DECIMAL(12, 2)
);
```

The fact table holds numbers and keys. Descriptive text lives in dimensions. This shape keeps the semantic model small and fast, and it matches the star schema taught in the modelling lesson.

Publishing a semantic model

A semantic model is the Power BI dataset in Fabric. You create it from the warehouse or lakehouse tables, add relationships and measures, and publish it to the workspace.

1. Open the warehouse and choose **New semantic model**.
2. Select the tables to include, for example `fact_sales`, `dim_customer` and `dim_product`.
3. Confirm the relationships. They should run from each dimension (one side) to the fact table (many side).
4. Write the DAX measures in the model, not in individual reports, so every report uses the same definition:

DAX

```dax
Total Sales = SUM ( fact_sales[sales_amount] )
Total Cost  = SUM ( fact_sales[cost_amount] )
Gross Margin % = DIVIDE ( [Total Sales] - [Total Cost], [Total Sales] )
```

5. Save the model. Reports now connect to it in Power BI and see the same measures.

Connecting a report

From Power BI Desktop, choose **Get data > Power BI semantic models** and select the published model. The report is built on top of the shared model: visuals and measures come from the model, while layout and filters belong to the report.

This separation is the key idea. A change to the model's measure updates every report that uses it, without editing each one. The reverse also matters: a report should not define its own revenue measure, or the numbers will drift.

Refresh and the chain of dependencies

Data flows through a chain: source, then dataflow or notebook, then lakehouse or warehouse, then semantic model, then report. A refresh must run in order. If the dataflow fails, the model shows yesterday's data, and the report shows yesterday's numbers without any visible error.

Practical rules:

- Schedule the upstream item first, and the semantic model refresh after it, with a short gap.
- Check the refresh history after each run, not only when a report looks wrong.
- Use a notification on failure so that the team hears about it first.

Governance basics in Fabric

- **Workspace roles** (Admin, Member, Contributor, Viewer) control who can change items. Give analysts Contributor or Member only where they need to build.
- **Endorsement** marks a semantic model as Promoted or Certified, so people know which model to trust.
- **Lineage view** shows how a report depends on the model, the warehouse and the source, which speeds up impact checks before a change.
- **Sensitivity labels** can classify data, for example as Confidential, and follow it through the chain.

::::key What to remember - Fabric keeps all data in one place, OneLake, and every item reads from it. - Use shortcuts instead of copies, and lakehouses for files and mixed data, warehouses for structured reporting. - Dataflows suit no-code cleaning; notebooks suit code; both write tables to OneLake. - Define measures once in the semantic model, and let every report use them. - Refresh in order, and check refresh history, because a silent failure shows old numbers. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is OneLake?

A A single report in Power BI

B The single logical data lake that all Fabric items read from

C A type of DAX measure

D A gateway for on-premises data

Answer: B. OneLake is the shared storage layer for Fabric.

2. When is a shortcut better than a copy?

A When you need an audit snapshot

B When you want the other location to read current data without duplicating storage

C When the source is offline

D Never; copies are always better

Answer: B. A shortcut points to the data, so it stays current.

3. Which item suits a team that writes T-SQL and needs stored procedures?

A A dataflow

B A warehouse

C A report

D A shortcut

Answer: B. Warehouses support full T-SQL, including stored procedures.

4. **Why define measures in the semantic model rather than in each report?**

A Reports cannot store measures

B So every report uses one definition, and a change is made once

C Measures run faster there only

D It hides the measures

Answer: B. One definition prevents numbers from drifting between reports.

5. **A dataflow fails overnight and nobody notices until a manager asks. What is the best fix?**

A Remove the dataflow

B Schedule refreshes in order and set a failure notification

C Refresh the report manually every hour

D Turn off the semantic model

Answer: B. Refresh order and failure alerts make silent failures visible.

EXERCISE

Design the chain

A company has order data in CSV files, customer and product lists in a CRM export, and three reports for finance, sales and operations. Draw the Fabric chain: which items you would use (lakehouse, warehouse, dataflow, notebook, semantic model, report), in what order, and where you would define the revenue and margin measures. Name one refresh rule and one governance control.

Show solution (worked answer)

One possible design: CSV files land in a lakehouse; a Dataflow Gen2 cleans and types them into tables; the tables are modelled as a star schema in a warehouse; one semantic model holds the relationships and the Total Sales and Gross Margin % measures; the three reports connect to that model. Refresh the dataflow first, then the model. Governance: endorse the model as Certified and limit edit rights to a named team.

POWER BI · LESSON 8

Databricks: Notebooks, Delta Lake, Spark Jobs and Unity Catalog

Work on a lakehouse with Databricks. Write notebooks in SQL and PySpark, store tables as Delta Lake with merges and time travel, run Spark jobs on a schedule, and govern data with Unity Catalog so that reports read trusted tables.

Gap-fill lesson | Databricks 2026 | Advanced | 150 min

By the end of this lesson you can

- Explain the Databricks workspace, clusters and notebooks
- Write a Delta table with SQL and PySpark, and explain why Delta adds reliability
- Use MERGE, time travel and schema checks on Delta tables
- Schedule a Spark job and understand how it differs from a local pandas script
- Organise data with Unity Catalog catalogs, schemas and permissions

What Databricks adds

Databricks is a data platform built on Apache Spark. Spark processes data in parallel across a cluster of machines, so it handles data too large for one computer. Databricks adds managed clusters, notebooks, scheduled jobs and a governance layer called Unity Catalog. Its storage format, **Delta Lake**, adds reliable updates and history to files stored in cloud object storage.

For a Power BI analyst the practical point is simple: many companies keep their reporting tables in Databricks, and reports connect to them through the SQL warehouse endpoint or through Power BI's Databricks connector.

CASE STUDY **Millions of rows, one daily refresh**

The sales file has grown to 80 million rows and no longer fits in a Power BI import. The data team keeps the detail in Delta tables in Databricks and builds a daily summary table with a scheduled Spark job. The report reads the summary, and the detail is queried only when someone drills down.

Workspace, clusters and notebooks

- **Workspace**: where notebooks, jobs and tables are organised.
- **Cluster** (or serverless compute): the machines that run your code. Clusters start, scale and stop; you pay while they run, so auto-termination matters.
- **Notebook**: a document of cells that mix Markdown, SQL, Python and PySpark. Each notebook attaches to a cluster and runs one cell at a time.

A notebook can switch language per cell with a magic command such as `%sql`, so one notebook can read a table in SQL and transform it in PySpark.

Reading data with PySpark

PySpark is the Python API for Spark. A DataFrame in Spark looks like a pandas DataFrame, but the work is split across the cluster and runs lazily: transformations are planned until a result is needed.

Python

```python
# In a Databricks notebook, `spark` is already created
sales = spark.read.table("retail.bronze.sales_raw")

summary = (
    sales
    .filter("order_date ≥ '2025-01-01'")
    .groupBy("region", "category")
    .agg({"sales": "sum", "profit": "sum"})
    .withColumnRenamed("sum(sales)", "total_sales")
    .withColumnRenamed("sum(profit)", "total_profit")
)

summary.write.mode("overwrite").saveAsTable("retail.gold.sales_summary_2025")
```

The names `retail.bronze.sales_raw` and `retail.gold.sales_summary_2025` follow the bronze, silver and gold layering: raw data, cleaned data and reporting data. This is the common pattern for lakehouse work.

::::note Lazy evaluation Calling `filter` or `groupBy` does not run anything. Spark runs the plan when you call an action such as `write`, `count` or `show`. A slow notebook is often a plan that recomputes data many times, which `cache()` or writing an intermediate table can fix. ::::

Delta Lake: reliable tables on files

A Delta table is a set of Parquet data files plus a transaction log that records every change. The log gives you:

- **ACID writes**: an update either completes or does not happen, so readers never see half-written data.
- **MERGE**: insert new rows and update changed ones in one statement, which is the upsert from the SQL incremental-loading lesson.
- **Time travel**: query the table as it was at an earlier version or time.
- **Schema enforcement**: writes with the wrong columns or types are rejected.

```sql
-- Upsert new and corrected order lines into the Delta table
MERGE INTO retail.silver.fact_sales AS target
USING retail.bronze.sales_today AS source
ON target.line_id = source.line_id
WHEN MATCHED THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *;

-- What did the table look like yesterday?
SELECT COUNT(*) AS lines, SUM(sales) AS total_sales
FROM retail.silver.fact_sales VERSION AS OF 12;

-- The history of changes
DESCRIBE HISTORY retail.silver.fact_sales;
```

Time travel is valuable for audits and for undoing a bad load: restore the previous version rather than rebuilding the table. Check retention settings first, because old versions are removed after a period.

::::warn MERGE needs a unique key If the source has two rows with the same `line_id`, MERGE fails or updates the same target row twice. Deduplicate the source before the merge, using the latest record for each key. ::::

Scheduled Spark jobs

A notebook becomes production work when it runs as a **job**:

1. Open the notebook, choose **Schedule** or create a job, and select a cluster or serverless compute.
2. Set the schedule (for example, every day at 5 a.m.) and the parameters.
3. Add a notification for failure, and set a retry for transient errors.
4. Check the run history after the first few runs, and compare the output totals with the source.

A job should be idempotent, like the loads in the SQL lesson: running it twice should not double the totals. Overwrite a summary table, or merge into a fact table, rather than appending.

Unity Catalog: governance across the platform

Unity Catalog is the central place to organise and secure data. Its three-level name is **catalog.schema.table**:

- **Catalog**: a top-level container, often one per environment or business unit (for example `retail`).
- **Schema**: a group of related tables (for example `gold` for reporting tables).
- **Table or view**: the data itself.

Permissions are granted on these objects, so a group can read `retail.gold` but not `retail.bronze`:

```sql
GRANT USE CATALOG ON CATALOG retail TO `analysts`;
GRANT USE SCHEMA ON SCHEMA retail.gold TO `analysts`;
GRANT SELECT ON SCHEMA retail.gold TO `analysts`;
```

Unity Catalog also records lineage, which shows which tables and notebooks produced a table. Use it to answer "where does this number come from?" before a report is trusted.

::::tip Give reports gold tables only Point Power BI at the gold layer, not at raw bronze tables. Gold tables are the cleaned, agreed definitions, so the report does not repeat cleaning logic and does not expose raw data to viewers. ::::

Connecting Databricks to Power BI

Power BI connects to a Databricks SQL warehouse through its Databricks connector, using the server hostname, HTTP path and an authentication method approved by the data team. Import mode suits daily summaries; DirectQuery suits large tables that must stay current. Choose the mode the same way as in the modes lesson: by data size, freshness and the number of users.

Comparing the platforms

Need	Fabric	Databricks
Storage	OneLake	Cloud storage with Delta tables
Large-scale code	Spark notebooks in a lakehouse	Spark notebooks and jobs
Governance	Workspaces, endorsement, lineage	Unity Catalog
Reporting	Native semantic models	Connect Power BI to a SQL warehouse

Both platforms run Spark, store tables in open formats, and feed Power BI. Learn the concepts once; the names change.

::::key What to remember - Databricks runs Spark on managed clusters; notebooks mix SQL and PySpark. - Delta Lake adds ACID writes, MERGE, time travel and schema checks to files. - Build summary tables as scheduled jobs, and make jobs safe to re-run. - Unity Catalog governs catalogs, schemas and tables with permissions and lineage. - Point reports at gold tables that hold agreed definitions. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is Delta Lake?

A A type of chart

B A table format that adds transaction logs, MERGE and time travel to files

C A Power BI gateway

D A programming language

Answer: B. Delta stores Parquet files with a log of every change.

2. What does time travel let you do?

A Make the cluster faster

B Query a table as it was at an earlier version

C Rename columns

D Skip a refresh

Answer: B. Time travel reads earlier table versions for audits and recovery.

3. Why must a MERGE source have unique keys?

A Keys are optional in Delta

B Duplicate keys in the source can make the merge fail or update one target row twice

C MERGE only works on text

D It speeds up the cluster

Answer: B. Deduplicate the source before merging.

4. In `retail.gold.sales_summary`, what is `gold`?

A The catalog

B The schema

C The table

D The cluster

Answer: B. The three-level name is catalog.schema.table.

5. Why should Power BI read gold tables rather than bronze?

A Bronze tables are faster to read

B Gold tables hold the cleaned, agreed definitions and avoid exposing raw data

C Gold tables cannot be queried

D Bronze tables are always deleted

Answer: B. Gold is the reporting layer.

EXERCISE **Design a daily job**

A retail team loads raw sales files each night into a bronze table. Design the daily job that produces a gold summary of sales by region and month for Power BI. Write the steps, state whether the summary is overwritten or merged and why, list one check to run before publishing, and name the Unity Catalog permission that lets the analysts read the summary but not the raw data.

Show solution (worked answer)

Steps: read the bronze table, deduplicate by line ID, filter the new period, aggregate sales and profit by region and month, write to the gold schema. Overwrite the summary because it is derived in full each run (re-runs give the same result). Check that the total sales in the summary equal the total in the cleaned silver table for the same period. Permission: `GRANT SELECT ON SCHEMA retail.gold TO analysts` with no grant on the bronze schema.

Skill Navora

TRACK 4

Tableau

Tableau Public and Desktop: connecting data, dimensions and measures, every core chart, calculated fields, table calculations and LOD expressions, filters, parameters and sets, data modelling, dashboards, stories and publishing.

6 lessons | Version 2026.10 | Updated October 2026

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Contents

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03 Calculations in Tableau: Calculated Fields, Table Calculations and LOD Expressions

04 Filters, Parameters, Sets, Groups and Hierarchies in Tableau

05 Data Modelling in Tableau: Relationships, Joins, Unions, Blending, Extracts and Tableau Prep

06 Dashboards, Actions, Stories and Publishing in Tableau (Capstone)

TABLEAU · LESSON 1

Getting Started with Tableau: Interface, Data Connections and Your First Views

Understand the Tableau product family, install Tableau Public, connect to the Superstore India data, read the Data Source page, learn dimensions vs measures and discrete vs continuous fields, master the Marks card and shelves, build and sort your first views, and save them to Tableau Public.

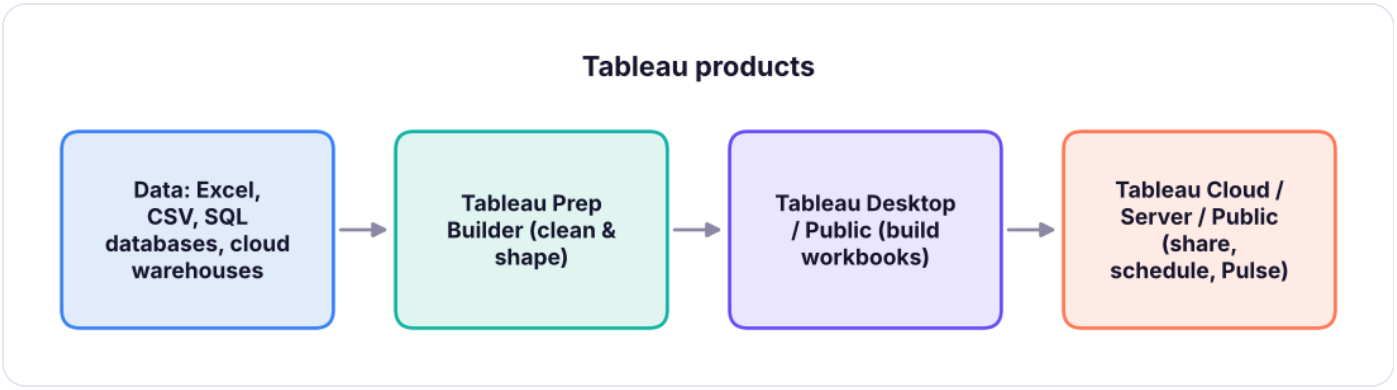
✦ Tableau Desktop / Tableau Public 2026 | Beginner | ⏱ 120 min

By the end of this lesson you can

- Describe Tableau Desktop, Public, Cloud, Server, Prep and Pulse, and when each is used
- Connect to a CSV, check data types and rename or hide fields on the Data Source page
- Explain dimensions vs measures and discrete (blue) vs continuous (green) fields
- Build bar, line and text-table views with the Columns, Rows, Marks, Filters and Pages shelves
- Sort, label, format and save a workbook to Tableau Public

What is Tableau?

Tableau is a visual analytics platform: you drag fields onto shelves and Tableau writes the query, aggregates the data and draws the chart. It is owned by Salesforce and, with Power BI, is one of the two BI tools most often named in Indian analyst job posts. Where Power BI leans on DAX formulas and the Microsoft stack, Tableau is known for fast visual exploration and highly customisable charts.



The Tableau product family. You build in Desktop (or Public), share on Cloud/Server/Public, and prepare data in Prep.

Product	What it is	Cost (2026)
Tableau Public (desktop app + website)	Free version of Desktop; workbooks are saved **publicly** on public.tableau.com	Free
Tableau Desktop	Full authoring tool; saves locally, connects to databases	Paid (Creator licence)
Tableau Cloud	Tableau's hosted service for sharing, refresh schedules and permissions	Paid, per user
Tableau Server	The same as Cloud, but installed on your company's own servers	Paid
Tableau Prep Builder	Visual tool to clean, join and reshape data before analysis	Included with Creator
Tableau Pulse / Tableau Agent	AI features: metric digests in plain language and an assistant that helps write calculations and build views	Cloud add-ons

TIP **Which one should you install?**

Learners should install **Tableau Public** (free). It has almost every authoring feature of Desktop. The two limits: workbooks are saved to your public profile (so never use confidential data) and it connects mainly to files (Excel, CSV, text, JSON, PDF, spatial) and Google Sheets rather than company databases. Your public profile also works as a portfolio for interviews.

Install and connect

STEP BY STEP **Install Tableau Public and connect to Superstore**

1. Go to public.tableau.com → **Create** → download **Tableau Public** for Windows or Mac and install it; sign up for a free profile.
2. Open Tableau Public. On the **Connect** pane choose **Text file** and pick `superstore_sales.csv` (for an Excel file choose **Microsoft Excel**).
3. The **Data Source page** opens: the table appears in the canvas and a preview grid below it.
4. Check each column's **data type icon** (Abc = string, # = number, calendar = date, globe = geographic). Click an icon to change it.
5. Click **Sheet 1** at the bottom to start building.

The data you connect to is the same Superstore India file used in the Excel and Power BI lessons. Check its size before you build anything:

Python

```python
import pandas as pd
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date", "ship_date"])
print(df.shape)
print(df.dtypes.astype(str).value_counts().to_dict())
print("Orders from", df.order_date.min().date(), "to", df.order_date.max().date())
df.head(3)
```

OUTPUT

```
(6323, 19)
{'object': 12, 'float64': 3, 'int64': 2, 'datetime64[ns]': 2}
Orders from 2023-01-01 to 2025-12-31
```

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name
0	1	IN-2023-00001	2023-01-01	2023-01-05	Second Class	C0765	Ishaan Gupta
1	2	IN-2023-00001	2023-01-01	2023-01-05	Second Class	C0765	Ishaan Gupta
2	3	IN-2023-00001	2023-01-01	2023-01-05	Second Class	C0765	Ishaan Gupta

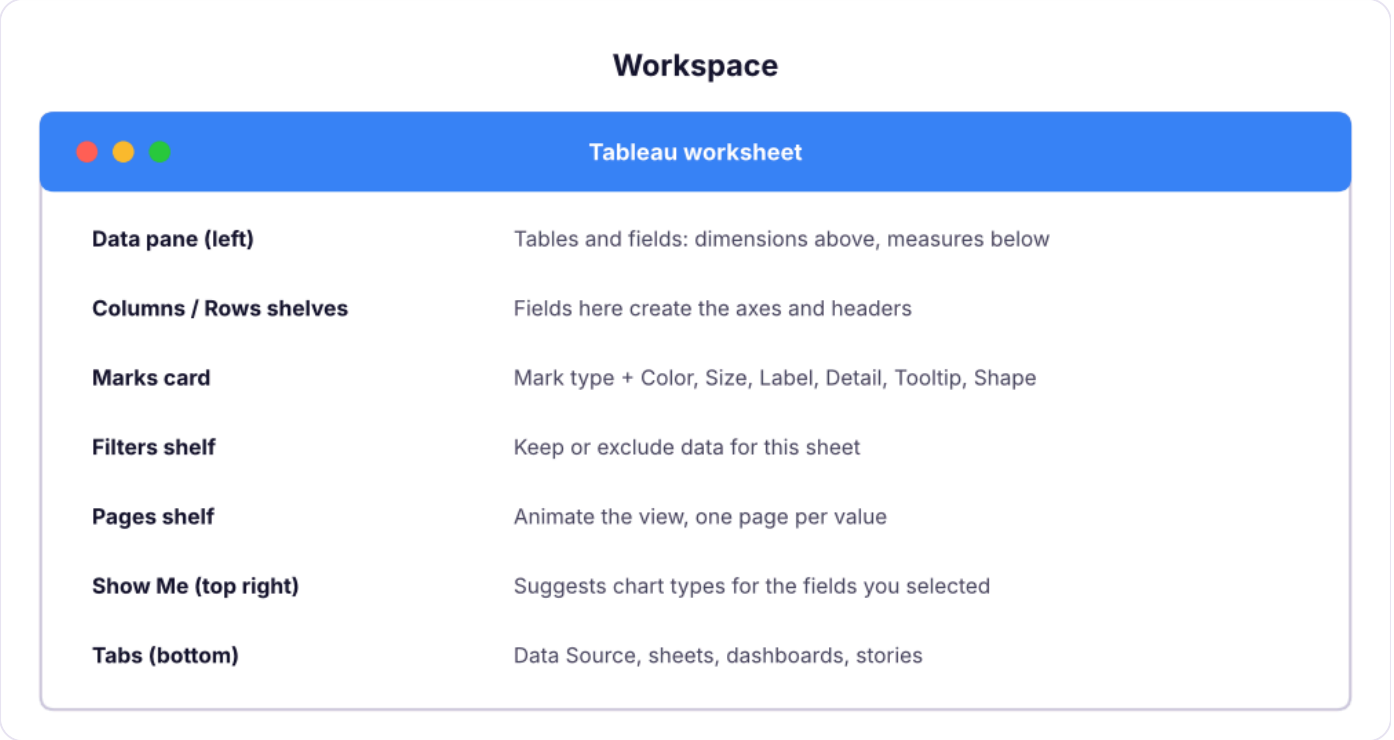
3 rows × 19 columns

6,323 rows (order lines) and 19 columns: Tableau will show these as **fields** in the Data pane.

The Data Source page

Feature	Where	Use
Rename a field	Click the column header → Rename	Friendly names such as "Sub-Category"
Hide a field	Column menu → Hide	Remove IDs you will never use
Change type / geographic role	Click the type icon	Make `state` a **State/Province** so it maps
Split	Column menu → Split	`customer_name` → first and last name
Pivot	Select columns → Pivot	Turn month columns into rows
Data Interpreter	Left pane checkbox	Clean messy Excel sheets with titles and merged cells
Live vs Extract	Top right	Extract (.hyper) is a fast snapshot; Live queries the source every time
Filters (data source)	Top right → Add	Keep only the rows the workbook needs

The workspace

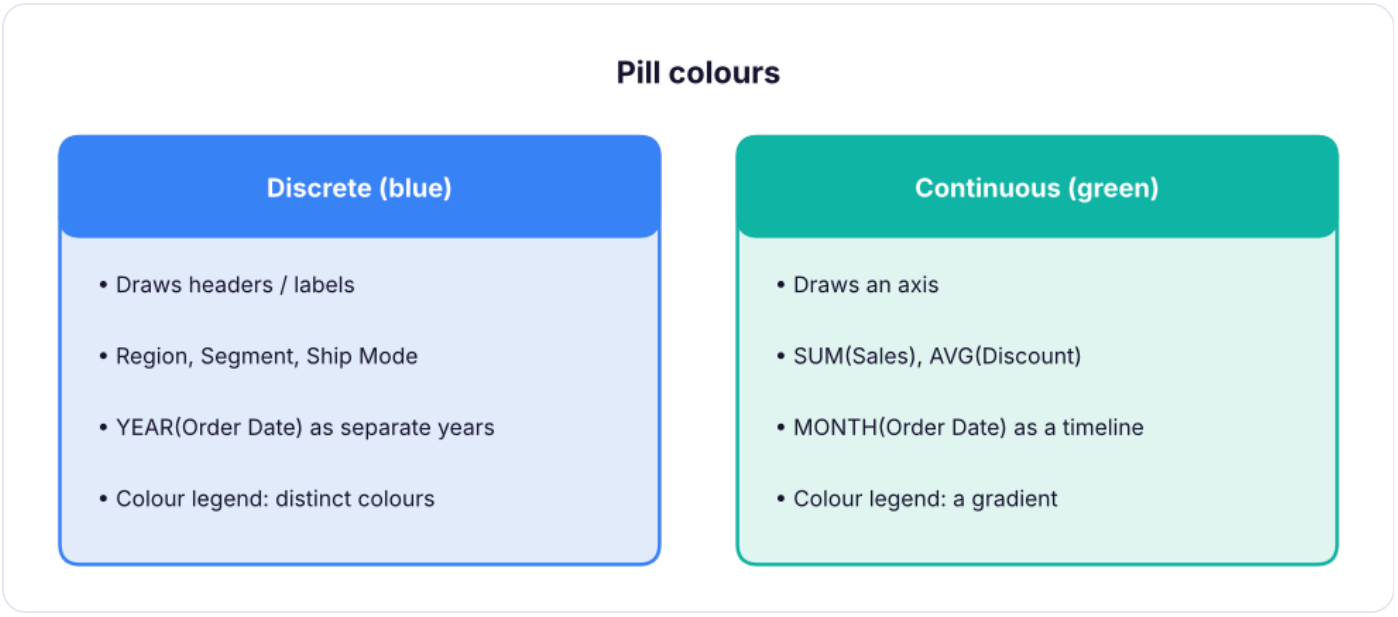


The Tableau worksheet layout. Drag fields from the Data pane onto shelves and the Marks card.

Dimensions vs measures, discrete vs continuous

Tableau sorts every field in two ways, and the colour of a pill tells you which:

	Meaning	Example	What it creates
Dimension	Categories you slice by	Region, Category, Customer Name	Groups (one mark per value)
Measure	Numbers you aggregate	Sales, Profit, Quantity	SUM, AVG, COUNT...
Discrete (blue pill)	Separate values	Region, YEAR(Order Date) as "2023, 2024"	**Headers**
Continuous (green pill)	An unbroken range	SUM(Sales), exact date	**Axes**



Blue (discrete) pills create headers; green (continuous) pills create axes.

A measure dropped into the view is **aggregated** (default SUM). Right-click the pill → **Measure** to switch to Average, Median, Count, Count (Distinct), Minimum, Maximum, Standard deviation and so on.

Your first views

Sales by region (bar chart)

Drag **Region** to Rows and **Sales** to Columns. Tableau writes the equivalent of `SELECT region, SUM(sales) FROM superstore GROUP BY region` and draws one bar per region:

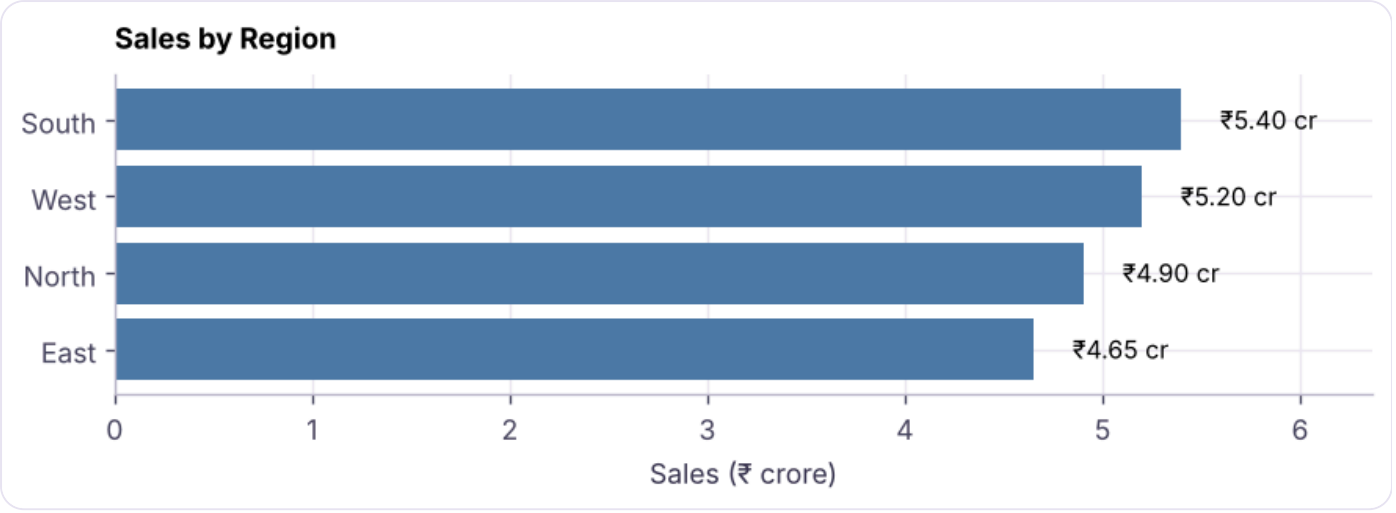
Python

```
by_region = (df.groupby("region")[["sales", "profit"]].sum() /
1e7).round(2).sort_values("sales", ascending=False)
by_region.columns = ["Sales (₹ crore)", "Profit (₹ crore)"]
by_region
```

OUTPUT

	Sales (₹ crore)	Profit (₹ crore)
region		
South	5.40	0.47
West	5.20	0.47
North	4.90	0.46
East	4.65	0.44

4 rows × 2 columns



The view you get: SUM(Sales) by Region, sorted descending, with labels.

The default Tableau colours (blue #4E79A7, orange #F28E2B, red #E15759…) are the **Tableau 10** palette used in the figures in this track.

STEP BY STEP **Sort, label and colour it**

1. Hover the Sales axis and click the **sort** icon, or click the toolbar sort descending button.
2. Click **Label** on the Marks card → tick *Show mark labels*.
3. Drag **Profit** onto **Color**: bars are shaded by profit, so sales and margin show together.
4. Right-click the Sales axis → **Format** → Numbers → Currency (Custom) → Display units: *Crores* is not built in, so use a calculated field `[Sales]/1e7` named "Sales (₹ cr)" (Lesson 3), or show *Millions*.
5. Double-click the sheet tab and rename it "Sales by Region".

Sales over time (line chart)

Drag **Order Date** to Columns: Tableau shows **YEAR(Order Date)** as a discrete header. Click the **+** on the pill to drill down to Quarter and Month. Right-click the pill and choose the **green** "Month" (continuous) to draw a single timeline, or the **blue** "Month" (discrete) to compare the same month across years.

Python

```
m = df.set_index("order_date").sales.resample("ME").sum() / 1e5
print("Months:", len(m), "| best month:", m.idxmax().strftime("%b %Y"), f"₹{m.max():.1f}
lakh", "| weakest:", m.idxmin().strftime("%b %Y"), f"₹{m.min():.1f} lakh")
yearly = (df.groupby(df.order_date.dt.year).sales.sum() / 1e7).round(2); yearly
```

OUTPUT

```
Months: 36 | best month: Nov 2025 ₹102.8 lakh | weakest: Jan 2023 ₹24.5 lakh
```

	sales
order_date	
2023	5.76
2024	6.54
2025	7.84

3 rows × 1 columns

Text table (crosstab)

Put **Category** on Rows, **YEAR(Order Date)** on Columns and **Sales** on **Text**. Add **Analysis → Totals → Show Row Grand Totals**. This is Tableau's pivot table:

Python

```
ct = pd.crosstab(df.category, df.order_date.dt.year, values=df.sales, aggfunc="sum",
margins=True, margins_name="Grand Total") / 1e5
ct.round(1)
```

OUTPUT

order_date	2023	2024	2025	Grand Total
category				
Furniture	99.5	121.3	127.6	348.4
Office Supplies	26.1	35.5	39.7	101.3
Technology	450.8	497.2	616.5	1,564.4
Grand Total	576.4	654.0	783.8	2,014.1

4 rows × 4 columns

Show Me

Show Me (top right) highlights the chart types that fit the fields you have selected and states what it needs, e.g. "*1 date, 1 or more dimensions, 1 or more measures*" for a line chart. It is a fast start, but learn to build each chart by hand: interviewers often ask *how* you would build a view.

Saving and sharing

Format	What it contains
`.twb` workbook	Sheets, dashboards and the connection, **no data**
`.twbx` packaged workbook	Workbook **plus** a copy of the data/extract: share this one
`.hyper` extract	Tableau's compressed, fast data snapshot
Tableau Public profile	**File → Save to Tableau Public**; anyone can view and download it

COMMON MISTAKES **Beginner mistakes**

- Saving confidential company data to Tableau Public: everything there is public.
- Sending a `.twb` file: the receiver cannot open it without your data. Send `.twbx`.
- Leaving a date as a string ("Abc"): you lose drill-down and date functions. Change the type on the Data Source page.
- Using a measure that should be a dimension (e.g. Postal Code as SUM): right-click → Convert to Dimension.

2026 UPDATE **Tableau in 2026**

- **Tableau Agent** (formerly Einstein Copilot for Tableau) suggests calculations and builds views from a typed question in Tableau Cloud.
- **Tableau Pulse** sends plain-language metric summaries ("Sales are up 8% this week, driven by the West region") to email and Slack.
- **Tableau Next** is Salesforce's agent-based analytics layer on top of Data Cloud. Desktop and Public remain the tools you learn first.

Summary

KEY TAKEAWAYS

- Tableau turns drag-and-drop into queries: shelves define the structure, the Marks card defines the look.
- Dimensions slice, measures aggregate; blue pills are discrete (headers), green pills are continuous (axes).
- Start with Tableau Public, connect to a file, fix data types on the Data Source page, then build.
- Share packaged workbooks (`.twbx`) or publish to Tableau Public/Cloud.

Quick quiz

6 questions · tap an answer to check it

1. What does a green pill on Columns create?

A A continuous axis

B A set of headers

C A filter

D A calculated field

Answer: A. Green = continuous = axis; blue = discrete = headers.

2. You drag Sales into the view. What does Tableau show by default?

- **A** SUM(Sales) for each combination of the dimensions in the view
- **B** Every row's sales value
- **C** The average sales
- **D** The count of rows

Answer: A. Measures are aggregated, and SUM is the default.

3. Which file should you email a colleague so they can open your work with the data?

- **A** A packaged workbook (.twbx)
- **B** A workbook (.twb)
- **C** A .hyper file only
- **D** A screenshot

Answer: A. .twbx bundles the workbook with its data.

4. What is the main limitation of Tableau Public?

- **A** Workbooks are saved publicly and it connects mainly to files
- **B** It cannot draw maps
- **C** It has no calculated fields
- **D** It only works on Mac

Answer: A. Never put confidential data on Tableau Public.

5. State is shown as "Abc" and will not draw a map. What do you do?

- **A** Assign the geographic role State/Province
- **B** Convert it to a measure
- **C** Delete the field
- **D** Use a dual axis

Answer: A. Geographic roles give a field latitude and longitude.

6. Which feature suggests chart types for the selected fields?

- **A** Show Me
- **B** Pages shelf
- **C** Data Interpreter
- **D** Extract

Answer: A. Show Me lists the chart types and what each needs.

EXERCISE **Practice**

Connect to `superstore_sales.csv` in Tableau Public and build three sheets: (1) Sales by Category sorted descending with labels; (2) a continuous monthly Sales line with Profit on Color; (3) a text table of Segment × Year with grand totals. Save as a packaged workbook and publish to your Tableau Public profile.

Show solution (worked answer)

Sheet 1: Category on Rows, SUM(Sales) on Columns, sort descending, Label on. Sheet 2: MONTH(Order Date) continuous (green) on Columns, SUM(Sales) on Rows, SUM(Profit) on Color. Sheet 3: Segment on Rows, YEAR(Order Date) on Columns, SUM(Sales) on Text, Analysis → Totals → Show Row and Column Grand Totals. File → Save to Tableau Public.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between a dimension and a measure? A dimension is a categorical field used to group the data (Region, Category); a measure is a numeric field that Tableau aggregates (SUM of Sales). Each mark represents one combination of the dimensions in the view.

Q: Discrete vs continuous? Discrete fields (blue) create headers and distinct colours; continuous fields (green) create axes and colour gradients. Dates can be either: discrete YEAR gives "2023, 2024" headers, continuous MONTH gives a timeline.

Q: .twb vs .twbx vs .hyper? .twb is the workbook definition without data; .twbx packages the workbook with its data or extract; .hyper is the extract file itself.

Q: Live connection or extract? Live always shows current data but depends on the source's speed; an extract is a fast, compressed snapshot that you refresh on a schedule. Use extracts for speed and offline work, live for real-time needs or very large governed warehouses.

TABLEAU · LESSON 2

Charts in Tableau: From Bar and Line to Maps, Dual Axis, Histograms and Pareto

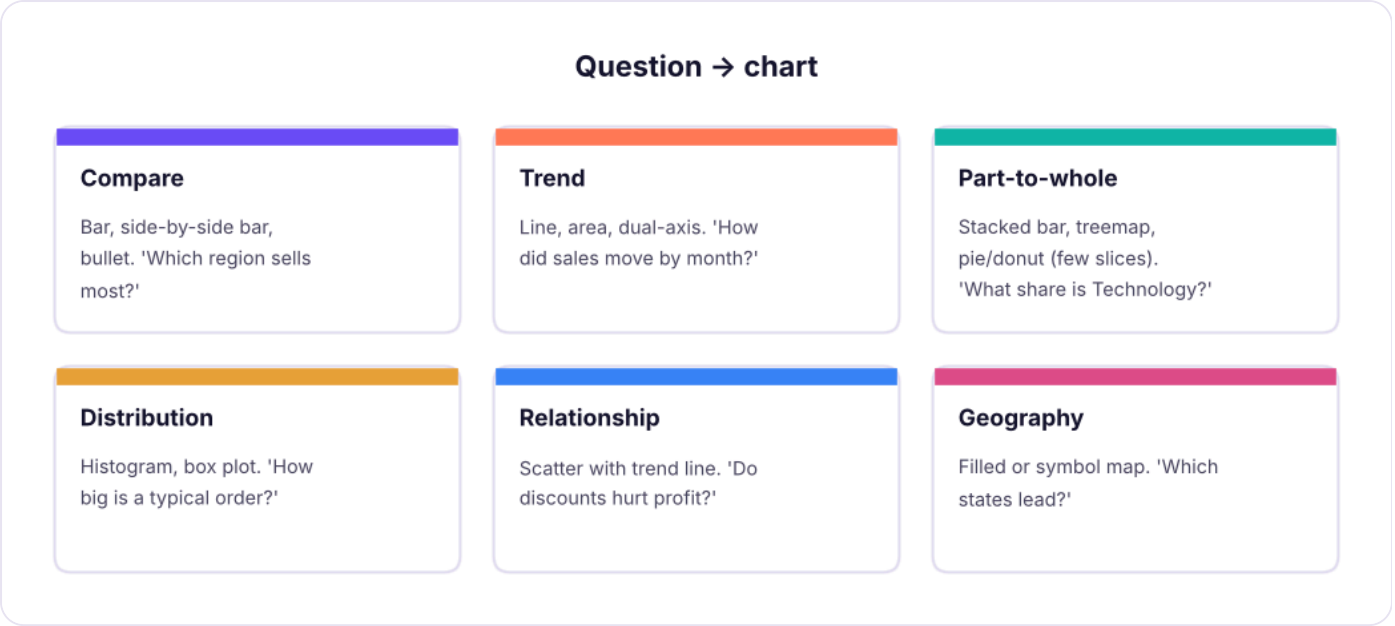
Build every core chart in Tableau and know when to use each: bar, stacked and side-by-side bars, line and area, pie and donut, treemap, highlight table, scatter plot, histogram with bins, box plot, filled and symbol maps, dual-axis combo charts, bullet charts and a Pareto chart, all on the Superstore India data.

✦ Tableau Desktop / Tableau Public 2026 | Beginner | ⏱ 150 min

By the end of this lesson you can

- Choose the right chart for comparison, trend, part-to-whole, distribution, relationship and geography questions
- Build bar, line, area, pie, donut, treemap and highlight-table views
- Create bins and histograms, box plots and scatter plots with trend lines
- Map states with geographic roles, and combine two measures with dual and blended axes
- Build a bullet chart and a Pareto (80/20) chart

Choose the chart from the question



Match the business question to the chart family.

All examples below use the Superstore India data. The tables show the numbers your Tableau view should display, computed here with pandas so you can check your work.

Python

```
import pandas as pd, numpy as np
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
df["year"] = df.order_date.dt.year
print(len(df), "order lines,", df.order_id.nunique(), "orders,", df.customer_id.nunique(),
"customers")
```

OUTPUT

```
6323 order lines, 3200 orders, 785 customers
```

Bar charts

Variant	Build	Use
Bar	Dimension on Rows, measure on Columns	Ranking categories
Stacked bar	Add a second dimension to **Color**	Total and composition
Side-by-side bar	Second dimension on Columns *before* the measure (or Show Me → side-by-side)	Comparing subgroups directly
100% stacked	Stacked bar + Quick Table Calculation → **Percent of Total** (compute using the colour dimension)	Mix comparisons

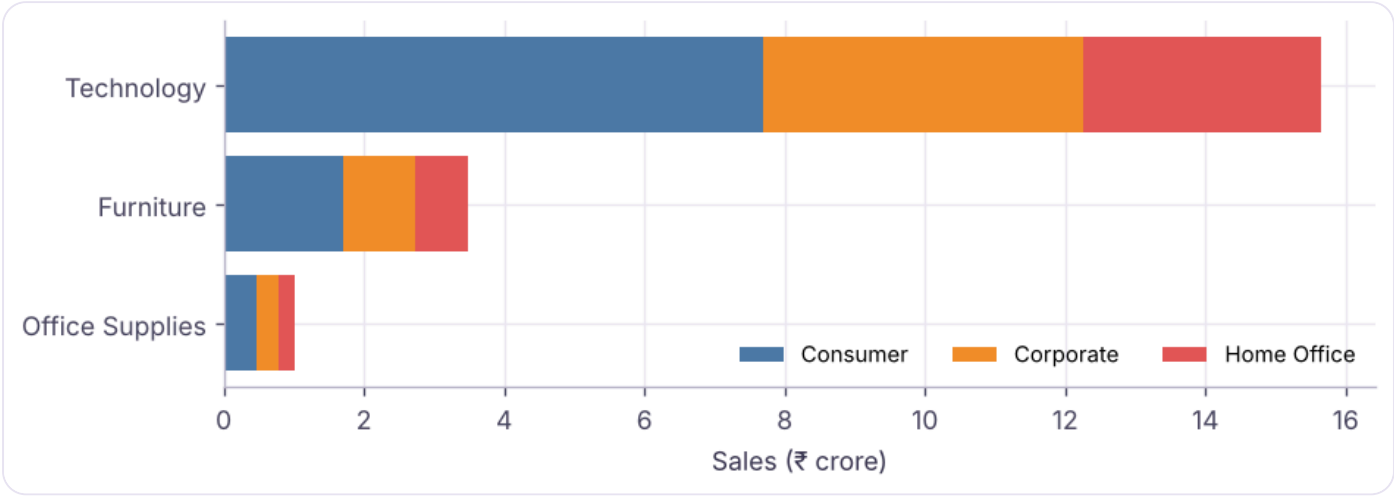
Python

```
seg = pd.crosstab(df.category, df.segment, values=df.sales, aggfunc="sum") / 1e7
seg.round(2)
```

OUTPUT

segment	Consumer	Corporate	Home Office
category			
Furniture	1.70	1.03	0.75
Office Supplies	0.47	0.32	0.23
Technology	7.69	4.56	3.39

3 rows × 3 columns



Stacked bar: Sales by Category, coloured by Segment (₹ crore).

Line and area charts

Put a **continuous** date (green MONTH(Order Date)) on Columns and Sales on Rows for one timeline. Put **YEAR** on Color and a **discrete** MONTH on Columns to compare the same months across years (seasonality).

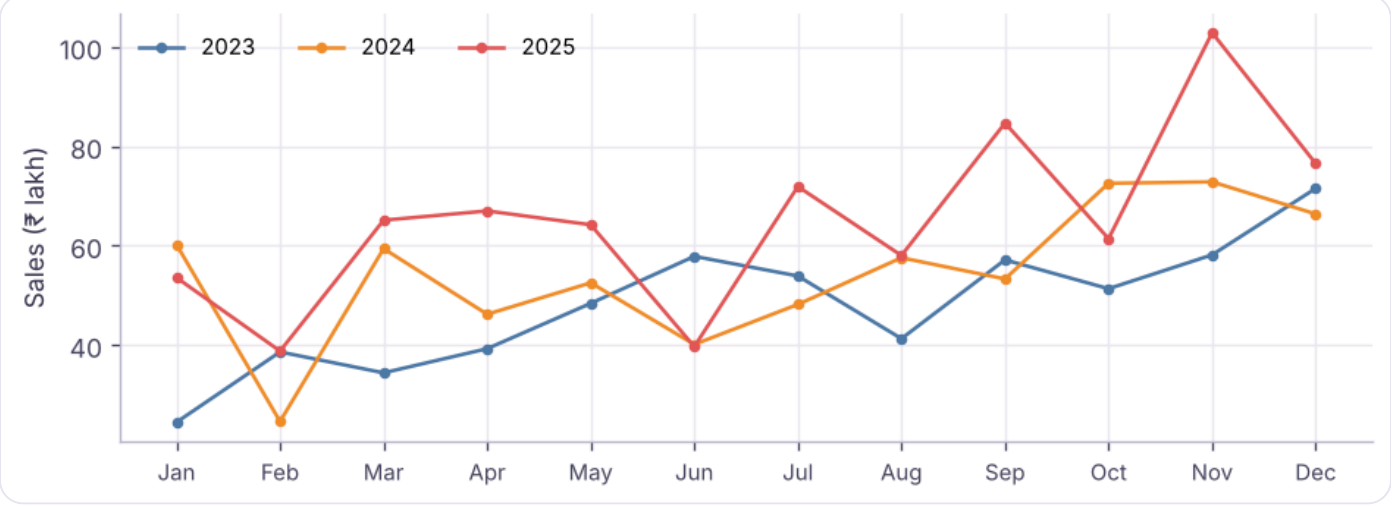
Python

```
pv = df.pivot_table(index=df.order_date.dt.month, columns="year", values="sales",
aggfunc="sum") / 1e5
pv.index.name = "month"
pv.round(1).head(6)
```

OUTPUT

year	2023	2024	2025
month			
1	24.5	60.2	53.6
2	38.6	24.6	38.8
3	34.4	59.4	65.1
4	39.3	46.2	67.0
5	48.4	52.5	64.2
6	57.9	40.1	39.7

6 rows × 3 columns



Discrete month on Columns, Year on Color: each year is a line, so seasonality is easy to compare.

An **area chart** is the same view with Mark type = Area; stack a dimension on Color to show how the total is built up.

Part-to-whole: pie, donut, treemap

- **Pie**: Mark type Pie, dimension on Color, measure on **Angle**. Use for 2–5 slices only.
- **Donut**: two pies on a dual axis. Put `AVG(0)` (type `avg(0)` on Rows) twice, make the second smaller and white, turn on dual axis, and put the total on the inner label.
- **Treemap**: dimension on **Detail/Label**, measure on **Size** and **Color**. Good for many categories (e.g. the 13 sub-categories).

Python

```
sub = (df.groupby("sub_category").agg(sales=("sales", "sum"), profit=("profit", "sum")) /
1e5).sort_values("sales", ascending=False)
sub["margin %"] = (sub.profit / sub.sales * 100).round(1)
sub.round(1).head(8)
```

OUTPUT

	sales	profit	margin %
sub_category			
Copiers	787.5	84.4	10.7
Machines	425.9	51.9	12.2
Phones	319.2	36.6	11.5
Tables	140.5	-8.2	-5.8
Bookcases	112.7	-4.2	-3.7
Chairs	80.9	3.4	4.2
Appliances	63.3	9.4	14.9
Accessories	31.8	3.7	11.6

8 rows × 3 columns

Highlight table (heat map)

Category on Rows, YEAR on Columns, Sales on **Text** and **Color**, Mark type **Square**. Tableau shades each cell, so high and low cells stand out without reading every number.

Distribution: histogram and box plot

STEP BY STEP **Histogram of order value**

1. Right-click **Sales** → Create → **Bins**; set the bin size (e.g. ₹10,000). A new dimension "Sales (bin)" appears.
2. Drag **Sales (bin)** to Columns and **CNT(Sales)** (or Number of Records) to Rows.
3. Or select Sales and pick the histogram in Show Me: it creates the bins for you.

Python

```
bins = pd.cut(df.sales, bins=[0, 5000, 10000, 20000, 50000, 100000, df.sales.max()],
right=False)
print(bins.value_counts().sort_index())
print("median line value: ₹", round(df.sales.median()), "| mean: ₹",
round(df.sales.mean()))
```

OUTPUT

```
sales
[0.0, 5000.0)           2530
[5000.0, 10000.0)        829
[10000.0, 20000.0)       896
[20000.0, 50000.0)       942
[50000.0, 100000.0)      546
[100000.0, 619290.0)     579
Name: count, dtype: int64
median line value: ₹ 8430 | mean: ₹ 31854
```

The mean is well above the median: a few large orders pull it up (right skew). A **box plot** (Show Me → box-and-whisker) shows the median, quartiles and outliers per category at a glance.

Relationship: scatter plot

Put SUM(Sales) on Columns, SUM(Profit) on Rows and **Sub-Category** on Detail: one dot per sub-category. Add **Analytics pane → Trend Line** (linear) and hover it to read R^2 and the p-value.

Python

```
s = df.groupby("sub_category")[["sales", "profit", "discount"]].agg({"sales": "sum",
"profit": "sum", "discount": "mean"})
print("correlation sales vs profit across sub-categories:", round(s.sales.corr(s.profit),
2))
print("correlation avg discount vs profit margin:", round(s.discount.corr(s.profit /
s.sales), 2))
```

OUTPUT

```
correlation sales vs profit across sub-categories: 0.95
correlation avg discount vs profit margin: -0.88
```

Maps

1. Make sure **State** has the geographic role *State/Province* (and the country is India: Map → Edit Locations).
2. Double-click State: Tableau adds Latitude/Longitude (generated) and draws a **symbol map**. Change Mark type to **Map** for a **filled (choropleth) map**.
3. Put Sales on Color for a filled map, or on Size for a symbol map. Use Profit on Color with a diverging palette (orange-blue) to show loss-making states.

Python

```
st = (df.groupby("state")[["sales", "profit"]].sum() / 1e5).round(1).sort_values("sales",
ascending=False)
st.head(6)
```

OUTPUT

	sales	profit
state		
Maharashtra	256.7	24.4
Delhi	156.6	17.3
Kerala	147.1	12.6
Madhya Pradesh	144.1	12.7
West Bengal	138.9	14.6
Karnataka	136.3	11.8

6 rows × 2 columns

> **TIP** **Indian maps**
>
> Tableau's built-in geocoding knows Indian states and many districts and cities. If a name does not match (old spellings, "Orissa" vs "Odisha"), use **Map → Edit Locations** to match it, or join a custom latitude/longitude file.

Dual-axis and combo charts

Show two measures with different scales, e.g. Sales as bars and Profit margin as a line:

1. Continuous MONTH(Order Date) on Columns; SUM(Sales) and the margin calculation on Rows.
2. Right-click the second pill → **Dual Axis**. Set the Marks card for Sales to **Bar** and for margin to **Line**.
3. Right-click the right axis → **Synchronize Axis** only when both measures share units (Sales and Profit); never synchronise ₹ with %.

A **blended axis** (drag a second measure onto the *existing* axis until the double-ruler icon shows) puts two measures with the same unit on one axis, using Measure Names on Color.

Bullet chart (actual vs target)

Select a measure and a target (e.g. Sales and a target field) → Show Me → **Bullet graph**. The bar is the actual, the reference line is the target, and the shaded bands are % of target (60%, 80%). If the bars and lines come out the wrong way round, right-click the axis → **Swap Reference Line Fields**.

Pareto chart (80/20)

> **STEP BY STEP** **Build a Pareto of products**
>
> 1. Product Name on Columns sorted by SUM(Sales) descending; SUM(Sales) on Rows as bars.
> 2. Drag SUM(Sales) to Rows again → Quick Table Calculation → **Running Total**, then Edit Table Calculation → *Add secondary calculation* → **Percent of Total**.
> 3. Make it dual axis; set the second mark to Line; add a reference line at 80%.
> 4. Optional: replace Product Name on Columns with a table calc `INDEX()/SIZE()` so the x-axis reads "% of products".

Python

```python
p = df.groupby("product_name").sales.sum().sort_values(ascending=False)
cum = p.cumsum() / p.sum()
n80 = int((cum < .8).sum()) + 1
print(f"{n80} of {len(p)} products ({n80 / len(p):.0%}) make 80% of sales")
```

OUTPUT

```
21 of 78 products (27%) make 80% of sales
```

Sales are concentrated: about a quarter of the products bring in 80% of sales, close to the classic 80/20 rule. Those top products deserve the best stock availability and marketing. A Pareto chart shows how concentrated a business really is, so you can judge whether a "focus on the top products" strategy makes sense.

Summary

KEY TAKEAWAYS

- Pick the chart from the question: compare, trend, part-to-whole, distribution, relationship, geography.
- Continuous dates draw timelines; discrete dates compare periods.
- Bins create histograms; the Analytics pane adds trend lines, reference lines and box plots.
- Assign geographic roles before mapping; use filled maps for rates and symbol maps for totals.
- Dual axis combines two marks (bar + line); synchronise only when units match.

Quick quiz

6 questions · tap an answer to check it

1. **Which chart best shows how a single order-value variable is distributed?**

- **A** Histogram
- **B** Pie chart
- **C** Treemap
- **D** Line chart

Answer: A. Histograms show distributions; create bins first.

2. **How do you get Profit as a line on top of Sales bars with its own scale?**

- **A** Put both on Rows, use Dual Axis, set one mark to Bar and the other to Line
- **B** Put Profit on Color
- **C** Use a blended axis
- **D** Synchronise the axes

Answer: A. Dual axis gives two independent axes and separate mark types.

3. **Which is the right way to show 13 sub-categories as a part-to-whole?**

- **A** Treemap or sorted bar chart
- **B** Pie chart with 13 slices
- **C** Scatter plot
- **D** Line chart

Answer: A. Pies stop being readable beyond about five slices.

4. You want each year as its own line to compare seasonality. What goes on Columns?

- **A** Discrete MONTH(Order Date)
- **B** Continuous MONTH(Order Date)
- **C** YEAR(Order Date) only
- **D** Sales

Answer: A. Discrete month gives Jan–Dec headers; Year goes on Color.

5. What does the Pareto line show?

- **A** The running total of sales as a percent of the grand total
- **B** Profit margin
- **C** Average sales
- **D** The number of orders

Answer: A. Running total + percent of total secondary calculation.

6. When should you synchronise a dual axis?

- **A** When both measures share the same unit and scale makes sense
- **B** Always
- **C** When one is in ₹ and the other in %
- **D** Never

Answer: A. Synchronising ₹ with % distorts the chart.

EXERCISE

Practice

Build: (1) a side-by-side bar of Sales by Region and Segment; (2) a filled map of Profit by State with a diverging palette; (3) a histogram of Sales with ₹10,000 bins; (4) a dual-axis chart of monthly Sales (bars) and Profit (line); (5) a Pareto chart of sub-categories with an 80% reference line.

Show solution (worked answer)

Use the steps in each section. For the map, set Profit on Color → Edit Colors → Orange-Blue Diverging, centre 0. For the Pareto use Running Total + Percent of Total, compute using Sub-Category, then Analytics → Reference Line → Constant 0.8 on the line axis.

INTERVIEW CORNER

Questions you may be asked

Q: Dual axis vs blended axis? A dual axis gives two independent axes (left and right) and allows different mark types; a blended axis puts several measures on one shared axis (same unit) using Measure Names/Values.

Q: How do you build a Pareto chart in Tableau? Sort the dimension by the measure, show bars, add the measure again with a Running Total table calculation and a secondary Percent of Total, make it dual axis as a line, and add an 80% reference line.

Q: Filled map or symbol map? Filled maps suit rates and ratios (profit margin by state) because area does not scale with value; symbol maps with size suit totals (sales by city).

TABLEAU · LESSON 3

Calculations in Tableau: Calculated Fields, Table Calculations and LOD Expressions

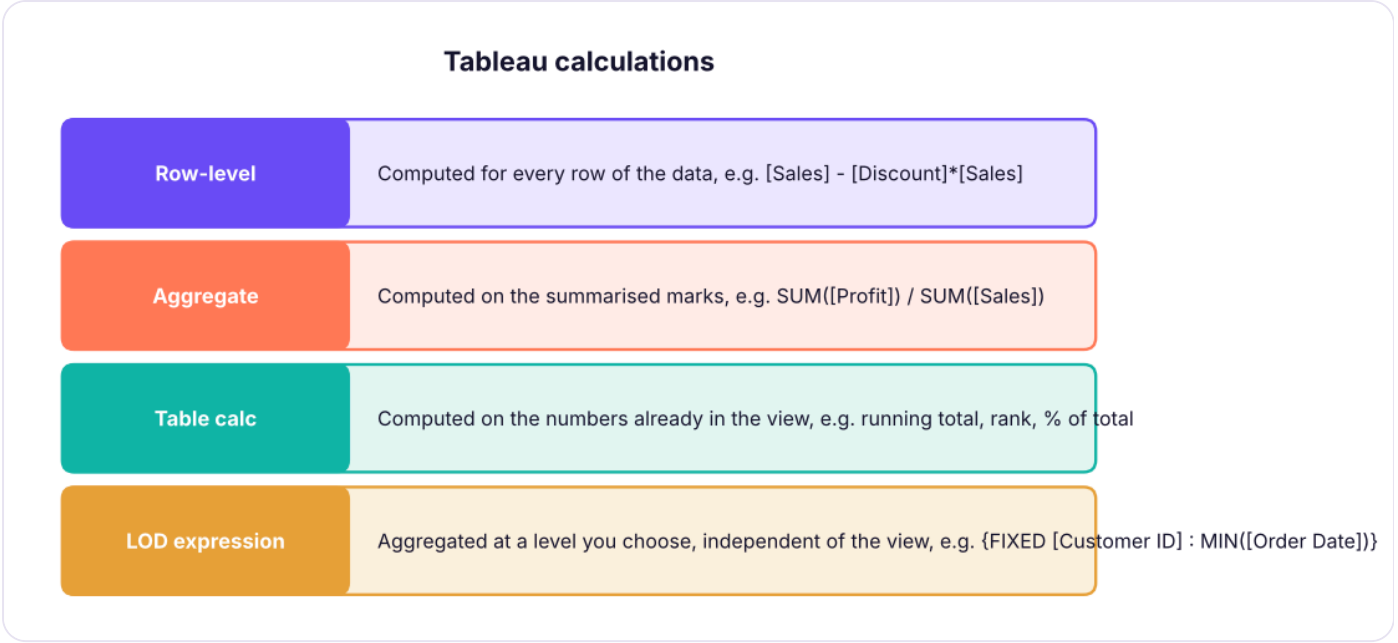
Write calculated fields (row-level and aggregate), use string, date, number and logical functions, handle nulls, then master table calculations (running total, percent of total, difference and growth, rank, moving average, and Compute Using) and Level of Detail expressions (FIXED, INCLUDE, EXCLUDE), with every result checked against pandas.

✦ Tableau Desktop / Tableau Public 2026 | Intermediate | ⏱ 150 min

By the end of this lesson you can

- Write row-level and aggregate calculated fields and know which one a question needs
- Use IF/ELSEIF, CASE, IIF, ZN, IFNULL, DATEDIFF, DATETRUNC and string functions
- Apply quick table calculations and control them with Compute Using
- Write FIXED, INCLUDE and EXCLUDE LOD expressions and explain where they sit in the order of operations
- Solve classic interview problems: profit margin, customer first-purchase cohorts and percent of total

Three kinds of calculation



Where each kind of calculation is computed.

Create one with **Analysis → Create Calculated Field** (or right-click a field → Create → Calculated Field). Field names go in square brackets; comments start with `//`.

Python

```
import pandas as pd, numpy as np
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date", "ship_date"])
print(df[["sales", "profit", "discount"]].describe().round(1).loc[["mean", "min", "max"]])
```

OUTPUT

```
          sales    profit  discount
mean    31854.1    2902.4       0.1
min        42.0  -75873.8       0.0
max    619290.0  149424.9       0.4
```

Row-level vs aggregate: the profit-margin trap

The most common beginner bug in Tableau (and Power BI) is averaging ratios.

Calculated field	Formula	What it means
Margin (row)	`[Profit] / [Sales]`	Margin of each order line; in the view it becomes SUM or AVG of those ratios: **wrong**
Margin (aggregate)	`SUM([Profit]) / SUM([Sales])`	Total profit ÷ total sales for whatever the mark represents: **right**

Python

```python
by_cat = df.groupby("category").apply(lambda g: pd.Series({
    "aggregate margin %": g.profit.sum() / g.sales.sum() * 100,
    "average of row margins %": (g.profit / g.sales).mean() * 100}), include_groups=False)
by_cat.round(2)
```

OUTPUT

	aggregate margin %	average of row margins %
category		
Furniture	-2.35	-2.38
Office Supplies	14.95	13.76
Technology	11.29	9.95

3 rows × 2 columns

The two columns differ because the average of row ratios gives a ₹500 order the same weight as a ₹5 lakh order. Format the aggregate field as a percentage (right-click → Default Properties → Number Format → Percentage).

Useful functions

Purpose	Tableau	Example
Logic	`IF … THEN … ELSEIF … ELSE … END`	`IF SUM([Profit]) < 0 THEN "Loss" ELSE "Profit" END`
Logic	`CASE [Region] WHEN "South" THEN … END`, `IIF(test, then, else)`	Short mappings
Nulls	`ZN([Profit])`, `IFNULL([x], 0)`, `ISNULL([x])`	Replace nulls with 0
Dates	`DATEDIFF('day', [Order Date], [Ship Date])`, `DATETRUNC('month', [Order Date])`, `DATEPART('weekday', …)`, `DATEADD`, `TODAY()`	Shipping days, month start
Strings	`LEFT`, `RIGHT`, `MID`, `SPLIT([Customer Name], " ", 1)`, `CONTAINS`, `UPPER`, `TRIM`, `REPLACE`	First name
Numbers	`ROUND`, `ABS`, `CEILING`, `FLOOR`, `INT`	Rounding
Type	`STR()`, `INT()`, `FLOAT()`, `DATE()`	Conversions
Aggregates	`SUM`, `AVG`, `COUNT`, `COUNTD`, `MIN`, `MAX`, `MEDIAN`, `ATTR`	`COUNTD([Order ID])` = orders

Python

```python
df["ship_days"] = (df.ship_date - df.order_date).dt.days          # DATEDIFF('day', [Order Date], [Ship Date])
df["first_name"] = df.customer_name.str.split(" ").str[0]         # SPLIT([Customer Name], " ", 1)
df["order_size"] = np.select([df.sales ⩾ 50000, df.sales ⩾ 10000], ["Large", "Medium"], "Small")   # IF/ELSEIF
print(df.groupby("ship_mode").ship_days.mean().round(1).sort_values())
print(df.order_size.value_counts().to_dict())
```

OUTPUT

```
ship_mode
Same Day          0.0
First Class       1.5
Second Class      3.0
Standard Class    5.6
Name: ship_days, dtype: float64
{'Small': 3359, 'Medium': 1838, 'Large': 1126}
```

Table calculations

A table calculation works on the **numbers already in the view**, so its result depends on the layout. Right-click a measure → **Quick Table Calculation**:

Quick table calc	Use
Running Total	Cumulative sales through the year
Difference / Percent Difference	Month-on-month or year-on-year change
Percent of Total	Share of each category
Rank	Top products
Percentile	Position in a distribution
Moving Average	Smooth noisy monthly data (Edit → previous 2 values + current)
YTD Total / YTD Growth / YoY Growth	Financial reporting

Compute Using (addressing and partitioning)

Compute Using tells Tableau the direction of the calculation. Fields it computes *along* are **addressing** fields; the remaining fields **partition** the view into separate calculations. Example: Category on Rows, MONTH on Columns, Running Total of Sales:

- Compute Using **Table (across)** → running total along months, restarting for each category (partitioned by Category).
- Compute Using **Table (down)** → running total down the categories within each month.

Python

```
m = df[df.order_date.dt.year == 2025].assign(month=lambda x:
x.order_date.dt.to_period("M"))
t = m.pivot_table(index="category", columns="month", values="sales", aggfunc="sum") / 1e5
running = t.cumsum(axis=1)              # Table (across): restarts per category
print(running.iloc[:, [0, 1, 2, -1]].round(1))
share = t.sum(axis=1) / t.values.sum() * 100   # Percent of Total, Table (down)
print((share).round(1).to_dict())
```

OUTPUT

```
month            2025-01  2025-02  2025-03  2025-12
category
Furniture           12.9     16.8     28.7    127.6
Office Supplies      2.6      4.8      9.4     39.7
Technology          38.0     70.8    119.4    616.5
{'Furniture': 16.3, 'Office Supplies': 5.1, 'Technology': 78.7}
```

Common table-calculation functions you write yourself: `RUNNING_SUM(SUM([Sales]))`, `WINDOW_AVG(SUM([Sales]), -2, 0)` (3-month moving average), `LOOKUP(SUM([Sales]), -1)` (previous value), `RANK(SUM([Sales]))`, `INDEX()`, `SIZE()`, `TOTAL(SUM([Sales]))`.

YoY growth as a calculation: `(SUM([Sales]) - LOOKUP(SUM([Sales]), -1)) / ABS(LOOKUP(SUM([Sales]), -1))` with YEAR(Order Date) in the view.

Python

```python
y = df.groupby(df.order_date.dt.year).sales.sum()
print(((y / y.shift(1) - 1) * 100).round(1).dropna().to_dict(), "% YoY")
```

OUTPUT

```
{2024: 13.5, 2025: 19.8} % YoY
```

Level of Detail (LOD) expressions

LODs compute an aggregate at a level **you** choose, regardless of the dimensions in the view.

Syntax	Meaning	Example
`{FIXED [Dim] : AGG([Measure])}`	Exactly at [Dim], ignoring the view (but after context filters)	`{FIXED [Customer ID] : MIN([Order Date])}` first purchase
`{INCLUDE [Dim] : AGG(...)}`	View's dimensions **plus** [Dim]	Average sales **per customer** by region
`{EXCLUDE [Dim] : AGG(...)}`	View's dimensions **minus** [Dim]	Region total next to each category, for % of region
`{ : SUM([Sales])}`	Table-scoped (grand total)	% of grand total without table calcs

FIXED: customer cohorts by first-purchase year

`[First Purchase] = {FIXED [Customer ID] : MIN([Order Date])}`, then `YEAR([First Purchase])` on Rows and `COUNTD([Customer ID])` gives new customers per year: an acquisition cohort.

Python

```python
first = df.groupby("customer_id").order_date.min().rename("first_purchase")
d2 = df.merge(first, on="customer_id")
cohort = d2.groupby([d2.first_purchase.dt.year.rename("cohort"),
d2.order_date.dt.year.rename("order year")]).customer_id.nunique().unstack(fill_value=0)
cohort
```

OUTPUT

order year	2023	2024	2025
cohort			
2023	540	397	415
2024	0	193	156
2025	0	0	52

3 rows × 3 columns

Read it row by row: customers acquired in 2023 (first row) and how many of them bought again in 2024 and 2025. This retention table is a favourite interview task.

INCLUDE: average sales per customer by region

`AVG({INCLUDE [Customer ID] : SUM([Sales])})` with Region in the view first sums sales per customer *within* each region, then averages those customer totals.

Python

```python
per_cust = df.groupby(["region", "customer_id"]).sales.sum().groupby("region").mean() / 1e3
print((per_cust).round(1).rename("avg sales per customer (₹ thousand)"))
```

OUTPUT

```
region
East     238.4
North    263.5
South    261.9
West     262.4
Name: avg sales per customer (₹ thousand), dtype: float64
```

EXCLUDE: share of the region

With Region and Category in the view, `SUM([Sales]) / SUM({EXCLUDE [Category] : SUM([Sales])})` gives each category's share of its region.

Python

```python
rc = df.groupby(["region", "category"]).sales.sum()
(rc / rc.groupby(level=0).transform("sum") * 100).round(1).unstack()
```

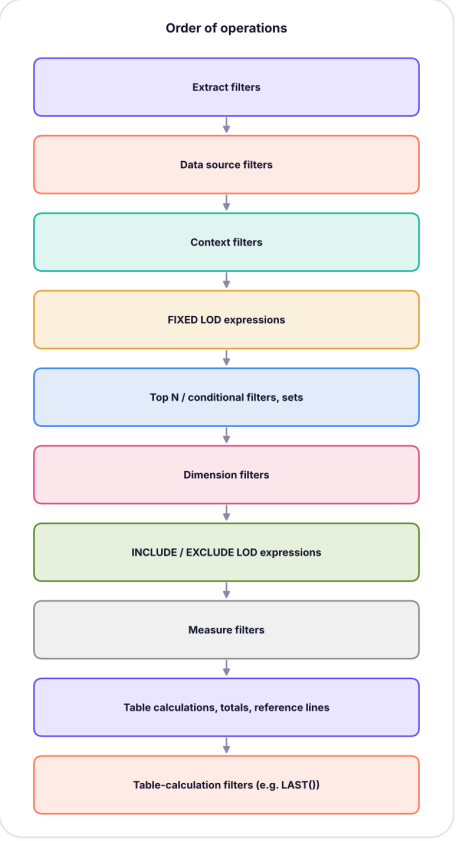
OUTPUT

category	Furniture	Office Supplies	Technology
region			
East	17.1	5.9	77.1
North	16.3	4.5	79.2
South	17.8	4.6	77.6
West	18.0	5.2	76.8

4 rows × 3 columns

Order of operations

Filters and calculations run in a fixed order. This explains most "why did my number not change?" questions:



Tableau's order of operations (simplified). FIXED LODs ignore normal dimension filters unless the filter is added to context.

TIP **Make a FIXED LOD respect a filter**

A FIXED LOD is computed before dimension filters, so filtering Region does not change `{FIXED [Customer ID] : MIN([Order Date])}`. To make it respect the filter, right-click the filter → **Add to Context**.

Summary

KEY TAKEAWAYS

- Row-level calculations run per row; aggregate calculations run on marks: compute margins as SUM/SUM.
- Table calculations work on the view's numbers; Compute Using sets the direction (addressing) and the restart points (partitioning).
- FIXED sets an exact level; INCLUDE adds a level; EXCLUDE removes one.
- FIXED LODs run before dimension filters; use context filters when they must respect a filter.

Quick quiz

6 questions · tap an answer to check it

1. Which formula gives the correct profit margin for a category?

A SUM([Profit]) / SUM([Sales])

B AVG([Profit] / [Sales])

C [Profit] / [Sales] summed

D MAX([Profit]) / MAX([Sales])

Answer: A. Divide the totals; do not average row ratios.

2. You want each customer's first order date regardless of the view. Which expression?

- **A** {FIXED [Customer ID] : MIN([Order Date])}
- **B** {INCLUDE [Customer ID] : MIN([Order Date])}
- **C** RUNNING_MIN([Order Date])
- **D** MIN([Order Date])

Answer: A. FIXED computes at exactly the customer level.

3. A running total restarts for every category. Why?

- **A** Category is partitioning the calculation (Compute Using is across months)
- **B** The data is wrong
- **C** It is a FIXED LOD
- **D** The filter is in context

Answer: A. Fields not used for addressing partition the calculation.

4. Your FIXED LOD ignores the Region filter. How do you fix it?

- **A** Add the Region filter to context
- **B** Use a measure filter
- **C** Change to a table calculation
- **D** Hide the field

Answer: A. Context filters run before FIXED LODs.

5. **Which table-calculation function returns the previous value in the partition?**

A LOOKUP(SUM([Sales]), -1)

B INDEX()

C SIZE()

D TOTAL(SUM([Sales]))

Answer: A. LOOKUP with offset −1 looks one mark back.

6. **Which expression gives each category's share of its region with Region and Category in the view?**

A SUM([Sales]) / SUM({EXCLUDE [Category] : SUM([Sales])})

B SUM([Sales]) / SUM([Sales])

C {FIXED [Category] : SUM([Sales])}

D RANK(SUM([Sales]))

Answer: A. EXCLUDE removes Category, leaving the region total.

EXERCISE

Practice

Create: (1) a "Profit status" field that labels each sub-category Loss or Profit and colour a bar chart with it; (2) a 3-month moving average of monthly sales; (3) a cohort table of customers by first-purchase year and order year; (4) a percent-of-region field with EXCLUDE; (5) a Top-N rank of products by profit.

Show solution (worked answer)

(1) `IF SUM([Profit]) < 0 THEN "Loss" ELSE "Profit" END` on Color. (2) `WINDOW_AVG(SUM([Sales]), -2, 0)` with continuous MONTH on Columns. (3) `{FIXED [Customer ID] : MIN([Order Date])}`, YEAR of it on Rows, YEAR(Order Date) on Columns, COUNTD([Customer ID]) on Text. (4) As in the lesson. (5) `RANK(SUM([Profit]))` computed along Product Name, then filter rank ≤ 10 (a table-calculation filter).

INTERVIEW CORNER

Questions you may be asked

Q: Explain FIXED, INCLUDE and EXCLUDE. FIXED aggregates at exactly the listed dimensions, ignoring the view; INCLUDE adds dimensions to the view's level (finer), then the result is re-aggregated; EXCLUDE removes dimensions from the view's level (coarser). FIXED runs before dimension filters; INCLUDE and EXCLUDE run after them.

Q: Table calculation vs LOD expression? A table calculation runs last, on the aggregated numbers in the view, and depends on layout (Compute Using). An LOD runs in the data source query at a level you define and can be used as a dimension, e.g. for cohorts.

Q: Why might a running total be wrong? Usually Compute Using: it is computing along the wrong field, or a dimension is partitioning it. Check Edit Table Calculation → Specific Dimensions and the sort order.

TABLEAU · LESSON 4

Filters, Parameters, Sets, Groups and Hierarchies in Tableau

Control what a view shows and let users interact with it. Learn every filter type and where it sits in the order of operations, context filters and Top N, groups, hierarchies, bins, sets (fixed, conditional, top) and combined sets, and parameters for dynamic Top N, measure swapping and what-if analysis.

✦ Tableau Desktop / Tableau Public 2026 | Intermediate | ⏱ 120 min

By the end of this lesson you can

- Use dimension, measure, date, context, data-source and table-calculation filters correctly
- Build hierarchies and groups, and explain groups vs sets
- Create fixed, conditional and Top N sets and use IN/OUT in calculations
- Create parameters for Top N, measure selection and what-if discount scenarios
- Combine filters, sets and parameters into interactive views

Filters

Filter	How	Notes
Dimension filter	Drag a dimension to Filters → General / Wildcard / Condition / Top tabs	e.g. Region = South
Measure filter	Drag a measure to Filters → choose aggregation → range	e.g. SUM(Profit) < 0 (loss-making items)
Date filter	Relative dates (last 3 months), range of dates, or discrete years/months	Relative dates update automatically
Context filter	Right-click a filter → Add to Context (grey pill)	Runs first; Top N and FIXED LODs respect it
Data source filter	Data Source page → Filters → Add	Applies to every sheet; good for security and speed
Extract filter	When creating an extract	Shrinks the extract itself
Table-calc filter	Filter on a table calc such as `LAST()=0` or rank	Hides marks without changing other calculations

Show Filter (right-click the pill → Show Filter) puts an interactive control on the sheet: single value list, dropdown, multiple values, slider, wildcard match. **Apply to Worksheets → All Using This Data Source / Selected Worksheets** makes one filter drive several sheets on a dashboard.

Python

```python
import pandas as pd, numpy as np
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
loss = df.groupby("sub_category").profit.sum().sort_values()
print("Loss-making sub-categories (measure filter SUM(Profit) < 0):")
print((loss[loss < 0] / 1e5).round(1))
```

OUTPUT

```
Loss-making sub-categories (measure filter SUM(Profit) < 0):
sub_category
Tables      -8.2
Bookcases   -4.2
Name: profit, dtype: float64
```

Top N and context

Top N is computed **after** context filters but **before** ordinary dimension filters. So "Top 5 products **in the South**" needs the Region filter in context; otherwise Tableau finds the national Top 5 first and

then keeps only those that happen to be in the South.

Python

```python
top_all = df.groupby("product_name").sales.sum().nlargest(5)
south = df[df.region == "South"]
top_south = south.groupby("product_name").sales.sum().nlargest(5)
print("National top 5:", list(top_all.index))
print("South top 5 (Region in context):", list(top_south.index))
print("Overlap:", len(set(top_all.index) & set(top_south.index)), "of 5")
```

OUTPUT

```
National top 5: ['Orbit Copier 746', 'Apex Copier 347', 'Vertex Copier 262', 'Zenith Copier 719', 'Zenith Machine 401']
South top 5 (Region in context): ['Orbit Copier 746', 'Vertex Copier 262', 'Zenith Machine 401', 'Apex Copier 347', 'Pioneer Copier 366']
Overlap: 4 of 5
```

Hierarchies, groups and bins

- **Hierarchy**: drag Sub-Category onto Category in the Data pane → name it "Product hierarchy". A **+** on the pill drills down. Date fields have a built-in Year → Quarter → Month → Day hierarchy.
- **Group**: select marks or right-click a field → Create → Group, e.g. group Indian states into "North-East". A group is a new dimension; its members are fixed.
- **Bins**: equal-width buckets of a measure (Lesson 2).

Sets

A **set** is a subset of a dimension that is either **IN** or **OUT**.

Set type	Created by	Example
Fixed (manual)	Select members	Six key accounts
Conditional	Condition tab, e.g. SUM(Profit) > 0	Profitable customers
Top	Top tab, e.g. Top 10 by SUM(Sales)	Top 10 customers
Combined	Two sets → Create Combined Set (union, intersection, difference)	Top customers who are also unprofitable

Use a set on **Color** to show IN vs OUT, on Filters to keep only members, or in calculations: `IF [Top Customers] THEN [Sales] END`. **Set actions** (Lesson 6) let users change a set by clicking marks.

Python

```python
cust = df.groupby("customer_id").agg(sales=("sales", "sum"), profit=("profit", "sum"))
top10 = cust.sales.nlargest(10).index
in_set = cust.index.isin(top10)
print(f"Top 10 customers (IN): {cust.sales[in_set].sum() / cust.sales.sum():.1%} of sales")
unprofitable = cust.index[cust.profit < 0]
print("Customers in Top 10 AND unprofitable (combined set, intersection):", len(set(top10)
& set(unprofitable)))
print("All unprofitable customers:", len(unprofitable), "of", len(cust))
```

OUTPUT

```
Top 10 customers (IN): 5.3% of sales
Customers in Top 10 AND unprofitable (combined set, intersection): 1
All unprofitable customers: 142 of 785
```

NOTE **Groups vs sets**

A **group** combines members into new categories (many buckets, fixed). A **set** splits a dimension into IN/OUT, can be dynamic (conditional or Top N) and can be changed by users with set actions.

Parameters

A **parameter** is a single user-controlled value (number, string, date or list) that does nothing until a calculation, filter or reference line uses it. Right-click → **Show Parameter**.

1. Dynamic Top N

1. Create parameter **Top N** (integer, range 5–20, step 5).
2. Product Name filter → Top tab → By field → Top **[Top N]** by SUM(Sales).
3. Show the parameter: users choose 5, 10, 15 or 20.

2. Measure swap

```
// Parameter "Choose measure": list of strings Sales, Profit, Quantity
CASE [Choose measure]
WHEN "Sales" THEN SUM([Sales])
WHEN "Profit" THEN SUM([Profit])
WHEN "Quantity" THEN SUM([Quantity])
END
```

Put this calculated field on the axis; one chart now shows whichever measure the user picks. Use a dynamic title: `<Parameters.Choose measure> by Region`.

3. What-if: extra discount

```
// Parameter "Extra discount" : float 0.00–0.20
[Sales] * (1 - [Extra discount])                    // projected sales
[Profit] - [Sales] * [Extra discount]               // projected profit
```

Python

```python
for extra in (0, .05, .10):
    proj_profit = (df.profit - df.sales * extra).sum()
    print(f"extra discount {extra:.0%}: projected profit ₹{proj_profit / 1e7:.2f} crore")
```

OUTPUT

```
extra discount 0%: projected profit ₹1.84 crore
extra discount 5%: projected profit ₹0.83 crore
extra discount 10%: projected profit ₹-0.18 crore
```

With margins around 9%, a further 10% discount turns the whole business loss-making. That is exactly the kind of insight a what-if parameter makes obvious to a manager.

2026 UPDATE **Dynamic parameters and zone visibility**

Parameters can refresh their list and default value from a field when the workbook opens (*When workbook opens → value of field*). Combined with **dynamic zone visibility** (show or hide a dashboard container based on a boolean field or parameter), they power menus and drill-in panels without bookmarks.

Summary

KEY TAKEAWAYS

- Pick the right filter level: data source → context → dimension → measure → table calc.
- Top N and FIXED LODs ignore normal dimension filters unless those filters are in context.
- Groups make fixed buckets; sets make IN/OUT subsets that can be dynamic and combined.
- Parameters are user inputs: wire them into filters, calculations, titles and reference lines.

Quick quiz

6 questions · tap an answer to check it

1. You need "Top 10 products within the selected region". What must you do?

A Add the Region filter to context

B Use a measure filter

C Use a group

D Hide the other regions

Answer: A. Top N runs before normal dimension filters.

2. Which object can a user change by clicking marks on a dashboard?

A A set (through a set action)

B A group

C A bin

D A hierarchy

Answer: A. Set actions update set membership.

3. **A parameter on its own does nothing. Why?**

- **A** It must be referenced by a calculation, filter or reference line
- **B** It needs an extract
- **C** It must be added to context
- **D** It only works on Tableau Server

Answer: A. Parameters are inputs that something must use.

4. **Which filter applies to every sheet using the data source and is good for row-level restrictions?**

- **A** Data source filter
- **B** Table-calculation filter
- **C** Measure filter
- **D** Page shelf

Answer: A. Data source filters run near the start for all sheets.

5. **Which set type keeps the 10 largest customers by sales, updating as data changes?**

- **A** Top set
- **B** Fixed set
- **C** Group
- **D** Combined set

Answer: A. The Top tab makes a dynamic set.

6. Which calculation swaps the measure shown in a chart?

- **A** A CASE on a string parameter returning SUM([Sales]), SUM([Profit]) or SUM([Quantity])
- **B** A FIXED LOD
- **C** A group
- **D** A data source filter

Answer: A. The parameter value chooses the branch.

EXERCISE **Practice**

Build: (1) a dynamic Top N products chart with a parameter and Region in context; (2) a Top 20 customers set shown on Color in a scatter of Sales vs Profit; (3) a measure-swap chart with a dynamic title; (4) a what-if discount parameter showing projected profit by Category.

Show solution (worked answer)

Follow the three parameter recipes. For (2): Customer Name on Detail, SUM(Sales) and SUM(Profit) on the axes, the Top 20 set on Color. For (4): a "Projected profit" calculated field `SUM([Profit]) - SUM([Sales]) * [Extra discount]` by Category with the parameter shown as a slider.

INTERVIEW CORNER **Questions you may be asked**

Q: Difference between a set and a group? A group merges members into new categories and is static; a set divides members into IN/OUT, can be conditional or Top N, can be combined with other sets and changed through set actions.

Q: What is a context filter and when do you use it? A filter that runs before other dimension filters, Top N filters and FIXED LODs. Use it for dependent Top N, to make FIXED LODs respect a filter, and to speed up large workbooks by reducing the data early.

Q: Parameter vs filter? A filter removes data; a parameter is a single input value that you can use anywhere: in calculations, Top N, reference lines, titles, or to swap measures and dimensions.

TABLEAU · LESSON 5

Data Modelling in Tableau: Relationships, Joins, Unions, Blending, Extracts and Tableau Prep

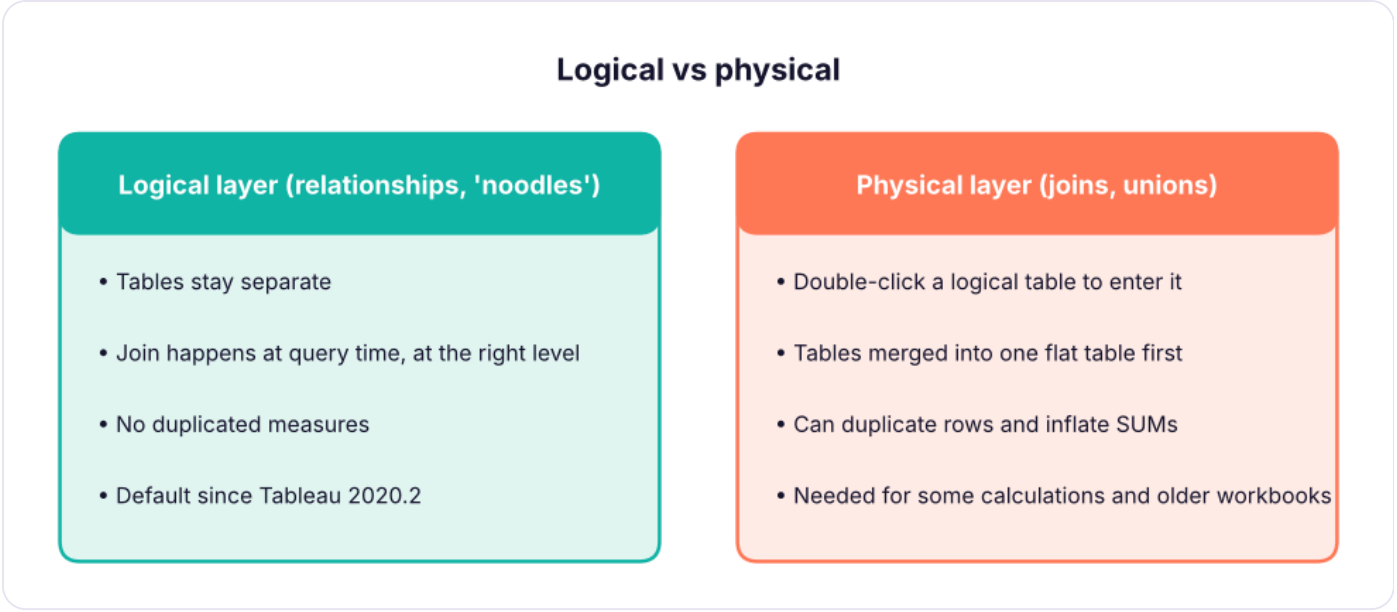
Combine and shape data in Tableau. Understand the logical layer (relationships) and the physical layer (joins), when to union or blend, how row duplication inflates totals, live connections vs extracts, data interpreter and pivoting, and how Tableau Prep Builder cleans data in repeatable flows, using the Adventure Cycles star schema.

✦ Tableau Desktop / Tableau Public 2026 | Intermediate | ⏱ 120 min

By the end of this lesson you can

- Build a relationships-based data model and explain why it avoids duplicated totals
- Choose between relationships, joins, unions and blends
- Pivot, split and clean data on the Data Source page and in Tableau Prep
- Decide between live connections and extracts, and plan refreshes
- Explain the logical vs physical layer in an interview

The two layers of a Tableau data source



Relationships live in the logical layer; joins and unions live inside a logical table (physical layer).

When you drag a second table onto the canvas, Tableau draws a **relationship** (a flexible "noodle") and asks for the matching fields. Tableau then generates the right join for each sheet, using only the tables that sheet needs and aggregating each table at its own level.

The Adventure Cycles star schema

The same star schema used in the Power BI lessons: one fact table of 9,000 sales lines and four dimension tables.

Python

```python
import pandas as pd
base = "adventure_cycles/"
fact = pd.read_csv(base + "fact_sales.csv")
dims = {n: pd.read_csv(base + f"dim_{n}.csv") for n in ["product", "territory", "customer",
"date"]}
print("fact_sales:", fact.shape, list(fact.columns))
for n, d in dims.items(): print(f"dim_{n}:", d.shape, list(d.columns)[:6])
```

OUTPUT

```
fact_sales: (9000, 9) ['order_number', 'order_date', 'product_key', 'customer_key',
'territory_key', 'quantity', 'unit_price', 'sales_amount', 'total_cost']
dim_product: (45, 6) ['product_key', 'product', 'subcategory', 'category',
'list_price', 'standard_cost']
dim_territory: (6, 3) ['territory_key', 'territory', 'country']
dim_customer: (1200, 5) ['customer_key', 'customer_name', 'gender', 'age_band',
```

```
'territory_key']
dim_date: (1096, 6) ['date', 'year', 'quarter', 'month_no', 'month', 'weekday']
```

STEP BY STEP **Build the model with relationships**

1. Connect to the folder of CSVs (Text file → choose `fact_sales.csv`). Drag it to the canvas.
2. Drag `dim_product.csv` next to it; Tableau suggests the relationship on the shared key (ProductKey). Check it in the relationship dialog.
3. Repeat for territory, customer and date.
4. Go to a sheet: the Data pane shows each table as a folder. Build Sales by Category: Tableau relates only fact and product for this view.

Why joins can inflate totals

A join merges tables **before** aggregation. If a table on one side has several matching rows, the other side's rows are repeated, and SUM counts them more than once. Relationships avoid this because each table is aggregated at its own level before combining.

Python

```python
# A targets table at territory + year level (one row per territory per year), related to line-level sales
fact["year"] = pd.to_datetime(fact.order_date).dt.year
targets = fact[["territory_key", "year"]].drop_duplicates().assign(target=10_00_00_000)   # ₹10 crore each
joined = fact.merge(targets, on=["territory_key", "year"])
# physical join at line level
print("territories x years:", len(targets), "| sales lines:", len(fact))
print("true total target: ₹", f"{targets.target.sum():,}")
print("SUM(target) after a row-level join: ₹", f"{joined.target.sum():,}")
print("inflation factor:", round(joined.target.sum() / targets.target.sum()), "x")
```

OUTPUT

```
territories x years: 18 | sales lines: 9000
true total target: ₹ 1,800,000,000
SUM(target) after a row-level join: ₹ 900,000,000,000
inflation factor: 500 x
```

The join repeats each territory's target once per sales line, so SUM(target) is wildly overstated. With a **relationship** between the sales table and a targets table on territory and year, Tableau aggregates targets at their own level and the total is correct.

Relationships, joins, unions and blends

Method	Combines	When to use
Relationship	Tables at different levels, related on keys	Default for star schemas and multi-level data
Join (inner, left, right, full)	Columns from tables into one flat table	Same grain, or when you need row-level calcs across tables
Union	Rows of tables with the same columns	Monthly files, one sheet per year
Blend	Two separate data sources, linked in a sheet on a common dimension (orange chain icon)	Sources that cannot be joined (e.g. a Google Sheet of targets with a SQL database); aggregated before linking

> **TIP** **Wildcard union**
>
> On the Data Source page, drag **New Union** onto a table and choose **Wildcard (automatic)** with a pattern such as `sales_2025_*.csv`. Every new monthly file in the folder is picked up on refresh, and Tableau adds a *Path* or *Sheet* column so you know where each row came from.

Cleaning on the Data Source page

Task	Feature
Messy Excel with titles, blank rows, merged cells	**Use Data Interpreter**
Months as columns (Jan, Feb, ...)	Select the columns → **Pivot** → rename "Pivot Field Names" to Month
"City, State" in one column	Column menu → **Split** / Custom Split
Wrong types	Click the type icon
Unneeded columns	Hide

Python

```
wide = pd.DataFrame({"region": ["East", "North"], "Jan": [120, 140], "Feb": [130, 150],
"Mar": [125, 160]})
long = wide.melt(id_vars="region", var_name="month", value_name="sales")   # Tableau Pivot
print(wide, "\n"); print(long)
```

OUTPUT

```
  region  Jan  Feb  Mar
0   East  120  130  125
1  North  140  150  160

  region month  sales
0   East   Jan    120
1  North   Jan    140
2   East   Feb    130
3  North   Feb    150
4   East   Mar    125
5  North   Mar    160
```

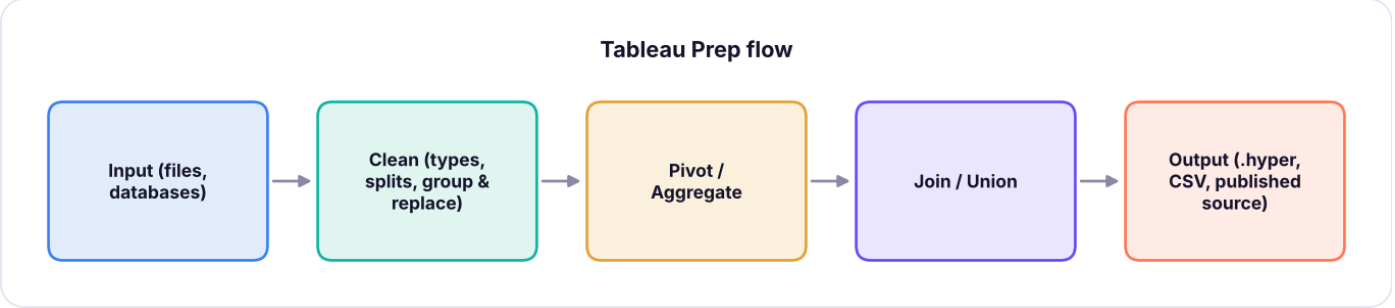
Live connection vs extract

	Live	Extract (.hyper)
Freshness	Always current	As of the last refresh
Speed	Depends on the database	Fast, columnar, compressed
Works offline	No	Yes
Good for	Real-time ops, governed warehouses with fast engines (Snowflake, BigQuery, Databricks)	Most dashboards, files, slow sources, Tableau Public
Refresh	Not needed	Full or incremental, scheduled on Tableau Cloud/Server

Extract options: keep only needed fields (**Hide all unused fields**), add extract filters, aggregate to visible dimensions, and use **incremental refresh** on a date or ID column for large tables.

Tableau Prep Builder

Prep is a separate app for repeatable cleaning flows. A flow is a left-to-right chain of steps:



A typical Tableau Prep flow.

Useful steps: **Clean** (profile pane shows each field's distribution; fix types, trim spaces, *Group and Replace* to merge spelling variants such as "Bengaluru"/"Bangalore"), **Aggregate**, **Pivot** (columns to rows and back), **Join** (with a join-clause summary that shows unmatched rows), **Union**, **New Rows** (fill missing dates) and **Script** (Python/R). Outputs can be scheduled with Tableau Prep Conductor on Cloud/Server.

Summary

KEY TAKEAWAYS

- Relationships (logical layer) are the default; they join at query time at the right level and avoid duplicated totals.
- Joins flatten tables first and can inflate SUMs when grains differ; unions stack rows; blends link separate sources per sheet.
- Use Data Interpreter, Pivot and Split for quick fixes; use Tableau Prep for repeatable flows.
- Extracts are fast snapshots; live connections are always current.

Quick quiz

6 questions · tap an answer to check it

1. Sales lines are joined to a territory-level targets table and SUM(Target) is far too high. Why?

- **A** The join repeated each target once per sales line
- **B** Targets are text
- **C** The extract is stale
- **D** A context filter is missing

Answer: A. Different grains duplicate rows; use a relationship.

2. Twelve monthly CSVs with identical columns should be combined. Which method?

- **A** Union (ideally a wildcard union)
- **B** Join
- **C** Blend
- **D** Relationship

Answer: A. Unions stack rows of same-shaped tables.

3. When is data blending the right choice?

- **A** When two sources cannot be joined in one data source and you link them on a common dimension in a sheet
- **B** For every star schema
- **C** To union files
- **D** To speed up extracts

Answer: A. Blends combine separate sources after aggregation.

4. Your Excel sheet has a title row and merged headers. What helps first?

- **A** Data Interpreter
- **B** A FIXED LOD
- **C** A parameter
- **D** A table calculation

Answer: A. Data Interpreter detects the real table.

5. Which is TRUE about extracts?

- **A** They are fast compressed snapshots that need refreshing
- **B** They always show real-time data
- **C** They cannot be filtered
- **D** They only work on Tableau Server

Answer: A. Schedule full or incremental refreshes.

6. Where do relationships live?

- **A** In the logical layer
- **B** In the physical layer
- **C** In Tableau Prep only
- **D** In the Marks card

Answer: A. Double-click a logical table to see its physical joins.

EXERCISE

Practice

Build a relationships model of the Adventure Cycles tables and a sheet of Sales by Category and Country. Then create a small targets table (Country, Year, Target) in Excel, relate it on Country and Year, and show Sales vs Target as a bullet chart. Finally, simulate the join problem: join the targets instead, and compare SUM(Target).

Show solution (worked answer)

Relate fact_sales to each dimension on its key; relate the targets table to the territory dimension on Country (and to the date table on Year). With relationships, SUM(Target) by Country equals the targets you typed. With a physical join to the fact table, each target repeats per sales line and the sum is inflated.

INTERVIEW CORNER

Questions you may be asked

Q: Relationships vs joins in Tableau? Relationships keep tables separate in the logical layer and Tableau generates the right joins per query, aggregating each table at its native level, so totals are not duplicated. Joins merge tables in the physical layer into one table before aggregation, which can duplicate rows when grains differ.

Q: When would you use data blending? When the data is in separate sources that cannot be related in one data source, such as a published data source plus a local spreadsheet of targets. Blends link on common dimensions and aggregate the secondary source first; they are slower and less flexible than relationships.

Q: How do you make a slow dashboard faster? Use an extract with only needed fields and rows, data source or context filters, fewer marks and quick filters, avoid high-cardinality filters showing all values, and check the Performance Recorder.

TABLEAU · LESSON 6

Dashboards, Actions, Stories and Publishing in Tableau (Capstone)

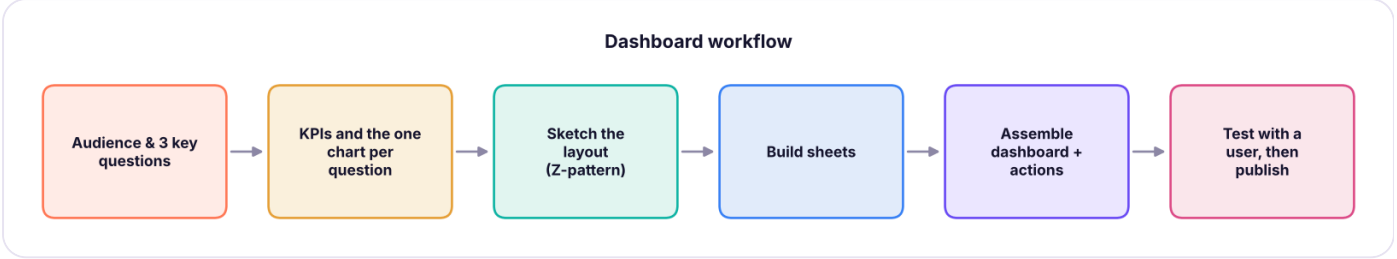
Turn sheets into an interactive dashboard and a data story. Plan the layout with containers, add filter, highlight, URL, parameter and set actions, design for phones, tell a story with Story Points, publish to Tableau Public or Cloud, and complete two capstones, a Superstore India sales dashboard and the video-game regional sales case study from class.

✦ Tableau Desktop / Tableau Public 2026 | Intermediate | ⏱ 180 min

By the end of this lesson you can

- Lay out a dashboard with tiled and floating objects, containers and device layouts
- Add filter, highlight, URL, go-to-sheet, parameter and set actions
- Build a Story with Story Points that explains an insight step by step
- Publish to Tableau Public or Cloud and set up refresh, permissions and subscriptions
- Deliver an executive dashboard and a written insight summary

Plan before you build



The order that keeps dashboards clear: questions first, layout last.

People read a screen in a **Z-pattern**: top-left first. Put the KPI cards across the top, the main trend or map in the middle, and the detail (tables, Top N) at the bottom. Keep to about 5–7 views and one colour meaning per dashboard (for example orange = loss everywhere).

Building the dashboard

Element	Use
Size	Fixed (e.g. 1200 × 800) for predictable layouts, Automatic or Range for flexible screens
Tiled vs floating	Tiled objects snap into a grid; floating objects sit on top (legends, logos). Prefer tiled.
Containers	Horizontal and vertical containers group objects; *Distribute Contents Evenly* keeps KPI cards equal
Objects	Text (titles), Image (logo), Web Page (not on Tableau Public), Blank (spacing), Navigation and Download buttons, Extension
Show/Hide button	Collapsible filter panel on a floating container
Device layouts	Device Preview → Phone: stack KPIs and one chart vertically

KPI cards

Each card is a sheet: the measure on **Text**, Mark type Text, a formatted label such as `₹<SUM(Sales)> cr`, and a small YoY calculation in the second line with ▲/▼ and colour.

Python

```python
import pandas as pd, numpy as np
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
y = df.groupby(df.order_date.dt.year).agg(sales=("sales", "sum"), profit=("profit", "sum"),
orders=("order_id", "nunique"), customers=("customer_id", "nunique"))
cur, prev = y.loc[2025], y.loc[2024]
for k in y.columns:
    unit = 1e7 if k in ("sales", "profit") else 1
    val = f"₹{cur[k] / unit:,.2f} cr" if unit > 1 else f"{cur[k]:,.0f}"
    print(f"{k:10s} 2025: {val:>12s}   YoY {cur[k] / prev[k] - 1:+.1%}")
print(f"margin     2025: {cur.profit / cur.sales:.1%}   (2024: {prev.profit /
prev.sales:.1%})")
```

OUTPUT

```
sales      2025:     ₹7.84 cr   YoY +19.8%
profit     2025:     ₹0.75 cr   YoY +22.1%
orders     2025:        1,232   YoY +13.5%
customers  2025:          623   YoY +5.6%
margin     2025: 9.6%   (2024: 9.4%)
```

The KPI row for the 2025 Superstore India dashboard (values from the data above).

Dashboard actions

Dashboard → Actions → Add Action:

Action	What happens	Example
Filter	Clicking a mark filters other sheets	Click a state on the map → the bar charts show only that state
Highlight	Matching marks stay bright, others fade	Hover a category to highlight it everywhere
Go to Sheet	Navigates to another dashboard	Click a KPI to open a detail page
URL	Opens a web page, with field values in the link	Open a product page or a Google search for the customer
Change Parameter	Clicking sets a parameter value	Click a region to set the "Selected region" parameter used in titles
Change Set Values	Clicking adds/removes marks from a set	Drill-in: show sub-categories only for the clicked category (set on a calculation)

Choose **Run action on: Select / Hover / Menu** and what happens **when clearing the selection** (show all values is the usual choice). Use *Use as Filter* (the funnel icon on a sheet) for the quick version.

> **TIP** **The "drill-in" pattern with a set action**
>
> Create a set on Category; a calculation `IF [Category Set] THEN [Sub-Category] ELSE [Category] END` on Rows; a Change Set Values action targeting the set. Clicking a category expands it to its sub-categories in the same chart, and clicking again collapses it.

Stories

A **Story** is a sequence of **Story Points**, each a captioned snapshot of a sheet or dashboard with its filters. Use it for a presentation: "1. Sales grew 20% in 2025 → 2. Growth came from Technology in the South → 3. Tables and Bookcases lose money → 4. Recommendation".

Publishing and sharing

Destination	How	Notes
Tableau Public	File → Save to Tableau Public	Free; public; you can allow viewers to download the workbook; embed code for your portfolio site
Tableau Cloud / Server	Server → Publish Workbook	Choose project, permissions, embedded or prompted credentials, refresh schedule
Subscriptions and alerts	On the published view	Email snapshots on a schedule; alerts when a measure crosses a threshold
Export	Worksheet/Dashboard → Export Image, PowerPoint, PDF, Crosstab	For offline reports
Embedding	Share → copy embed code	Place a dashboard inside a website or portal

Before publishing: give every sheet a clear title, hide unused sheets (right-click the dashboard tab → Hide All Sheets), add tooltips in plain language, set the default filters, and test on the phone layout.

Capstone 1: Superstore India executive dashboard

Build a one-page dashboard for the Head of Sales:

1. **KPI row**: Sales, Profit, Margin, Orders for the selected year with YoY change.
2. **Trend**: monthly Sales (bars) and Margin (line) on a dual axis.
3. **Map**: filled map of profit margin by state (diverging palette).
4. **Category → sub-category** bars with the drill-in set action; loss-makers in orange.
5. **Top 10 customers** table with a Top N parameter.
6. Filters: Year (single value), Region, Segment, applied to all sheets.

Python

```
d25 = df[df.order_date.dt.year == 2025]
sub = d25.groupby("sub_category").agg(sales=("sales", "sum"), profit=("profit", "sum"))
sub["margin %"] = (sub.profit / sub.sales * 100).round(1)
print("2025 loss-making sub-categories:"); print(sub[sub.profit < 0].round(0))
st = d25.groupby("state").agg(sales=("sales", "sum"), profit=("profit", "sum")); st["margin
%"] = st.profit / st.sales * 100
print("\nHighest and lowest margin states:", st["margin %"].idxmax(), f"{st['margin
%'].max():.1f}%", "|", st["margin %"].idxmin(), f"{st['margin %'].min():.1f}%")
```

OUTPUT

```
2025 loss-making sub-categories:
                  sales    profit  margin %
sub_category
Bookcases     3918268.0 -169018.0      -4.0
Tables        5297064.0 -286188.0      -5.0

Highest and lowest margin states: West Bengal 13.7% | Telangana 7.5%
```

Write three insight sentences under the dashboard, for example: what grew, what lost money, and what you recommend (discount caps on loss-making sub-categories, focus on the highest-margin states).

Capstone 2: Regional video-game sales (the class case study)

The same business case you solved in Excel in class: *how do video-game sales vary across North America, Europe, Japan and other regions, and where are the opportunities?*

Python

```
vg = pd.read_csv("video_game_sales.csv")
print(vg.shape, "| years", vg.year.min(), "to", vg.year.max())
regions = ["na_sales", "eu_sales", "jp_sales", "other_sales"]
best_genre = vg.groupby("genre")[regions].sum().idxmax()
print("Best genre by region:", best_genre.to_dict())
vg["decade"] = (vg.year // 10 * 10).astype(str) + "s"
print((vg.groupby("decade")[regions].sum()).round(1))
```

OUTPUT

```
(2400, 11) | years 2010 to 2025
Best genre by region: {'na_sales': 'Sports', 'eu_sales': 'Sports', 'jp_sales': 'Role-
Playing', 'other_sales': 'Sports'}
        na_sales  eu_sales  jp_sales  other_sales
decade
```

```
2010s      312.5     249.4     136.8        108.9
2020s      188.1     148.9      83.8         62.6
```

Tableau tasks: 1. **Pivot** the four regional sales columns into Region / Sales on the Data Source page (select the columns → Pivot), so Region becomes one dimension. 2. Top 5 games per region: Region on Rows, Name on Rows, SUM(Sales) on Columns, a **Top 5 by Sales computed per region** using `RANK(SUM([Sales]))` with Compute Using = Name, filtered ≤ 5. 3. Regional sales by decade: a calculated field `STR(INT(YEAR([Year])/10)*10) + "s"` (or `([Year] // 10) * 10` if Year is a number) and a stacked bar coloured by Region. 4. Best genre in each region: highlight table of Genre × Region with the maximum per region marked using `WINDOW_MAX`. 5. A stacked bar of regional distribution over time with a clear title, then assemble all four into a dashboard with a Region filter action.

Summary

KEY TAKEAWAYS

- Plan the questions and the layout before building; use containers and a Z-pattern.
- Actions make dashboards interactive: filter, highlight, go to sheet, URL, parameter and set actions.
- Stories present insights step by step.
- Publish to Tableau Public for portfolios and to Cloud/Server for work, with refresh schedules and permissions.

Quick quiz

6 questions · tap an answer to check it

1. Which action makes clicking a state on a map filter every other chart?

A A filter action (or Use as Filter)

B A highlight action

C A URL action

D A story point

Answer: A. Filter actions pass the selected values to target sheets.

2. Which feature expands a category into its sub-categories in place when clicked?

A A set action with a calculation that switches between Category and Sub-Category

B A Story

C A blend

D A data source filter

Answer: A. Change Set Values drives the drill-in pattern.

3. Where should the KPI cards usually go?

A Across the top-left of the dashboard

B Bottom right

C In a hidden sheet

D In a tooltip only

Answer: A. Readers start top-left (Z-pattern).

4. What is a Story in Tableau?

A A sequence of captioned story points showing sheets or dashboards

B A type of chart

C An extract

D A calculated field

Answer: A. Stories present an analysis step by step.

5. Which is NOT available on Tableau Public?

- **A** Keeping your workbook private on the server
- **B** Saving to your public profile
- **C** Dashboards with actions
- **D** Calculated fields

Answer: A. Everything saved to Tableau Public is public.

6. How do you make one Region filter apply to all sheets on a dashboard?

- **A** Apply to Worksheets → All Using This Data Source (or Selected Worksheets)
- **B** Duplicate the filter on every sheet
- **C** Use a parameter only
- **D** Add it to context

Answer: A. One filter can drive many sheets.

EXERCISE **Capstone deliverables**

Publish both dashboards to Tableau Public. For each, write: the audience, three questions it answers, three insights with numbers, and two recommendations. Add the links to your resume and LinkedIn Featured section.

Show solution (worked answer)

Superstore: KPI row, dual-axis trend, margin map, drill-in category chart, Top N customers, three filters applied to all sheets, filter actions from the map and the category chart. Video games: pivoted regions, Top 5 per region (RANK table calc), decade stacked bar, genre × region highlight table, filter action on Region. Insights should quote numbers from your own views.

INTERVIEW CORNER

Questions you may be asked

Q: How do you design a dashboard for executives? Start from their decisions: 3–5 KPIs with comparisons at the top, one trend, one breakdown, one exception list; consistent colours; filters applied to all sheets; tooltips that explain; a phone layout; and a short written insight.

Q: Filter action vs highlight action? A filter action removes non-selected data from target sheets; a highlight action keeps all data but fades the non-matching marks, preserving context.

Q: How do you share a Tableau dashboard securely? Publish to Tableau Cloud or Server with project permissions, use row-level security (user filters or entitlement tables with USERNAME()), schedule extract refreshes, and share via links, subscriptions or embedding with authentication. Never use Tableau Public for confidential data.

Skill Navora

TRACK 5

AI for Data Analysis

Generative AI for analysts: prompting, AI vs GenAI vs agentic AI, AI over SQL (Metabase, MinusX), Copilot and Gemini in Excel, and office agents.

4 lessons | Version 2026.10 | Updated October 2026

Created by **Varun Kumar**
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Contents

Every entry is a clickable link. The PDF also has bookmarks for each lesson and section.

01 Generative AI for Analysts: How It Works, Prompting & AI vs GenAI vs Agentic AI

02 AI with SQL: Text-to-SQL, Docker, Metabase & MinusX

03 AI in Excel & Office: Copilot, Gemini, Claude, Python in Excel & Agents

04 The AI-Assisted Analyst Workflow, Agents & Governance (Capstone)

AI FOR DATA ANALYSIS · LESSON 1

Generative AI for Analysts: How It Works, Prompting & AI vs GenAI vs Agentic AI

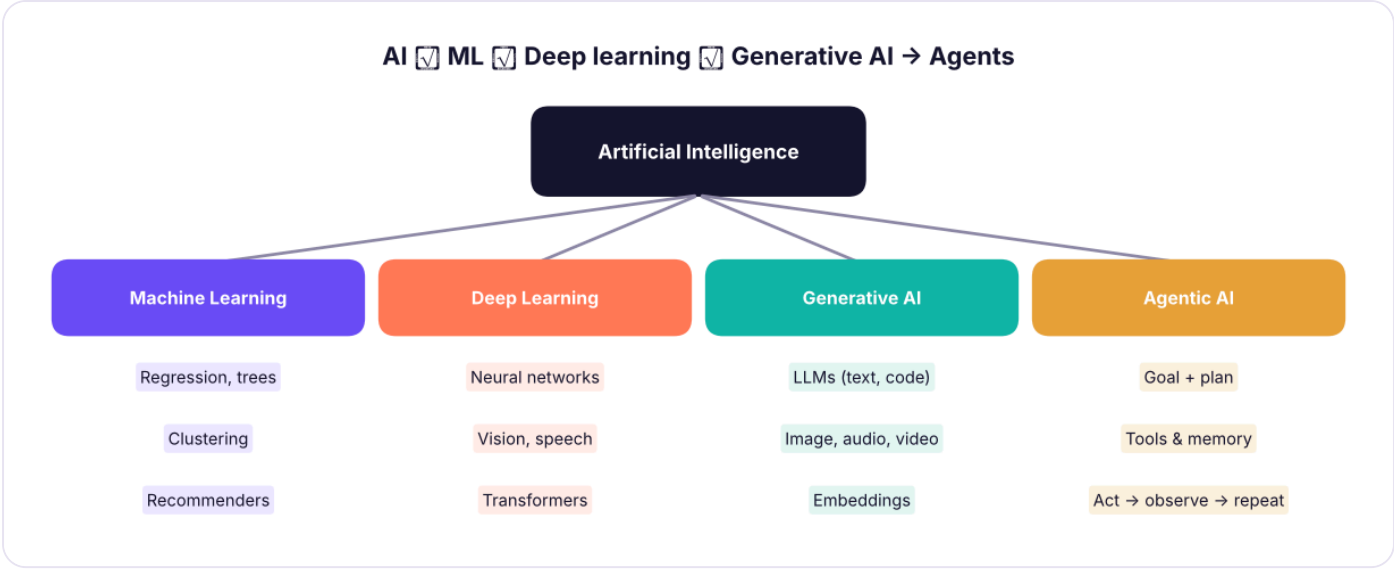
Understand what generative AI is and how large language models behave (tokens, context, temperature, hallucination), how AI, machine learning, deep learning, GenAI and agentic AI differ, and how to write prompts that give analysts accurate, checkable results, with reusable templates and privacy rules.

DVA with AI · class 1 | ✦ Models and tools as of 2026 | Beginner | ⏱ 110 min

By the end of this lesson you can

- Explain the relationship between AI, ML, deep learning, generative AI and agentic AI
- Describe in plain language how an LLM generates text and why it can be confidently wrong (hallucination)
- Write prompts using role, context, task, format and constraints, with examples and step-by-step reasoning
- Use prompt templates for common analyst tasks and know what never to paste into an AI tool
- Choose a suitable tool and check an AI answer before using it

The AI family tree



How the terms relate. Each ring is a subset of the one before it. Agentic AI uses generative models as its brain but adds goals, tools and loops.

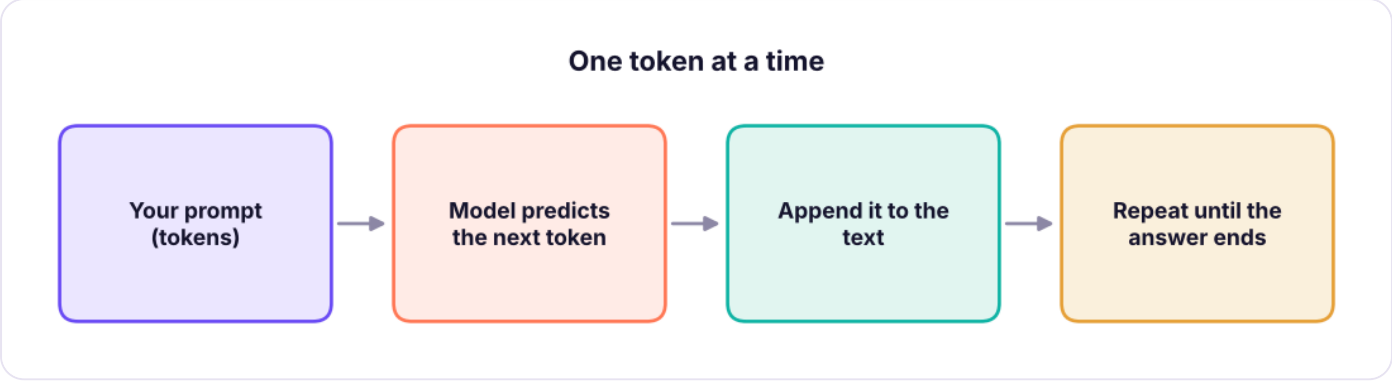
Term	What it means	Example
AI	Machines doing tasks that normally need human intelligence	Rule-based chess engine, spam filter
Machine learning (ML)	Systems that **learn patterns from data** instead of following hand-written rules	Predict loan default from past loans
Deep learning	ML with many-layered neural networks	Image recognition, speech-to-text
Generative AI (GenAI)	Models that **create new content** (text, images, code, audio)	ChatGPT, Claude, Gemini, Copilot, Midjourney
LLM	Large language model: the GenAI engine for text and code	GPT, Claude, Gemini, Llama
Agentic AI	An LLM **plus goals, tools and a loop**: it plans, calls tools (run SQL, search, edit files), checks results and continues until the goal is met	An analyst agent that explores a database and writes the report
AGI	Hypothetical AI at human level across tasks	Not here yet; ignore the hype

AI vs GenAI vs agentic AI

	Traditional ML	Generative AI	Agentic AI
Output	A prediction (number, label)	New content (text, code, image)	A completed multi-step task
Input	Structured data	A prompt (text, files, images)	A goal in plain language
Control	You train and deploy a model	You ask once, get one answer	The system decides the next step
Example	"Will this customer churn?" → 0.82	"Summarise these reviews" → paragraph	"Find why sales fell in Q3 and draft a memo" → runs queries, charts, writes
Main risk	Bias, drift	Hallucination	Hallucination **plus actions**: wrong tool calls, runaway cost, security

How an LLM works (without maths)

An LLM is trained on vast amounts of text to predict **the next token** (a word or word piece). Repeating that prediction one token at a time produces sentences, code and tables. It has learnt statistical patterns of language, facts and reasoning styles, but it does **not look things up** unless it is given a tool or documents.



Generating an answer: the prompt is split into tokens; the model repeatedly predicts the next token from everything so far.

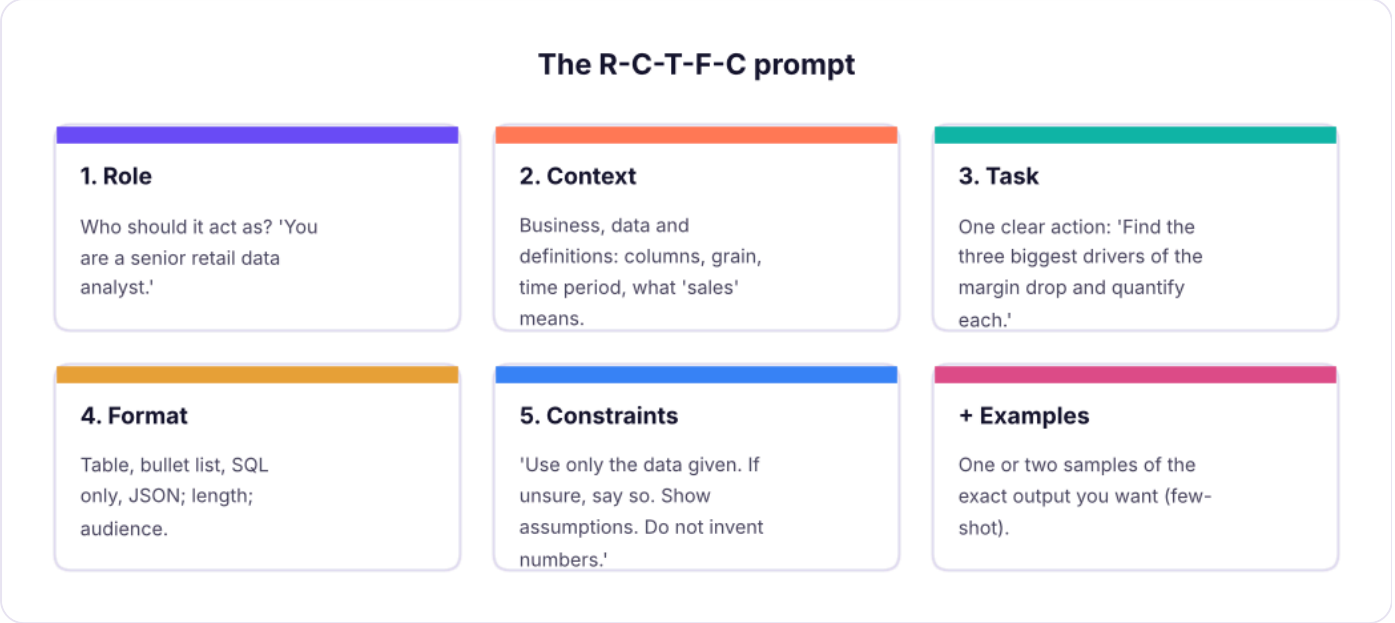
Concept	Meaning	Why analysts care
Token	Roughly ¾ of an English word (about 4 characters)	Limits and costs are measured in tokens
Context window	How much text the model can consider at once (prompt + answer)	A long file may not fit; the model forgets what is outside
Temperature	Randomness of word choice	Low (0–0.3) for SQL, maths and extraction; higher for brainstorming
System prompt	Hidden instructions that set behaviour	Custom GPTs/Projects store your standing instructions
Training cut-off	The model's knowledge stops at some date	It may not know recent tools or prices; give it documents
Grounding / RAG	Supplying real documents or data in the prompt (Retrieval-Augmented Generation)	Cuts hallucination: "answer only from this file"
Tool use / function calling	The model asks the app to run a calculator, database query or code	Makes answers computed instead of guessed
Multimodal	Accepts images, PDFs, audio as well as text	Paste a chart screenshot and ask what it shows

WATCH OUT **Hallucination**

Because an LLM generates *plausible* text, it can state false things with complete confidence: invented statistics, non-existent functions, wrong column names, fake citations. **Never use an AI answer as a fact until you have checked it**: run the code, recompute the number, open the source.

Prompt engineering for analysts

A prompt is an instruction plus the material the model needs. The most reliable structure has five parts:



A prompt recipe: Role, Context, Task, Format, Constraints.

Weak vs strong prompt

Weak: *"Analyse my sales data."*

Strong:

```text
Role: You are a senior retail data analyst for an Indian omnichannel retailer.
Context: Table superstore(order_id, order_date, region, category, sub_category, sales,
discount, profit),
         one row per order line, Jan 2023–Dec 2025, amounts in rupees. Margin = profit /
sales.
Task: Explain why overall margin is only about 9%. Look at category, sub-category and
discount band.
Format: (1) a 3-bullet summary for the CFO, (2) a table of the 5 worst sub-categories
(sales, profit, margin),
        (3) the SQL Server queries you used.
Constraints: Use only the columns listed. State any assumption. If a number cannot be
computed from the table, say so.
```

The strong prompt names the grain, the metric definition, the output format and the rules for uncertainty. The result is far more likely to be right *and* checkable.

Core techniques

Technique	How	Use for
Be specific	Name columns, periods, units, audience	Everything
Few-shot examples	Show 2–3 input→output pairs	Classification, extraction, tone matching
Step by step	"First list the assumptions, then compute, then summarise"	Multi-step analysis; makes errors visible
Ask for checks	"Then list three ways this answer could be wrong"	Self-critique
Structured output	"Return JSON with keys..."	Feeding results into code
Delimiters	Put data in `"""triple quotes"""` or XML tags	Stops the model confusing instructions with data
Iterate	Treat the first answer as a draft; refine with follow-ups	Always
Give it the schema, not the data	Column names and 5 sample rows, not the whole file	Privacy, token limits

Prompt templates for common analyst jobs

Job	Template (replace brackets)
Explain a formula	"Explain this Excel formula step by step for a beginner and give a simpler alternative: [formula]"
Write SQL	"Dialect: SQL Server. Tables: [schema with keys and grain]. Write a query that [question]. Return only SQL, then list assumptions."
Debug code/SQL	"This [language] query gives [wrong result / this error]. Expected: [x]. Query: [code]. Find the cause and show the fix."
Clean data	"Here are 10 sample values of column [x]: [list]. Propose normalisation rules and a mapping table; flag ambiguous cases."
Interpret results	"Here is a result table: [table]. Write 3 insights for a non-technical manager, each with a number, and 2 caveats."
Choose a chart	"I want to show [message] to [audience] with [data]. Recommend a chart and what to label."
Draft a report	"Turn these findings into a one-page memo with: situation, key findings, recommendation, risks. Keep it under 250 words."
Interview practice	"Act as an interviewer for a data analyst role; ask one SQL question at a time and give feedback on my answer."

EXAMPLE

From vague to checkable

Prompt: *"Explain this formula:* `=IFERROR(INDEX($C:$C,MATCH(1,($A:$A=F2)*($B:$B=G2),0)),"Not found")` *"* A good answer says it looks up C where A equals F2 **and** B equals G2 (a two-condition lookup), returns "Not found" if none, and notes it needs array evaluation in older Excel. You can verify by testing it on 3 rows.

Choosing and using AI tools

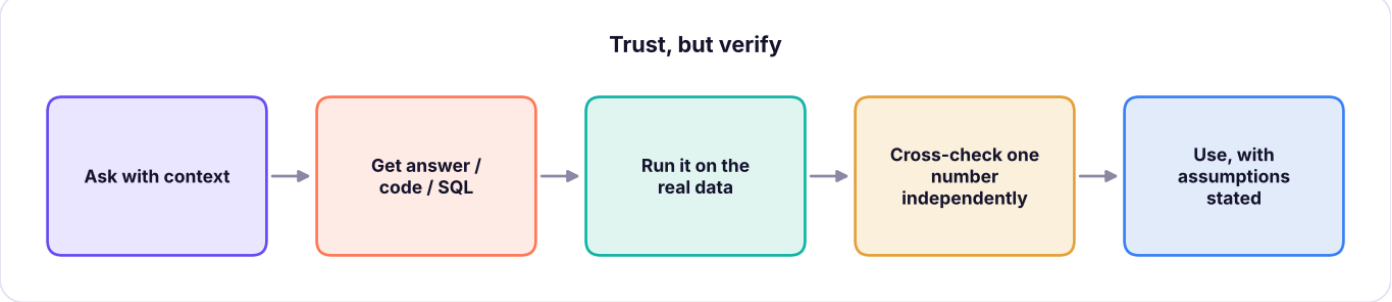
Tool type	Examples	Good for	Watch out
General chat assistants	Claude, ChatGPT, Gemini, Copilot, Grok, Perplexity	Explaining, drafting, code, brainstorming	Data privacy; hallucination
Search-grounded	Perplexity, Gemini with search, Copilot with web	Fresh facts with citations	Check the cited pages; sources vary in quality
Document/notebook tools	NotebookLM, Projects, Custom GPTs	Answering **only from your files**	Quality depends on the files
Embedded in apps	Copilot in Excel/Power BI, Gemini in Sheets, Metabase/ MinusX, Claude in Excel	Work where the data is	Permissions; see next lessons
Coding assistants and agents	Claude Code, Cursor, GitHub Copilot, Codex	Writing and running code, analysis scripts	Review diffs; run tests; secrets

Never paste

1. **Personal data** (names with phone numbers, emails, Aadhaar/PAN, health, salary of identifiable people) unless your company's approved tool and policy allow it.
2. **Confidential business data** (unreleased financials, customer lists, contracts, source code, credentials, API keys) into public tools.
3. **Anything you are not allowed to share**: client data under NDA.

Use **enterprise versions** with data-protection terms, **mask or sample** the data (replace names with IDs, share the schema and 5 fake rows), and follow your organisation's AI policy. In India, personal data is also covered by the Digital Personal Data Protection Act, 2023.

A verification habit for every AI answer



The analyst's rule: generate with AI, verify with data.

Here is why verification matters. Suppose an AI writes the following query for "average order value per region". It looks right, runs without error and returns plausible numbers, but it averages order **lines**, not **orders**:

Python

```
import duckdb, pandas as pd
con = duckdb.connect()
con.execute("CREATE TABLE superstore AS SELECT * FROM
read_csv_auto('superstore_sales.csv')")
wrong = con.execute("SELECT region, ROUND(AVG(sales), 0) AS avg_order_value FROM superstore
GROUP BY region ORDER BY region").df()
right = con.execute("""SELECT region, ROUND(AVG(order_total), 0) AS avg_order_value
                       FROM (SELECT region, order_id, SUM(sales) AS order_total FROM
superstore GROUP BY region, order_id) GROUP BY region ORDER BY region""").df()
pd.DataFrame({"region": wrong.region, "AI query (per line)": wrong.avg_order_value,
"Correct (per order)": right.avg_order_value})
```

OUTPUT

	region	AI query (per line)	Correct (per order)
0	East	30,521	60,684
1	North	32,038	61,892
2	South	32,580	64,925
3	West	32,192	64,067

4 rows × 3 columns

The AI's version understates order value by roughly half, because an average order has about two lines. A one-line sanity check (total sales ÷ number of orders) would have caught it:

Python

```
tot = con.execute("SELECT SUM(sales) / COUNT(DISTINCT order_id) FROM
superstore").fetchone()[0]
print(f"Total sales / distinct orders = {tot:,.0f}")
```

OUTPUT

```
Total sales / distinct orders = 62,942
```

COMMON MISTAKES **Common AI-assistant failure modes**

1. **Wrong grain**: counting lines as orders, double-counting after a join (Lesson 8 in SQL).
2. **Invented columns or functions**: the code references a field that does not exist.
3. **Silent assumptions**: it picks a date range, currency or definition without saying.
4. **Plausible but unverified numbers**: especially when asked to "estimate" or "recall" data it cannot see.
5. **Over-agreeing**: if you assert something wrong, it may follow you. Ask it to challenge your assumption.
6. **Stale knowledge**: new tool versions, prices, laws.

Responsible use

- **Accountability stays with you**: you sign off on the analysis, not the model.
- **Bias**: models mirror training data; check outputs for unfair assumptions about people or regions.
- **Transparency**: tell stakeholders when AI helped, and keep the prompt and verification notes.
- **Copyright and attribution**: do not present generated text or images as someone else's work; check licences.
- **Security**: treat prompts and uploaded files as data leaving your machine; beware prompt injection when an AI reads untrusted documents or web pages.

DO IT WITH AI **Do it with AI**

Pick a task you did earlier in this course (for example the SQL discount analysis). Ask an AI assistant to produce the answer from the schema, run its SQL, and compare with the result in this handbook. Where it differs, work out *why* (grain? date range? definition?). This single exercise teaches more about AI limits than any lecture.

Summary

KEY TAKEAWAYS

- AI ⊃ ML ⊃ deep learning ⊃ generative AI; **agentic AI** = an LLM with goals, tools and a loop.
- LLMs predict the next token; they do not look things up unless given documents or tools. They can **hallucinate**, so verify every answer.
- Good prompts have **role, context, task, format and constraints**, use examples and step-by-step reasoning, and give the schema (not the sensitive data).
- Use templates for SQL, formulas, cleaning, insights, memos and interview practice; iterate.
- Never paste personal or confidential data into public tools; mask or sample, and follow your company policy.
- The analyst's loop: ask with context → run on real data → cross-check one number → state assumptions.

Quick quiz

6 questions · tap an answer to check it

1. Which statement best describes agentic AI?

A A model that only classifies images

B An LLM-based system that plans, uses tools and repeats steps to achieve a goal

C Any chatbot

D A rule-based macro

Answer: B. Agents add goals, tools and a decision loop on top of a language model.

2. Why can an LLM state wrong facts confidently?

A It lacks memory

B It generates plausible text from patterns and does not verify facts unless given sources or tools

C It is connected to the internet

D It always copies its training data

Answer: B. Fluency is not accuracy.

3. Which setting is best for generating SQL?

A High temperature

B Low temperature

C No context

D Maximum length

Answer: B. Low temperature reduces random variation for precise tasks.

4. What makes a prompt more reliable for analysis?

A Shorter wording

B Role, context (schema and definitions), a clear task, an output format and constraints

C Asking for "everything"

D Hiding the table names

Answer: B. Specifics reduce guesswork and make the answer checkable.

5. **A teammate wants to paste a customer list with phone numbers into a public chatbot. Best advice?**

- **A** Go ahead
- **B** Do not; mask or sample the data, or use an approved enterprise tool
- **C** Delete the email column only
- **D** Convert it to PDF first

Answer: B. Personal data should not go into tools not approved for it.

6. **An AI query returns an "average order value" half the size of total sales ÷ orders. The likely error is:**

- **A** Wrong chart
- **B** Averaging order lines instead of order totals (wrong grain)
- **C** Too many decimals
- **D** A missing index

Answer: B. Always sanity-check results against a simple independent calculation.

EXERCISE **Prompt rewrite**

Rewrite this weak prompt into a strong one and list three checks you will run on the answer: *"Tell me about churn in my telecom data."* The data is `telecom_churn.csv` with columns customer_id, senior_citizen, tenure_months, contract, internet_service, payment_method, monthly_charges, total_charges, support_calls, churn (Yes/No).

Show solution (worked answer)

Strong prompt: "Role: senior telecom data analyst. Context: table telecom_churn(...columns...), one row per subscriber, churn = Yes/No. Task: find which contract types, tenure bands and support-call counts have the highest churn rate and quantify each. Format: a table (group, customers, churners, churn %), three bullets for a retention manager, and the SQL/pandas code used. Constraints: use only these columns, state assumptions, do not infer causes you cannot test, flag small groups (< 50 customers)." Checks: (1) recompute the overall churn rate by hand (count of Yes ÷ rows) and match it; (2) confirm each group's customer counts add up to the total; (3) spot-check one group with an independent query.

INTERVIEW CORNER **Questions you may be asked**

Q: Difference between AI, ML and generative AI? AI is the broad field; ML learns patterns from data to predict or classify; generative AI (a deep-learning subset) creates new content such as text, code and images.

Q: What is a hallucination and how do you reduce it? Confident but false output. Reduce it by grounding answers in provided documents or data (RAG), using tools that compute results, lowering temperature, asking for sources and assumptions, and verifying independently.

Q: How do you write a good prompt for a data task? State the role, context (schema, grain, definitions), a single clear task, the required output format, and constraints (use only given data, state assumptions); give examples; iterate.

Q: What is agentic AI? Systems where an LLM plans and executes multi-step tasks using tools, observes results and continues until a goal is met, with human oversight for risky actions.

AI FOR DATA ANALYSIS · LESSON 2

AI with SQL: Text-to-SQL, Docker, Metabase & MinusX

Use AI to write, explain and debug SQL safely. Learn how text-to-SQL works and where it fails, give the model the right schema, protect your database with read-only access, install Docker, run Metabase on your own data, connect an AI assistant such as MinusX, and validate every generated query.

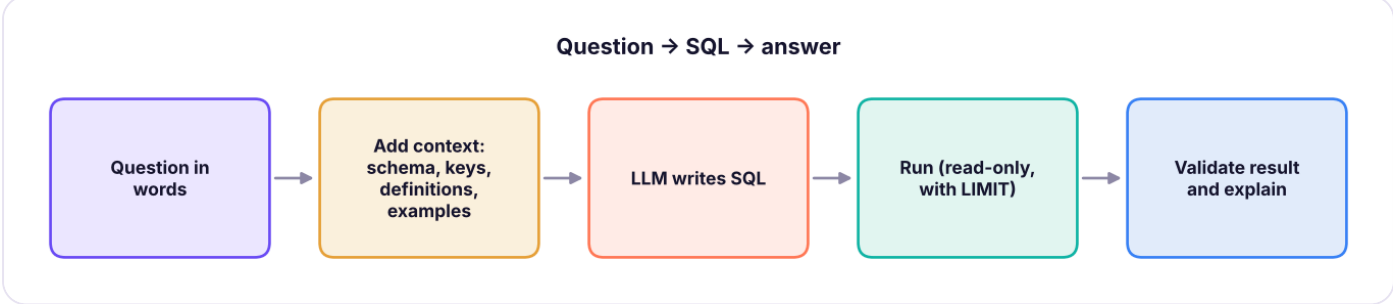
DVA with AI · class 2 | ✦ Docker, Metabase and AI assistants, 2026 | Intermediate | ⏱ 130 min

By the end of this lesson you can

- Explain how text-to-SQL works and why context (schema, definitions, examples) decides quality
- Write a schema-aware prompt and review generated SQL for the usual errors
- Install Docker and run Metabase locally; connect a database; build questions, models and dashboards
- Describe how an in-browser AI assistant such as MinusX works with Metabase, and its privacy trade-offs
- Apply guardrails: read-only roles, row limits, query review and result validation

What is text-to-SQL?

Text-to-SQL turns a business question in plain language into a SQL query: "Which region had the highest profit in 2025?" → `SELECT region, SUM(profit) …`. LLMs are good at SQL syntax, but **correct SQL for *your* database** needs your context.



Text-to-SQL pipeline. Quality depends mostly on the context you supply, not on the model alone.

What goes wrong without context

Problem	Example	Prevention
Wrong tables/columns	Uses `customer_name` in `orders`, where it does not exist	Provide the schema (tables, columns, types, keys)
Wrong grain / fan-out	Joins orders to items and then `COUNT(*)` for "orders"	State the grain: "one row per order line"; add "count distinct order_id" rule
Wrong definitions	"Active customer", "net sales", "this year"	Provide a metric dictionary
Dialect mismatch	`LIMIT` instead of `TOP` in SQL Server	Name the database and version
Dates	Uses `YEAR(date) = 2025` (slow) or the wrong fiscal year	State calendar vs fiscal year and preferred patterns
Hallucinated values	`WHERE region = 'Northern'` when data says `North`	Provide sample values for categorical columns
Unsafe actions	`DELETE` / `DROP` suggestions	Read-only connection; review before running

A schema prompt that works

Text

```
You are a senior analytics engineer. Write SQL for Microsoft SQL Server 2022.

SCHEMA (all in dbo):
customers(customer_id PK, customer_name, segment, city, state, region)
  - segment in ('Consumer','Corporate','Home Office'); region in
('North','South','East','West')
orders(order_id PK, order_date DATE, ship_date DATE, ship_mode, customer_id FK →
customers)
order_items(item_id PK, order_id FK → orders, product_id FK → products, quantity,
discount, sales, profit)
  - one row per product line; sales and profit in rupees
products(product_id PK, product_name, category, sub_category, list_price)

DEFINITIONS:
- "Orders" = COUNT(DISTINCT orders.order_id). "Margin %" = SUM(profit) / SUM(sales).
- Financial year runs April–March. Use half-open date ranges (≥ start AND < next start).

QUESTION: For FY2025 (Apr 2025–Mar 2026), show margin % by category and by discount band
(0, 1–20%, >20%).

RULES: Return only SQL; add comments; no SELECT *; list any assumptions after the query.
```

Notice the pieces: dialect, schema with keys, valid values, grain, definitions, date convention and output rules. This is the difference between a lucky answer and a reliable one.

> **TIP** **Keep a reusable "context file"**
>
> Save your schema, definitions and sample queries in one text file (or in an AI **Project** / custom instructions). Paste it into every session. Many BI tools (Metabase, Power BI Copilot, Snowflake Cortex, Looker) call this a **semantic layer** or "AI context": descriptions, synonyms and verified queries.

Reviewing AI-generated SQL: a checklist

1. **Does every table/column exist?** (Run it; fix any "invalid column name".)
2. **Is the grain right?** After each join, did rows multiply? Are distinct counts used for orders and customers?
3. **Are filters correct?** Date ranges half-open? Status filters? NULL handling?
4. **Do aggregates match the definition?** `SUM(profit)/SUM(sales)` not `AVG(margin)`.
5. **Does the total reconcile?** Compare with a simple query: `SELECT SUM(sales) FROM order_items`.
6. **Is it safe and cheap?** No writes; a `TOP` / `LIMIT` while testing; no accidental cross join.
7. **Is it readable?** CTEs, aliases and comments you could explain in an interview.

Let's see this in practice. A typical AI answer to "profit margin by discount band" that forgets `SUM(profit)/SUM(sales)` and averages row margins instead:

SQL

```sql
SELECT CASE WHEN discount = 0 THEN '0%' WHEN discount ⩽ 0.2 THEN '1-20%' ELSE '>20%' END
AS discount_band,
       ROUND(AVG(profit / sales) * 100, 1) AS ai_margin_pct          -- average of row
margins (unweighted)
FROM order_items
GROUP BY CASE WHEN discount = 0 THEN '0%' WHEN discount ⩽ 0.2 THEN '1-20%' ELSE '>20%' END
ORDER BY discount_band;
```

RESULT

discount_band	ai_margin_pct
0%	18.2
1–20%	2.0
>20%	-20.5

3 rows × 2 columns

SQL

```sql
SELECT CASE WHEN discount = 0 THEN '0%' WHEN discount ⩽ 0.2 THEN '1-20%' ELSE '>20%' END
AS discount_band,
       ROUND(100.0 * SUM(profit) / SUM(sales), 1) AS correct_margin_pct     -- weighted:
total profit / total sales
FROM order_items
GROUP BY CASE WHEN discount = 0 THEN '0%' WHEN discount ⩽ 0.2 THEN '1-20%' ELSE '>20%' END
ORDER BY discount_band;
```

RESULT

discount_band	correct_margin_pct
0%	17.5
1–20%	1.2
>20%	-20.6

3 rows × 2 columns

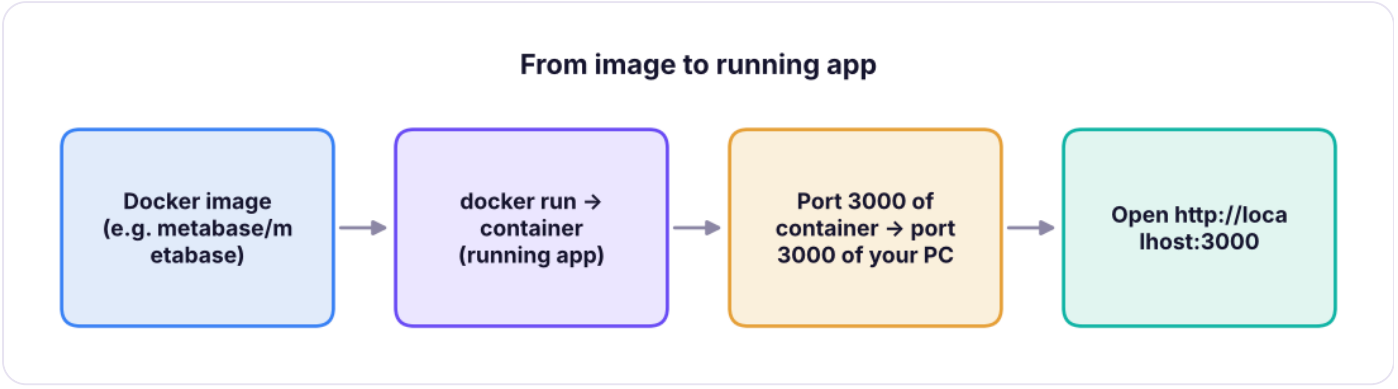
The two queries differ in every band (18.2% vs 17.5% and 2.0% vs 1.2% for the first two) because large order lines matter more in the weighted version than small ones. Which one the business wants must be written in the **definitions** part of your prompt. The checklist question "does the aggregate match the definition?" catches this.

Guardrails for letting AI near a database

Guardrail	How
Read-only login	Create a database user with `SELECT` only on approved views (`GRANT SELECT ON …`); never give AI tools `db_owner`
Expose views, not raw tables	A `vw_sales_for_ai` view with clean names, no personal data columns
Row and time limits	Timeouts, `TOP 1000`, query cost limits
Masking	Hide phone, email, ID columns; use dynamic data masking
Human review	Show the SQL before running it; log queries
Verified questions	Keep a library of approved queries for key metrics
No secrets in prompts	Connection strings and passwords stay in the tool's secret store

Docker: running tools in containers

Docker packages an application and everything it needs into a **container**, so it runs the same on any machine. Instead of installing Java, a database and an app by hand, you run one command.



An image is a read-only template; a container is a running instance of it. Many containers can run from one image.

STEP BY STEP **Install Docker Desktop**

1. Download **Docker Desktop** for Windows, Mac or Linux from the Docker website and install it. On Windows it uses **WSL 2** (Windows Subsystem for Linux); the installer enables it, and you may need to restart.
2. Start Docker Desktop and wait until it says "Engine running".
3. Open a terminal (PowerShell, Terminal) and check: `docker --version` and `docker run hello-world`.
4. Note your machine needs virtualisation enabled in BIOS and about 4 GB of free RAM for small tools.

Command	Meaning
`docker pull metabase/metabase:latest`	Download an image
`docker run -d -p 3000:3000 --name metabase metabase/metabase`	Run it in the background (`-d`), map port 3000, name it
`docker ps` / `docker ps -a`	List running / all containers
`docker logs -f metabase`	Follow its logs (see start-up progress)
`docker stop metabase` / `docker start metabase`	Stop / start again
`docker rm metabase`	Delete the container (not the image)
`docker images`	List downloaded images
`docker compose up -d`	Start everything defined in a `compose.yaml` file

Metabase: open-source BI for everyone

Metabase is an open-source business-intelligence tool: connect a database, ask questions with a point-and-click **question builder** (no SQL needed), or write SQL, then save charts into **dashboards**. It is popular in start-ups because non-technical colleagues can self-serve.

STEP BY STEP **Run Metabase and connect your data**

1. Start it: `docker run -d -p 3000:3000 --name metabase metabase/metabase`. First start takes a minute or two (check `docker logs -f metabase`).
2. Open **http://localhost:3000**, create the admin account, and choose **Add your data**.
3. Pick your database type (PostgreSQL, MySQL, SQL Server, BigQuery, Snowflake, SQLite files via drivers, CSV upload in newer versions), enter host, port, database name, username and password. **Use the read-only user.** If the database also runs on your PC, use the host `host.docker.internal` from inside the container.
4. Metabase scans the tables. Browse them under **Browse data**.
5. **New → Question**: choose a table, pick columns, add **Filter**, **Summarize** (`Sum of sales` by `region`), choose a visualisation, **Save**.
6. **New → Dashboard**: add saved questions, add filters (date, region) and connect them to the question fields.

For anything you want to keep, run Metabase with a proper application database instead of the default embedded one, and mount a volume, for example:

YAML

```yaml
# compose.yaml
services:
  metabase:
    image: metabase/metabase:latest
    ports: ["3000:3000"]
    environment:
      MB_DB_TYPE: postgres
      MB_DB_DBNAME: metabase
      MB_DB_PORT: 5432
      MB_DB_USER: metabase
      MB_DB_PASS: ${METABASE_DB_PASSWORD}      # from a .env file, never hard-coded
      MB_DB_HOST: postgres
    depends_on: [postgres]
  postgres:
    image: postgres:16
    environment:
      POSTGRES_USER: metabase
      POSTGRES_DB: metabase
      POSTGRES_PASSWORD: ${METABASE_DB_PASSWORD}
    volumes: ["pgdata:/var/lib/postgresql/data"]
volumes:
  pgdata:
```

Metabase concepts

Concept	Meaning
Question	A saved query (builder or SQL) with a visualisation
Model	A curated, cleaned table (a saved query with documented columns) for people to build on
Metric	A named, reusable calculation such as "Net revenue"
Dashboard	A page of questions with shared filters
Collection	Folders with permissions
Segment / filters	Reusable row filters
Permissions	Which groups see which databases and tables; row-level sandboxing (paid plans)
Metabot / AI features	Built-in AI assistance for questions and SQL in recent versions (availability depends on the edition and plan; check Metabase's current documentation)

Try it on the retail data: connect to your SQL Server `RetailDB`, then build three questions: *Sales by region*, *Monthly sales* (summarise by `order_date: Month`), and *Top 10 customers*. Then verify each one against this handbook's tables:

```sql
SELECT region, ROUND(SUM(sales), 0) AS sum_of_sales FROM superstore GROUP BY region ORDER BY sum_of_sales DESC;
```

RESULT

region	sum_of_sales
South	53,952,769
West	51,958,521
North	49,018,161
East	46,484,119

4 rows × 2 columns

MinusX: an AI assistant inside your BI tool

MinusX is an open-source AI assistant delivered as a **Chrome extension** that adds a chat sidebar to tools such as **Metabase** and **Jupyter notebooks**. Instead of copying your data into a chatbot, you work in Metabase and ask the sidebar things like *"Why did sales drop in March?"* or *"Write a query for monthly profit by category"*. According to its public description, it reads a simplified view of the page (metadata such as table and column names), then operates the tool by clicking, typing and writing queries, and lets you accept or reject its edits.

Aspect	How it behaves	What you should still do
Context	Sees the page, selected tables/columns and the query editor	Make sure the right database and tables are selected
Actions	Writes and runs SQL, edits questions, explains results	Read the SQL before accepting
Models	Supports several LLM providers (you may supply your own API key)	Check the provider's data policy
Privacy	Vendor says raw data is not sent; metadata and your prompts are	Verify the current policy; never expose sensitive tables
Cost	Free and paid plans; model API usage may cost	Check current pricing

STEP BY STEP

Using an AI assistant with Metabase

1. Install the Chrome extension from the Chrome Web Store (search "MinusX") and pin it; open your Metabase tab and open the sidebar.
2. Add your model/API key in settings if required.
3. Select the database/table in Metabase so the assistant knows the context.
4. Ask a precise question ("Monthly sales and profit by category for 2025, as a line chart; sales = SUM(sales), profit = SUM(profit)").
5. Review the generated SQL, run it, check totals against a simple query, then save it as a question.
6. Ask follow-ups: "Explain the biggest drop", "Break it down by region", "Why is Furniture negative?"

WATCH OUT

AI assistants and sensitive data

Anything the assistant can see, the model provider may receive. Use a **read-only role limited to non-sensitive views**, keep personal columns out of the connected schema, use an enterprise model agreement for company data, and keep a record of which prompts were run. If your company bans browser extensions that read pages, ask IT before installing any.

Other ways to use AI with SQL

Use	How
Explain a query	Paste the query (not the data) and ask for a line-by-line explanation and the business meaning
Optimise	Paste query + execution plan + table sizes; ask for rewrites and indexes; test each
Convert dialects	"Convert this SQL Server query to PostgreSQL; list functions that differ"
Generate test data	"Create INSERT statements for 20 fictional customers across four regions" (no real data)
Write documentation	Column descriptions, data dictionaries, comments
Data quality checks	"Write queries to find nulls, duplicates, orphans and out-of-range values for these tables"
Platform AI	Snowflake Cortex Analyst, BigQuery Gemini, Databricks Genie, SSMS Copilot, Azure Data Studio/VS Code Copilot, Fabric Copilot: all follow the same principle: context in, SQL out, you verify

2026 UPDATE **Trend: the semantic layer**

Teams that succeed with text-to-SQL do not rely on the raw schema: they define metrics (net sales, active customer) once in a **semantic layer** (dbt Semantic Layer, Cube, LookML, Power BI semantic models, Metabase models/metrics) and let AI query *that*. The AI then picks from governed metrics instead of reinventing definitions, which removes most wrong-number errors.

Summary

KEY TAKEAWAYS

- Text-to-SQL works when you provide **dialect, schema with keys, valid values, grain, metric definitions and date rules**.
- Review every query: columns exist, **grain** is right, filters correct, aggregates match the definition, totals reconcile, nothing writes.
- Protect the database with **read-only logins on curated views**, row limits and human review; never expose secrets or personal columns.
- **Docker** runs tools in containers (`docker run -d -p 3000:3000 metabase/metabase`); **Metabase** gives self-service questions, models and dashboards.
- An assistant such as **MinusX** adds an AI sidebar to Metabase; treat it as a fast junior analyst whose SQL you review, and check privacy terms.
- A governed **semantic layer** makes AI answers consistent.

Quick quiz

6 questions · tap an answer to check it

1. Why does giving the AI your schema and metric definitions improve results?

A It makes the model larger

B The model can use real table and column names and your definitions instead of guessing

C It speeds up the database

D It removes the need to review the SQL

Answer: B. Context is the main driver of text-to-SQL accuracy; review is still required.

2. Which command starts Metabase in a container on port 3000?

A `docker build metabase`

B `docker run -d -p 3000:3000 --name metabase metabase/metabase`

C `docker stop metabase`

D `metabase install`

Answer: B. `-d` runs detached and `-p 3000:3000` maps the port.

3. Which login is safest for an AI tool?

A The database owner

B A read-only user restricted to curated views

C The admin account

D An anonymous login

Answer: B. Least privilege limits damage if the AI makes a mistake.

4. A generated query averages `profit/sales` row by row to get a margin. Why is that risky?

A It is invalid SQL

B It is an unweighted average of ratios and may not match the business definition SUM(profit)/SUM(sales)

C It is slower

D It drops NULLs

Answer: B. Weighted vs unweighted aggregates differ; define the metric in the prompt.

5. What is a semantic layer?

A A type of chart

B A governed set of definitions for tables, metrics and relationships that BI and AI tools use

C A SQL function

D A backup system

Answer: B. It standardises metrics so every query uses the same logic.

6. What should you do before saving an AI-generated Metabase question?

A Nothing

B Read the SQL and reconcile at least one total against a simple independent query

C Share it with everyone

D Delete the database

Answer: B. Validation protects you from plausible but wrong numbers.

EXERCISE

Schema prompt and review

(1) Write a schema prompt for the `employees` table (HR data: emp_id, gender, age, department, job_title, education, hire_date, years_at_company, monthly_income, job_satisfaction, overtime, performance_rating, city, attrition) including at least three definitions. (2) Ask any AI assistant for a query that returns attrition % by department and overtime. (3) Review the result with the checklist and reconcile the overall attrition % (25.9%) with a simple query.

Show solution (worked answer)

Definitions to include: "Attrition %" = employees with attrition = 'Yes' ÷ all employees; `monthly_income` is in rupees; "tenure band" = 0–2, 3–5, 6+ years. Reconciliation query: `SELECT ROUND(100.0 * SUM(CASE WHEN attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 1) FROM employees;` should return 25.9. The weighted total from the grouped result (sum of leavers ÷ sum of employees across all groups) must also equal 25.9.

INTERVIEW CORNER

Questions you may be asked

Q: How would you use AI to speed up SQL work safely? Provide schema, definitions and dialect; generate and explain queries; review for grain, filters and definitions; run read-only with limits; reconcile totals; keep verified queries in a library.

Q: What is Docker and why is it useful for data tools? A container platform that packages apps with their dependencies so the same setup runs anywhere; useful to spin up Metabase, databases or Jupyter in one command and share reproducible environments.

Q: What are the risks of connecting an AI assistant to a database? Data exposure to a model provider, destructive queries, wrong answers presented as facts, and prompt injection; mitigate with read-only access to curated views, masking, review and logging.

Q: What is Metabase? An open-source BI tool where users can ask questions with a visual builder or SQL, build dashboards and share them, with permission control by group.

AI FOR DATA ANALYSIS · LESSON 3

AI in Excel & Office: Copilot, Gemini, Claude, Python in Excel & Agents

Use AI where analysts work every day. Work with Copilot in Excel (formulas, insights, charts, the COPILOT function), Gemini in Google Sheets and Excel add-ins, Claude and other assistants for formulas and files, Python in Excel, and office agents that perform multi-step tasks, and learn how to prompt, verify and stay safe.

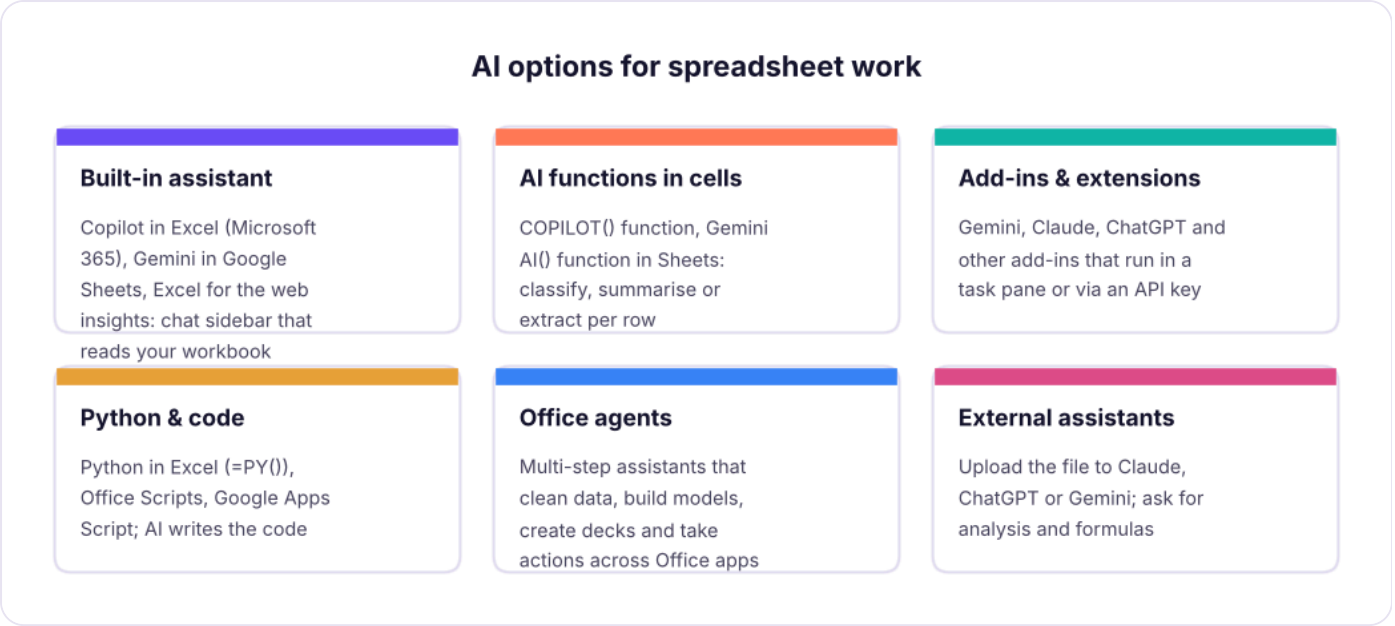
● DVA with AI · class 3 ✦ Microsoft 365 Copilot, Gemini and agent features as of 2026

Intermediate ⏱ 120 min

By the end of this lesson you can

- Prepare a workbook so AI tools understand it (Tables, clear headers, no merged cells)
- Use Copilot and Gemini-style assistants to explain, write and debug formulas, create pivots and charts, and summarise data
- Use the Excel COPILOT function and Python in Excel for text, classification and analysis inside cells
- Describe what office agents do and what controls they need
- Verify AI-generated formulas and numbers with independent checks

The AI-in-Excel landscape



Four ways AI shows up in spreadsheets.

Step 0: make your workbook AI-ready

AI assistants can only reason about data they can *see and parse*. The same rules that make pivot tables work (Excel Lesson 1) make AI work:

1. **Convert data to an Excel Table** (`Ctrl + T`) with a clear name (`tblSales`) and one header row.
2. **Descriptive column names** (`order_date` , `sales_inr`) instead of `Col1` ; add units.
3. No merged cells, blank rows or notes inside the data; keep totals outside.
4. One sheet per dataset; a small **Data dictionary** sheet that defines columns and metrics (AI reads it too).
5. Remove or mask sensitive columns before connecting an AI service.
6. Save to **OneDrive/SharePoint** (Copilot needs cloud-saved files with AutoSave on).

Copilot in Excel

Needs a Microsoft 365 Copilot licence and your data formatted as a Table or range with headers.

Task	Example prompt
Explain	"Explain what column H does and how it is calculated"
Formulas	"Add a column that shows profit margin %" / "Write a formula that looks up the manager's name from the Managers table"
Insights	"Show me insights about sales by region" → Copilot adds pivot tables and charts
Highlight	"Highlight rows where discount > 20% and profit is negative" (conditional formatting)
Sort / filter	"Sort by sales descending and filter to Technology"
Pivot & charts	"Create a pivot table of sales by category and year with YoY %"
Clean up	"Fix the inconsistent city names" / "Split full name into first and last name"
Analyse with Python	"Run a correlation between discount and profit and plot it" (uses Python in Excel)

STEP BY STEP

A safe Copilot workflow

1. Open your workbook (saved in OneDrive) → **Home → Copilot**.
2. Select the Table (or let Copilot detect it).
3. Ask one thing at a time, with column names: *"Add a column margin_pct = profit / sales, formatted as a percentage."*
4. **Read the proposed formula** and its explanation before clicking **Insert column**.
5. Check three rows by hand, and one total against a plain `SUM`.
6. Undo (`Ctrl + Z`) if wrong, and rephrase with more detail.

The COPILOT() function

Microsoft 365 also provides a `COPILOT` **function** that sends text and cell ranges to the model and returns results into cells that **spill and recalculate**:

Excel formula

```
=COPILOT("Classify this customer review as Positive, Neutral or Negative", D2:D100)
=COPILOT("Summarise the main complaint in 6 words or fewer", D2)
=COPILOT("Extract the city name from this address", A2)
```

Use it for text tasks (classify, summarise, extract, translate, tag) over **small to moderate** ranges, not for arithmetic: use normal formulas for numbers. Results can change between recalculations and

each call consumes model capacity; availability, limits and licensing change, so check Microsoft's current documentation. Convert important results to values (copy → paste values) once checked.

Here is the same classify-sentiment idea applied to the product reviews dataset, with the (non-AI) ground-truth labels already in the file, so you can see how to *measure* an AI's accuracy:

Python

```
import pandas as pd
rev = pd.read_csv("product_reviews.csv")
print(rev.shape)
rev[["review_id", "product", "rating", "review_text", "sentiment"]].head(6)
```

OUTPUT

```
(1200, 5)
```

	review_id	product	rating	review_text	sentiment
0	1	Speaker	4	Very comfortable to use all day	positive
1	2	Smartwatch	5	Very comfortable to use all day	positive
2	3	Smartphone	4	Delivery was quick and packaging was neat. great value for money, totally worth it	positive
3	4	Earbuds	1	Battery drains too fast	negative
4	5	Laptop	2	Delivery was late and the box was damaged	negative
5	6	Laptop	2	Refund process is very slow. heats up a lot while charging	negative

6 rows × 5 columns

Python

```python
# How would you evaluate an AI classifier? Compare its labels with a trusted label set.
# Here a deliberately weak keyword rule plays the role of "the AI" so you can see the evaluation method.
# (These reviews are synthetic and templated, so real reviews will be much harder to classify.)
neg_words = ("stopped", "late", "poor", "crashing")          # a deliberately small word list: an imperfect classifier
pos_words = ("excellent", "great", "amazing", "premium")
def guess(t):
    t = t.lower()
    n = sum(w in t for w in neg_words); p = sum(w in t for w in pos_words)
    return "negative" if n > p else ("positive" if p > n else "neutral")
rev["pred"] = rev.review_text.apply(guess)
acc = (rev.pred == rev.sentiment).mean()
print(f"Agreement with trusted labels: {acc:.1%}\n")
print(pd.crosstab(rev.sentiment, rev.pred, rownames=["actual"], colnames=["predicted"]))
```

OUTPUT

```
Agreement with trusted labels: 56.0%

predicted  negative  neutral  positive
actual
negative        204      206         0
neutral           0      175         0
positive          0      322       293
```

The cross-tab (a **confusion matrix**, ML track Lesson 15) shows *where* a classifier goes wrong. Do the same with an AI column: take 100 reviews you labelled yourself, compare, and only then trust it on 10,000.

Gemini in Google Sheets and with Excel

- **Gemini in Google Sheets** (Workspace plans): a side panel that can *create tables, write formulas, summarise, generate charts and analyse trends*, plus an **`AI()` function** to run a prompt per row (classify, summarise, extract) similar to `COPILOT()`.
- **Gemini and other LLM add-ins for Excel**: some marketplaces offer add-ins or custom functions (for example `=GEMINI("prompt", A2)`) that call a model through **your own API key**. They are useful for batch text tasks, but you must protect the key, watch the **cost per call**, and avoid sending sensitive data.
- **Apps Script / Office Scripts**: AI can write the script; you review and run it.

COMMON MISTAKES **API-key hygiene for add-ins**

Never paste an API key into a shared workbook or a cell others can read; use the add-in's secure settings, set a **spending limit** on the key, and rotate it if exposed. A key in a shared file is a billing and security incident waiting to happen.

Claude, ChatGPT and other assistants with your files

Upload a CSV or XLSX to a general assistant that can **run code** (analysis/code-execution tools). It reads the file, writes Python, runs it, and shows charts and tables: closer to an analyst than a chatbot that only talks.

A good first message with a file:

Text

```
I've attached superstore_sales.csv (one row per order line, Jan 2023-Dec 2025, amounts in
rupees).
1) Profile the data: row count, columns and types, missing values, duplicates, date range.
2) Compute total sales, profit and margin by year and category.
3) Find the three biggest drivers of low margin; quantify each.
4) Show your Python code, and tell me which assumptions you made.
Do not guess: if something cannot be computed from the file, say so.
```

Typical good behaviour: it prints the profile, shows code, makes charts. What you still do: spot-check numbers, make sure it used the right columns, and ask for the code so you can re-run it yourself. Assistants that run code produce **reproducible** analysis; those that only "think" may fabricate numbers.

Assistant strengths	Check
Writing and explaining formulas (even LET, LAMBDA, XLOOKUP, array formulas)	Test on sample rows
Cleaning plans and Power Query M code	Preview the result; keep a backup
Building DAX measures	Validate against a pivot
Summarising and drafting commentary	Verify each number quoted
Designing dashboards and chart choices	Compare with your audience's needs

Python in Excel

=PY(turns a cell into a Python editor (Formulas → Insert Python). It uses **pandas, matplotlib, seaborn, statsmodels and scikit-learn**, runs in Microsoft's secure cloud (no local Python needed), and reads Excel ranges with `xl()`.

Python

```
df = xl("tblSales[#All]", headers=True)          # Excel Table → pandas DataFrame
df.groupby("region")["profit"].sum().sort_values(ascending=False)
```

Ask Copilot: *"Use Python to show the correlation between discount and profit and draw a scatter plot."* Behind the scenes it writes the =PY() code. Check what it computed against this handbook's result:

Python

```
sup = pd.read_csv("superstore_sales.csv")
print(f"Correlation discount vs profit margin (lines): {sup.discount.corr(sup.profit / sup.sales):.2f}")
print(sup.groupby(sup.discount.map(lambda d: f"{d:.0%}")).apply(lambda g: round(g.profit.sum() / g.sales.sum() * 100, 1)).rename("margin_%").to_string())
```

OUTPUT

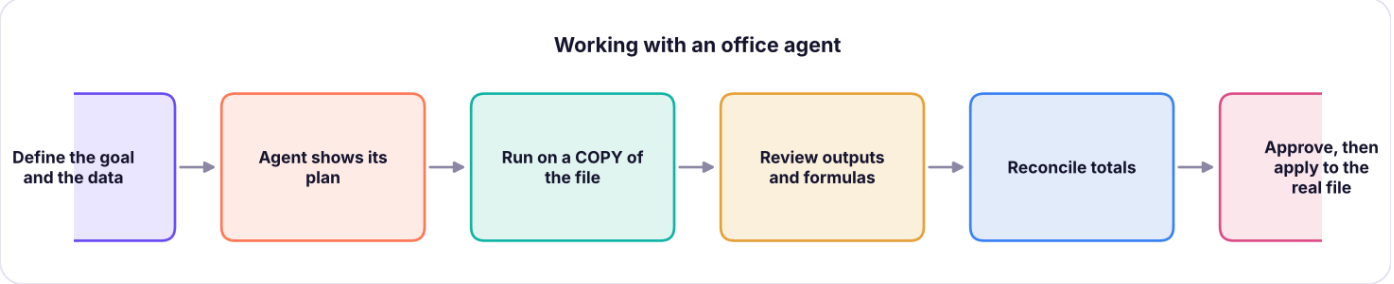
```
Correlation discount vs profit margin (lines): -0.91
discount
0%     17.5
10%     5.9
15%     0.9
20%    -4.6
30%   -19.0
40%   -26.3
```

A strong negative correlation, and margin turning negative from a 20% discount upwards: the same story as the SQL and Power BI lessons, arrived at a third way. Cross-checking tools against each other is how you build trust.

Office agents: the next step

Agents go beyond answering: given a goal ("Clean this export, build a pivot by region, add a chart, and draft a three-slide summary"), they **plan, use the spreadsheet or Office apps as tools, check results and iterate**, then present the outcome for your approval. Microsoft, Google, Anthropic, OpenAI and others have all released agent modes for office work; names and capabilities change quickly, so evaluate any specific product (including whichever "Frontier"-style agent your organisation provides) against the questions below.

Question to ask about any agent	Why it matters
What can it **read** and **write**? (files, mail, calendar, web)	Scope of risk
Does it **show its plan and steps** and ask before acting?	Oversight
Can I **undo** or review changes (version history)?	Recoverability
Where does data **go** (tenant, region, retention, training use)?	Compliance
How does it handle **untrusted content** (prompt injection in emails/web pages)?	Security
Does it produce **formulas and sources** or just pasted values?	Auditability
What does it **cost per task**?	Budget



A safe agent workflow: small steps, visible plan, human approval at risky points, independent checks.

What AI does well and badly in spreadsheets

Strong	Weak / risky
Explaining unfamiliar formulas and files	Large-scale precise calculation (use formulas/Python)
Drafting formulas, DAX, SQL, M code, VBA	Knowing your **business rules** unless told
Text tasks: classify, summarise, extract, translate	Consistent results every time (non-deterministic)
Suggesting charts and structure	Hidden assumptions (date ranges, definitions)
Debugging errors (`#N/A`, `#SPILL!`)	Anything involving **sensitive data** in the wrong tool
Documentation and commentary	Replacing review and sign-off

A checklist before you trust an AI-built spreadsheet

1. Totals reconcile with the source (`SUM` of raw vs summary).
2. Row counts match (no lost or duplicated rows).
3. Formulas are **visible** (not pasted values) and use absolute references correctly.
4. Spot-check five random rows by hand.

5. Edge cases: blanks, zero, negative, text in number columns, duplicates.
6. Units and currency are labelled; dates are real dates.
7. Assumptions are written on the sheet.
8. A second tool or person confirms the headline number.

Summary

KEY TAKEAWAYS

- Prepare data as **Tables** with clear headers, no merged cells, and a data dictionary; save in the cloud for Copilot.
- **Copilot** explains and writes formulas, creates pivots, charts, highlights and cleans data; `COPILOT()` and Sheets' `AI()` apply prompts per row for text tasks.
- General assistants that can **run code** make reproducible analyses from uploaded files; always request the code and assumptions.
- **Python in Excel** (`=PY()`) brings pandas into cells; AI can write it, you verify it.
- **Agents** plan and act across apps: work on a copy, require visible plans, approvals, undo, audit trail and clear data-handling terms.
- Verify with reconciliation, spot checks, edge cases and a second tool; protect keys and sensitive data.

Quick quiz

6 questions · tap an answer to check it

1. What should you do first to help an AI understand your Excel data?

A Merge headers

B Convert it to a clearly named Excel Table with descriptive headers and no merged cells

C Colour the cells

D Hide the columns

Answer: B. Structured, well-named data is easier for AI (and pivot tables) to use.

2. The COPILOT() function is best used for:

A Precise financial arithmetic

B Text tasks such as classifying, summarising or extracting over a range

C Replacing SUM

D Sorting

Answer: B. Use normal formulas for numbers; use AI functions for language tasks.

3. Why ask an assistant to show its Python code?

A To make the answer longer

B So the analysis is reproducible and can be checked

C Because code is always correct

D To avoid cleaning data

Answer: B. Visible code lets you verify and re-run the work.

4. Which is a sensible control for an office agent?

A Let it edit the live file without review

B Run it on a copy, review its plan and outputs, and keep version history

C Give it all passwords

D Disable logging

Answer: B. Human oversight and recoverability are essential.

5. An AI classifier labels reviews; how do you measure its quality?

A Read three results

B Compare its labels with a trusted labelled sample and look at accuracy and the confusion matrix

C Use temperature 1.0

D Ask it if it is right

Answer: B. Evaluate against ground truth, not self-reports.

6. Where should an API key for an Excel add-in be stored?

A In a visible cell

B In the add-in's secure settings with a spending limit

C In the sheet name

D In an email

Answer: B. Keys in shared files can be abused and billed to you.

EXERCISE

AI-assisted analysis, verified

Using `superstore_sales.csv`: (1) ask an AI assistant with code execution to profile the data and compute margin by category and discount band; (2) write the pivot table version yourself; (3) compare the two and list any differences; (4) create three `COPILOT()`-style prompts (or Gemini `AI()` prompts) for a column of customer comments, and say how you would evaluate them on 50 labelled examples.

Show solution (worked answer)

Expected margins by discount band: 0% → 17.5%, 1–20% → 1.2%, >20% → −20.6% (overall 9.1%). Differences usually come from averaging margins instead of total profit ÷ total sales, or from using order lines versus orders. Prompts: "Classify as Positive, Neutral or Negative (one word)"; "Extract the product mentioned (or 'none')"; "Summarise the complaint in under 8 words". Evaluation: label 50 comments yourself, run the prompt, compute accuracy and a confusion matrix, inspect errors, refine the prompt, and re-test on a fresh sample.

INTERVIEW CORNER

Questions you may be asked

Q: How would you use AI tools to speed up Excel analysis responsibly? Prepare structured data, ask specific questions, review generated formulas and pivots, reconcile totals, document assumptions, keep sensitive data out of unapproved tools, and keep a human sign-off.

Q: What is Python in Excel? A feature that runs Python (pandas, matplotlib, scikit-learn) in cells via `=PY()`, in a secure cloud, reading Excel ranges with `xl()`, useful for analysis and charts that are hard with formulas.

Q: What is an AI agent in an office context and what are its risks? A system that plans and executes multi-step tasks using apps and files; risks include wrong actions, data leakage, prompt injection and lack of audit; mitigate with scoped permissions, approvals, copies and logs.

Q: How do you check an AI-generated formula? Test on a few rows including edge cases, compare totals with a simple independent calculation, read the formula's explanation, and make sure references and units are right.

AI FOR DATA ANALYSIS · LESSON 4

The AI-Assisted Analyst Workflow, Agents & Governance (Capstone)

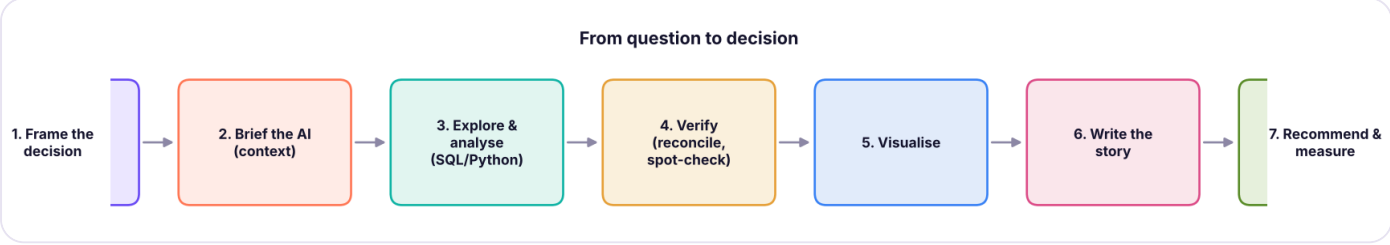
Combine everything into one repeatable workflow: frame a question, brief the AI, analyse with code, verify, visualise and write the story. See how an analyst agent works (plan, tools, observe, repeat) with a runnable toy example, learn MCP and tool-calling basics, and finish with governance, ethics and a telecom-churn capstone.

DVA with AI · wrap-up | ✦ 2026 | Intermediate | ⏱ 140 min

By the end of this lesson you can

- Apply a seven-step AI-assisted analysis workflow from question to recommendation
- Explain the agent loop, tool calling and the Model Context Protocol (MCP) at a conceptual level
- Build a tiny agent-style loop that plans, runs SQL tools and reports, and describe where an LLM would sit
- Design human-in-the-loop controls, logging and evaluation for AI analytics
- Complete an end-to-end churn analysis and explain it to a business audience

The AI-assisted analyst workflow



Seven steps. AI accelerates several of them, but the analyst owns the first and the last two.

Step	What you do	Where AI helps	Your responsibility
1. Frame	Decision, audience, metric definitions, constraints	Brainstorm questions, critique your framing	Own the decision question
2. Brief	Schema, definitions, sample rows, rules	Turn notes into a clean brief	No sensitive data; correct definitions
3. Analyse	Profile, clean, aggregate, model	Draft SQL/Python, explain code, suggest checks	Run it; read it
4. Verify	Reconcile totals, test edge cases, second method	Write test queries, find anomalies	Sign off the numbers
5. Visualise	Choose and label charts	Suggest chart types, write plotting code	Honest, readable charts
6. Write	Insight, caveat, action	Draft commentary, simplify language	Every claim traceable to a number
7. Recommend	Action, owner, metric, date	Pre-mortem: "how could this fail?"	Business judgement

Case: why do telecom customers leave?

CASE STUDY **Brief**

A telecom company has 3,000 subscribers in `telecom_churn.csv` (tenure, contract type, internet service, payment method, charges, support calls, churn Yes/No). The retention head asks: **Who is most likely to churn, and what should we do about it?** We will follow the workflow and use AI-style assistance where it helps, checking everything.

Steps 1–2: frame and brief

Text

```
Decision: choose which customers get a retention offer next quarter (budget: 300 offers).
Metric: churn rate = churners / customers in the group. Report groups with at least 50
customers only.
Schema: telecom_churn(customer_id, senior_citizen 0/1, tenure_months, contract {Month-to-
month, One year, Two year},
        internet_service {Fiber, DSL, No internet}, payment_method, monthly_charges,
total_charges, support_calls, churn {Yes, No})
Rules: use only this table; state assumptions; do not claim causes; show code.
```

Step 3: profile and analyse

Python

```
import pandas as pd, numpy as np
tc = pd.read_csv("telecom_churn.csv")
tc["churned"] = (tc.churn == "Yes").astype(int)
print(f"{len(tc):,} customers | overall churn {tc.churned.mean():.1%} | missing values:
{int(tc.isna().sum().sum())} | duplicate IDs: {int(tc.customer_id.duplicated().sum())}")
tc.describe(include="all").T[["count", "unique", "top", "mean", "min",
"max"]].round(1).head(10)
```

OUTPUT

```
3,000 customers | overall churn 25.7% | missing values: 0 | duplicate IDs: 0
```

	count	unique	top	mean	min	max
customer_id	3000	3000	T00001	NaN	NaN	NaN
senior_citizen	3,000	NaN	NaN	0.17	0	1
tenure_months	3,000	NaN	NaN	36.06	0	72
contract	3000	3	Month-to-month	NaN	NaN	NaN
internet_service	3000	3	Fiber	NaN	NaN	NaN
payment_method	3000	4	Card	NaN	NaN	NaN
monthly_charges	3,000	NaN	NaN	67.07	20	131.74
total_charges	3,000	NaN	NaN	2,418.32	19.58	8,724.35
support_calls	3,000	NaN	NaN	1.28	0	6
churn	3000	2	No	NaN	NaN	NaN

10 rows × 6 columns

Churn by one factor at a time (the same GROUP BY thinking as SQL Lesson 7):

Python

```
def churn_by(col, bins=None, min_n=50):
    x = tc[col] if bins is None else pd.cut(tc[col], bins)
    g = tc.groupby(x, observed=True).agg(customers=("churned", "size"), churners=
("churned", "sum"))
    g["churn_%"] = (g.churners / g.customers * 100).round(1)
    return g[g.customers ⩾ min_n]
print(churn_by("contract").to_string()); print();
print(churn_by("internet_service").to_string()); print()
print(churn_by("support_calls").to_string()); print(); print(churn_by("tenure_months",
bins=[-1, 6, 12, 24, 48, 72]).to_string())
```

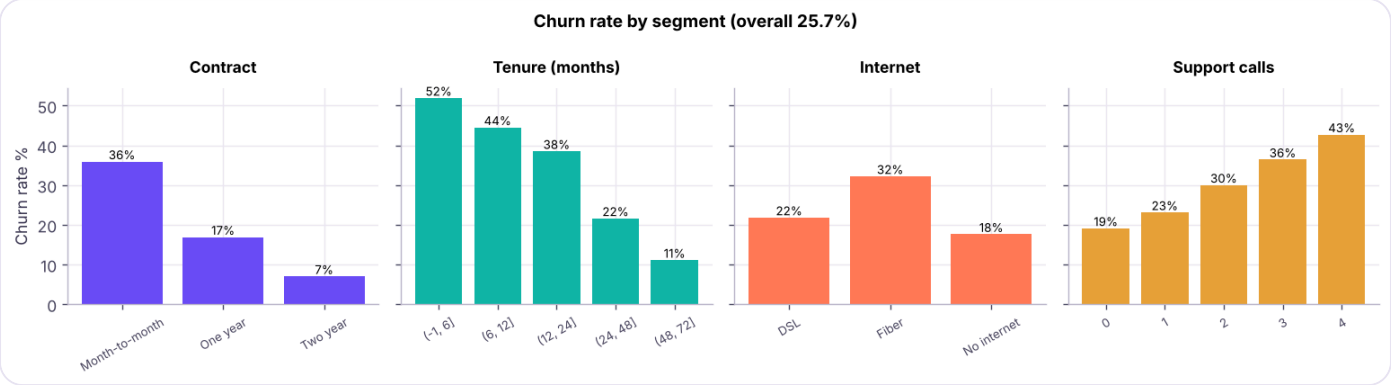
OUTPUT

```
                customers  churners  churn_%
contract
Month-to-month       1690       607     35.9
One year              739       125     16.9
Two year              571        40      7.0

                  customers  churners  churn_%
internet_service
DSL                    1047       228     21.8
Fiber                  1368       441     32.2
No internet             585       103     17.6

               customers  churners  churn_%
support_calls
0                    822       157     19.1
1                   1101       254     23.1
2                    649       195     30.0
3                    294       107     36.4
4                    110        47     42.7

               customers  churners  churn_%
tenure_months
(-1, 6]              318       165     51.9
(6, 12]              238       106     44.5
(12, 24]             452       174     38.5
(24, 48]            1011       218     21.6
(48, 72]             981       109     11.1
```



Churn rate by contract type, tenure band, internet service and support calls. Month-to-month customers, new customers, fibre users and frequent callers churn most.

Step 4: verify before you believe

Python

```python
# 1. Reconcile: group counts add up; 2. a second method gives the same answer; 3. check small groups
g = churn_by("contract", min_n=0)
print("Group customers sum to total :", g.customers.sum() == len(tc))
print("Group churners sum to total  :", g.churners.sum() == tc.churned.sum())
import duckdb
sql = duckdb.sql("SELECT contract, COUNT(*) n, ROUND(100.0*AVG(CASE WHEN churn='Yes' THEN 1 ELSE 0 END),1) churn_pct FROM tc GROUP BY contract ORDER BY contract").df()
print("SQL cross-check:\n", sql.to_string(index=False))
print("\nCombinations with fewer than 50 customers are excluded from rankings; check:", (tc.groupby(['contract','internet_service']).size() < 50).sum(), "such groups")
```

OUTPUT

```
Group customers sum to total : True
Group churners sum to total  : True
SQL cross-check:
       contract    n  churn_pct
Month-to-month 1690       35.9
      One year  739       16.9
      Two year  571        7.0

Combinations with fewer than 50 customers are excluded from rankings; check: 0 such groups
```

Associations are **not causes**: month-to-month customers may also be newer or on fibre. To look at several factors together, combine them:

Python

```
tc["tenure_band"] = pd.cut(tc.tenure_months, [-1, 12, 72], labels=["≤12 months", ">12 months"])
combo = tc.groupby(["contract", "tenure_band"], observed=True).agg(customers=("churned", "size"), churn_pct=("churned", lambda s: round(s.mean() * 100, 1)))
combo[combo.customers ⩾ 50]
```

OUTPUT

		customers	churn_pct
contract	**tenure_band**		
Month-to-month	**≤12 months**	306	67.0
	>12 months	1384	29.0
One year	**≤12 months**	151	33.8
	>12 months	588	12.6
Two year	**≤12 months**	99	15.2
	>12 months	472	5.3

6 rows × 2 columns

Steps 5–7: prioritise and recommend

With a budget of 300 offers, focus on the **highest-risk group that is also large enough to matter**. A simple **risk × size** table:

Python

```
seg = tc.groupby(["contract", "internet_service"], observed=True).agg(customers=("churned", "size"), churn_pct=("churned", "mean"), avg_charges=("monthly_charges", "mean"))
seg["expected_churners"] = (seg.customers * seg.churn_pct).round(0)
seg["churn_pct"] = (seg.churn_pct * 100).round(1); seg["avg_charges"] = seg.avg_charges.round(0)
seg = seg[seg.customers ⩾ 50].sort_values("expected_churners", ascending=False)
seg.head(6)
```

OUTPUT

		customers	churn_pct	avg_charges	expected_churn
contract	**internet_service**				
Month-to-month	**Fiber**	743	43.5	87	323
	DSL	617	30.8	61	190
	No internet	330	28.5	31	94
One year	**Fiber**	341	26.7	86	91
	DSL	252	11.1	61	28
Two year	**Fiber**	284	9.5	87	27

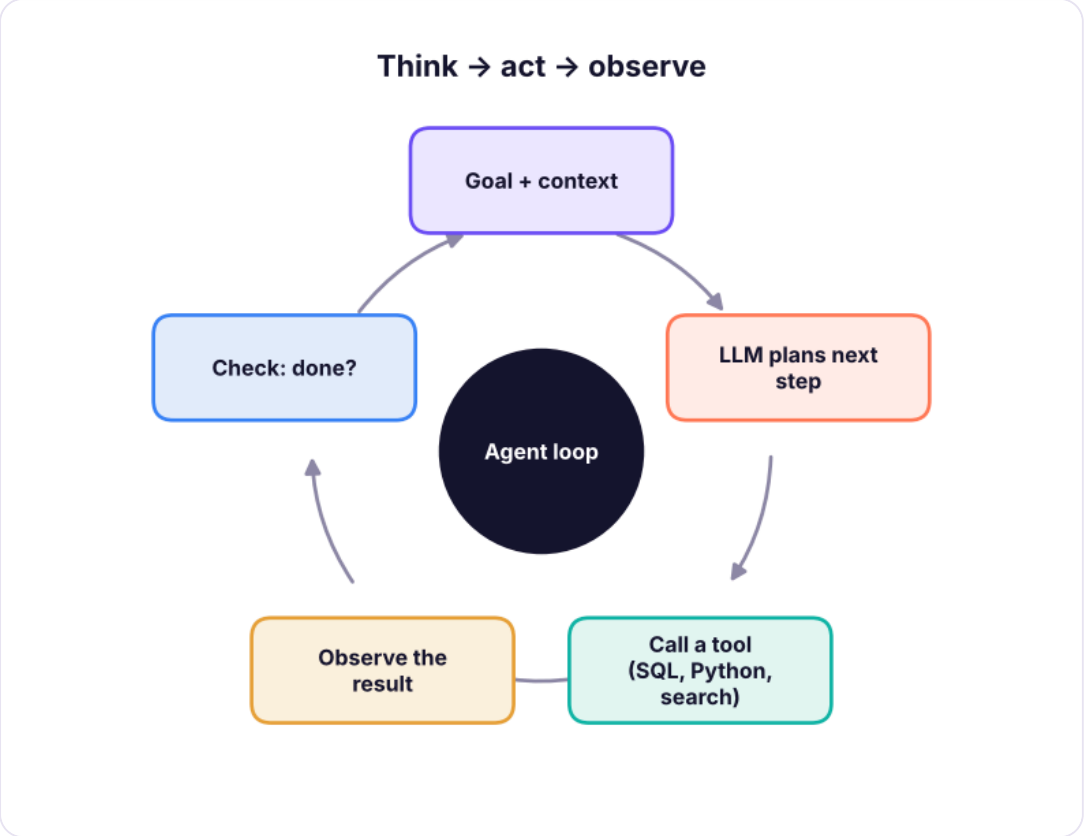
6 rows × 4 columns

KEY TAKEAWAYS **A recommendation a manager can act on**

1. **Target** the top segments by *expected churners*, not only by churn rate: they hold most of the preventable loss.
2. **Offer** a discounted switch to a 1-year contract for month-to-month customers in their first year, and a service check for fibre customers with 2+ support calls.
3. **Measure** with an A/B test: random half of the target list gets the offer; compare churn after 90 days (the test design is in the ML track, hypothesis-testing lessons).
4. **Limit**: this analysis is descriptive; it does not prove that contracts *cause* churn or estimate the offer's effect.

How an AI agent does analysis

An agent turns the workflow above into a loop that an LLM drives:



The agent loop. The model decides the next action, a tool executes it, and the result goes back into the model's context.

Piece	Meaning	Analytics example
Goal	What to achieve	"Find the top churn drivers and draft a memo"
Tools	Functions the model can call	`run_sql(query)`, `run_python(code)`, `plot(spec)`, `search_docs(text)`
Memory / context	What it has seen so far	Previous queries and results
Planner (LLM)	Chooses the next action	"Check churn by contract next"
Guardrails	Limits	Read-only DB, max queries, no network, human approval for writes
Evaluation	How you know it worked	Unit tests on metrics, golden questions, human review

A toy agent you can run (no LLM)

To make the loop concrete, here is a miniature "analyst agent" in which a hard-coded planner stands in for the LLM. It has **one tool** (read-only SQL) and a rule: stop when it has found the top three churn drivers.

Python

```python
import duckdb
db = duckdb.connect(); db.execute("CREATE TABLE tc AS SELECT * FROM
read_csv_auto('telecom_churn.csv')")

def run_sql(q):                                  # TOOL: read-only SQL with a row limit
    assert q.strip().lower().startswith("select"), "only SELECT statements are allowed"
    return db.execute(q).df().head(20)

def planner(memory):                             # stands in for the LLM: decides the next
tool call
    todo = [c for c in ["contract", "internet_service", "payment_method", "support_calls"]
if c not in memory]
    if todo and len(memory) < 4:
        c = todo[0]
        return f"SELECT {c} AS grp, COUNT(*) AS n, ROUND(100.0*AVG(CASE WHEN churn='Yes'
THEN 1 ELSE 0 END),1) AS churn_pct FROM tc GROUP BY {c} HAVING COUNT(*) ⩾ 50", c
    return None, None

memory, log = {}, []
while True:
    q, key = planner(memory)
    if q is None: break
    res = run_sql(q); memory[key] = res; log.append(f"ran query on '{key}' → best group:
{res.sort_values('churn_pct').iloc[-1].grp} ({res.churn_pct.max()}%)")
print("\n".join(log))
spread = {k: v.churn_pct.max() - v.churn_pct.min() for k, v in memory.items()}
print("\nDrivers ranked by spread in churn rate (percentage points):")
for k, v in sorted(spread.items(), key=lambda x: -x[1]): print(f"  {k:18s} {v:5.1f}")
```

OUTPUT

```
ran query on 'contract' → best group: Month-to-month (35.9%)
ran query on 'internet_service' → best group: Fiber (32.2%)
ran query on 'payment_method' → best group: Net banking (28.5%)
ran query on 'support_calls' → best group: 4.0 (42.7%)

Drivers ranked by spread in churn rate (percentage points):
  contract            28.9
  support_calls       23.6
  internet_service    14.6
  payment_method       5.0
```

A real agent replaces `planner` with an LLM call that reads the memory and writes the next SQL, adds many more tools, and decides when to stop. Everything else (read-only guard, row limit, log) is the same safety scaffolding you should demand from any agent.

Tool calling and MCP

- **Tool (function) calling**: the model returns a structured request such as `{"tool": "run_sql", "arguments": {"query": "SELECT …"}}`; your code runs it and sends the result back.
- **Model Context Protocol (MCP)** is an open standard for connecting AI applications to **tools and data sources** through a common interface, so one connector (a "server") for your database, files or SaaS app works with many AI clients. For analysts it means an assistant can query a **governed** database or semantic layer through an approved server instead of you pasting data around.
- **RAG (retrieval-augmented generation)** gives the model *documents* (metric definitions, past reports) to ground its answers (GenAI track).
- **Multi-agent systems** split work among specialised agents (a SQL agent, a chart agent, a reviewer); more moving parts mean more to test.

Evaluating AI analytics

What to evaluate	How
Correctness of numbers	A set of "golden questions" with known answers (total sales by year, churn by contract) run on every change
SQL quality	Execution accuracy (does the result match the reference result?), not just text similarity
Faithfulness of narrative	Every number in the text must appear in the tables; automatic check by extracting numbers
Safety	Attempted writes blocked, personal data not exposed, prompt-injection tests
Cost and latency	Tokens, tool calls and time per task
Usefulness	Do users make better decisions? Track adoption and feedback

Governance, ethics and your career

- **Policy**: know your organisation's approved tools, data-classification rules and retention.
- **Human in the loop** for decisions that affect people (credit, hiring, pricing, health): AI assists; accountable humans decide.
- **Fairness**: test whether results differ unfairly across groups (gender, age, region) before acting.
- **Explainability**: record the data, prompt, code and checks behind each insight; be able to re-run it.
- **Skills that stay valuable**: asking the right question, domain knowledge, data modelling, SQL/Python fluency to *read and verify* AI output, statistics and experimentation, storytelling and stakeholder management. The analysts who thrive are those who use AI for speed **and** can prove their answers.

2026 UPDATE

What is changing in 2026

Analytics agents are moving into the BI tools themselves (Copilot in Power BI/Fabric, Gemini in Looker and BigQuery, Snowflake Cortex, Databricks Genie, Metabase AI features). Vendors compete on accuracy with governed semantic layers, so expect "ask a question of certified metrics" to become a standard feature. Learn the concepts here and re-check each product's current capabilities, pricing and data-handling terms before adopting it.

Summary

KEY TAKEAWAYS

- Follow a **seven-step workflow**: frame, brief, analyse, verify, visualise, write, recommend; AI accelerates but you own framing, verification and judgement.
- Always **verify**: reconcile totals, cross-check with a second method, exclude tiny groups, and remember associations are not causes.
- An **agent** = goal + LLM planner + tools + memory + guardrails in a loop; tool calling and **MCP** connect models to tools and governed data.
- Evaluate AI analytics with golden questions, execution accuracy, faithfulness checks, safety tests and cost tracking.
- Keep humans accountable, log the work, check fairness and follow data policy.

Quick quiz

6 questions · tap an answer to check it

1. In the agent loop, which part executes a SQL query?

A The planner

B A tool

C The memory

D The prompt

Answer: B. The LLM plans; tools perform actions such as running SQL.

2. What is MCP?

A A chart type

B An open protocol for connecting AI applications to tools and data sources

C A database engine

D A prompt format

Answer: B. The Model Context Protocol standardises how AI clients use external tools and data.

3. Why report churn only for groups with at least 50 customers?

A To shorten the report

B Small groups give unstable percentages that can mislead

C Because SQL requires it

D To hide data

Answer: B. Small samples produce noisy rates.

4. Which statement about our churn findings is correct?

A Contracts cause churn

B Month-to-month customers churn more, but this is an association that needs testing before claiming cause

C Fibre causes churn

D Support calls prove dissatisfaction

Answer: B. Descriptive analysis shows patterns, not causes; use an experiment to test an intervention.

5. What is a "golden question" set?

A Questions that cost the most

B Questions with known correct answers used to test an AI analytics system after each change

C A list of dashboards

D Training data

Answer: B. They enable regression testing of answers.

6. Which control best limits damage if an agent writes a bad query?

A A longer prompt

B A read-only database role with row limits and logging

C A bigger model

D A louder warning

Answer: B. Technical guardrails beat instructions alone.

EXERCISE

Run the workflow on a new question

Using `superstore_sales.csv`, follow the seven steps for: *"Which customer segment × region combinations should sales focus on next quarter?"* Write the brief, run the analysis (use groupby or SQL), verify with two methods, produce one chart and write a five-sentence recommendation that includes one limitation.

Show solution (worked answer)

Brief: define "focus" (for example profit, not just sales; minimum 100 order lines). Analysis: group by segment and region, compute sales, profit, margin and lines; filter groups with ≥ 100 lines. Verify: group totals equal the overall totals; repeat with SQL; check one group manually. Chart: bubble or bar of profit by segment × region. Recommendation example: "Focus on [top profit segment-region] (profit ₹X, margin Y%); protect [healthy group]; reduce discounts in [low-margin group]. Limitation: three years of data, no cost-to-serve or competitor information; profits include discounts that may not repeat."

INTERVIEW CORNER

Questions you may be asked

Q: Walk me through how you use AI in your analysis workflow. I frame the decision, give the AI a clear brief with schema and definitions, use it to draft and explain SQL or Python, run everything myself, reconcile totals and cross-check with a second method, use AI for chart and narrative drafts, then I validate each number and take responsibility for the recommendation.

Q: What is an AI agent and how is it different from a chatbot? A chatbot responds once to a prompt; an agent pursues a goal across multiple steps by calling tools, observing results and deciding what to do next, so it needs stronger guardrails and evaluation.

Q: How do you stop an AI analytics tool from giving wrong numbers? Governed semantic layer, read-only access to curated data, golden-question tests, execution-based evaluation, human review of important outputs, and logging.

Q: A model says "contract type drives churn". What do you say? The data shows an association; to claim causation we would need an experiment or a causal method, controlling for tenure and other factors.